



Consumer preference and perception of specific categories of tobacco and related products

Request for Service Chafea/2017/Health/34 under
Framework Contract Chafea/2015/CP/01

Specific contract number 2017 85 07

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DG SANTE

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TABLE OF CONTENTS

EXECUTIVE SUMMARY	7
1 INTRODUCTION	14
1.1 BACKGROUND	14
1.2 OBJECTIVES AND SCOPE	17
1.3 OVERALL APPROACH	19
1.3.1 <i>Systematic review and gap analysis</i>	21
1.3.2 <i>Focus groups</i>	24
1.3.3 <i>Online survey</i>	25
1.4 RATIONALE OF THE REPORT	30
2 WATERPIPE TOBACCO	31
2.1 BACKGROUND	31
2.2 MARKET DEVELOPMENTS	35
2.2.1 <i>Socio-demographic characteristics</i>	35
2.2.2 <i>Pattern of use</i>	39
2.2.3 <i>Dual use with other tobacco and related products</i>	48
2.3 CONSUMER PERCEPTION	52
2.4 HEALTH, SOCIAL AND ECONOMIC IMPACTS	61
2.5 CONCLUDING REMARKS ON WATERPIPE TOBACCO	63
3 SLIM CIGARETTES	65
3.1 BACKGROUND	65
3.2 MARKET DEVELOPMENTS	70
3.2.1 <i>Socio-demographic characteristics</i>	70
3.2.2 <i>Pattern of use</i>	74
3.2.3 <i>Parallel use with other tobacco and related products</i>	83
3.3 CONSUMER PERCEPTION	87
3.4 HEALTH, SOCIAL AND ECONOMIC IMPACTS	93
3.5 CONCLUDING REMARKS ON SLIM CIGARETTES	95
4 SMALL CIGARILLOS	97
4.1 BACKGROUND	97
4.2 MARKET DEVELOPMENTS	100
4.2.1 <i>Socio-demographic characteristics</i>	100
4.2.2 <i>Pattern of use</i>	104
4.2.3 <i>Parallel use with other tobacco and related products</i>	115
4.3 CONSUMER PERCEPTION	121
4.4 HEALTH, SOCIAL AND ECONOMIC IMPACTS	128
4.5 CONCLUDING REMARKS ON SMALL CIGARILLOS	129
5 NOVEL TOBACCO PRODUCTS	131
5.1 BACKGROUND	131
5.2 MARKET DEVELOPMENTS	135
5.2.1 <i>Socio-demographic characteristics</i>	135
5.2.2 <i>Pattern of use</i>	139
5.2.3 <i>Parallel use with other tobacco and related products</i>	149
5.3 CONSUMER PERCEPTION	153
5.4 HEALTH, SOCIAL AND ECONOMIC IMPACTS	169

5.5	CONCLUDING REMARKS ON NOVEL TOBACCO PRODUCTS.....	172
6	ELECTRONIC CIGARETTES	174
6.1	BACKGROUND.....	174
6.2	MARKET DEVELOPMENTS.....	184
6.2.1	<i>Socio-demographic characteristics.....</i>	<i>184</i>
6.2.2	<i>Pattern of use.....</i>	<i>188</i>
6.2.3	<i>Parallel use with other tobacco and related products.....</i>	<i>200</i>
6.3	CONSUMER PERCEPTION	207
6.4	HEALTH, SOCIAL AND ECONOMIC IMPACTS	219
6.5	CONCLUDING REMARKS ON ELECTRONIC CIGARETTES.....	221
7	COMPARATIVE ANALYSIS OF TOBACCO AND RELATED PRODUCTS.....	224
7.1	MARKET DEVELOPMENTS.....	224
7.2	CONSUMER PERCEPTION	230
7.3	HEALTH, SOCIAL AND ECONOMIC IMPACTS	232
8	CONCLUSION.....	238
9	REFERENCES	243
10	ANNEXES.....	251
10.1	BACKGROUND.....	251
10.2	WATERPIPE TOBACCO PRODUCTS	257
10.3	SLIM CIGARETTES	268
10.4	NOVEL TOBACCO PRODUCTS	275
10.5	ELECTRONIC CIGARETTES.....	285
10.6	COMPARATIVE ANALYSIS.....	289
11	LIST OF FIGURES.....	294
12	LIST OF TABLES.....	301

EXECUTIVE SUMMARY

The European Union (EU) and Member States (MS) have implemented a number of measures to reduce the consumption of tobacco and related products. These initiatives were designed to support the functioning of the internal market and to ensure a high level of health protection. The revised Tobacco Products Directive 2014/40/EC (TPD) acts as one of the main pillars of tobacco control in the EU. In terms of geographic scope, the study focused on the 27 EU Member States and the United Kingdom (UK). Despite the UK leaving the EU and the transition period ending on 31 December 2020, the UK will be considered and referred to as a Member State for the purposes of this report. Moreover, conditionally to the results of the ongoing negotiations, the TPD may continue to apply in Northern Ireland also after 2020.

The current study is a contribution to the Commission's report to the European Parliament and the Council by 20 May 2021 on the application of the Directive, as per Article 28(1) of the TPD. The study covers five categories of tobacco and related products, namely waterpipe tobacco, slim cigarettes, small cigarillos, novel tobacco products and electronic cigarettes. The market developments, consumer perceptions, and the health, social and economic impacts for each product were investigated. The study comprised a systematic literature review and analysis of existing data with the objective of identifying informational gaps. To investigate these gaps, focus group discussions were held in five Member States (France, Germany, Bulgaria, Spain and Greece) to collect qualitative insights into current trends. An online survey was designed to assemble quantitative evidence about consumer perceptions and reported behaviours. To ensure geographical coverage the following Member States were surveyed, Bulgaria, France, Germany, Greece, Italy, Latvia, Poland, Slovakia, Spain and the UK. In each country two samples were recruited. The first sample was 600 smokers and people who had quit in the last 12 months, aged 26 and above. Prompted by concerns about the use of tobacco and related products among young people, the second sample was comprised of 600 people between the age of 18-25. A total of 12,000 respondents were surveyed. The following paragraphs present generic and product specific findings from the research.

Risk perception. Conventional cigarettes were perceived as the most harmful, about 20% less harmful were small cigarillos and slim cigarettes, and at a further 15% less harmful were e-cigarettes, HTPs and waterpipe tobacco. Current users of e-cigarettes rated them as about 50% less harmful than conventional cigarettes. Users of a particular product consistently rated the health risk to be lower than the rest of the sample but still rated the product as harmful to health. The older age group (26 years +) reported lower risk perception of tobacco and related products than the younger age group.

Quitting the use of tobacco and related products. A small percentage of current users of the tobacco and related products featured in this study reported quitting all tobacco products. Many more had tried to quit particular products, sometimes on a number of occasions, motivated by concerns about their health, health warning messages and worry about the health of others. Price was also a factor. Those who were unsuccessful attributed the failure to stress, the difficulty of stopping, their social environment, or concluded that what they were trying to quit was not that harmful or that it was less harmful than other products. Clearly, addiction makes quitting a painful challenge.

Dual use. The majority of users of particular tobacco products were **dual users** of tobacco products with conventional cigarette smoking being most frequently cited. For example, while only 8% of current e-cigarette users in the 18-25 age group used no other product, 74% were dual users of whom 82% reported smoking conventional cigarettes. In the 26+ age group 78% of the current e-cigarette users reported dual use; 90% cited smoking conventional cigarettes.

Another common thread across the range of products was **under age initiation in the use** of tobacco products.

Regulation. The majority of the survey respondents thought that the current tax regime and regulations on conventional tobacco products should be applied to e-cigarettes and HTPs. They also agreed that the current prohibitions on smoking cigarettes at work, in public transport and in bars and restaurants should also apply to e-cigarettes and HTPs.

Waterpipes. In the 18-25 age group 27% had tried waterpipe tobacco, 15% were past users and 11% current users. Spain, Bulgaria and Greece had the highest percentages of current users. Use of waterpipe tobacco varied from daily to monthly with sessions between 5 and 30 minutes. Industry marketing targeted younger people with flavoured tobacco blends and in parallel social media provided a new channel for indirect advertisements via user-generated content. Waterpipe use appealed to both men and women who cited the attractions of the social ritual, affordability, low health risk perception, and feelings of zen, pleasure, and relaxation. A significant majority in both age groups were dual users; cigarettes, e-cigarettes and slim cigarettes were popular choices. The use of waterpipes in premises such as shisha bars, where products are unpacked by staff, may limit users' exposure to health warnings. Compared to other tobacco and related products, waterpipe tobacco is subject to variable and often lower tax rates across the EU, potentially distorting the internal market.

Slim cigarettes. Since 2000, the market share for slim cigarettes has grown significantly in a declining EU cigarette market. Slim cigarettes with the thinner packaging were marketed as a high-status product with an image of feminine elegance, exclusivity and modernity. The survey found that about 40% of current users were male. Compared to conventional cigarettes, they were perceived as 'lighter', giving off less odour, more discrete and socially acceptable. While respondents acknowledged the risks to health, slim cigarettes were seen as less harmful to users and bystanders than conventional cigarettes. 70% of slim cigarette users also reported using other tobacco and related products.

Small cigarillos. The tobacco industry has promoted small cigarillos as a luxury, up- market product, a small cigar as enjoyed by successful people in business and connoisseurs. In both age groups the current users reported themselves as high socio-economic status. Small cigarillos contribute to special occasions and social gatherings. In both age groups enjoyment and socialising led people to smoke cigarillos. Over 90% of current smokers reported dual use and of these 90% reporting the smoking of conventional cigarettes. Among the current smokers there were twice as many men than women. In total 6% of the 18-25 age group were current users with 7% in the 26+ age group. Above average consumption was found in the younger generation in the UK, Spain and France, and in the older generation in Italy, Spain and Poland. Low prices in some Member States and the introduction of flavours making inhalation easier and more pleasant (some TPD restrictions do not apply to small cigarillos) were mentioned as attractive features.

Novel tobacco products. The current study focused on heated tobacco products (HTPs) of which the market-leading brand IQOS was introduced in Europe in 2017. The EU Member States have adopted different position on HTPs with respect to taxes, resulting in different interpretations of how to implement the TPD, and large variations in prices across European countries. Studies funded by the tobacco industry have claimed to show significant reductions in the exposure to harmful constituents compared to conventional cigarettes, a claim that independent studies have called into question. In the focus groups there was some confusion between HTPs and e-cigarettes, about what was inhaled and whether they would be safer than other tobacco products. Some thought that they might be less satisfying than conventional cigarettes and offer only a 'cold experience'. In the 18-25 age group 8% were current HTP

smokers, 7% past users and 16% had tried HTPs once or twice. The UK, Italy and Bulgaria had above average numbers of current users. Marketing and promotion of HTPs has included alleviating health concerns through claims that HTPs are reduced-risk products; designing HTP systems to boost the user's social image; using a "bait and hook" pricing strategy; employing social marketing techniques and brand ambassadors; selling and marketing via multiple channels, including dedicated stores, retail establishments and e-commerce websites. Younger respondents were twice as likely to have heard about the product through social media. In the older generation there were 11% current users, 6% past users and 10% had tried them. In this 26+ age group Italy and Spain had above average numbers of current users. In both age groups 'enjoyment' was mentioned by about three in ten as the reason for taking up HTPs. For the 26+ age group 41% cited 'trying to quit or reduce consumption of another tobacco product', a reason mentioned by 29% of the younger age group. In a word association task, the most frequent associations with HTPs for the 18-25 age group were 'unhealthy', 'modern and technological', and 'costly'. The 26+ age group selected a similar list of associations – 'unhealthy', 'modern and technological', and 'costly'. More than 80% of current HTP users in both age groups were dual users with 8 out of 10 reporting conventional cigarette use. The majority of respondents agreed that HTPs should be regulated as conventional cigarettes and many supported restrictions in the use of HTPs in workplaces, public transport and bars and restaurants.

Electronic cigarettes. E-cigarettes have gained increased popularity in recent years. In parallel policy initiatives promoting e-cigarettes as a risk reduction mechanism for current cigarette smokers versus the concern that e-cigarette use could open a gateway for young people to take up smoking conventional cigarettes or develop addictive behaviours are hotly debated. In the two age groups about half of the respondents had tried e-cigarettes. 15% of the 18-25 age group were current users as were 33% in the 26+ age group. Of current users in the 18-25 age group 74% were dual users (82% citing cigarettes). In the 26+ age group 78% were dual users (90% citing cigarettes). Between the two age groups there was a striking difference in the reasons for using e-cigarettes. Marginally more current users in the younger age group cited 'enjoyment', but almost double the number in the older age group compared to the younger age group cited 'to quit or reduce consumption of other tobacco products'. Of the past users of e-cigarettes 24% in the 18-25 age group and 23% in the 26+ age group were now not using any tobacco or related products. However, in the younger age group 23% had progressed from e-cigarettes to other tobacco products, mainly conventional cigarettes, as had 31% of the older age group, suggestive of the gateway scenario. Should both scenarios be shown to have some validity it will present a dilemma for public policy.

The findings of this study highlight the need for increased and centralised efforts to monitor and assess marketing strategies employed by the tobacco industry (e.g. via internet and social media) to adequately protect the public, and young people in particular, from exposure to online marketing of novel tobacco products. The online environment should be monitored for sales control given that many e-cigarettes and e-liquids are advertised and purchased online. Various measures could be implemented to reduce the rate of uptake of e-cigarettes by younger people e.g. plain packaging has been suggested as an effective measure to reduce the marketing advantages of certain products. To educate the public, exchange of best practices on MS-level or EU-wide campaigns should be encouraged. Finally, research is needed to produce robust evidence on the health impacts of NTPs as well as the potential impact of different policy interventions.

RÉSUMÉ ANALYTIQUE

L'Union Européenne (UE) et les Etats membres ont mis en place un nombre de mesures pour réduire la consommation des produits du tabac et des produits connexes. Ces initiatives ont été conçues pour soutenir le fonctionnement du marché intérieur et pour garantir un niveau de protection élevé en matière de santé. La Directive révisée sur les produits du tabac 2014/40/UE (« la Directive ») constitue l'un des piliers principaux du contrôle du tabac dans l'UE. En termes de périmètre géographique, l'étude portait sur les 27 Etats membres de l'UE et le Royaume-Uni. Malgré le retrait du Royaume-Uni de l'UE et la période de transition se terminant le 31 décembre 2020, le Royaume-Uni sera considéré et mentionné comme Etat membre dans le cadre du présent rapport. De plus, en fonction des résultats des négociations en cours, la Directive pourrait continuer de s'appliquer en Irlande du Nord aussi après 2020.

La présente étude contribue au rapport de la Commission au Parlement Européen et au Conseil d'ici le 20 mai 2021 sur l'application de la Directive, conformément à l'article 28(1) de la Directive. L'étude porte sur cinq catégories de produits du tabac et produits connexes, à savoir le tabac à pipe à eau, les cigarettes fines, les petits cigarillos, les nouveaux produits du tabac et les cigarettes électroniques. Les évolutions du marché, les perceptions des consommateurs, et les impacts sociaux, économiques et en matière de santé de chaque produit ont été étudiés. L'étude a compris un examen documentaire et une analyse des données existantes dans l'objectif d'identifier les lacunes en matière d'information. Pour enquêter sur ces lacunes, des groupes de discussions ont été conduits dans cinq Etats membres (France, Allemagne, Bulgarie, Espagne, Grèce) afin de recueillir des informations qualitatives sur les tendances actuelles. Une enquête en ligne a été réalisée pour rassembler des données quantitatives sur les perceptions et les comportements déclarés des consommateurs. Pour garantir la représentation géographique l'enquête a été réalisée en Bulgarie, France, Allemagne, Grèce, Italie, Lettonie, Pologne, Slovaquie, Espagne et au Royaume-Uni. Dans chaque pays, deux échantillons ont été recrutés. Le premier échantillon contenait 600 fumeurs et personnes qui ont arrêté de fumer dans les 12 derniers mois, âgés de 26 ans et plus. Poussé par les préoccupations concernant la consommation de tabac et de produits connexes chez les jeunes, le second échantillon était composé de 600 personnes âgées de 18 à 25 ans. Au total, 12 000 personnes ont été interrogées. Les paragraphes suivants présentent les résultats généraux et spécifiques à chaque produit de notre recherche.

Perception du risque. Les cigarettes classiques sont perçues comme les plus nocives, les petits cigarillos et les cigarettes fines environ 20 % moins nocifs, et les cigarettes électroniques, tabac chauffé et tabac à pipe à eau encore 15% moins nocifs. Les utilisateurs actuels de cigarettes électroniques les perçoivent environ 50% moins nocives que les cigarettes classiques. Les utilisateurs d'un produit particulier ont à chaque fois estimé que le risque pour la santé était inférieur au reste de l'échantillon, mais ils ont tout de même estimé que le produit était nocif pour la santé. Le groupe plus âgé (26 ans et plus) a montré une perception du risque lié aux produits du tabac et produits connexes plus faible que le groupe plus jeune.

Arrêter l'usage des produits du tabac et produits connexes. Un faible pourcentage des utilisateurs actuels des produits du tabac et produits connexes présentés dans cette étude ont déclaré arrêter de fumer tous les produits du tabac. Bien plus ont essayé d'arrêter des produits spécifiques, parfois à plusieurs reprises, motivés par des préoccupations concernant leur santé, les avertissements de santé et des inquiétudes concernant la santé d'autrui. Le coût était aussi un facteur. Ceux qui n'ont pas réussi ont attribué l'échec au stress, à la difficulté d'arrêter, à leur environnement social, ou ont conclu que ce qu'ils essayaient d'arrêter n'était pas si nocif ou était moins nocif que d'autres produits. Il est clair que la dépendance rend l'arrêt du tabac d'autant plus difficile.

Double usage. La majorité des utilisateurs de produits particuliers du tabac sont des **doubles utilisateurs** de produits du tabac, la consommation de cigarettes classiques étant la plus souvent citée. Par exemple, alors que seulement 8% des fumeurs actuels de cigarettes électroniques dans le groupe de 18-25 ans n'utilisait aucun autre produit, 74% étaient des doubles utilisateurs dont 82% ont déclaré fumer des cigarettes classiques. Dans le groupe de 26 ans et plus, 78% des fumeurs actuels de cigarettes électroniques ont déclaré un double usage ; 90% déclarant fumer des cigarettes classiques. Un autre point commun à toute la gamme de produits était **l'initiation avant l'âge à la consommation de produits du tabac**.

Réglementation. La majorité des participants à l'enquête trouvent que les réglementations et le régime fiscal actuels sur les produits traditionnels du tabac devraient être appliqués aux cigarettes électroniques et aux produits du tabac chauffés. Ils s'accordent aussi à dire que les interdictions actuelles de fumer des cigarettes au travail, dans les transports publics et dans les bars et restaurants devraient aussi s'appliquer aux cigarettes électroniques et aux produits du tabac chauffés.

Tabac à pipe à eau (« waterpipe »). Dans le groupe de 18-25 ans, 27% avaient essayé le tabac à pipe à eau, 15% étaient d'anciens utilisateurs et 11% des utilisateurs actuels. L'Espagne, la Bulgarie et la Grèce ont montré les plus hauts pourcentages d'utilisateurs actuels. L'usage du tabac à pipe à eau variait de quotidien à mensuel avec des sessions entre 5 et 30 minutes. Le marketing de l'industrie a ciblé les jeunes avec des mélanges de tabacs aromatisés et en parallèle les réseaux sociaux ont fourni un nouveau vecteur de publicités indirectes via le contenu généré par les utilisateurs. Le tabac à pipe à eau attire autant les hommes que les femmes qui ont cité l'attrait du rituel social, l'accessibilité financière, la perception d'un risque faible pour la santé, et la sensation de zen, plaisir, et relaxation. Une grande majorité des deux groupes d'âges sont des doubles utilisateurs ; les cigarettes, cigarettes électroniques et cigarettes fines sont les principaux choix. L'usage du tabac à pipe à eau dans des lieux tels que les bars à chicha, où les produits sont déballés par le personnel, peut limiter l'exposition des utilisateurs aux avertissements de santé. Comparé aux autres produits du tabac et produits connexes, le tabac à pipe à eau est soumis à des taux d'imposition variables et souvent inférieurs à travers l'UE, susceptibles de distordre le marché intérieur.

Cigarettes fines (« slim »). Depuis 2000, la part de marché des cigarettes fines a considérablement augmenté dans un marché européen des cigarettes en déclin. Les cigarettes fines avec leur emballage plus fin ont été commercialisées comme un produit de prestige avec une image d'élégance féminine, d'exclusivité et de modernité. L'enquête a montré que environ 40% des utilisateurs actuels étaient des hommes. Comparées aux cigarettes traditionnelles, elles sont perçues comme « plus légères », produisant une odeur moins forte, plus discrètes et acceptables socialement. Bien que les participants reconnaissent les risques pour la santé, les cigarettes fines sont considérées comme moins nocives pour les fumeurs et les personnes qui se trouvent autour que les cigarettes classiques. 70% des utilisateurs de cigarettes fines ont déclaré aussi utiliser d'autres produits du tabac et produits connexes.

Petits cigarillos. L'industrie du tabac a mis en avant les petits cigarillos comme un produit de luxe, haut de gamme, un petit cigare apprécié par les hommes et femmes d'affaires prospères et les connaisseurs. Dans les deux groupes d'âge les utilisateurs actuels se sont déclarés comme venant d'un milieu socio-économique élevé. Les petits cigarillos contribuent à des occasions spéciales et à des rassemblements sociaux. Dans les deux groupes d'âge, le plaisir et la socialisation ont amené les gens à fumer des cigarillos. Plus de 90% des fumeurs actuels ont déclaré un double usage et parmi eux 90% ont déclaré fumer des cigarettes classiques. Il y avait deux fois plus d'hommes que de femmes parmi les fumeurs actuels. Au total 6% du groupe de 18-25 ans étaient des utilisateurs actuels et 7% dans le groupe de 26 ans et plus. Une

consommation au-dessus de la moyenne a été trouvée dans la population jeune du Royaume-Uni, de l'Espagne, de la France, et dans la population plus âgée en Italie, Espagne et Pologne. Les prix bas dans certains Etats membres et l'introduction d'arômes rendant l'inhalation plus facile et plus agréable (certaines restrictions de la Directive ne s'appliquent pas aux petits cigarillos) ont été mentionnés comme des caractéristiques attrayantes.

Nouveaux produits du tabac. La présente étude porte sur les produits du tabac chauffés (« heated tobacco products » ou HTP), dont la marque IQOS, leader du marché, a été introduite en Europe en 2017. Les Etats membres de l'UE ont adopté des positions différentes sur les HTP en termes de taxes, menant à différentes interprétations sur comment appliquer la Directive, et à de grandes variations de prix entre pays européens. Des études financées par l'industrie du tabac ont prétendu montrer des réductions significatives de l'exposition aux composants nocifs par rapport aux cigarettes classiques, une affirmation que des études indépendantes ont remise en question. Dans les groupes de discussion, il y a eu une certaine confusion entre les HTP et les cigarettes électroniques, sur ce qui était inhalé et si les HTPs seraient plus sûrs que les autres produits du tabac. Certains ont pensé qu'ils pourraient être moins satisfaisants que les cigarettes classiques et offrir une moindre expérience. Dans le groupe des 18-25 ans 8% étaient des fumeurs actuels de HTP, 7% d'anciens utilisateurs et 16% avaient essayé des HTP une ou deux fois. Le Royaume-Uni, l'Italie et la Bulgarie avaient des nombres d'utilisateurs actuels supérieurs à la moyenne. La commercialisation et la promotion des HTP ont entre autres visé à atténuer les préoccupations de santé en affirmant que les HTP sont des produits à risque réduit ; à concevoir des systèmes de HTP pour renforcer l'image sociale de l'utilisateur ; à utiliser une stratégie de tarification "appât et hameçon" ; à employer des techniques de marketing social et des ambassadeurs de marque ; à vendre et à commercialiser via de multiples canaux, notamment des magasins spécialisés, des établissements de vente au détail et des sites web de e-commerce. Les plus jeunes participants avaient deux fois plus de chance d'avoir entendu parler du produit par les réseaux sociaux. Dans la génération plus âgée, 11% étaient des utilisateurs actuels, 6% d'anciens utilisateurs et 10% en avaient essayé. Dans ce groupe de 26 ans et plus, l'Italie et l'Espagne avaient des nombres d'utilisateurs actuels supérieurs à la moyenne. Dans les deux groupes d'âge, le « plaisir » a été mentionné par environ trois personnes sur dix comme étant la raison de l'adoption des HTP. Pour le groupe de 26 ans et plus, 41% ont cité « essayer d'arrêter ou de réduire sa consommation d'un autre produit du tabac », une raison mentionnée par 29% du plus jeune groupe d'âge. Lors d'un exercice d'association des mots, les plus fréquentes associations avec les HTP pour les 18-25 ans étaient « mauvais pour la santé », « moderne et technologique », et « coûteux ». Le groupe de 26 ans et plus a choisi une liste similaire d'associations – « mauvais pour la santé », « moderne et technologique », et « coûteux ». Plus de 80% des utilisateurs actuels de HTP dans les deux groupes d'âge étaient des doubles utilisateurs avec 8 personnes sur 10 déclarant utiliser des cigarettes classiques. La majorité des répondants s'est accordée à dire que les HTP devraient être réglementés comme les cigarettes classiques et beaucoup se sont montrés en faveur de restrictions dans l'usage des HTP dans les lieux de travail, les transports publics et les bars et restaurants.

Cigarettes électroniques. Les cigarettes électroniques ont gagné en popularité ces dernières années. En parallèle, les initiatives politiques visant à promouvoir les cigarettes électroniques en tant que mécanisme de réduction des risques pour les fumeurs de cigarettes actuels et la crainte que l'utilisation des cigarettes électroniques n'ouvre aux jeunes la possibilité de commencer à fumer des cigarettes classiques ou de développer des comportements de dépendance font l'objet d'un débat animé. Dans les deux groupes d'âge environ la moitié des répondants avaient essayé la cigarette électronique. 15% des 18-25 ans étaient des utilisateurs actuels ainsi que 33% des 26 ans et plus. Parmi les utilisateurs actuels dans les 18-25 ans 74% étaient des doubles utilisateurs (82% citant la cigarette). Dans les 26 ans et plus 78% étaient

des doubles utilisateurs (90% citant la cigarette). Entre les deux groupes d'âge on note une différence frappante dans les raisons d'utiliser les cigarettes électroniques. Marginalement plus d'utilisateurs actuels dans le plus jeune groupe d'âge ont cité « le plaisir », alors que presque le double de personnes dans le groupe plus âgé comparé au plus jeune groupe ont cité « arrêter ou réduire la consommation d'un autre produit du tabac ». Parmi les anciens utilisateurs de cigarettes électroniques 24% des 18-25 ans et 23% des 26 ans n'étaient plus utilisateurs d'aucun produit du tabac ou de produits connexes actuellement. Cependant, dans le plus jeune groupe 23% ont évolué des cigarettes électroniques vers d'autres produits du tabac, surtout les cigarettes classiques, tout comme 31% du groupe plus âgé, suggérant un scénario de porte d'entrée. Si les deux scénarios devaient s'avérer valables, cela posera un dilemme pour les politiques publiques.

Les conclusions de cette étude soulignent la nécessité d'accroître et de centraliser les efforts de suivi et d'évaluation des stratégies de marketing employées par l'industrie du tabac (par exemple via internet et les réseaux sociaux) pour protéger de manière adéquate le public, et les jeunes en particulier, contre l'exposition à la commercialisation en ligne de nouveaux produits du tabac. L'environnement en ligne devrait être surveillé pour le contrôle des ventes étant donné que de nombreuses cigarettes et liquides électroniques sont vendus et achetés en ligne. Diverses mesures pourraient être mises en œuvre pour réduire le taux d'adoption des cigarettes électroniques par les jeunes, par exemple, l'emballage neutre a été suggéré comme une mesure efficace pour réduire les avantages commerciaux de certains produits. Pour éduquer le public, il convient d'encourager l'échange de bonnes pratiques sur les campagnes menées au niveau des États membres ou de l'UE. Enfin, plus de recherche est nécessaire pour produire des preuves solides sur les effets des nouveaux produits du tabac sur la santé ainsi que sur l'impact potentiel des différentes interventions politiques.

1 Introduction

1.1 Background

Tobacco consumption is the single largest avoidable health risk and is accountable for nearly 700,000 deaths each year in the European Union (EU) alone (EC, 2020). In fact, compared to the rest of the world, the World Health Organisation (WHO) reported that the European region had one of the highest proportions of deaths attributable to tobacco (Glahn et al., 2018). The consumption of tobacco and related products puts both consumers and passive (or second-hand) users at higher risks of developing health problems, such of cancer, cardiovascular and respiratory diseases. Consumption by younger generations has also been particularly concerning, as 93% of consumers reported taking up the product use before they turned 26 years (EC, 2020).

Tobacco control efforts have been advancing worldwide, and the EU and its Member States (MS) have implemented a number of legal, political and financial measures to reduce the consumption of tobacco and related products. The aim of these initiatives was to facilitate the functioning of the internal market while protecting public health. Specifically, in terms of consumer protection, they sought to protect the general public from the harmful effects of tobacco consumption, as well as to assist current users to quit, and to discourage others – notably younger generations – from taking up tobacco use. The EU has implemented a number of tobacco legislations¹, including the **Tobacco Products Directive (2014/40/EU) (TPD)**. This entered into force on 19 May 2014 to replace the 2001 Directive (2001/37/EC) and became applicable in EU Member States (MS) on 20 May 2016.²

The TPD has been the backbone of regulation around the manufacturing, presentation and sale of tobacco products in the EU. It was developed in response to the emergence of new scientific evidence around effective control measures, new tobacco and related products entering the market, as well as the EU's ratification of the WHO's Framework Convention on Tobacco Control (WHO FCTC) in 2005. It lays down rules governing the manufacture, presentation and sale of tobacco and related products. Some of the key provisions of the Directive include, but are not limited to, requirements around packaging and labelling, reporting tobacco product ingredients, prohibiting characterising flavours in cigarettes and roll-your-own tobacco, banning promotional and misleading elements on tobacco products, and tracking and tracing measures to combat illicit trade. Exemptions from the rules were authorised for some products but warranted a close monitoring of the market developments. These exemptions must be removed if a '*substantial change of circumstances*'³ is detected on the market, as per Art. 2(28) of the TPD. The TPD thus effectively aimed to maximise health protection and consumer awareness among EU citizens. In this context, objective and up-to-date market data on recent market trends, product perceptions and uptake of products by young people are essential for the operationalisation of the Directive.

Partly owing to these measures, there has been considerable progress in consumer protection related to tobacco and related products in the EU. The Eurobarometer 458 revealed an overall

¹ <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32003L0033>
<https://eur-lex.europa.eu/legal-content/EN/ALL/?uri=CELEX:32010L0013>
<https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELLAR:220cd0f2-82d0-4734-8a7f-e0dff6c8fc5d>
² https://ec.europa.eu/health/sites/health/files/tobacco/docs/dir200137ec_tobaccoproducts_en.pdf

³³ A substantial change of circumstances is defined as an increase of at least 10 % in at least five Member States, based on sales data or an increase of the level of prevalence of use in the under 25 years of age consumer group by at least five percentage points in at least five Member States for the respective product category (based on the Special Eurobarometer 385 report of May 2012) or equivalent prevalence studies.

increase in the awareness of negative impacts of conventional cigarettes, as well as a parallel decrease in the use of tobacco products in the EU between 2012 and 2017. Decreases have been seen in 20 of the 28 countries considered, while it increased or remained stable in 7.⁴ However, the last few years have seen the introduction to the EU market and the rise in popularity of new tobacco and related products (Regan, 2012), with products such as the electronic cigarettes (e-cigarettes) and heated tobacco products (HTPs). The EU has closely monitored the developments of the market to adequately tackle current and upcoming challenges and to ensure that regulations adequately respond to the market developments. For instance, e-cigarettes have recently come under intense public scrutiny following a series of e-cigarette-related deaths and lung injuries in the United States (USA) in summer of 2019. While the phenomenon appeared to be unique to the USA, renewed research and policy discussions have been triggered on the safety of these products and their regulatory provisions.

In the TPD five product categories were identified for close monitoring in the EU: waterpipe tobacco, slim cigarettes, small cigarillos, electronic cigarettes, and novel tobacco products. The latter products were broadly defined to encompass new developments in the tobacco market. Recently, heated tobacco products have emerged on the market and the present report focusses on this product to address issues and developments linked to novel tobacco products. Table 1 provides an overview of the consumption rates of these five product categories across EU Member States and reflects the range of “*national consumption ecologies*” according to 2017 data, where data was available. The highest share of consumers of waterpipe tobacco was found in Austria (4%), followed by Latvia (3%). The countries with the largest share of slim cigarettes consumers were Bulgaria (9%), Lithuania (7%) and Latvia (7%). For small cigarillos, France (2%) and Belgium (2%) were the most relevant markets across EU Member States. Among the countries for which data was available on electronic cigarettes, the countries with the largest share of users were the UK (5%), France (4%) and Belgium (4%).

⁴ Source: Eurobarometer 458. EU average for 2012 and 2015 was calculated for 27 EU members. No data were available for Croatia prior to 2017 and therefore a trend could not be implied.

Table 1. Current Consumption by product and by Member State

	Smokers ⁵	Novel tobacco products ⁶	Small cigarillos ⁷	Slim cigarettes ⁸	Electronic cigarettes ⁹	Water pipe tobacco ¹⁰	Average starting age ¹¹
EU28	26%		1.04%	1.03%	2%	1%	17.8
AT	28%	Yes	1.96%	1.80%	3%	4%	17.7
BE	19%	Yes	2.09%	0.68%	4%	2%	17
BG	36%		0.36%	9.33%	-	1%	18.8
CZ	29%	Yes	1.16%	2.90%	1%	1%	18.3
DK	19%	Yes	1.33%	0.16%	2%	1%	17.1
DE	25%	Yes	1.50%	0.37%	2%	1%	18.1
EE	23%	Yes	0.69%	2.96%	1%	1%	18.8
IE	19%	Yes	0.00%	0.26%	2%	0%	17.3
EL	37%		0.37%	1.71%	3%	0%	19.1
ES	28%	Yes	0.56%	0.23%	1%	0%	17.8
FR	36%	Yes	2.16%	0.24%	4%	0%	17.8
HR	35%		0.35%	1.91%	-	0%	18.6
IT	24%	Yes	0.48%	1.75%	-	1%	17
CY	28%		1.40%	1.68%	3%	2%	19.2
LV	32%		0.32%	6.62%	1%	3%	19.7
LT	29%	Yes	0.29%	7.16%	1%	1%	19.9
LU	21%	Yes	0.84%	1.07%	2%	2%	18.2
HU	27%		0.27%	0.80%	1%	1%	17.4
MT	24%		1.20%	0.43%	2%	0%	17.1
NL	19%	Yes	1.14%	0.12%	2%	0%	17.3
PL	30%	Yes	0.60%	4.32%	1%	1%	18.9
PT	26%	Yes	0.52%	0.48%	1%	0%	17.6
RO	28%	Yes	-	1.56%	-	0%	18.9
SI	28%		0.28%	2.17%	1%	1%	18.4
SK	26%		0.26%	1.91%	-	1%	17.9
FI	20%	Yes	1.60%	0.17%	1%	0%	17.4
SE	7%	Yes	0.42%	-	-	1%	17.2
UK	17%	Yes	0.51%	0.38%	5%	1%	17.1

Note: Cells in **Red** reveal high consumption across the market while those in **Green** show low consumption.

Source: OE elaboration of Eurobarometer 458 (2017) and WHO data available at https://www.who.int/tobacco/publications/prod_regulation/heated-tobacco-products/en/

⁵ Percentage of people currently interviewed smoking cigarettes, cigars, cigarillos or pipe. This percentage excludes e-cigarettes and novel tobacco products (Eurobarometer 458)

⁶ Countries where HTPs are marketed as of September 2017 (WHO)

⁷ Percentage of people interviewed currently smoking cigarillos. OE elaboration of Eurobarometer 458

⁸ Percentage of people interviewed currently smoking slim cigarettes. OE elaboration of Eurobarometer 458

⁹ Percentage of people interviewed currently using electronic cigarettes. Eurobarometer 458

¹⁰ Percentage of people interviewed currently using water pipe tobacco products. Eurobarometer 458

¹¹ Average starting age of people interviewed smoking cigarettes, cigars, cigarillos or pipe. This percentage excludes e-cigarettes and novel tobacco products (Eurobarometer 458)

1.2 Objectives and scope

The present study is a contribution to the European Commission's monitoring of the use of tobacco and related products. It aims to support the Commission by assessing how specific product categories are perceived by the public, what impacts they may have, and the mapping of consumer preferences and use patterns of these products. The study is also relevant to the 2021 Report on the Application of the Tobacco Products Directive 2014/40/EU (TPD), which the Commission will submit to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions. As previously mentioned, the five product categories of tobacco and related products that fall within the scope of this study are presented:

- **Waterpipe tobacco.** A "tobacco product that can be consumed via a waterpipe. For the purpose of this study, and in line with the TPD, waterpipe tobacco is deemed to be a tobacco product for smoking. If a product can be used both via waterpipes and as roll-your-own tobacco, it shall be deemed to be roll-your-own tobacco" [Art.2(13)].
- **Slim cigarette.** A tobacco stick with an outer wrapper that is not made of natural tobacco leaves and that can be consumed via a combustion process and has a diameter of less than 7.5 millimetres [Art. 28(e)].
- **Small cigarillos.** In line with references to in Article 8(1) of Council Directive 2007/74/EC and Article 4(1) of Directive 2011/64/EU, cigarillos are rolls of tobacco with an outer wrapper of natural tobacco of a maximum weight of 3 grams each and that may come with or without filter. For the purpose of this study a reference is made to "small" cigarillos in the sense of their likeness with factory-made cigarettes (typically of a weight of 1 gram with a filter) [Art. 2(12)].
- **Novel tobacco product.** A tobacco product that "(a) does not fall into any of the following categories: cigarettes, roll-your-own tobacco, pipe tobacco, waterpipe tobacco, cigars, cigarillos, chewing tobacco, nasal tobacco or tobacco for oral use; and (b) was placed on the market after 19 May 2014" [Art. 2(14)]. Current examples include the heated tobacco products (HTPs).
- **Electronic cigarette.** A "product that can be used for consumption of nicotine-containing vapour via a mouthpiece, or any component of that product, including a cartridge, a tank and the device without cartridge or tank. Electronic cigarettes can be disposable or refillable by means of a refill container and a tank, or rechargeable with single use cartridges." [Art. 2(16)]. E-liquid is typically distributed in cartridges or refill containers – the latter is a receptacle that contains a nicotine-containing liquid, which can be used to refill an electronic cigarette.

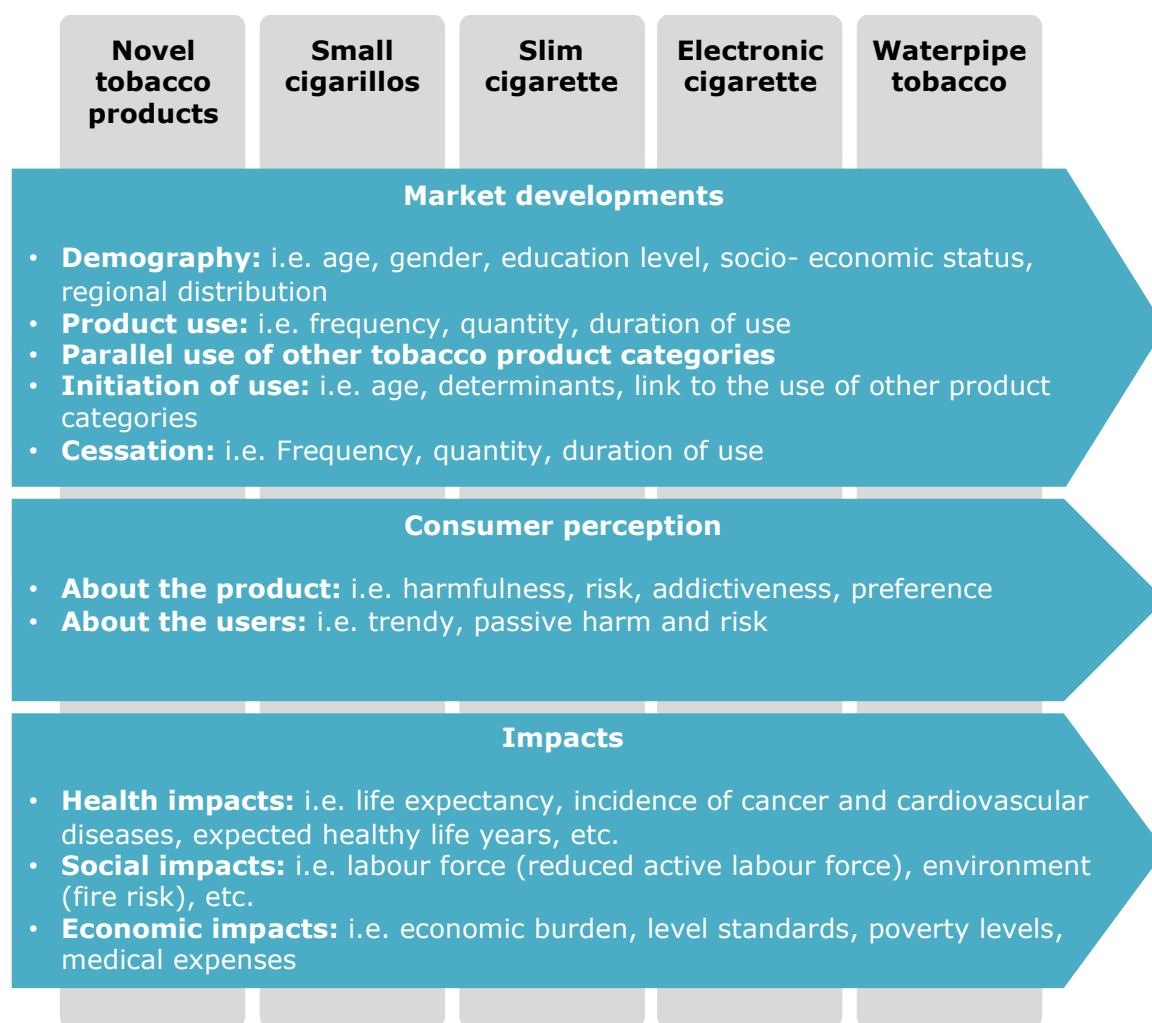
The research aimed to address three general objectives:

- to capture the market developments concerning different tobacco and related products and to understand what the main drivers of these developments are;
- to determine the extent to which these changes are driven by consumer perceptions;
- to assess the health, social and economic impacts associated with these developments.

Specifically, the **market developments** focussed on various key aspects for each tobacco and related product category at the EU level and (see Figure 1.). It aimed at better understanding the current landscape of the EU market of tobacco and related products and the way in which EU citizens were using these products. **Consumer perceptions** focussed on what users and past users and never users think about the harmfulness, risk, addictiveness, regulation and preferences of the five product categories. Given the different characteristics of each product category, specific issues were analysed in greater detail. With regards to novel tobacco products, uptake in trends by age profile was identified as key indicator to monitor changes in the tobacco market. Consumer perception and branding of slim cigarettes were also important, especially

with a focus on differences between genders. On e-cigarettes issues of interest included reasons for initiating their use, whether they have functioned as an aid to cessation of use of conventional cigarettes and whether, for young people, they act as a gateway to the use of conventional cigarettes. Finally, for both waterpipe tobacco and e-cigarette users there was the question of the using of flavouring and how these have affected user choices. The **impacts** were categorised in three areas: health impacts, social impacts and economic impacts. The figure below sketches the scope of the study.

Figure 1. Scope of the study



Source: Own elaboration

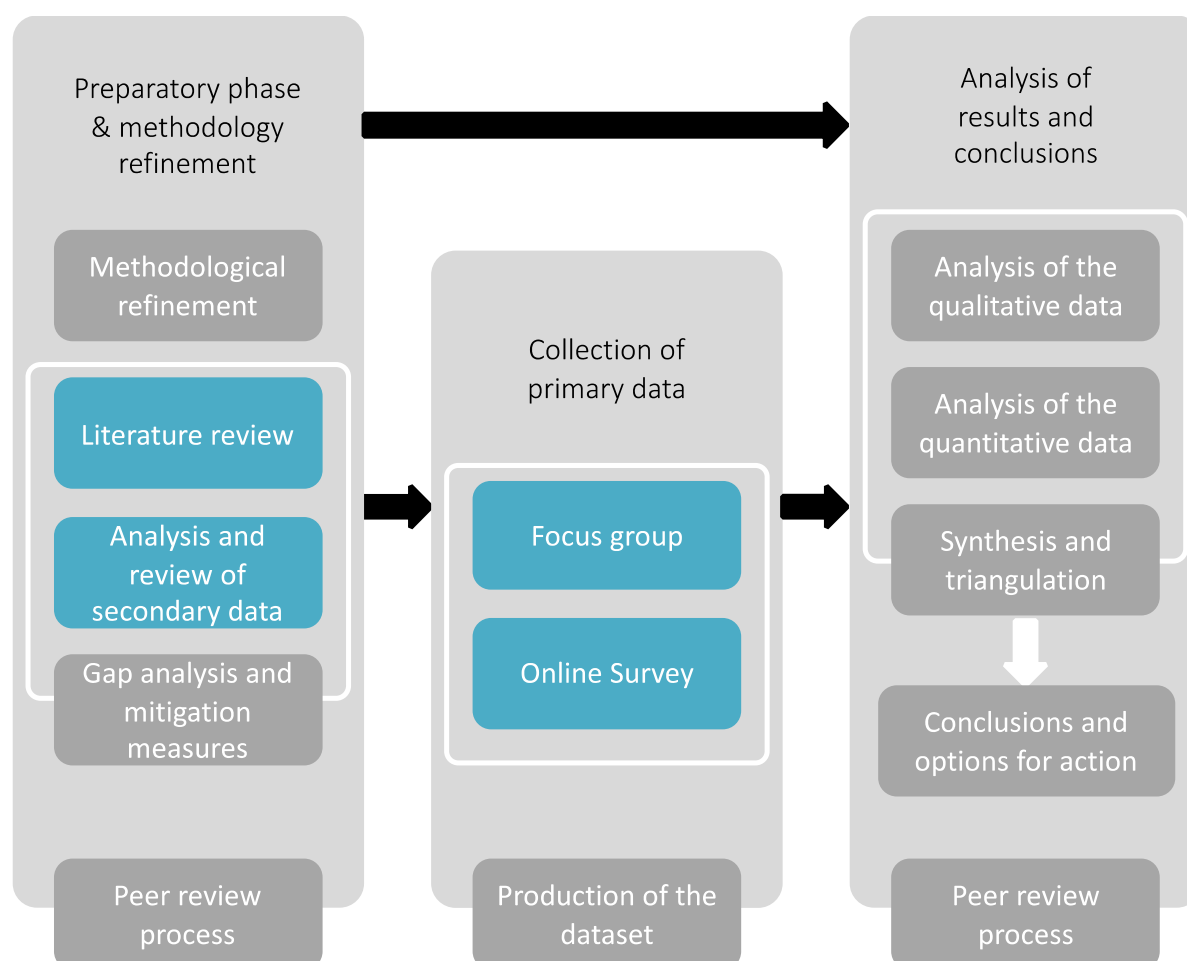
In terms of geographic scope, the study focused on the 27 EU Member States and the United Kingdom (UK). Despite the UK leaving the EU and the transition period ending on 31 December 2020, the UK will be considered and referred to as a Member State for the purposes of this report. Moreover, conditionally to the results of the ongoing negotiations, the TPD may continue to apply in Northern Ireland also after 2020.¹²

¹² General background https://ec.europa.eu/commission/presscorner/detail/en/QANDA_20_104. Notice on the withdrawal of the UK and EU rules on tobacco and related products: https://ec.europa.eu/info/sites/info/files/brexit_files/info_site/tobacco-products_en_0.pdf

1.3 Overall approach

To address the objectives and scope of this study a **mixed-methods approach** was adopted in which different methods/instruments and different sources (primary and secondary data) were collected and subsequently triangulated. Figure 2. summarises the overall approach to the study.

Figure 2. Overall approach



Source: Own elaboration

For the analysis of market developments, a variety of available data sources were drawn upon, including the Eurobarometer 458 survey (op. cit.), market information from commercial sources, and data submitted by the tobacco industry as part of regulatory requirements. The analysis was further enriched by the results of a systematic literature review. In terms of monitoring changes in market and consumer perceptions, the challenge was that (i) the market penetration of the products within scope was low (seldom more than 2%), that (ii) there was relatively little extant research on them, and (iii) existing data pointed to national/cultural differences in preferences across the different product categories. The two primary data collection methods, focus groups and an online survey, were designed to complement the existing sources and, in particular, to focus on the gaps identified in the literature. Information on the social, health and economic impacts drew mainly on secondary sources.

Table 2 maps the research areas of the study against the different methodologies used.

Table 2. Mixed methods approach

Research questions	Main information source	Support information source
Market developments - Demography: i.e. age, gender, education level, socio-economic status, regional distribution - Product use: i.e. frequency, quantity, duration of use - Parallel use of other tobacco product categories - Initiation of use: i.e. age, determinants, link to the use of other product categories - Cessation: i.e. frequency, quantity, duration of use	Secondary data gathered & online survey	Literature review
Consumers perceptions - Product perception - Potential appeal and harm to youth - Risk and harm to active and passive users	Literature review & focus groups	Secondary data gathered & online survey
Impacts - Health impacts: i.e. life expectancy, incidence of cancer, expected healthy life years, diseases, etc. - Social impacts: i.e. social acceptability, exposition, social smoking, social disruption, etc. - Economic impacts: i.e. economic burden, level standards, poverty levels, medical expenses, labour force, etc.	Secondary data gathered	Focus group & literature review

Source: Own elaboration

In terms of **geographic coverage**, the preparatory phase focused on all EU Member States and, when necessary, relevant literature beyond the EU was also considered. A robust secondary data search was conducted (i.e. a literature review and quantitative information review) to define the state-of-the-art in available knowledge on the five products in scope. It also looked to identify the main information gaps to be addressed in primary data collection phase in order to effectively answer the study's research questions. Country coverage of primary data collection focussed on a representative subgroup of countries based on the following criteria:

- geographic distribution and representativeness of the countries;
- national tobacco consumption profile, including a diverse selection of countries with high, medium and low prevalence of tobacco consumption;
- extent of introduction of novel products such as heated tobacco products into the national market since at least September 2017;
- countries with particularly high and low market penetration of small cigarillos, slim cigarettes, electronic cigarettes and waterpipe tobacco.

These consumption characteristics were explored through the analysis of secondary data gathered from the 2017 Eurostat survey, Passport data and WHO data.¹³ With this information, the 2017 EU market (see Table 1) was reconstructed to aid in the selection of countries for the fieldwork in order to reflect the range of “*national consumption ecologies*”. Focus groups and an online survey in a representative subgroup of EU Member States were used to collect qualitative

¹³ Eurobarometer 458

and quantitative data to fill informational gaps identified during the preparatory phase of the study. In the following sections, a brief description of the main methods used is provided. More detailed information can also be found in Technical Appendices.

1.3.1 Systematic review and gap analysis

An extensive literature review was conducted in order to gain an up to date and comprehensive picture on what is known about market developments and consumer perceptions of different tobacco product categories. For this purpose, a total of 685 records were initially identified through database searching. This list was reduced to 171 items comprising both the scientific and the grey literature after title and abstract screening. The final list of 171 records was gathered, read in detail and summarised to reconstruct the state-of-the-art on the tobacco and related products under scope. Of the 171 articles, 67 were from the US and Canada, 45 from Member States of the European Union and 59 from other countries. Technical Appendix I provides the full methodology and protocol used to carry out the systematic review.

As part of the literature review, different qualitative information was collected for the various products within scope. Table 3 summarises the areas where information was retrieved, and those where no information was found. Where the cell is striped, the information available was partial and collected via additional desk research. Exhaustive information was found for waterpipe tobacco products and electronic cigarettes. Only partial information was found for small cigarillos, novel tobacco products and slim cigarettes. Further additional desk research filled some of the gaps, especially for small cigarillos and slim cigarettes, while only minor additional information was found for novel tobacco products. Some of the difficulties in discovering articles related to small cigarillos and slim cigarettes refer to the fact that they are often considered as subcategories of cigarettes and cigars, and as such they are more rarely the focus of ad-hoc studies. As a result, no specific articles were found on the impacts of these two products.

Table 3. Literature review information found by product

	Small cigarillos	Waterpipe tobacco	Electronic cigarettes	Novel tobacco products	Slim cigarette
Market Developments					
Availability and promotion					
Patterns of use					
Consumer perceptions					
Trends & preferences					
Perceived risk & harmfulness					
Impacts					
Outcomes					
Policy implications					



Available information

Partial information available via additional desk research

Source: Own elaboration

In terms of novel tobacco products, little information was available due to their recent introduction in the market. Information on the patterns of use were limited to early results from Italy. No information was found on the perceived risk and harmfulness, as well as the policy implications of these products. In describing HTPs, one of the major producers stated that they

were “designed as an alternative to cigarettes for current, adult smokers”¹⁴, but data on the patterns of use have not been collected as yet. Data on the consumer perception of the perceived risk and harmfulness of novel tobacco products were also lacking up to the date of the review. The data gathering activity was conducted to gain an up to date and comprehensive picture on the available data on market developments and consumer perceptions of different tobacco product categories. This research was carried out using inter alia: official statistics, statistical portals and customised Google searches. The data collected was mapped following the dimensions specified in the Technical Specifications.

Table 4. Results by tobacco product category

Tobacco product category	Number of data sources collected
Novel Tobacco Products	3
Small cigarillos	5
Slim cigarettes	4
Electronic cigarettes	4
Waterpipe	5

Source: Own elaboration

Technical Appendix II provides a summary of the data mapping activity carried out. Table 5 summarises the areas where quantitative information was retrieved, and those where no information was found. In the striped cells, the information available was partial in terms of geographic scope and quality.

Table 5. Quantitative information found by product

	Small cigarillos	Waterpipe tobacco	Electronic cigarettes	Novel tobacco products	Slim cigarette
Market Developments					
Market size					
Pricing					
Consumer population					
Initiation age					
Quit ratio and attempts					
Import-Export data					
Market Share					
Consumer perceptions					
Health perception					
Advertising impact					
Consumer perception					
Impacts					
Taxation					
Expenditure & affordability					
Third party exposure					
Health impact					



Available information

Partial information in terms of geographic scope and quality

Source: Own elaboration

¹⁴ <https://cot.food.gov.uk/sites/default/files/tox2017-25.pdf>

For market developments, no data was found on the demographics of the users of novel tobacco products. Data on consumer perception of small cigarillos and waterpipe was extracted from the Global Tobacco Surveillance System (GTSS) database, while no data was found for novel tobacco products. Although the data gathered from the GTSS database provided a starting point for the analysis of consumer perceptions of these product categories, the available data was restricted to three EU Member States and, as such, the geographical coverage of the analysis was limited. With regard to the analysis of impacts, the study focused on the health, economic and social dimensions. In terms of economic impacts, the analysis was limited to taxes for all products due to the availability of data. In terms of social impacts, the data for small cigarillos and waterpipes was limited to the GTSS database and therefore has the above-mentioned limitations. Limited data was found regarding health impacts of novel tobacco products.

Data gaps were found to be particularly significant for novel tobacco products. However, HTPs had been introduced in a few Member States that provided an appropriate setting to collect information on these products. An in-depth review of the available market data also allowed us to identify and select countries in which the use of these products were more common.

Table 6. Summary of the main information gaps

Literature review gaps
• There was very limited information about pattern of use of novel tobacco products • There was no information about the perceived risk & harmfulness of novel tobacco products • Information related to the socio-economic and health impacts of small cigarillos, slim cigarettes and novel tobacco products was limited
Quantitative data gaps
• From a geographical point of view, data for Cyprus and Luxembourg was limited • Data for HTPs was limited. There was little to no data on: • Social impacts of HTPs for all countries • Initiation / cessation of HTPs • Demography of population using HTPs • Use of HTPs

Source: Own elaboration

1.3.2 Focus groups

The focus groups were designed to provide a qualitative approach to people's knowledge of and attitudes towards the different tobacco and related products, and to better formulate the survey questions in a clear and understandable manner. The focus groups had a standard duration of 120 minutes. They targeted consumers of tobacco and related products or those that quit in the last 12 months, while taking into account gender, product usage and age balance. A total of 41 participants aged 20 to 69 years old took part in the activities. The countries in which focus groups were conducted are listed in table 7 below and the rationale for selecting them is presented in the following paragraph.

Table 7. Focus groups geographical scope

Country	n	Age range
Bulgaria (E)	9	24 - 69
Germany (C)	8	22 - 63
France (W)	8	20 - 66
Greece (S)	8	20 - 58
Spain (S)	8	20 - 57

Source: Own elaboration

Bulgaria was a country showing high consumption of tobacco and the highest consumption of slim cigarettes across the EU. Germany showed high rates of small cigarillos consumption and had a significant waterpipe tobacco market, according to Passport data. France was also found to be a very relevant market since it showed an overall high consumption of tobacco products, and specifically of small cigarillos and electronic cigarettes. Greece was also the country with the highest percentage of tobacco consumers, and Spain was flagged by the Catalan Institute of Oncology (Institut Català d'Oncologia, ICO) experts due to its early adoption of HTPs and the growing trend identified in the literature (see section 5.1).

The primary data collection activities focussed on covering the identified gaps. The focus groups were a useful tool to collect qualitative information and to provide an in-depth understanding of representations of consuming different products, as well as consumption careers and transitions. Table 8 shows the key topics addressed in the focus group discussions.

Table 8. Data gaps addressed by the focus group

Tool / methodology	Issues Addressed
Focus Group	• The pattern of use of the different tobacco products; • Perception of risk and harmfulness; • Health, social and economic impacts; and, • Opinions and ideas on regulation.

Source: Own elaboration

Technical Appendix V provides a more detailed description of the methodology used to carry out the Focus Group. Technical Appendix VI also provides the guidelines applied by the moderators to carry out this research activity.

1.3.3 Online survey

The online survey was designed to collect quantitative data to fill the information gaps identified in the systematic review. Overall, 12,000 respondents took part and in addition to the five countries in which the focus groups were conducted, a further five EU Member States were selected to achieve geographic representativeness of the sample and to cover markets that were relevant for particular tobacco and related products. The final list of countries is shown in Table 9. The respondents were selected on the basis of quotas in order to (i) survey sufficient numbers of users of the five product categories and (ii) and on account of the concerns about young people's use of tobacco and related product to over-sample people aged 18-25. For each of the ten countries the samples were as follows and displayed in table 9:

- 600 users of tobacco and related products (including those who had quit in the last 12 months) aged 26 years and above.
- 600 young people aged 18 to 25 (including users and non-users)

Socio-demographic profiles of the survey samples can be found in Table 12 and Table 13.

Table 9. Online survey: Geographic scope and quota breakdown by country (N=12,000)

Country	Youth respondents (aged 18 to 25)	Consumers of tobacco and related products ¹⁵	Total # respondents
Bulgaria (E)	610	777	1200
France (W)	606	634	1200
Germany (C)	603	625	1200
Greece (S)	610	718	1200
Italy (S)	601	664	1200
Latvia (N)	603	617	1200
Poland (E)	609	696	1200
Slovakia (C)	643	626	1200
Spain (S)	605	759	1200
United Kingdom (W)	600	644	1200

Source: Own elaboration

Latvia was chosen due to its medium-high consumption rates as well as its consumption of slim cigarettes and waterpipe tobacco, according to the latest Eurobarometer. Poland was the most relevant market for slim cigarettes given its size and relatively high usage. Slovakia was included since the national authorities had recently reported high rates of forecast growth for HTPs. Increasing trends in volume sales were also observed for Slovakia in the Passport data. The UK was the country with the highest share of e-cigarette users. Italy was an early adopter of HTPs (Liu et al., 2018) and was included as it was important to have a country in which HTPs had been present in the market for a longer time.

Table 10 maps the way in which the online survey was used to fill in the identified information gaps (see Technical Appendix VIII for the full online questionnaire).

¹⁵ Including those that had quit in the last 12 months

Table 10. Data gaps addressed by the Online survey

Tool / methodology	Issues Addressed
Online survey	• Consumption history; • Initiation, switching and cessation of different tobacco products; • Perceived harmfulness of the different products; • Awareness and perceptions of emerging products; • Consumer demographics.

Source: Own elaboration

The online survey used questions and response alternatives that are typical of Eurobarometer surveys. An exception was the adoption of a novel form of questioning about health risk in health research. To assess the perceived harmfulness of the 5 tobacco products, a 'swing' methodology was used. The procedure is as follows. Respondents were asked to think about the potential 'damage to health' of the five products under scope, as well as conventional cigarettes, and to select the product that they considered most damaging to health. They were then told that this product has been allocated 100 points on a 'damage to health' scale' running from 100 (high damage) to 0 (no damage). Respondents were then asked to assess the five remaining tobacco and related products on the 100-point scale, giving a relative damage to health score for all products. So, for example, if a respondent rated slim cigarettes as the most damaging to health of the tobacco and related products, slim cigarettes were rated at 100. If the respondent then rates e-cigarettes at 50, that implies that the respondent thinks e-cigarettes are half as damaging to health as are conventional cigarettes. Whether the size of slim cigarettes reduces the perceived 'damage to health' is one of the interesting questions for the analyses of the survey. The following question focused on the 'potential damage to health of other people' and used the same method to elicit judgements of the relative extent of damage to health of 'passive smoking' from the five products.

Table 11 maps how the survey addressed for each product category the research aspects identified in the gap analysis, for each product category.

Table 11. Mapping of survey questionnaire by research question

	Small cigarillos	Waterpipe tobacco	Electronic cigarettes	Novel tobacco products	Slim cigarettes
Market developments					
Demography of user population					
Age	Q5.2	Q5.2	Q5.2	Q5.2	Q5.2
Gender	Q5.1	Q5.1	Q5.1	Q5.1	Q5.1
Education level	Q5.3	Q5.3	Q5.3	Q5.3	Q5.3
Socioeconomic status	Q5.6	Q5.6	Q5.6	Q5.6	Q5.6
Regional Distribution	Q5.4	Q5.4	Q5.4	Q5.4	Q5.4
Pattern of use					
Frequency	Q1.3; Q1.6	Q1.3; Q1.6	Q1.3; Q1.6	Q1.3; Q1.6	Q1.3; Q1.6
Quantity	Q1.4.a	Q1.4.a	Q1.4.b; Q1.4.c	Q1.4.a	Q1.4.a
Duration of use	Q1.2.a; Q1.2.b; Q5.2	Q1.2.a; Q1.2.b; Q5.2	Q1.2.a; Q1.2.b; Q5.2	Q1.2.a; Q1.2.b; Q5.2	Q1.2.a; Q1.2.b; Q5.2
Parallel use of other tobacco products	Q1.1	Q1.1	Q1.1	Q1.1	Q1.1
Initiation of use					
Age	Q1.2.a; Q1.2.c	Q1.2.a	Q1.2.a	Q1.2.a	Q1.2.a
Determinants	Q2.1.a; Q2.1.b	Q2.1.a; Q2.1.b	Q2.1.a; Q2.1.b	Q2.1.a; Q2.1.b	Q2.1.a; Q2.1.b
Link to use of other products	Q1.1; Q2.6; Q2.7; Q2.8	Q1.1; Q2.6; Q2.7; Q2.8	Q1.1; Q2.6; Q2.7; Q2.8; Q4.9	Q1.1; Q2.6; Q2.7; Q2.8; Q4.3	Q1.1; Q2.6; Q2.7; Q2.8
Cessation					
Frequency	Q1.5; Q2.3	Q1.5; Q2.3	Q1.5; Q2.3	Q1.5; Q2.3	Q1.5; Q2.3
Short-term results	Q2.2; Q2.3; Q2.4	Q2.2; Q2.3; Q2.4	Q2.2; Q2.3; Q2.4	Q2.2; Q2.3; Q2.4	Q2.2; Q2.3; Q2.4
Long-term results	Q1.2.b; Q2.4	Q1.2.b; Q2.4	Q1.2.b; Q2.4	Q1.2.b; Q2.4	Q1.2.b; Q2.4
Consumer perceptions					
Harmfulness	Q3.1.a; Q3.1.b; Q3.1.c; Q3.2.a; Q3.2.b	Q3.1.a; Q3.1.b; Q3.1.c; Q3.2.a; Q3.2.b	Q3.1.a; Q3.1.b; Q3.1.c; Q3.2.a; Q3.2.b	Q3.1.a; Q3.1.b; Q3.1.c; Q3.2.a; Q3.2.b	Q3.1.a; Q3.1.b; Q3.1.c; Q3.2.a; Q3.2.b
Awareness and perceptions of emerging products		Q4.15; Q4.16; Q4.17; Q4.18	Q4.9; Q4.10; Q4.11; Q4.12; Q4.13; Q4.14	Q4.1; Q4.2; Q4.3; Q4.4; Q4.5; Q4.6; Q4.7; Q4.8; Q4.9; Q4.10	

Source: Own elaboration

Table 12 and Table 13 provide a description of the sample of respondents from the online survey. The first table describes the 18 to 25 age sub-group (hereinafter referred to as *18-25 age group*), while the second table focusses on the sample group of people aged 26 years and above (henceforth *26+ age group*). Both tables provide values for each of the countries in percentage terms, as well as the absolute number of respondents in the first row. The sub-samples are described by gender, education level, employment status, residency and self-reported socio-economic status. A detailed breakdown, in number of respondents and shares, can also be found in the Annex 10.1.

Table 12. Socio-demographic breakdown of online survey (18-25 age group)

Percentage and number of respondents in the 18-25 age group

18-25 age group	Group total	BG	DE	EL	ES	FR	IT	LV	PL	SK	UK
Total number (in absolute values)	(6090)	(610)	(603)	(610)	(605)	(606)	(601)	(603)	(609)	(643)	(600)
Gender											
Male %	47	47	43	49	49	50	50	39	49	43	50
Female %	53	53	57	51	51	50	50	61	51	57	50
Education											
Low %	6	5	3	2	3	9	4	10	4	9	5
Medium %	63	63	72	56	59	46	70	60	73	66	63
High %	32	32	24	42	38	46	25	31	22	25	32
Employment Status											
Self-employed %	9	14	4	13	11	9	7	11	6	5	9
Managerial position %	6	8	4	2	3	5	2	8	7	6	14
Other white collar %	18	20	17	15	20	15	15	25	19	12	17
Manual worker %	13	16	22	4	10	13	5	9	17	13	18
Unemployed %	9	8	3	13	9	5	12	10	7	8	11
Retired %	0	0	0	0	0	0	0	0	0	0	0
Student %	42	31	46	50	45	42	55	31	42	51	25
Non-employed %	4	2	3	2	1	10	4	5	2	4	5
Residency											
Urban %	65	87	54	76	73	59	55	73	67	59	46
Suburban %	19	8	19	16	17	21	30	16	17	8	43
Rural %	16	5	27	8	9	21	16	11	16	33	11
Self-reported socio-economic status											
Low %	18	22	11	20	12	21	14	20	20	20	26
Medium %	65	65	65	67	68	57	67	64	70	69	57
High %	17	13	25	13	20	22	19	16	10	12	17

Base: all respondents in the 18-25 age group (N=6,090)

Notes: The group total category refers to all respondents within the age group. The orange column represents the share of respondents in the age group that have this trait, i.e. it is the socio-demographic breakdown for the whole age group. The white columns represent the country breakdown. The first row provides the absolute number of respondents per country.

Source: OE elaboration from online survey data

Table 13. Socio-demographic breakdown of online survey (26+ age group)

Percentage and number of respondents in the 26+ age group)

26+ age group	Group total	BG	DE	EL	ES	FR	IT	LV	PL	SK	UK
Total number (in absolute values)	(5910)	(590)	(597)	(590)	(595)	(594)	(599)	(597)	(591)	(557)	(600)
Gender											
Male %	52	51	57	50	50	47	49	62	49	57	49
Female %	48	49	43	50	50	53	51	38	51	43	51
Education											
Low %	5	1	14	1	2	14	9	5	2	2	2
Medium %	46	32	61	32	40	37	50	46	43	63	54
High %	49	66	25	67	58	49	41	49	55	35	44
Employment status											
Self-employed %	14	18	9	19	16	12	18	15	11	11	9
Managerial position %	13	15	13	7	13	4	7	16	16	11	24
Other white collar %	30	37	22	41	36	27	35	28	35	24	18
Manual worker %	16	17	18	7	17	20	7	17	19	25	12
Unemployed %	7	4	7	6	6	6	10	6	3	4	6
Retired %	12	6	23	6	5	17	6	9	11	19	19
Student %	2	1	2	2	2	1	4	1	2	1	1
Non-employed %	6	2	6	2	4	13	12	7	3	5	10
Residency											
Urban %	66	94	52	80	80	46	64	72	78	64	27
Suburban %	17	4	19	13	12	17	22	13	11	9	50
Rural %	17	3	28	6	8	37	14	15	11	28	23
Self-reported socio-economic status											
Low %	23	27	19	20	14	26	19	29	23	27	30
Medium %	64	65	59	68	71	60	66	62	66	62	59
High %	13	8	22	12	15	14	15	9	12	11	12

Base: all respondents in the 26+ age group (N=5,910)

Notes: The group total category refers to all respondents within the age group. The orange column represents the share of respondents in the age group that have this trait, i.e. it is the socio-demographic breakdown for the whole age group. The white columns represent the country breakdown. The first row provides the absolute number of respondents per country.

Source: OE elaboration from online survey data

1.4 Rationale of the report

This report is structured into 8 chapters. In the **introductory chapter**, an overview of the state of the art is provided together with the overall objectives and scope of the study. The chapter also describes the design and approach used to answer the different research questions by highlighting which information source was used and how it was collected.

The following five chapters (chapter 2 to 6) describe in a structured way the results found for each of the **five tobacco and related products** within scope. Each chapter starts with a background section that summarises the state-of-the-art developed during the preparatory phase (systematic review). The initial section is followed by three sections which detail the main findings for each of the overall study objectives:

- Market developments
- Consumer perceptions
- Health, social and economic impacts

Findings from the online survey have been structured to highlight information for two different age sample groups: the '18-25 age group' and the '26+ age group'. The chapters on individual products conclude with a triangulation of the different sources and a discussion of the key points identified for each product.

Common terminology is used throughout the text to refer to the types of user identified. Never users refer to respondents in the online survey who had never used a particular product (e.g. "never user of e-cigarettes") at the time of the survey. People that tried the product refers to respondents that said they had only tried a specific product once or twice. Past users respondents that had consumed a specific product more than once or twice in the past but have not used the product in the 12 months prior to the survey. Current users refer to survey respondents that had consumed a specific product more than once or twice in the last 12 months prior to the survey. Exclusive users refer to respondents that only used one particular product at the time of the survey. Dual users refers to those who, at the time of the survey were using more than one of the tobacco and related products.

Chapter 7 provides a **comparative analysis** of the primary data collected across the different products as well as traditional cigarettes. The chapter is divided into three sections, one for each of the overall study objectives. Chapter 8 provides the **conclusions** of the study.

The Annexes present additional data tables that are complementary to the core related analysis of the five tobacco and related products covered in this study. The Technical Appendices are found in a separate document and gather the detailed methodologies used to carry out the activities under the present study as well as data sources produced in the context of the report.

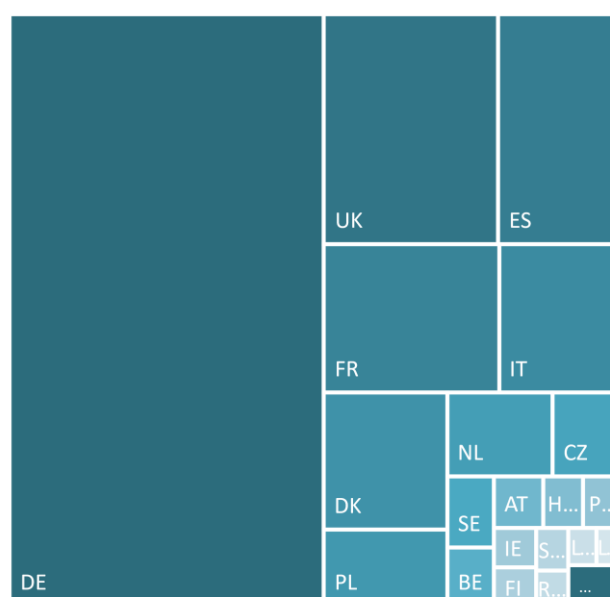
2 Waterpipe tobacco

2.1 Background

Over the past 30 years the consumption of waterpipe tobacco has become more prevalent particularly among young adults and college students (EC, 2010). In parallel, there has been a global increase in the waterpipe tobacco market. Most of the literature on waterpipe tobacco has a focus on North America and the Middle East, paying special attention to the product use by young people (Al Moamary, 2011, Saudi Arabia; Abdullah, 2016, Canada, N=20.703; Rahman, 2012, USA, N=478; Minaker, 2015, Canada, N=27.404). As identified by a recent study, there was a notable scarcity of data on the trade and consumption of waterpipe tobacco products in the EU,¹⁶ but the recent increase in popularity of the product signalled the need for more information on the product. In a systematic review, Jawad et al. (2018, global) marked Europe as one of two regions with the highest prevalence of waterpipe tobacco use amongst youth.¹⁷ He had previously noted a particularly high use of waterpipe products among those aged 13 to 15 years in parts of Eastern Europe (Latvia 22.7%, the Czech Republic 22.1%, Estonia 21.9%) through a secondary analysis of the Global Youth Tobacco Survey¹⁸ (Jawad, 2015, EU).

Market developments. According to Passport data, Germany was the largest market of waterpipe tobacco¹⁹ in the EU in 2017 with 368 million euros (m EUR) in retail volumes, representing over half of the whole European market. It was followed by the UK (79 m EUR), Spain (56 m EUR) and France (52 m EUR). It is important to note that, in the Passport data, waterpipe tobacco includes cut tobacco sold in packaged format for smoking in both pipes and waterpipes. While this is likely to overestimate waterpipe consumption, it is the best proxy for the market.

Figure 3. Market size of waterpipe & pipe tobacco products in million EUR in 2017



Note: Category of the waterpipe tobacco includes cut tobacco sold in packaged format for smoking in pipes and water pipe tobacco. The market size is estimated total sales in million euros for 2017, without

¹⁶ European Commission (2017): Study on Council Directive 2011/64/EU on the structure and rates of excise duty applied to manufactured tobacco.

¹⁷ The other region was the Middle East, for both youth and adults.

¹⁸ A nationally representative study of students aged 13-15 years

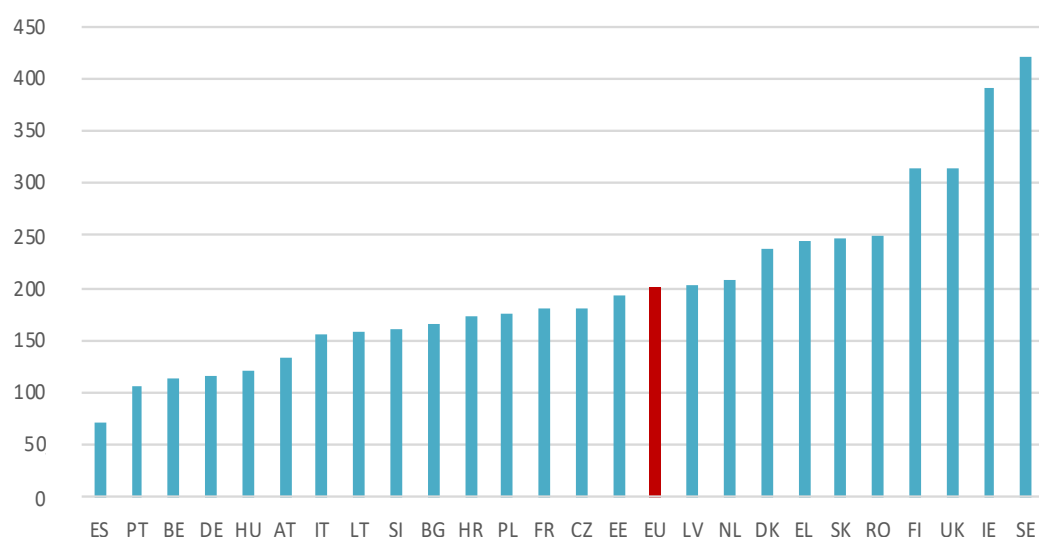
¹⁹ Category of the waterpipe tobacco includes cut tobacco sold in packaged format for smoking in pipes and water pipe tobacco.

population adjustments. It is displayed as a tree map graph that consists of series of nested rectangles of sizes proportional to the corresponding data value.

Source: OE elaboration of Passport, 2017

A European Commission study²⁰ revealed that tax evasion was particularly high in the case of pipe and waterpipe tobacco, resulting in the creation of a black market for the product and revenue loss. The wide range of prices for waterpipe tobacco in the Member States is likely to have been a key driver for cross-border selling in the EU. These developments have hindered the effectiveness of tobacco control measures. A recent EU-level study (EC, 2020) suggested that the impact of taxation on dissuading youth from tobacco consumption has been lower than expected, due *inter alia* to the substitution of cigarettes with less expensive products, including waterpipe tobacco.²¹ Indeed, for excise duty purposes, waterpipe tobacco falls within the residual category of “other smoking tobacco” for excise duty purposes²² (see art. 2.1(c) in the Directive 2011/64 (‘Tobacco Excise Directive’ – TED²³). As depicted in Figure 4., the highest prices for pipe and waterpipe tobacco were found in Sweden and Ireland, two markets that were not observed as particularly relevant for waterpipe tobacco products (see Figure 3.). On the contrary, in the largest market, Germany, prices (€115 per kg) were below the EU average of €201 per kg.

Figure 4. Price of waterpipe and pipe tobacco products across EU in 2017, euros per kg



Note: EU average is calculated for 25 EU members (excluding Cyprus, Luxembourg and Malta). Category of the waterpipe tobacco includes cut tobacco sold in packaged format for smoking in pipes and water pipe tobacco.

Source: OE elaboration of Passport, 2017

The market dynamics over the period 2012-2017 showed a small decrease in the sales of pipe and waterpipe tobacco for a majority of EU Member States, with the notable exceptions of Romania, Germany, Slovenia, Bulgaria and Portugal, where market volumes increased considerably (see Figure 5.). The most significant increase compared to the year 2012 occurred in Romania, where the market size increased almost by a ten-fold in terms of both retail sale price (RSP) and real volume.

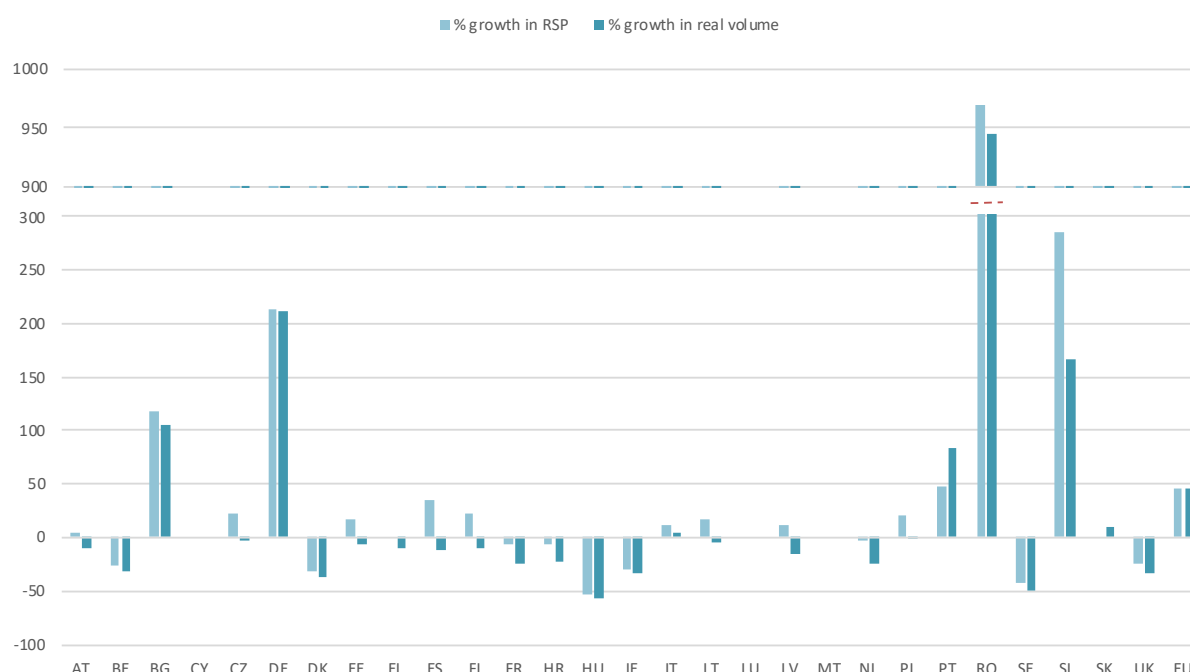
²⁰ European Commission (2017): Study on Council Directive 2011/64/EU on the structure and rates of excise duty applied to manufactured tobacco

²¹ https://ec.europa.eu/taxation_customs/sites/taxation/files/10-02-2020-tobacco-taxation-report.pdf

²² In standard economic theory, excise duty increases the price of a product and therefore is expected to decrease demand.

²³ Directive 2011/64/EU on the structure and rates of excise duty applied to manufactured tobacco. Available at: <https://eur-lex.europa.eu/legal-content/en/ALL/?uri=CELEX%3A32011L0064>

Figure 5. Market size trends of waterpipe & pipe tobacco products: 2012-2017



Note: RSP stands for retail selling price in the graph (current prices). The EU average is calculated for 25 EU MS only (excluding Cyprus, Luxembourg & Malta - no data were available). Category of the waterpipe tobacco includes cut tobacco sold in packaged format for smoking in pipes and water pipe tobacco.

Source: OE elaboration of Passport, 2017

There has been evidence of the growing popularity of waterpipe tobacco consumption (Kilibarda et al., 2019; EC, 2017), although the extent of use varies greatly across EU Member States (Jawad, 2018²⁴). For instance, Jawad (2018) estimated that there were many fewer 'ever users' of waterpipe tobacco in the UK (9.9%) than in Germany (28.9%) and Finland (15.3%). A considerable surge in popularity has particularly been noted in younger generations. Several studies found an alarmingly high prevalence of waterpipe tobacco recurrent users specifically among high school students (Akl, 2011, global; Jawad et al., 2013, UK; Jawad, 2016, global; Jawad, 2018, UK). One study estimated that the prevalence of waterpipe recurrent use was 1.5 times higher than cigarette smoking amongst school students in Europe, a concerning development as it may encourage the use of tobacco and related products (Akl, 2011). Data from the Eurobarometer 458 found that 28% of young people aged 15-24 reported having tried waterpipe tobacco products – over twice as many as in the overall sample (13%). In a meta-analysis, Jawad (2015, EU) estimated higher odds of waterpipe tobacco smoking amongst males, cigarette users, those whose parents or peers smoked and those whose parents had higher educational attainment. To this list, Grant et al. (2014) added adults with a higher socio-economic status, while Balogh (2018, Germany & Hungary, N=2,935) showed that there was no association between the financial situation of university students and waterpipe tobacco consumption. In terms of patterns of use, data suggested that most consumers of waterpipe tobacco only use the product occasionally, often in social settings (EC, 2010). Many studies have explored the dual use of waterpipe tobacco with other tobacco products and documented that being a concurrent user of cigarettes was strongly correlated with waterpipe tobacco use (Akl, 2011; Jawad et al., 2018; Balogh, 2018) (see section 2.4 for health impacts of dual use). For instance, over 10% prevalence of dual waterpipe tobacco and cigarette use was reported in Eastern European countries (Jawad, 2015). Al

²⁴ Data was used from four different survey waves: N=2,062 (2013); N=1,952 (2014); N=1,943 (2015); N=2,019 (2016).

Oweini et al. (2018) analysed waterpipe tobacco consumption in light of the gateway hypothesis and found that those using waterpipe tobacco products had double the odds of starting to consume cigarettes later.

Consumer perception. The increase in the popularity of waterpipe tobacco has been attributed to the social culture of waterpipe use, to the lack of awareness about the harmful health effects of waterpipe tobacco use, and to aggressive marketing by the industry through various social channels (Balogh et al., 2018; Maziak et al., 2015; Singh, 2017) – specifically targeting younger consumer groups. Several studies, mainly from the USA, have looked at the effect of the mass media and social network environments on young people’s perception of waterpipe tobacco products. Nevertheless, due to the global access of online content, the findings may also be considered relevant to Europeans. In a systematic literature review, Singh et al. (2017) analysed the factors that drove youth to waterpipe smoking initiation. The factors were categorised as: (i) individual factors, including curiosity, perception/attitude towards waterpipe smoking and pleasure seeking; (ii) interpersonal influences, including the influence of family and friends; (iii) cigarette and alcohol use, as waterpipe can be used as a substitute for conventional cigarettes, and (iv) media influences. Of these, interpersonal factors were the most influential.

Primack (2012, USA) analysed the way in which establishments dedicated to hookah tobacco smoking advertised their services on the Internet. He reported that their promotional material aimed to cultivate an image of highly social, cultural and fun places for young people, with relatively few mentions of health warnings. Jackler (2019, USA) found that tobacco marketing was still prevalent on Facebook, despite efforts from the platform to ban it. Brand-sponsored pages were often found to use content strategies to engage users, for example, through memes, sponsored events, product innovations and contests to attract “likes”. These commercial practices allowed tobacco brands to escape the scrutiny of regulations on tobacco promotion, while potentially shaping the views of vulnerable groups. In Europe, the waterpipe industry, including waterpipe cafes, has been said to operate in an almost completely unregulated market and to employ deceptive marketing techniques to attract new users (Jawad, 2013, UK). Cultural and commercial factors – such as the social acceptability of the shisha lounge culture or the introduction of flavoured tobacco blends – were believed to be key drivers behind the recent growth in the market in the region.²⁵ A recent systematic review found that the most cited reasons for waterpipe tobacco use amongst university students in the global north – namely, Europe, United States and Australia – were curiosity, peer or social influence, and wanting to overcome the social anxiety associated with new situations whilst still engaging in a legal activity (Arshad et al., 2019). Research has shown evidence that youth were particularly attracted to the product due to the range of “appealing” flavours available (Qasim et al., 2019, global). In terms of perception of harmfulness, qualitative research around the world has evidence lower harm perception compared to cigarettes (Ali & Jawad, 2017; Jawad et al., 2013; Cornacchione et al., 2016). Jorge (2018, N=1,770) explored the perceptions of waterpipe tobacco products among Spanish teenagers, aged 12-18 years. In general, the teenagers viewed waterpipe use as less harmful and less addictive than conventional cigarettes. In a more recent global study, Arshad et al. (2019) found that although students knew about the harms associated with waterpipe tobacco, it did not seem to deter them from using the product – a finding that differs from global studies on adult waterpipe tobacco users (Akl et al., 2015).

Impacts. Overall, most studies on health impacts agreed that users of waterpipe tobacco products were exposed to many of the same toxic compounds and by-products as conventional cigarette users, and potentially at much higher levels. The health, economic and social impacts are further developed in section 2.4.

²⁵ https://ec.europa.eu/taxation_customs/sites/taxation/files/study_on_directive-2011_64_main_text_en.pdf

2.2 Market developments

2.2.1 Socio-demographic characteristics

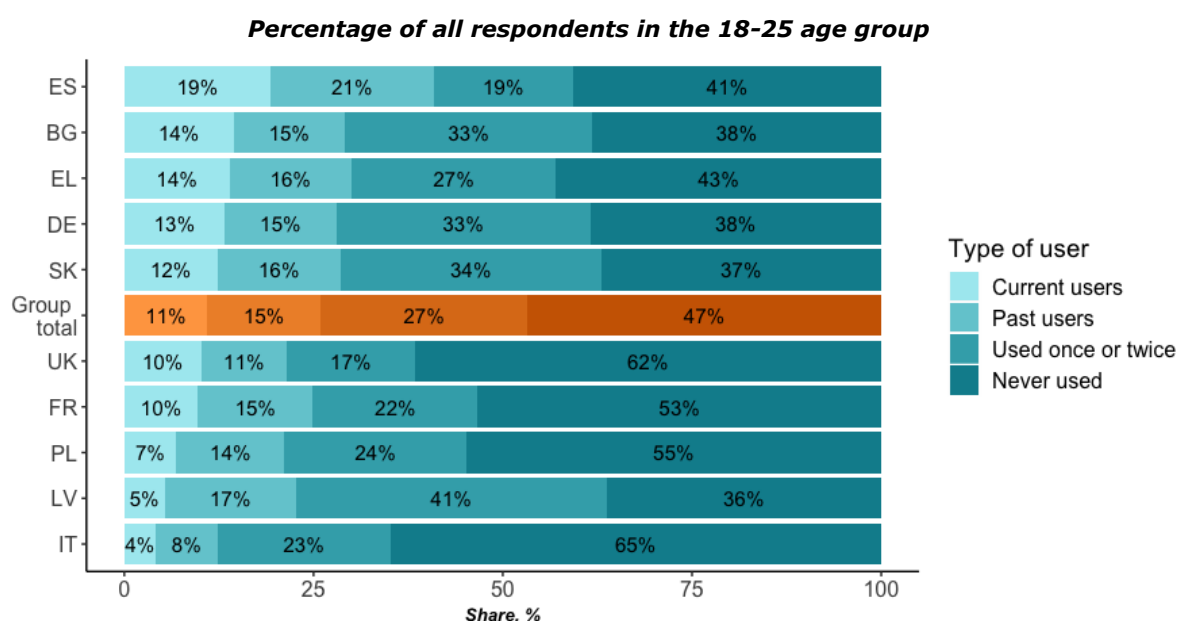
The online survey found that there was a high prevalence of waterpipe tobacco use in the **18-25 age group**. Over half of the respondents (53%) had used the product at least once or twice. Spain showed a particularly high consumption rate with one fifth (19%) of respondents reporting current waterpipe use in the 18-25 age group. There were considerably fewer current users in Latvia (5%) and Italy (4%), although two fifth (41%) of respondents in Latvia reported having tried the product once or twice. Across the Member States included in the survey, the percentages of those who had tried waterpipe tobacco was typically higher than current users, suggesting that many respondents in the 18-25 age group had experimented with the product but did not continue using them. The findings of the online survey were echoed in the focus groups in which participants from Spain, France and Germany commented on a recent surge in waterpipe tobacco popularity.

In the **26+ age group**, there were 6% of current users on average across the MS studied. Spain was once again the leading market for waterpipe tobacco consumption with 9% of reporting current use. The UK and France were particularly small markets with 83% and 79% respectively reporting having never used waterpipes. Latvia showed a similar pattern as the one noted in the 18-25 age group. While it had one of the lowest shares of current users (4%), two fifths (40%) of respondents had tried the product once or twice. Overall, waterpipe use in the 26+ age group was lower than in the younger age group. There was a large majority of never users in the 26+ age group (65%) while less than half of respondents in the 18-25 group (47%) were never users of waterpipe tobacco. The findings from the focus group confirmed that the product was mainly associated to younger generations:

"Well if I were with elders, I wouldn't take out a hookah." (M, 31, ES)

"Many minors smoke it." (M, 20, ES)

Figure 6. Use of waterpipe: country breakdown (18-25 age group)



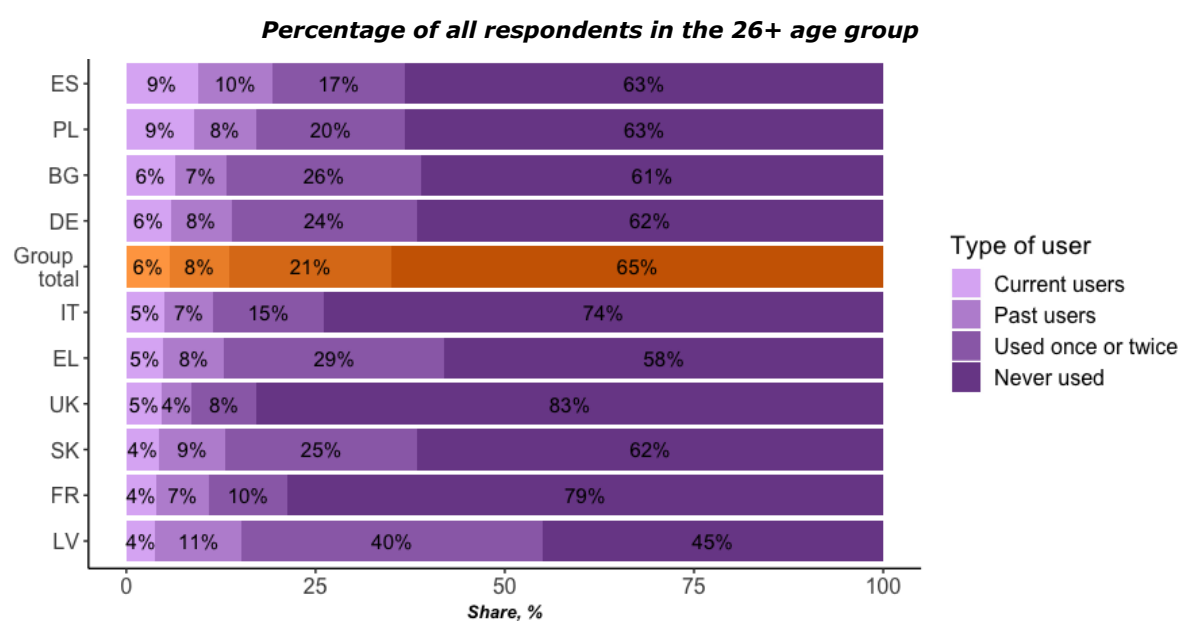
Base: all respondents in the 18-25 age group (N=6,090)

Question: Q.1.1. Have you ever used waterpipe tobacco?

Note: The group total category refers to all respondents within the age group.

Source: OE elaboration from online survey data

Figure 7. Use of waterpipe: country breakdown (26+ age group)



Base: all respondents in the 26+ age group (N=5,910)

Question: Q.1.1. Have you ever used waterpipe tobacco?

Note: The group total category refers to all respondents within the age group.

Source: OE elaboration from online survey data

Table 14 and Table 15 present a socio-demographic breakdown for each age group. For the 18-25 age group the differences were marginal in terms of gender and education level. Users were more likely to come from urban settings and to have reported higher socio-economic status. A similar pattern was observed for the 26+ age group. Urban residency and high reported socio-economic status were associated with waterpipe use.

Table 14. Waterpipe use socio-demographic breakdown (18-25 age group)

Percentage and number of respondents in the age group 18 to 25

	Group total % (n)	Current users %	Past users %	Used once or twice %	Never used %
Use of product	100 (6090)	11	15	27	47
Gender					
Male	47 (2869)	13	16	24	47
Female	53 (3221)	9	14	31	46
Education level²⁶					
1-2 (low)	5 (335)	13	16	21	50
3-4 (medium)	63 (3829)	11	14	27	49
5-6 (high)	32 (1926)	12	17	29	42
Residency					
Urban	65 (3949)	12	16	28	44
Suburban	19 (1173)	9	15	24	51
Rural	16 (968)	7	12	27	54
Self-reported socio-economic status					
1-4 (low)	18 (1126)	8	13	27	53
5-7 (medium)	65 (3954)	10	15	29	46
8-10 (high)	17 (1010)	17	19	21	43

Base: all respondents in the 18-25 age group (N=6,090)

Notes: The group total column refers to all respondents within the age group, i.e. it is a within column percentage. It provides the breakdown of each socio-demographic variable, including absolute numbers. The white cells in the four right columns represent the breakdown by user type, i.e. they are within row percentages (white cells in each row sum to 100).

* The sub-categories of the education level were the following: (1) No official education; (2) Primary school; (3) High school; (4) Some years of university (not completed); (5) University degree completed; (6) Post-graduate (master, PhD, other)

Source: OE elaboration from online survey data

²⁶ Note: (1) No official education; (2) Primary school; (3) High school; (4) Some years of university (not completed); (5) University degree completed; (6) Post-graduate (master, PhD, other)

Table 15. Waterpipe use socio-demographic breakdown (26+ age group)

Percentage and number of respondents over the age of 26

	Group total % (n)	Current users %	Past users %	Used once or twice %	Never used %
Use of product	100 (5910)	6	8	21	65
Gender					
Male	52 (3074)	7	8	22	63
Female	48 (2836)	5	7	21	67
Education level²⁷					
1-2 (low)	5 (308)	6	6	14	74
3-4 (medium)	46 (2704)	4	6	19	71
5-6 (high)	49 (2898)	7	9	25	58
Residency					
Urban	66 (3879)	7	9	23	61
Suburban	17 (1010)	5	7	20	69
Rural	17 (1021)	2	5	16	77
Self-reported socio-economic status					
1-4 (low)	23 (1374)	3	5	20	72
5-7 (medium)	64 (3769)	5	8	22	65
8-10 (high)	13 (767)	15	12	22	51

Base: all respondents in the 26+ age group (N=5,910)

Notes: The group total column refers to all respondents within the age group, i.e. it is a within column percentage. It provides the breakdown of each socio-demographic variable, including absolute numbers. The white cells in the four right columns represent the breakdown by user type, i.e. they are within row percentages (white cells in each row sum to 100).

* The sub-categories of the education level were the following: (1) No official education; (2) Primary school; (3) High school; (4) Some years of university (not completed); (5) University degree completed; (6) Post-graduate (master, PhD, other)

Source: OE elaboration from online survey data

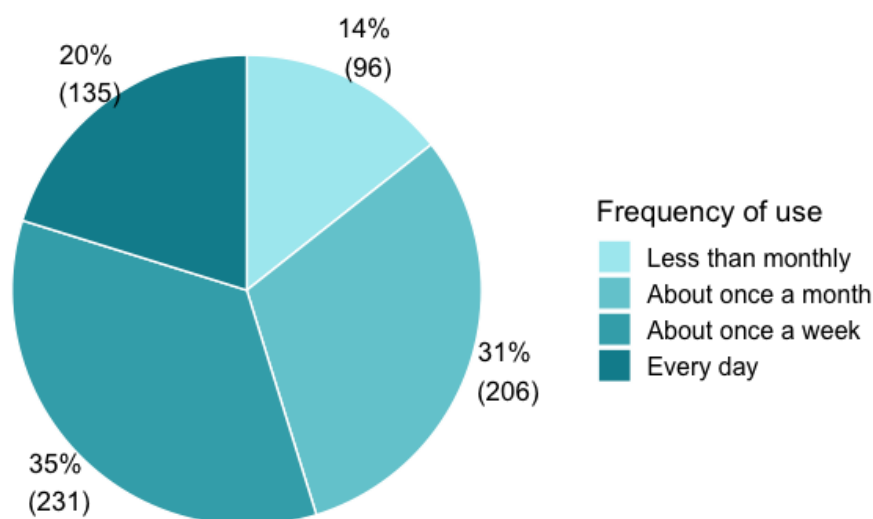
²⁷ Note: (1) No official education; (2) Primary school; (3) High school; (4) Some years of university (not completed); (5) University degree completed; (6) Post-graduate (master, PhD, other)

2.2.2 Pattern of use

Figure 8 shows that in the 18-25 age group, one fifth of current users (20%) reported daily use of the product, and 35% reported weekly usage. Although there were fewer current users in the 26+ age group (see Figure 9) those that did used the product appeared to do so more often: a quarter (25%) reported waterpipe use daily and close to two fifths (37%) reported using waterpipe tobacco weekly.

Figure 8. Frequency of use of current waterpipe users (18-25 age group)

Percentage and number of current users in the 18-25 age group



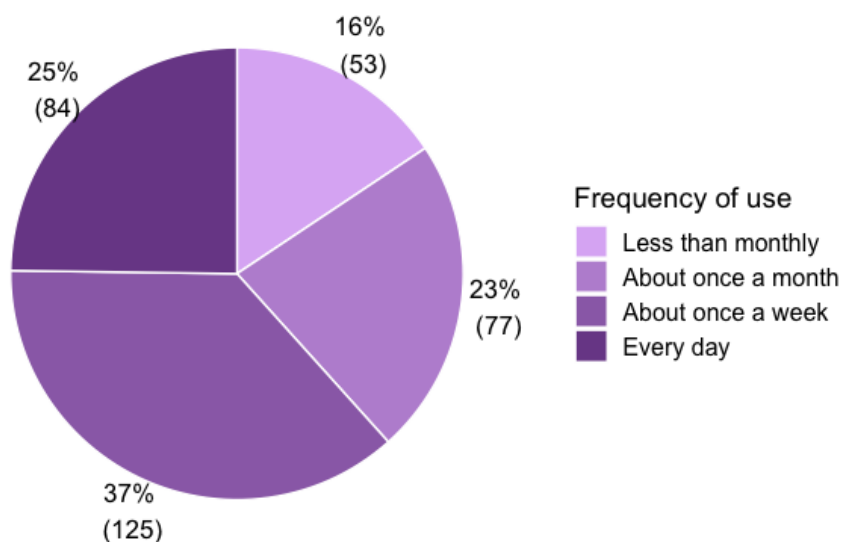
Base: current users of waterpipe tobacco in the 18-25 age group (N=668)

Question: Q.1.3. How often do you use waterpipe (i.e. shisha & narghile) tobacco?

Source: OE elaboration from online survey data

Figure 9. Frequency of use of current waterpipe users (26+ age group)

Percentage and number of current users in the 26+ age group



Base: current users of waterpipe tobacco in the 26+ age group (N=339)

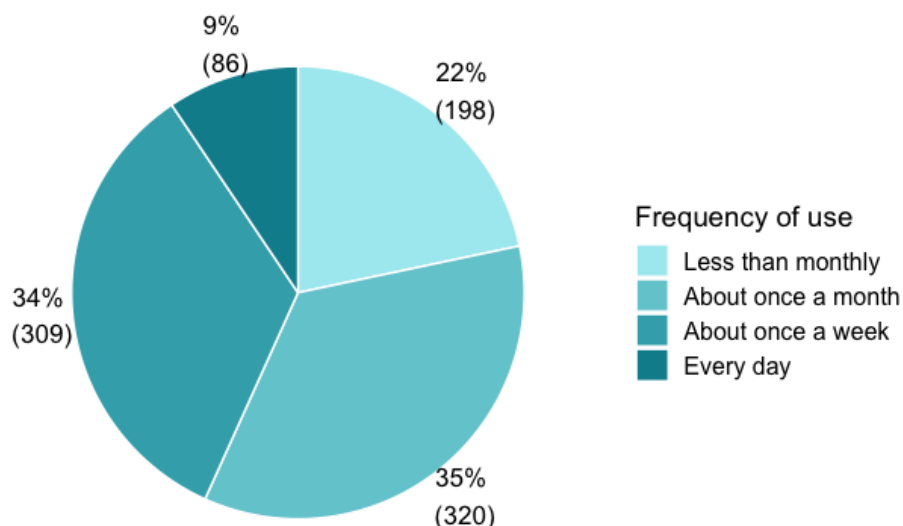
Question: Q.1.3. How often do you use waterpipe (i.e. shisha & narghile) tobacco?

Source: OE elaboration from online survey data

Figure 10 and Figure 11 show that, in both age groups, only about one in ten past users reported daily use of waterpipes, considerably fewer than current users. On the criterion of use 'every day and about once a week' there were 55% of users in the 18-25 age group and 62% in the 26+ age group. This compares with 43% and 51% among the past users an indication of the increasing popularity of waterpipe tobacco.

Figure 10. Frequency of use of past waterpipe users (18-25 age group)

Percentage and number of current users in the 18-25 age group



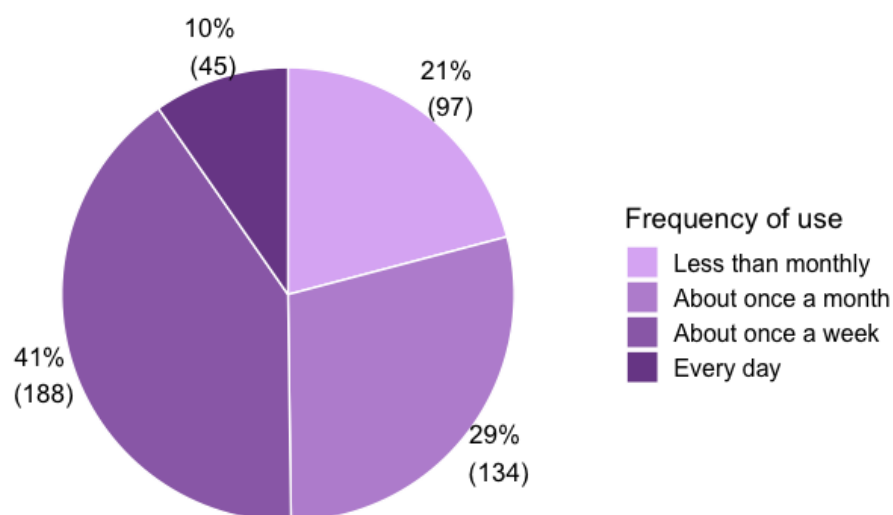
Base: past users of waterpipe tobacco in the 18-25 age group (N=913)

Question: Q.1.6. How often did you use waterpipe (i.e. shisha & narghile) tobacco?

Source: OE elaboration from online survey data

Figure 11. Frequency of use of past waterpipe users (26+ age group)

Percentage and number of current users in the 26+ age group



Base: past users of waterpipe tobacco in the 26+ age group (N=464)

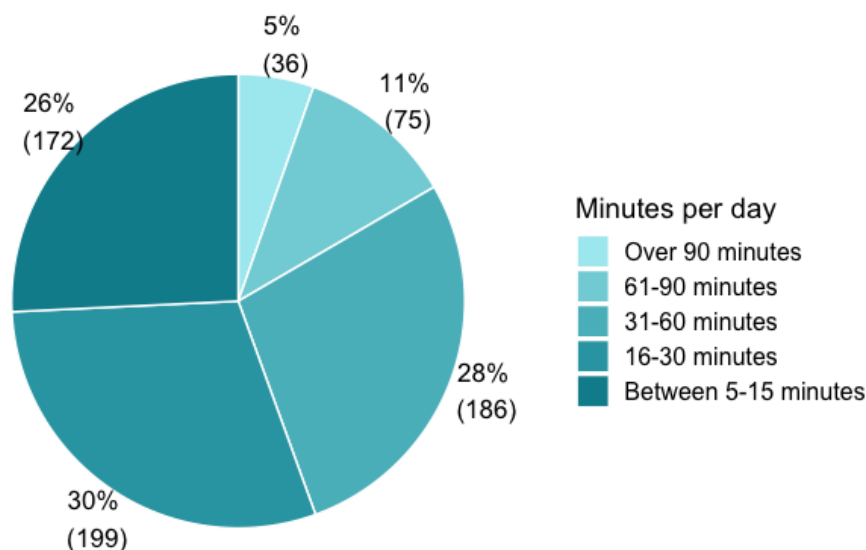
Question: Q.1.6. How often did you use waterpipe (i.e. shisha & narghile) tobacco?

Source: OE elaboration from online survey data

Although users in the 26+ age group consumed waterpipe tobacco more often, the users in the 18-25 age group reported using the product more intensively. Over two fifths (44%) of current users in the 18-25 age group reported consuming it for over 30 minutes per day, compared to 30% in the 26+ age group.

Figure 12. Average number of minutes per day of waterpipe use (18-25 age group)

Percentage and number of current users in the 18-25 age group



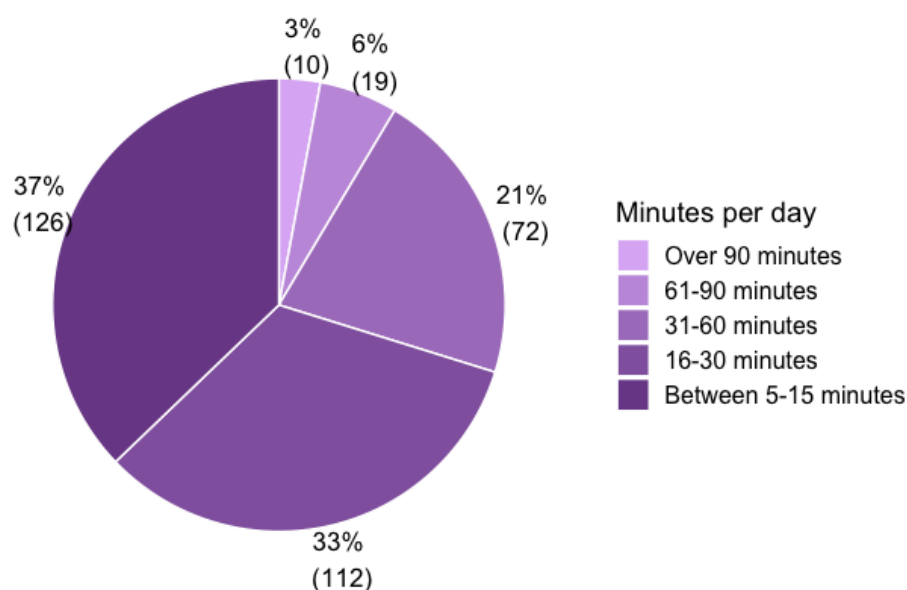
Base: current users of waterpipe tobacco in the 18-25 age group (N=668)

Question: Q.1.4.a. On the days that you smoke/use, about how much do you smoke/use waterpipe (i.e. shisha or narghile) tobacco?

Source: OE elaboration from online survey data

Figure 13. Average number of minutes per day of waterpipe use (26+ age group)

Percentage and number of current users in the 26+ age group



Base: current users of waterpipe tobacco in the 26+ age group (N=339)

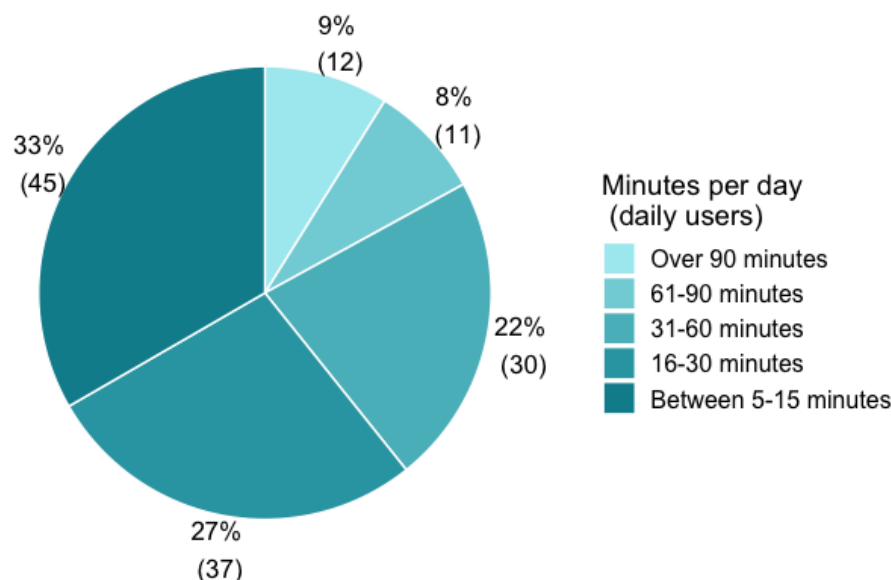
Question: Q.1.4.a. On the days that you smoke/use, about how much do you smoke/use waterpipe (i.e. shisha or narghile) tobacco?

Source: OE elaboration from online survey data

Figure 14 and Figure 15 report how much time daily users spend on waterpipes. A majority (60%) of the 18-25 years group reported less than 31 minutes per day. The 26+ age group reported the similar figure (64%).

Figure 14. Average number of minutes per day for daily waterpipe users (18-25 age group)

Percentage and number of daily users in the 18-25 age group



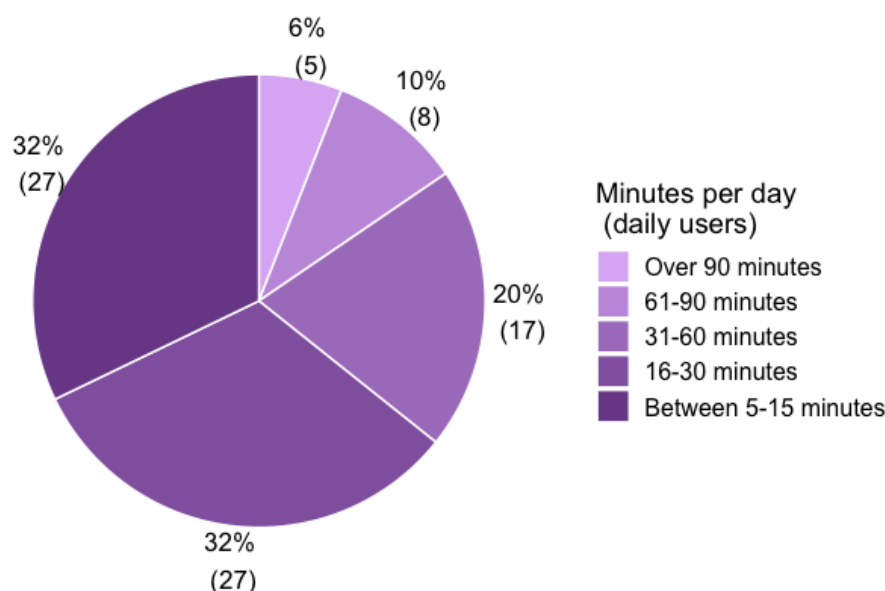
Base: daily current users of waterpipe tobacco in the 18-25 age group (N=135)

Question: Q.1.4.a. On the days that you smoke/use, about how much do you smoke/use waterpipe (i.e. shisha or narghile) tobacco?

Source: OE elaboration from online survey data

Figure 15. Average number of minutes per day, for daily waterpipe users (26+ age group)

Percentage and number of daily users in the 26+ age group



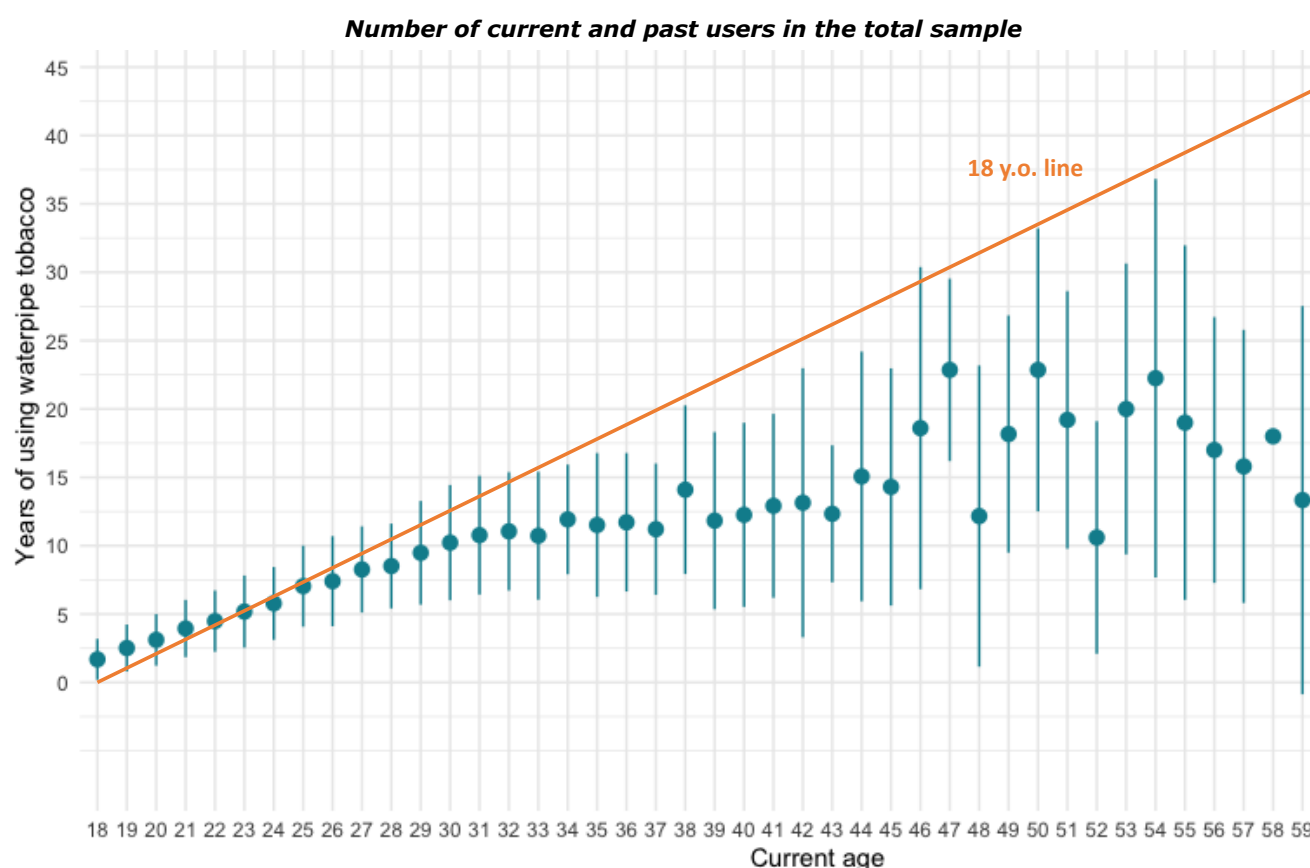
Base: daily current users of waterpipe tobacco in the 26+ age group (N=84)

Question: Q.1.4.a. On the days that you smoke/use, about how much do you smoke/use waterpipe (i.e. shisha or narghile) tobacco?

Source: OE elaboration from online survey data

On average, respondents that were under the age of 22 years had on average begun using waterpipe tobacco as minors as did other under the age of 32 years. However, on average users between the age of 30 and 45 has started using about ten years ag, while the age of initiation for those over 45 years was more varied. A possible explanation is that waterpipes have become more popular, available and socially acceptable in the last 10-15 years.

Figure 16. Initiation of use for current and past users (waterpipe; total sample)



Base: current and past users of waterpipe tobacco in the total sample ($N=2,384^{28}$)

Question: Q.1.2.a. At about what age did you start using waterpipe tobacco? & Q.5.2. Age

Note: The figure illustrates the approximate starting age of waterpipe tobacco use among all respondents in the online survey, where the x-axis represents the current age of the respondent (Q.5.2) and the y-axis is the average numbers of years of product consumption (calculated by subtracting the current age from the starting age). The orange line represents the 45° or 18 y.o. line. Each blue point represents the average starting age for respondents of that age; points above the 18 y.o. line reveal those initiating consumption of waterpipe tobacco before the age of 18, while those below the line reveal age brackets starting consumption after the age of 18 on average. The blue lines represent the mean standard deviation, i.e. it does not show the full range of values for each age.

Source: OE elaboration from online survey data

Respondents in the 18-25 age group had on average used the product for 3 years as compared to an average of 7 years in the 26+ age group (see Figure 176 and Figure 177). Many reported using the product for 7 to 11 years in the 18-25 age group, which confirms the findings above that younger respondents had begun using the product as minors.

Enjoyment and sociability play a key role in the initiation of waterpipe tobacco use across age groups. Table 16 reports that 60% of the 18-25 age group cited 'for enjoyment' as the reason for taking up waterpipe smoking. Other reasons included 'to socialise' and 'it was trendy' (35%) and 'everybody

²⁸ NA and misreported values are excluded.

else is smoking' (18%). Table 17 shows that this pattern of responses held for both current users and those who had tried waterpipes once or twice. There is a remarkably similar pattern among the 26+ age group with 58% of current and former users citing 'enjoyment' and 32% 'socialising and trendy'.

Table 16. Reasons for starting to use or trying waterpipe (18-25 age group)

Percentages by user profile in the 18-25 age group

Reasons	Current & past users, %	Used once or twice, %
For enjoyment	60	58
To socialise / it was trendy	35	42
I was trying to quit or reduce consumption of another product	9	3
I have parents, adults or guardians who smoke*	6	4
To show independence or defiance	8	6
Everyone else is smoking	18	18
Tobacco advertisements	7	5
Base (n)	(1,581)	(1,622)

Base: the base of each column is provided in the last row of the table.

Question: Q.2.1.a. Why did you start using waterpipe tobacco? Please select up to 2 options.

Q.2.1.b. Why did you try waterpipe tobacco? Please select up to 2 options.

Note: Each column focusses on a specific consumer profile (i.e. current & past users or used once or twice) for the 18-25 age group and provides a breakdown of the reasons identified for starting to use waterpipe tobacco. Note that the percentages do not add up to 100% because respondents could select more than one term.

* The type of product 'smoked' was not specified in the survey options.

Source: OE elaboration from online survey data

Table 17. Reasons for starting to use or trying waterpipe (26+ age group)

Percentage by user profile in 26+ age group

Reasons	Current & past users, %	Used once or twice, %
For enjoyment	58	58
To socialise / it was trendy	32	38
I was trying to quit or reduce consumption of another product	11	4
I have parents, adults or guardians who smoke*	6	3
To show independence or defiance	11	6
Everyone else is smoking	14	11
Tobacco advertisements	10	6
Base (n)	(803)	(1,268)

Base: the base of each column is provided in the last row of the table.

Question: Q.2.1.a. Why did you start using waterpipe tobacco? Please select up to 2 options. Q.2.1.b. Why did you try waterpipe tobacco? Please select up to 2 options.

Note: Each column focusses on a specific consumer profile (i.e. current & past users or used once or twice) for the 26+ age group and provides a breakdown of the reasons identified for starting to use waterpipe tobacco. Note that the percentages do not add up to 100% because respondents could select more than one term.

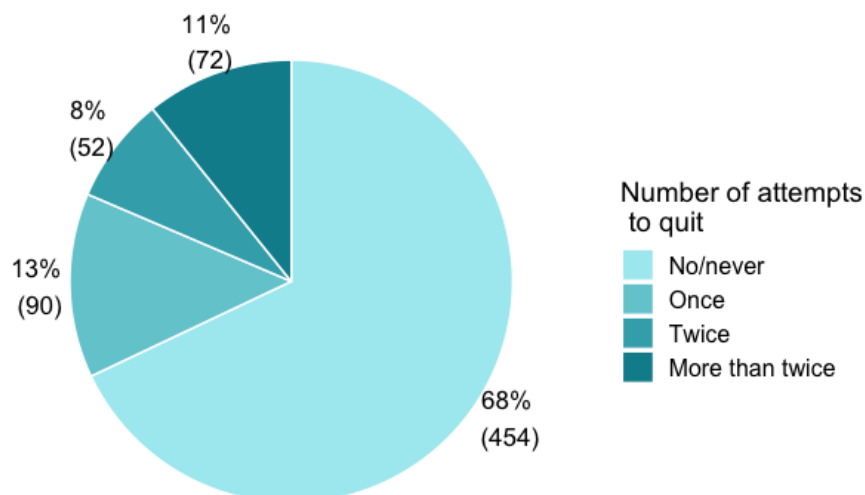
* The type of product 'smoked' was not specified in the survey options.

Source: OE elaboration from online survey data

Figure 17 and Figure 18 show that 68% of current users in the 18-25 age group had not attempted to quit waterpipe tobacco use compared to 51% of the 26+ age group.

Figure 17. Cessation: attempts to quit waterpipe (18-25 age group)

Percentage and number of current users in the 18-25 age group



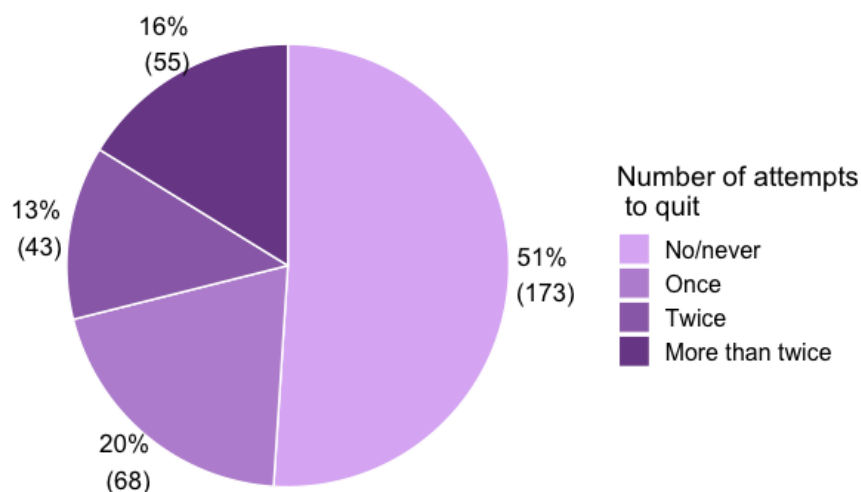
Base: current users of waterpipe tobacco in the 18-25 age group (N=668)

Question: Q.1.5. Have you ever tried to quit waterpipe tobacco? If yes, how many times?

Source: OE elaboration from online survey data

Figure 18. Cessation: attempts to quit waterpipe (26+ age group)

Percentage and number of current users in the 26+ age group



Base: current users of waterpipe tobacco in the 26+ age group (N=339)

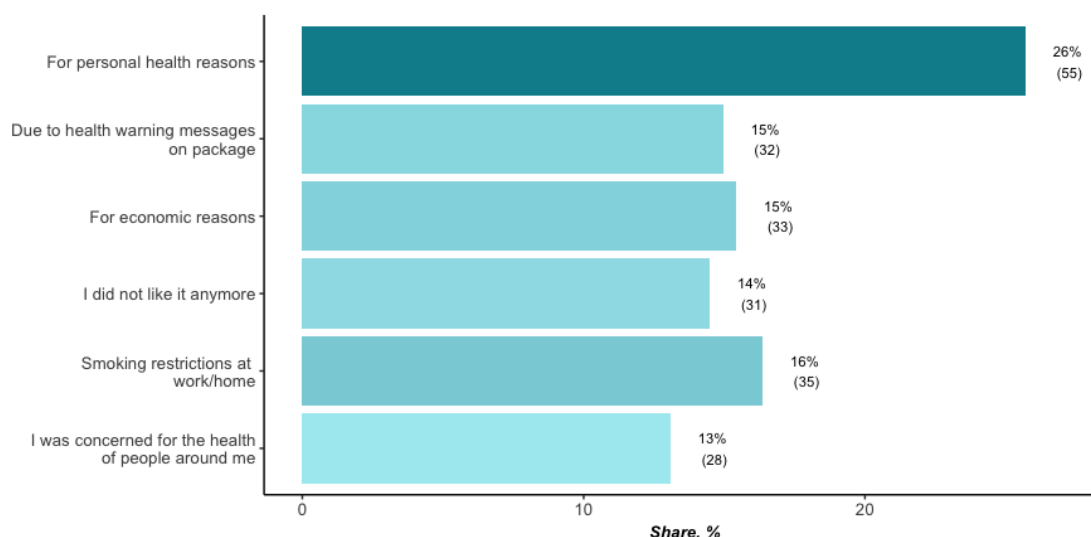
Question: Q.1.5. Have you ever tried to quit waterpipe tobacco? If yes, how many times?

Source: OE elaboration from online survey data

About a quarter of respondents in both age categories who had attempted to quit waterpipe tobacco use had done so for 'personal health reasons'. The second most cited reason for attempting to quit in the 18-25 age group was 'smoking restrictions at work/home', while it was 'due to health warning messages on the package' for the 26+ age group.

Figure 19. Cessation: reasons for trying to quit waterpipe use (18-25 age group)

Percentage and number of current users in the 18-25 age group



Base: current users of waterpipe tobacco that tried to quit in the 18-25 age group (N=214)

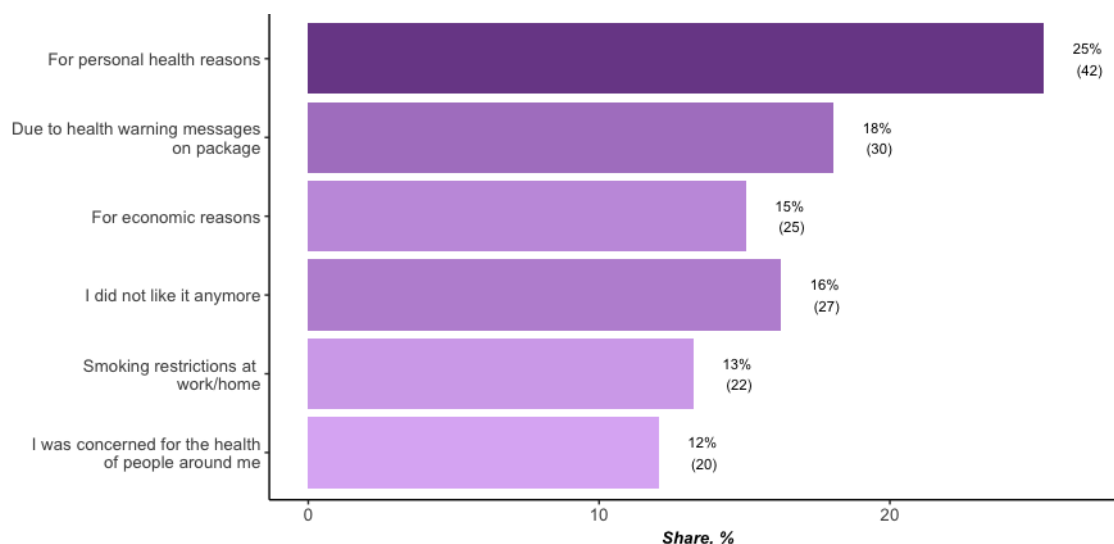
Question: Q2.2. What made you try to quit waterpipe tobacco? Please select the most fitting answer (single answer).

Note: Each bar represents the proportion of respondents that selected the option within the age group.

Source: OE elaboration from online survey data

Figure 20. Cessation: reasons for trying to quit waterpipe use (26+ age group)

Percentage and number of current users in 26+ age group



Base: current users of waterpipe tobacco that tried to quit in the 26+ age group (N=166)

Question: Q2.2. What made you try to quit waterpipe tobacco? Please select the most fitting answer (single answer).

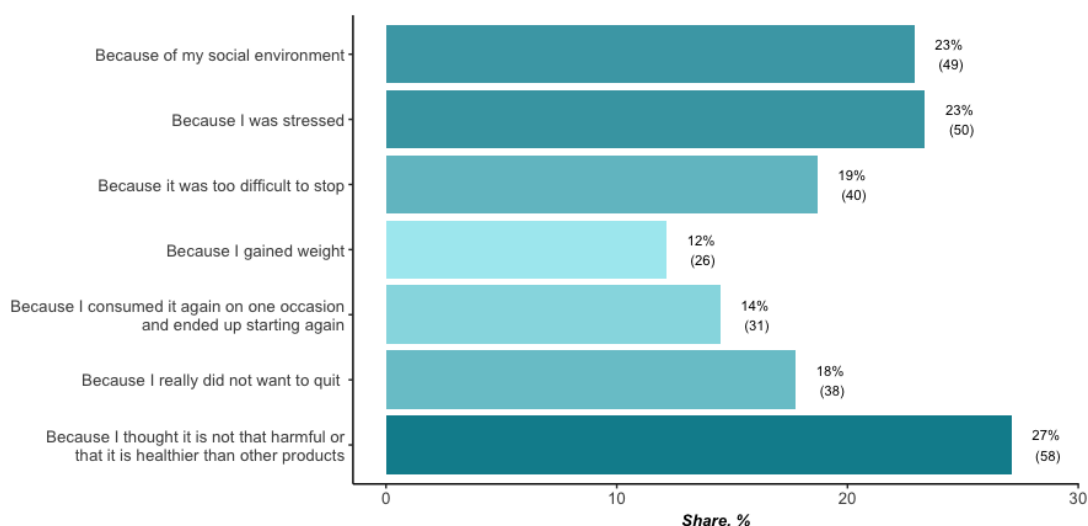
Note: Each bar represents the proportion of respondents that selected the option within the age group.

Source: OE elaboration from online survey data

For respondents in the 18-25 age group, the most cited reason for starting to use waterpipe tobacco again after an attempt to quit was because they believed that waterpipe tobacco was not that harmful or that it was healthier than other products (27%) followed closely by 'stress' and 'my social environment'. In the 26+ age group, the 'social environment' was the main reason for starting waterpipe smoking again (28%) followed by 'stress' (24%)

Figure 21. Reasons for starting to use waterpipe again (18-25 age group)

Percentage and number of current users in the 18-25 age group



Base: current users of waterpipe tobacco that tried to quit in the 18-25 age group (N=214)

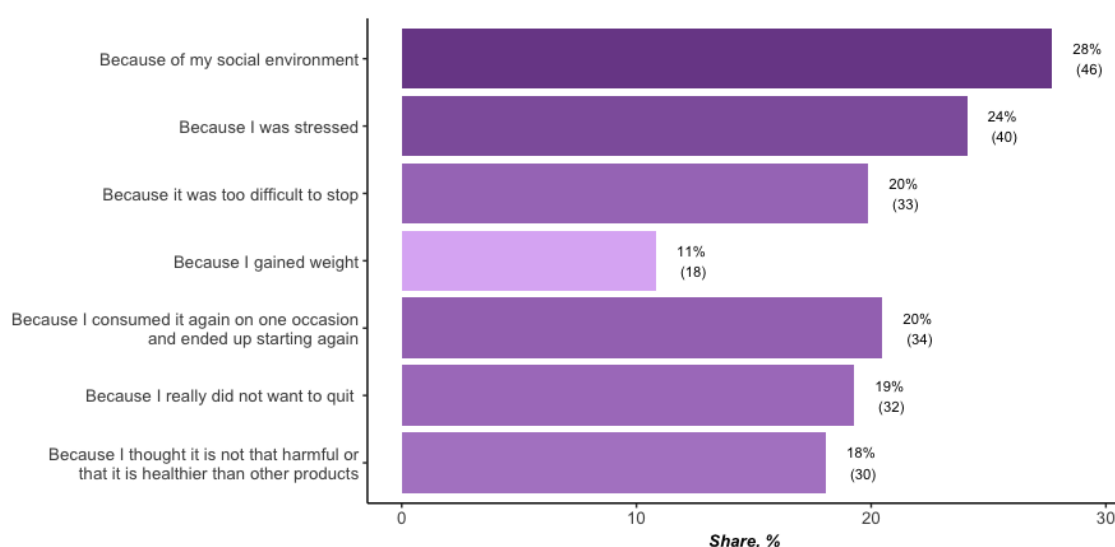
Question: Q.2.3. Why did you start to use waterpipe tobacco again? Select up to 2 options.

Note: Each bar represents the proportion of respondents that selected the option within the age group. The percentages do not add up to 100% because respondents could select more than one answer; therefore, the bars must be interpreted separately.

Source: OE elaboration from online survey data

Figure 22. Reasons for starting to use waterpipe again (26+ age group)

Percentage and number of current users in the 26+ age group



Base: current users of waterpipe tobacco that tried to quit in the 26+ age group (N=166)

Question: Q.2.3. Why did you start to use waterpipe tobacco again? Select up to 2 options.

Note: Each bar represents the proportion of respondents that selected the option within the age group. The percentages do not add up to 100% because respondents could select more than one answer; therefore, the bars must be interpreted separately.

Source: OE elaboration from online survey data

2.2.3 Dual use with other tobacco and related products

Over two thirds (67%) of current users of waterpipe tobacco in the **18-25 group** consumed other tobacco products. One in six current users (17%) indicated that they had previously used other products but were now only using waterpipe tobacco products. A similar share (16%) had been exclusive users of waterpipe tobacco (see Table 18). A large majority of current users (87%) in the **26+ group** were current dual users while the share of those that were only consuming waterpipe tobacco now was much lower in this age group (see Table 19). Figure 182 and Figure 183 in Annex 0 provide an indicative visualisation of the user scenarios amongst current consumers of waterpipe tobacco in the 18-25 group and the 26+ age group, respectively.

The pattern of use can be partially explained by the perception of the product. Focus group participants tended to distinguish between waterpipe tobacco and other tobacco products. As the waterpipe was not expected to satisfy tobacco related cravings, users typically did not regard its use as “real smoking”.

Table 18. Career flows of use for current waterpipe users (18-25 age group)
Percentage and number of current users in the 18-25 age group

			Number	%
Total			668	100
Never users of other products			107	16
Exclusive current user of waterpipe tobacco & past user of other product(s)			113	17
<i>Out of them, switching from (one or more products) *</i>				
Cigarettes	(n=81)	72% *		
E-cigarettes	(n=46)	41% *		
Slim cigarettes	(n=42)	37% *		
Small cigarillos	(n=19)	17% *		
HTPs	(n=19)	17% *		
Current user of waterpipe & both current and past user of other product(s)			448	67
<i>Other products consumed in parallel now *</i>				
Cigarettes	(n=337)	75% *		
E-cigarettes	(n=281)	63% *		
Slim cigarettes	(n=226)	50% *		
Small cigarillos	(n=188)	42% *		
HTPs	(n=219)	49% *		

Base: current users of waterpipe tobacco in the 18-25 age group (N=668);
For the rows in blue and italics, the base is the respondents that are current (exclusive) users of the product and that used other products only in the past (N=113)

Question: Q.1.1. Have you ever used waterpipe tobacco?

Note: Patterns of use and consumption careers are explored for different consumer profiles of current users. A breakdown is provided for the two variables of interest, i.e. current parallel use patterns. The last row only provides the breakdown of current dual use as this is the variable of interest; it does not include a breakdown that takes account past consumption of products to avoid confusion (from too many permutations).

(*) Percentages add up to more than 100% on account of the prevalence of dual use.

Source: OE elaboration from online survey data

Table 19. Career flows of current waterpipe users (26+ age group)
Percentage and number of current users in the 26+ age group

			Number	%
Total			339	100
Never users of other products			20	6
Exclusive current user of waterpipe tobacco & past user of other product(s)			25	7
<i>Out of them, switching from (one or more products) *</i>				
Cigarettes	(n=19)	76% *		
E-cigarettes	(n=12)	48% *		
Slim cigarettes	(n=11)	44% *		
Small cigarillos	(n=5)	20% *		
HTPs	(n=4)	16% *		
Current user of waterpipe & both current and past user of other product(s)			294	87
<i>Other products consumed in parallel now *</i>				
Cigarettes	(n=246)	84% *		
E-cigarettes	(n=202)	67% *		
Slim cigarettes	(n=173)	59% *		
Small cigarillos	(n=162)	55% *		
HTPs	(n=182)	62% *		

Base: current users of waterpipe tobacco in the 26+ age group (N=339)
 For the rows in purple and italics, the base is the respondents that are current (exclusive) users of the product and that used other products only in the past (N=25)

Question: Q.1.1. Have you ever used waterpipe tobacco?

Note: Patterns of use and consumption careers are explored for different consumer profiles of current users. A breakdown is provided for the two variables of interest, i.e. current parallel use patterns. The last row only provides the breakdown of current dual use as this is the variable of interest; it does not include a breakdown that takes account past consumption of products to avoid confusion (from too many permutations).

Source: OE elaboration from online survey data

Table 20 shows that 44% of past users of waterpipe tobacco in the **18-25 age group**, were no longer consuming any tobacco or related product (18% had used waterpipe tobacco exclusively and were no longer consuming it and 26% were dual past user and not using any products now). Furthermore, 14% of past users were exclusive past users of waterpipe tobacco and had transitioned to other tobacco and related product(s) – mainly conventional cigarettes.

Within the **26+ age group** (see Table 21), 58% of past waterpipe tobacco users were also current users of other tobacco products, and 15% had been exclusive past users who had transitioned to other products, mainly conventional cigarettes.

Table 20. Career flow of past waterpipe users (18-25 age group)
Percentage and number of current users in the 18-25 age group

	Number	%
Total	913	100
Never users of other products	161	18
Dual past user & not using any products now	241	26
Exclusive past user of waterpipe tobacco & current user of other products (but not waterpipe tobacco)	132	14
<i>Out of them, switching to (one or more products) *</i>		
Cigarettes (n=120)	91% *	
E-cigarettes (n=46)	35% *	
Slim cigarettes (n=27)	20% *	
HTPs (n=19)	14% *	
Small cigarillos (n=17)	13% *	
Dual past user & current user of other products (but not waterpipe tobacco)	379	42
<i>Other products currently consumed *</i>		
Cigarettes (n=260)	69% *	
E-cigarettes (n=123)	32% *	
Small cigarillos (n=42)	11% *	
Slim cigarettes (n=82)	22% *	
HTPs (n=84)	21% *	

Base: past users of waterpipe tobacco in the 18-25 age group (N=913);

Question: Q.1.1. Have you ever used waterpipe tobacco?

Note: Patterns of use and consumption careers are explored for different consumer profiles of past users. A breakdown is provided for the variables of interest, i.e. those who continued using other tobacco products after using e-cigarettes. The last row only provides the breakdown of current consumption because this is the variable of interest; it does not include a breakdown that takes account past consumption of products to avoid confusion (from too many permutations).

(*) Percentages add up to more than 100% on account of the prevalence of dual use.

Source: OE elaboration from online survey data

Table 21. Career flows of past waterpipe users (26+ age group)
Percentage and number of current users in the 26+ age group

	Number	%
Total	464	100
Never users of other products	25	5
Dual past user & not using any products now	142	31
Exclusive past user of waterpipe tobacco & current user of other products (but not waterpipe tobacco)	70	15
<i>Out of them, switching to (one or more products) *</i>		
Cigarettes (n=63)	90% *	
E-cigarettes (n=31)	44% *	
HTPs (n=19)	27% *	
Slim cigarettes (n=18)	26% *	
Small cigarillos (n=12)	17% *	
Dual past user & current user of other products (but not waterpipe tobacco)	227	49
<i>Other products currently consumed in parallel *</i>		
Cigarettes (n=159)	70% *	
E-cigarettes (n=115)	51% *	
Small cigarillos (n=40)	18% *	
Slim cigarettes (n=63)	28% *	
HTPs (n=64)	28% *	

Base: past users of waterpipe tobacco in the 26+ age group (N=464)

Question: Q.1.1. Have you ever used waterpipe tobacco?

Note: Patterns of use and consumption careers are explored for different consumer profiles of past users. A breakdown is provided for the variables of interest, i.e. those who continued using other tobacco products after using e-cigarettes. The last row only provides the breakdown of current consumption because this is the variable of interest; it does not include a breakdown that takes account past consumption of products to avoid confusion (from too many permutations).

(*) Percentages add up to more than 100% on account of the prevalence of dual use.

Source: OE elaboration from online survey data

2.3 Consumer perception

According to focus group participants, the consumption of waterpipe tobacco was an emerging trend in Europe especially among younger generations. It was also perceived as a timeless and classic product due to its long presence on the market. Waterpipe tobacco was considered a non-conventional tobacco product that evoked the ideas of “zen,” “pleasure,” and “relaxation” amongst participants. Moreover, the use of waterpipe tobacco was sometimes perceived as endemic to specific cultures (e.g. Turkish people in Germany).

“Shisha is being hyped by the rap scene. The youth thinks they are cool using this.”
(M, 24, DE)

“(…) Waterpipe tobacco, nowadays it is a new fashion.” (M, 28, ES)

“Timeless. It has been used for generations” (M, 36, EL)

In the online survey, respondents in the **18-25 age group** (see Table 22) most often associated waterpipe tobacco with ‘nice flavours (e.g. apple, mint)’ with 42% of respondents recording this answer. This was a common view among those who had used the product once or twice. In addition, 37% of respondents in this age group characterised the product as ‘enjoyable’. Once again, this response was evident among those who had used waterpipes once or twice, as well as users of the product. The more negative terms (e.g. ‘complicated’, ‘unhealthy’ and ‘addictive’) were mainly cited by never users. Very few respondents (7%) considered the product to be ‘practical’.

Table 22. Perception of waterpipe tobacco (18-25 age group)
Percentage and number of current users in the 18-25 age group

	Group total % (n)	Current users %	Past users %	Used once or twice %	Never used %
Use of product	100 (6090)	11	15	27	47
Complicated	20 (1212)	7	12	22	58
Unhealthy	35 (2135)	6	11	26	57
Costly	24 (1440)	9	14	23	54
Addictive	24 (1470)	9	11	21	59
Practical	7 (416)	20	19	23	38
Nice flavours (e.g. apple, mint)	42 (2557)	13	18	36	34
Elegant/trendy/fashionable	28 (1707)	14	16	31	38
Enjoyable	37 (2237)	15	19	37	28

Base: respondents in the 18-25 age group (N=6,090)

Question: Q.4.15. Which of the following words best captures what you think about waterpipe (e.g. shisha & narghile) tobacco? Please tick as many boxes as you would like.

Note: The group total column refers to all respondents within the age group, i.e. it is a within column percentage, e.g. 7% of respondents in this age group selected “practical” to describe waterpipe. The white cells (in the four right columns) represent the breakdown by user type, i.e. they are within row percentages (white cells in each row sum to 100).

Source: OE elaboration from online survey data

In the **26+ age group**, 39% of respondents characterised the product as '*unhealthy*'; this was the most often cited term in this age group (see Table 23). '*Nice flavours*' were also often cited, with 31% of respondents polling this response; those that had tried the product once or twice were overrepresented in the group that provided this response. In short, flavours appeared to be strongly linked to waterpipe tobacco use, especially among the younger age group, but respondents also seemed aware of the unhealthy aspect of waterpipe use.

Table 23. Perception of waterpipe tobacco; 26+ age group)
Percentage and number of current users in the 26+ age group

Use of product	Group total % (n)	Current users %	Past users %	Used once or twice %	Never used %
Use of product	100 (5910)	6	8	21	65
Complicated	26 (1565)	4	5	20	72
Unhealthy	39 (2330)	2	5	17	76
Costly	20 (1165)	5	6	19	70
Addictive	27 (1576)	4	6	15	75
Practical	5 (293)	20	15	20	45
Nice flavours (e.g. apple, mint)	31 (1842)	7	11	33	48
Elegant/trendy/fashionable	19 (1106)	8	11	26	55
Enjoyable	27 (1619)	9	14	34	43

Base: respondents in the 26+ age group (N=5,910)

Question: Q.4.15. Which of the following words best captures what you think about waterpipe (e.g. shisha & narghile) tobacco? Please tick as many boxes as you would like.

Note: The group total column refers to all respondents within the age group, i.e. it is a within column percentage. The white cells (in the four right columns) represent the breakdown by user type, i.e. they are within row percentages (white cells in each row sum to 100).

Source: OE elaboration from online survey data

The focus group participants' views complemented the online survey findings. The range of flavours added to the enjoyment of waterpipe tobacco consumption:

"At the start with cigarettes there were more choices with flavours but now with all the regulations we just have mint whereas with narghile we are more open to choices." (M, 33, FR)

"When I see things in several flavours, I feel like trying them." (M, 20, FR)

"[The flavours] They are available everywhere. Yes, it is an advantage." (M, 25, BG)

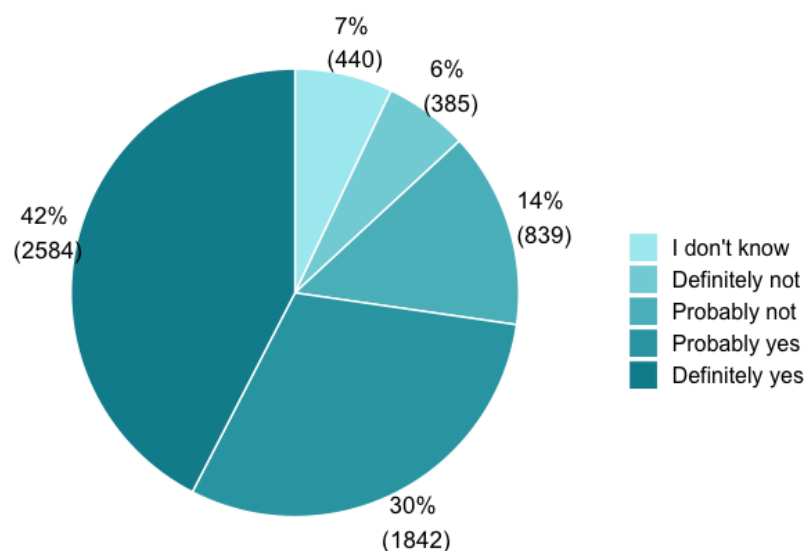
While flavours were often mentioned as an enticing factor, participants in the focus groups also expected the social ritual surrounding waterpipe tobacco consumption and the seemingly lower prices to influence younger generations use of the product. Indeed, youth was expected to have less economic resources and therefore, the intermittent use of the product (as reported in Figure 8 and

Figure 9) leads the use of the product to be interpreted as a cheaper habit when compared to other tobacco and related products. Indeed, attempts to quit consumption of this product were rarely motivated by economic reasons.

Respondents in the online survey were asked whether they thought that waterpipe tobacco might be particularly appealing to younger people. The figures below report that 72% of the 18-25 age group agreed as did 64% of the 26+ age group.

Figure 23. Perceived appeal of waterpipe to youth (waterpipe; 18-25 age group)

Percentage and number of current users in the 18-25 age group



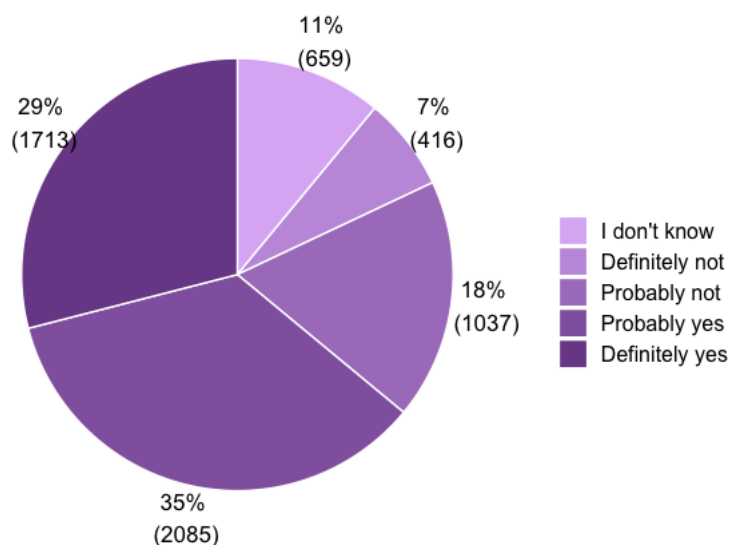
Base: respondents in the 18-25 age group (N=6,090)

Question: Q.4.16. Do you think that young people might be particularly attracted to waterpipe (e.g. shisha & narghile) tobacco, or not?

Source: OE elaboration from online survey data

Figure 24. Perceived appeal of waterpipe to youth (waterpipe; 26+ age group)

Percentage and number of current users in the 26+ age group



Base: respondents in the 26+ age group (N=5,910)

Question: Q.4.16. Do you think that young people might be particularly attracted to waterpipe (e.g. shisha & narghile) tobacco, or not?

Source: OE elaboration from online survey data

Younger participants in the focus group (aged 20 to 30 y.o.) characterised waterpipes as an experimental and innovative product. They viewed it as a social ritual and believed that it was better than other tobacco and related products.

"Young girls and young boys [from 15 to 25] who don't smoke much" on those that are likely to experiment with waterpipe tobacco; (M, 32, ES)

"I've noticed that, when I was 16 or 17, when I told my mother that I'm going to a shisha bar to celebrate a birthday, she said okay. [...] It's more tolerated than smoking cigarettes." (M, 24, DE)

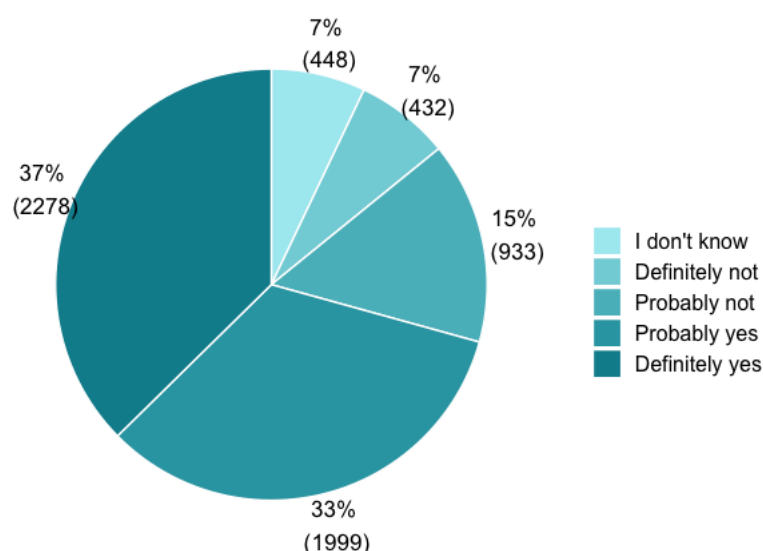
"A student, at 5 in the morning, dawn, the party is over, and they sit around, the last ones hang out, they smoke a narghile" on the perceived ritual of waterpipe tobacco consumption; (M, 27, EL)

"[Waterpipe smokers] are mostly young." (M, 20, FR)

Respondents of the online survey were also concerned for the younger generations. Seven in ten respondents in both age groups agreed that youth might underestimate the risk of using waterpipe tobacco and this belief was relatively equal across user types.

Figure 25. Extent to which respondents believed that youth underestimate risks of using waterpipe tobacco (18-25 age group)

Percentage and number of current users in the 18-25 age group



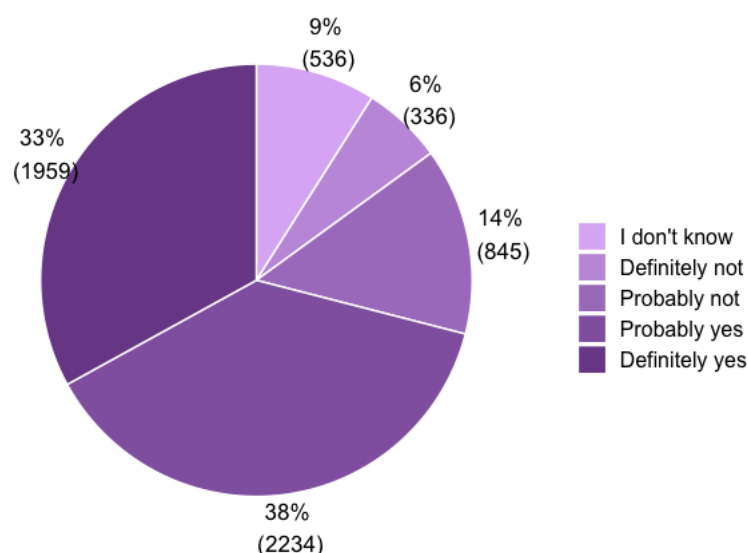
Base: respondents in the 18-25 age group (N=6,090)

Question: Q.4.17. Do you think that young people might underestimate the risk of using to waterpipe (e.g. shisha & narghile) tobacco, or not?

Source: OE elaboration from online survey data

Figure 26. Extent to which respondents believed that youth underestimate risks of using waterpipe tobacco (26+ age group)

Percentage and number of current users in 26+ age group



Base: respondents in the 26+ age group (N=5,910)

Question: Q.4.17. Do you think that young people might underestimate the risk of using to waterpipe (e.g. shisha & narghile) tobacco, or not?

Source: OE elaboration from online survey data

While the previous tables evidenced concerns about the risk of waterpipe use, in the focus groups we heard a range of views.

"For me the waterpipe recently has started to be considered harmful." (M, 24, BG)

"Ok, so what is most harmful is the shisha/narghile." (M, 20, FR)

"The IQOS and the waterpipe, as they work with heating, they are considered less harmful (than normal tobacco) although there are still not enough data." (M, 25, BG)

"Someone told me if you pull a smoke from the shisha it's equivalent to one or two cigarettes. However, I would say it's even worse than a cigarette [the waterpipe]." (M, 24, DE)

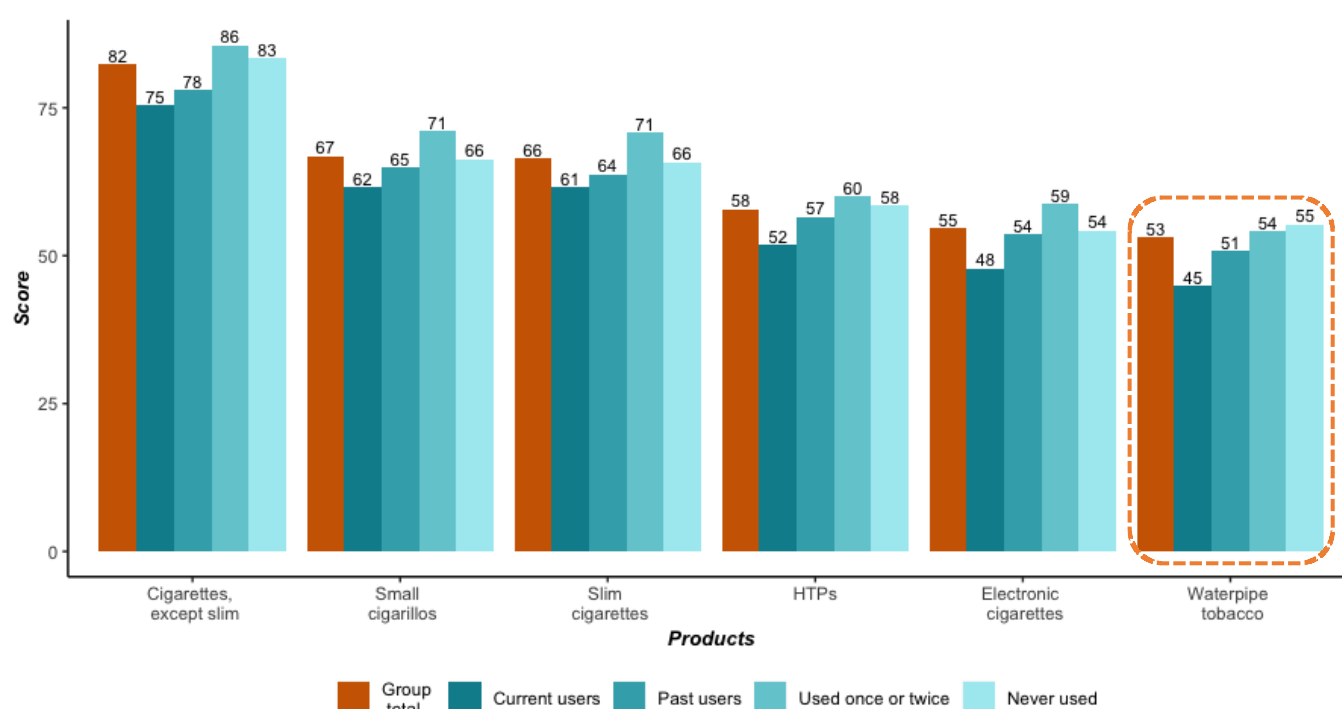
Furthermore, apart from German respondents, the focus group participants did not regard waterpipe tobacco as addictive.

The 'swing methodology' was used to assess the perceived harmfulness (risk perception) of the five tobacco and related products, plus conventional cigarettes. As described in the section 1.3.3 respondents were asked to choose the product that they thought was most damaging to the health of users. This product was given a score of 100 against which the other products were awarded scores between 0 – 100. the average scores for the other five products were computed (see Figure 23) in which the rank ordering of products runs from the 'most damaging to health of users' on the left hand side to the least damaging on the right hand side.

Waterpipe tobacco was perceived as the least harmful product by the **18-25 age group** (see Figure 27), with an average score of 53 out of 100. With a group average score of 86 for conventional cigarettes, waterpipe use carries about 35% less risk of damage to the health of users. It is notable that current users of waterpipes rate the health risk at 45.

Respondents in the **26+ age group** also saw conventional cigarettes as most harmful with an average score of 82 (see Figure 28). The full sample scored waterpipes at 47 and current users at 41. While, the survey had found that respondents thought it likely that younger people might underestimate the risks of waterpipe use, the swing method of assessing risk perception suggests that such concerns should be extended to all users regardless of age.

Figure 27. Perceived harmfulness of waterpipe tobacco to users (18-25 age group)
(average score)



Base: respondents in the 18-25 age group (N=6,090)

Question: Q.3.1.a.²⁹ & Q.3.1.b. You have chosen (e.g. cigarettes, except slims³⁰) as likely to be the most damaging to the health of users. We will give (e.g. cigarettes, except slims) 100 points on a scale of 0-100 for damaging users' health. Please assess the other products' damage on a scale of 0 to 100. If you think a product is about half as damaging to the health of users, you would give that product 50 points. If you think another product is just as damaging as (e.g. cigarettes, except slims), you would give it 100 points.

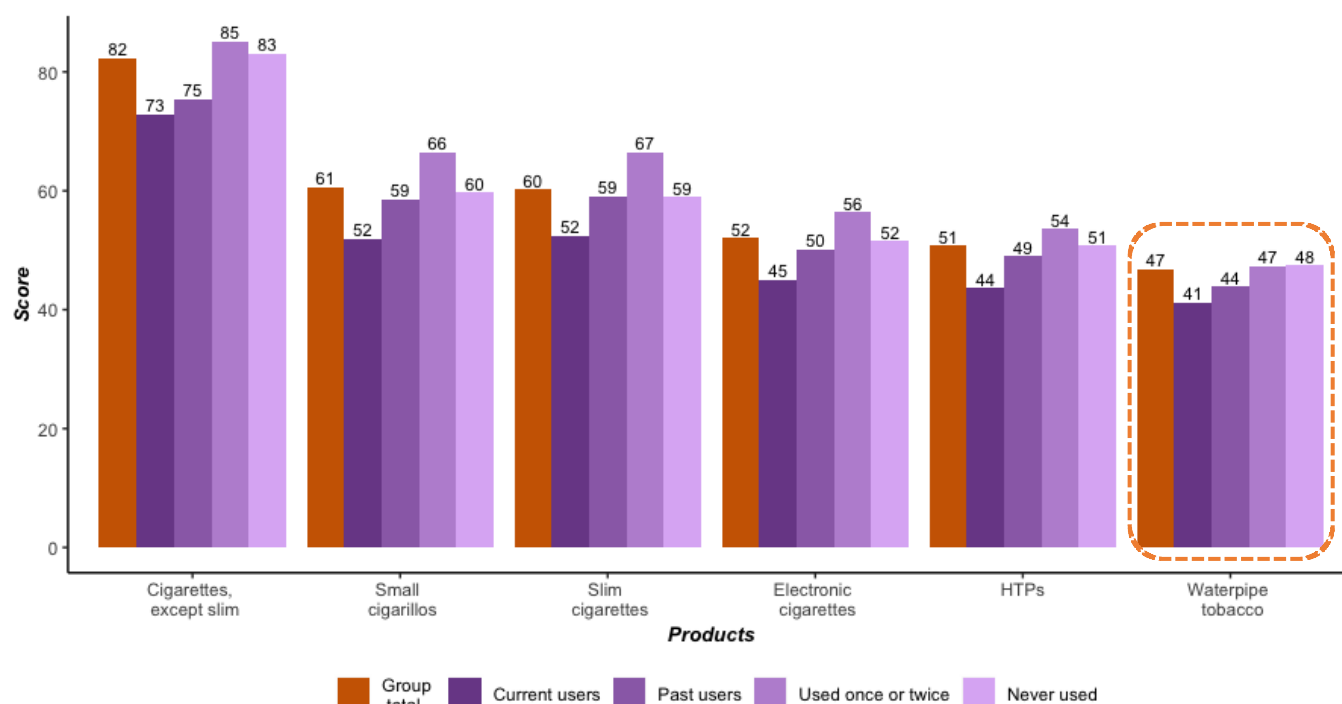
Note: The graph is based on the swing method. The scores were constructed using the responses to the two abovementioned questions. The bars in the graph are the average score of harm to the user provided by respondents for products under the scope of this study. For this chapter, the breakdown is provided for waterpipe tobacco user profiles, and therefore differs from the other chapter. The products are sorted in descending order of the average score provided by the "group total," i.e. amongst all respondents of that age group. The dotted orange box identifies the product in focus.

Source: OE elaboration from online survey data

²⁹ Q.3.1.a. Which of the six tobacco products listed below do you think is likely to be most damaging to the health of users?

³⁰ The product in parentheses would change in the online survey based on individual responses to Q.3.1.a.

Figure 28. Perceived harmfulness of waterpipe to users (26+ age group)
(average score)



Base: respondents in the 26+ age group (N=5,910)

Question: Q.3.1.a.³¹ & Q.3.1.b. You have chosen (e.g. cigarettes, except slims³²) as likely to be the most damaging to the health of users. We will give (e.g. cigarettes, except slims) 100 points on a scale of 0-100 for damaging users' health. Please assess the other products' damage on a scale of 0 to 100. If you think a product is about half as damaging to the health of users, you would give that product 50 points. If you think another product is just as damaging as (e.g. cigarettes, except slims), you would give it 100 points.

Note: The graph is based on the swing method. The scores were constructed using the responses to the two abovementioned questions. The bars in the graph are the average score of harm to the user provided by respondents for products under the scope of this study. For this chapter, the breakdown is provided for waterpipe tobacco user profiles, and therefore differs from the other chapter. The products are sorted in descending order of the average score provided by the "group total," i.e. amongst all respondents of that age group. The dotted orange box identifies the product in focus.

Source: OE elaboration from online survey data

The focus group participants reported that they lacked information and/or read inconclusive research findings about the health impacts of waterpipe tobacco use. It was mentioned that consumers may not actually see the product warnings because the product issued in special premises. Yet, a modest share of respondents reported trying to stop or reduce consumption of waterpipe tobacco on account of tobacco health warnings, with 15% and 18% reporting this in the 18-25 and 26+ age group, respectively.³³

"There isn't enough research on both of them yet (e-cigarette and waterpipe tobacco)." (M, 22, DE)

"I don't think [...] waterpipe are health friendly, I'm sceptic about that. I think studies continue." (F, 42, BG)

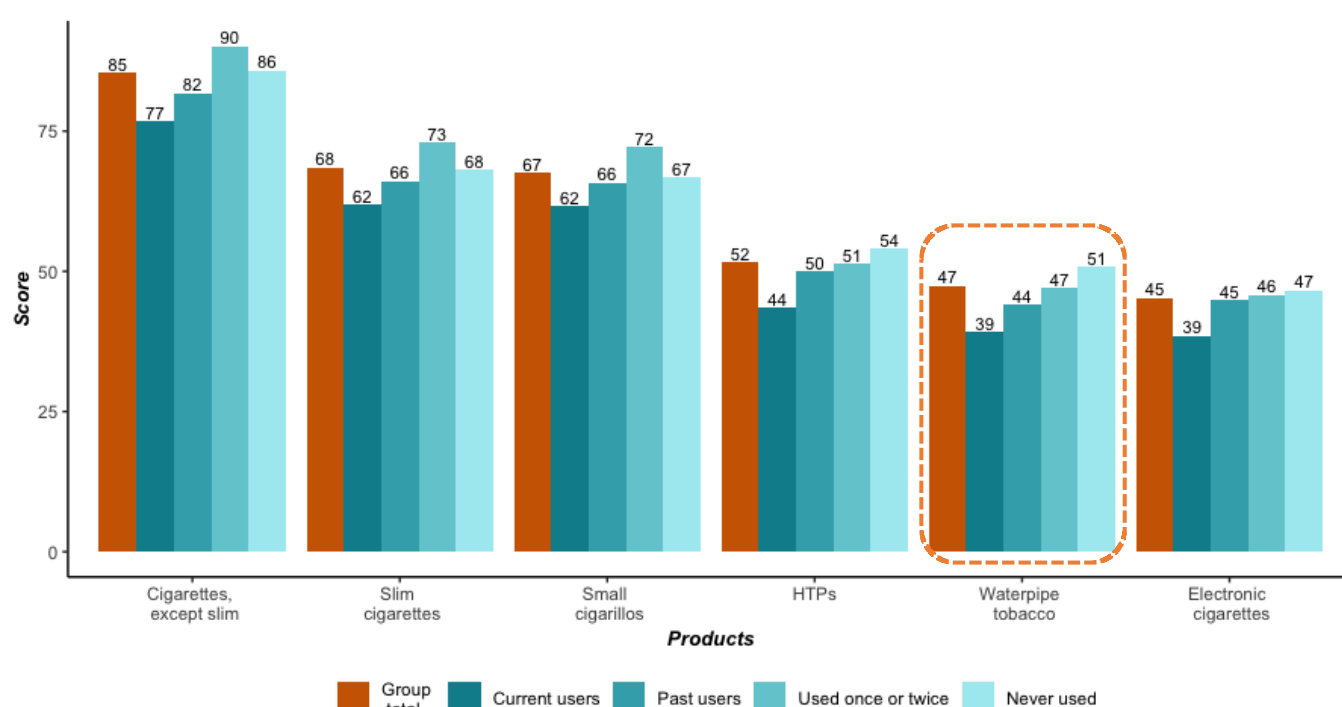
³¹ Q.3.1.a. Which of the six tobacco products listed below do you think is likely to be most damaging to the health of users?

³² The product in parentheses would change in the online survey based on individual responses to Q.3.1.a.

³³ Note that the type of tobacco advertisement was not taken into account.

The perception of damage to the health of bystanders exposed to second-hand smoke from waterpipe tobacco was also examined in the online survey. In both age groups, respondents believed that the product was more harmful to bystanders than e-cigarettes, but less harmful than the other products. However, in the 18-25 age group (see Figure 29), current users believed it was just as harmful as e-cigarettes to the health of bystanders, while current users in the 26+ age group (see Figure 30) considered it to be the least harmful product, even less than e-cigarettes. Current users in the 26+ age group provided a low score of 38 out of 100 points in terms of harm to the passive user. There was an overall misperception on the harms of the product; focus group participants dissociated it from other tobacco products due to the difference in consumption pattern (i.e. smoked in social settings and less frequently). This factor could be in part reflected in their perception of harmfulness.

Figure 29. Perceived harmfulness of waterpipe to bystanders (18-25 age group)
(average score)



Base: respondents in the 18-25 age group (N=6,090)

Question: Q.3.2.a.³⁴ & Q.3.2.b. Q.3.2.b. You have chosen (e.g. cigarettes, except slims³⁵) as likely to be the most damaging to the health of passive users. We will give (e.g. cigarettes, except slims) 100 points on a scale of 0-100 for damaging passive users' health. Please assess the other products' damage on the health of passive users on a scale of 0 to 100. If you think a product is about half as damaging to the health of passive users, you would give it 50 points. But if you think another product is just as damaging as (e.g. cigarettes, except slims), you would give it 100 points

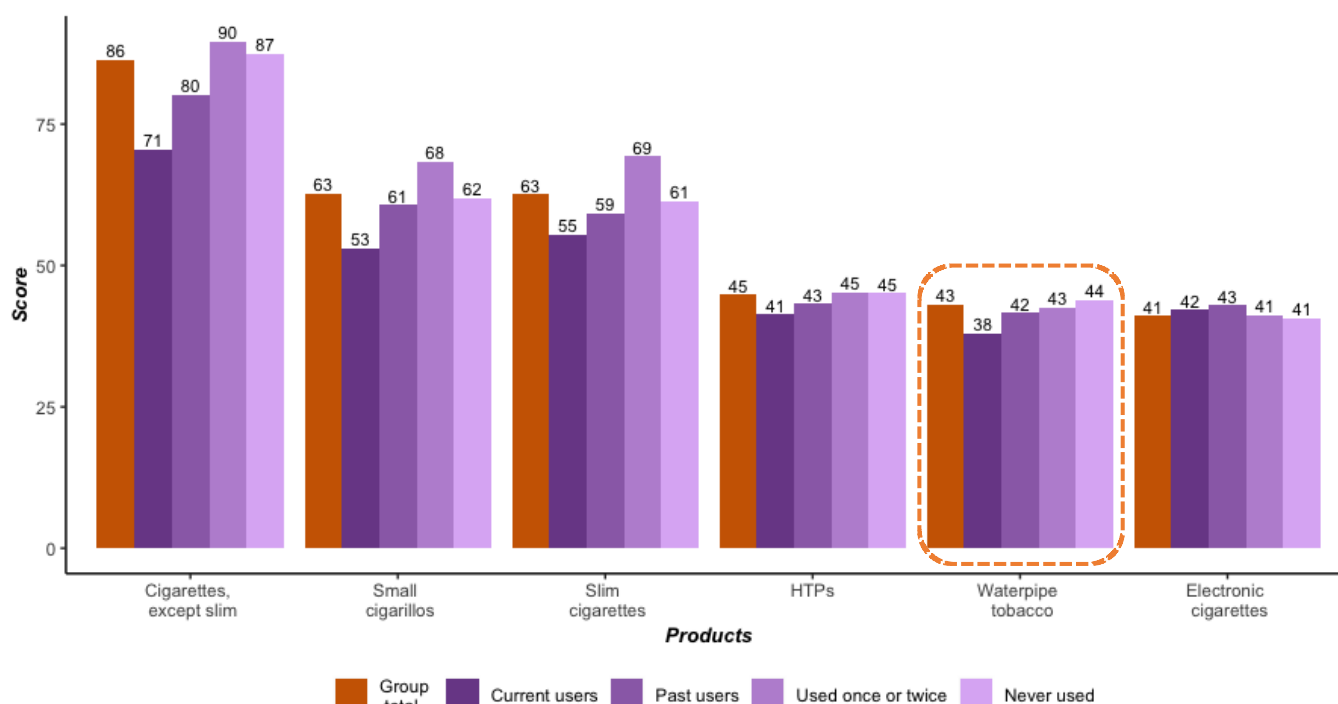
Note: The graph is based on the swing method. The scores were constructed using the responses to the two abovementioned questions. The bars in the graph are the average score of harm to the passive user provided by respondents for products under the scope of this study. For this chapter, the breakdown is provided for waterpipe tobacco user profiles, and therefore differs from the other chapter. The products are sorted in descending order of the average score provided by the "group total," i.e. amongst all respondents of that age group. The dotted orange box identifies the product in focus.

Source: OE elaboration from online survey data

³⁴ Q.3.2.a. Which of the six tobacco products listed below is likely to be the most damaging to the health of other people from passive use?

³⁵ The product in parentheses would change in the online survey based on individual responses to Q.3.1.a.

Figure 30. Perceived harmfulness of waterpipe to bystanders (26+group)
(average score)



Base: respondents in the 26+ age group (N=5,910)

Question: Q.3.2.a.³⁶ & Q.3.2.b. Q.3.2.b. You have chosen (e.g.; cigarettes, except slims³⁷) as likely to be the most damaging to the health of passive users. We will give (e.g.; cigarettes, except slims) 100 points on a scale of 0-100 for damaging passive users' health. Please assess the other products' damage on the health of passive users on a scale of 0 to 100. If you think a product is about half as damaging to the health of passive users, you would give it 50 points. But if you think another product is just as damaging as (e.g.; cigarettes, except slims), you would give it 100 points

Note: The graph is based on the swing method. The scores were constructed using the responses to the two abovementioned questions. The bars in the graph are the average score of harm to the passive user provided by respondents for products under the scope of this study. For this chapter, the breakdown is provided for waterpipe tobacco user profiles, and therefore differs from the other chapter. The products are sorted in descending order of the average score provided by the "group total," i.e. amongst all respondents of that age group. The dotted orange box identifies the product in focus.

Source: OE elaboration from online survey data

Finally, in terms of consumer perception, the online survey findings noted a general tendency to believe all tobacco and related products should be taxed equally in order not to favour any product in particular. Participants in the focus group were critical of the insufficiency information on the health effects of waterpipe tobacco and that such information was not readily available. The conviction that the product was inadequately regulated stemmed from the easy accessibility for underage users in "shisha" bars. Participants claimed that there should be more transparency about the potential negative effects of using waterpipe tobacco, and particularly the ingredients it contained. They suggested that the launch of public campaigns similar to those on the consequences of conventional tobacco the establishment of "Consumer Advice Centre" specialising in waterpipe tobacco products could be effective ways to inform consumers.

³⁶ Q.3.2.a. Which of the six tobacco products listed below is likely to be the most damaging to the health of other people from passive use?

³⁷ The product in parentheses would change in the online survey based on individual responses to Q.3.1.a.

"Quality standards and declaration. All ingredients should be listed and the consequences as well. People should be made aware of the risks. I personally didn't know when I started out. If people had more information on this then they might think twice." (M, 30, DE)

"I think it [waterpipe tobacco] is not regulated enough." (M, 34, ES)

Moreover, some people assumed that bars allowing indoor smoking of waterpipe products should be more regulated to strengthen the restrictions on access to premises offering waterpipe tobacco products.

2.4 Health, social and economic impacts

Despite the fact that waterpipe tobacco products have existed for a very long time, research on the effects of waterpipe tobacco on health are still limited. Investigations of waterpipe use must consider in detail the many variables associated with the product's use, e.g. the type of equipment, the smoking patterns, and tobacco quality. Overall, most studies on health impacts agree on the fact that users of waterpipe tobacco products are exposed to many of the same toxic compounds and by-products as conventional cigarette users. A major source of tobacco addiction is nicotine – and the levels of nicotine in waterpipe tobacco are highly variable as they depend on the type of tobacco used. Neergaard et al. (2007) conducted a review of evidence on the exposure to nicotine stemming from waterpipe tobacco use. The reviewed smoking-machine studies indicated that the nicotine yield of a 45-minute session of waterpipe tobacco use (2.94 mg) greatly exceeded the nicotine yield from smoking a conventional cigarette (0.88 mg). This finding is especially concerning that users, and notably younger users, typically perceive waterpipe consumption as less harmful than other tobacco products (see section 2.1 for more on perceptions of impact to health from the literature). Another well-known harmful component of tobacco is tar. A single waterpipe tobacco session of 45 minutes reached levels of tar equal to 802 mg, representing a level that is 36.5 times the one found with a single conventional cigarette use (Qasim, 2019).

Waterpipe tobacco has a similar toxicant/chemical profile to conventional cigarettes but at much higher levels, which may produce detrimental impacts on the user (Qasim et al., 2019). According to the WHO (2015), laboratory analyses of waterpipe smoke have revealed measurable levels of carcinogens as well as toxicants (e.g. nitric oxide). In addition, the burning charcoal was found to generate high levels of carbon monoxide (CO) and other carcinogenic substances. Data from biomarker assays have measured the levels of toxicant and carcinogen in blood and urine tests of users and found that they were significantly exposed to these compounds (St Helen et al., 2014³⁸). As a result, waterpipe use may result in long-term adverse health outcomes, such as strokes, cancers (e.g. lung, oral, gastric and oesophageal), low birth weight and other pregnancy complications, periodontal disease, lung disease, cardiovascular diseases, chronic obstructive pulmonary disease, and other medical conditions (Akl et al., 2011; Akl et al., 2016; Jawad, 2013; Kim, 2016; DKFZ, 2018, Germany).

A recent study on dual use (Khan et al., 2020) found that waterpipe tobacco altered several biomarkers in users and evidence of aggravated pulmonary and cardiovascular conditions in users of waterpipe tobacco and conventional cigarettes. In other words, dual use was found to be more harmful than exclusive use of one of the two products. A recent medical study also investigated biomarkers of toxicants associated with waterpipe use and passive waterpipe exposure among employees at waterpipe venues (Kaplan et al., 2019). Waterpipe tobacco users and non-smoking employees at waterpipe venues were found to have higher urinary concentrations of several toxic metals (e.g.

³⁸ As reported in the WHO (2015) Factsheet.

manganese and cobalt) and were therefore likely to be at an increased risk of health problems due to first-hand or second-hand exposure alike.

At the time of the discussions surrounding the 2014 TPD revision, certain niche tobacco products could be granted an exemption from certain labelling requirements (i.e. cigars). It was established that the full regime should be applied to waterpipe tobacco products in order to fight the misconceptions of the product being less harmful than other tobacco and related products.³⁹ Ali & Jawad (2017) suggested that health warning messages should focus on health outcomes specific to waterpipe tobacco, e.g. the potential for CO poisoning and the risk of infection transmission when sharing the waterpipe. In line with this, Arshad et al. (2019) concluded that interventions and policy measures to portray a message of increased harm of waterpipe tobacco use amongst students are required. In general, students were quite dismissive of policies employed to reduce waterpipe use, stating they had no access to public health campaigns or that current public health campaigns surrounding waterpipe tobacco use were not very good.⁴⁰ Social media was also flagged as a channel for waterpipe tobacco advertising and promotion (Jackler, 2019, USA; Ali & Jawad, 2017). Social media may have a role to play in directly addressing social norms towards waterpipe tobacco to combat the positive normative perception that has been observed online. Indeed, from a random sample of 5,000 waterpipe-related tweets in 2014, 87% were deemed to normalise waterpipe tobacco use by making it seem common, normal to use, and portraying it positively (Krauss et al., 2015).

With respect to economic impacts, the waterpipe tobacco consumed on the EU market is overwhelmingly manufactured outside of the EU – with very limited manufacturing reported in Germany and Poland (EC, 2017). In the EU the product is mainly smoked in ‘shisha lounges’, with approximately 80% of overall use occurring in lounges and 20% bought for private consumption online or from retailers (EC, 2017; Ebrahimi Kalan et al, 2020). There were an estimated 8,000 lounges in Germany, about 8,000 in France, and some 3,000 in the UK in 2017 (EC, 2017). Currently, waterpipe tobacco falls under the “other tobacco product” category in the Tobacco Excise Directive and the structure for taxing this category is different than for cigarettes. Member States can choose between applying a specific component, an ad valorem component or a mixture of the two. As a result, tax rates applied to waterpipe tobacco varied greatly across EU Member States as of 2017 (see Figure 175). As noted in section 2.1, this may encourage cross-border selling as well as the consumption of the product in some Member States that sell the product for cheaper than tobacco alternatives (EC, 2020). The flexibility granted to Member States for the taxation under the other smoking tobacco category may divert efforts to reduce tobacco consumption by almost incentivising youth to switch from conventional cigarettes to waterpipe tobacco. However, a majority of MS did not consider that the issue with waterpipe tobacco was of sufficient severity to justify the establishment of a separate tax category and were not ready to bear the related adaptation costs (EC, 2017).

Jawad (2014, global) studied global tobacco control policies prior to the implementation of the TPD. After reviewing the legislation of 62 countries, he found that most regulations did not address waterpipe smoking explicitly. The review concluded that, based on harmful exposure alone, there was sufficient justification to commence health awareness campaigns and incorporate waterpipe products into existing tobacco control efforts in the UK. The TPD was implemented in 2014, and precisely addressed these issues. Jawad et al. (2015) analysed the impact of waterpipe tobacco pack health warning labels among regular users in London. They found that perceptions of harmfulness stemming from health warnings on waterpipe tobacco packs were “dose-responsive”. In other words, the response became more negative as the health warning increased in size. The evidence argues in support of the need for warning labels on waterpipe tobacco products, as well as associated

³⁹ https://ec.europa.eu/taxation_customs/sites/taxation/files/study_on_directive-2011_64_main_text_en.pdf

⁴⁰ This is notably from studies conducted in the United Kingdom.

accessories and waterpipe-serving premises (Jawad, 2016). Most social users of the product report an overall lack of exposure to health warnings on waterpipe tobacco packs due to the inherent nature of waterpipe smoking, that is, smoking in a café with the apparatus already prepacked by staff. Health warnings at the point of consumption had reportedly more impact than health warnings at the point of sale (Jawad et al., 2015).

2.5 Concluding remarks on waterpipe tobacco

This section summarises the main findings that emerged from the research conducted on waterpipe tobacco products. The systematic review revealed that:

- Waterpipe users are exposed to toxicants from both the tobacco and the charcoal used to heat it. Toxicant yield studies have consistently found substantial levels of tar, nicotine, CO and cancer-causing chemicals in waterpipe tobacco smoke. Emerging evidence suggests that waterpipe tobacco has harmful long term effects on health, similar to those of other forms of smoked tobacco. Growing evidence shows the risk of second-hand waterpipe tobacco smoke, particularly in waterpipe cafes, where the product is mainly smoked.
- As found in the Eurobarometer 458 (2017), the prevalence of waterpipe tobacco current use in the EU was negligible among adults but it was 2% in the 15-24 age group. However, in this same group, the lifetime prevalence was 28%, while in the overall population it was 13%. It is difficult to estimate the total sales of waterpipe tobacco because the available data reviewed merges waterpipe and pipe tobacco. However, according to the Eurobarometer 458 (2017), the highest lifetime prevalence of use was in Nordic and Baltic countries of the EU. There is very limited manufacturing in the EU, with some reported in Germany and Poland.
- Currently, waterpipe tobacco falls under the “other tobacco product” category in the Tobacco Excise Directive (TED) and the structure for taxing this category is slightly different than for cigarettes. MS can choose between applying a specific component, an ad valorem component or a mixture of the two. As a result, tax rates applied to waterpipe tobacco varied greatly across EU Member States as of 2017.
- Most social users of the product report an overall lack of exposure to health warnings on waterpipe tobacco packs due to smoking in social settings where the products are already unpacked by staff. Health warnings at the point of consumption had reportedly more impact than health warnings at the point of sale.

The empirical work (focus groups and online survey) revealed the following user profile:

- Respondents in the 18-25 age group were flagged as an important consumer group for waterpipe tobacco, with the majority (53%) in the Member States included in this study having at least tried the product.
- The profile of the current waterpipe users is characterised by the variability of frequency (at least once a month) and duration of use (5-60 minutes per session). Younger users started using as minors, while older users started as adults, which opens the question if the age of initiation is rapidly changing. A great majority of users started consuming waterpipe tobacco for enjoyment. However, socialising is another frequently mentioned reason. At least 8 in 10 current waterpipe users were or are using other tobacco and related products concurrently.
- Current waterpipe users were evenly split between those that had never attempted quitting use of the product and those who tried at least once and failed in the 18-25 group. In the 26+ age group, almost 2 in 3 had never tried to quit. Among failed quitters, the main reason to attempt to quit was for personal health. A distant second reason was to quit in response to

tobacco control measures - either existing restrictions to smoke at work or home in the 18-25 age group or due to health warnings on packages in the 26+ age group.

- The profile of the past waterpipe users is typified by the variability of frequency (from daily to at least once a month) and duration of use (5-60 minutes per session).). The great majority of past users were dual users of tobacco products. However, their evolution seems to split the group into those who quit all tobacco products and those who progress to other tobacco products. About 1 in 2 of past waterpipe users in the 18-25 group and 1 in 3 in the 26+ group, were abstaining completely from using any tobacco and related product. By contrast, about 1 in 3 former waterpipe tobacco users in the 18-25 group and 1 in 2 in the 26+ group were now using other tobacco and related products.

Lastly, the following paragraphs highlight the *Product and Regulatory perceptions remarks*

- *Perception of marketing practices.* Consumption of waterpipe tobacco was perceived as an emerging novel product, especially among younger generations, as well as a timeless and classic product due to its long presence on the market and in certain ethnic groups in Europe. It evoked the idea of "zen," "pleasure," and "relaxation". The more negative terms (e.g. 'complicated', 'unhealthy' and 'addictive' were mainly cited by never users. In the 26+ age group, 39% of respondents characterised the product as 'unhealthy' as the most often cited term. In the 18-25 age group, the term associated most often with waterpipe tobacco was "nice flavours".
- Between 2 in 3 and 3 in 4 respondents, depending on age, believed that smoking waterpipe tobacco would be particularly appealing to youth. The flavours, social ritual (i.e. rarely used alone), perceived affordability, lower risk perception than other tobacco products were the main elements driving the appeal of waterpipe tobacco products according to focus group participants.
- *Perception of risk.* Waterpipe tobacco was perceived as the least harmful product to users and the second least harmful to bystanders (after e-cigarettes) of the studied products. It was also perceived as less addictive. The reduced perception of harmfulness of waterpipe might be partly related to the difference in consumption pattern with respect to other tobacco products, i.e. smoked in social settings and less frequently. The behavioural intent of waterpipe tobacco use may be shaped by social pressures and through social media, where both advertising and promotion occurs.
- *Regulatory support.* There was a perception that the product was hardly regulated. It stemmed from the easy accessibility for underage users in "shisha" bars. There was a general tendency to believe all tobacco and related products should be taxed equally in order not to favour any product. Participants of the focus groups criticised the insufficient and not easily available information regarding health effects of waterpipe tobacco. It was mentioned that consumers may not actually see the product warnings because the product is mainly used in special premises where the product is served unpacked.

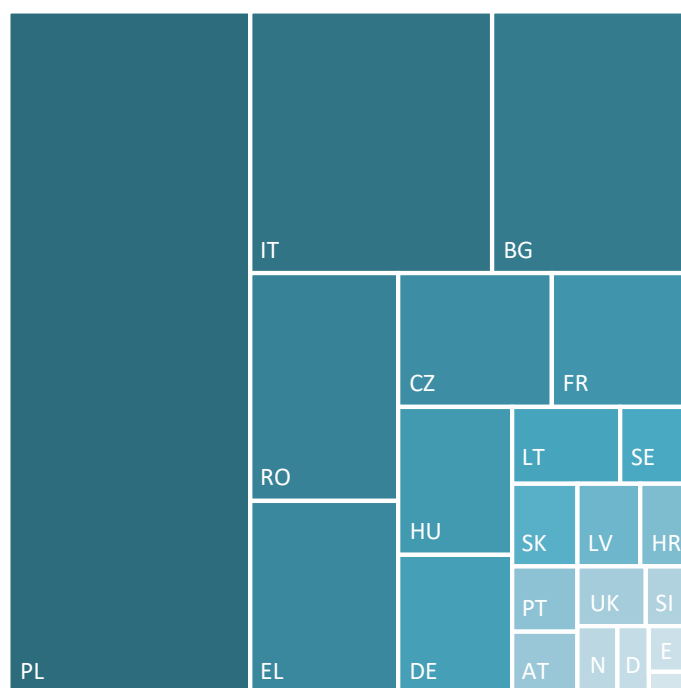
3 Slim cigarettes

3.1 Background

Slim cigarettes are one of the oldest products on the EU market among those within the scope of the study. They were historically marketed with the positive connotations of femininity and elegance as the tobacco companies tried to attract new consumer groups, namely young women. Since 2000, the market share of slim cigarettes has grown significantly in an overall declining EU cigarette market.⁴¹ In December 2012, the EU Commission proposed a ban on the sale of flavoured and slim cigarettes across the EU through a revision of the Tobacco Products Directive. However, following resistance from some Member States the ban was dropped in final draft and statement was made that market developments and perceptions of the product should be monitored.

Market development. The figure below represents the structure of the EU market of slim cigarettes in absolute values (million units) by country for 2017.⁴² The largest markets in terms of retail volume of slim cigarettes were Poland, Italy, Bulgaria, Romania, Greece and Czech Republic – the product was especially popular in Eastern Europe.

Figure 31. Market size of slim cigarettes in million units in 2017



Note: The market size is estimated total sales in million units for 2017, without population adjustments. Note that this is different than other products as the information in million euros was not available for this product. It is displayed as a tree map graph that consists of series of nested rectangles of sizes proportional to the corresponding data value.

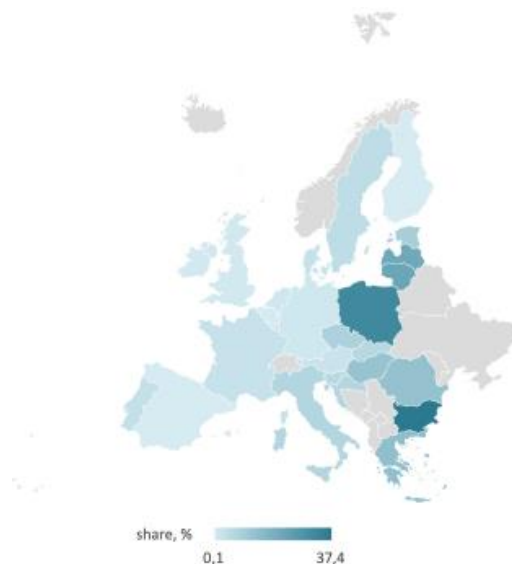
Source: OE elaboration from Passport, 2017

⁴¹ Slim cigarette fact sheet. https://ec.europa.eu/health/sites/health/files/tobacco/docs/fs_slim_cigarettes_en.pdf

⁴² Data on slim cigarettes data was available for the market size in million units, but not in retail selling price (RSP).

The map below provides a visual representation of the share of slim cigarettes by national tobacco markets across the EU.⁴³ Among the Member States, the consumption of slim cigarettes was particularly high in Bulgaria in 2017, representing a share of almost two fifth of the entire tobacco market, followed by Poland (close to 32%), and Latvia and Lithuania with close to one fifth.

Figure 32. Share of slim cigarettes within the national total tobacco market, per country (2017)

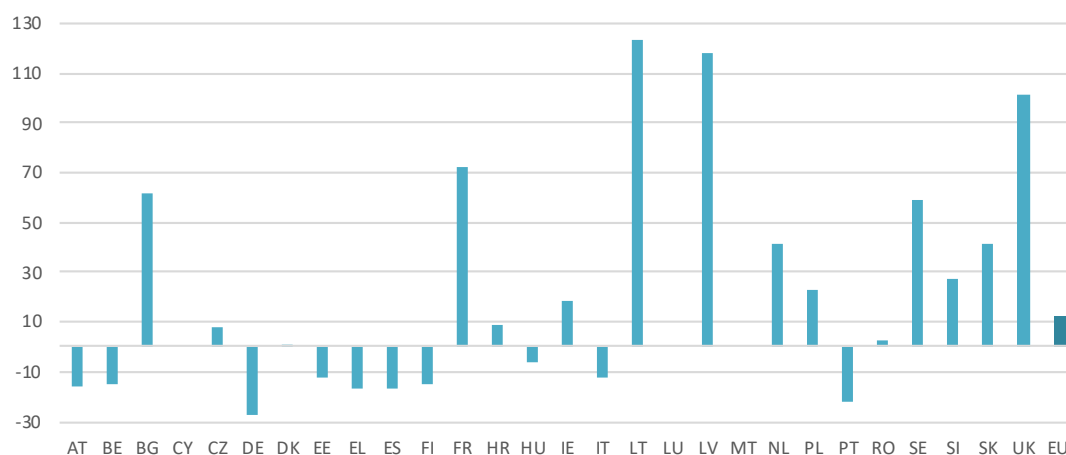


Note: Share of slim cigarettes were estimated with respect to the whole tobacco market in the country in million euros.

Source: OE Elaboration from Passport, 2017

The data showed that the market development trend in the slim cigarettes varied greatly across the EU (see Figure 33.). In the past years, retail volume of slim cigarettes market increased significantly in Lithuania, Latvia, the UK, France, Bulgaria and Sweden. It also decreased in several Member States, notably Germany and Portugal, but by a much smaller magnitude.

Figure 33. Market size growth of slim cigarettes, in retail volume in 2017 as a baseline of 2012



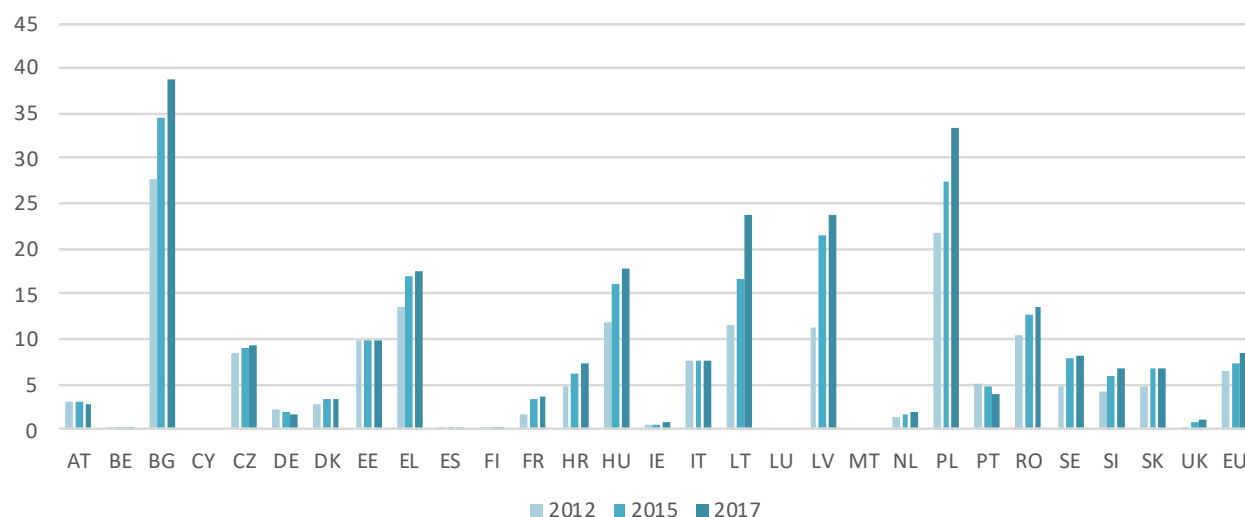
Note: EU average is calculated for 25 EU MS (excluding Cyprus, Luxembourg & Malta - no data were available). The figure looks at the market growth in retail volume of slim cigarettes between 2012 and 2017.

⁴³ In this case, the figure is based on the share of slim cigarettes available in terms of real retail volume and calculated the share in RSP values with respect to the whole tobacco market.

Source: OE elaboration of Passport, 2017

The dynamics of historical data are illustrated in Figure 34. Most countries saw an increase in the share of slim cigarettes on the cigarette market over the past years. In Bulgaria and Poland, the increase and the share of slim cigarettes were highest, with 38% and 33% respectively, in 2017. Slim cigarettes were also in high demand in Latvia, Lithuania, Hungary, Greece and Romania.

Figure 34. Share of slim cigarettes in total market of cigarettes, in retail volume: 2012-2017



Note: EU average is calculated for 25 EU members (excluding Cyprus, Luxembourg and Malta - no data were available). The dynamics of time series data was constructed from the share of the slim cigarettes in the market of all types of cigarettes (i.e. not in terms of the whole tobacco market, as above).

Source: OE elaboration from Passport, 2017

Several studies have found that there was a greater prevalence of slim cigarette use among female smokers (Milcarz et al., 2017, Poland). While there is limited data available from EU studies, Canadian studies show that tobacco companies increasingly targeted females and young people through slim cigarette design features and packaging (Minaker, 2017⁴⁴). However, while the rate of slim cigarette use was higher for females than males among school grades 9–12 smokers, the difference was not statistically significant. In line with these results, Slater (2016, global) found no evidence for greater slim cigarette smoking by females.

⁴⁴ The study was based on secondary data from the Youth Smoking Survey 2008/2009 (N=29,296); Youth Smoking Survey 2010/2011 (N=31,396); Youth Smoking Survey 2012/2013 (N=27,404); and the Cancer Risk Assessment in Youth Survey 2015 (N=12,110).

Consumer perceptions. The marketing of slim cigarette towards women began in the 1990s to resist the 'good girl' image and to be part of the transition from childhood into adulthood. The tobacco industry made slim cigarettes a symbol of gender equality, sophistication and upward mobility. Smoking was made to look fun, glamorous, healthy, liberating, relaxing, sexually attractive, slimming and sophisticated. Coloured packs were also developed to attract children (European Institute of Women's Health⁴⁵). The image propagated by the industry is still relevant. Young people have been shown to associate slim cigarettes with femininity and elegance (Moodie et al., 2015).

The Eurobarometer 458 highlighted the positive perception of slim cigarettes especially in the Czech Republic (46%), Belgium (42%), Slovenia (41%) and Greece (39%). 18% of all EU potential consumers perceiving the product as attractive, feminine and elegant in 2017 (see Table 28), suggesting that part of their attraction is aesthetic.⁴⁶ 7% agreed that slim cigarettes '*are less harmful than normal cigarettes*', with Member States varying in the level of agreement from 2% to 17%. Research has reported that the industry misled consumers by giving a false impression of reduced harm from slim cigarettes in comparison to conventional cigarettes (Mutti, 2011, Canada, US and UK, N=6,075). However, there was no evidence for this claim (Mladjenovic, 2012, Canada). That people perceived slim cigarettes as less harmful than traditional cigarettes motivated discussions on a ban of the product in the European Union in 2013 (Braun et al., 2019).

Moodie (2015, UK) studied the perceptions of slim cigarettes, using focus groups with female non-smokers and tobacco users aged 12 to 24 years. Participants ranked different types of cigarettes (slim, coloured, aromatised and capsule) on the criteria of appeal, taste and harm. Slim cigarettes generated significant interest among participants and were perceived as some of the most appealing, pleasant tasting and least harmful products. In similar vein, Gallopel-Morvan (2018, France, N=998) found that 15 to 30 year old tobacco smokers and non-smokers viewed slim cigarettes as less harmful as well as more appealing than conventional cigarettes.

⁴⁵ <https://eurohealth.ie/wp-content/uploads/2013/05/SmokingPBFIN.pdf>

⁴⁶ GTSS data cover small cigarillos, e-cigarettes and waterpipe products. Unfortunately, the data are limited and only available for three countries (Greece, Poland and Romania).

Table 24. "Which of the following statements, if any, do you think apply to slim cigarettes?" (2017)

	They help you lose weight (%)	They are attractive (e.g. feminine, elegant) (%)	They are easier to smoke than normal cigarettes (%)	They are less harmful than normal cigarettes (%)	None (%)	Do not know (%)
EU28	1	18	6	7	47	25
AT	3	19	9	10	41	23
BE	7	42	17	15	15	12
BG	1	33	10	7	30	27
CY	2	22	8	4	31	37
CZ	1	46	14	8	22	18
DE	0	6	2	2	67	25
DK	2	22	5	2	39	29
EE	1	30	5	3	38	26
EL	0	39	16	8	36	11
ES	0	8	3	2	57	29
FI	1	10	3	5	56	26
FR	1	27	4	4	42	25
HR	1	32	8	8	36	20
HU	3	32	11	11	33	17
IE	2	12	7	8	41	31
IT	2	22	10	16	36	19
LT	2	31	12	6	45	12
LU	2	27	3	7	45	20
LV	1	30	10	7	43	14
MT	2	14	3	4	48	31
NL	1	12	8	14	26	41
PL	2	21	7	9	38	27
PT	2	15	5	7	38	35
RO	2	18	14	10	37	25
SE	1	13	6	6	33	45
SI	1	41	4	6	44	8
SK	2	29	12	10	25	27
UK	1	6	2	2	67	23

Base: all respondents of Eurobarometer 458 (representative sample of EU), N=27,901

Note: Each row represents a different country and provides the proportion of respondents within that country that selected the response option. Note that multiple answers were allowed. The table was based on QB.8 of the Eurobarometer survey.

Source: Eurobarometer 458, 2017

Impacts. Braun et al. (2019) assessed the exposure risk of particulate matter to passive smokers for super-slim size cigarettes compared to a king size brand. They concluded that the particulate matter pollution generated by slim-size tobacco products were substantial and sometimes higher than those produced by king size tobacco products.

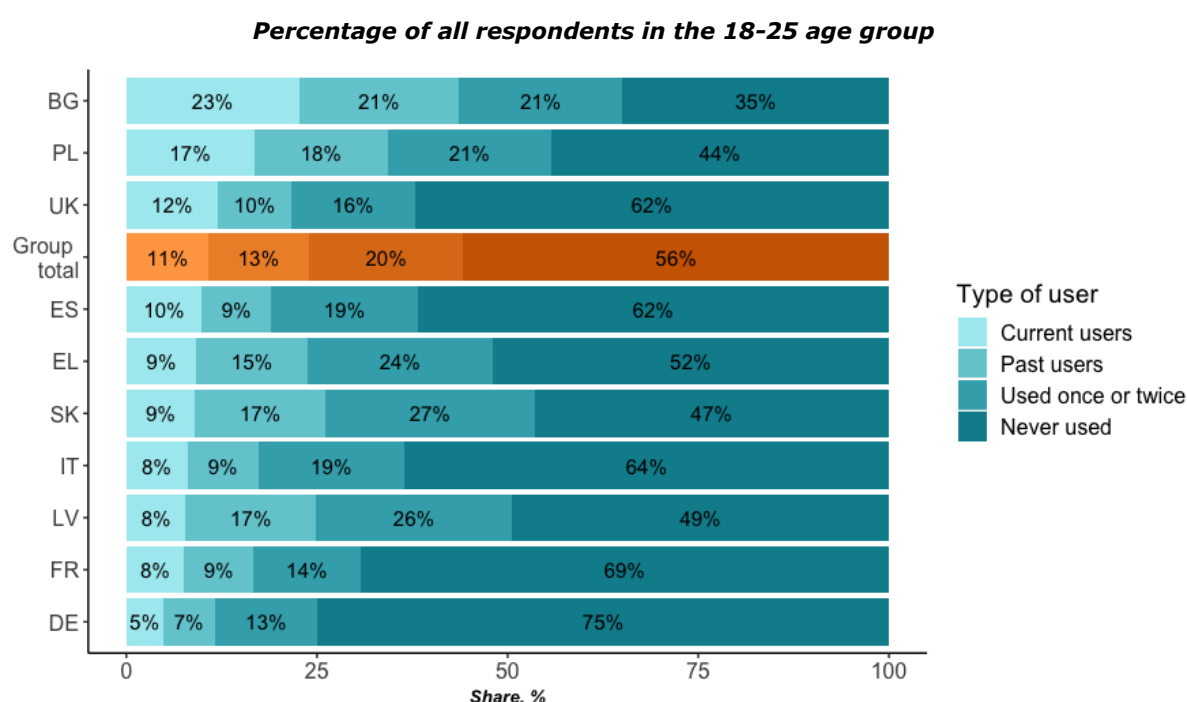
An analysis of Vogue cigarettes found no significant difference in the content of tar of the classic cigarette sizes (KSC⁴⁷ Vogue La Cigarette) and slim cigarettes (SSL⁴⁸ Vogue BLEUE). The nicotine content of slim cigarettes (0.60 mg) was higher than the standard size (0.40 mg), but this difference was not found to be significant. The analysis revealed that slim cigarette particular matter (PM) levels were higher than in conventional cigarettes. It was concluded that the smaller-sized cigarettes were potentially more harmful to smokers and passive smokers (Kant et al., 2016).

3.2 Market developments

3.2.1 Socio-demographic characteristics

The online survey showed current usage of slim cigarettes at 11% for the **18-25 age group** (see Figure 35) and 12% for the **26+ age group** (see Figure 36). Bulgaria and Poland had high prevalence of slim cigarette users in both age groups. In the **18-25 age group**, 23% of respondents reported current use of slim cigarettes in Bulgaria, with only 35% of never users. By contrast, 75% of respondents in Germany had never tried the product (i.e. never users). In the **26+ age group**, a majority had never tried slim cigarettes, but this varied significantly across the countries studied, ranging from the 35% to 75% of never users.

Figure 35. Use of slim cigarettes: country breakdown (18-25 age group)



Base: all respondents in the 18-25 age group (N=6,090)

Question: Q.1.1. Have you ever used slim cigarettes?

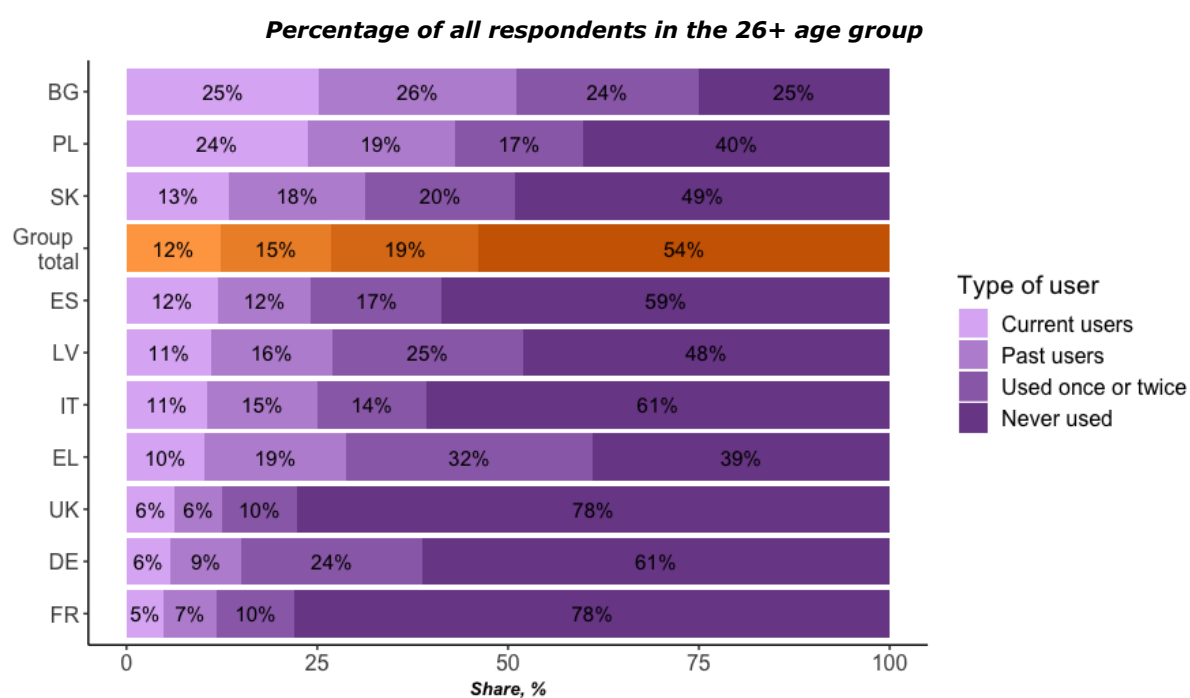
Note: The group total category refers to all respondents within the age group.

Source: OE elaboration from online survey data

⁴⁷ King Size Cigarette

⁴⁸ Super Slim Size

Figure 36. Use of slim cigarettes: country breakdown (26+ age group)



Base: all respondents in the 26+ age group (N=5,910)

Question: Q.1.1. Have you ever used slim cigarettes?

Note: The group total category refers to all respondents within the age group.

Source: OE elaboration from online survey data

Table 25 and Table 26 report the socio-demographic profiles of current, past, occasional and never users of slim cigarettes. In the **18-25 age group**, current users were marginally more likely to be female, with lower education, living in urban settings and with self-reported high socio-economic status. In the **26+ age group** more females (15%) than males (10%) were current smokers. Higher education, from urban settings and high socio-economic status were also characteristics of current smokers. In both age groups slim cigarettes had low penetration in rural areas.

Table 25. Slim cigarette use socio-demographic breakdown (18-25 age group)

Percentage and number of all respondents in the 18-25 age group

	Group total % (n)	Current users %	Past users %	Used once or twice %	Never users %
Use of product	100 (6090)	11	13	20	56
Gender					
Male	47 (2869)	10	11	20	59
Female	53 (3221)	12	15	21	53
Education level *					
1-2 (low)	5 (335)	14	10	22	54
3-4 (medium)	63 (3829)	10	12	21	57
5-6 (high)	32 (1926)	11	16	19	54
Residency					
Urban	65 (3949)	12	14	22	52
Suburban	19 (1173)	9	11	19	61
Rural	16 (968)	7	11	16	66
Self-reported socio-economic status					
1-4 (low)	18 (1126)	10	12	20	58
5-7 (medium)	65 (3954)	10	13	21	56
8-10 (high)	17 (1010)	15	15	17	53

Base: all respondents in the 18-25 age group (N=6,090)

Notes: The group total column refers to all respondents within the age group, i.e. it is a within column percentage. It provides the breakdown of each socio-demographic variable, including absolute numbers. The white cells in the four right columns represent the breakdown by user type, i.e. they are within row percentages (white cells in each row sum to 100).

* The sub-categories of the education level were the following: (1) No official education; (2) Primary school; (3) High school; (4) Some years of university (not completed); (5) University degree completed; (6) Post-graduate (master, PhD, other)

Source: OE elaboration from online survey data

Table 26. Slim cigarette use socio-demographic breakdown (26+ age group)

Percentage of all respondents in the 26+ age group

	Group total % (n)	Current users %	Past users %	Used once or twice %	Never users %
Use of product	100 (5910)	12	15	19	54
Gender					
Male	52 (3074)	10	13	21	56
Female	48 (2836)	15	16	18	51
Education level *					
1-2 (low)	5 (308)	10	7	15	68
3-4 (medium)	46 (2704)	11	13	19	58
5-6 (high)	49 (2898)	14	17	20	49
Residency					
Urban	66 (3879)	15	17	21	47
Suburban	17 (1010)	9	11	17	63
Rural	17 (1021)	6	8	16	69
Self-reported socio-economic status					
1-4 (low)	23 (1374)	10	13	17	59
5-7 (medium)	64 (3769)	12	15	20	54
8-10 (high)	13 (767)	19	17	19	46

Base: all respondents in the 26+ age group (N=5,910)

Notes: The group total column refers to all respondents within the age group, i.e. it is a within column percentage. It provides the breakdown of each socio-demographic variable, including absolute numbers. The white cells in the four right columns represent the breakdown by user type, i.e. they are within row percentages (white cells in each row sum to 100).

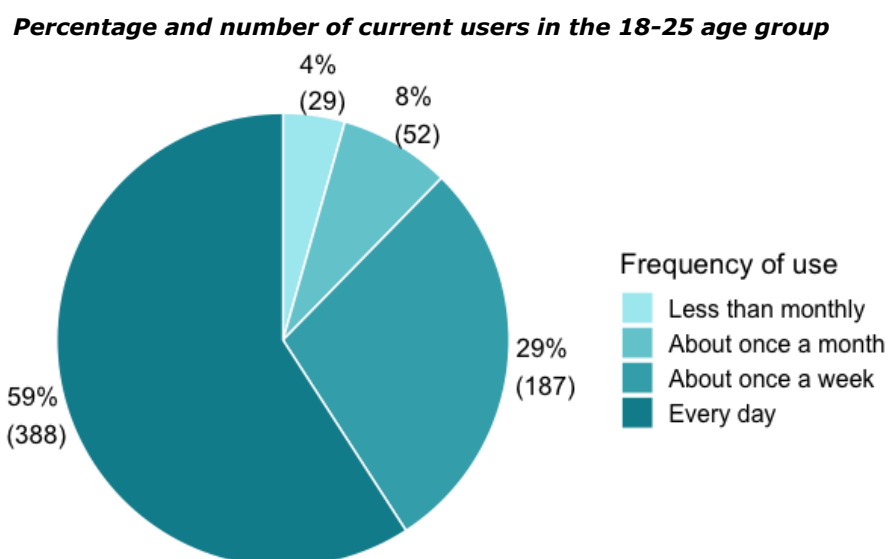
* The sub-categories of the education level were the following: (1) No official education; (2) Primary school; (3) High school; (4) Some years of university (not completed); (5) University degree completed; (6) Post-graduate (master, PhD, other)

Source: OE elaboration from online survey data

3.2.2 Pattern of use

Among current users of slim cigarettes in the 18-25 age group, 59% smoked them daily and an additional 29% smoked them about once a week. Similar percentages were found in the 26+ age group, at 55% and 30% respectively (see Figure 37 and Figure 38).

Figure 37. Frequency of use of current slim cigarette users (18-25 age group)

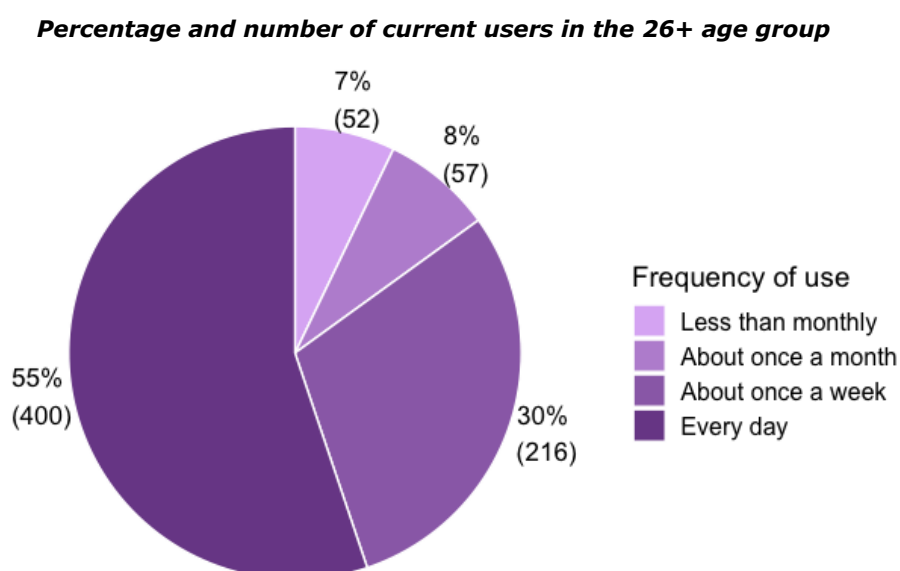


Base: current users of slim cigarettes in the 18-25 age group (N=656)

Question: Q.1.3. How often do you use slim cigarettes?

Source: OE elaboration from online survey data

Figure 38. Frequency of use of current slim cigarette users (26+ age group)



Base: current users of slim cigarettes in the 26+ age group (N=802)

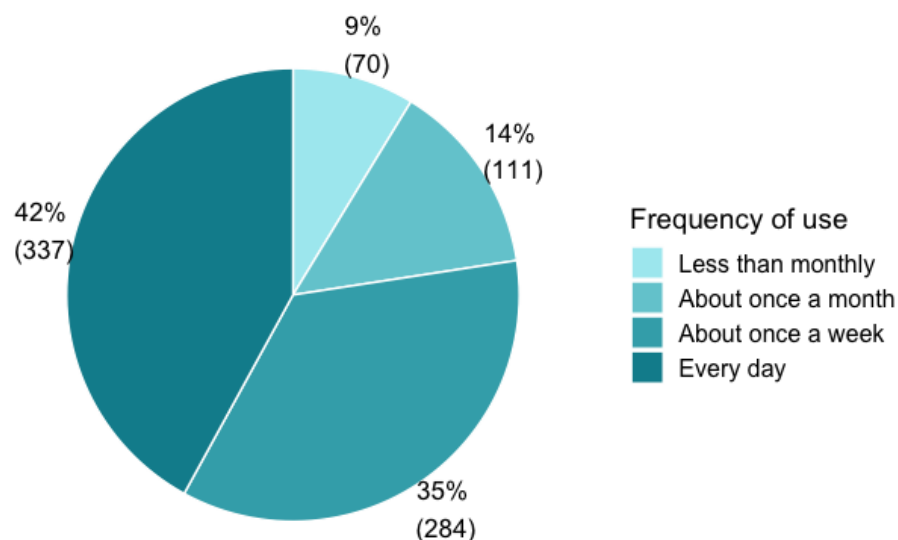
Question: Q.1.3. How often do you use slim cigarettes?

Source: OE elaboration from online survey data

Past slim cigarette users in the 18-25 age group reported smoking less frequently. 42% had used the product daily and 35% weekly. In the 26+ age group, 51% had smoked the product daily and 29% weekly. Past users in the 26+ age group were much more regular users than the younger age group (see Figure 39 and Figure 40).

Figure 39. Frequency of use of past slim cigarette users (18-25 age group)

Percentage and number of past users in the 18-25 age group



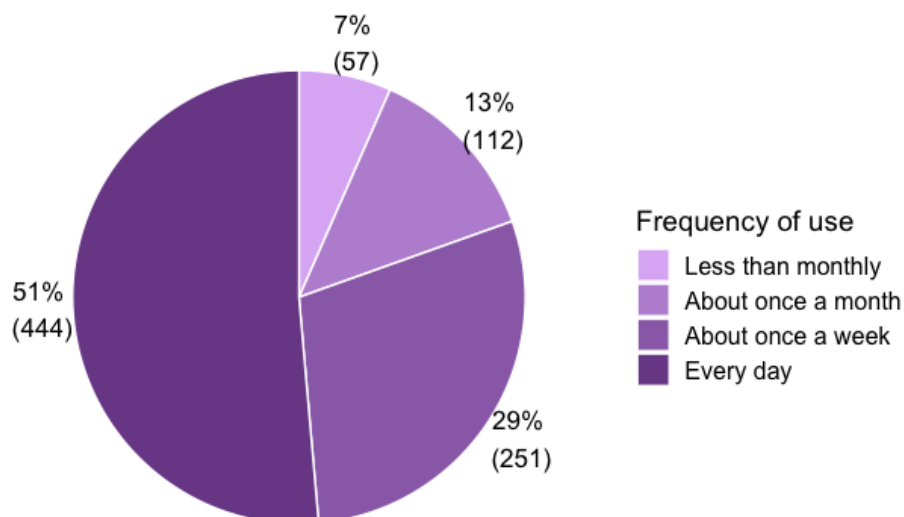
Base: past users of slim cigarettes in the 18-25 age group (N=536)

Question: Q.1.6. How often did you use slim cigarettes?

Source: OE elaboration from online survey data

Figure 40. Frequency of use of past slim cigarette users (26+ age group)

Percentage and number of past users in the 26+ age group



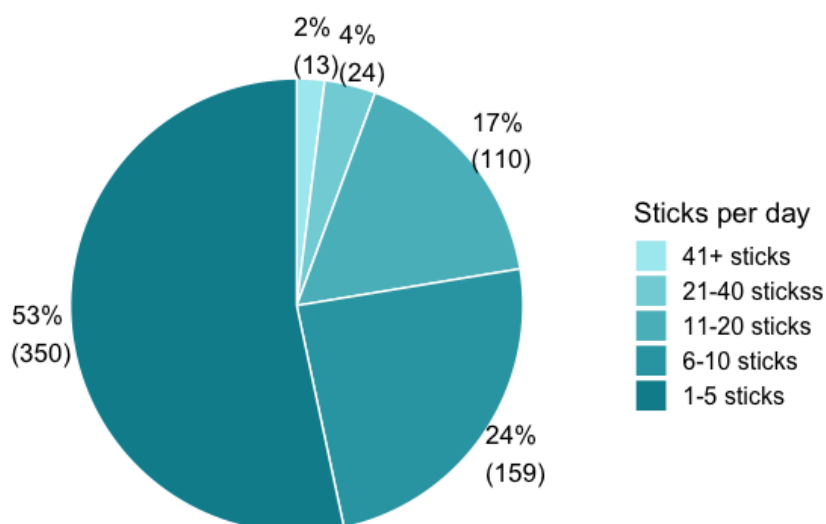
Base: past users of slim cigarettes in the 26+ age group (N=864)

Question: Q.1.6. How often did you use slim cigarettes?

Source: OE elaboration from online survey data

Figure 41 and Figure 42 report intensity of use – how many sticks used on days when they had smoked. Current users in the 26+ age group reported being more intense use of slim cigarettes than the 18-25 age group. 53% of respondents in the 18-25 age group reported smoking 1 to 5 slim cigarettes per day of use, against 47% in the 26+ age group. In both age groups, close to one in four current users smoked over 10 slim cigarettes per day of use. Daily smokers also revealed that they had a more intensive pattern of use, especially in the 26+ age group (see Figure 186. and Figure 187. in Annex). For example, 36% of daily users reported smoking over 10 cigarettes per day of use, compared to 26% in the overall current users' sub-group.

Figure 41. Average number of slim cigarettes per day of use (18-25 age group)
Percentage and number of current users in the 18-25 age group

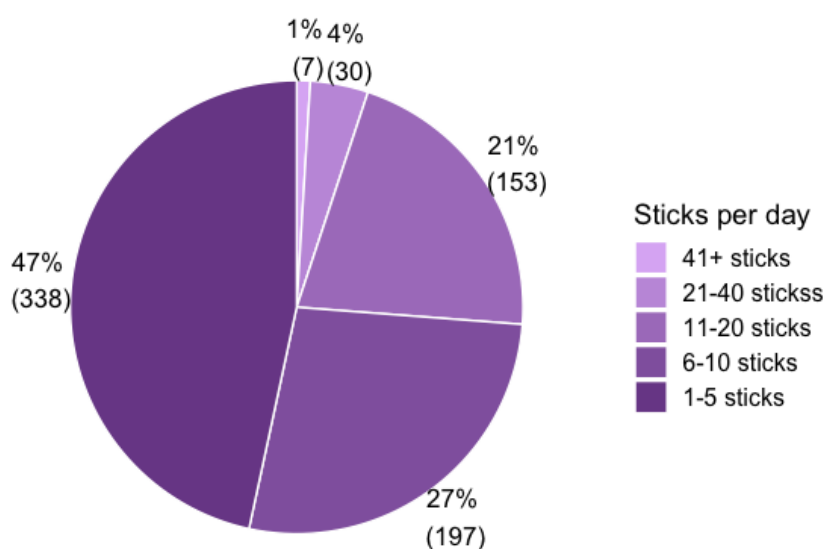


Base: current users of slim cigarettes in the 18-25 age group (N=656)

Question: Q.1.4.a. On the days that you use slim cigarettes, about how much do you use?

Source: OE elaboration from online survey data

Figure 42 Average number of slim cigarettes per day of use (26+ age group)
Percentage and number of current users in the 26+ age group



Base: current users of slim cigarettes in the 26+ age group (N=725)

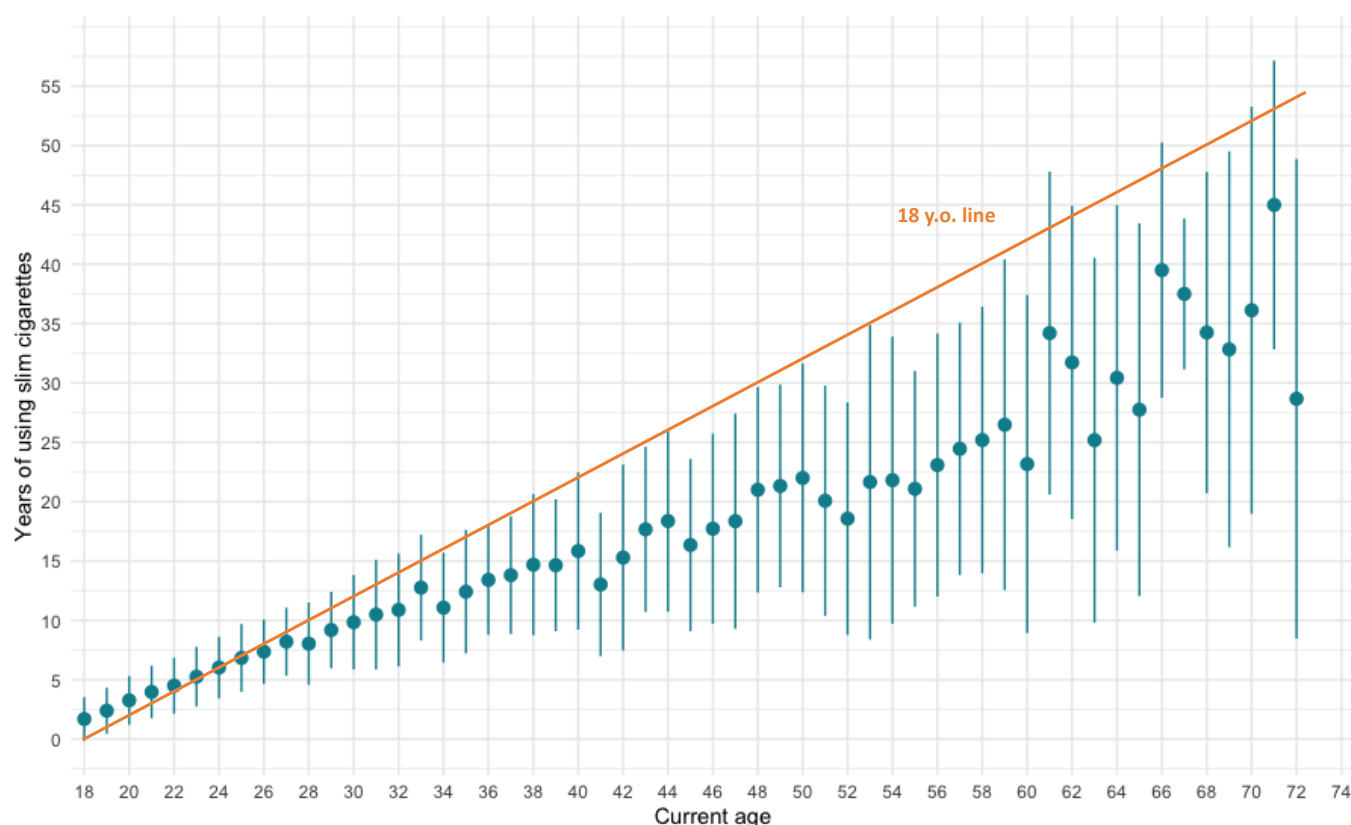
Question: Q.1.4.a. On the days that you use slim cigarettes, about how much do you use?

Source: OE elaboration from online survey data

Respondents under the age of 23 at the time of the study were on average likely to have started as a minor. Several respondents between the ages of 23 and 33 also began prior to turning 18 years of age. While the majority of respondents between 34 and 60 did not, several respondents over the age of 60 reported having started before turning 18.

Figure 43. Initiation of use for current and past users (slim cigarettes; total sample)

(number of current and past users in the overall sample)



Base: current and past users of slim cigarettes in the total sample (N=2,933⁴⁹)

Question: Q.1.2.a. At about what age did you start using the products below? & Q.5.2. Age

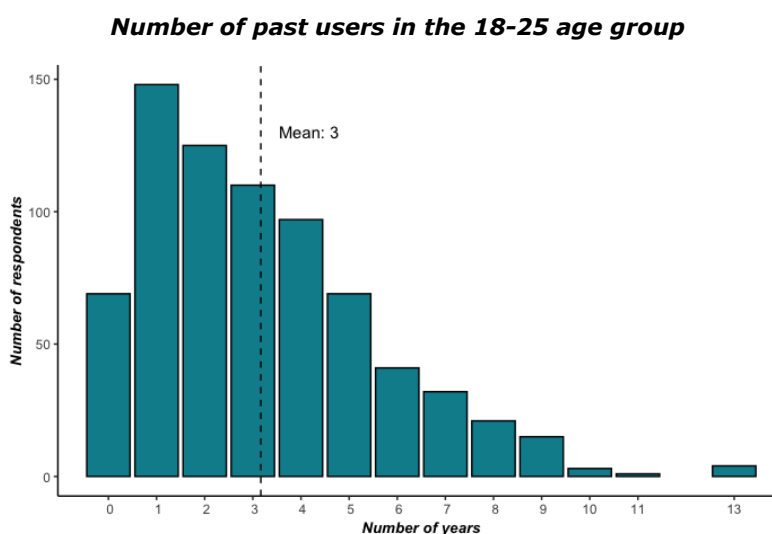
Note: The figure illustrates the approximate starting age of slim cigarette use among all respondents of the online survey, where the x-axis represents the current age of the respondent (Q.5.2) and the y-axis is the average numbers of years of product consumption (calculated by subtracting the current age from the starting age). The orange line represents the 45° or 18 y.o. line. Each blue point represents the average starting age for respondents of that age; points above the 18 y.o. line reveal those initiating consumption of slim cigarettes before the age of 18, while those below the line reveal age brackets starting consumption after the age of 18 on average. The blue lines represent the mean standard deviation, i.e. it does not show the full range of values for each age.

Source: OE elaboration from online survey data

⁴⁹ NA and misreported values are excluded.

Past users in the 18-25 age group had consumed slim cigarettes for 3 years on average; the average for those in the 26+ age group was 9 years (see Figure 44 and Figure 45)

Figure 44. Duration of use of past slim cigarette users (18-25 age group)



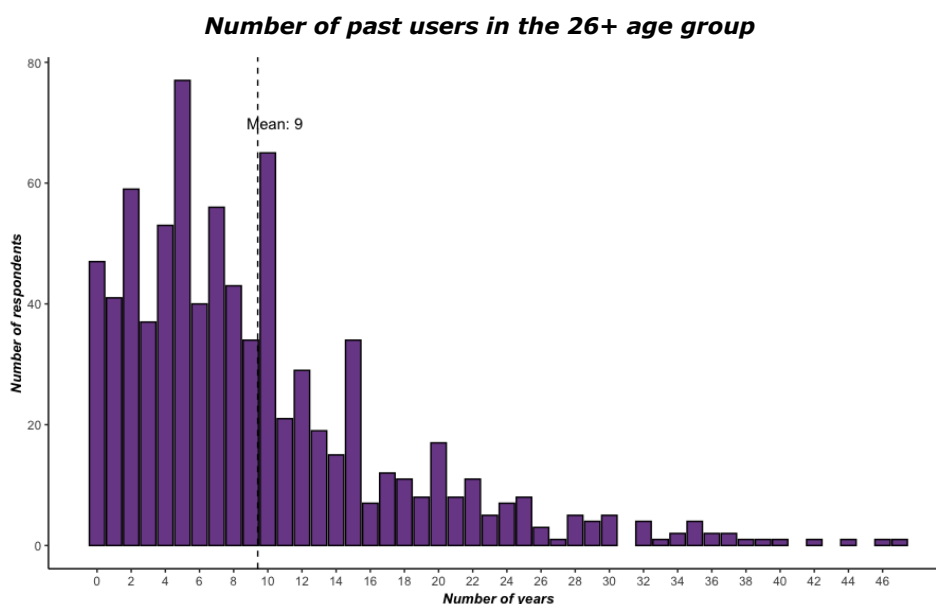
Base: past users of slim cigarettes in the 18-25 age group (N=503⁵⁰)

Question: Q.1.2.a. At about what age did you start using slim cigarettes? & Q.1.2.b. At about what age did you stop using slim cigarettes?

Note: The duration was calculated by subtracting Q.1.2.a. from Q.1.2.b for each individual that was a past user of the product. The mean shows the average duration of use.

Source: OE elaboration from online survey data

Figure 45. Duration of use of past slim cigarette users (26+ age group)



Base: past users of slim cigarettes in the 26+ age group (N=804⁵¹)

Question: Q.1.2.a. At about what age did you start using slim cigarettes? & Q.1.2.b. At about what age did you stop using slim cigarettes?

Note: The duration was calculated by subtracting Q.1.2.a. from Q.1.2.b for each individual that was a past user of the product. The mean shows the average duration of use.

Source: OE elaboration from online survey data

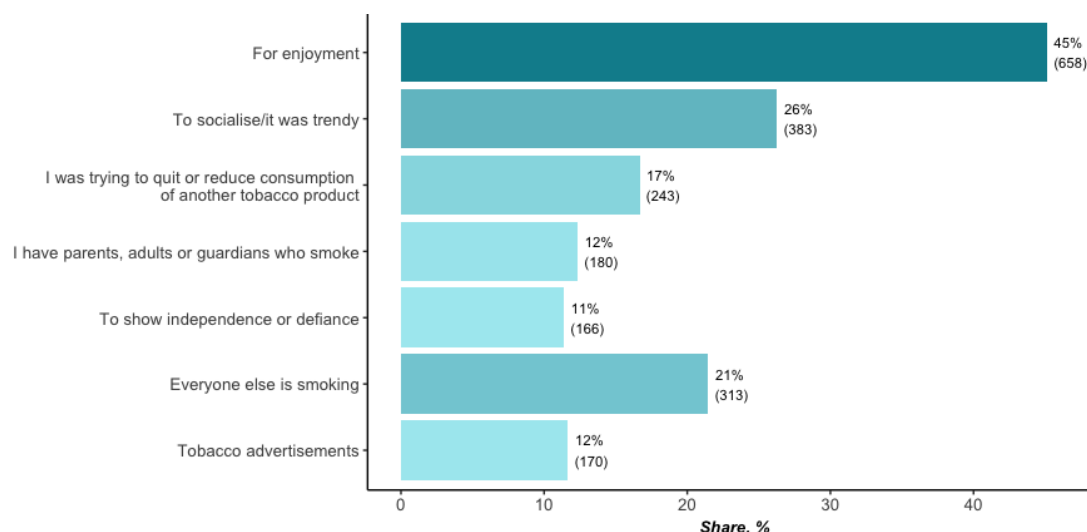
⁵⁰ NA and misreported values are excluded.

⁵¹ NA and misreported values are excluded.

Figure 46 and Figure 47 report the reasons why people started to smoke slim cigarettes. In the 18-25 age group, 45% of users said it was for enjoyment, 26% to socialise or because it was trendy and 21% because everyone else was smoking. A similar pattern was observed in the 26+ age group, with 47%, 28% and 16% respectively. In both age groups about one in five said it was to quit or to reduce consumption of other tobacco products.

Figure 46. Reasons for initiation of slim cigarette use (18-25 age group)

Percentage and number of current and past users in the 18-25 age group)



Base: current and past users of slim cigarettes in the 18-25 age group (N=1,458)

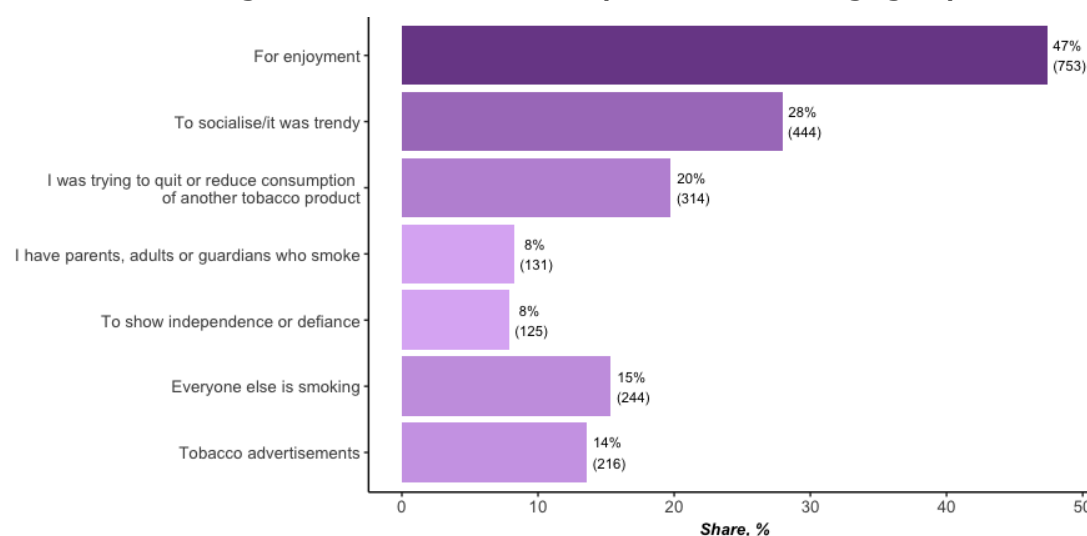
Question: Q.2.1.a. Why did you start using slim cigarettes? Please select up to 2 options.

Note: Each bar represents the proportion of respondents that selected the option within the age group. The percentages do not add up to 100% because respondents could select more than one term. For the response options including "smoking", the type of product 'smoked' was not specified.

Source: OE elaboration from online survey data

Figure 47. Reasons for initiation of slim cigarette use (26+ age group)

Percentage and number of current & past users in 26+ age group



Base: current and past users of slim cigarettes in the 26+ age group (N=1,589)

Question: Q.2.1.a. Why did you start using slim cigarettes? Please select up to 2 options.

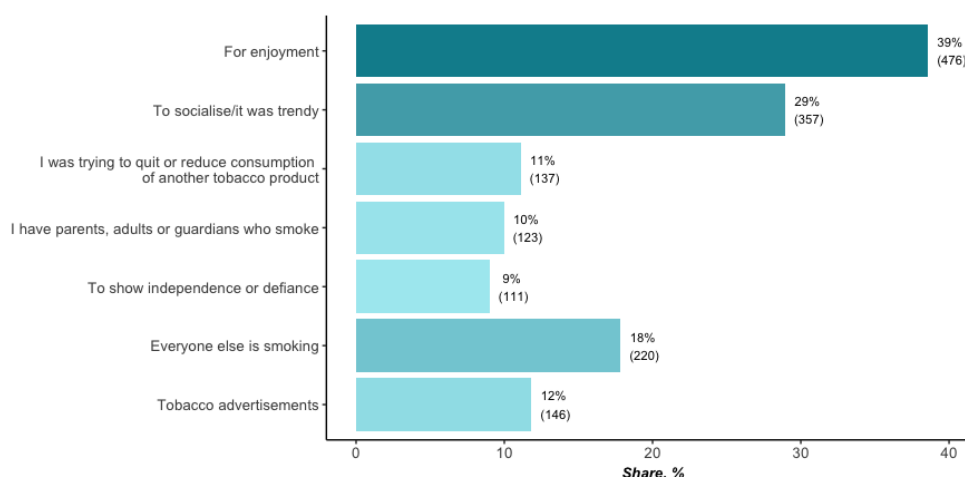
Note: Each bar represents the proportion of respondents that selected the option within the age group. The percentages do not add up to 100% because respondents could select more than one term. For the response options including "smoking", the type of product 'smoked' was not specified.

Source: OE elaboration from online survey data

Figure 48 and Figure 49 report on the reasons for starting to smoke slim cigarettes among those who had tried them 'once or twice'. Enjoyment, socialising and trendy were the main reasons for both age groups. It is of note that, in the 26+ age group, tobacco advertisements were the third most cited reason (16%). The advertisements must have been very memorable given the regulations restricting tobacco advertising and sponsorships in Directive 2003/33/EC.

Figure 48. Initiation of slim cigarette use: reasons for trying (18-25 age group)

Percentage and number of those who tried once or twice in the 18-25 age group



Base: those who tried to use slim cigarettes in the 18-25 age group (N=1,234)

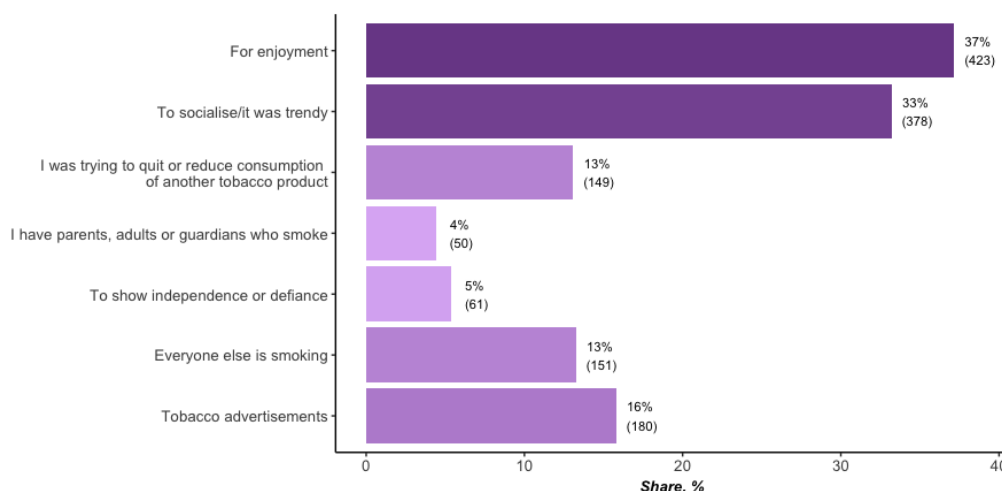
Question: Q.2.1.b. Why did you try slim cigarettes? Please select up to 2 options.

Note: Each bar represents the proportion of respondents that selected the option within the age group. The percentages do not add up to 100% because of the fact that respondents could select more than one term – therefore, the bars must be interpreted separately. For the response options including "smoking", the type of product 'smoked' was not specified.

Source: OE elaboration from study survey data

Figure 49. Initiation of slim cigarette use: reasons for trying (26+ age group)

Percentage and number of those who tried once or twice in the 26+ age group



Base: those who tried to use slim cigarettes once or twice in the 26+ age group (N=1,138)

Question: Q.2.1.b. Why did you try slim cigarettes? Please select up to 2 options.

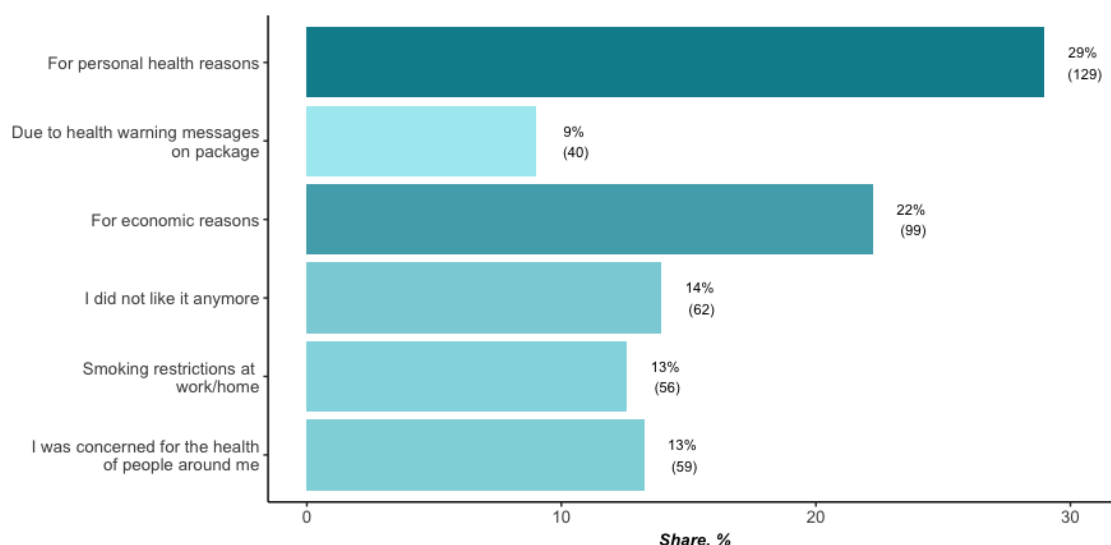
Note: Each bar represents the proportion of respondents that selected the option within the age group. The percentages do not add up to 100% because of the fact that respondents could select more than one term – therefore, the bars must be interpreted separately. For the response options including "smoking", the type of product 'smoked' was not specified.

Source: OE elaboration from online survey data

Figure 50 and Figure 51 chart the reasons for attempting to quit slim cigarette smoking. In the 18-25 age group, 68% of current slim cigarette smokers reported having tried to quit at least once (see Figure 188 in Annex). The most cited reason for attempts to quit were personal health (29%), followed by price (economic reasons) (22%). In the 26+ age group, 71% of current users had attempted to quit, and 30% having tried to quit more than twice (see Figure 189 in Annex). Their reasons were much the same as the younger age group: 35% for health and 21% on account of price. As with the younger age group, health warning messages on the packages were the least cited reason.

Figure 50. Reasons for trying to quit slim cigarette use (18-25 age group)

Percentage and number of current users in the 18-25 age group who tried to quit



Base: current users of slim cigarettes that tried to quit in the 18-25 age group (N=445)

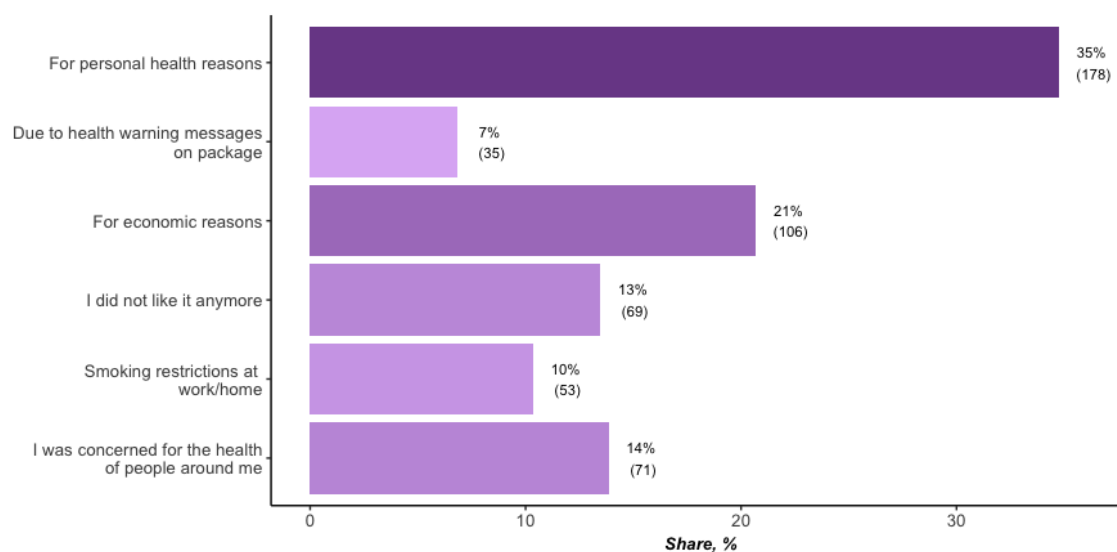
Question: Q.2.2. What made you try to quit slim cigarettes? Select the most fitting answer.

Note: Each bar represents the proportion of respondents that selected the option within the age group.

Source: OE elaboration from online survey data

Figure 51. Reasons for trying to quit slim cigarette use (26+ age group)

Percentage and number of current users in the 26+ age group who tried to quit



Base: current users of slim cigarettes that tried to quit in the 26+ age group (N=512)

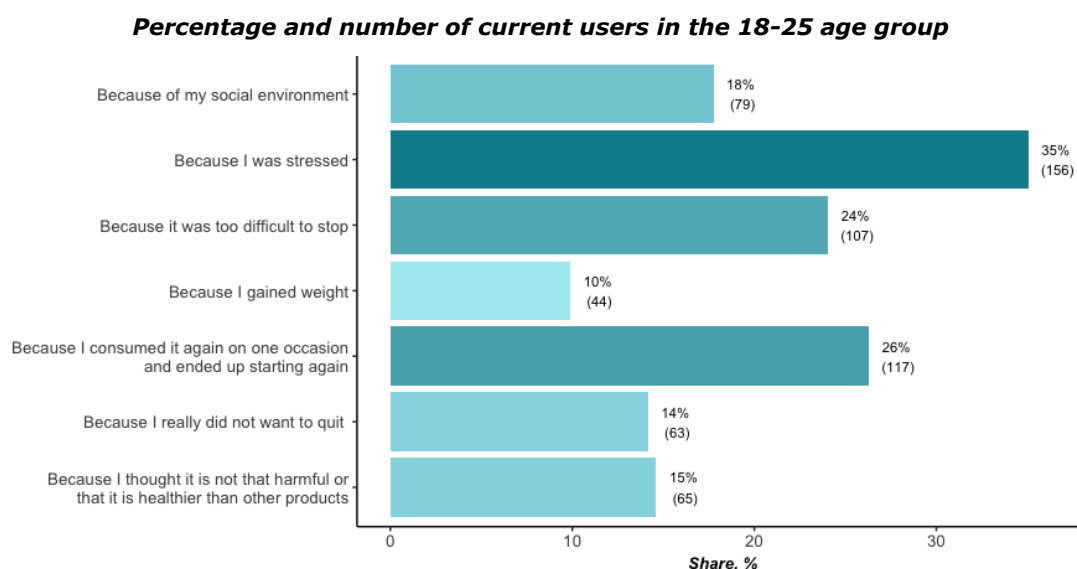
Question: Q.2.2. What made you try to quit slim cigarettes? Select the most fitting answer.

Note: Each bar represents the proportion of respondents that selected the option within the age group.

Source: OE elaboration from online survey data

Figure 52 and Figure 53 report the reasons for starting to smoke slim cigarettes again after trying to quit. The reasons were both personal and social. The 18-25 age group cited 'stress' (35%), 'temptation' (26%) and it was 'too difficult' (24%). Almost one in five cited their 'social environment'. A similar pattern was evident 26+ age group; 'stress' (32%), 'too difficult' (31%) and 'temptation' (24%). It appeared that for both age groups addiction made quitting an almost unsurmountable challenge.

Figure 52. Reasons for starting to use slim cigarettes again (18-25 age group)



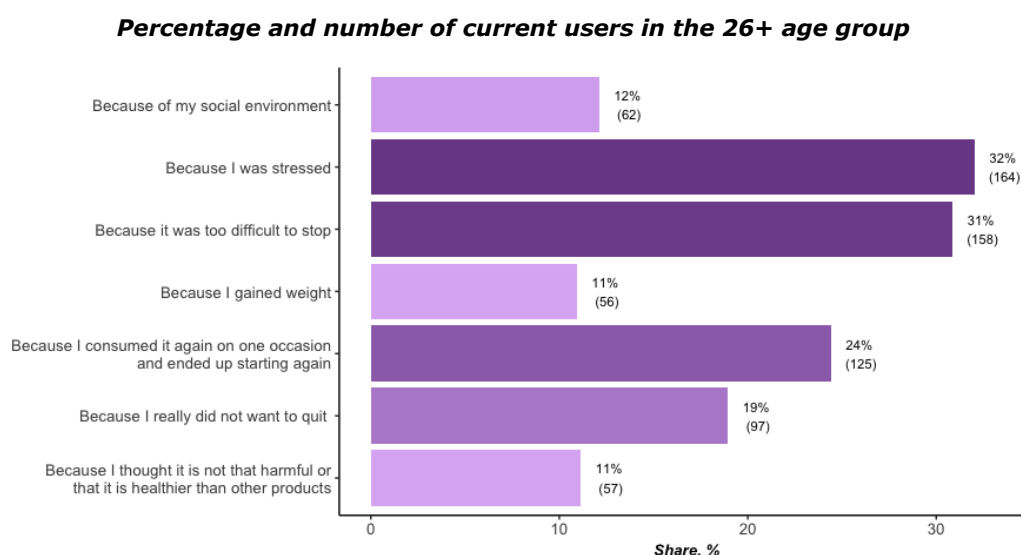
Base: current users of slim cigarettes that tried to quit in the 18-25 age group (N=445)

Question: Q.2.3. Why did you start to use slim cigarettes again? Please select up to 2 options.

Note: Each bar represents the proportion of respondents that selected the option within the age group. The percentages do not add up to 100% because of the fact that respondents could select more than one term – therefore, the bars must be interpreted separately.

Source: OE elaboration from online survey data

Figure 53. Reasons for starting to use slim cigarettes again (26+ age group)



Base: current users of slim cigarettes that tried to quit in the 26+ age group (N=512)

Question: Q.2.3. Why did you start to use slim cigarettes again? Please select up to 2 options.

Note: Each bar represents the proportion of respondents that selected the option within the age group. The percentages do not add up to 100% because of the fact that respondents could select more than one term – therefore, the bars must be interpreted separately.

Source: OE elaboration from online survey data

3.2.3 Parallel use with other tobacco and related products

The survey found that a large majority of current of slim cigarettes smokers were dual users i.e. using another tobacco product in parallel. In the 18-25 age group, 81% used slim cigarettes with other tobacco products, mainly conventional cigarettes (also see Figure 190 in Annex). The focus group participants did not see much difference between these two products with the choice between them a matter of personal preference:

"I am speaking about the cigarettes and the slims. They are the easiest to consume because you don't need to roll them, all you need is a lighter and that is all." (M, 20, FR)

"These are in small quantities, because it's true that when we take the normal cigarette, we may not smoke it until the end because we get sick of it after a while. Whereas with these slim ones, it's just what you need." (F, 49, FR)

"I smoke slims when I want a lighter cigarette." (M, 24, BG)

Table 27 and Table 28 chart the career flows of current slim cigarette users. In the 18-25 age group 14% of current users had transitioned from other products, mainly conventional cigarettes to consume slim cigarettes exclusively: 5% were 'only ever' slim cigarette smokers and 89% of current slim cigarette smokers were dual users with 85% reporting the smoking of conventional cigarettes. A similar pattern is evidenced for the 26+ age group: 17% transitioned from other tobacco products, 5% were only ever slim cigarette smokers and 78% reported dual use with 95% of them smoking conventional cigarettes in parallel.

Table 27. Career flows of slim cigarette use for current users (18-25 age group)

Percentage and number of current users in the 18-25 age group

			Number	%
Total			656	100
Never users of other products			34	5
Exclusive current user of slim cigarettes & past user of other product(s)			89	14
<i>Out of them, switching from (one or more products) *</i>				
Cigarettes	(n=76)	85% *		
Waterpipe tobacco	(n=21)	24% *		
Electronic cigarettes	(n=14)	16% *		
HTPs	(n=14)	16% *		
Small cigarillos	(n=6)	7% *		
Current user of slim cigarettes & both current and past user of other product(s)			533	81
<i>Other products consumed in parallel now *</i>				
Cigarettes	(n=452)	85% *		
E-cigarettes	(n=274)	51% *		
HTPs	(n=228)	43% *		
Waterpipe tobacco	(n=226)	42% *		
Small cigarillos	(n=216)	41% *		

Base: current users of slim cigarettes in the 18-25 age group (N=656)

Question: Q.1.1. Have you ever used slim cigarettes?

Note: Patterns of use and consumption careers are explored for different consumer profiles of current users. A breakdown is provided for the two variables of interest, i.e. current parallel use patterns. The last row only provides the breakdown of current dual use as this is the variable of interest; it does not include a breakdown that takes account past consumption of products to avoid confusion (from too many permutations).

(*) Percentages add up to more than 100% on account of the prevalence of dual use.

Source: OE elaboration from online survey data

Table 28. Career flows of current slim cigarette users (26+ age group)

Percentage and number of current users in the 26+ age group

			Number	%
Total			725	100
Never users of other products			34	5
Exclusive current user of slim cigarettes & past user of other product(s)			124	17
<i>Out of them, switching from (one or more products) *</i>				
Cigarettes	(n=113)	91% *		
Electronic cigarettes	(n=19)	15% *		
Small cigarillos	(n=18)	14% *		
HTPs	(n=18)	14% *		
Waterpipe tobacco	(n=9)	7% *		
Current user of slim cigarettes & both current and past user of other product(s)			567	78
<i>Other products consumed in parallel now *</i>				
Cigarettes	(n=503)	95% *		
E-cigarettes	(n=284)	50% *		
HTPs	(n=251)	44% *		
Small cigarillos	(n=237)	42% *		
Waterpipe tobacco	(n=173)	31% *		

Base: current users of slim cigarettes in the 26+ age group (N=725)

Question: Q.1.1. Have you ever used slim cigarettes?

Note: Patterns of use and consumption careers are explored for different consumer profiles of current users. A breakdown is provided for the two variables of interest, i.e. current parallel use patterns. The last row only provides the breakdown of current dual use as this is the variable of interest; it does not include a breakdown that takes account past consumption of products to avoid confusion (from too many permutations).

(*) Percentages add up to more than 100% on account of the prevalence of dual use.

Source: OE elaboration from online survey data

Table 29 and Table 30 depict the career flows for past slim cigarette smokers. In the 18-25 age group, of all past users of slim cigarette, 14% had been used the product exclusively in the past and had now switched to another product, mainly traditional cigarettes. Furthermore, 29% had been dual users and had now quit all tobacco products and 53% were former slim cigarettes smokers who were now using other tobacco and related products, mainly conventional cigarettes.

In the 26+ age group, 42% of past slim cigarette smokers now used other tobacco products. 39% of dual users had quit all tobacco products and 42% were past and current dual users of whom 68% reported smoking traditional cigarettes.

Table 29 Career flow of past slim cigarette users (18-25 age group)

Percentage and number of past users in the 18-25 age group

	Number	%
Total	802	100
Never user of other products	31	4
Dual past user & not using any products now	236	29
Exclusive past user of slim cigarettes & current user of other products (but not slim cigarettes)	112	14
<i>Out of them, switching to (one or more products) *</i>		
Cigarettes (n=98)	87% *	
Electronic cigarettes (n=22)	20% *	
HTPs (n=22)	20% *	
Waterpipe tobacco (n=21)	19% *	
Small cigarillos (n=11)	19% *	
Dual past user & current user of other products (but not slim cigarettes)	423	53
<i>Other products currently consumed in parallel (one or more products) *</i>		
Cigarettes (n=251)	59% *	
E-cigarettes (n=152)	36% *	
HTPs (n=122)	29% *	
Waterpipe tobacco (n=94)	22% *	
Small cigarillos (n=50)	12% *	

Base: past users of slim cigarettes in the 18-25 age group (N=802)

Question: Q.1.1. Have you ever used slim cigarettes?

Note: Patterns of use and consumption careers are explored for different consumer profiles of past users. A breakdown is provided for the variables of interest, i.e. those who continued using other tobacco products after using e-cigarettes. The last row only provides the breakdown of current consumption because this is the variable of interest; it does not include a breakdown that takes account past consumption of products to avoid confusion (from too many permutations).

(*) Percentages add up to more than 100% on account of the prevalence of dual use.

Source: OE elaboration from online survey data

Table 30 Career flows of past slim cigarette users (26+ age group)

Percentage and number of past users in the 26+ age group

	Number	%
Total	864	100
Never users of other products	24	3
Dual past user & not using any products now	335	39
Exclusive past user of slim cigarettes & current user of other products (but not slim cigarettes)	141	16
<i>Out of them, switching to (one or more products) *</i>		
Cigarettes (n=191)	96% *	
Electronic cigarettes (n=40)	20% *	
HTPs (n=38)	19% *	
Small cigarillos (n=15)	10% *	
Waterpipe tobacco (n=11)	6% *	
Dual past user & current user of other products (but not slim cigarettes)	364	42
<i>Other products currently consumed (one or more products) *</i>		
Cigarettes (n=248)	68% *	
E-cigarettes (n=174)	48% *	
HTPs (n=114)	31% *	
Waterpipe tobacco (n=59)	16% *	
Small cigarillos (n=44)	12% *	

Base: past users of slim cigarettes in the 26+ age group (N=864)

Question: Q.1.1. Have you ever used slim cigarettes?

Note: Patterns of use and consumption careers are explored for different consumer profiles of past users. A breakdown is provided for the variables of interest, i.e. those who continued using other tobacco products after using e-cigarettes. The last row only provides the breakdown of current consumption because this is the variable of interest; it does not include a breakdown that takes account past consumption of products to avoid confusion (from too many permutations).

(*) Percentages add up to more than 100% on account of the prevalence of dual use.

Source: OE elaboration from online survey data

3.3 Consumer perception

While the literature pointed to the promotion of slim cigarettes as elegant, high status and appealing to women, the survey data showed an almost even prevalence of slim cigarette smoker in women and men. However, the focus group participants repeatedly mentioned femininity in association with slim cigarettes:

"I would say they are feminine." (M, 34, ES)

"I would rather say those long cigarettes, the cigarettes are for ladies." (F, 27, DE)
referring to slim cigarettes

"Back in the 20's it was acceptable to smoke them. They had sticks to make them longer as they were wearing white gloves, to prevent them from turning yellow. So, they had something to help them hold it." (M, 49, DE)

"Right, feminine. I couldn't imagine a man smoking these." (M, 30, DE)

"To me it's a fad, I've never smoked them." (F, 54, DE)

Another association for the participants reflected wealth and status. Participants considered that tobacco products were costly, especially in terms of taxes, but this was not considered a major drawback for a status symbol.

"They have a higher economic status." (M, 34, ES)

"I started with conventional cigarettes and when I started earning more money, I started using the slims." (M, 20, ES)

"Slim cigarettes are expensive." (F, 41, EL)

In the survey economic resources were an important factor in the decision to consume slim cigarettes or not. The second most cited reason for attempting to quit smoking slim cigarettes was for economic reasons, after personal health reasons (see Figure 50). The focus group participants suggested that slim cigarettes were no longer considered a trend, but rather a timeless fashion. They were associated with professional and business-oriented environments and were known to be particularly popular in Russia. The product evoked a universe of exclusivity, of higher status, and modernity. Accordingly, it was considered an expensive product that conveyed an image of sophistication, luxury and frivolity. The thinner packaging was also perceived as more elegant and handier.

"It (slim cigarette) is more elegant." (F, 43, EL); when asked why she chose slim cigarettes over conventional ones

"Yes, we like it when it's ergonomic, compact and small. Something that will not be cumbersome." (M, 20, FR)

"More elegant and sophisticated." (M, 32, ES)

"The attribute fancy-schmancy." (F, 63, DE)

Slim cigarettes were also associated with hedonism, bon-vivant and a pleasure-oriented atmosphere, imagery most likely based on the portrayal of the higher classes of the 1920s with their long cigarette holders, the Charleston and not "real" smokers.

The online survey showed current usage of slim cigarettes at 11% for the **18-25 age group** (see Figure 35) and 12% for the **26+ age group** (see Figure 36). Bulgaria and Poland had high prevalence of slim cigarette users in both age groups. In the 18-25 age group, 23% of respondents reported current use of slim cigarettes in Bulgaria, with only 35% of never users. By contrast, 75% of respondents in Germany had never tried the product (i.e. never users). In the 26+ age group, a

majority had never tried slim cigarettes, but this varied significantly across the countries studied, ranging from the 35% to 75% of never users.

Slim cigarettes were also perceived as more “discreet” and “lighter”, with apparently less smell than conventional cigarettes. However, the act of smoking slim cigarettes was also sometimes experienced as too short or fast and that it might lead consumers to smoke a greater quantity of slim cigarettes to feel as satiated as compared to a traditional cigarette.

"We get the impression that it's milder and so less harmless but it's just as harmful."
(M, 28, FR)

"(...) I need three of them to get the effect of one cigarette, I'm not sure why. It's weird." (M, 22, DE)

"You might think they are light, but you end up smoking a lot more of them. It's pseudo healthy." (M, 49, DE)

Due to this characteristic, it was sometimes falsely assumed that the slim cigarettes were less harmful than conventional cigarettes because they contained less tobacco and had a “lighter” taste. The opinion was particularly shared in Greece, where slim cigarettes were associated with brands that produced low tar and low nicotine cigarettes. The misconception of reduced harmfulness meant that participants sometimes assumed the product could be used as a transitional product to quit smoking and were more likely to infer that the product was more acceptable within society.

"I started smoking this because, even though it is slightly longer than a cigarette, less tobacco is smoked." (F, 54, ES)

"I don't like to bother, also it is lighter, I smoke slim occasionally, when I want to relax." (F, 41, EL)

"It is thinner and, in theory, it has less quantity of tobacco." (M, 20, ES)

"We can say that they are for people that wish to quit cigarettes." (M, 28, FR)

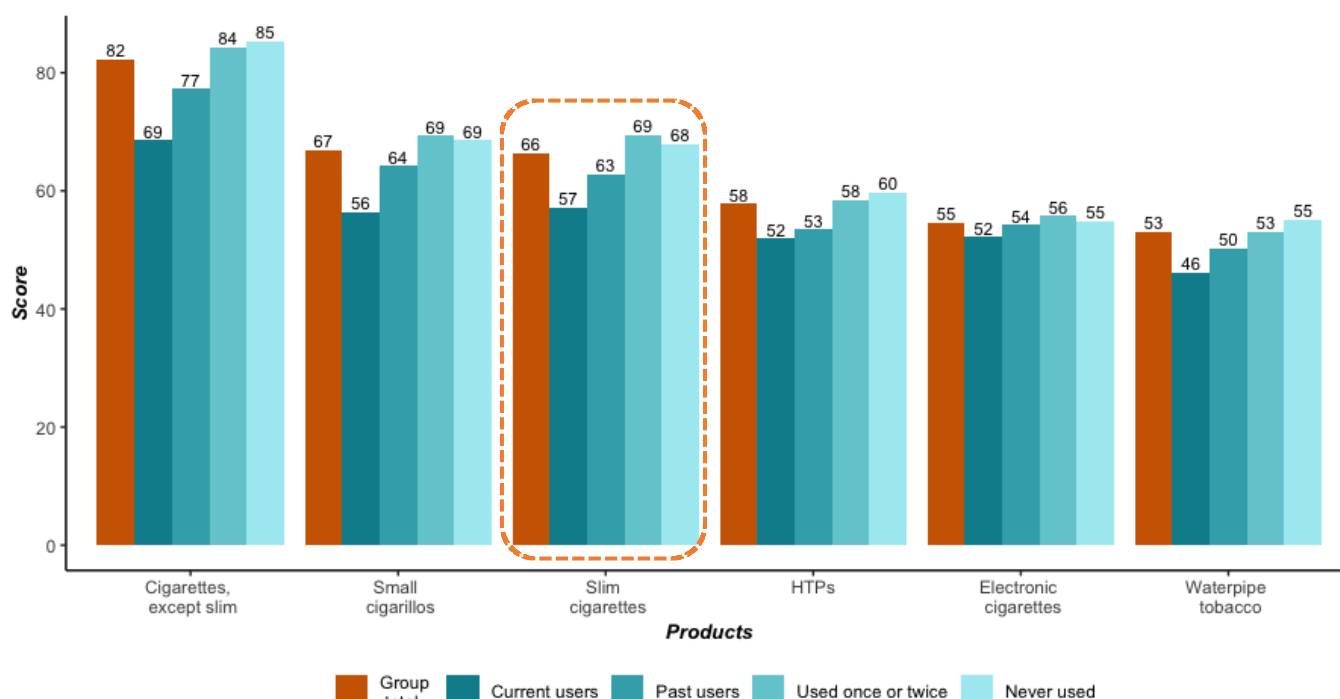
Figure 54 and Figure 55 show the results of the ‘swing method’ to assess risk perceptions of slim cigarettes and the other tobacco and related products included in this study plus conventional cigarettes. The swing method is explained in section 1.3.3. In brief, the respondent is asked which of the six products they think is most damaging to a user’s health. This product is given 100 points on a scale from 100 (most harmful) to 0 (not harmful). Then using this scale, the respondent is asked to rate each of the other five products on its harmfulness to health of users.

In the **18-25 age group** (see Figure 54), conventional cigarettes at a score of 82 are seen as the most harmful to health of the six products. Slim cigarettes ranked third highest in terms of harmfulness to the users’ health about the same as small cigarillos). Current users perceived slim cigarettes as less harmful than ‘past’ ‘once or twice’ and ‘never’ users. Current users gave a score of 57 out of 100 (where 100 represents the most damage to the users’ health) to e-cigarettes, in contrast to 68 for ‘never’ users.

The findings on perceived harmfulness in the **26+ age group** (see Figure 55) are broadly similar to those of the 18-25 age group (see Figure 166 in the Annex for further details). Conventional cigarettes at 82 were seen as most harmful. Slim cigarettes at 60 are seen as equally harmful as small cigarillos. Current e-cigarette users of gave a score of 55 compared to 59 for never users.

In the focus groups participants commented on current trends towards healthier lifestyles, notably for the younger generations. That the 26+ age group were more aware of the harmful effects associated with slim cigarettes than the 18-25 age group confirms that view.

Figure 54. Perceived harmfulness of slim cigarettes to users (18-25 age group)
(average score)



Base: respondents in the 18-25 age group (N=6,090)

Question: Q.3.1.a.⁵² & Q.3.1.b. You have chosen (e.g. cigarettes, except slims⁵³) as likely to be the most damaging to the health of users. We will give (e.g. cigarettes, except slims) 100 points on a scale of 0-100 for damaging users' health. Please assess the other products' damage on a scale of 0 to 100. If you think a product is about half as damaging to the health of users, you would give that product 50 points. If you think another product is just as damaging as (e.g. cigarettes, except slims), you would give it 100 points.

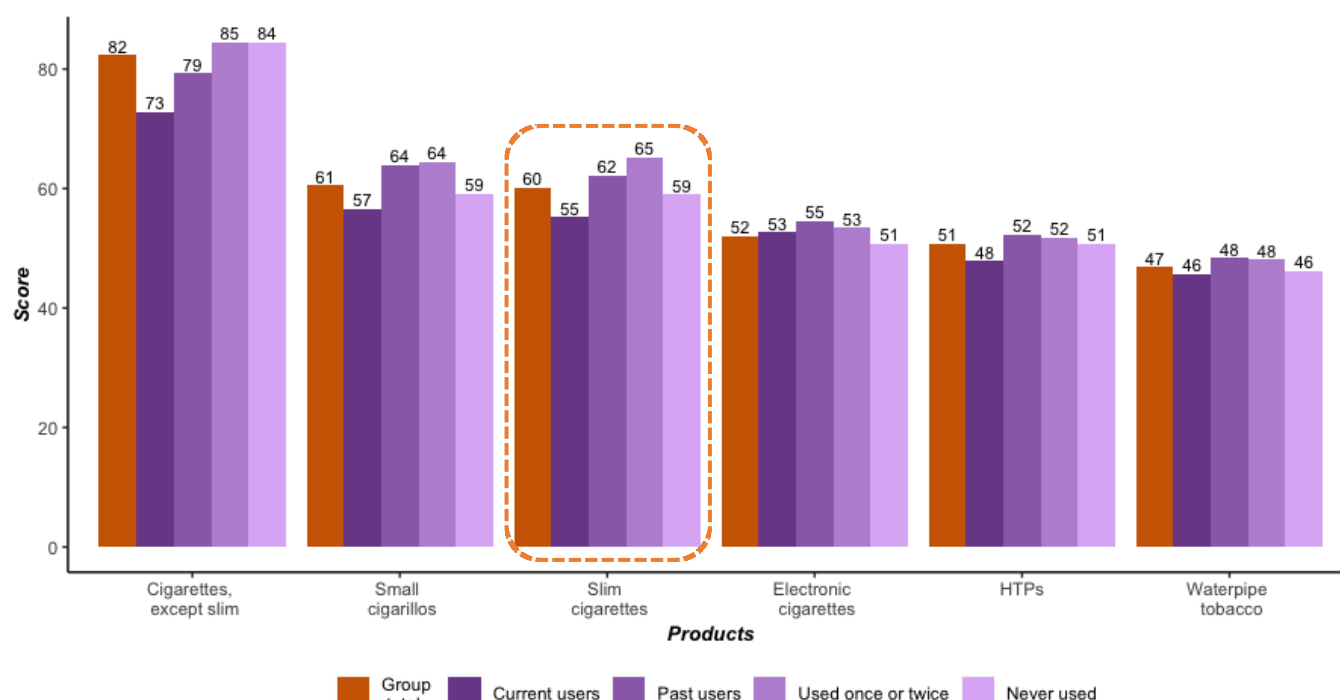
Note: The graph is based on the swing method. The scores were constructed using the responses to the two abovementioned questions. The bars in the graph are the average score of harm to the user provided by respondents for products under the scope of this study. For this chapter, the breakdown is provided for small cigarillos user profiles, and therefore differs from the other chapter. The products are sorted in descending order of the average score provided by the "group total," i.e. amongst all respondents of that age group. The dotted orange box identifies the product in focus.

Source: OE elaboration from online survey data

⁵² Q.3.1.a. Which of the six tobacco products listed below do you think is likely to be most damaging to the health of users?

⁵³ The product in parentheses would change in the online survey based on individual responses to Q.3.1.a.

Figure 55. Perceived harmfulness of slim cigarettes to users (26+ age group)
(average score)



Base: respondents in the 26+ age group (N=5,910)

Question: Q.3.1.a.⁵⁴ & Q.3.1.b. You have chosen (e.g. cigarettes, except slims⁵⁵) as likely to be the most damaging to the health of users. We will give (e.g. cigarettes, except slims) 100 points on a scale of 0-100 for damaging users' health. Please assess the other products' damage on a scale of 0 to 100. If you think a product is about half as damaging to the health of users, you would give that product 50 points. If you think another product is just as damaging as (e.g. cigarettes, except slims), you would give it 100 points.

Note: The graph is based on the swing method. The scores were constructed using the responses to the two abovementioned questions. The bars in the graph are the average score of harm to the user provided by respondents for products under the scope of this study. For this chapter, the breakdown is provided for slim cigarette user profiles, and therefore differs from the other chapter. The products are sorted in descending order of the average score provided by the "group total," i.e. amongst all respondents of that age group. The dotted orange box identifies the product in focus.

Source: OE elaboration from online survey data

Slim cigarettes were considered as a harmful product that closely resembled conventional cigarettes because it had the same paper, filters and tobacco – which provides a potential explanation for why current users in the 18-25 age group ranked slim cigarettes as slightly more harmful than small cigarillos.

"Well yes, it's the slim. It has all the chemicals." (F, 48, FR); in response to which was the most harmful product under the scope

"It tries to look different from normal tobacco and, in the end, you are smoking the same thing." (M, 31, ES)

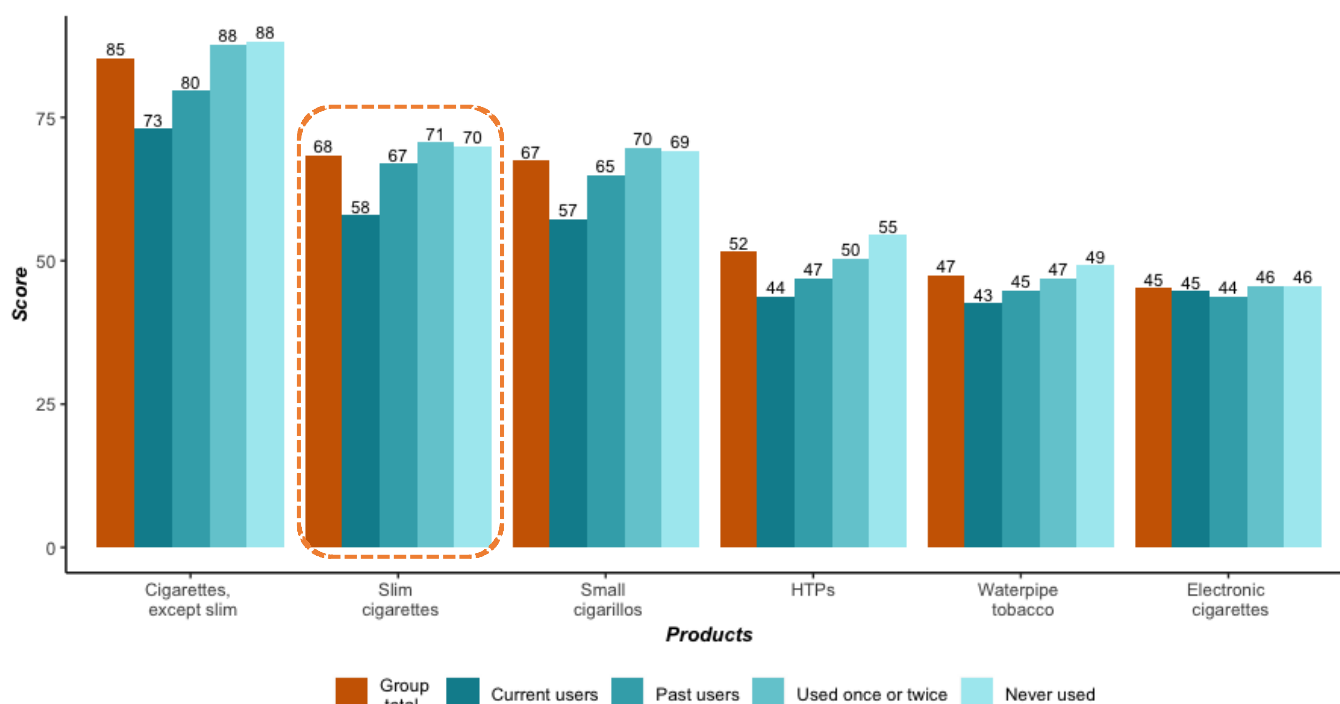
"That tobacco is rolled in paper which has several things we don't know." (F, 54, ES)

⁵⁴ Q.3.1.a. Which of the six tobacco products listed below do you think is likely to be most damaging to the health of users?

⁵⁵ The product in parentheses would change in the online survey based on individual responses to Q.3.1.a.

Figure 56 and Figure 57 focus on the passive smoking health risk of slim cigarettes. The 18-25 age group (see Figure 56) assessed slim cigarettes as the most harmful product to bystanders after traditional cigarettes and just as harmful to bystanders as to smokers. While harm the risk to others was widely acknowledged, it was not cited as a common reason for quitting the use of this product (see Figure 50). The 26+ age group (see Figure 57) assessed the passive smoking risk as the same as small cigarillos. As with the younger age group, the risk of harm to bystanders was seen as about the same as the risk to users.

Figure 56. Perceived harmfulness of slim cigarettes to bystanders (18-25 age group)
(average score)



Base: respondents in the 18-25 age group (N=6,090)

Question: Q.3.2.a.⁵⁶ & Q.3.2.b. Q.3.2.b. You have chosen (e.g.; cigarettes, except slims⁵⁷) as likely to be the most damaging to the health of passive users. We will give (e.g.; cigarettes, except slims) 100 points on a scale of 0-100 for damaging passive users' health. Please assess the other products' damage on the health of passive users on a scale of 0 to 100. If you think a product is about half as damaging to the health of passive users, you would give it 50 points. But if you think another product is just as damaging as (e.g.; cigarettes, except slims), you would give it 100 points.

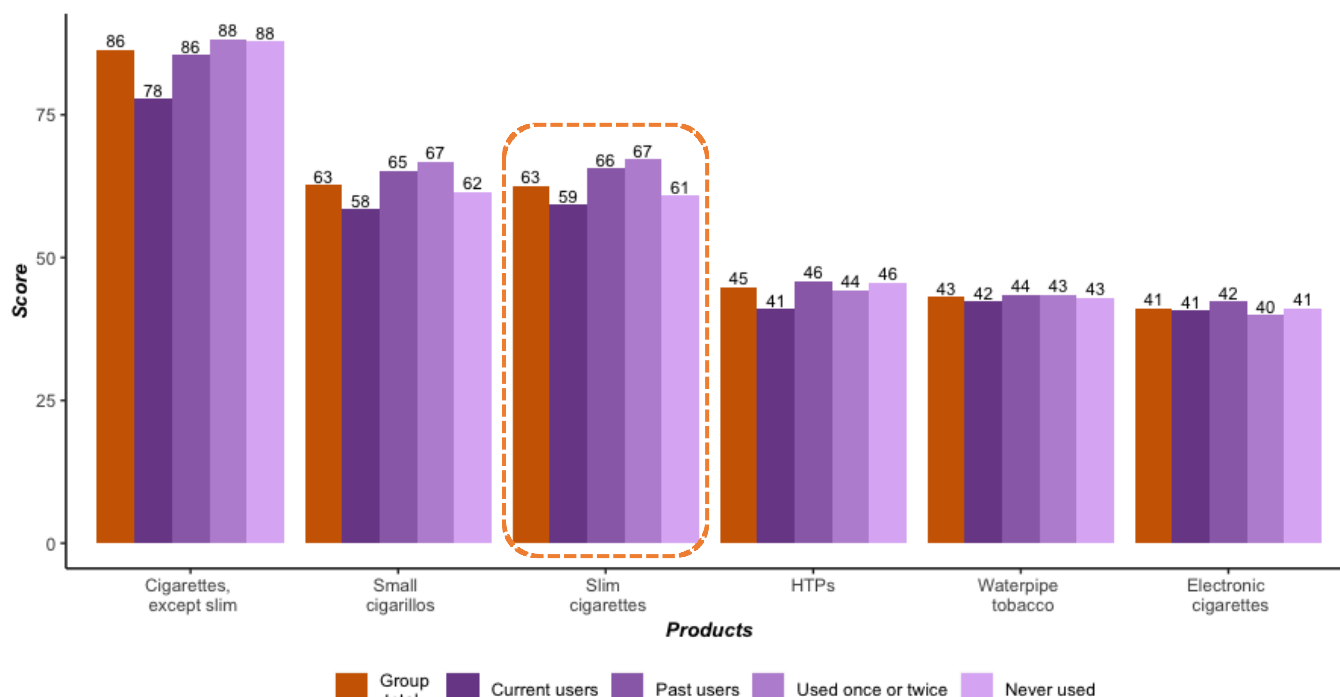
Note: The graph is based on the swing method. The scores were constructed using the responses to the two abovementioned questions. The bars in the graph are the average score of harm to the passive user provided by respondents for products under the scope of this study. For this chapter, the breakdown is provided for small cigarillo user profiles, and therefore differs from the other chapter. The products are sorted in descending order of the average score provided by the "group total," i.e. amongst all respondents of that age group. The dotted orange box identifies the product in focus.

Source: OE elaboration from online survey data

⁵⁶ Q.3.2.a. Which of the six tobacco products listed below is likely to be the most damaging to the health of other people from passive use?

⁵⁷ The product in parentheses would change in the online survey based on individual responses to Q.3.1.a.

Figure 57. Perceived harmfulness of slim cigarettes to bystanders (26+ group)
(average score)



Base: respondents in the 26+ age group (N=5,910)

Question: Q.3.2.a.⁵⁸ & Q.3.2.b. Q.3.2.b. You have chosen (e.g.; cigarettes, except slims⁵⁹) as likely to be the most damaging to the health of passive users. We will give (e.g.; cigarettes, except slims) 100 points on a scale of 0-100 for damaging passive users' health. Please assess the other products' damage on the health of passive users on a scale of 0 to 100. If you think a product is about half as damaging to the health of passive users, you would give it 50 points. But if you think another product is just as damaging as (e.g.; cigarettes, except slims), you would give it 100 points.

Note: The graph is based on the swing method. The scores were constructed using the responses to the two abovementioned questions. The bars in the graph are the average score of harm to the passive user provided by respondents for products under the scope of this study. For this chapter, the breakdown is provided for slim cigarette user profiles, and therefore differs from the other chapter. The products are sorted in descending order of the average score provided by the "group total," i.e. amongst all respondents of that age group. The dotted orange box identifies the product in focus.

Source: OE elaboration from online survey data

⁵⁸ Q.3.2.a. Which of the six tobacco products listed below is likely to be the most damaging to the health of other people from passive use?

⁵⁹ The product in parentheses would change in the online survey based on individual responses to Q.3.1.a.

3.4 Health, social and economic impacts

The impacts of conventional tobacco smoking, including slim cigarettes, have been studied in greater depth than the other products featured in this study. Most research includes slim cigarettes with other types of cigarette-like products; therefore, the present section relies in part on findings that are not exclusive to slim cigarettes. The economic costs of tobacco use are substantial and include significant healthcare costs for treating tobacco-related diseases as well as the loss of human capital due to tobacco-attributable morbidity and mortality. It has been suggested that about 15% of the aggregate health care expenditure in high-income countries can be attributed to smoking (Epku & Brown, 2015). Smoking also poses considerable indirect costs to society as it may affect second-hand smokers, it can cost employers in the form of loss of productivity and absenteeism of smokers due to illnesses, and it pollutes the environment (Epku & Brown, 2015).

Cigarette-like products are known to be highly toxic and can exert multiple effects on the pulmonary system, such as inflammation, DNA breaks as well as deranging the mucus clearance system (Ghosh et al., 2017). Some super-slim brands have been found to contain more dangerous tobacco chemicals than their larger counterparts (Ford et al., 2013). However, other studies have noted that slim cigarettes may reduce exposure to tar and other chemical substances found in cigarettes due to the reduced amount of tobacco within the cigarette stick. For instance, a study by Talhout et al. (2018) found reduced emissions of some but not all smoke components. The level of intoxication is not only dependent on the format, but also on the way in which the product is inhaled and the type of tobacco smoked. Reducing the circumference of the rod also reduces the "smoke velocities", i.e. the time for the smoke to pass from the coal to the mouth end of the cigarette, which in turn decreases filtration and retention of harmful chemicals in the filter (Talhout et al., 2018). Furthermore, the study found that there may be evidence for compensatory behaviour linked to the design of slim cigarettes. It should be noted that the study was based on smoke emissions measured from a machine, which do not translate directly to actual human exposure.⁶⁰ In terms of harm to passive users, second-hand smoke has been found to cause serious cardiovascular and respiratory diseases, including coronary heart disease and lung cancer. In pregnant women, it causes pregnancy complications and low birth weight, while it also raises the risk of sudden infant death syndrome for young children.⁶¹

While the harmfulness of tobacco is widely known, many studies have demonstrated that slim cigarette users misinterpret the small circumference of the product for reduced harmfulness and that the design increases appeal among vulnerable groups (see section 3.1). In terms of social impact, some studies have also shown that there is a high prevalence of smoking among disadvantaged population groups (Milcarz et al., 2017, Poland, N=1,817), who may not have the resources to sustain such a habit but have developed an addiction for it.⁶² In fact, tobacco use has been said to contribute to poverty by diverting economic resources from basic needs (e.g. food and shelter) to tobacco. This spending behaviour is difficult to curb because tobacco is so addictive.

In terms of excise duty applied to manufactured tobacco, the Council adopted the Tobacco Excise Directive (2011/64/EU), which updated the common fiscal legislation on tobacco products in the EU. Following the directive, slim cigarettes are taxed as cigarettes.

⁶⁰ In other words, they do not account for individual use behaviours.

⁶¹ <https://www.who.int/news-room/fact-sheets/detail/tobacco>

⁶² Most of the respondents in the study showed willingness to quit.

Table 31. Tax provisions applicable to manufactured tobacco

Product	Tax structure
Slim cigarettes (same as normal cigarettes)	• Mandatory mixed structure including both an ad valorem excise duty and a specific excise duty, which must be between 7.5% and 76.5% of total tax burden (since Jan. 2014). • At least 60% of weighted average price (WAP) and no less than EUR 90 per 1,000 cigarettes; or EUR 115 per 1,000 cigarettes. • A minimum excise duty (MED) may apply (i.e. a fixed monetary amount per quantity applicable if the amount of the excise duty falls below a minimum floor).

Source: Own elaboration

The regulation and tax measures were aligned with those of conventional cigarettes. Passport data showed that the highest specific tax on slim cigarettes was applied in Ireland, the UK and the Netherlands as of 2017 (see Figure 175). Increasing prices of tobacco products through taxes has been shown to decrease smoking prevalence as well as the quantity of cigarette sticks smoked per day (Ekpu & Brown, 2015; Ranabhat et al., 2019). Talhout et al. (2019) examined the physical cigarette design features and how they affected smoke emissions, product appeal and smoking behaviours. The study found that slim cigarettes were usually perceived as relatively safe and high-quality products, which translated into a stronger appeal, particularly for women. One study also showed that smokers often perceived longer cigarettes to be attractive and of high quality (Borland & Savvas, 2013), while another (although based in Australia) suggested that standard cigarette sticks were perceived as the highest quality and the most attractive product in terms of length/diameter for men, and that they were more inclined to rate a slim stick as less attractive (Borland, 2012, N=160). In addition, Ford et al. (2013, UK) showed that that slim and super-slim cigarettes with white filters and decorative features were viewed most favourably across gender and socio-economic groups. The smaller diameter motivated a perception of lower harm among (mostly) non-smoking adolescents. In general, the view was that because they contain less tobacco, they must, therefore, be less harmful. To reduce the possibility of cigarette appearance misleading consumers, the European Commission's draft Tobacco Products Directive proposed banning cigarettes <7.5 mm in diameter. However, this ban has not been included in the final Tobacco Products Directive,⁶³ but should be closely monitored in the future.

⁶³ European Parliament. European Tobacco Products Directive 2014/40/EU. [Accessed August 25, 2017]; Available at: http://ec.europa.eu/health/tobacco/docs/dir_201440_en.pdf

3.5 Concluding remarks on slim cigarettes

This section summarises the main findings that emerged from the research conducted on slim cigarettes. The systematic review of secondary sources revealed:

- Slim cigarettes are tobacco products in which their length and circumference, is modified to increase their appeal. However, slim cigarettes have the equal or very similar harm potentials as other cigarettes.
- The tobacco industry made slim cigarettes a symbol of gender equality, sophistication and upward mobility. Smoking was made to look fun, glamorous, healthy, liberating, relaxing, sexually attractive, slimming and sophisticated. Coloured packs were also developed to attract children.
- The slim cigarettes' market development varies greatly across the EU in absolute terms. Some have grown substantially while others have seen a decrease. However, the share of slim cigarettes in the cigarette market has grown over the past years in most countries and particularly in Eastern Europe (Passport, 2017).
- The regulation and tax measures for slim cigarettes were aligned with those on conventional cigarettes in the EU.
- Eurobarometer 458 found that the prevalence of slim cigarette use was 10% among females and 2% among males in 2017. Almost 1 in 5 respondents of the Eurobarometer believed that slim cigarettes were attractive.

The empirical work (focus groups and online survey) revealed the following user profile:

- The online survey found high consumption of slim cigarettes in Bulgaria and Poland.
- While more women than men were users of slim cigarettes the difference at 2% was relatively small for the 18-25 age group it rose to 5% for the 26+ age group.
- A typical current user consumes slim cigarettes daily and between 1 and 10 sticks every day of use. The majority started smoking slim cigarettes for enjoyment or to socialise. The younger respondents were more likely to have started as minors. A large majority of current users of slim cigarettes were also using other tobacco and related products.
- About 7 in 10 current users had tried to quit for personal health reasons or economic reasons. However, they failed to do so because of stress, they had used them on one occasion and ended up starting again, or it was too difficult to stop.
- A typical past user smoked at least once a week and between 1 and 10 sticks every day that he or she smoked. About 80% of former slim smokers were dual users. In the 18-25 age group, 37% have quit using all tobacco and related products under study, while 63% continued using them. These figures were 45% and 55%, respectively, in the 26 group.

Lastly, the following paragraphs highlight the *Product and Regulatory perceptions remarks*

- *General perception, including price.* Slim cigarettes have a connotation of femininity; however, this perception is less strongly rooted than it has been in the past. Slim cigarettes and their thinner packaging are also perceived as a product related to higher economic status, elegance, exclusivity and modernity. Accordingly, it was considered an expensive product. On the other hand, it was also sometimes conveyed a hedonistic, bon-vivant and pleasure-oriented atmosphere. This could also be associated with status-seekers who are rather frivolous and not "real" smokers.

- *Perception of harmfulness.* Slim cigarettes were perceived as “lighter”, with less smell than conventional cigarettes. It was also considered not as rewarding as conventional cigarettes. Users judged them to be less harmful to their health and to bystanders than conventional cigarettes. However, they were considered one of the most harmful products, along with small cigarillos. Never-users, compared to current users, perceived slims as more dangerous.
- *Perception of marketing practices.* The design of the product and its packaging have been found to impact the perception of the product. The product was historically marketed to target young women. Marketing practices on this type of product have been heavily regulated at the EU level. Accordingly, not many mentions were made to advertisements in the online survey nor in focus group discussions.
- *Regulatory support.* Slims are recognised as conventional tobacco products from the regulatory point of view.

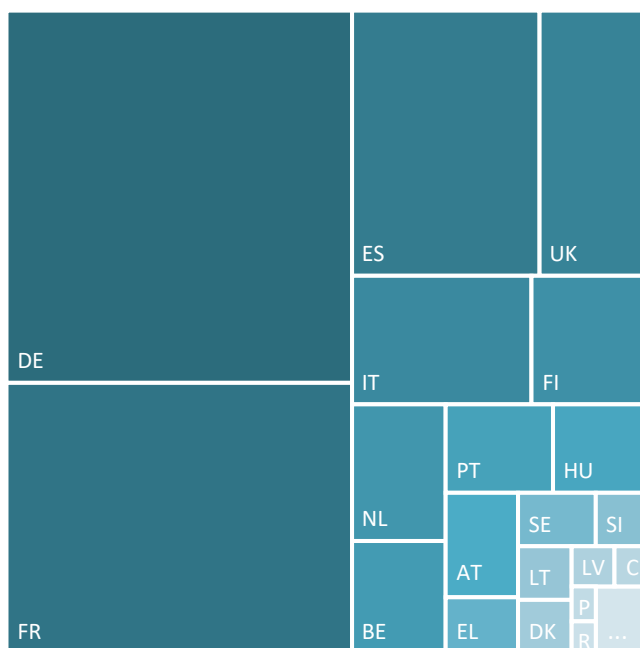
4 Small cigarillos

4.1 Background

The systematic literature review found very few articles on the perception and consumption of small cigarillos in the EU. However, information from the USA was widely available, in part due to the growing popularity of the product among specific population subgroups. A study by RAMBOLL (2014) found that the consumption of small cigarillos had increased in most EU Member States in the early 2010s. For instance, Latvia reported a very substantial increase in the consumption of cigars and cigarillos between 2009 and 2010, while cigarette consumption decreased by 23%. The increase in consumption in Latvia was almost entirely attributed to the release of cigarillos by the brand, Partner. According to the Eurobarometer 458, the prevalence of regular and former users of cigarillos remained relatively stable between 2014 and 2017 in the EU – with only 3% of respondents reporting regular current use in 2017. The Eurobarometer 458 also noted a slight decrease in the number of respondents that had tried cigarillos and/or cigars. 12% of respondents reporting consuming the product in December 2014 compared to 8% in March 2017.⁶⁴

Market developments. The figure below presents the structure of the EU market of small cigarillos in absolute values (million euros) by country. Germany (659 m EUR) and France (483 m EUR) had the largest market for small cigarillos in the EU in 2017, accounting for almost half of the entire EU small cigarillos market. They were followed by Spain (256 m EUR), the UK (153 m EUR) and Italy (120 m EUR).

Figure 58. Market size of small cigarillos in million euro in 2017



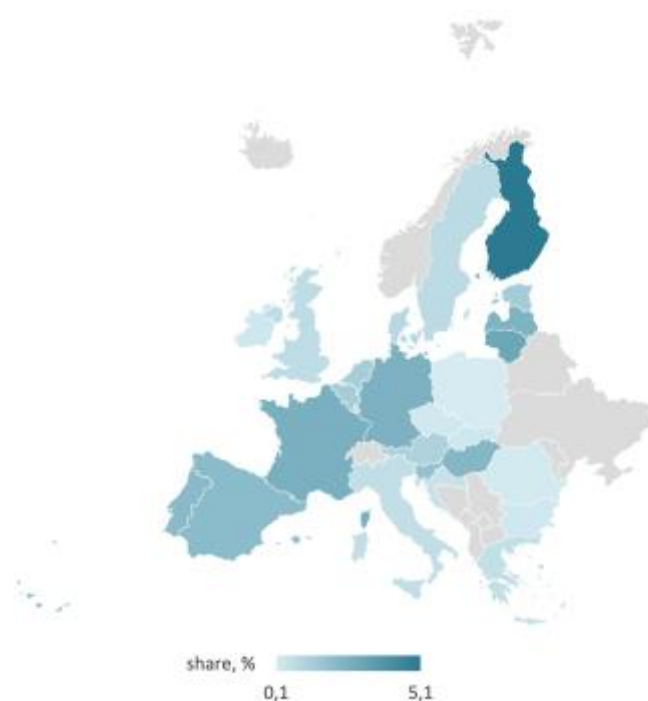
Note: The market size is estimated total sales in million euros for 2017, without population adjustments. It is displayed as a tree map graph that consists of series of nested rectangles of sizes proportional to the corresponding data value.

Source: OE elaboration of Passport, 2017

⁶⁴ There was a slight modification in the Eurobarometer 458 which clarified the definition of the two items (i.e. clearly distinguished the two products).

The map below presents the share of small cigarillos in the national tobacco markets across EU countries. The share of small cigarillos in the Finnish tobacco market was particularly high in comparison to other Member States, reaching over 5%. France, Germany, Lithuania and Latvia also appeared to be significant markets – all reaching a share of at least 2.5% in the national tobacco market.

Figure 59. Share of small cigarillos within the national total tobacco market, per country (2017)

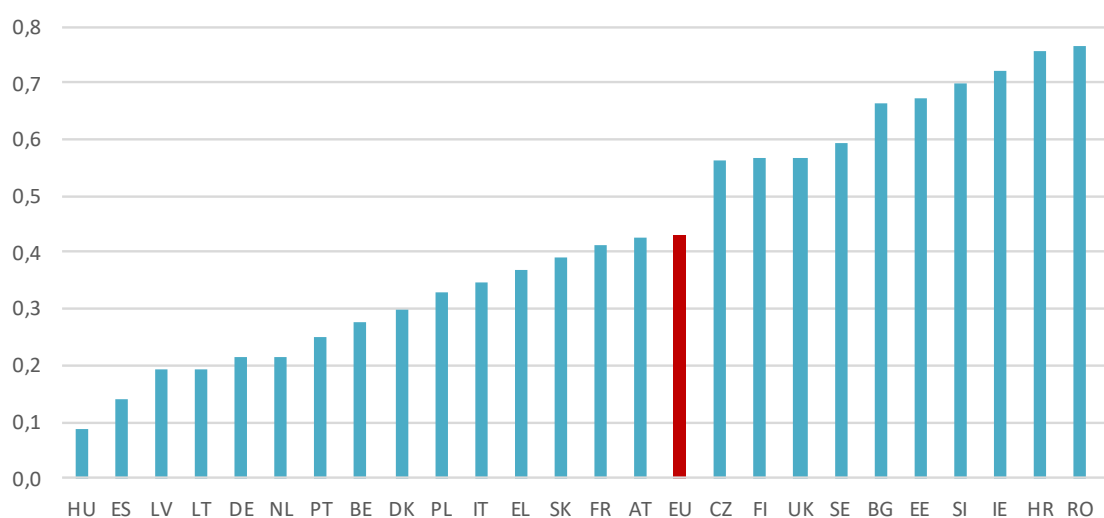


Note: Percentage share of small cigarillos were estimated with respect to the whole tobacco market in the country in million euros.

Source: OE elaboration of Passport, 2017

The distribution of the prices on small cigarillos in 2017 showed significant variation across Member States (see Figure 60). The prices were highest in Romania and Croatia. Based on the available data, the lowest prices were found in Hungary and Spain.

Figure 60. Price of small cigarillos in 2017, euros per unit (stick)

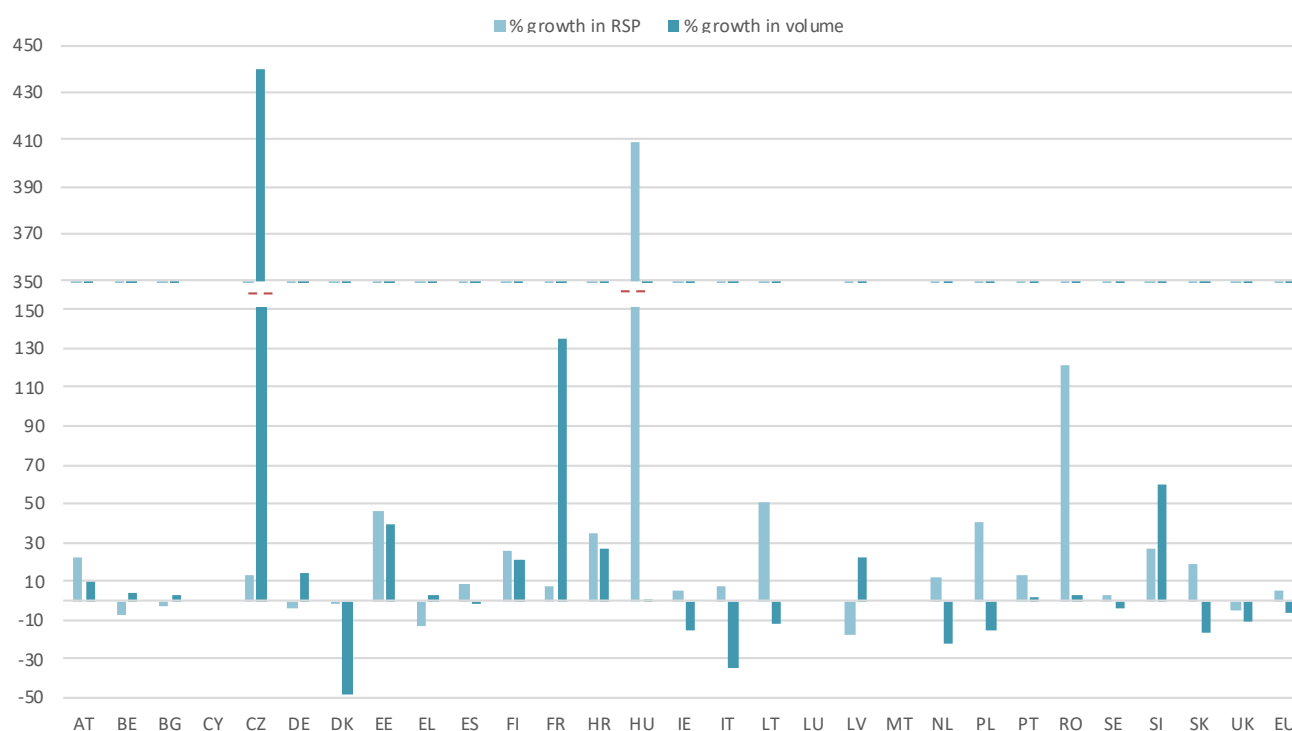


Note: EU average is calculated for 25 EU MS (excluding Cyprus, Luxembourg & Malta)

Source: Passport, 2017

From 2012 to 2017 the small cigarillo market followed an upward trend in terms of current retail selling price (RSP) across a majority of EU countries (see Figure 61). The most significant growth in terms of retail selling price occurred in Hungary with a four-fold increase in retail sales price between 2012 and 2017. The Czech Republic saw a four-fold increase in real volume while Denmark saw the largest decrease. It is also important to note that there has been a decline in the growth in volume of small cigarillos in Europe overall.

Figure 61. Market size trends of small cigarillos over 2012-2017



Note: EU average is calculated for 25 EU MS (excluding Cyprus, Luxembourg & Malta)

Source: Passport, 2017

In the USA, Regan (2012, USA, N=1,556) reported that the marketing strategies of the industry targeted young adults, that use of flavoured small cigarillos was much more common among the youth, and that certain demographic groups were more likely to be influenced by tobacco marketing, including males, young adults, and people with low income or education. A study conducted by Harrell (2017, USA, N=TATAMS (2483); M-PACT (4326)) also found evidence that flavoured tobacco products had a high appeal among young people (aged 12-17) and young adults (aged 18-29). While these studies are specific to the USA, the marketing strategies have been used online, to which European citizens may be exposed.

Consumer perception. Kostygina et al. (2016, USA) argued that tobacco companies had engaged in efforts to blur the line between little cigars and cigarettes to increase appeal, and that the use of flavours had facilitated these efforts. They found that the addition of flavours masked the harshness of tobacco products and made it easier for users to inhale the smoke, which in turn made the product appear less harmful and more palatable to younger users. Cornacchione et al. (2016, USA) also found that participants generally had positive associations with little cigars and cigarillos (LLCs). They could be used in social gatherings, they came in various flavours, and they could be used to perform smoke tricks.

Sterling et al. (2015, USA, N=90) conducted focus group interviews to gain a better understanding of the perceptions and behaviours of young adult smokers aged 18-34 years old with respect to flavoured

LCC products. The participants were excited and fascinated by them and agreed that flavoured tobacco made smoking LCCs more palatable. The design of the packaging also influenced the young adults' liking and the consumption patterns, stimulating the switch from cigarettes to flavoured LCCs.

In a US survey few adults were found to believe that LLCs were less harmful to their health than cigarettes (Regan, 2012). In Europe, LLCs were marketed as cigars, i.e. with a lower excise duty, in some markets including Spain, Portugal, Latvia and Lithuania (RAMBOLL, 2014). Nevertheless, they were perceived as a substitute for cigarettes.

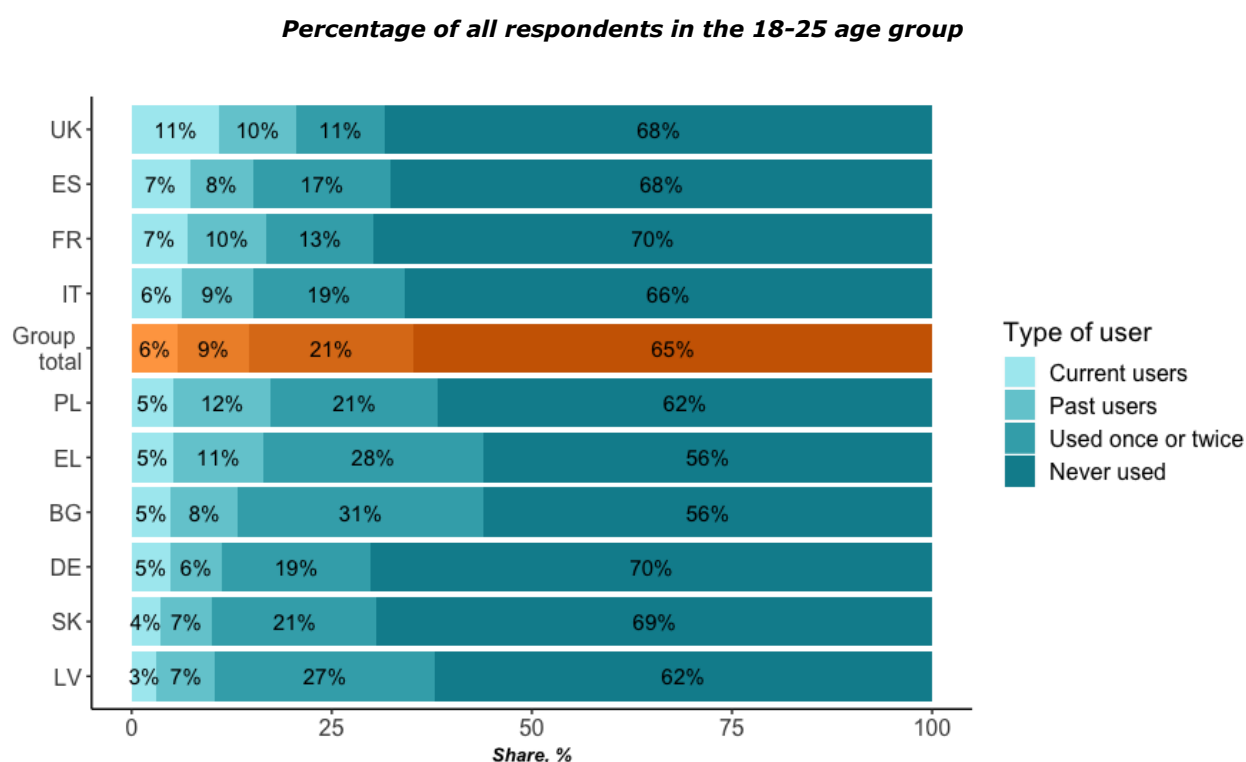
Impacts. Research on small cigarillos typically has not distinguished the product from cigars and other cigar-like products. However, small cigarillos have been assumed to lead to similar health problems to conventional cigarettes. The industry has increasingly developed small cigarillos as a cigarette substitute, as the product attracted lower taxation than traditional tobacco products and was exempt from bans on flavourings. The increasing global reach of social media and its role in indirect product promotion has been noted in the USA (see section 4.4 for more information on the health, social and economic impacts of small cigarillos).

4.2 Market developments

4.2.1 Socio-demographic characteristics

Across the countries surveyed in the **18-25 age group** 6% were current users of small cigarillos, 9% past users and 65% never users (see Figure 62). The highest percentage of current users was found in the United Kingdom (11%), followed by Spain (7%) and France (7%). Across all surveyed countries, the share of former users was higher than current users for this age group.

Figure 62. Use of small cigarillos: country breakdown (18-25 age group)



Base: all respondents in the 18-25 age group (N=6,090)

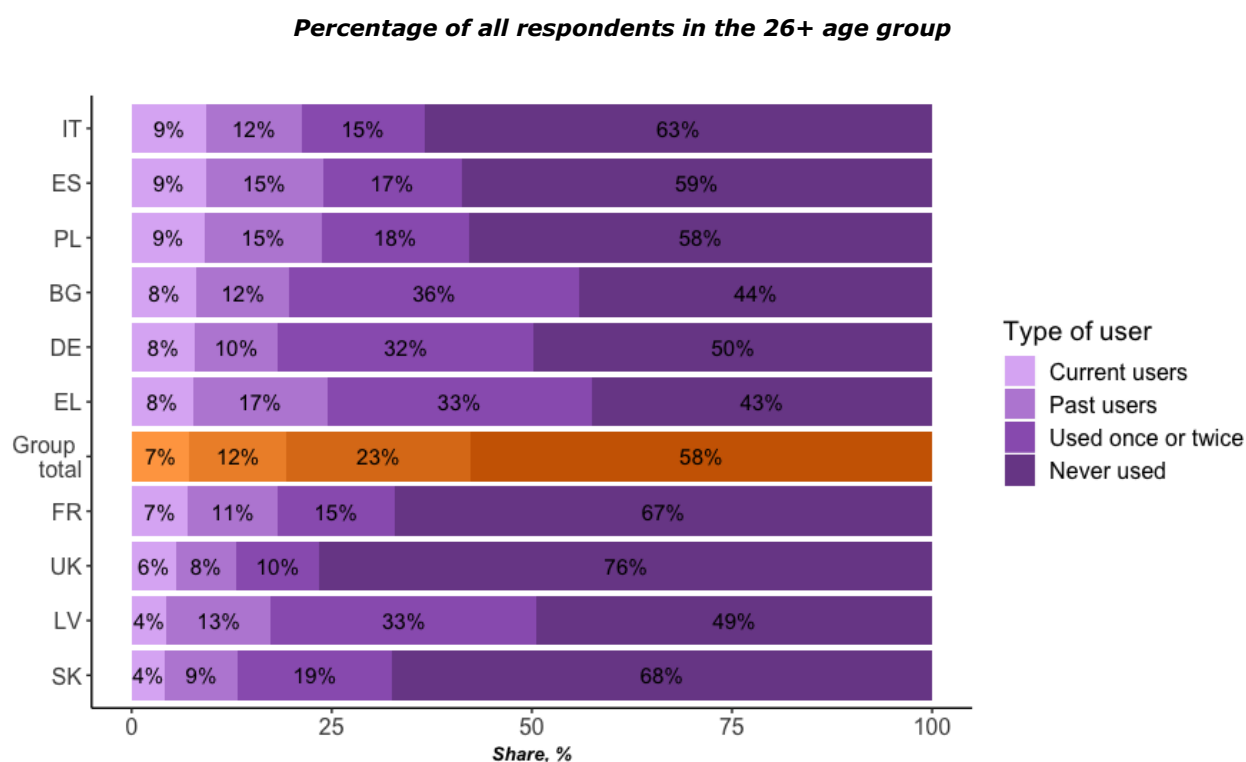
Question: Q.1.1. Have you ever used small cigarillos?

Note: The group total category refers to all respondents within the age group.

Source: OE elaboration from online survey data

Figure 63. charts the pattern of use for small cigarillos in the **26+ age group**. Of the total sample 58% reporting never use of small cigarillos, 23% had tried the product once or twice and 7% were current users. Similar to the 18-25 age group, the percentage of past users was consistently higher than the current users. Of note is that the UK had the highest percentage of current users never users in the 18-25 age groups and the highest percentage of never users in the 26+ age group.

Figure 63. Use of small cigarillos: country breakdown (26+ age group)



Base: all respondents in the 26+ age group (N=5,910)

Question: Q.1.1. Have you ever used small cigarillos?

Note: The group total category refers to all respondents within the age group.

Source: OE elaboration from online survey data

Table 32 and Table 33 report the socio-demographic breakdown in relation to small cigarillos. In both age groups there was a substantially high percentage of male over female current smokers of small cigarillos, with a larger gender gap in the 18-25 age group. In the 26+ age group, 68% of females were never users as compared against 48% of males. Another similarity across the two age groups was that more current users reported high socio-economic status and fewer never users.

Table 32. Small cigarillo use socio-demographic breakdown (18-25 age group)

Percentage and number of all respondents in the 18-25 age group

	Group total % (n)	Current users %	Past users %	Used once or twice %	Never users %
Use of product	100 (6090)	6	9	21	65
Gender					
Male	47 (2869)	9	12	22	58
Female	53 (3221)	3	6	20	70
Education level *					
1-2 (low)	5 (335)	11	13	19	58
3-4 (medium)	63 (3829)	5	8	21	66
5-6 (high)	32 (1926)	7	9	21	63
Residency					
Urban	65 (3949)	6	9	23	62
Suburban	19 (1173)	6	11	18	65
Rural	16 (968)	4	6	16	74
Self-reported socio-economic status					
1-4 (low)	18 (1126)	5	8	19	69
5-7 (medium)	65 (3954)	5	8	22	66
8-10 (high)	17 (1010)	12	13	19	57

Base: all respondents in the 18-25 age group (N=6,090)

Notes: The group total column refers to all respondents within the age group, i.e. it is a within column percentage. It provides the breakdown of each socio-demographic variable, including absolute numbers. The white cells in the four right columns represent the breakdown by user type, i.e. they are within row percentages (white cells in each row sum to 100).

* The sub-categories of the education level were the following: (1) No official education; (2) Primary school; (3) High school; (4) Some years of university (not completed); (5) University degree completed; (6) Post-graduate (master, PhD, other)

Source: OE elaboration from online survey data

Table 33. Small cigarillo use socio-demographic breakdown (26+ age group)

Percentage and number of all respondents in the 26+ age group

	Group total % (n)	Current users %	Past users %	Used once or twice %	Never users %
Use of product	100 (5910)	7	12	23	58
Gender					
Male	52 (3074)	10	15	27	48
Female	48 (2836)	5	9	19	68
Education level *					
1-2 (low)	5 (308)	9	12	19	59
3-4 (medium)	46 (2704)	7	10	22	61
5-6 (high)	49 (2898)	7	14	24	55
Residency					
Urban	66 (3879)	8	13	25	53
Suburban	17 (1010)	8	11	18	63
Rural	17 (1021)	4	9	19	69
Self-reported socio-economic status					
1-4 (low)	23 (1374)	4	9	23	64
5-7 (medium)	64 (3769)	7	12	23	58
8-10 (high)	13 (767)	16	17	22	46

Base: all respondents in the 26+ age group (N=5,910)

Notes: The group total column refers to all respondents within the age group, i.e. it is a within column percentage. It provides the breakdown of each socio-demographic variable, including absolute numbers. The white cells in the four right columns represent the breakdown by user type, i.e. they are within row percentages (white cells in each row sum to 100).

* The sub-categories of the education level were the following: (1) No official education; (2) Primary school; (3) High school; (4) Some years of university (not completed); (5) University degree completed; (6) Post-graduate (master, PhD, other)

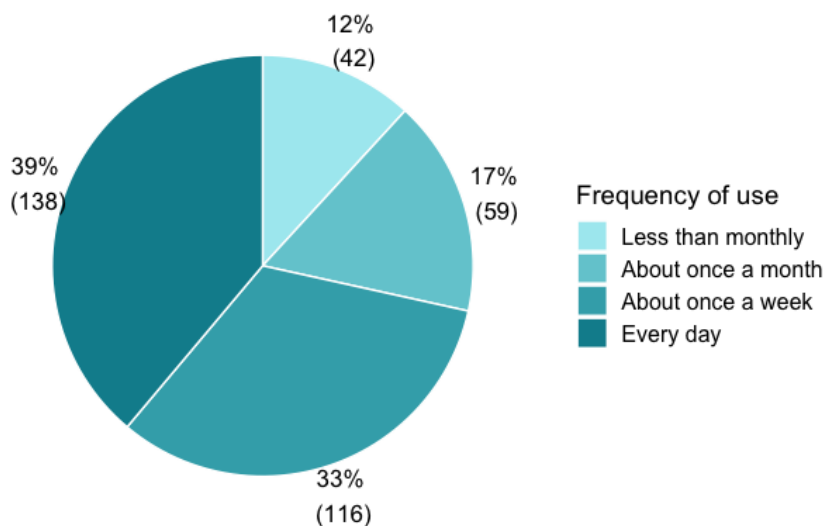
Source: OE elaboration from online survey data

4.2.2 Pattern of use

Figure 64. and Figure 65. report the frequency of smoking small cigarillos. In the **18-25 age group** 39% were smoking daily and 33% once a week. These figures translate to approximately 2% of the overall sample of 6,090 (see Figure 64). A similar pattern was observed in the **26+ age group** with 33% smoking daily and 40% weekly, translating to approximately 3% of the overall sample of 5,910 (see Figure 65).

Figure 64. Frequency of use of current small cigarillo users (18-25 age group)

Percentage and number of current users in the 18-25 age group



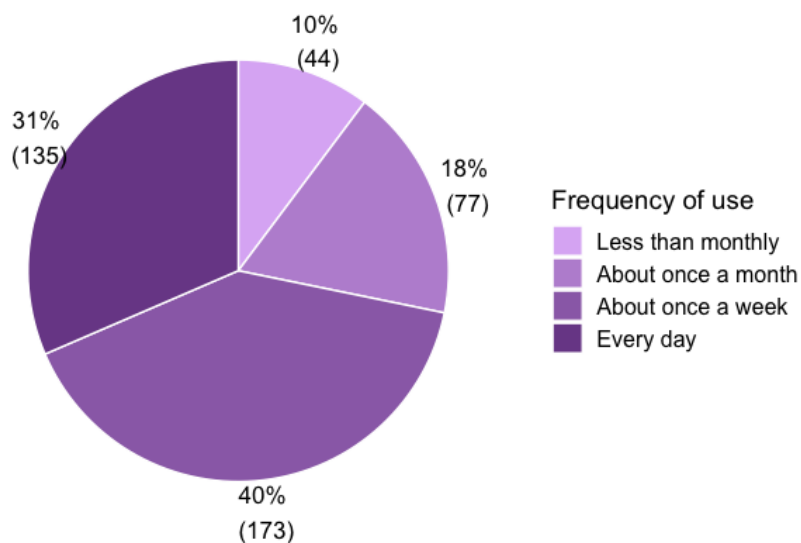
Base: current users of small cigarillo in the 18-25 age group (N=355)

Question: Q.1.3. How often do you use small cigarillos?

Source: OE elaboration from online survey data

Figure 65. Frequency of use of current small cigarillo users (26+ age group)

Percentage and number of current users in the 26+ age group



Base: current users of small cigarillo in the 26+ age group (N=429)

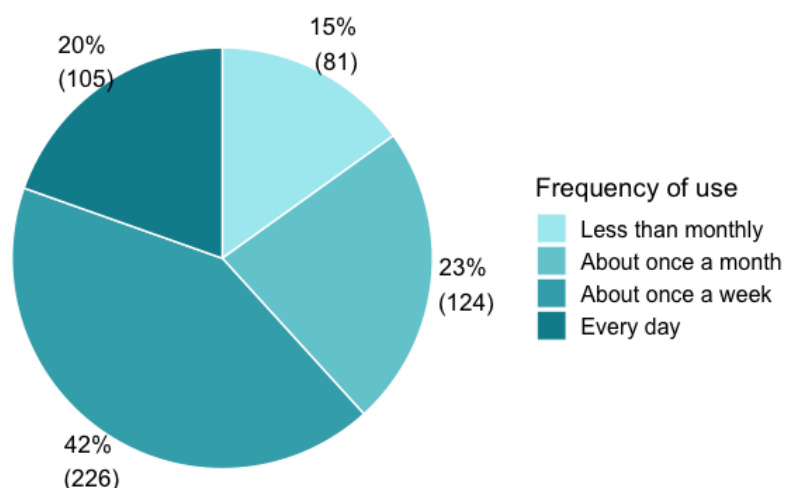
Question: Q.1.3. How often do you use small cigarillos?

Source: OE elaboration from online survey data

Figure 66 and Figure 67 report the frequency of use among past small cigarillo users. In the 18-25 age group 20% were smoking daily and 42% weekly. Similar percentages were found in the 26+ age group, 21% daily and 39% weekly. Past users of small cigarillos appeared to be less frequent users than current user peers for both age groups respectively.

Figure 66. Frequency of use of past small cigarillo users (18-25 age group)

Percentage and number of past users in the 18-25 age group



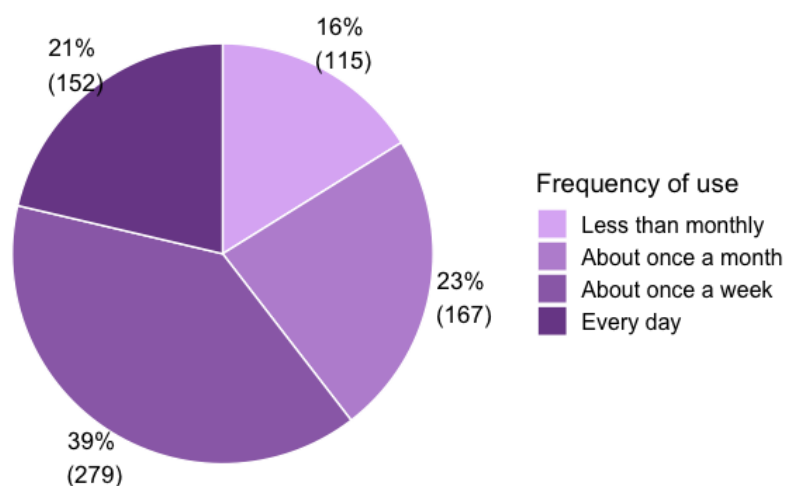
Base: past users of small cigarillos in the 18-25 age group (N=536)

Question: Q.1.6. How often did you use small cigarillos?

Source: OE elaboration from online survey data

Figure 67 Frequency of use of past small cigarillo users (26+ age group)

Percentage and number of past users in the 26+ age group



Base: past users of small cigarillos in the 26+ age group (N=713)

Question: Q.1.6. How often did you use small cigarillos?

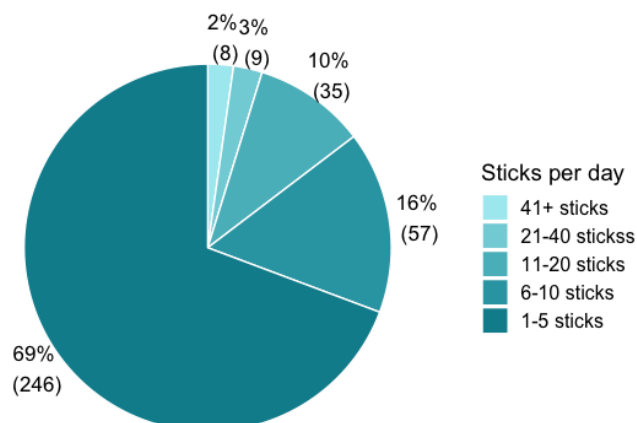
Source: OE elaboration from online survey data

Figure 68 and

Figure 69 chart the average number of small cigarillos (sticks) smoked by current users on days when they smoke. In the **18-25 age group**, 69% of current users reported 1-5 sticks a day, 16% 6-10 sticks and 15% more than 11 (see Figure 68). In the **26+ age group**, 75 reported 1-5 sticks a day, 17% 6-10 and 8% more than 11 (see

Figure 69).

Figure 68. Average number of small cigarillo sticks per day of use (18-25 age group)
Percentage and number of current users in the 18-25 age group

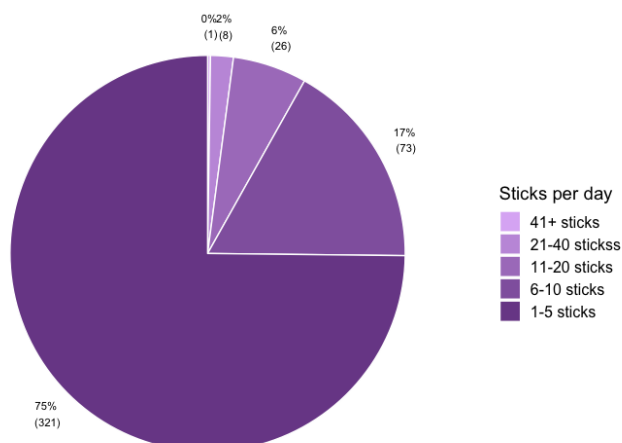


Base: current users of small cigarillos in the 18-25 age group (N=355)

Question: Q.1.4.a. On the days that you smoke/use, about how much do you smoke/use?

Source: OE elaboration from online survey data

Figure 69. Average number of small cigarillo sticks per day of use (26+ age group)
Percentage and number of current users in the 26+ age group



Base: current users of small cigarillos in the 26+ age group (N=429)

Question: Q.1.4.a. On the days that you smoke/use, about how much do you smoke/use?

Source: OE elaboration from online survey data

The focus group participants thought that small cigarillos were consumed less frequently than conventional cigarettes. Many believed that the product had a strong taste and was a satiating experience when compared to conventional cigarettes. As such it was not necessary to smoke as many.

"A replacement for cigarettes because they are cheaper and you don't need as many. The taste is more intense." (M, 49, DE)

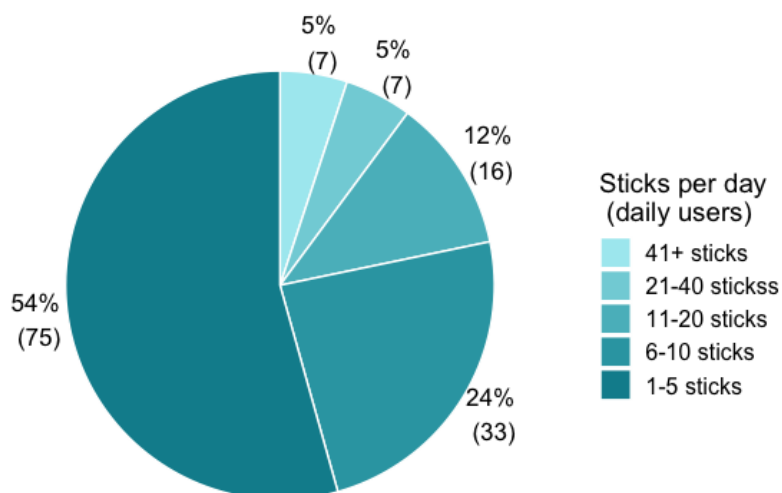
"It's occasional as well, we don't do that every day!" (F, 53, FR)

"1-2 small cigarillos a day are better than 20 cigarettes." (M, 38, BG)

Figure 70 and Figure 71 focus on the number of sticks smoked by daily users of small cigarillos. It is worth bearing in mind that the percentage of respondents in this category was relatively small; 2% of the 18-25 age group and 3% of the 26+ age group. In both age groups one in two daily users smoked 1-5 sticks a day. In the **18-25 age group**, the share of intensive daily users – defined by those that consume more than 10 sticks per day of use – was higher at 22% than in the overall sample of current users at 15%. The same pattern was found in the **26+ age group**, with 18% of current daily users consuming more than 10 sticks per day, as compared to 8% of current users.

Figure 70. Average number of small cigarillo sticks per day, for daily users (18-25 age group)

Percentage and number of current daily users in the 18-25 age group



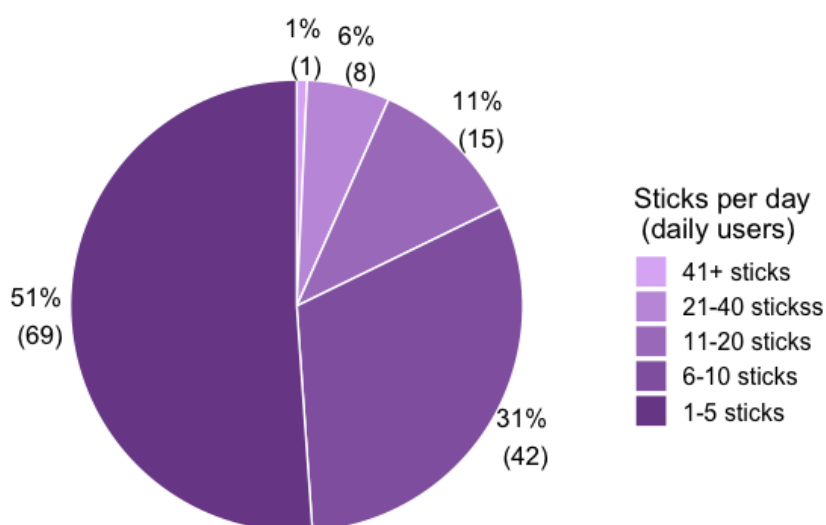
Base: daily current users of small cigarillos in the 18-25 age group (N=138)

Question: Q.1.4.a. On the days that you smoke/use, about how much do you smoke/use?

Source: OE elaboration from online survey data

Figure 71. Average number of small cigarillo sticks per day, for daily users (26+ age group)

Percentage and number of current daily users in the 26+ age group



Base: daily current users of small cigarillos in the 26+ age group (N=135)

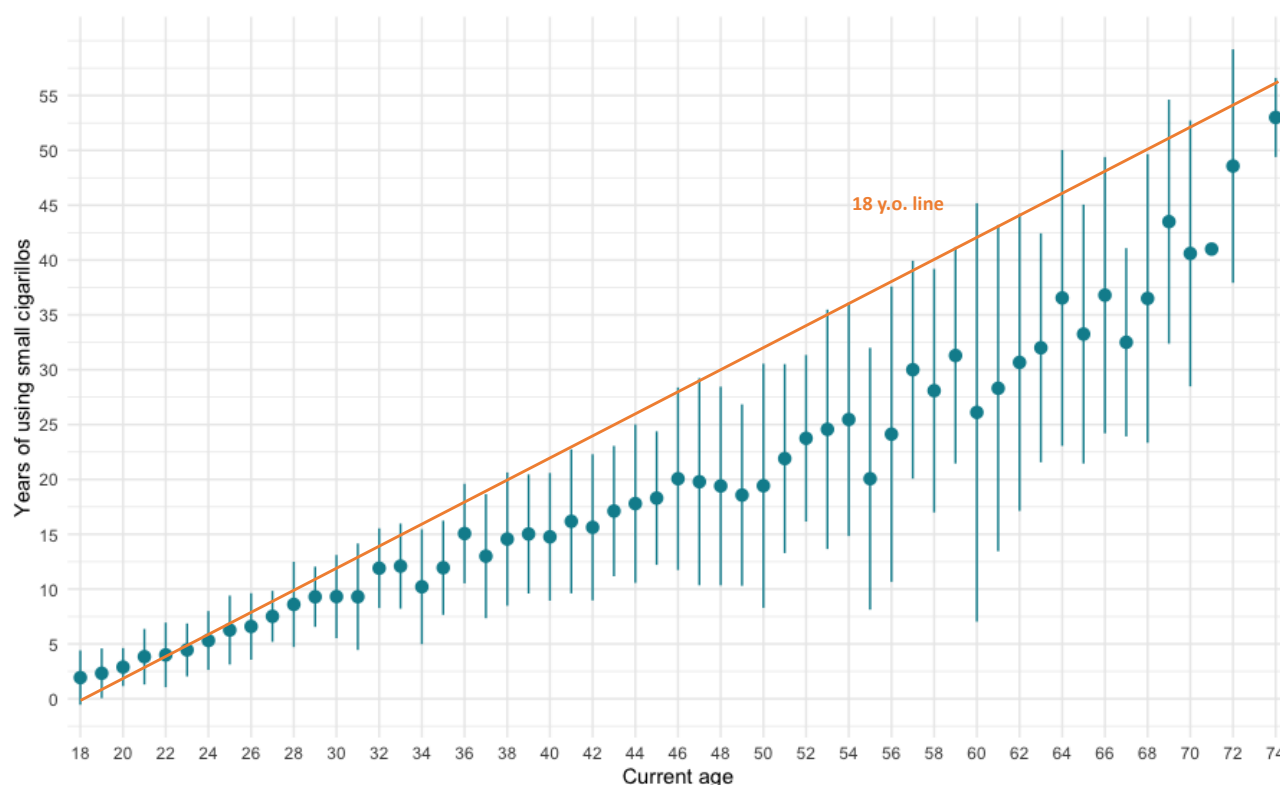
Question: Q.1.4.a. On the days that you smoke/use, about how much do you smoke/use?

Source: OE elaboration from online survey data

On average, respondents that were younger than 23 years old in the 18-25 age group were more likely have started smoking small cigarillos under the age of 18 (see Figure 72). Those who were 18 years of age had been consuming 2 years on average.

Figure 72. Initiation of use for current and past small cigarillo users (total sample)

Number of current and past users in the total sample



Base: current and past users of small cigarillos in the total sample (N=1,919⁶⁵)

Question: Q.1.2.a. At about what age did you start using the products below? & Q.5.2. Age

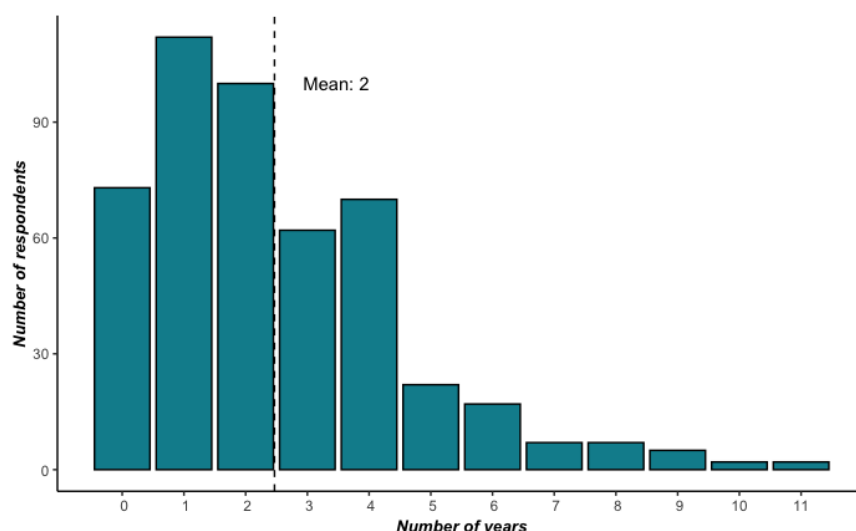
Note: The figure illustrates the approximate starting age of small cigarillos use among all respondents of the online survey, where the x-axis represents the current age of the respondent (Q.5.2) and the y-axis is the average numbers of years of product consumption (calculated by subtracting the current age from the starting age). The orange line represents the 45° or 18 y.o. line. Each blue point represents the average starting age for respondents of that age; points above the 18 y.o. line reveal those initiating consumption of small cigarillos before the age of 18, while those below the line reveal age brackets starting consumption after the age of 18 on average. The blue lines represent the mean standard deviation, i.e. it does not show the full range of values for each age.

Source: OE elaboration from online survey data

The average duration of use for past small cigarillo users in the **18-25 age group** was 2 years, as displayed in Figure 73. In the **26+ age group**, the average duration of use for former users was 10 years (see Figure 74.). In both groups, the histograms were slightly skewed, suggesting that typically people used the product for a shorter period of time. Participants in the 26+ age group may have used the product for a longer time due to the addictive nature of the product. They may also have had more difficulty quitting.

⁶⁵ NA and misreported values are excluded.

Figure 73. Duration of use of past small cigarillo users (18-25 age group)

Number of past users in the 18-25 age group

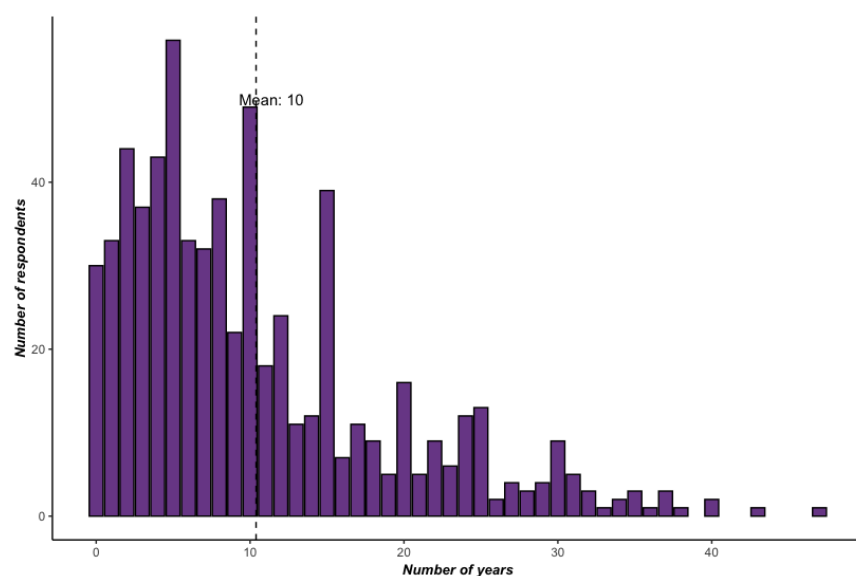
Base: past users of small cigarillos in the 18-25 age group (N=479⁶⁶)

Question: Q.1.2.a. At about what age did you start using small cigarillos? & Q.1.2.b. At about what age did you stop using small cigarillos?

Note: The duration was calculated by subtracting Q.1.2.a. from Q.1.2.b for each individual that was a past user of the product. The mean shows the average duration of use.

Source: OE elaboration from online survey data

Figure 74. Duration of use of past small cigarillo users (26+ age group)

Number of past users in the 26+ age group

Base: past users of small cigarillos in the 26+ age group (N=660⁶⁷)

Question: Q.1.2.a. At about what age did you start using small cigarillos? & Q.1.2.b. At about what age did you stop using small cigarillos?

Note: The duration was calculated by subtracting Q.1.2.a. from Q.1.2.b for each individual that was a past user of the product. The mean shows the average duration of use.

Source: OE elaboration from online survey data

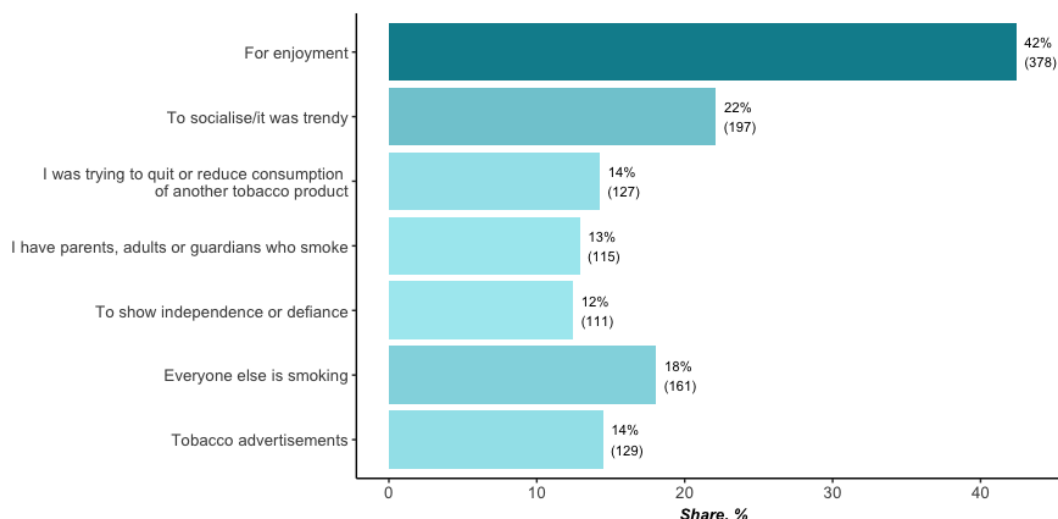
⁶⁶ NA and misreported values are excluded.

⁶⁷ NA and misreported values are excluded.

Figure 75. show the reasons for starting to smoke small cigarillos. The primary reason for taking up small cigarillos in the **18-25 age group** (see Figure 75.) was 'for enjoyment', 42% reported this reason. This was followed by 'to socialise / it was trendy' at 22% and 'everyone else was smoking' at 18%. The **26+ age group** (see Figure 76.) reported similar reasons; 51% 'for enjoyment' and 26% 'to socialise / it was trendy'.

Figure 75. Initiation of small cigarillo use: reasons (18-25 age group)

Percentage and number of current and past users in the 18-25 age group



Base: current and past users of small cigarillos in the 18-25 age group (N=891)

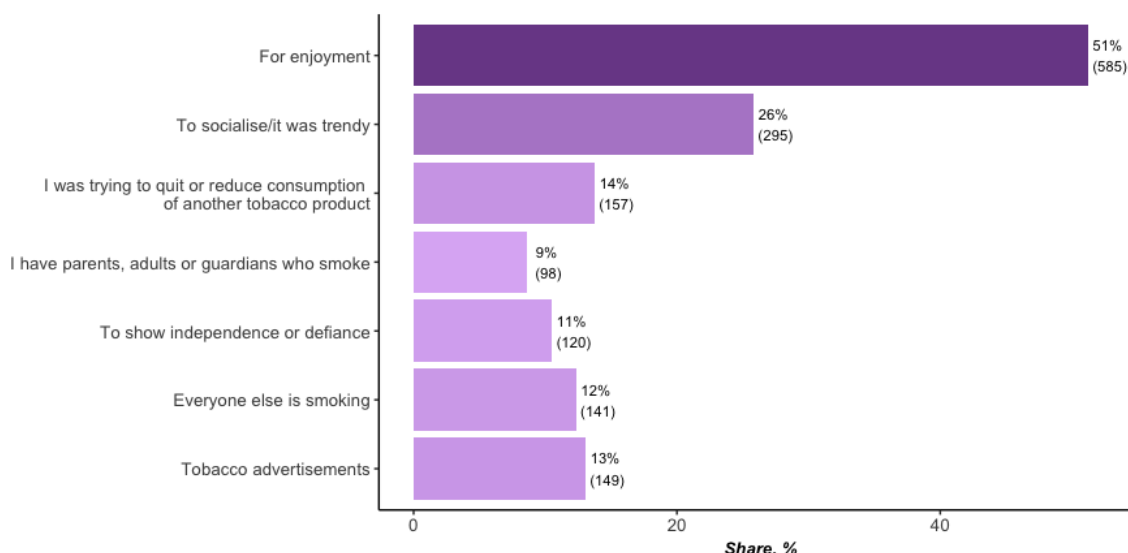
Question: Q.2.1.a. Why did you start using small cigarillos? Please select up to 2 options.

Note: Each bar represents the share of respondents that selected the option within the age group. The percentages do not add up to 100% because respondents could select more than one term. For the response options with "smoking", the product 'smoked' was not specified.

Source: OE elaboration from online survey data

Figure 76. Initiation of small cigarillo use: reasons (26+ age group)

Percentage and number of current & past users in 26+ age group



Base: current and past users of small cigarillos in the 26+ age group (N=1,142)

Question: Q.2.1.a. Why did you start using small cigarillos? Please select up to 2 options.

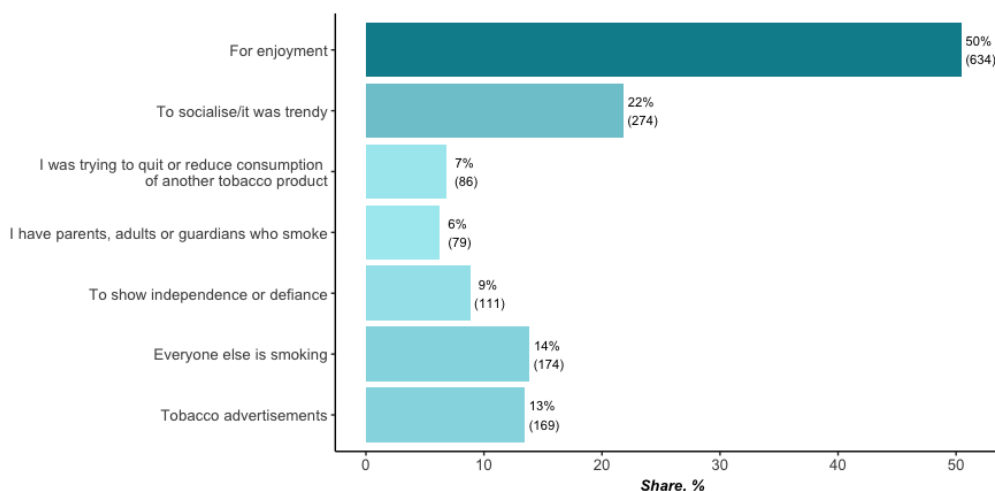
Note: Each bar represents the share of respondents that selected the option within the age group. The percentages do not add up to 100% because respondents could select more than one term. For the response options including "smoking", the product 'smoked' was not specified.

Source: OE elaboration from online survey data

Figure 77 and Figure 78 look at those respondents who had only tried small cigarillos once or twice. 50% or the 18-25 age group reported 'for enjoyment' and 22% 'to socialise/it was trendy'. On these two reasons the 26+ age group reported higher percentages 54% and 26% respectively. More than one in 10 in both age groups cited 'tobacco advertisements' which may in part have contributed to the trendy association.

Figure 77. Initiation of small cigarillo use: reasons for trying small cigarillos (18-25 age group)

Percentage and number of those who tried once or twice in the 18-25 age group



Base: those who tried to use small cigarillos in the 18-25 age group (N=1,257)

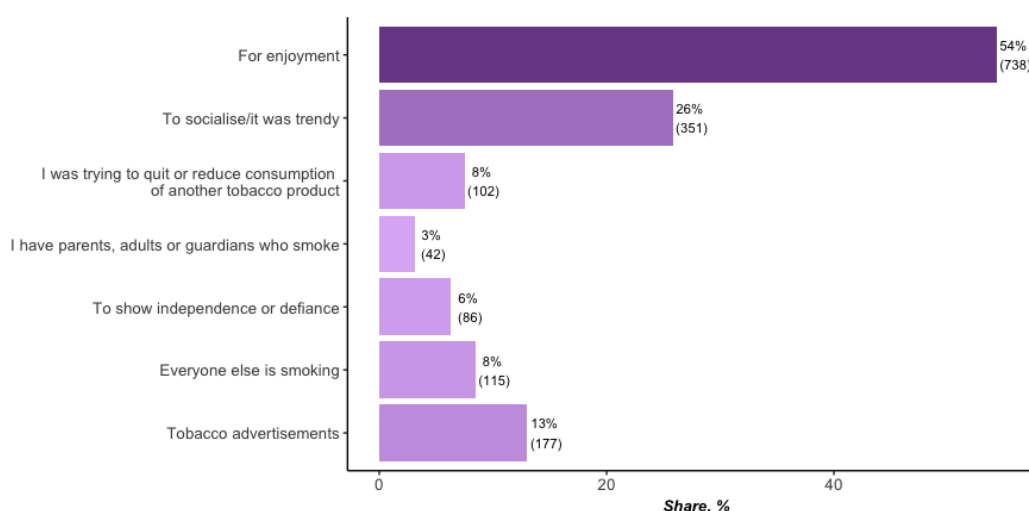
Question: Q.2.1.b. Why did you try small cigarillos? Please select up to 2 options.

Note: Each bar represents the share of respondents that selected the option within the age group. The percentages do not add up to 100% because of the fact that respondents could select more than one term – therefore, the bars must be interpreted separately. For the response options with "smoking", the type of product 'smoked' was not specified.

Source: OE elaboration from study survey data

Figure 78. Initiation of small cigarillo use: reasons for trying small cigarillos (26+ age group)

Percentage and number of those who tried once or twice in the 26+ age group



Base: those who tried to use small cigarillos once or twice in the 26+ age group (N=1,359)

Question: Q.2.1.b. Why did you try small cigarillos? Please select up to 2 options.

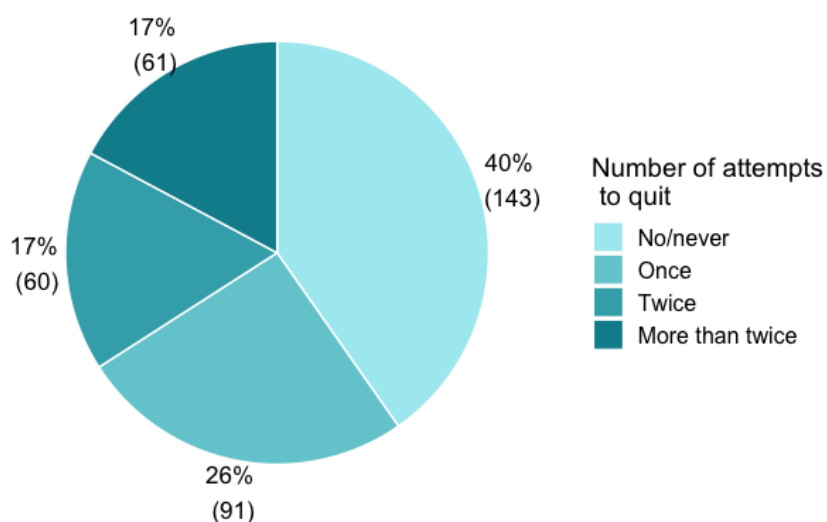
Note: Each bar represents the share of respondents that selected the option within the age group. The percentages do not add up to 100% because of the fact that respondents could select more than one term – therefore, the bars must be interpreted separately. For the response options with "smoking", the type of product 'smoked' was not specified.

Source: OE elaboration from online survey data

Figure 79 and Figure 80 chart the number of attempts by current smokers to quit small cigarillos. While the majority of current small cigarillo users reported attempting to quit at least once in both the 18-25 and the 26+ age groups, 40% in both age groups had never attempted to quit.

Figure 79. Cessation: attempts to quit small cigarillos (18-25 age group)

Percentage and number of current users in the 18-25 age group



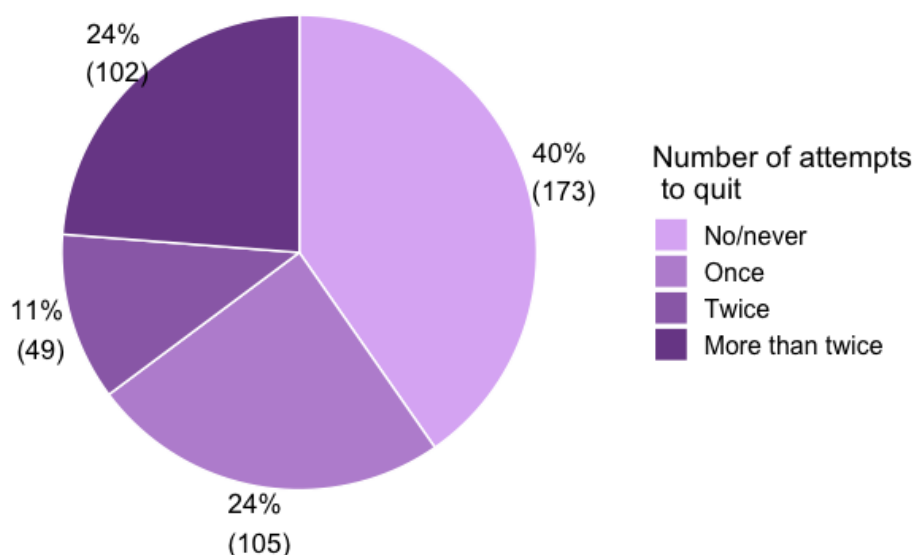
Base: current users of small cigarillos in the 18-25 age group (N=355)

Question: Q.1.5. Have you ever tried to quit small cigarillos? If yes, how many times?

Source: OE elaboration from online survey data

Figure 80. Cessation: attempts to quit small cigarillos (26+ age group)

Percentage and number of current users in the 26+ age group



Base: current users of small cigarillos in the 26+ age group (N=429)

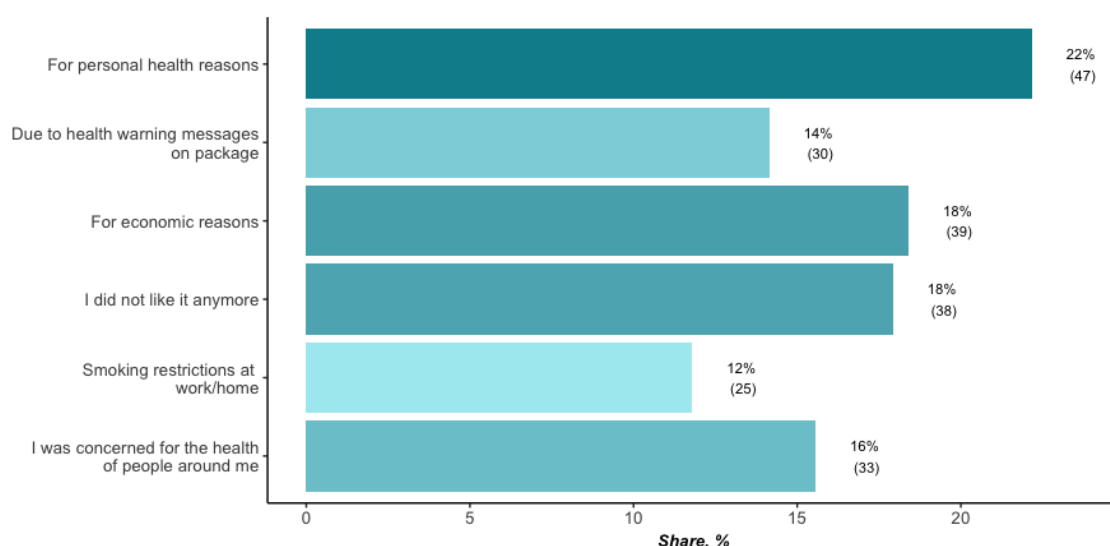
Question: Q.1.5. Have you ever tried to quit small cigarillos? If yes, how many times?

Source: OE elaboration from online survey data

The reasons for trying to quit smoking small cigarillos are shown in Figure 81 and Figure 82. In both age groups reasons around health are the dominant motivation of trying to quit. For the **18-25 age groups** 22% cited 'personal health', 14% 'health warnings on packages' and 16% 'health of people around me'. For the **26+ age group** the respective percentages on these three health related criteria were 25%, 10% and 14%. In aggregate one in two users of small cigarillos are worried about health impacts. The charts also show that one in five users tried to quit for economic reasons.

Figure 81. Cessation: reasons for trying to quit small cigarillos (18-25 age group)

Percentage and number of current users in the 18-25 age group who tried to quit



Base: current users of small cigarillos that tried to quit in the 18-25 age group (N=212)

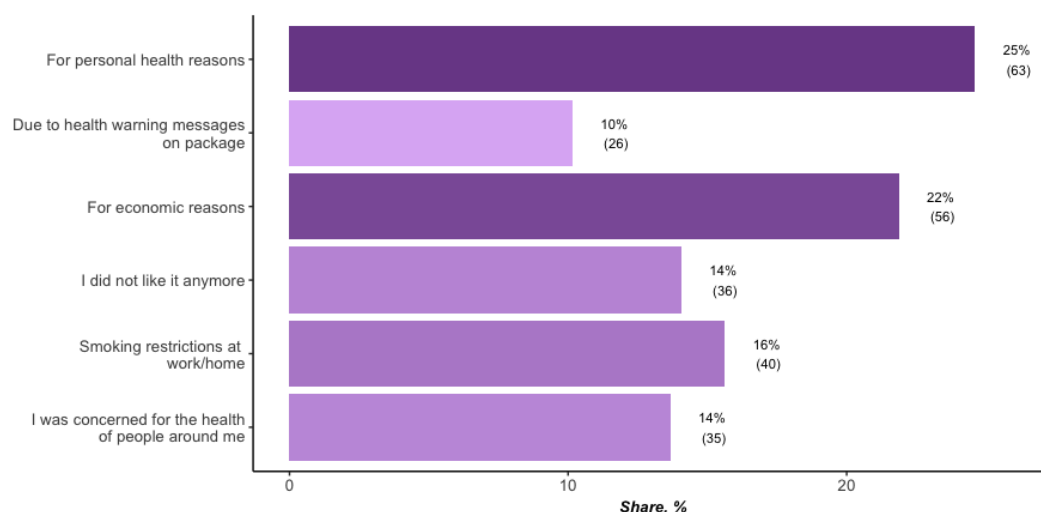
Question: Q.2.2. What made you try to quit small cigarillos? Please select the most fitting answer.

Note: Respondents were asked to provide only one option. Therefore, each bar represents the share of respondents that selected the option within the age group.

Source: OE elaboration from online survey data

Figure 82. Cessation: reasons for trying to quit small cigarillos (26+ age group)

Percentage and number of current users in the 26+ age group who tried to quit



Base: current users of small cigarillos that tried to quit in the 26+ age group (N=256)

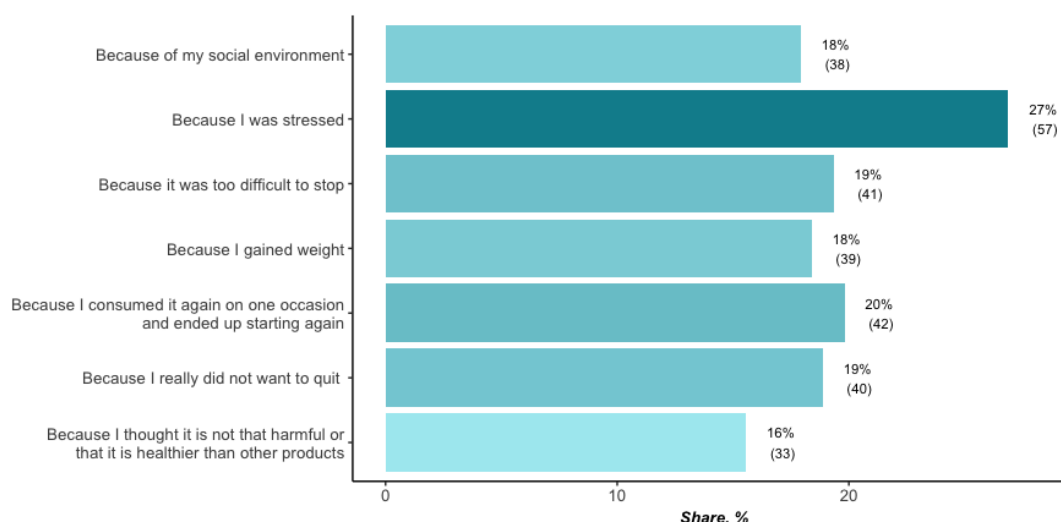
Question: Q.2.2. What made you try to quit small cigarillos? Please select the most fitting answer.

Note: Respondents were asked to provide only one option. Therefore, each bar represents the share of respondents that selected the option within the age group.

Source: OE elaboration from online survey data

The reasons for taking up smoking of small cigarillos again after an attempt to quit are shown in Figure 83 and Figure 84. What is reported in both age groups looked like the consequence of addiction. In the **18-25 age group** the reasons cited were 'stress' 27%, 'difficult to stop' 19%, 'I had one and started again' 20% and 'I didn't really want to stop' 19%. In the **26+ age group** the percentages reporting these reasons were all marginally higher.

Figure 83. Reasons for starting to use small cigarillos again (18-25 age group)
Percentage and number of current users that had attempted to quit in the 18-25 age group



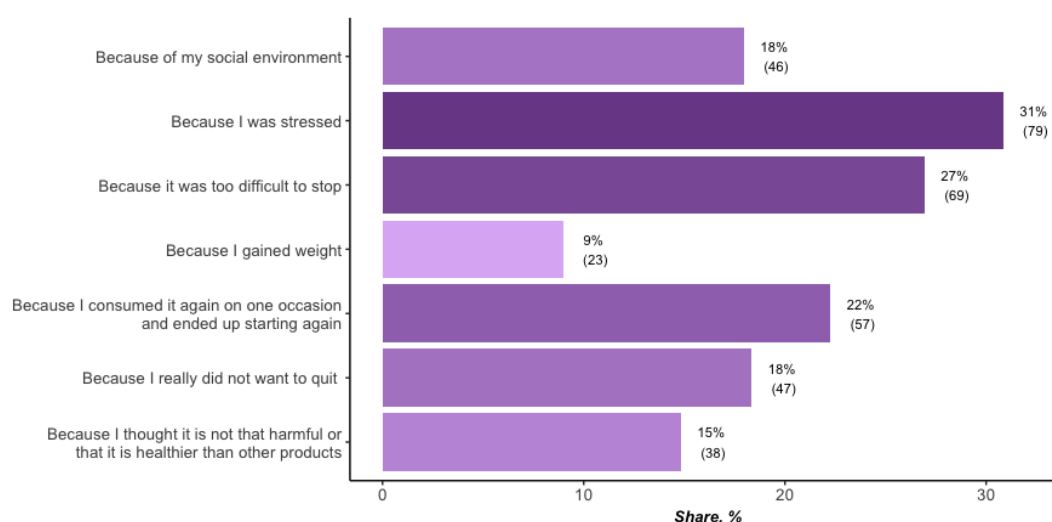
Base: current users of small cigarillos that tried to quit in the 18-25 age group (N=212)

Question: Q.2.3. Why did you start to use small cigarillos again? Please select up to 2 options.

Note: Each bar represents the share of respondents that selected the option within the age group. The percentages do not add up to 100% because of the fact that respondents could select more than one term – therefore, the bars must be interpreted separately.

Source: OE elaboration from online survey data

Figure 84. Reasons for starting to use small cigarillos again (26+ age group)
Percentage and number of current users that had attempted to quit in the 26+ age group



Base: current users of small cigarillos that tried to quit in the 26+ age group (N=256)

Question: Q.2.3. Why did you start to use small cigarillos again? Please select up to 2 options.

Note: Each bar represents the share of respondents that selected the option within the age group. The percentages do not add up to 100% because of the fact that respondents could select more than one term – therefore, the bars must be interpreted separately.

Source: OE elaboration from online survey data

4.2.3 Parallel use with other tobacco and related products

Over 90% of current consumers in both age groups smoked small cigarillos in parallel with other tobacco and related products. Very few transitioned from using another type of tobacco product to the exclusive use of small cigarillos, 6% in the 18-25 age group and 7% in the 26+ age group. Only 1% were never users of other products in the 18-25 age group and 2% in the 26+ age group (see Table 34 and Table 35).

Of the 330 current small cigarillo users in the 18-25 age group (see Table 34), dual use of other tobacco and related products could be more accurately described as multiple use. Almost all were smoking conventional cigarettes and high percentage smoking other tobacco products. The same pattern of multiple use was seen in the 26+ age group (see Table 35) with somewhat lower percentages using waterpipe tobacco and HTPs.

Table 34. Career flows of small cigarillo use for current users (18-25 age group)

Percentage and number of current users in the 18-25 age group

			Number	%
Total			355	100
Never user of other products			3	1
Exclusive current user of small cigarillos & past user of other product(s)			22	6
Out of them, switching from (one or more products) *				
Cigarettes	(n=18)	82% *		
Electronic cigarettes	(n=13)	59% *		
Waterpipe tobacco	(n=8)	36% *		
HTPs	(n=8)	36% *		
Slim cigarettes	(n=7)	32% *		
Current user of small cigarillos & both current and past user of other product(s)			330	93
Other products consumed in parallel now (one or more products) *				
Cigarettes	(n=296)	90% *		
E-cigarettes	(n=218)	66% *		
Slim cigarettes	(n=216)	65% *		
Waterpipe tobacco	(n=188)	57% *		
HTPs	(n=185)	56% *		

Base: current users of small cigarillos in the 18-25 age group (N=355);

Question: Q.1.1. Have you ever used small cigarillos?

Note: Patterns of use and consumption careers are explored for different consumer profiles of current users. A breakdown is provided for the two variables of interest, i.e. current parallel use patterns. The last row only provides the breakdown of current dual use as this is the variable of interest; it does not include a breakdown that takes account past consumption of products to avoid confusion (from too many permutations).

(*) Percentages add up to more than 100% on account of the prevalence of dual use.

Source: OE elaboration from online survey data

Table 35. Career flows of current small cigarillo users for current users (26+ age group)

Percentage and number of current users in the 26+ age group

			Number	%
Total			429	100
Never user of other products			8	2
Exclusive current user of small cigarillos & past user of other product(s)			32	7
Out of them, switching from (one or more products) *				
Cigarettes	(n=29)	91% *		
Slim cigarettes	(n=8)	25% *		
Electronic cigarettes	(n=58)	16% *		
Waterpipe tobacco	(n=2)	6% *		
HTPs	(n=5)	16% *		
Current user of small cigarillos & both current and past user of other product(s)			389	91
Other products consumed in parallel now (one or more products) *				
Cigarettes	(n=352)	90% *		
Slim cigarettes	(n=237)	61% *		
E-cigarettes	(n=231)	59% *		
HTPs	(n=182)	47% *		
Waterpipe tobacco	(n=162)	42% *		

Base: current users of small cigarillos in the 26+ age group (N=429)

Question: Q.1.1. Have you ever used small cigarillos?

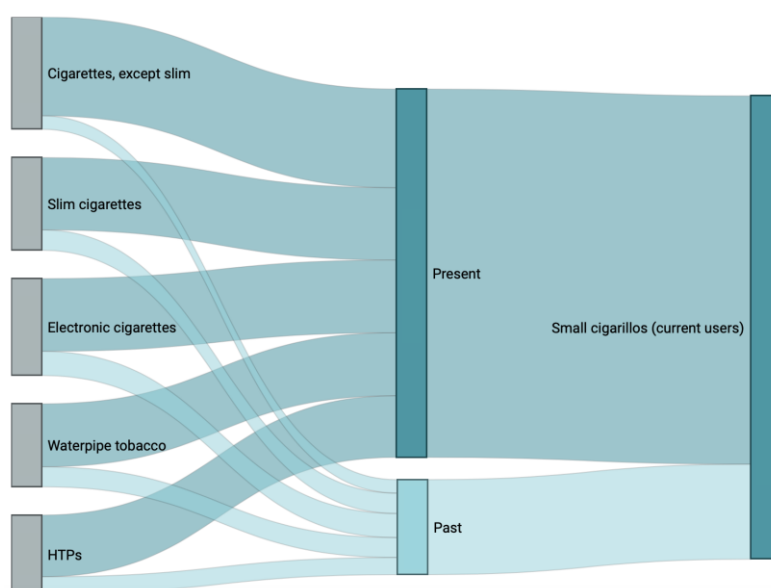
Note: Patterns of use and consumption careers are explored for different consumer profiles of current users. A breakdown is provided for the two variables of interest, i.e. current parallel use patterns. The last row only provides the breakdown of current dual use as this is the variable of interest; it does not include a breakdown that takes account past consumption of products to avoid confusion (from too many permutations).

(*) Percentages add up to more than 100% on account of the prevalence of dual use.

Source: OE elaboration from online survey data

The next figures present a visual representation of Table 34 and Table 35 with a focus on dual use patterns, hence never users of other products were excluded from the figures. The thickness of the tail is only indicative when interpreting the figure due to double counting (see note below the figure). A majority of current small cigarillo users in the **18-25 age group** (see Figure 85) were dual users of other tobacco and related products. It corroborates the hypothesis that small cigarillos are often used for festive occasions. The career flow of current small cigarillo users in the **26+ age group** (see Figure 86) is similar to the younger age group.

Figure 85. Representation of career flows for current small cigarillo users (18-25 age group)



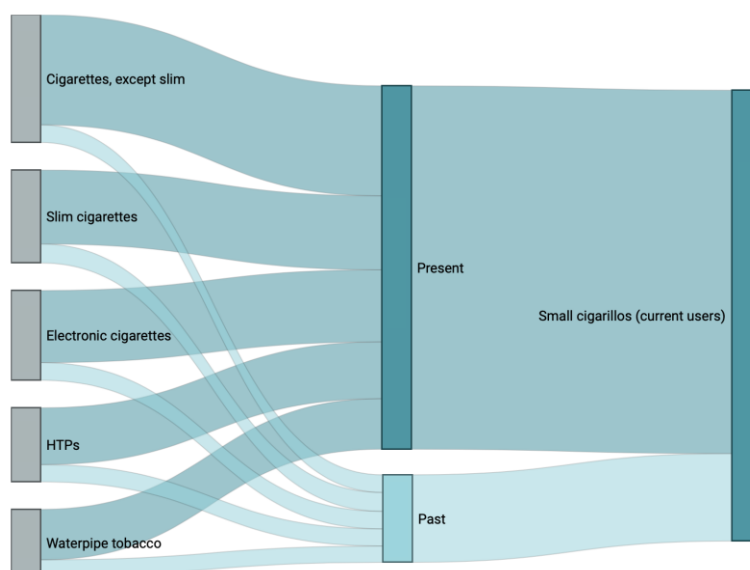
Base: current users of small cigarillos (except never users of other products) in the 18-25 age group (N=842)

Question: Q.1.1. Have you ever used small cigarillos?

Note: The diagram is a simplified version of the career flows of dual current small cigarillo users, i.e. it highlights dual use patterns as well as transitions to/from other products; therefore, never users of other products (i.e. only used small cigarillos in their lifetime) were excluded. The thickness/width of the tails, as well as the percentages included, are only indicative. Some individuals may be represented twice (or more) if they use/have used more than one product at a time.

Source: OE elaboration from online survey data

Figure 86. Representation of career flows for current small cigarillo users (26+ age group)



Base: current users of small cigarillos (except never users of other products) in the 26+ age group (N=1,010)

Question: Q.1.1. Have you ever used small cigarillos?

Note: The diagram is a simplified version of the career flows of dual current small cigarillo users, i.e. it highlights dual use patterns as well as transitions to/from other products; therefore, never users of other products (i.e. only used small cigarillos in their lifetime) were excluded. The thickness/width of the tails, as well as the percentages included, are only indicative. Some individuals may be represented twice (or more) if they use/have used more than one product at a time.

Source: OE elaboration from online survey data

Table 36 and Table 37 report the consumption careers of past users of small cigarillos. In the **18-25 age group** (see Table 36), a majority of former small cigarillo consumers (61%) were consuming other tobacco and related products at the time of the survey. Of all past users, 14% had been exclusive users of small cigarillos in the past but had switched to using other tobacco products.

In the **26+ age group** (see Table 37), 32% had quit all tobacco use, 15% were exclusive users of small cigarillos who had switched to other tobacco products, mainly conventional cigarettes, and 51% were dual users in the past who quit small cigarillos and were now dual users of mainly conventional cigarettes.

Table 36. Career flow of past small cigarillo users (18-25 age group)

Percentage and number of past users in the 18-25 age group

	Number	%
Total	536	100
Never user of other products	13	2
Dual past user & not using any products now	122	23
Exclusive past user of small cigarillos & current user of other products (but not small cigarillos)	76	14
<i>Out of them, switching to (one or more products) *</i>		
Cigarettes (n=68)	89% *	
Electronic cigarettes (n=36)	47% *	
Slim cigarettes (n=28)	37% *	
Waterpipe tobacco (n=18)	24% *	
HTPs (n=15)	20% *	
Dual past user & current user of other products (but not small cigarillos)	325	61
<i>Other products currently consumed in parallel (one or more products) *</i>		
Cigarettes (n=225)	69% *	
E-cigarettes (n=126)	39% *	
HTPs (n=77)	24% *	
Waterpipe tobacco (n=73)	22% *	
Slim cigarettes (n=63)	19% *	

Base: past users of small cigarillos in the 18-25 age group (N=536)

Question: Q.1.1. Have you ever used small cigarillos?

Note: Patterns of use and consumption careers are explored for different consumer profiles of past users. A breakdown is provided for the variables of interest, i.e. those who continued using other tobacco products after using e-cigarettes. The last row only provides the breakdown of current consumption because this is the variable of interest; it does not include a breakdown that takes account past consumption of products to avoid confusion (from too many permutations).

(*) Percentages add up to more than 100% on account of the prevalence of dual use.

Source: OE elaboration from online survey data

Table 37. Career flows of past small cigarillo users (26+ age group)

Percentage and number of past users in the 26+ age group

	Number	%
Total	713	100
Never user of other products	12	2
Dual past user & not using any products now	230	32
Exclusive past user of small cigarillos & current user of other products (but not small cigarillos)	107	15
<i>Out of them, switching to (one or more products) *</i>		
Cigarettes (n=103)	96% *	
Electronic cigarettes (n=50)	47% *	
Slim cigarettes (n=34)	32% *	
Waterpipe tobacco (n=23)	21% *	
HTPs (n=10)	9% *	
Dual past user & current user of other products (but not small cigarillos)	364	51
<i>Other products currently consumed in parallel (one or more products) *</i>		
Cigarettes (n=252)	69% *	
E-cigarettes (n=180)	49% *	
HTPs (n=113)	31% *	
Slim cigarettes (n=81)	22% *	
Waterpipe tobacco (n=57)	16% *	

Base: past users of small cigarillos in the 26+ age group (N=713)

Question: Q.1.1. Have you ever used small cigarillos?

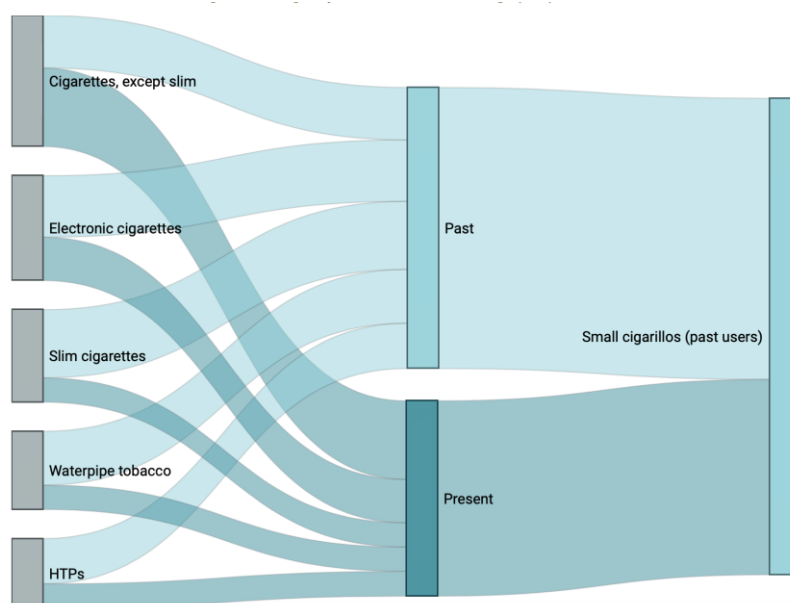
Note: Patterns of use and consumption careers are explored for different consumer profiles of past users. A breakdown is provided for the variables of interest, i.e. those who continued using other tobacco products after using e-cigarettes. The last row only provides the breakdown of current consumption because this is the variable of interest; it does not include a breakdown that takes account past consumption of products to avoid confusion (from too many permutations).

(*) Percentages add up to more than 100% on account of the prevalence of dual use.

Source: OE elaboration from online survey data

As previously explained, the figures below provide a visual representation of dual product user behaviours. It should be interpreted in the same way as the previous figures. Many of the former consumers of small cigarillos in the **18-25 age group** (see Figure 87) and in the **26+ age group** (see Figure 88) had also been users of other tobacco and related products in the past. But many had also switched between products. Overall, the study showed that many users of small cigarillos use the product in parallel with another product, i.e. the 'present' tail is thicker for current users and the 'past' tail is thicker amongst past users of the product.

Figure 87. Representation of career flows for past small cigarillo users (18-25 age group)



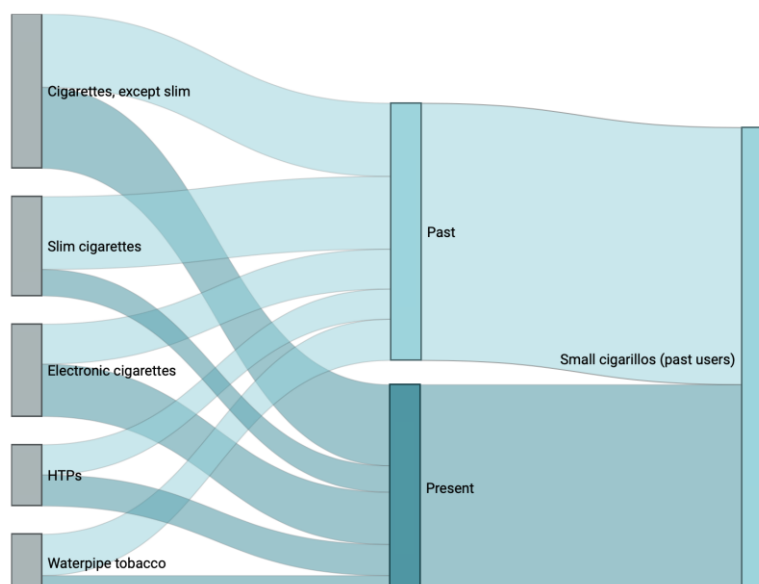
Base: past users of small cigarillos (excluding never users of other products) in the 18-25 age group (N=917)

Question: Q.1.1. Have you ever used small cigarillos?

Note: The diagram is a simplified version of the career flows of dual past users of small cigarillos, i.e. it highlights dual use patterns as well as transitions to/from other products; therefore, never users of other products (i.e. only used small cigarillos in their lifetime) were excluded. The thickness/width of the tails, as well as the percentages included, are only indicative. Some individuals may be represented twice (or more) if they use/have used more than one product at a time.

Source: OE elaboration from online survey data

Figure 88. Representation of career flows for past small cigarillo users (26+ age group)



Base: past users of small cigarillos (excluding never users of other products) in the 26+ age group (N=628)

Question: Q.1.1. Have you ever used small cigarillos?

Note: The diagram is a simplified version of the career flows of dual past users of small cigarillos, i.e. it highlights dual use patterns as well as transitions to/from other products; therefore, never users of other products (i.e. only used small cigarillos in their lifetime) were excluded. The thickness/width of the tails, as well as the percentages included, are only indicative. Some individuals may be represented twice (or more) if they use/have used more than one product at a time.

Source: OE elaboration from online survey data

4.3 Consumer perception

Participants in the focus group sessions thought small cigarillos satisfied emotional rather than functional needs. They were viewed as old-fashioned by many, which could be either positive or negative. Some believed it made the product authentic, *"the real thing,"* while others viewed it as *"disgusting"* due to the strong taste and smell which participants equated to low quality and *"rough."* For some, the product evoked cosiness and friendliness in a relaxed and easy-going atmosphere. But younger participants in Spain associated the product with Farias, an old and low-quality cigar brand. In Greece in particular, the focus group found that users of the product differentiated themselves from other tobacco and related product users because the product was not as common as cigarettes. However, it is important to note that few participants were actual users.

The online survey showed that 'enjoyment' and 'socialising' were the most cited reasons for starting or trying to smoke small cigarillos (e.g. see Figure 75.), which provides an explanation for the more intermittent use of the product (see Figure 64. and Figure 65.). Focus group participants explained this pattern of use by clarifying that small cigarillos were often a feature of special occasions.

"The cigarette is rather everyday use and then when we are at a party, we are in a more convivial situation, so we change the way we smoke (and have a small cigarillo)." (F, 53, FR)

"I don't associate it so much with smoking as a routine... I associate it more with something you smoke after a meal... I don't see it as something you would smoke when you get out of work." (M, 34, ES)

"My lung can't handle them. They've been pressed too hard or I don't know. I can't pull a smoke." (F, 37, DE)

Focus group participants also described the ritual of consuming small cigarillos as enjoyable and the product as widely available. These were motivating factors for the use of the product. Participants reported that recent developments in the small cigarillo market included the launch of a wide range of flavours (vanilla, chocolate, etc.) which could provide extra pleasure to the user and encourage non-users to experiment with them. Flavours were mentioned a number of times in the consumer research, such as *"masculine"* flavours (e.g. brandy) which had recently been launched. There were contrasting views in regard to the taste: the satiating character of the product was usually associated with masculinity, while many believed the pleasantness came from the milder taste and the aromas.

"I smoke cigarillos. My favourite brand is Gluk Master, mini. I prefer vanilla. I have tried other flavours too. It is not difficult to find them, at least in Sofia. I do experiment, but I like vanilla the most." (F, 42, BG)

"It gives a good aroma" (M, 36, EL)

"I associate them (small cigarillos) with flavours." (F, 42, BG)

"I smoke a lot less with this. If I smoked cigarettes, I would need to smoke five to be satisfied instead of one of these." (M, 52, ES)

"I smoke them because it is milder than a cigarette, it lasts longer, and to me it also tastes better." (M, 31, ES)

"Strong and with nasty taste" (M, 24, BG)

Participants in the group sessions pointed out that the image of the product had historically been one of masculinity. The online survey findings supported this image as there were more male than female users of small cigarillos, especially in the 18-25 age group (see Table 33). Small cigarillos were usually perceived as traditional, basic, old, adult-oriented, and masculine. They were seen to closely resemble traditional cigars but of a smaller size, a type of *"cheap"* cigar that may or may not have a filter. The

focus group respondents expected the product to be consumed by adults over the age of 35-40 but thought it could also be a gateway product into the tobacco market for younger men.

"The cigar and cigarillo are traditional." (F, 53, FR)

"Cigarillos are more for men; you will not see women." (F, 27, EL)

"Nobody wants to smoke from my cigarillos because they are heavy, and they smell. Smell of man. The scent is heavy. They have a kind of heavy presence. I like this. [...] The classic one, with the well-known scent of cigar." (M, 38, BG)

Yet, in the online survey, the 18-25 age group appeared to follow a similar pattern as the 26+ age group, with 21% of young respondents having already tried the product, as reported in section 4.2.2. It suggested that while the belief is that the product is traditional, for older consumers and for males exclusively, the results of the online survey paint a different picture. Females appeared to make up a considerable share in an otherwise male-dominated market, and younger generations not only followed a similar pattern of consumption as their older counterparts but also appeared to be starting use at younger ages.

In the online survey users of small cigarillos self-reported socio-economic status (see section 4.2.1). In the focus groups the link to cigars and the connection with businesspeople and connoisseurs was mentioned - particularly in Bulgaria. The variety of views across focus group sessions may in part be due to the prices of small cigarillos relative to cigarettes; small cigarillos can be cheaper or more expensive than cigarettes depending on the country (see 4.1). In the online survey, "economic reasons" were the second most cited reason for attempting to quit consuming small cigarillos (see Figure 81. and Figure 82.).

"In more expensive categories (cigars) they are a luxury, but this one is ordinary." (M, 27, EL)

"(Smoking a small cigarillo) It underlines the fact that something special is going on." (M, 58, EL)

"Cigarillos smell, ok, cigarettes have an unpleasant, heavy smell. The cigarillos I use, the brandy flavour, smell somewhat good and I need fewer of them." (M, 36, EL)

"Cigars and cigarillos are of higher quality, more expensive – a metric for status of the consumers. Recently a friend of mine came back from Cuba and brought cigars that are very expensive here. This product is something different. You don't even smoke it, you sense it." (M, 38, BG)

Furthermore, participants sometimes confused cigars and cigarillos, which could explain the range of perceptions regarding the product. Another possibility is that the perception of the product may have changed over the years. Companies have put in further efforts to change the image of the product and to promote it as one of luxury.⁶⁸ Further, with the introduction of the revised TPD which prohibited the production of most flavours in cigarettes and RYO, producers turned to using flavours in small cigarillos.

"That is the cigar we have known all our lives; it can just be smaller or bigger." (F, 54, ES)

In the focus groups, small cigarillos were typically viewed as a more natural version of cigarettes because the product is less processed and wrapped in a tobacco leaf rather than paper. This natural image was falsely interpreted as containing less harmful ingredients and as such, overall it was a less harmful tobacco product. However, this idea of lower harm decreased when there was no filter.

⁶⁸ <https://www.simplycigars.co.uk/guide/cigarillos-short-small-2349/>

Small cigarillos were expected to have higher levels of tar and nicotine than other products under study (e.g. e-cigarette) because it is consumed by combustion and has a stronger taste. Nicotine was also usually thought to be the substance that provokes the (physical) addiction, but consumers saw it as a habit that goes beyond a physical addiction and into a psychological dependence which is much more complex. The act of smoking has traditionally been associated with a social activity to meet new people and to bond with other users. The ritual around the consumption sometimes made the user feel less “addicted” than if they used other tobacco-related products that do not have such a ritual (i.e. snus, chewing tobacco and sniffing tobacco). Furthermore, small cigarillos were considered to satisfy cravings more thoroughly and therefore are smoked less frequently.

Similar to the slim cigarettes, small cigarillos were classified as a conventional tobacco product in the focus groups; therefore, they were widely recognised as an unhealthy habit which could cause various damage and diseases to both consumers and passive consumers. In fact, the main reasons stated by focus group participants for giving up or attempting to quit conventional tobacco products were: experiencing some annoying or threatening symptom (e.g. chronic cough, dysphonia or low physical performance due to breathlessness); being told to quit by a Doctor; disliking the smell of tobacco; feeling uncomfortable for disturbing others; or, to prevent eventual health problems over the long term. Concerns over personal health were also the most stated reason for attempting to quit in the online survey. Small cigarillos, just like slim cigarettes, had a tendency to be considered as the most harmful products across the present study’s consumer research.

“They are healthier because they don’t have paper that burns, it’s the same tobacco leaf.” (M, 52, ES)

“The slim is worse (than small cigarillos) ... that tobacco is rolled in paper which has several things we don’t know.” (F, 54, ES)

“I have the impression, because I was smoking both, that it is cleaner, e.g. causes less of a problem for lungs than smoking a cigarette.” (M, 36, EL)

“A colleague of mine smoked this (small cigarillo), I tried it and I found it better than a normal cigarette, so I started smoking it.” (M, 31, ES)

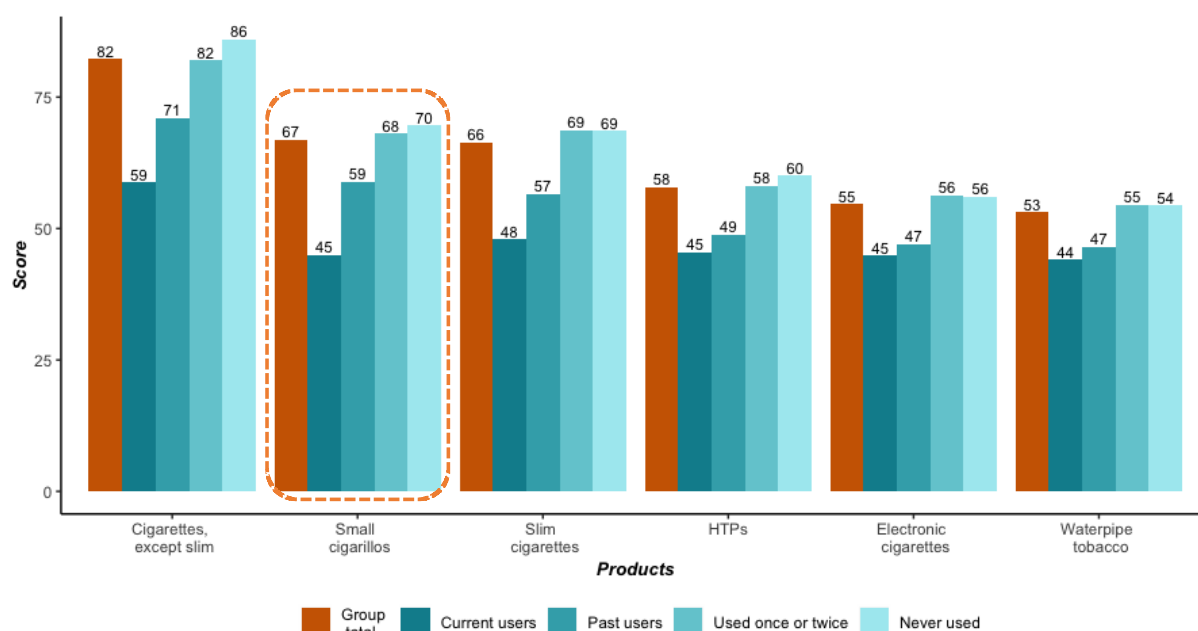
“You just don’t use it like cigarettes. You inhale cigarettes fully when you smoke, while this one you take down to your throat and I believe one should not use it a lot because there is some sensitivity afterwards.” (F, 27, EL)

Figure 89 and Figure 90 show the results of the ‘swing method’ to assess risk perception of small cigarillos and the other tobacco and related products included in this study plus conventional cigarettes. The swing method is explained in section 1.3.3. In brief, the respondent is asked which of the six products they think is most damaging to a user’s health. This product is given 100 points on a scale from 100 (most harmful) to 0 (not harmful). Then using this scale, the respondent is asked to rate each of the other five products on its harmfulness to health of users.

Overall, the **18-25 year group** (see Figure 89) rated the harmfulness to the users health of small cigarillos at 67 out of 100, 15 points lower than conventional cigarettes and about the same as slim cigarettes. Current users rated them at 45 compared to never users at 70.

The **26+ age group** (see Figure 90) rated them at 61, 21 point lower than conventional cigarettes. The rating for current users was 51 compared to 60 for never users.

Figure 89. Perceived harmfulness of small cigarillos to users (18-25 age group)
(average score)



Base: respondents in the 18-25 age group (N=6,090)

Question: Q.3.1.a.⁶⁹ & Q.3.1.b. You have chosen (e.g.; cigarettes, except slims⁷⁰) as likely to be the most damaging to the health of users. We will give (e.g.; cigarettes, except slims) 100 points on a scale of 0-100 for damaging users' health. Please assess the other products' damage on a scale of 0 to 100. If you think a product is about half as damaging to the health of users, you would give that product 50 points. If you think another product is just as damaging as (e.g.; cigarettes, except slims), you would give it 100 points.

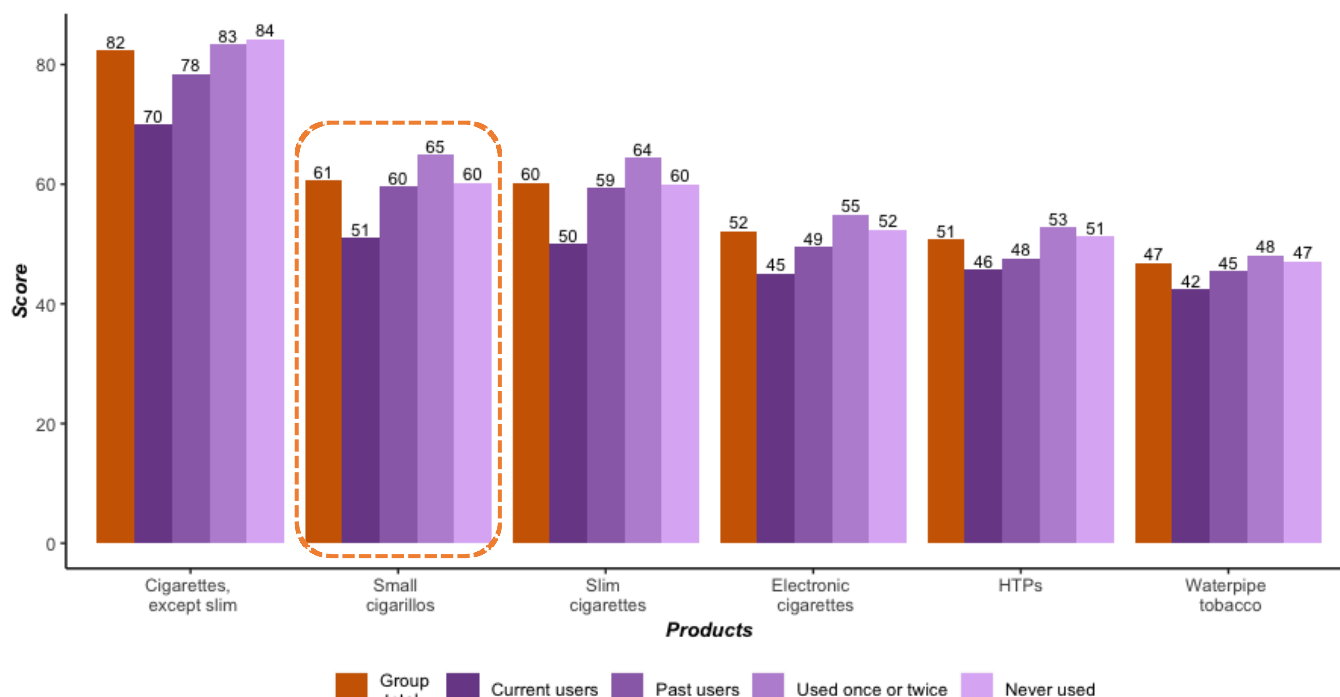
Note: The graph is based on the swing method. The scores were constructed using the responses to the two abovementioned questions. The bars in the graph are the average score of harm to the user provided by respondents for products under the scope of this study. For this chapter, the breakdown is provided for small cigarillo user profiles, and therefore differs from the other chapter. The products are sorted in descending order of the average score provided by the "group total," i.e. amongst all respondents of that age group. The dotted orange box identifies the product in focus.

Source: OE elaboration from online survey data

⁶⁹ Q.3.1.a. Which of the six tobacco products listed below do you think is likely to be most damaging to the health of users?

⁷⁰ The product in parentheses would change in the online survey based on individual responses to Q.3.1.a.

Figure 90. Perceived harmfulness of small cigarillos to users (26+ age group)
(average score)



Base: respondents in the 26+ age group (N=5,910)

Question: Q.3.1.a.⁷¹ & Q.3.1.b. You have chosen (e.g.; cigarettes, except slims⁷²) as likely to be the most damaging to the health of users. We will give (e.g.; cigarettes, except slims) 100 points on a scale of 0-100 for damaging users' health. Please assess the other products' damage on a scale of 0 to 100. If you think a product is about half as damaging to the health of users, you would give that product 50 points. If you think another product is just as damaging as (e.g.; cigarettes, except slims), you would give it 100 points.

Note: The graph is based on the swing method. The scores were constructed using the responses to the two abovementioned questions. The bars in the graph are the average score of harm to the user provided by respondents for products under the scope of this study. For this chapter, the breakdown is provided for small cigarillo user profiles, and therefore differs from the other chapter. The products are sorted in descending order of the average score provided by the "group total," i.e. amongst all respondents of that age group. The dotted orange box identifies the product in focus.

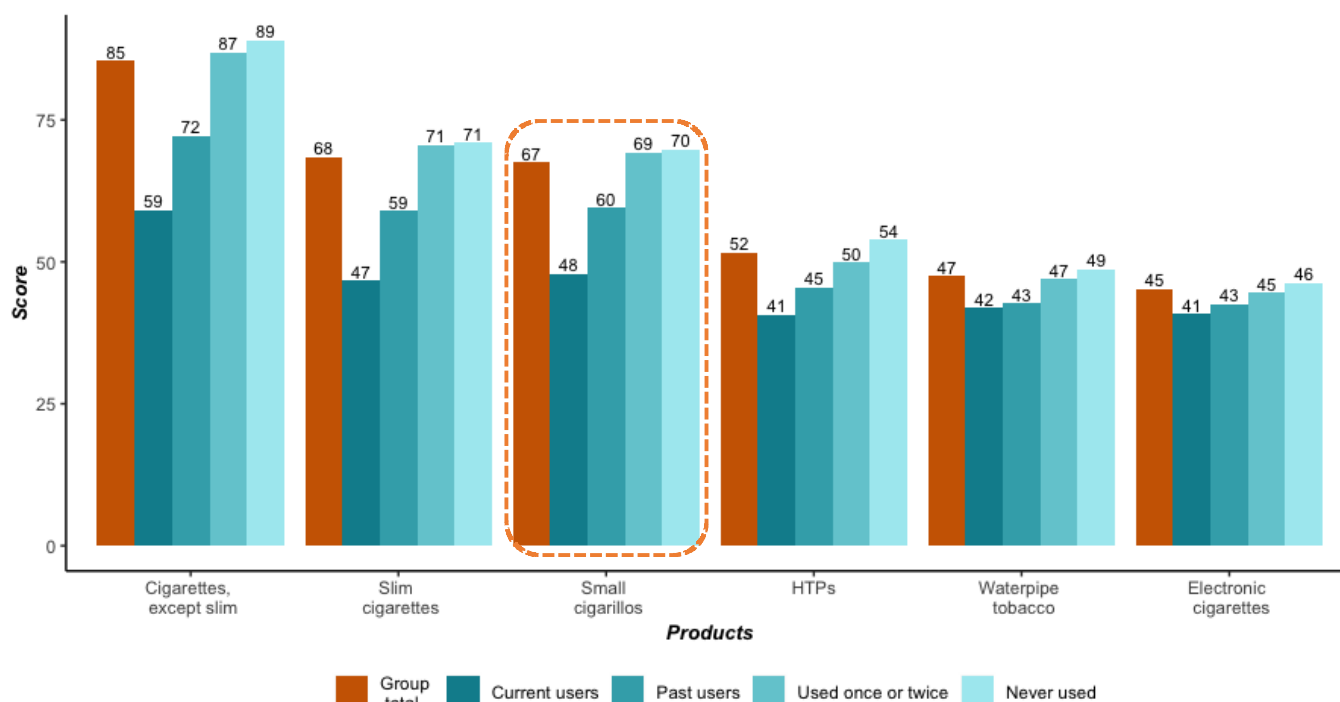
Source: OE elaboration from online survey data

⁷¹ Q.3.1.a. Which of the six tobacco products listed below do you think is likely to be most damaging to the health of users?

⁷² The product in parentheses would change in the online survey based on individual responses to Q.3.1.a.

Figure 91 and Figure 92 chart the perception of harmfulness to bystanders through the passive smoking of small cigarillos. The **18-25 age group** (see Figure 91) rated small cigarillos at 67 out of 100 and posing about the same risk to bystanders as slim cigarettes. Consistent with the findings for other tobacco products, current users rated the risk to bystanders at only 48, much lower than never users and once or twice users who rated the risk at 70 and 69 respectively. Overall, the **26+ age group** (see Figure 92) gave a marginally lower assessment than the younger age groups of 63 out of 100 to the harmfulness of the passive smoking. However, current users rated the risk to bystanders at 54 out of 100, some 6 points higher than the younger age group.

Figure 91. Perceived harmfulness of small cigarillos to bystanders (18-25 age group)
(average score)



Base: respondents in the 18-25 age group (N=6,090)

Question: Q.3.2.a.⁷³ & Q.3.2.b. You have chosen (e.g.; cigarettes, except slims⁷⁴) as likely to be the most damaging to the health of passive users. We will give (e.g.; cigarettes, except slims) 100 points on a scale of 0-100 for damaging passive users' health. Please assess the other products' damage on the health of passive users on a scale of 0 to 100. If you think a product is about half as damaging to the health of passive users, you would give it 50 points. But if you think another product is just as damaging as (e.g.; cigarettes, except slims), you would give it 100 points

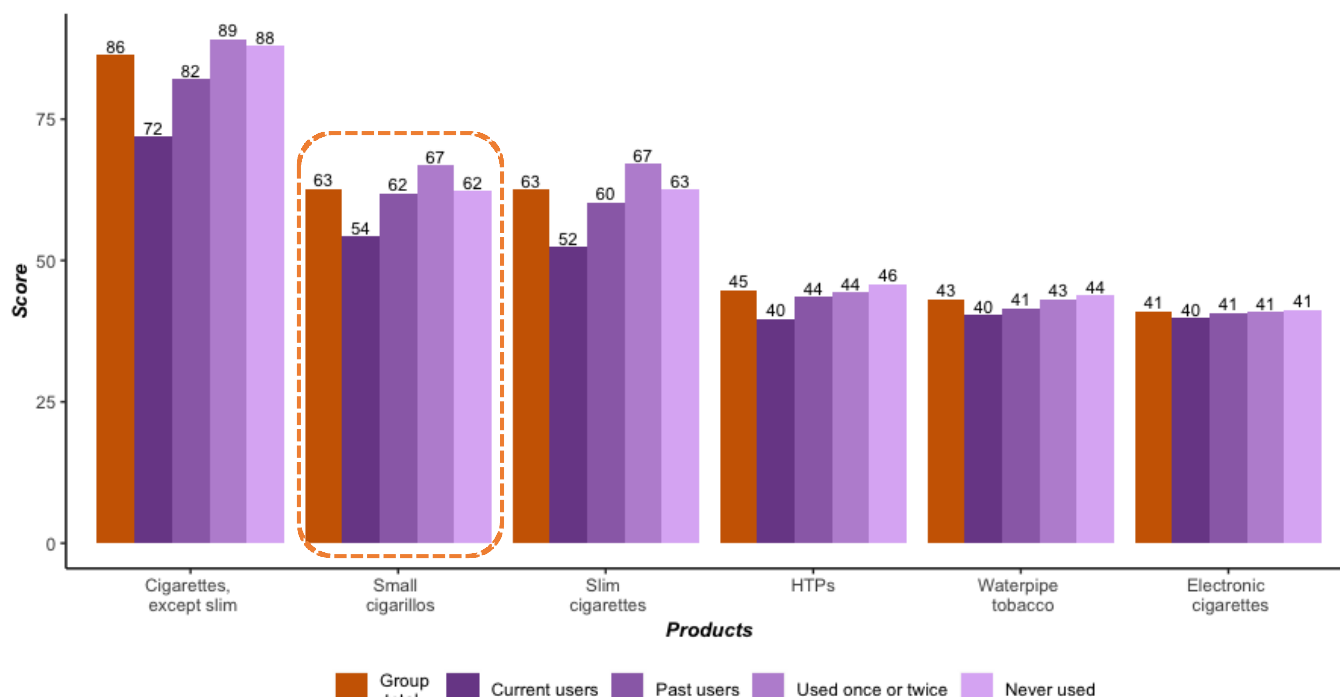
Note: The graph is based on the swing method. The scores were constructed using the responses to the two abovementioned questions. The bars in the graph are the average score of harm to the passive user provided by respondents for products under the scope of this study. For this chapter, the breakdown is provided for small cigarillo user profiles, and therefore differs from the other chapter. The products are sorted in descending order of the average score provided by the "group total," i.e. amongst all respondents of that age group. The dotted orange box identifies the product in focus.

Source: OE elaboration from online survey data

⁷³ Q.3.2.a. Which of the six tobacco products listed below is likely to be the most damaging to the health of other people from passive use?

⁷⁴ The product in parentheses would change in the online survey based on individual responses to Q.3.1.a.

Figure 92. Perceived harmfulness of small cigarillos to bystanders (26+ age group)
(average score)



Base: respondents in the 26+ age group (N=5,910)

Question: Q.3.2.a.⁷⁵ & Q.3.2.b. You have chosen (e.g.; cigarettes, except slims 76) as likely to be the most damaging to the health of passive users. We will give (e.g.; cigarettes, except slims) 100 points on a scale of 0-100 for damaging passive users' health. Please assess the other products' damage on the health of passive users on a scale of 0 to 100. If you think a product is about half as damaging to the health of passive users, you would give it 50 points. But if you think another product is just as damaging as (e.g.; cigarettes, except slims), you would give it 100 points

Note: The graph is based on the swing method. The scores were constructed using the responses to the two abovementioned questions. The bars in the graph are the average score of harm to the passive user provided by respondents for products under the scope of this study. For this chapter, the breakdown is provided for small cigarillo user profiles, and therefore differs from the other chapter. The products are sorted in descending order of the average score provided by the "group total," i.e. amongst all respondents of that age group. The dotted orange box identifies the product in focus.

Source: OE elaboration from online survey data

⁷⁵ Q.3.2.a. Which of the six tobacco products listed below is likely to be the most damaging to the health of other people from passive use?

⁷⁶ The product in parentheses would change in the online survey based on individual responses to Q.3.1.a.

4.4 Health, social and economic impacts

While small cigarillos are sometimes considered to be less harmful than regular cigarettes (e.g. see section 4.3), this belief is not supported by the scientific evidence. Cigar smoke has been proven to contain the same toxic and carcinogenic constituents found in cigarette smoke and can cause oral, laryngeal, oesophageal, lung cancer, heart disease, aortic aneurysm, etc. Users often inhale smoke from little cigars and cigarillos (Kostygina et al., 2016). Ghosh et al. (2017) examined the biological effects of little cigar use and found that it caused significantly greater adverse effects than cigarettes (e.g. on airway epithelial function). The online survey confirms other research that small cigarillos were widely used in combination of with other tobacco, particularly cigarettes (Sterling, 2016).

In terms of social impacts, much of the research specific to cigarillos has been conducted in the USA. Cohn (2015, N=1.609) found that a strong but variable association between new tobacco products and alcohol and marijuana use among adolescents (aged 18-24) in the USA. Alcohol use was correlated with little cigars and cigarillos, but to a lesser extent than other tobacco and related products. By contrast, marijuana use was strongly correlated to little cigars and cigarillos use (Cohn, 2015). Studies have demonstrated that Twitter was a major information-sharing and marketing platform for little cigars and cigarillos (Kostygina et al., 2016). The global reach of the internet means that impressionable and vulnerable people may be exposed to indirect product promotion or marketing practices from outside the EU.

In the United States, Delnevo (2017, N=909) argued that tobacco manufacturers had responded to increased cigarette regulation and taxation by exploiting policy loopholes. The industry offered little cigars and filtered cigars as an alternative to cigarettes to be exempt them from regulatory oversight. Furthermore, Cantrell et al. (2013, USA) found that the availability, advertising, and price in the point-of-sale environment of LLCs encouraged switching or initiation of use in minority and young adult neighbourhoods. The TPD's restrictions on flavours do not apply to small cigarillos and the TED applies a lower tax rate to this product. These measures have already motivated producers to increasingly produce small cigarillos, including those with flavours. The consumption patterns of potential consumers must be closely monitored as it is susceptible to change. It is especially concerning as the use of additives as flavouring agents has been established to increase the appeal of tobacco products and to veil their harmful potential (DKFZ, 2010).

In Europe, RAMBOLL (2014) led a study for the Impact Assessment of a possible revision of Council Directive 2011/64/EU on the structure and rates of excise duty applied to manufactured tobacco. Fourteen Member States reported that cigars/cigarillos were potentially attempting to abuse the definitions of Directive 2011/64/EU with the purpose of obtaining preferential tax treatment. They expressed significant concern that the preferential rate structure created substantial incentives for economic operators to market products as cigarette substitutes but fulfilling the minimum requirements of the definitions in order to be classified as one with a lower tax rate. While some MS expressed concern over the development of "borderline" cigarillos to take advantage of the lower tax rates on the cigar and cigarillos,⁷⁷ other states argued that there was no real demand on the market for this type of product. However the report also found an increase in consumption of cigars and cigarillos between 2008 and 2012, notably in France, the UK, Germany, Spain and Italy.⁷⁸ A more recent European Commission study (2017)⁷⁹ found that cigarillos became popular in some EU countries a few years ago, but have been declining in most EU markets due to a combination of:

⁷⁷ Finland, Germany, Lithuania, Latvia, Slovakia, Hungary, the Netherlands, Greece, Portugal, Czech Republic, Spain, Italy, Poland and Sweden.

⁷⁸ https://ec.europa.eu/taxation_customs/sites/taxation/files/docs/body/ramboll-tobacco-study.pdf

⁷⁹ European Commission (2017): Study on Council Directive 2011/64/EU on the structure and rates of excise duty applied to manufactured tobacco

- a revised product definition;
- the end of certain derogations for Germany and Hungary, and
- the adoption, in various countries, of tax structures and rates that reduced the incentives for low price products.

The revision of the definition in 2008 clarified the tax schemes to be applied to this product through the predecessor of the current Directive 2011/64/EU (i.e. Directive 2002/10/EU). The Minimum Excise Duty (MED) is used to refer to the minimum amount of excise duty (per quantity of product) that Member States can levy on certain tobacco products when the sum of specific and ad-valorem duties falls below an established threshold that is defined nationally. MED consists in a tax 'floor' and was designed to prevent the risk of too low taxation (hence too low prices) and 'race to bottom' in the cheaper segments of the market.⁸⁰ For instance:

- Germany introduced in 2012 a 'dynamic' total tax of 5,760 Cent per piece (minus the VAT of the taxed cigar/cigarillo).
- Denmark more than doubled its fully specific rate between 2014 and 2016 (from € 26.5 to € 67.0 per 1,000 pieces).
- Spain introduced a minimum excise duty (MED) of €41.5 per 1,000 units in 2013.
- Austria increased the MED to €100 per 1,000 pieces in 2013.
- France switched to a mix structure in 2013, establishing a specific rate of € 17.5 per 1,000 pieces in addition to the pre-existing MED.

These measures were largely effective in closing the regulatory loopholes that had incentivised the development of 'borderline' cigarillos but there is still a lack of harmonisation between Member States (see Figure 175). For instance, small cigarillos were found to be relatively cheaper in Germany than in neighbouring countries, which may explain the observed mismatch between consumption and sales volumes. Since the Scandinavian neighbours tended to sell small cigarillo at a higher retail selling prices (RSP), consumers may source themselves in Germany.

4.5 Concluding remarks on small cigarillos

This section summarises the main findings that emerged from the research conducted on small cigarillos. The systematic review of secondary sources revealed:

- Users often inhale smoke from small cigarillos which is expected to cause equal or similar health problems to conventional cigarettes.
- According to the Eurobarometer 458 (2017), the prevalence of regular and past users of cigarillos remained relatively stable between 2014 and 2017 in the EU – with only 3% of respondents reporting current use of the product in 2017. The EU market of small cigarillos was 2.1 billion euros in 2017, with Germany and France accounting for almost half of the entire EU market.
- Social-networking sites have enabled indirect product promotion, through user-generated content which influences public opinion on tobacco control policy decisions.
- Tobacco companies have endeavoured to blur the distinction between little cigars and conventional cigarettes to take advantage of the lighter regulation of cigarillos and cigars. Cigarillos are now flavoured to increase their palatability and appeal, since TPD's restrictions on flavours do not apply to small cigarillos. The industry also circumvents the higher tax rates applied to cigarettes by promoting a tax-induced substitution of cigarettes with 'borderline' cigarillos that are taxed at a lower rate. The difference in prices of small cigarillos in 2017

⁸⁰ See art14(1a) and (2b) Council Directive 2011/64/EU of 21 June 2011

varied by 8-fold between MS with the highest prices (Romania and Croatia) and those with the lowest (Hungary and Spain) (see Figure 175), due to a certain extent to the different tax approaches to cigarillos by MS.

The empirical research (focus groups and online survey) revealed the following user profile:

- The average current user of small cigarillo smoked them at least once a week, between 1 and 5 sticks every day of use. Most users started smoking cigarillos for enjoyment. However, socialising was also mentioned. Close to 9 in 10 current cigarillo users are using other tobacco and related products concurrently.
- About 60% of current users had attempted to quit. Among failed quitters, the main reasons to try quitting centred on health – personal health, having seen health warnings and the impact on passive smokers. The main reason cited for failing to quit was 'stress' and 'difficulty', in all probability responses to addiction.
- A typical past user used the product at least once a week. Most were dual users of tobacco and related products. Past users were divided between those who managed to entirely quit tobacco products, and those who transitioned to other tobacco products, mainly conventional cigarettes. Between 60% and 70% of former cigarillo users, depending on the age group, were using now other tobacco products, mainly conventional cigarettes. Between 25% and 35% of past users had completely quit the use of all the tobacco products.

Lastly, the following paragraphs highlight the *Product and Regulatory perceptions remarks*

- Small cigarillos have traditionally been associated with masculinity. However, this view is changing with more young females using them. In the focus groups cigarillos were said to evoke cosiness and friendliness and to be a contribution to special occasions and social gatherings. Cigarillos were also considered by many as a high-class product due to the association with cigars, and as such associated to businesspeople and connoisseurs. They were viewed as old-fashioned and authentic, but also "disgusting" due to the strong taste and smell by others. The variety of views may in part be due to three factors. First, the prices of small cigarillos can be higher or lower than cigarettes, depending on the country. Second, cigarillos are sometimes confounded with cigars. And third, the tobacco industry has endeavoured to change the image of the cigarillo into a luxury product.
- Small cigarillos are exempt from the ban on flavours in the revision of the TPD. Such flavours have been found to be an appeal to never users, and young people in particular. The online survey identified the UK as having the highest share of small cigarillo users (at 11%) in the 18-25 age group but the highest share of never users (76%) in the 26+ age group.
- *Perception of harmfulness.* Small cigarillos, like slim cigarettes, were considered as the most harmful products to users and passive across the present study products, after conventional cigarettes. Never users perceived cigarillos as much more harmful to both smokers and passive smokers. The perception of small cigarillos as less harmful than conventional cigarettes could be due to the wrapping in a tobacco leaf rather than paper and therefore perceived as more "natural". Another possibility is that flavouring reduces the perception of risk (see section 4.4).
- *Perception of marketing practices.* Participants reported the emergence of flavoured cigarillos.
- *Regulatory support.* Cigarillos are recognised as conventional tobacco products.

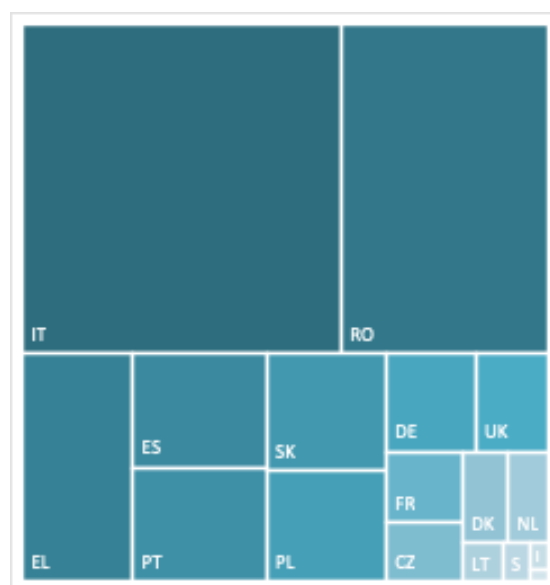
5 Novel tobacco products

5.1 Background

There has been a proliferation of novel tobacco products entering the EU market. Of particular significance to the TPD are heated tobacco products (HTPs). They have been marketed in over 40 countries worldwide as of July 2019 (WHO, 2020), with at least 22 countries from the WHO European region (Lugo et al., 2018). In the EU, the brand leader is the Philip Morris IQOS system, comprising the IQOS device that heats a tobacco stick called Heetstick.⁸¹ It was launched in 2014 in Milan as a pilot test for the European market and was expanded to the whole Italian market in 2015 (Lugo et al., 2018). In 2017, tobacco companies began aggressive worldwide promotion of HTPs (PMI, 2017). As of February 2020, the product was available in 20 Member States.⁸² In 2018, the WHO published an information sheet on HTPs, providing information on how the products were marketed in countries where they are available and on the manufactures' strategies manufacturers to extend their reach to other markets.

Market developments. In this chapter, only heated tobacco products are covered under the 'novel tobacco product' category. According to the Spanish Ministry of Health (2019, Spain), there is little evidence on the use and prevalence of heated tobacco products, especially in Europe. However, several studies have pointed to a rapid rise in consumption and market growth for this product (PHE, 2018). Figure 93 depicts the structure of the EU market in volume (millions euros) for HTPs according to 2017 Passport data.

Figure 93. Market size of heated tobacco products (HTPs) in million euros in 2017



Note: The market size is estimated total sales in million euros for 2017, without population adjustments. It is displayed as a tree map graph that consists of series of nested rectangles of sizes proportional to the corresponding data value.

Source: OE elaboration of Passport, 2017

In terms of retail volume, the biggest market for HTPs was Italy, as the first EU country in which the product was introduced. Relatively large markets of HTPs were also observed in Romania and Greece. Similar findings were reported by the Tobacco Market Commission of the Spanish Ministry of Health

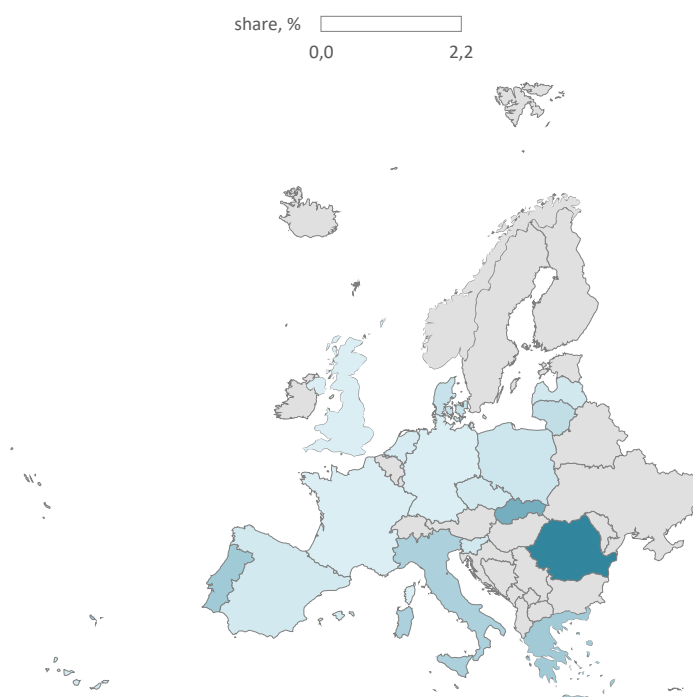
⁸¹ Note there is also currently a new product in some EU countries as well, called *glo* of British American Tobacco (BAT). https://www.bat.com/group/sites/UK__9D9KCY.nsf/vwPagesWebLive/DOAWUGNJ

⁸² <https://www.pmi.com/smoke-free-products/iQOS-our-tobacco-heating-system>, as of February 2020

(2019). In Italy there has been a significant rise in sales of HTPs from sales of 11 tons in 2015 and to 519 tons in 2017. In Spain, there was also a nine-fold rise in monthly sales between January 2017 and July 2018. In most Member States HTPs have only recently been available.

The share of HTPs in the national tobacco markets in 2017 is visually represented for EU markets in Figure 94, for countries where data were available.⁸³ The share of HTPs in the national tobacco market was particularly high in Romania, Slovakia, Italy, Greece, Slovenia and Portugal.

Figure 94. Share of HTPs within the national total tobacco market, per country (2017)



Note: Percentage share of heated tobacco products were estimated with respect to the whole tobacco market in the country in million euros.

Source: OE elaboration of Passport, 2017

Since younger generations are increasingly less likely to smoke conventional tobacco products, tobacco companies have developed new products which are marketed as “safer” alternatives. Hair (2018) reported on the outcomes of expert interviews in which it was said that HTPs appeal to tobacco users because the product is associated with cleanliness, exclusivity and a high-tech appearance. IQOS would be appealing to youth because it would sit in the “control space (cleanliness, official branding, not ‘hackable’ or flexible in terms of flavour).”

Overall, it is too early to forecast the market developments of HTPs with any accuracy given the large variation in market presence across Member States. For example, according to Public Health England (PHE), awareness and ever use of heated tobacco products were still rare in Great Britain as of 2017 (PHE, 2018), while a study in Italy highlighted a rapid increase between 2016 and 2017, although it represented only 0.7% of the total market of tobacco products in the country in 2017 (Lugo et al., 2018).

⁸³ In this case we used the share of heated tobacco products for the real retail volume and calculated the share in RSP values with respect to the whole tobacco market.

Consumer perception. Research on the consumer perception of HTPs was limited at the time of this study, and most of the available evidence came from industry studies. A study commissioned in the UK (PHE, 2018) emphasised the need for evidence on the appeal of HTPs to smokers and non-smokers alike, with a particular focus on the appeal to youth. A review conducted by McKelvey et al. (2018) found that there are currently no studies on the potential appeal of HTPs appeal to people not using any tobacco products.

HTPs were designed to mimic the behaviour of conventional cigarette smoking without the combustion component, which has been acknowledged to be one of the most harmful factors of conventional tobacco use. According to the WHO, tobacco companies have employed non-traditional marketing and product distribution strategies for the latest generation of HTPs to attract customers and increase sales, such as:⁸⁴

- Reducing consumer health concerns through (uncertain) claims that HTPs are reduced-risk products.
- Designing HTP systems to fulfil the role cigarettes once held to boost the user's social image. For example, the IQOS device is available in different colours, limited edition designs and increases consumers' convenience through reduced odour, rapid charging, and similar features.
- Employing the "bait and hook" pricing strategy, where a base device (e.g. IQOS device and its charging station) is typically sold at a discounted price, with a recurrent price being charged for HTPs (e.g. HeetSticks). This increases customer retention rates through a discounted upfront switching cost and lifetime value of consumers as a result of repeat buying.
- Using social marketing techniques involving the development of community activists and brand ambassadors who promote HTPs through various channels, ranging from individual and community engagement to social media marketing (e.g. social media platforms, promotional videos).
- Advertising by acknowledging the health risks of traditional cigarettes and describing the new products as cleaner alternatives, and the shift in tobacco companies' corporate image towards promoting alternative tobacco products.
- Selling and marketing on multiple channels, including dedicated HTP stores, retail establishments and e-commerce websites.

The industry has targeted its communication strategy to closely resemble the experience of cigarettes, as the sensory characteristic is an important part of the product (WHO, 2019b). The marketing efforts were designed to highlight the product as a new concept of traditional smoking, to evoke familiarity and nostalgia for smoking. HTP technology – and notably the IQOS system – has become increasingly popular in recent years, with tobacco sticks available in multiple flavours (Jankowski et al., 2019). Hair (2018) suggested that IQOS had been marketed and advertised with four key message themes: cleanliness, customisation, comparisons with combustible smoking and sociability. The analysis of the product packaging also identified eight themes specific to the device: clinical purity, a closed system, premium design, sensory invitations, nostalgia for combustible tobacco smoking, stability, familiar technology and normalisation. The packaging has been said to resemble iPhones and other high-end smartphones, where the device and parts are neatly placed in moulded plastic trays inside a glossy white box (McKelvey et al., 2018). There is concern that this similarity in appearance to popular electronic devices could increase appeal among younger generations, as the most frequent users of such tech-based devices.

⁸⁴ https://www.who.int/tobacco/publications/prod_regulation/https-marketing-monitoring/en/

A pilot study with a small convenience sample (n=60) and a majority of smokers was conducted in Italy to validate the SECRHET⁸⁵ questionnaire on knowledge and behaviours related to cigarettes and HTP use among the youth. The data showed that less than half of participants considered HTPs to be harmful to the health of users and that 34% would accept one of these products if offered by a friend (La Torre et al., 2019). Some findings have suggested that there exists some dissatisfaction with the product as a longer-term substitute for smoking over time (WHO, 2019b).

Impacts. The tobacco industry has repeatedly promoted HTPs as a less harmful and a safer alternative for conventional tobacco product consumers. There has been considerable debate around the absence of combustion and the effects that this product may have on the user and passive users. Studies led by the industry claimed a 90%-95% reduction in harmful and potentially harmful substances and toxicity.⁸⁶ Other studies have found other toxicants not present in conventional cigarettes (e.g. Leigh et al., 2018). Manufacturers have been careful to qualify their claims or include a waiver since the research on the health impacts were few and inconclusive (WHO, 2019). As a result, both positive and negative information has been circulated. A summary of the impacts is provided in section 5.4.

⁸⁵ Smoking E-CigaRette and HEat-noT-burn products (SECRHET)

⁸⁶ https://www.pmiscience.com/resources/docs/default-source/library-documents/presentation_maira_gilchrist_gtnf_2015a.pdf

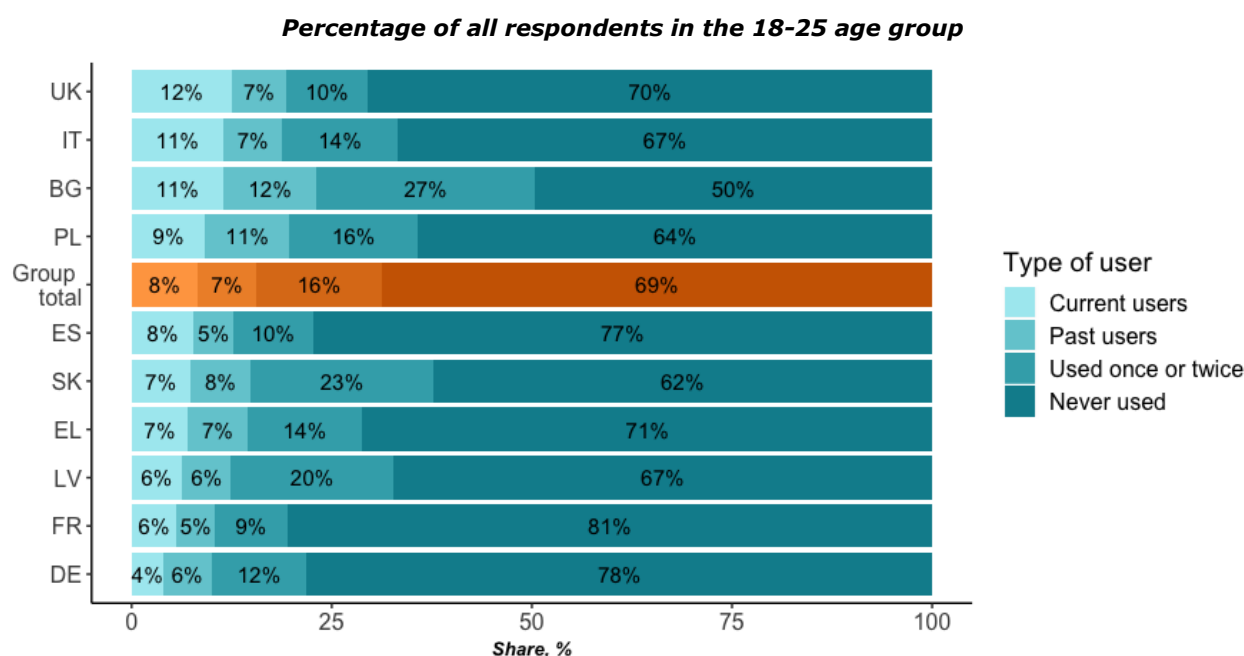
5.2 Market developments

5.2.1 Socio-demographic characteristics

Participants of the focus groups confirmed that the IQOS system was the most well established brand in Europe.⁸⁷ HTPs were usually referred to by the brand name, IQOS, except in France where they were also referred to as “tobacco-to-heat” and in Germany where they were called “vaporisers” (as with e-cigarettes in the other countries). The product brand was particularly well viewed in Greece because it was said that the government supported Philip Morris due to their substantial investments in the country during the economic crisis.

Figure 95 and show the use of HTPs across the ten selected EU Member States. Italy, the first country to see a European launch, had the highest share of current users (27%) in the 26+ age group. The UK, Italy and Bulgaria have double digit percentages (above 10% among the 18-25 age group); and Italy, Spain Poland, Greece and Bulgaria in the 26+ age group.

Figure 95. Use of HTPs: country breakdown (18-25 age group)



Base: all respondents in the 18-25 age group (N=6,090)

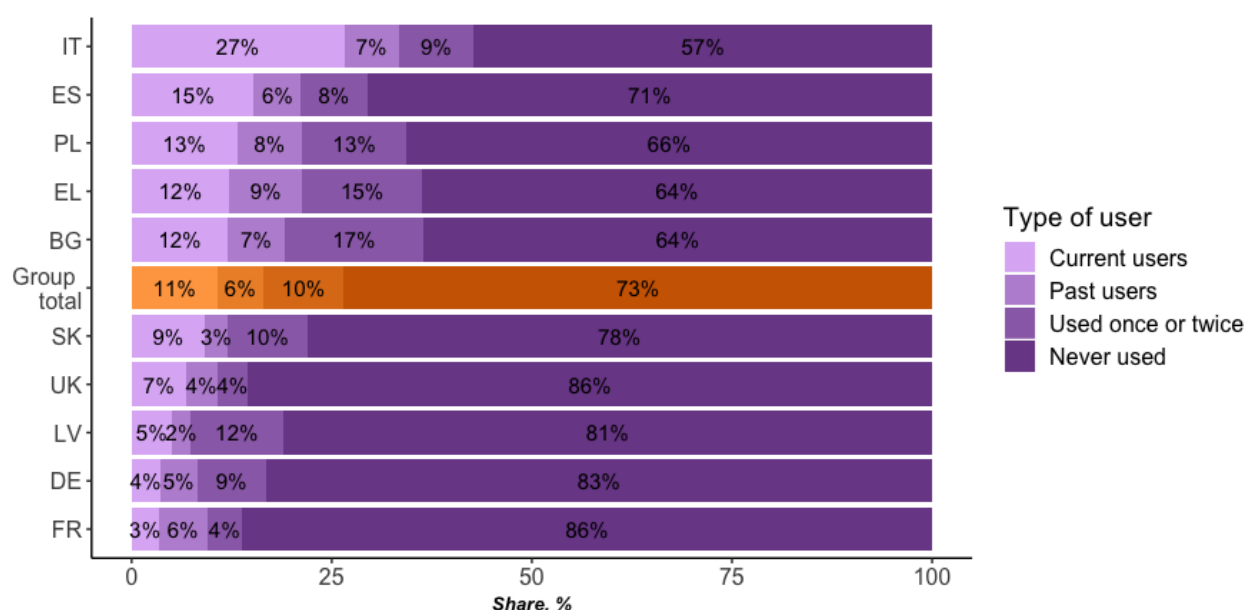
Question: Q.1.1. Have you ever used HTPs?

Note: The group total category refers to all respondents within the age group.

Source: OE elaboration from online survey data

⁸⁷ In the focus group, HTPs were usually referred to by the brand name of the leader in the market, IQOS, except in France where they were also called “tobacco to heat” and in Germany, where they were also referred to as “vaporisers” (like e-cigarettes in the other countries). IQOS was reportedly known to belong to Philip Morris by some participants who thought that this company owned the patent. IQOS was particularly well viewed in Greece due to the government propagating that Philip Morris had been doing a great investment in the country during the economic crisis.

Figure 96. Use of HTPs: country breakdown (26+ age group)

Percentage of all respondents in the 26+ age group

Base: all respondents in the 26+ age group (N=5,910)

Question: Q.1.1. Have you ever used HTPs?

Note: The group total category refers to all respondents within the age group.

Source: OE elaboration from online survey data

Table 38 and Table 39 present the distributions of some socio-demographic characteristics for the age group samples and by HTP user type namely, current user, past user, used once or twice, or never used. Among the current users there were marginally more males than females. Current users of HTPs in the 26+ age group were much more likely to have a high education level, but not in the 18-25 age group in which a percentage of respondents were still in education. Current users were more likely to live in urban areas and to consider themselves as having high socio-economic status. Given the recent introduction of HTPs they may not have extended their marketing to rural areas, and with an upfront cost for the device at circa €40 and the sticks priced at the same as conventional cigarettes, they may appeal only to wealthier people.

Table 38. HTP use socio-demographic breakdown (18-25 age group)

Percentage and number of respondents in the 18-25 age group

	Group total %(n)	Current users %	Past users %	Used once or twice %	Never users %
Use of product	100 (6090)	8	7	16	69
Gender					
Male	47 (2869)	10	9	17	64
Female	53 (3221)	7	6	14	73
Education level *					
1-2 (low)	5 (335)	13	10	17	60
3-4 (medium)	63 (3829)	8	7	16	70
5-6 (high)	32 (1926)	9	8	15	68
Residency					
Urban	65 (3949)	9	8	17	66
Suburban	19 (1173)	8	8	15	70
Rural	16 (968)	5	5	13	77
Self-reported socio-economic status					
1-4 (low)	18 (1126)	8	6	14	73
5-7 (medium)	65 (3954)	7	7	16	70
8-10 (high)	17 (1010)	14	12	16	58

Base: all respondents in the 18-25 age group (N=6,090)

Notes: The group total column refers to all respondents within the age group, i.e. it is a within column percentage. It provides the breakdown of each socio-demographic variable, including absolute numbers. The white cells in the four right columns represent the breakdown by user type, i.e. they are within row percentages (white cells in each row sum to 100).

* The sub-categories of the education level were the following: (1) No official education; (2) Primary school; (3) High school; (4) Some years of university (not completed); (5) University degree completed; (6) Post-graduate (master, PhD, other)

Source: OE elaboration from online survey data

Table 39. HTP use socio-demographic breakdown (26+ age group)

Percentage and number the of respondents in the 26+ age group)

	Group total % (n)	Current users %	Past users %	Used once or twice %	Never user %
Use of product	100 (5910)	1111	6	1010	73
Gender					
Male	52 (3074)	12	6	11	71
Female	48 (2836)	9	5	10	76
Education level*					
1-2 (low)	5 (308)	8	6	5	81
3-4 (medium)	46 (2704)	8	4	9	79
5-6 (high)	49 (2898)	14	7	12	67
Residency					
Urban	66 (3879)	13	7	12	68
Suburban	17 (1010)	8	5	8	79
Rural	17 (1021)	5	2	5	88
Self-reported socio-economic status					
1-4 (low)	23 (1374)	6	3	9	83
5-7 (medium)	64 (3769)	11	6	11	73
8-10 (high)	13 (767)	20	11	10	58

Base: all respondents in the 26+ age group (N=5,910)

Notes: The group total column refers to all respondents within the age group, i.e. it is a within column percentage. It provides the breakdown of each socio-demographic variable, including absolute numbers. The white cells in the four right columns represent the breakdown by user type, i.e. they are within row percentages (white cells in each row sum to 100).

* The sub-categories of the education level were the following: (1) No official education; (2) Primary school; (3) High school; (4) Some years of university (not completed); (5) University degree completed; (6) Post-graduate (master, PhD, other)

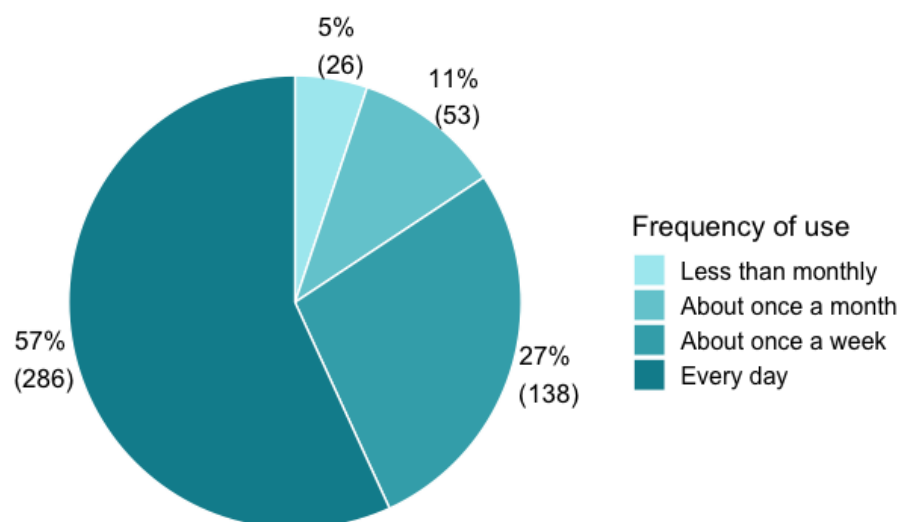
Source: OE elaboration from online survey data

5.2.2 Pattern of use

Figure 97 and Figure 98 report the current frequency of use of HTP users. In the 18-25 age group 57% are daily users and 27% use about once a week. In the 26+ age group, the frequency of use is marginally higher with 59% daily users and 31% about once a week.

Figure 97. Frequency of use of current HTP users (18-25 age group)

Percentage and number of current users in the 18-25 age group



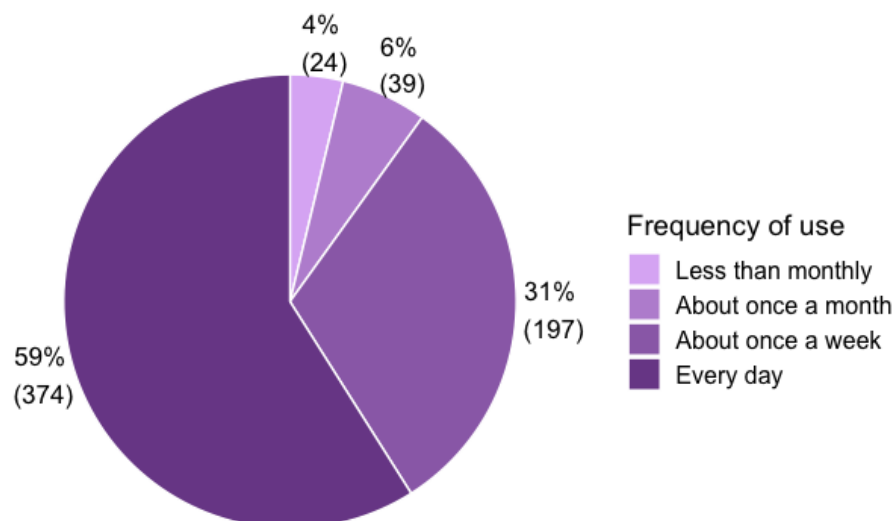
Base: current users of HTPs in the 18-25 age group (N=503)

Question: Q.1.3. How often do you use HTPs?

Source: OE elaboration from online survey data

Figure 98. Frequency of use of current HTP users (26+ age group)

Percentage and number of current users in the 26+ age group



Base: current users of HTPs in the 26+ age group (N=634)

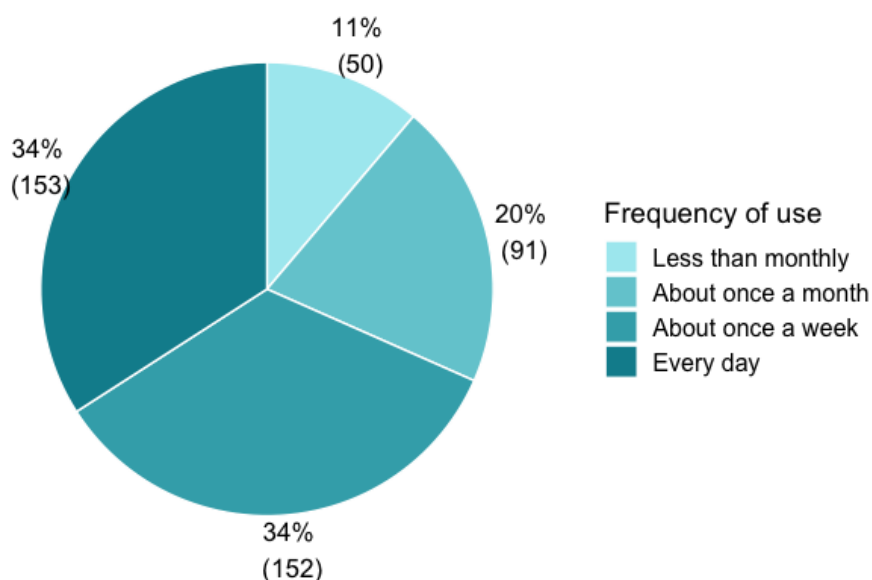
Question: Q.1.3. How often do you use HTPs?

Source: OE elaboration from online survey data

The frequency of use among past users is shown in Figure 99 and Figure 100. In both age groups the percentage of past daily users was much lower than the current users. In the 18-25 age group, 34% were past daily users, 23% fewer than current users. In the 26+ age group the past daily users at 33% were 26% fewer than current users.

Figure 99. Frequency of use of past HTP users (18-25 age group)

Percentage and number of past users in the 18-25 age group



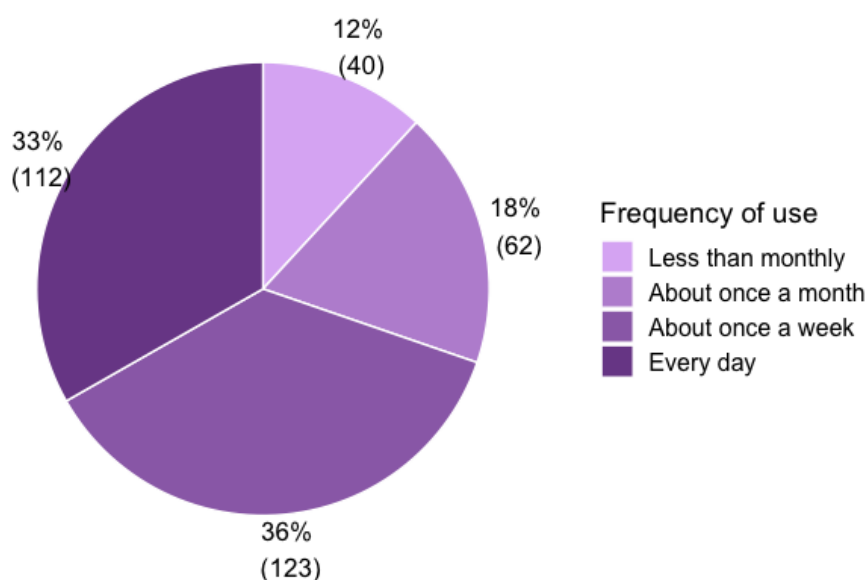
Base: past users of HTPs in the 18-25 age group (N=446)

Question: Q.1.6. How often did you use HTPs?

Source: OE elaboration from online survey data

Figure 100. Frequency of use of past HTP users (26+ age group)

Percentage and number of past users in the 26+ age group



Base: past users of HTPs in the 26+ age group (N=337)

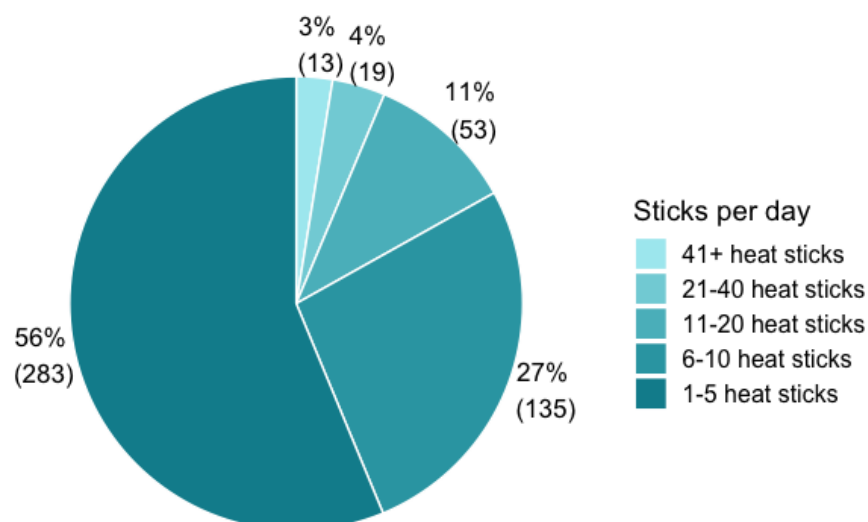
Question: Q.1.6. How often did you use HTPs?

Source: OE elaboration from online survey data

Figure 101 and Figure 102 report the number of sticks per day of use of HTPs. The two age groups report almost identical figures. In the 18-25 age group 56% report 1 to 5 sticks and 27% 6 to 10 sticks. The respective percentages for the 26+ age group were 56% and 17%.

Figure 101. Average number of HTP sticks a day for current users (18-25 age group)

Percentage and number of current users in the 18-25 age group



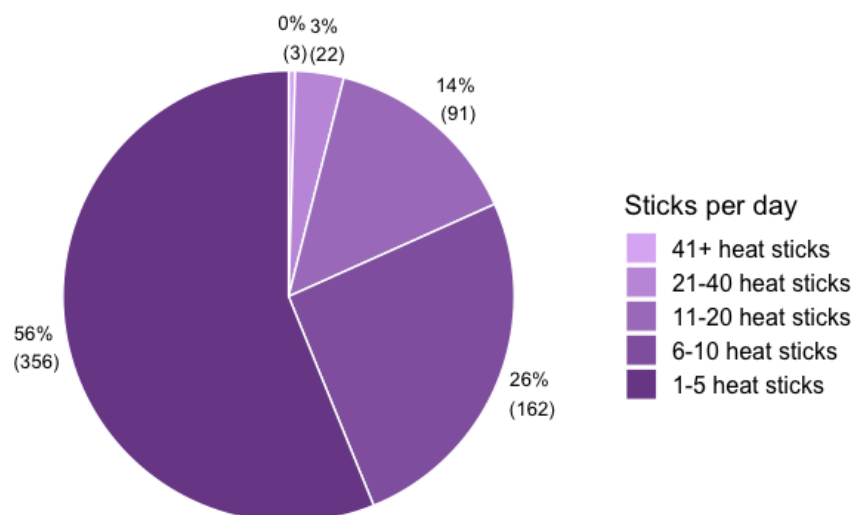
Base: current users of HTPs in the 18-25 age group (N=503)

Question: Q.1.4.a. On the days that you smoke/use, about how much do you smoke/use?

Source: OE elaboration from online survey data

Figure 102. Average number of HTP sticks a day for current users (26+ age group)

Percentage and number of current users in the 26+ age group



Base: current users of HTPs in the 26+ age group (N=634)

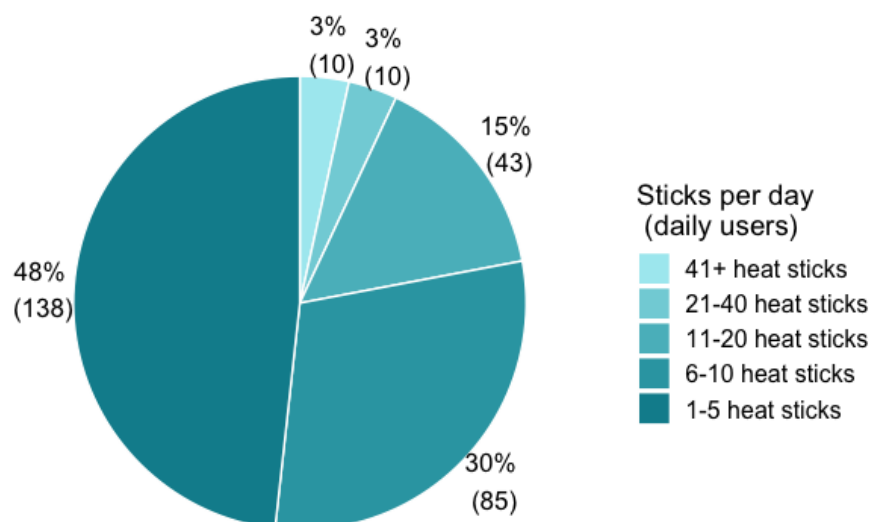
Question: Q.1.4.a. On the days that you smoke/use, about how much do you smoke/use?

Source: OE elaboration from online survey data

Figure 103 and Figure 104 report the number of sticks per day for the daily users of HTPs. A majority of daily users in both age groups reported consuming 6 or more sticks per day. 51% in the 18-25 age group of whom 21% used 11 or more sticks a day and 57% in the 26+ age group of whom 25% used more than 11 sticks a day.

Figure 103. Average number of HTP sticks a day, for daily users (18-25 age group)

Percentage and number of current daily users in the 18-25 age group



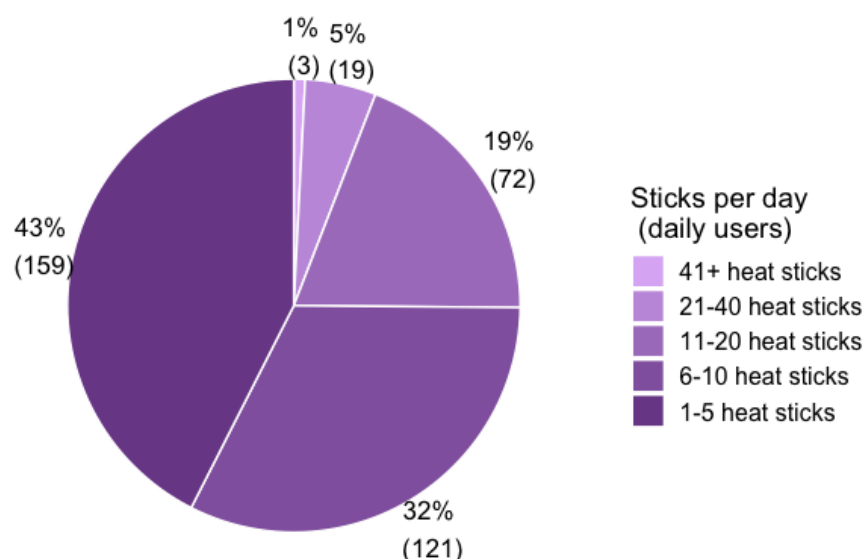
Base: daily current users of HTPs in the 18-25 age group (N=286)

Question: Q.1.4.a. On the days that you smoke/use, about how much do you smoke/use?

Source: OE elaboration from online survey data

Figure 104. Average number of HTP sticks a day, for daily users (26+ age group)

Percentage and number of current daily users in the 26+ age group



Base: daily current users of HTPs in the 26+ age group (N=374)

Question: Q.1.4.a. On the days that you smoke/use, about how much do you smoke/use?

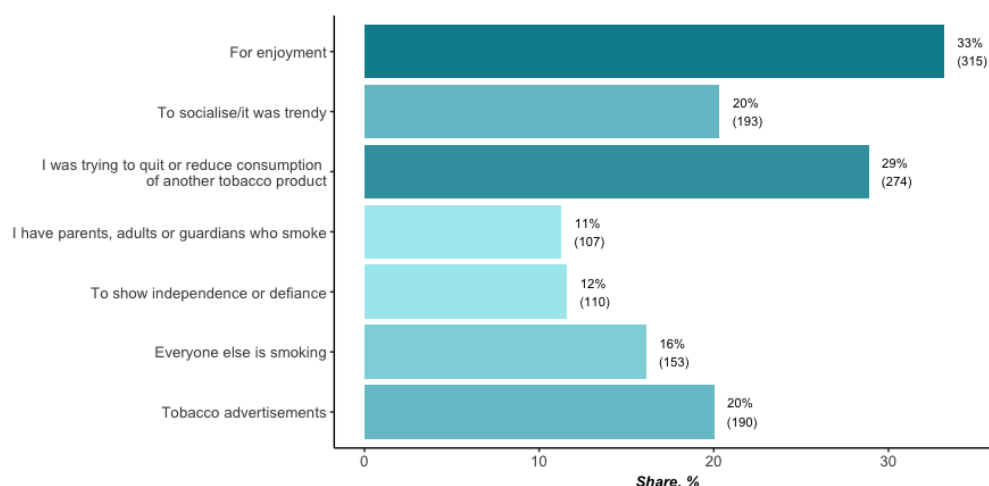
Source: OE elaboration from online survey data

HTPs have been on the market for a relatively short time, therefore no starting age trends and duration of use could yet be observed (see

Figure 194 and Figure 195 in the annex). Figure 105 and Figure 106 report the reasons why people started to use HTPs. In the **18-25 age group** the main reasons were 'enjoyment' at 33%, 'trying to quit or reduce consumption of another tobacco product' 29%, with 'to socialise/it was trendy' and 'tobacco advertisements' both 20%. Among the **26+ age group** 'trying to quit or reduce consumption of another tobacco product' at 41% was the dominant reason, followed by 'enjoyment' 34%, 'tobacco advertisements' 18% and 'to socialise/it was trendy' 17%. For both age groups the industry's claim that HTPs carry lower health risks has made an impact.

Figure 105. Initiation of HTP use: reasons (18-25 age group)

Percentage and number of current & past users in the 18-25 age group



Base: current and past HTP users in the 18-25 age group (N=949)

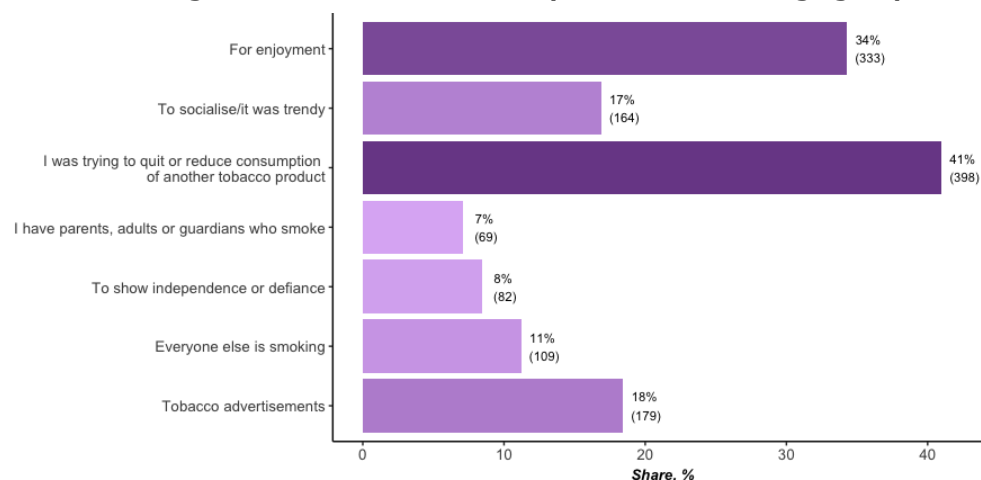
Question: Q.2.1.a. Why did you start using HTPs? Please select up to 2 options.

Note: Each bar represents the share of respondents that selected the option within the age group. The percentages do not add up to 100% because of the fact that respondents could select more than one term – therefore, the bars must be interpreted separately.

Source: OE elaboration from online survey data

Figure 106. Initiation of HTP use: reasons (26+ age group)

Percentage and number of current & past users in 26+ age group



Base: current and past HTP users in the 26+ age group (N=971)

Question: Q.2.1.a. Why did you start using HTPs? Please select up to 2 options.

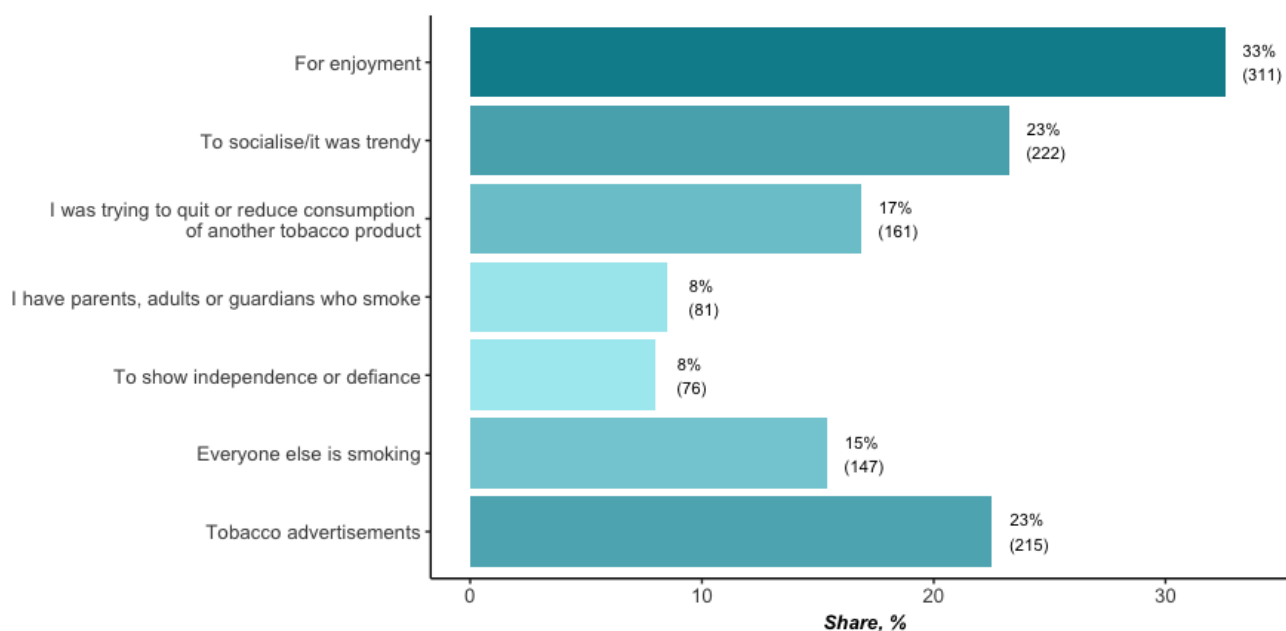
Note: Each bar represents the share of respondents that selected the option within the age group. The percentages do not add up to 100% because of the fact that respondents could select more than one term – therefore, the bars must be interpreted separately.

Source: OE elaboration from online survey data

For those who had used HTPs once or twice, the reasons for trying them are shown in Figure 107 and Figure 108. The **18-25 age group** reported experimenting with HTPs for the following reasons 'enjoyment' 33%, 'to socialise/it was trendy' 23%, and 'tobacco advertisements' 23%. In the **26+ age group**, 33% cited 'trying to quit or reduce consumption of another tobacco product', 'enjoyment' 26%, tobacco advertisements 25% and 'to socialise/it was trendy' 18%. A possible explanation for some of these reasons was provided by focus group participants. They reported that free trials of HTPs offered at points of sales were believed to be a key factor in making the products more user-friendly and inviting for new customers. Might the salespersons have promoted HTPs as an aid to healthier smoking or an aid to quitting other tobacco products?

Figure 107. Initiation of HTP use: reasons for trying (18-25 age group)

Percentage and number of those who tried once or twice in the 18-25 age group



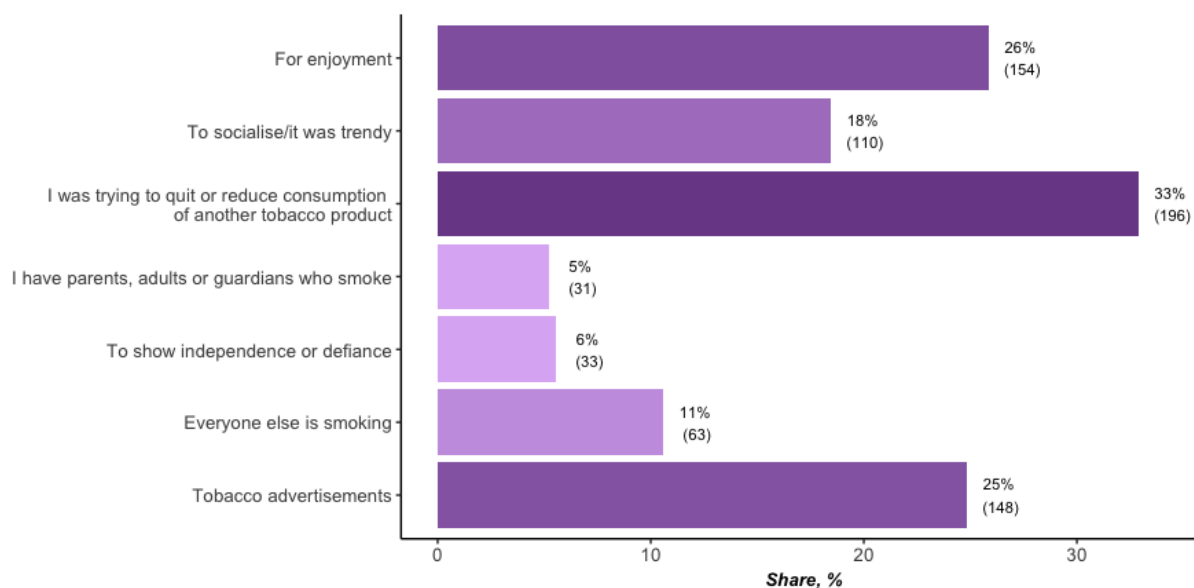
Base: those who tried to use HTPs in the 18-25 age group (N=955)

Question: Q.2.1.b. Why did you try HTPs? Please select up to 2 options.

Note: Each bar represents the share of respondents that selected the option within the age group. The percentages do not add up to 100% because of the fact that respondents could select more than one term – therefore, the bars must be interpreted separately.

Source: OE elaboration from study survey data

Figure 108. Initiation of HTP use: reasons for trying (26+ age group)

Percentage and number of those who tried once or twice in the 26+ age group

Base: those who tried to use HTPs once or twice in the 26+ age group (N=596)

Question: Q.2.1.b. Why did you try HTPs? Please select up to 2 options.

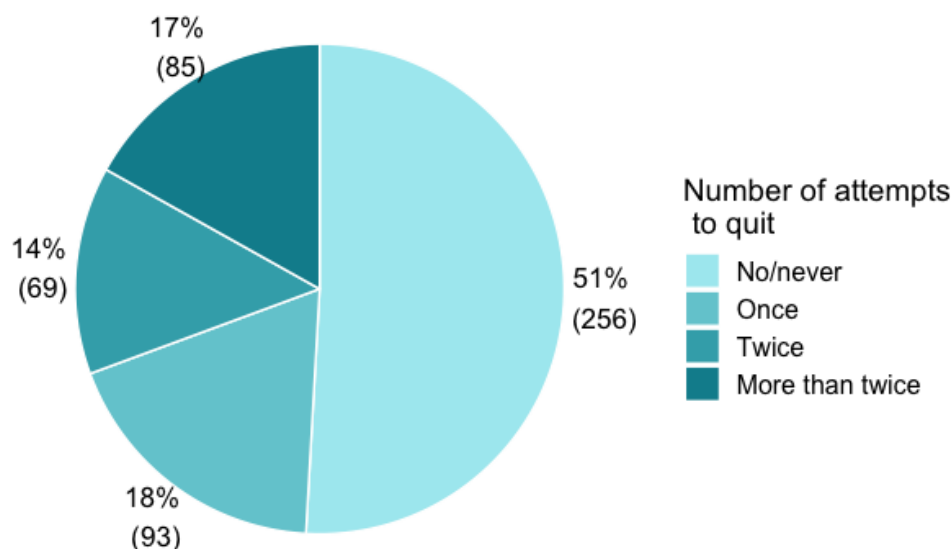
Note: Each bar represents the share of respondents that selected the option within the age group. The percentages do not add up to 100% because of the fact that respondents could select more than one term – therefore, the bars must be interpreted separately.

Source: OE elaboration from online survey data

Figure 109 and Figure 110 report on attempts to quit HTPs. The majority of current users reported never having attempted to quit HTP use; 62% in the 26+ age group and 51% in the 18-25 age group. Among the latter age group 31% had tried to quit more than once, compared to 21% in the older age group.

Figure 109. Cessation: attempts to quit HTPs (18-25 age group)

Percentage and number of current users in the 18-25 age group



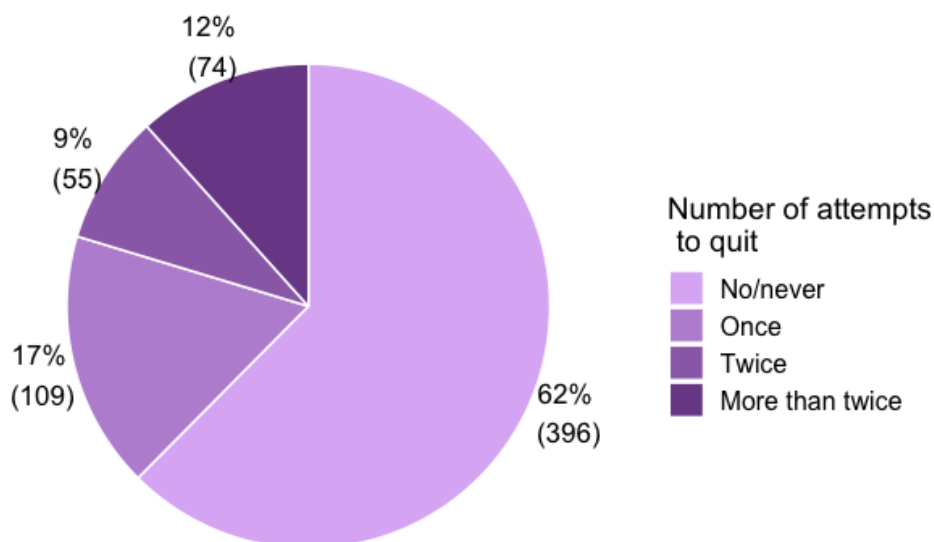
Base: current HTP users in the 18-25 age group (N=503)

Question: Q.1.5. Have you ever tried to quit HTPs? If yes, how many times?

Source: OE elaboration from online survey data

Figure 110. Cessation: attempts to quit HTPs (26+ age group)

Percentage and number)- of current users in the 26+ age group



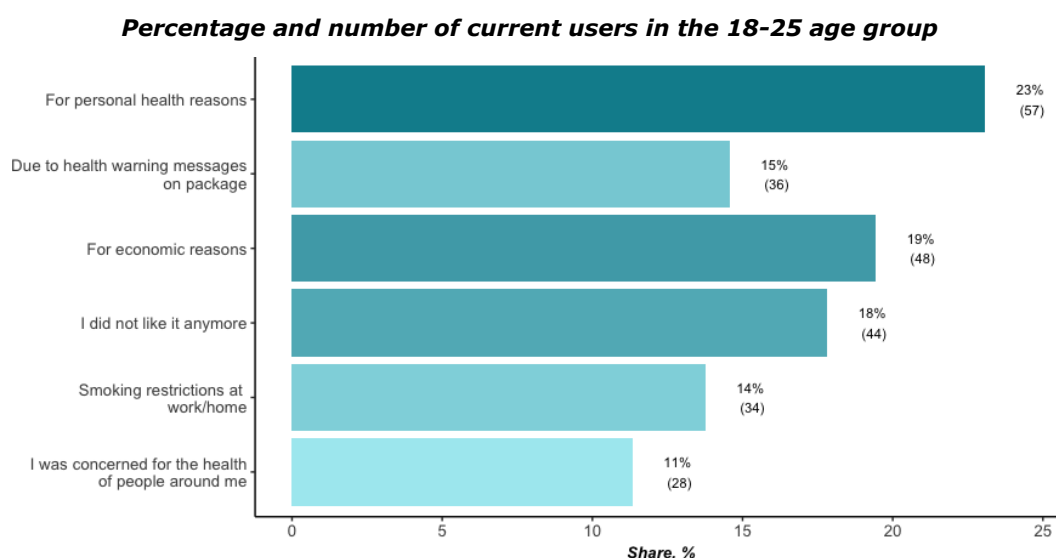
Base: current HTP users in the 26+ age group (N=634)

Question: Q.1.5. Have you ever tried to quit HTPs? If yes, how many times?

Source: OE elaboration from online survey data

The reasons for trying to quit are shown in Figure 111 and Figure 112. In the **18-25 age group**, respondents that had attempted to quit reported doing so mainly due to 'personal health reasons' (23%), for 'economic reasons' (19%); or due to 'not liking it anymore' (18%). Aggregating the reasons related to health – personal, health warnings and passive smoking – one in two current users were concerned about health issues. In the **26+ age group**, current users mainly stated that their attempts to quit were motivated by their concern for their 'personal health' (24%), the 'health of others' (21%) and 'economic reasons' (20%). While 'health warnings messages' were cited by only 10%, the aggregate of health concerns at 55% is marginally higher than the younger age group.

Figure 111. Cessation: reasons for trying to quit HTP use (18-25 age group)



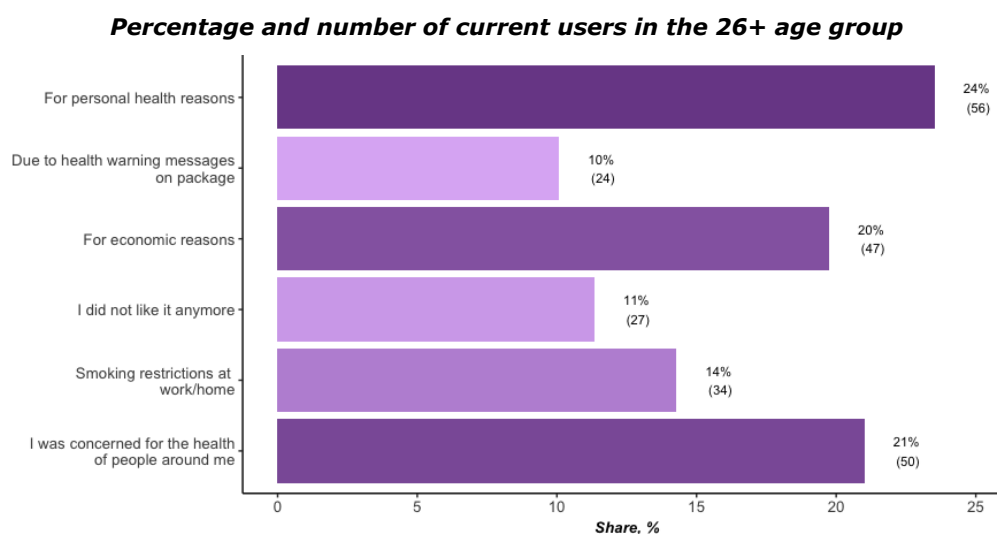
Base: current users of HTPs that tried to quit in the 18-25 age group (N=247)

Question: Q2.2. What made you try to quit HTPs? Please select the most fitting answer.

Note: Respondents were asked to provide only one option. Therefore, each bar represents the share of respondents that selected the option within the age group.

Source: OE elaboration from online survey data

Figure 112. Cessation: reasons for trying to quit HTP use (26+ age group)



Base: current users of HTPs that tried to quit in the 26+ age group (N=238)

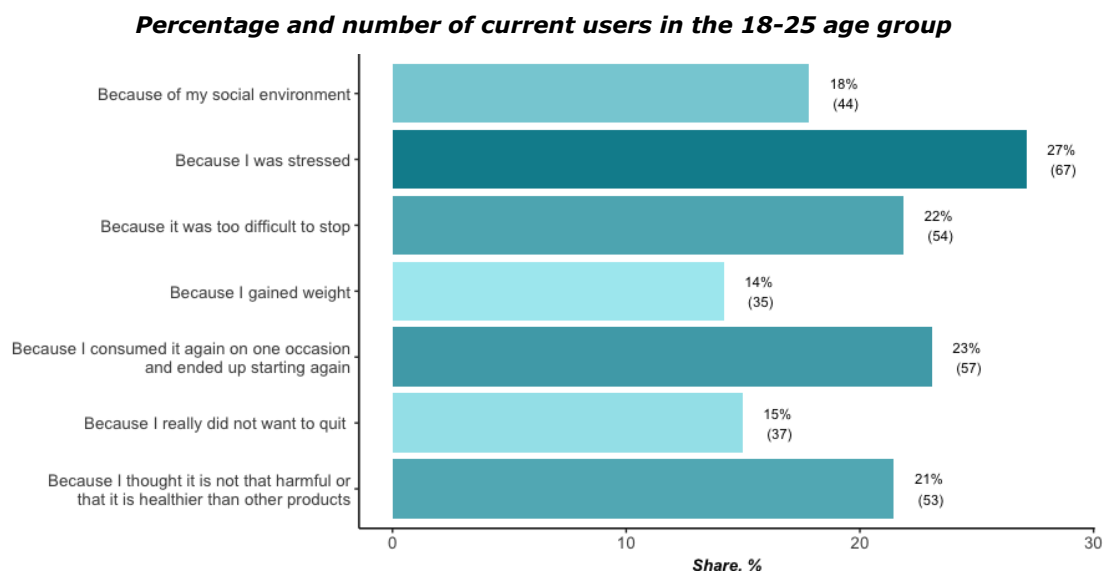
Question: What made you try to quit HTPs? Please select the most fitting answer.

Note: Respondents were asked to provide only one option. Therefore, each bar represents the share of respondents that selected the option within the age group.

Source: OE elaboration from online survey data

Figure 113 and Figure 114 report the reasons why those who tried to quit HTPs started using them again. In the 18-25 age group, 27% of current users who had attempted to quit had started consuming again on account of 'stress'. 23% gave in to temptation 'because I consumed it again on one occasion and ended up starting again' and 22% said it was 'too difficult to stop'. In the 26+ age group, 25% said it was 'too difficult to stop' and 25% cited 'stress'. In both age groups one in five respondents cited 'Because I thought it is not that harmful or that it is healthier than other products'.

Figure 113. Reasons for starting to use HTPs again (18-25 age group)



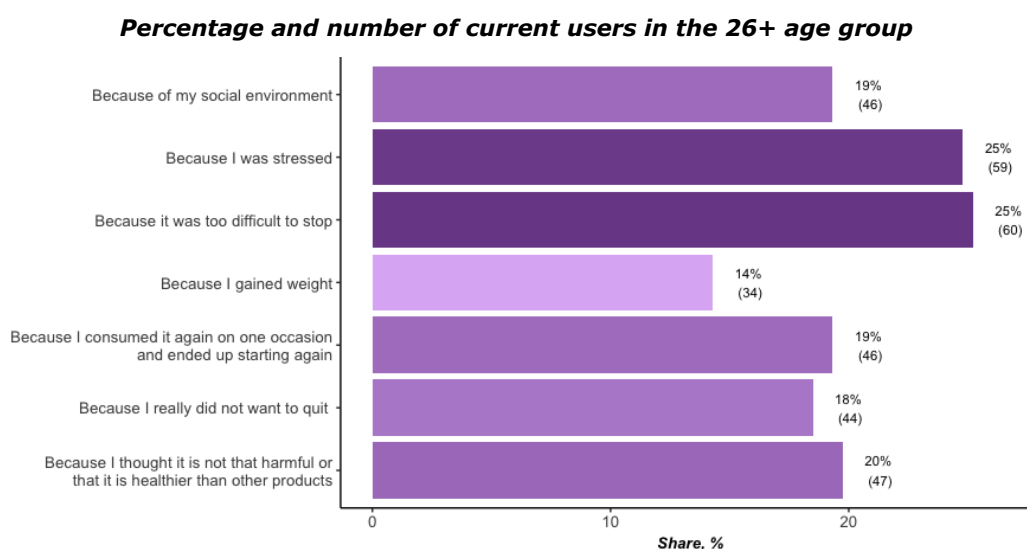
Base: current users of HTPs that tried to quit in the 18-25 age group (N=247)

Question: Q.2.3. Why did you start to use HTPs again? Please select up to 2 options.

Note: Each bar represents the share of respondents that selected the option within the age group. The percentages do not add up to 100% because of the fact that respondents could select more than one term – therefore, the bars must be interpreted separately.

Source: OE elaboration from online survey data

Figure 114. Reasons for starting to use HTPs again (26+ age group)



Base: current users of HTPs that tried to quit in the 26+ age group (N=238)

Question: Q.2.3. Why did you start to use HTPs again? Please select up to 2 options.

Note: Each bar represents the share of respondents that selected the option within the age group. The percentages do not add up to 100% because of the fact that respondents could select more than one term – therefore, the bars must be interpreted separately.

Source: OE elaboration from online survey data

5.2.3 Parallel use with other tobacco and related products

Only 1% in both age groups had only ever used HTPs (i.e. never users of other products). In the **18-25 age group** (see Table 40), 14% of current users had switched to HTP use from cigarettes (78%), slim cigarettes (64%) and e-cigarettes (31%). Dual use was reported by 84% of current HTP users.

A similar pattern was reported by the **26+ age group** (see Table 41). 12% had switched from other tobacco products (85% mentioned conventional cigarettes) and 86% were dual users who, on average were using two other products. The most common were cigarettes (85%) and e-cigarettes (66%).

While HTPs have been advertised as a tool to reduce or quit other tobacco products, it was clear from the survey that the actual use of the product did not reflect this purpose, with extensive parallel use of HTPs with other tobacco and related products across all consumer groups. Furthermore, it is seen from the visual representation that dual use is quite prevalent, as depicted by the thick “present” tail (see Figure 196 and Figure 197 in Annex).

Table 40. Career flows of use for current HTP users (18-25 age group)
Percentage and number of current users in the 18-25 age group

			Number	%
Total			503	100
Never user of other products			5	1
Exclusive current user of HTP & past user of other product(s)			73	14
Out of them, switching from (one or more products) *				
Cigarettes	(n=57)	78% *		
Slim cigarettes	(n=47)	64% *		
E-cigarettes	(n=23)	31% *		
Waterpipe tobacco	(n=22)	30% *		
Small cigarillos	(n=10)	14% *		
Current user of HTP & both current and past user of other product(s)			425	84
Other products consumed in parallel now (one or more products) *				
Cigarettes	(n=348)	82% *		
E-cigarettes	(n=277)	65% *		
Slim cigarettes	(n=228)	54% *		
Waterpipe tobacco	(n=219)	52% *		
Small cigarillos	(n=185)	44% *		

Base: current users of HTPs in the 18-25 age group (N=503);

Question: Q.1.1. Have you ever used HTPs?

Note: Patterns of use and consumption careers are explored for different consumer profiles of current users. A breakdown is provided for the two variables of interest, i.e. current parallel use patterns. The last row only provides the breakdown of current dual use as this is the variable of interest; it does not include a breakdown that takes account past consumption of products to avoid confusion (from too many permutations).

(*) Percentages add up to more than 100% on account of the prevalence of dual use.

Source: OE elaboration from online survey data

Table 41. Career flows of current HTP users (26+ age group)
Percentage and number of current users in the 26+ age group

			Number	%
Total			634	100
Never user of other products			7	1
Exclusive current user of HTP & past user of other product(s)			79	12
Out of them, switching from (one or more products) *				
Cigarettes	(n=72)	91% *		
Slim cigarettes	(n=25)	32% *		
E-cigarettes	(n=13)	16% *		
Small cigarillos	(n=12)	15% *		
Waterpipe tobacco	(n=7)	9% *		
Current user of HTP & both current and past user of other product(s)			548	86
Other products consumed in parallel now (one or more products) *				
Cigarettes	(n=466)	85% *		
E-cigarettes	(n=362)	66% *		
Slim cigarettes	(n=251)	46% *		
Small cigarillos	(n=125)	33% *		
Waterpipe tobacco	(n=182)	33% *		

Base: current users of HTPs in the 26+ age group (N=634)

Question: Q.1.1. Have you ever used HTPs?

Note: Patterns of use and consumption careers are explored for different consumer profiles of current users. A breakdown is provided for the two variables of interest, i.e. current parallel use patterns. The last row only provides the breakdown of current dual use as this is the variable of interest; it does not include a breakdown that takes account past consumption of products to avoid confusion (from too many permutations).

(*) Percentages add up to more than 100% on account of the prevalence of dual use.

Source: OE elaboration from online survey data

Table 42 and Table 43 report the career flow for past HTP users. In the **18-25 age group** (see Table 42), very few (1%) of the former users had only ever tried HTP and no other tobacco and related products: 23% were past dual users who reported that they had now quit using any tobacco products and 13% reported having used only HTP in the past and switched to other products. These findings point to the gateway potential of HTPs for other tobacco products. Most of these respondents had switched to conventional cigarettes (61%) with around 40% taking up slim cigarettes, e-cigarettes or waterpipe tobacco after quitting HTP use: 64% of past dual users who reported quitting HTP were now using conventional cigarettes (71%) and about 3 in 10 reported using e-cigarettes or slim cigarettes.

In the **26+ age group** (see Table 43), 18% of past users were no longer consuming any tobacco products. Of the 13% that had used HTP only in the past and had transitioned to other products, 89% had switched to conventional cigarettes and 59% to e-cigarettes. 66% of past HTP users reported continued dual use of conventional cigarettes (72%) and e-cigarettes (50%). It is important to note that while the above percentages are high, the absolute numbers are quite low.

Table 42. Career flow of past HTP users (18-25 age group)
Percentage and number of past users in the 18-25 age group

d			Number	%
Total			446	100
Never user of other products			3	1
Dual past user & not using any products now			101	23
Exclusive past user of HTPs & current user of other products (but not HTPs)			57	13
<i>Out of them, switching to (one or more products) *</i>				
Cigarettes	(n=35)	61% *		
Slim cigarettes	(n=24)	42% *		
E-cigarettes	(n=24)	42% *		
Waterpipe tobacco	(n=20)	35% *		
Small cigarillos	(n=11)	19% *		
Dual past user & current user of other products (but not HTPs)			285	64
<i>Other products currently consumed in parallel (one or more products) *</i>				
Cigarettes	(n=201)	71%		
E-cigarettes	(n=97)	34%		
Slim cigarettes	(n=80)	28%		
Waterpipe tobacco	(n=64)	22%		
Small cigarillos	(n=40)	14%		

Base: past users of HTPs in the 18-25 age group (N=446)

Question: Q.1.1. Have you ever used HTPs?

Note: Patterns of use and consumption careers are explored for different consumer profiles of past users. A breakdown is provided for the variables of interest, i.e. those who continued using other tobacco products after using e-cigarettes. The last row only provides the breakdown of current consumption because this is the variable of interest; it does not include a breakdown that takes account past consumption of products to avoid confusion (from too many permutations).

(*) Percentages add up to more than 100% on account of the prevalence of dual use.

Source: OE elaboration from online survey data

Table 43 Career flows of past HTP users (26+ age group)
Percentage and number of past users in the 26+ age group

	Number	%
Total	337	100
Never user of other products	3	1
Dual past user & not using any products now	57	17
Exclusive past user of HTPs & current user of other products (but not HTPs)	56	17
<i>Out of them, switching to (one or more products) *</i>		
Cigarettes (n=50)	89% *	
E-cigarettes (n=33)	59% *	
Slim cigarettes (n=18)	32% *	
Small cigarillos (n=12)	21% *	
Waterpipe tobacco (n=5)	9% *	
Dual past user & current user of other products (but not HTPs)	221	66
<i>Other products currently consumed in parallel (one or more products) *</i>		
Cigarettes (n=159)	72% *	
E-cigarettes (n=110)	50% *	
Slim cigarettes (n=59)	27% *	
Small cigarillos (n=42)	19% *	
Waterpipe tobacco (n=38)	17% *	

Base: past users of HTPs in the 26+ age group (N=337)

Question: Q.1.1. Have you ever used HTPs?

Note: Patterns of use and consumption careers are explored for different consumer profiles of past users. A breakdown is provided for the variables of interest, i.e. those who continued using other tobacco products after using e-cigarettes. The last row only provides the breakdown of current consumption because this is the variable of interest; it does not include a breakdown that takes account past consumption of products to avoid confusion (from too many permutations).

(*) Percentages add up to more than 100% on account of the prevalence of dual use.

Source: OE elaboration from online survey data

5.3 Consumer perception

In the focus groups discussions HTPs were the least known of the products featured in this study and there was some confusion between HTPs and the electronic cigarette, as both were considered non-conventional tobacco products, requiring an electronic device.

Table 44 and Table 45 focus on where the survey respondents first saw or heard about HTPs. In the 18-25 age group, 36% reported hearing about the product from friends and family, 14% from marketing in shops and tobacco stores, 12% from social media and 10% from product promotions in stores, stations or airports. In the 26+ age group friends and family were the primary source for 37%, followed by marketing in shops and tobacco stores (17%).

Table 44. Perception of HTPs (18-25 age group)
Percentage and number of respondents in the 18-25 age group

	Group total % (n)	Current users %	Past users %	Used once or twice %	Never used %
Use of product	100 (4167)	12	11	23	54
Facebook, Instagram or other social media networks	12 (487)	15	15	20	49
Other online advertisement	7 (274)	16	11	26	47
Traditional media (print, TV, radio, billboards)	7 (300)	14	9	23	54
Marketing in shops or tobacco stores	14 (598)	12	10	23	55
Observed in the street	13 (538)	11	10	21	58
Friends and family	36 (1492)	10	8	23	59
Product promotions in stores, stations or airports etc.	10 (398)	16	15	27	42
Other	2 (80)	12	11	21	55

Base: respondents in the 18-25 age group that had heard of HTPs prior to the survey (N=4,167)

Question: Q4.2. Where did you first see or hear about heated tobacco products (e.g. iQOS, glo)? [SINGLE ANSWER]

Note: The group total column refers to all respondents within the age group, i.e. it is a within column percentage. It provides the breakdown of the response options (e.g. 12% of respondents in this age group first saw or heard about HTPs through Facebook, Instagram or other social media networks). The white cells (in the four right columns) represent the breakdown by user type, i.e. they are within row percentages (white cells in each row sum to 100).

Source: OE elaboration from online survey data

Table 45 Perception of HTPs (26+ age group)
Percentage and number of respondents in the 26+ age group

	Group total % (n)	Current users %	Past users %	Used once or twice %	Never used %
Use of product	100	11	6	10	73
Facebook, Instagram or other social networks	6 (243)	26	10	12	51
Other online advertisement	8 (296)	19	9	16	56
Traditional media (print, TV, radio, billboards)	10 (366)	14	9	16	61
Marketing in shops or tobacco stores	17 (633)	19	10	16	55
Observed in the street	11 (399)	15	11	10	65
Friends and family	37 (1384)	15	8	17	61
Product promotions in stores, stations or airports etc.	10 (384)	19	10	21	51
Other	2 (74)	5	5	16	73

Base: respondents in the 26+ age group that had heard of HTPs prior to the survey (N=3,779)

Question: Q4.2. Where did you first see or hear about heated tobacco products (e.g. iQOS, glo)? [SINGLE ANSWER]

Note: The group total column refers to all respondents within the age group, i.e. it is a within column percentage. It provides the breakdown of the response options (e.g. 6% of respondents in this age group first saw or heard about HTPs through Facebook, Instagram or other social media networks). The white cells (in the four right columns) represent the breakdown by user type, i.e. they are within row percentages (white cells in each row sum to 100).

Source: OE elaboration from online survey data

From the focus group discussions eight words or word combinations that people used to describe tobacco and related products were selected; four were positive associations and four negative. The associations were complicated, unhealthy, costly, addictive, practical, modern/technological, elegant/trendy/fashionable and enjoyable.

In the online survey respondents were asked which of the words best captured what they thought about HTPs. The results are shown in Table 46 and Table 47. For the **18-25 age group** as a whole the most frequent associations with HTPs were 'unhealthy' 43%, 'modern and technological' 41%, costly 31%, and 'elegant, trendy, fashionable' 26%. The **26+ age group** selected a very similar list of associations 'unhealthy' 45%, 'modern and technological' 36%, costly 30%, and 'elegant, trendy, fashionable' 21%.

Current users, particularly those in the 26+ age group selected more positive associations. Unsurprisingly, they were overrepresented in the group of respondents that selected the term 'enjoyable' to describe HTPs.

Table 46. Perception of the HTPs (18-25 age group)
Percentage and number of respondents in the 18-25 age group

	Group total % (n)	Current users %	Past users %	Used once or twice %	Never used %
Use of product	100 (6090)	8	7	16	69
Complicated	11 (688)	8	6	15	72
Unhealthy	43 (2644)	3	4	12	80
Costly	31 (1892)	6	6	15	73
Addictive	33 (1995)	5	5	13	77
Practical	19 (1138)	14	11	19	56
Modern/technological	41 (2506)	8	7	16	69
Elegant/trendy/fashionable	26 (1591)	11	9	21	59
Enjoyable	11 (655)	24	12	20	44

Base: respondents in the 18-25 age group (N=6,090)

Question Q.4.4. Which of the following words best captures what you think about heated tobacco products (e.g. iQOS, glo)? Please tick as many boxes as you would like.

Note: Respondents had the option to choose as many terms as they wanted. The group total column refers to all respondents within the age group, i.e. it is a within column percentage, e.g. 11% of respondents perceived the product as complicated. The white cells (in the four right columns) represent the breakdown by user type, i.e. they are within row percentages (white cells in each row sum to 100).

Source: OE elaboration from online survey data

Table 47 Perception of HTPs (26+ age group)
Percentage and number of respondents in the 26+ age group

	Group total % (n)	Current users %	Past users %	Used once or twice %	Never used %
Use of product	100 (5909)	11	6	10	73
Complicated	13 (767)	5	5	10	81
Unhealthy	45 (2646)	3	3	7	87
Costly	30 (1799)	7	6	12	76
Addictive	33 (1962)	6	5	8	82
Practical	16 (941)	27	8	13	52
Modern/technological	36 (2155)	15	5	12	68
Elegant/trendy/fashionable	21 (1266)	18	8	15	59
Enjoyable	10 (585)	34	14	14	38

Base: respondents in the 26+ age group (N=5,909)

Question Q.4.4. Which of the following words best captures what you think about heated tobacco products (e.g. IQOS, glo)? Please tick as many boxes as you would like.

Note: Respondents had the option to choose as many terms as they wanted. The group total column refers to all respondents within the age group, i.e. it is a within column percentage, e.g. 13% of respondents perceived the product as complicated. The white cells (in the four right columns) represent the breakdown by user type, i.e. they are within row percentages (white cells in each row sum to 100).

Source: OE elaboration from online survey data

The focus group sessions took the position that although HTPs conveyed a stylish, exclusive, adult-like and futuristic image, but the hype around this product was believed to be temporary.

"The IQOS is the trendiest one and I think it's a product that is mainly consumed by people who want to be trendy and I associate the pink IQOS with the pink iPhone." (M, 28, BG)

"They are very trendy. People buy them for gifts." (F, 43, BG)

"It's rather a fashion. Or a substitute for people who work in offices and do not have breaks." (M, 24, BG)

"I bought the e-cigarette and the IQOS because I saw them in the tobacconist, and they looked very cute and nice." (F, 54, ES)

"Most of the people, 80 – 90% who smoke IQOS smoke cigarettes too and won't stay with the IQOS only." (F, 25, BG)

Participants in the focus groups did not expect HTPs to be particularly appealing to non-smokers but rather only to experienced adult smokers. HTPs allowed for the nostalgia of conventional tobacco products without some of the disadvantages associated with traditional cigarettes.

"Well, it is innovative. It is different from all the rest... It is different because you are still smoking a cigarette, not a vaper." (F, 54, ES)

"What convinced me was the fact that it is real tobacco. It's important because that way the flavour and the satisfying effect are greater." (M, 32, ES)

"This one (IQOS) contains the concept of cigarettes, in the sense that there is a device but what reaches your mouth is something of a reminder of cigarettes. It is the filter, etc." (M, 26, EL)

"It remains a cigarette and I tasted this cigarette and it's true that we have the same sensations that we do with a normal cigarette. Which is something I cannot find with the electronic cigarette! It's a way of smoking that is completely different." (F, 49, FR)

"Smokers might use it to avoid stinking up the room." (F, 27, EL)

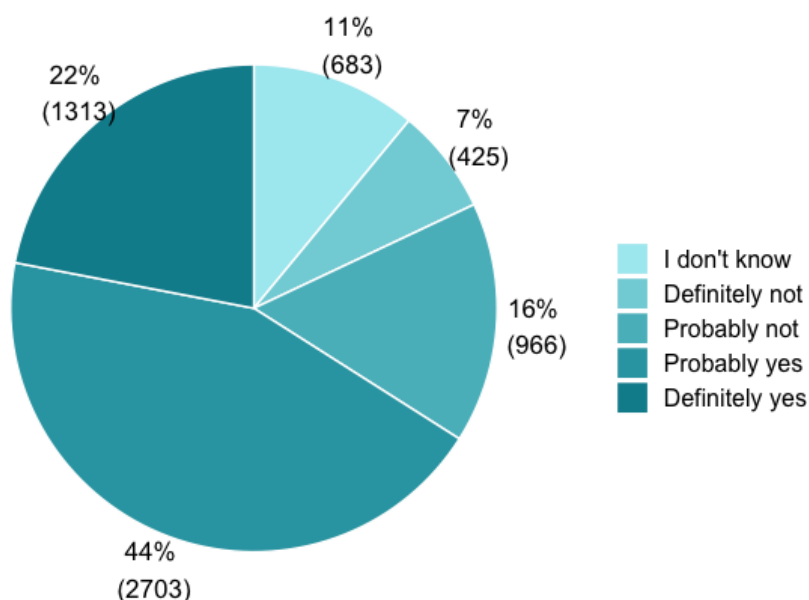
The focus groups commented on the marketing tactics of the industry which had been to promote the product as an experience a close to cigarette smoking and to convey an image of a sophisticated and innovative product for users. The heated tobacco stick contains "real" tobacco and conventional tobacco consumers considered HTPs as a more suitable substitute than other products (notably, e-cigarettes). However, overall participants did not believe that it was as satisfying as conventional tobacco products. Rather it was experienced as "colder" emotionally speaking. It was repeatedly reported as bothersome and/or frustrating because the battery required charging, so that users might not have the product topped up when the craving strikes. These features were off-putting for some participants in the focus groups.

"I don't find it elegant at all. Personally no, frankly it already seems impractical to prepare and I can't see what occasion this could be used in, I don't know." (F, 66, FR)

Figure 115 and Figure 116 report the survey respondents' opinions on the 'gateway' potential of HTPs. In the 18-25 age group, 66% agreed that HTPs might definitely or probably encourage people to start smoking other tobacco products, compared to 23% who thought it would probably or definitely not. 60% of the 26+ age group held a similar view on the gateway effect with 23% disagreeing.

Figure 115. Perceived gateway effect of HTPs (18-25 age group)

Percentage and number of the 18-25 age group



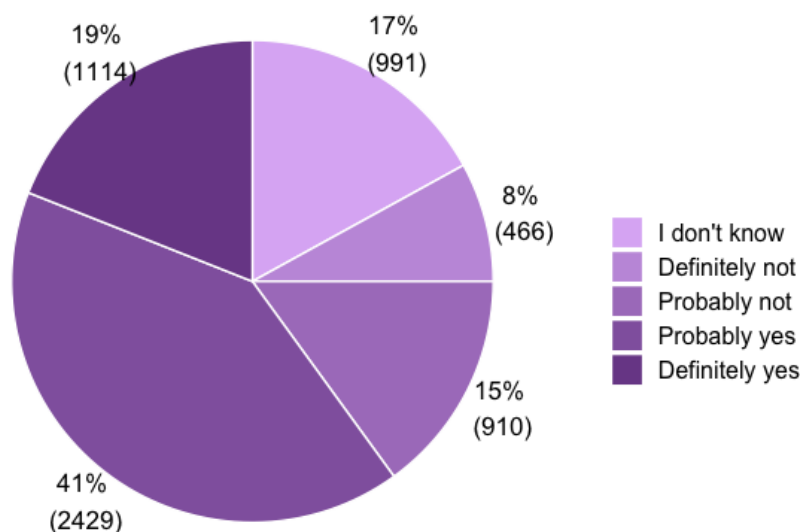
Base: respondents in the 18-25 age group (N=6,090)

Question: Q4.3. Do you think using heated tobacco products (e.g. iQOS, glo) might encourage people to start smoking/using other tobacco products, or not? [SINGLE ANSWER]

Source: OE elaboration from online survey data

Figure 116 Perceived gateway effect of HTPs (26+ age group)

Percentage and number of the 26+ age group



Base: respondents in the 26+ age group (N=5,910)

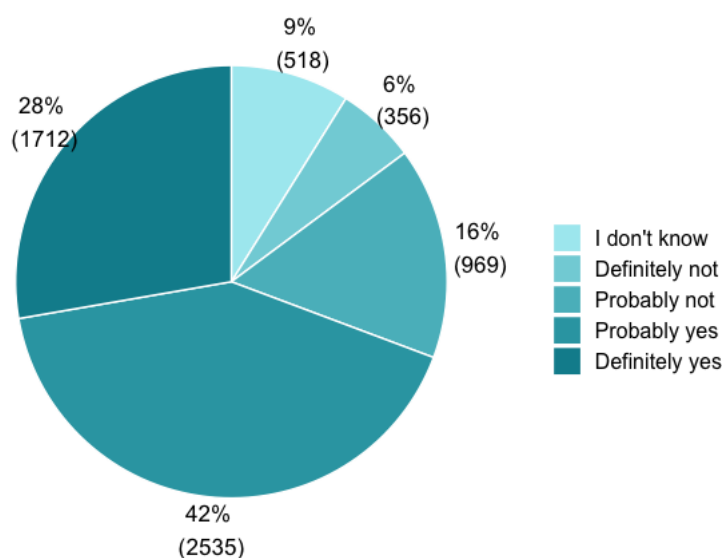
Question: Question: Q4.3. Do you think using heated tobacco products (e.g. iQOS, glo) might encourage people to start smoking/using other tobacco products, or not? [SINGLE ANSWER]

Source: OE elaboration from online survey data

Figure 117 and Figure 118 report the survey respondents on whether HTPs would be attractive to young people. In the 18-25 age groups 70% said probably or definitely yes. 22% took the opposite view. Much the same opinion was held by the 26+ age group. 66% said probably or definitely yes, HTP would be attractive to young people; 20% had the opposite view.

Figure 117 Perceived appeal of HTPs to youth (18-25 age group)

Percentage and number of 18-25 age group



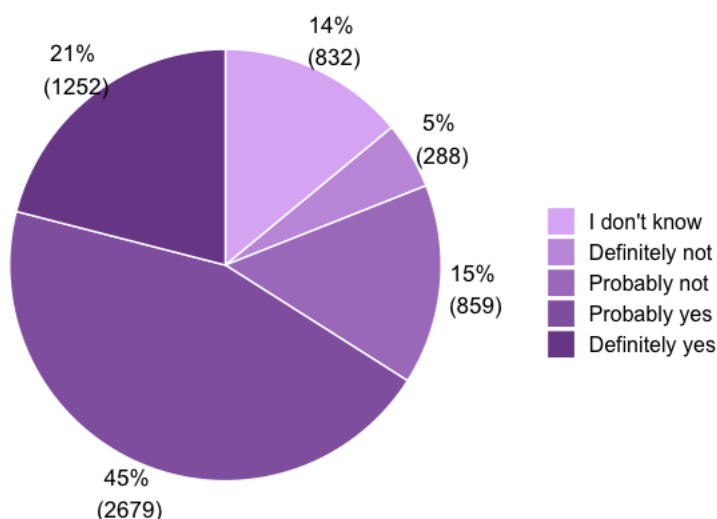
Base: respondents in the 18-25 age group (N=6,090)

Question: Q.4.5. Do you think that young people might be particularly attracted to heated tobacco products (e.g. iQOS, glo), or not? [SINGLE ANSWER]

Source: OE elaboration from online survey data

Figure 118 Perceived appeal of HTPs to youth (26+ age group)

Percentage and number of 26+ age group



Base: respondents in the 26+ age group (N=5,910)

Question: Q.4.5. Do you think that young people might be particularly attracted to heated tobacco products (e.g. iQOS, glo), or not? [SINGLE ANSWER]

Source: OE elaboration from online survey data

A rather different picture from the focus groups emerged. The focus groups did not see this as a concern as HTPs were considered too expensive (except in Germany) as there was a charge for both the device and the heated tobacco sticks. (e.g. IQOS device and HeetSticks). On this basis the participants did not expect the product to become popular, especially among the younger people. The participants believed that the product would fail to interest this age group due to its price and perceived exclusive character.⁸⁸ Rather, participants associated the product with adults and/or professionals between the ages of 25 and 50.

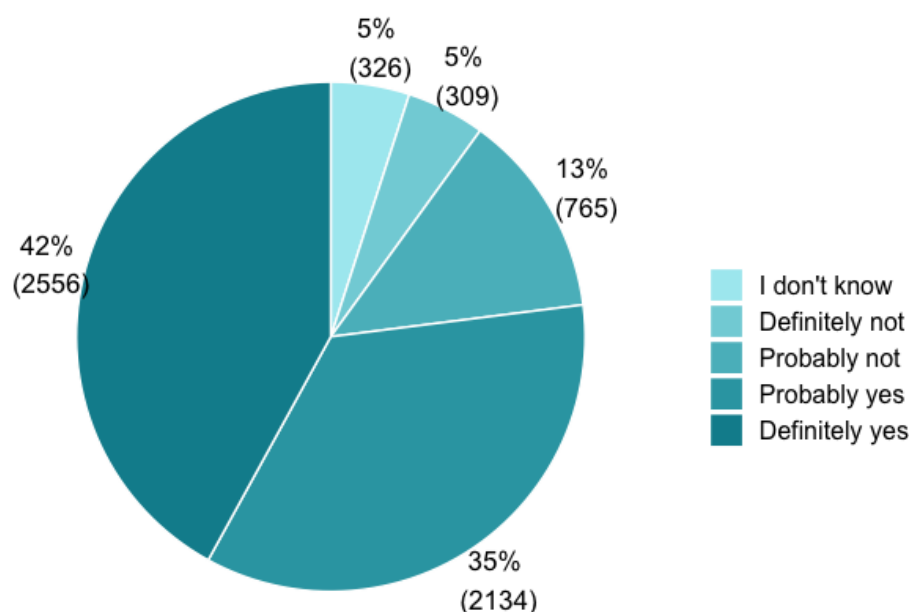
"I think price could play a role on the last one (IQOS). (...) the heat not burn tobacco is quite expensive..." (M, 28, FR)

"If after a year or two, I tell myself that I'm sick of smoking and want to stop then I will get an e-cigarette again. It will certainly not be that heat not burn ones just because of the price." (M, 20, FR)

Figure 119 and Figure 120 look at another concern about young people, that they are inclined to misperceive the risk of smoking. The survey respondents were asked "Do you think that young people might underestimate the risk of consuming heated tobacco products". The figures depict a widespread consensus across both age groups. In the 18-25 age group, 77% agree that young people probably or definitely underestimate the risk of consuming HTPs and 18% said that young people probably or definitely do not underestimate the risk (see Figure 119). The respective percentages in the 26+ age group at 76% and 16% (see Figure 120).

Figure 119 Extent to which respondents believed that youth underestimated risks of using HTPs

Percentage and number of respondents in the 18-25 age group



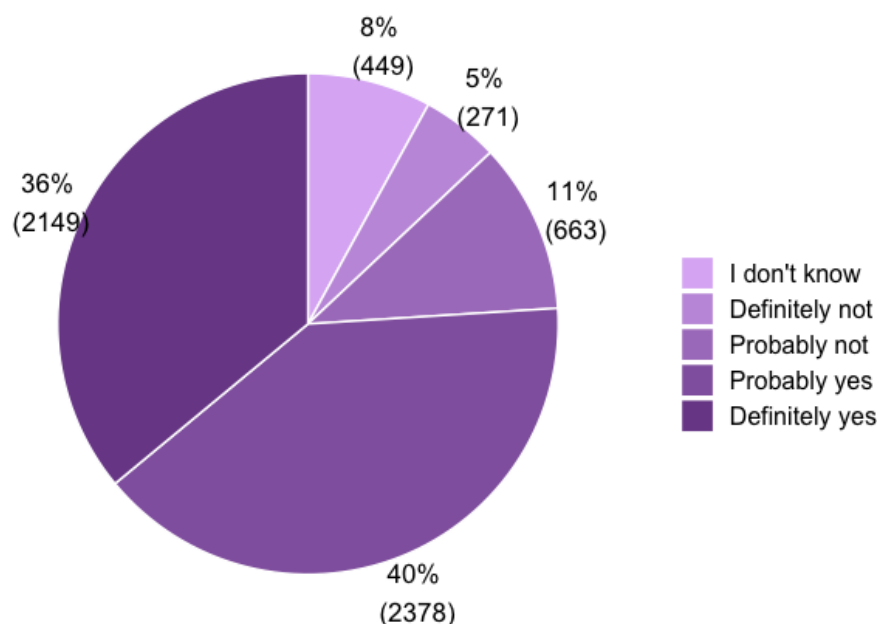
Base: respondents in the 18-25 age group (N=6,090)

Question: Q.4.6. Do you think that young people might underestimate the risk of consuming heated tobacco products (e.g. IQOS, glo), or not? [SINGLE ANSWER]

Source: OE elaboration from online survey data

⁸⁸ Note that this is only the perception of focus group participants and that this study's fieldwork did not cover those under the age of 18.

Figure 120 Extent to which respondents believed that youth underestimate risks of HTPs
Percentage and number of 26+ age group)



Base: respondents in the 26+ age group (N=5,910)

Question: Q.4.6. Do you think that young people might underestimate the risk of consuming heated tobacco products (e.g. IQOS, glo), or not? [SINGLE ANSWER]

Source: OE elaboration from online survey data

On the issue of the health risks of HTPs the focus group participants commented on the unhealthy characteristics of the product but also acknowledged that long-term effects were still unknown. Some believed that HTPs may be as harmful as conventional tobacco products because they also contain tobacco and paper (as well as the plastic filter); others thought that HTPs might lead to different types of harm due to new harmful substances. Participants were also concerned about the environment because HTPs contain tobacco, papers and plastic filters which may also pollute the environment like conventional cigarettes.

"Maybe the fact that we do not burn the tobacco makes it less harmful." (F, 49, FR)

"IQOS because it is heated does not bring out the tar, I know this." (M, 58, EL)

"IQOS. In my view, this is the most harmful product among all we are reviewing... I have friends who have made a research. And I have unwrapped such a cigarette and have studied it before and after smoking. The filter inside is made of nylon and melts during smoking. I think the most harmful effects come from this." (M, 24, BG)

"For me IQOS is the least harmful because you do not inhale smoke and solid particles. And when I smoke more IQOS I feel better than when I smoke conventional cigarettes." (F, 25, BG)

"I have heard that it (IQOS) has become legal and that it is healthy and has some certification in contrast to the electronic cigarette." (F, 27, EL)

"It's not natural (e-cigarette) so we try to go backwards with the heated tobacco because the tobacco is natural" (F, 49, FR)

Another view was that heated tobacco sticks may be less harmful than conventional tobacco products because they are not burned but rather heated with no combustion and no smoke. The use of an electronic device was perceived as more up to date and that people might assume it was healthier and research and development of the industry should deliver better products over time – not worse. In general, participants were unsure about the working of HTPs (e.g. whether it is vapour or smoke that is inhaled), but most believed that the product was not completely harmless, and they expected it to be addictive given the presence of nicotine.

"We already know that what is bad about the cigarette is the combustion! Well, at least that is what is worst about it." (M, 28, FR)

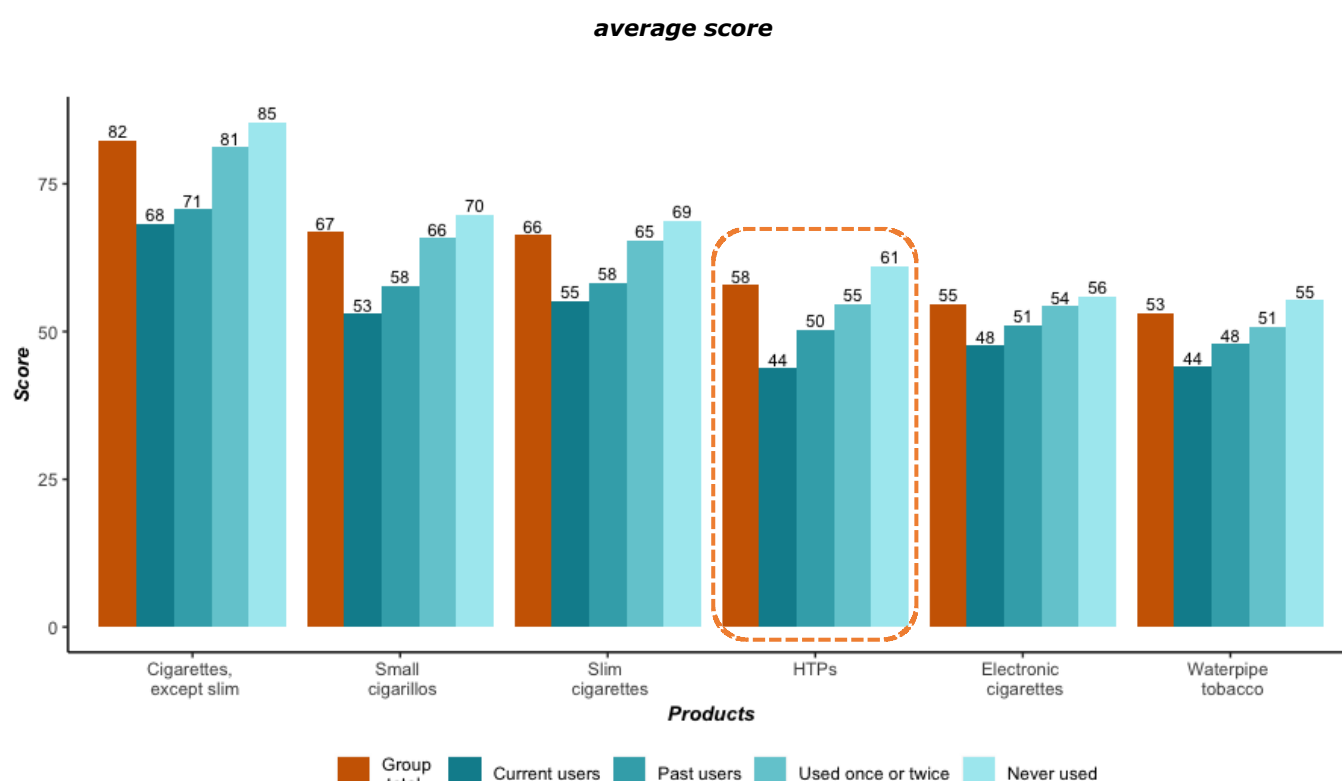
"It resembles an e-cigarette. But, from what was said, the e-cigarette was great because it took away all the harmful things of tobacco. And there (the HTP), not at all. We still have tobacco and we have no hindsight at all. It's just a new gimmick." (F, 48, FR)

"IQOS is a technology with no direct burning, with lower temperature, producing less harmful substances." (M, 28, BG)

In the online survey the swing methodology was employed to assess the relative harmfulness to users of HTP compared to the other products considered in this study plus conventional cigarettes. The swing method is explained in section 1.3.3. In brief, the respondent is asked which of the six products they think is most damaging to a user's health. This product is given 100 points on a scale from 100 (most harmful) to 0 (not harmful). Then using this scale, the respondent is asked to rate each of the other five products on its harmfulness to health of users.

Figure 121 and Figure 122 show the perceived harmfulness to the health of HTP users. In the 18-25 age group conventional cigarettes were considered most harmful with a rating of 82. HTPs were rated at 58. Current users of HTPs rated them at 44 and never users at 61. While respondents in both age groups thought that young people might underestimate the risks of HTPs, when asked directly about the harm to users' health the 18-25 age group (see Figure 121) perceived HTPs as more harmful than the 26+ age group (see Figure 122). A heightened view of the health risk was also evident among users of HTPs in the 18-25 age group.

Figure 121. Perceived harmfulness of HTPs to users (18-25 age group)



Base: respondents in the 18-25 age group (N=6,090)

Question: Q.3.1.a.⁸⁹ & Q.3.1.b. You have chosen (e.g.; cigarettes, except slims⁹⁰) as likely to be the most damaging to the health of users. We will give (e.g.; cigarettes, except slims) 100 points on a scale of 0-100 for damaging users' health. Please assess the other products' damage on a scale of 0 to 100. If you think a product is about half as damaging to the health of users, you would give that product 50 points. If you think another product is just as damaging as (e.g.; cigarettes, except slims), you would give it 100 points.

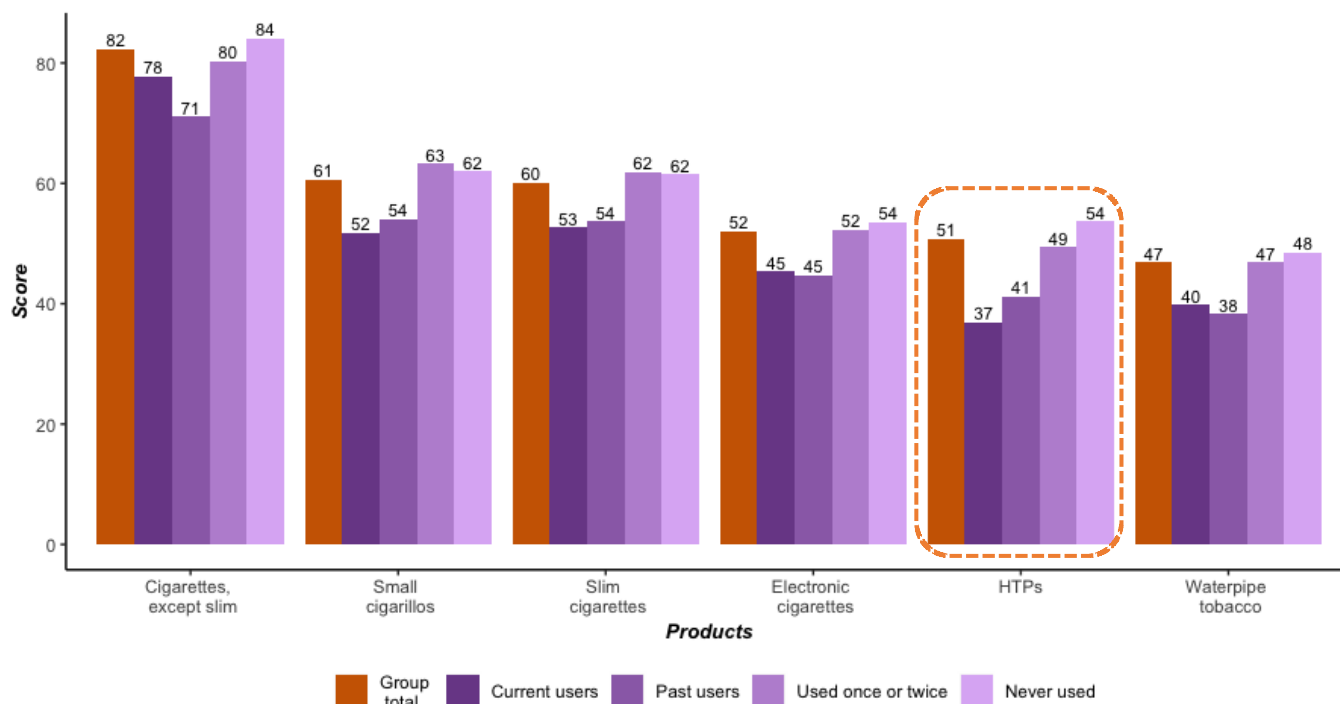
Note: The graph is based on the swing method. The scores were constructed using the responses to the two abovementioned questions. The bars in the graph are the average score of harm to the user provided by respondents for products under the scope of this study. For this chapter, the breakdown is provided for HTP user profiles, and therefore differs from the other chapter. The products are sorted in descending order of the average score provided by the "group total," i.e. amongst all respondents of that age group. The dotted orange box identifies the product in focus.

Source: OE elaboration from online survey data

⁸⁹ Q.3.1.a. Which of the six tobacco products listed below do you think is likely to be most damaging to the health of users?

⁹⁰ The product in parentheses would change in the online survey based on individual responses to Q.3.1.a.

Figure 122. Perceived harmfulness of HTPs to users (26+ age group)

average score

Base: respondents in the 26+ age group (N=5,910)

Question: Q.3.1.a.⁹¹ & Q.3.1.b. You have chosen (e.g.; cigarettes, except slims⁹²) as likely to be the most damaging to the health of users. We will give (e.g.; cigarettes, except slims) 100 points on a scale of 0-100 for damaging users' health. Please assess the other products' damage on a scale of 0 to 100. If you think a product is about half as damaging to the health of users, you would give that product 50 points. If you think another product is just as damaging as (e.g.; cigarettes, except slims), you would give it 100 points.

Note: The graph is based on the swing method. The scores were constructed using the responses to the two abovementioned questions. The bars in the graph are the average score of harm to the user provided by respondents for products under the scope of this study. For this chapter, the breakdown is provided for HTP user profiles, and therefore differs from the other chapter. The products are sorted in descending order of the average score provided by the "group total," i.e. amongst all respondents of that age group. The dotted orange box identifies the product in focus.

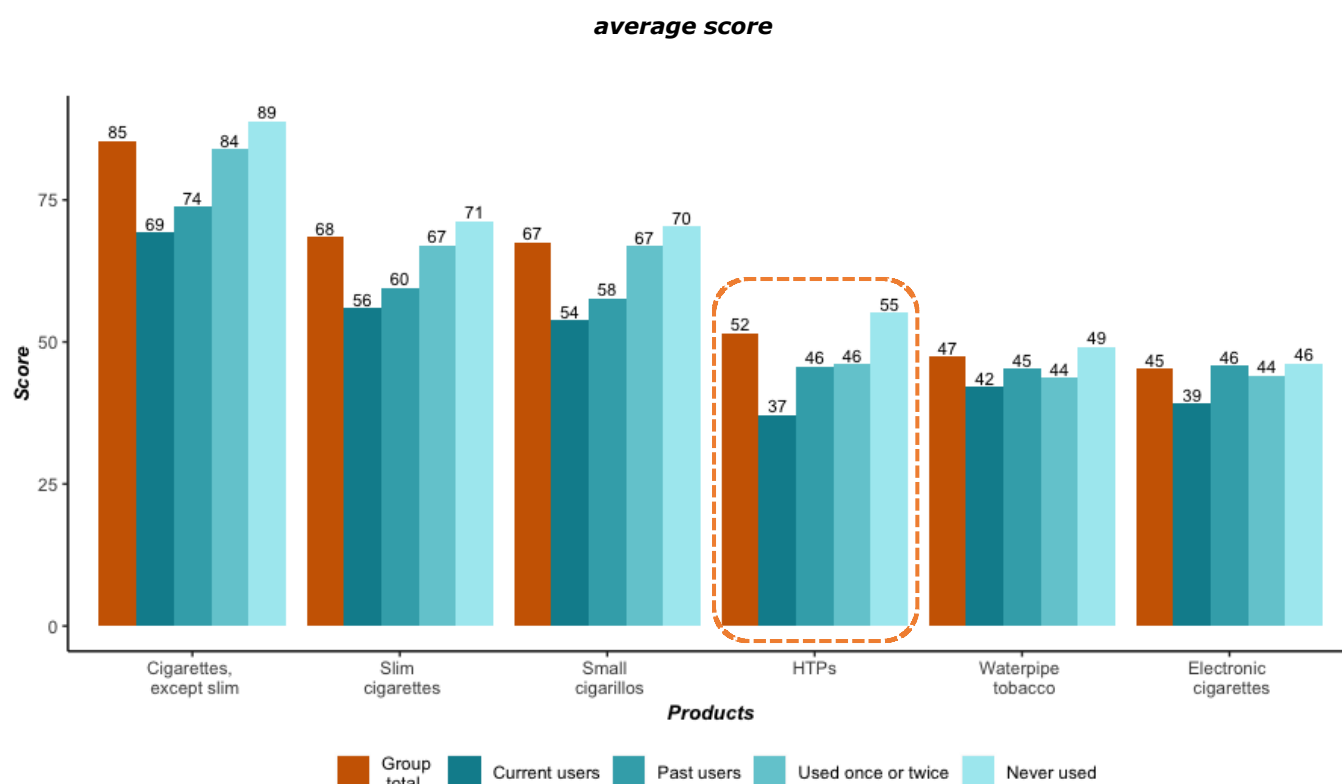
Source: OE elaboration from online survey data

⁹¹ Q.3.1.a. Which of the six tobacco products listed below do you think is likely to be most damaging to the health of users?

⁹² The product in parentheses would change in the online survey based on individual responses to Q.3.1.a.

Figure 123 and Figure 124 report on the perceived harm of HTPs to bystanders. In the 18-25 age group, the passive smoking risk from HTPs was rated lower than conventional and slim cigarettes, and small cigarillos and above waterpipe tobacco and e-cigarettes. The 26+ age group showed a similar ordering of the perceived harm of passive smoking. As with the risk to the health of users, the 18-25 age group rated the passive smoking risk higher (52) than the 26+ age group (45).

Figure 123. Perceived harmfulness of HTPs to bystanders (18-25 age group)



Base: respondents in the 18-25 age group (N=6,090)

Question: Q.3.2.a.⁹³ & Q.3.2.b. Q.3.2.b. You have chosen (e.g.; cigarettes, except slims⁹⁴) as likely to be the most damaging to the health of passive users. We will give (e.g.; cigarettes, except slims) 100 points on a scale of 0-100 for damaging passive users' health. Please assess the other products' damage on the health of passive users on a scale of 0 to 100. If you think a product is about half as damaging to the health of passive users, you would give it 50 points. But if you think another product is just as damaging as (e.g.; cigarettes, except slims), you would give it 100 points.

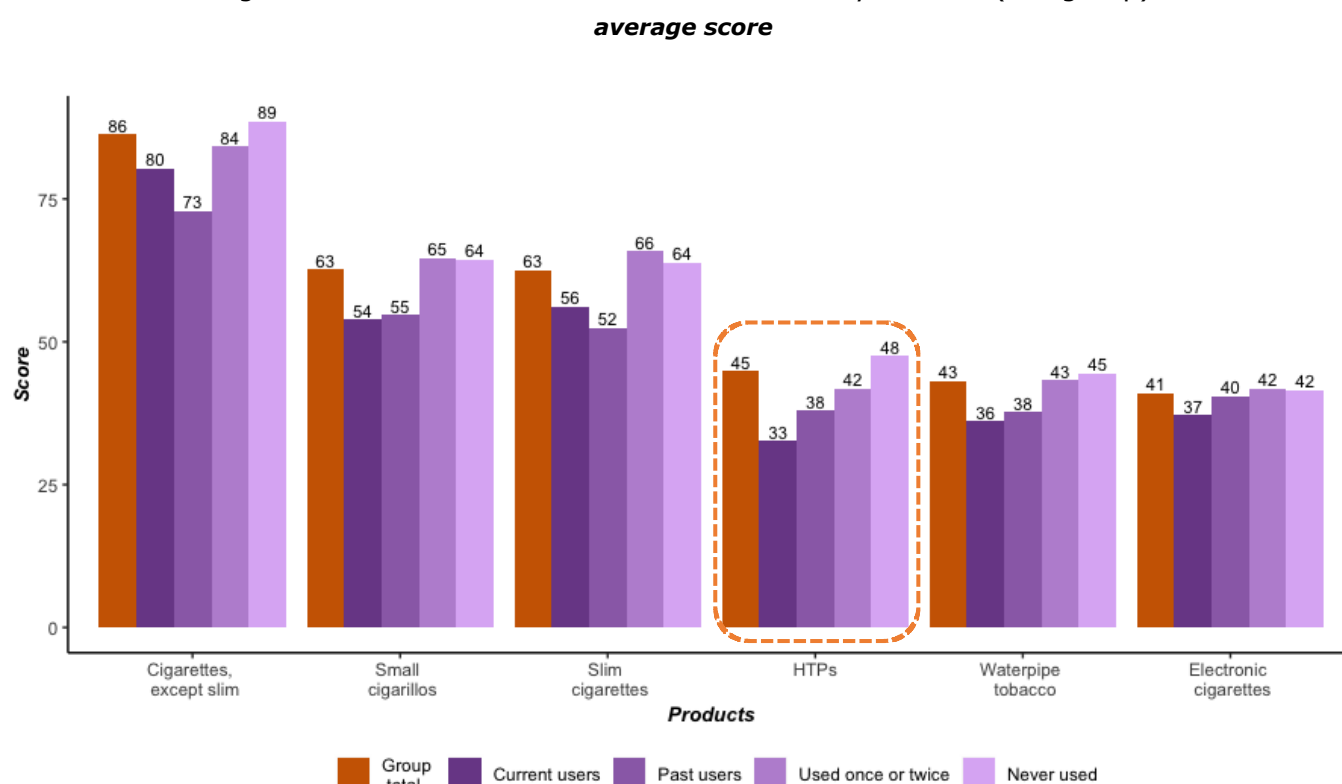
Note: The graph is based on the swing method. The scores were constructed using the responses to the two abovementioned questions. The bars in the graph are the average score of harm to the passive user provided by respondents for products under the scope of this study. For this chapter, the breakdown is provided for HTP user profiles, and therefore differs from the other chapter. The products are sorted in descending order of the average score provided by the "group total," i.e. amongst all respondents of that age group. The dotted orange box identifies the product in focus.

Source: OE elaboration from online survey data

⁹³ Q.3.2.a. Which of the six tobacco products listed below is likely to be the most damaging to the health of other people from passive use?

⁹⁴ The product in parentheses would change in the online survey based on individual responses to Q.3.1.a.

Figure 124. Perceived harmfulness of HTPs to bystanders (26+group)



Base: respondents in the 26+ age group (N=5,910)

Question: Q.3.2.a.⁹⁵ & Q.3.2.b. Q.3.2.b. You have chosen (e.g.; cigarettes, except slims⁹⁶) as likely to be the most damaging to the health of passive users. We will give (e.g.; cigarettes, except slims) 100 points on a scale of 0-100 for damaging passive users' health. Please assess the other products' damage on the health of passive users on a scale of 0 to 100. If you think a product is about half as damaging to the health of passive users, you would give it 50 points. But if you think another product is just as damaging as (e.g.; cigarettes, except slims), you would give it 100 points.

Note: The graph is based on the swing method. The scores were constructed using the responses to the two abovementioned questions. The bars in the graph are the average score of harm to the passive user provided by respondents for products under the scope of this study. For this chapter, the breakdown is provided for HTP user profiles, and therefore differs from the other chapter. The products are sorted in descending order of the average score provided by the "group total," i.e. amongst all respondents of that age group. The dotted orange box identifies the product in focus.

Source: OE elaboration from online survey data

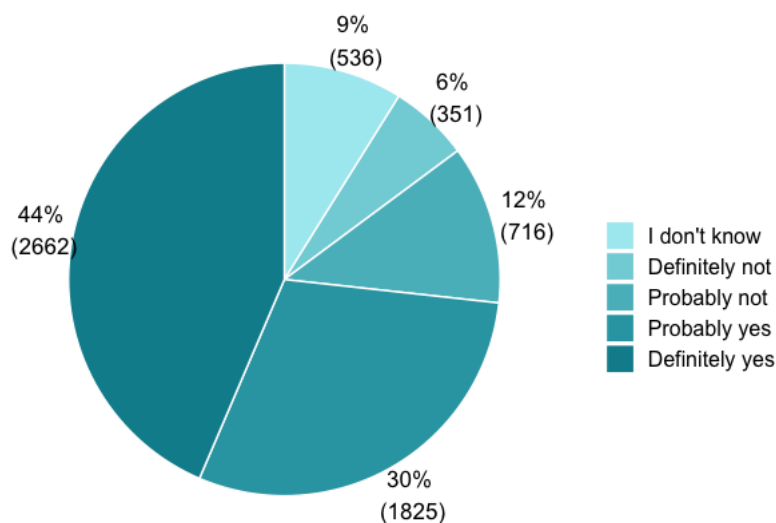
On the issue of regulation, the focus groups were clear that HTP should be regulated as other conventional tobacco products. They noted that HTPs contained tobacco and as such should be taxed accordingly. The online survey results confirmed these qualitative insights. Figure 125 and Figure 126 report the survey respondents' views on whether HTPs should be regulated like cigarettes or not: 74% of the 18-25 age groups responded probably (44%) or definitely (30%) yes. 79% of the 26+ age group said yes, definitely (48%) and probably (31%). Overall, a public endorsement for the regulation of HTPs.

⁹⁵ Q.3.2.a. Which of the six tobacco products listed below is likely to be the most damaging to the health of other people from passive use?

⁹⁶ The product in parentheses would change in the online survey based on individual responses to Q.3.1.a.

Figure 125. Extent of agreement with regulating HTPs like cigarettes (18-25 age group)

Percentage and number of 18-25 age group



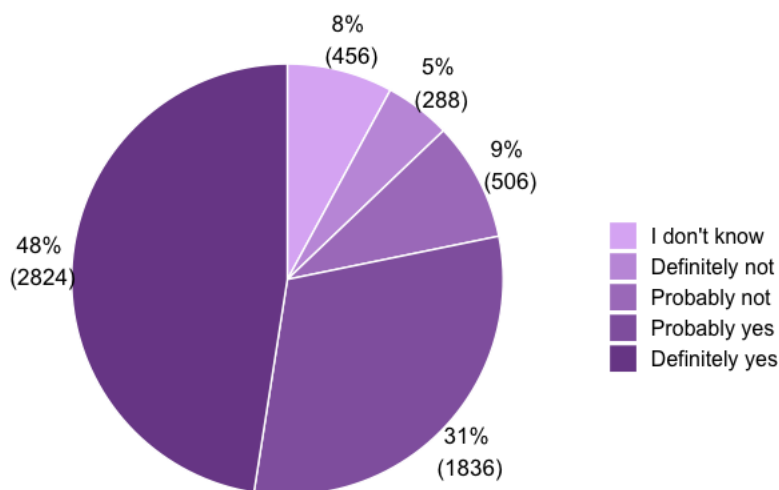
Base: respondents in the 18-25 age group (N=6,090)

Question: Q.4.7. Do you think that heated tobacco products should be regulated like cigarettes, or not?
[SINGLE ANSWER]

Source: OE elaboration from online survey data

Figure 126. Extent of agreement with regulating HTPs like cigarettes (26+ age group)

Percentage and number of 26+ age group



Base: respondents in the 26+ age group (N=5,910)

Question: Q.4.7. Do you think that heated tobacco products should be regulated like cigarettes, or not?
[SINGLE ANSWER]

Source: OE elaboration from online survey data

Table 48 and Table 49 report on the locations where the use of HTPs should be allowed or restricted. A wide range of views from the very highly restrictive to the libertarian were evidenced. But the weight of opinion for both age groups was that HTP should not be allowed at work, on public transport and in restaurants and bars.

Table 48. Views on settings where HTPs should be prohibited (18-25 age group)

Percentage and number of respondents in the 18-25 age group

	Group total % (n)	Current users %	Past users %	Used once or twice %	Never used %
Use of product	100 (6090)	8	7	16	69
Open outdoor spaces (e.g. bus stops, playgrounds)	23 (1429)	8	6	14	72
Work	51 (3119)	5	6	14	74
Public transport	63 (3808)	6	6	15	73
Restaurants/ bars indoors	56 (3308)	5	6	15	74
Restaurants/ bars outdoor terraces	18 (1078)	7	7	13	73
Private transportation	32 (1935)	5	5	14	76
All of the above	10 (620)	3	4	10	83
None	7 (400)	13	6	17	64
Other	3 (172)	8	7	12	73

Base: respondents in the 18-25 age group (N=6,090)

Question: Q.4.14. In which settings, if any, do you think that the consumption of HTPs should be prohibited?

Note: Respondents could provide multiple answers, except if they selected "All the above" or "None". Hence, a special consideration must be made when analysing the responses. Although 23% reported "open outdoor spaces", 33% of respondents in the age group were in favour of prohibiting product use in open outdoor spaces (i.e. 23% + 10% that answered "all of the above"). The group total category refers to the total respondents in that age group. The orange column represents the proportion of respondents in the age group that selected the response option (i.e. 3% of respondents in the 18-25 age group responded "other"). The white cells (in the four right columns) represent the breakdown by user type, i.e. they are within row percentages (white cells in each row sum to 100).

Source: OE elaboration from online survey data

Table 49. Views on settings where HTPs should be prohibited (26+ age group)
Percentage and number of respondents in the 26+ age group)

Reasons	Group total % (n)	Current users %	Past users %	Used once or twice %	Never used %
Use of product	100 (5910)	11	6	10	73
Open outdoor spaces (e.g. bus stops, playgrounds)	22 (1296)	10	5	8	76
Work	53 (3133)	10	5	10	76
Public transport	62 (3661)	10	5	10	75
Restaurants/ bars indoors	54 (3198)	9	5	9	77
Restaurants/ bars outdoor terraces	18 (1088)	10	5	10	75
Private transportation	30 (1755)	9	5	9	77
All of the above	16 (921)	4	2	8	86
None	8 (484)	11	5	12	72
Other	1 (88)	9	2	6	83

Base: respondents in the 26+ age group (N=5,910)

Question: Q.4.14. In which settings, if any, do you think that the consumption of HTPs should be prohibited?

Note: Respondents could provide multiple answers, except if they selected "All the above" or "None". Hence, a special consideration must be made when analysing the responses. Although 22% reported "open outdoor spaces", 38% of respondents in the age group were in favour of prohibiting product use in open outdoor spaces (i.e. 22% + 16% that answered "all of the above"). The group total category refers to the total respondents in that age group. The orange column represents the proportion of respondents in the age group that selected the response option (i.e. 1% of respondents in the 26+ age group responded "other"). The white cells (in the four right columns) represent the breakdown by user type, i.e. they are within row percentages (white cells in each row sum to 100).

Source: OE elaboration from online survey data

5.4 Health, social and economic impacts

As a consequence of an increased awareness of combustion-related health risks, HTPs have been marketed as reduced-toxicant and less harmful alternatives to conventional tobacco products (Pemberton, 2018). Even though the number of toxicants assessed in peer-review articles does not cover the full range of harmful and potentially harmful constituents (HPHCs) of interest, both tobacco-industry and independently funded studies have found a reduction in toxicants and exposure to some HPHCs relative to standard cigarettes (e.g. UK COT, 2019, Slade et al, 2002⁹⁷; Liu et al., 2019). In 2017, the UK Committee on Toxicology reported that HTPs are less harmful than traditional cigarettes

⁹⁷ Note that this product is an older version of heated tobacco products.

and that HTPs reduce the harmful and potentially harmful compounds of conventional cigarettes by 50 to 90%.⁹⁸ However, the Committee admitted being unable to fully assess the absolute risk of HTPs given the nature of the data available, but suggested that there would likely be a reduction in risk for conventional cigarette users deciding to use HTPs instead of smoking cigarettes, as well as for bystanders (PHE, 2018).

A systematic review of published peer-reviewed papers showed that the levels of analysed toxicants were at least 62% lower than in cigarette smoke and particulate matter (PM) was 75% lower than in conventional cigarette smoke (Simonavicius et al., 2019). According to the German Cancer Research Centre (DKFZ) (2018b), while acknowledging conflicts of interest and limited study quality, currently available studies showed that HTPs contained less HPHCs than conventional tobacco smoke, and that the consumer was exposed to less pollutants. They acknowledged that there was nevertheless not an inconsiderable risk and that it was still unclear to what extent the lower pollutant load was translated into a reduced health risk. The relationship between exposure and health effects is complex and the finding does not mean that HTPs are harmless, especially given that the long-term health risk potentials are currently unknown.

The mainstream emission from several HTP devices appeared to deliver less nicotine per stick than a conventional cigarette. Nicotine delivery measurements were similar in the tobacco industry and independently funded studies. Studies with humans that measured plasma nicotine levels after use of HTPs indicate that nicotine delivery of HTPs varies by brand but is systematically lower than that supplied by a conventional cigarette, with some exceptions (WHO, 2019b). Further, the mouth-level exposure was found to be lower with some HTPs (e.g. glo from British American Tobacco, BAT) than with conventional cigarettes, especially for nicotine (Gee et al., 2018). However, the overall information on product composition is still limited. The reports submitted by PMI to the FDA include levels of 57 other constituents that are not included in the FDA's list of HPHCs. The level of 56 of them was higher in IQOS emissions than in conventional cigarettes. The levels were double those in the reference conventional cigarettes for 22 compounds and more than 10 times higher for seven. It appears that IQOS reduces exposure to some toxicants but elevates exposure to other substances. A number of these substances belong to chemical classes that are known to have significant toxicity, but in general, there is limited information on the toxicity of many of them (WHO, 2019b).

In Europe, toxicological assessments of tobacco products have aimed at excluding elevated risks in relation to conventional products, but not to confirm less hazardous product properties (Mallock et al., 2019). The Spanish Ministry of Health (2019, Spain) stated that the compounds of HTPs are similar to conventional cigarettes and that these compounds can be toxic with or without combustion. Laboratory-based findings have suggested lung damage, although these do not translate to reduced risks in humans and still very little is known on the exact health effects of HTPs on humans (Psinger et al, 2019). Experimental animal studies showed pulmonary inflammation, while in vitro studies showed no improvement of lung function from switching from combustible to HTPs. Nicotine and tar were also detected at almost identical levels to those found in a conventional cigarette.⁹⁹ Several studies have also detected potentially toxic substances in HTP emissions. Independent assessment of industry data found over 20 harmful and potentially harmful constituents (HPHCs) in emissions that were significantly higher than in conventional cigarette smoke (St Helen et al, 2018), some of which carcinogen. Another study found that potentially carcinogenic substance had almost 3 times the level of those found in conventional cigarettes, and highly toxic compounds could also be released (namely,

⁹⁸ https://cot.food.gov.uk/sites/default/files/heat_not_burn_tobacco_summary.pdf

⁹⁹ ERS Position Paper on Heated Tobacco Products: A statement prepared by the ERS Tobacco Control Committee and approved by the ERS Advocacy Council, Science Council and Executive Committee. Available at: <https://www.ersnet.org/the-society/news/ers-position-paper-on-heated-tobacco-products>

formaldehyde cyanohydrin) - a great concern due to its high toxicity at very low concentrations (Leigh et al., 2018). These findings were also corroborated by the German Cancer Research Centre, who concluded that toxic compounds are not completely removed from the HTP aerosol and these products were not risk-free (DKFZ, 2018b).

In terms of health impacts WHO 2020 reported that currently there is insufficient evidence to determine the harm associated with HTPs in relation to conventional tobacco products. In November 2019, the Standing Committee of European Doctors (CPME) raised concerns on the health risks of these emerging tobacco products, highlighting that they fell in a dangerous regulatory gap. Based on submissions to some regulatory agencies, the industry has not been able to demonstrate that these products will reduce tobacco-related disease (WHO, 2020). For example, PMI reported in its submission to the United States Food and Drug Administration (FDA) on 40 of the 93 HPHCs in IQOS mainstream emission recommended by the agency.¹⁰⁰ The levels of the missing 53 HPHCs, of which 50 are carcinogenic, are unknown (WHO, 2019b). Additional independent studies will be required to substantiate any claims of reduced risk/harm relative to conventional cigarettes. More evidence is also needed on the effects of HTPs to passive users (WHO, 2020; McKelvey et al., 2018). The WHO (2018) has emphasised that all forms of tobacco use are harmful, including HTPs, and have recommended that HTPs be subject to the same policy and regulatory measures applied to all other tobacco products, in line with the WHO Framework Convention on Tobacco Control (WHO FCTC).

As per Article 13, Member States should ban misleading elements, notably any suggestions that a particular tobacco product is less harmful than others. As further research throws light on the precise health impacts of HTPs, the market developments should be closely monitored to avoid the renormalisation of tobacco consumption. For example, there has been concern in the UK over the potential for non-smokers, and especially younger generations, to take up using HTPs because they may believe the product is less harmful. In addition, there is still no evidence that HTPs are efficient as smoking cessation aid.¹⁰¹ Furthermore, as reported in the Study on Council Directive 2011/64/EU on the structure and rates of excise duty applied to manufactured tobacco, the HTPs national fiscal policies are fragmented across Europe:

“For instance, Italy has adopted a product-specific approach by establishing an equivalence of time consumption to conventional cigarettes under the same puffing conditions and applying a 50% reduction (that will significantly reduce in the course of 2019); in Portugal there is a mixed approach with an ad valorem component, a specific component and a minimum excise; Hungary has a fully specific non harmonised tax per items (and not per weight); six MS (including DE, FR and UK) consider it a harmonised product and apply the tax regime of other smoking tobacco (per weight in Kg), with the UK including the entire stick (with filter etc.) in the tax base; Slovenia applies the harmonised regime of FCT (per weight in Kg)”¹⁰²

The novelty of the policies and the product make it difficult to interpret how they affect consumption of the product. However, in terms of economic factors, HTPs (and IQOS in particular) were regarded

¹⁰⁰ In compliance with the Tobacco Control Act, versions of the application are available for public review and comment on the website of the FDA (<https://www.fda.gov/tobaccoproducts/labeling/marketingandadvertising/ucm546281.htm>), but much of the technical information has been redacted for confidentiality.

¹⁰¹ Statement on the toxicological evaluation of novel heat-not-burn tobacco products, Committees on Toxicity, Carcinogenicity and Mutagenicity of Chemicals in Food, Consumer Products and the Environment (COT, COC and COM), available at: <https://cot.food.gov.uk/cotstatements/cotstatementsyrs/cot-statements-2017/statement-on-heat-not-burn-tobacco-products>, 11 December 2017.

¹⁰² In the European Commission (2020): Study on Council Directive 2011/64/EU on the structure and rates of excise duty applied to manufactured tobacco, HTPs are grouped under ‘Other smoking tobacco’.

as very expensive by participants in the present study, except in Germany, with respect to both the heating device and the tobacco sticks.

Table 50. Overview of non-harmonised taxes for heated tobacco products

Country	Heated Tobacco Products
PT	Ad valorem 16% of RSP and specific € 80/kg. Minimum Excise: €169/kg
IT	Depending on SKU: € 63.25 to € 63.36/ 1,000 sticks
RO	RON 384/Kg (ca. € 86.5/93.2)
SI	€88/kg
LV	€62/kg
HU	HUF 10,000 / per 1,000 sticks (€32.46)
EL	€ 156.7 / kg
HR	HRK 600 / kg (ca. €80.7) (
SK	€73.9/kg

Source: European Commission (2020)

5.5 Concluding remarks on novel tobacco products

The main findings on novel tobacco products are summarised below. Heated tobacco products (HTPs) were selected as the most recent and most widely known novel tobacco products (NTPs) in the EU. The novelty of HTPs inevitably carries the consequence of a limited research base. The systematic review of secondary sources revealed:

- Since younger generations are increasingly less likely to smoke conventional tobacco products, tobacco companies have developed new products which are marketed as “safer” alternatives. There has been a proliferation of novel tobacco products entering the EU market. Of particular significance to the TPD are heated tobacco products (HTPs). They have been marketed in over 40 countries worldwide as of July 2019, with at least 22 countries from the WHO European region.
- Heated tobacco products (HTPs) are tobacco products that produce an emission containing nicotine and other chemicals, which is then inhaled by users. They have been marketed as so-called potentially reduced risk exposure products, or even as modified-risk tobacco products. Currently, there is insufficient evidence to conclude that HTPs are less harmful than conventional cigarettes. There are concerns that while they may expose users to lower levels of some toxicants than conventional cigarettes, they also expose users to higher levels of other toxicants. It is still unclear how this toxicological profile translates into short- and long-term health effects.
- *Market size and prospect.* According to Passport data (2017), HTP sales were 769 million euros. The leading markets were Italy and Romania. Although there is no time series trend data on HTP sales or HTP use, the market size for HTP is expected to grow soon but there is still a big gap before it reaches the size of the market for conventional tobacco cigarettes (EC, 2017).
- *Marketing HTPs.* Marketing and promotion has included alleviating health concerns through claims that HTPs are reduced-risk products; designing HTP systems to boost the user’s social image; using “bait and hook” pricing strategy; employing social marketing techniques and

brand ambassadors; selling and marketing via multiple channels, including dedicated stores, retail establishments and e-commerce websites.

- According to the few existing population surveys, the prevalence of HTP current use among adults is less than 0.5% in the general population of European Union MS.
- Member States have adopted different definitions of HTPs with respect to taxes, resulting in substantial fragmentation of the EU internal market and resulting in a large variation in prices across European countries in which the product is available.

The empirical research (focus groups and online survey) revealed that HTPs were the least known product among the products under study. HTPs and electronic cigarettes were sometimes confused, considered to be similar products, and believed to not be as satisfying as conventional tobacco products. The study revealed the following user profile:

- Everyday use of HTPs with 1 to 10 sticks was typical for current users. Initiation of use was motivated by either enjoyment or trying to quit or reduce smoking of conventional tobacco products. About 85% were dual users.
- Current HTP users were evenly split between those than had never attempted quitting and those who tried at least once and failed in the 18-25 group. In the 26+ group, almost two out of three had never tried to quit. Among failed quitters, the main reasons to try quitting were for 'personal health' and 'economic reasons'. In the 26+ group, another reason frequently mentioned was the 'concern for the health of the people around them'. The main reason cited for failing to quit was being 'stressed'. In the 26+ group, another important reason was that they found quitting HTPs 'too difficult'.
- The majority of past users of HTPs were still using other tobacco products at the time of the survey.

Lastly, the following paragraphs highlight the *Product and Regulatory perceptions remarks*

- *Source of awareness of HTP.* A formal source of advertising and promotion of the product (38% in the 18-25 group, 45% in the 26+ group) was the most common introduction to HTPs. Word-of-mouth was a close second source.
- *Perception of harm.* Conventional cigarettes were perceived to be the most harmful to the health of users and bystanders. Comparing the risk perception of users of conventional cigarettes with the users of HTPs, the 18-25 age group rated HTPs as 35% less harmful than conventional cigarettes to users and 46% less harmful to bystanders. In the 26+ age group HTPs were perceived to be 52% less harmful to users and 58% less harmful to bystanders.
- *Regulatory perceptions.* About 75% of respondents agree that HTPs should be regulated as conventional cigarettes and a many support restrictions in use in workplaces, public transport and the hospitality sector.

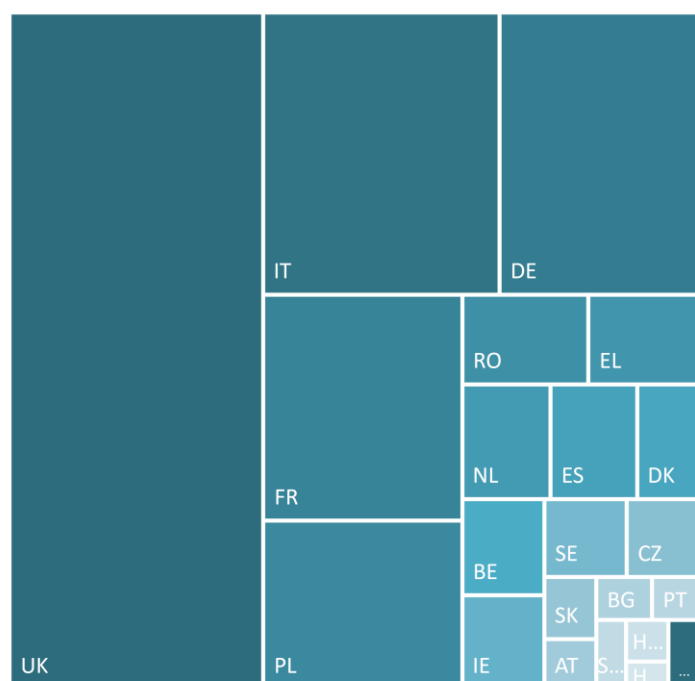
6 Electronic cigarettes

6.1 Background

According to the World Health Organisation (WHO), there was a 50% increase in the use of e-cigarettes among both adults and adolescents between 2008 and 2012, with evidence that this trend would continue (Schneider & Diehl, 2016). In 2012, approximately 29.3 million European Union adults had tried e-cigarettes (Vardavas, 2015, EU, N=26,566). Various types of e-cigarette devices are used and have co-existed on the market. However, open tanks¹⁰³ and modular systems¹⁰⁴ have grown in popularity in Europe over the past years and account for the majority of the market (between 60% and 90%, depending on the source), whereas disposable devices¹⁰⁵ and cig-a-likes¹⁰⁶ have declined (EC, 2016; EC, 2017).

Market development. The structure of the EU market of electronic cigarettes in 2017 is depicted in the figure below. It was mainly concentrated in Western Europe, with the largest market in the UK (1,538 m EUR), followed by Italy (597 m EUR), Germany (524 m EUR) and France (402 m EUR).

Figure 127. Market size of electronic cigarettes in million euros in 2017



Note: The market size is estimated total sales in million euros for 2017, without population adjustments. It is displayed as a tree map graph that consists of series of nested rectangles of sizes proportional to the corresponding data value.

Source: OE elaboration of Passport, 2017

¹⁰³ Tank systems are second-generation e-cigarettes. They are typically shaped like pens and feature a transparent reservoir to hold the liquid. There are "open" and "closed"-tank systems.

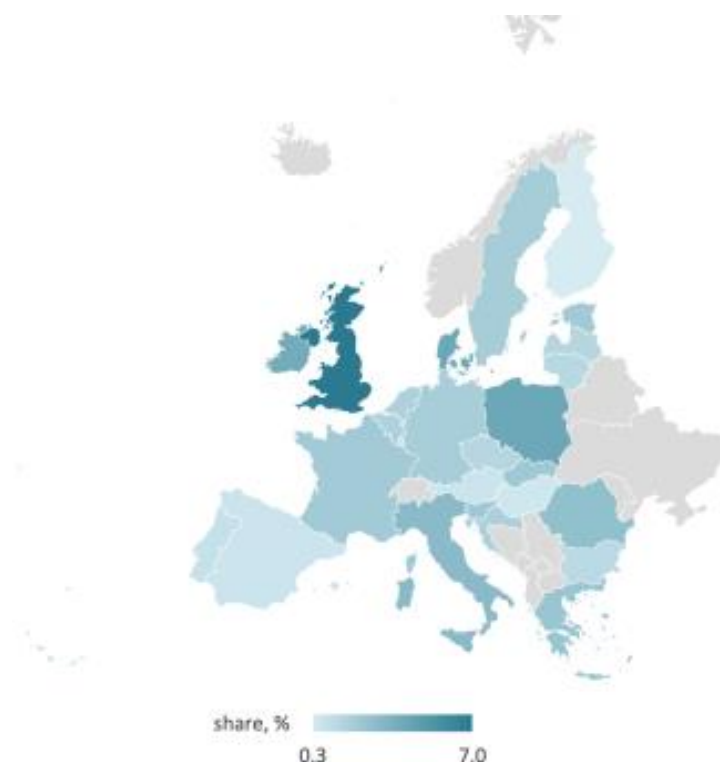
¹⁰⁴ Modular systems are the latest generation on the market, aesthetically departing from the cigarette-like shape. They allow various degrees of customization of component parts and allow users to regulate the power delivery and other settings.

¹⁰⁵ Refers to the "closed tank" systems.

¹⁰⁶ Cig-a-likes are the first-generation of electronic cigarettes and, as suggested by their name, often resemble cigarettes (although not necessarily). They can be disposable or rechargeable using prefilled cartridges.

The UK was not only the largest market of e-cigarettes in absolute values, but it also had the highest share of e-cigarettes of the total national market (in m. EUR) compared to the other EU countries – reaching almost 7% (see Figure 128). The e-cigarette market in Europe saw a continuous expansion since 2008.¹⁰⁷ According to the Eurobarometer 458 (2017), 15% of respondents had at least tried e-cigarettes, an increase for 3% since 2014. The proportion of current users has remained stable at 2%. The growth in past years has been partly attributable to the strong marketing practices of the industry. The Eurobarometer found that 44% of respondents had seen advertisements for e-cigarettes, with 7% reported seeing them 'often'. The UK actively promoted the use of e-cigarettes as an effective tobacco cessation aid (PHE, 2018¹⁰⁸). The Eurobarometer found that British respondents were exposed to more advertisement for e-cigarettes than the average EU citizen (65% in the UK with an EU average of 44%) (Eurobarometer, 2017).

Figure 128. Share of electronic cigarettes within the national total tobacco market, per country (2017)



Note: Percentage share of electronic cigarettes were estimated with respect to the whole tobacco market in the country in million euros.

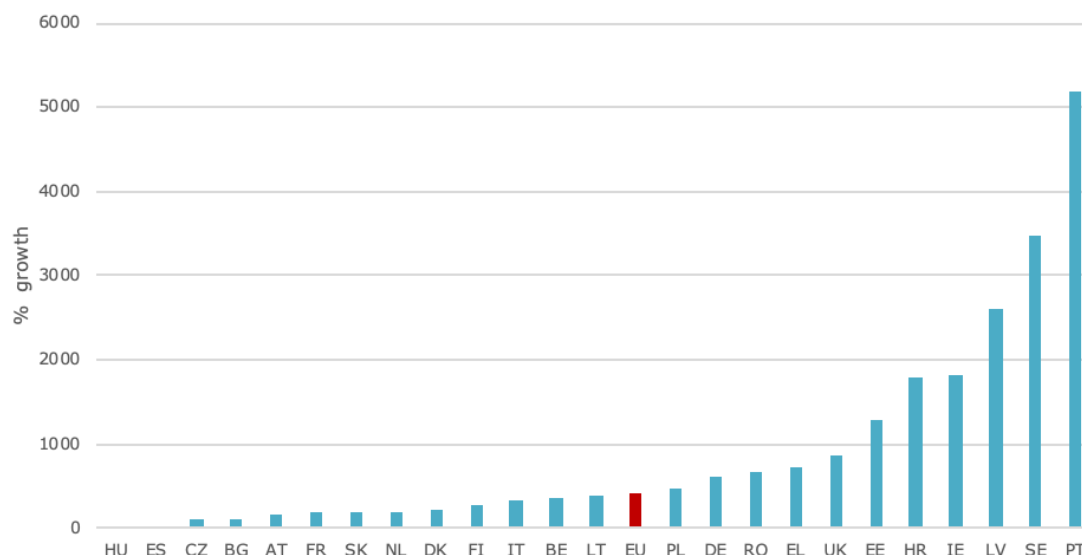
Source: OE elaboration of Passport, 2017

¹⁰⁷ https://ec.europa.eu/health/sites/health/files/tobacco/docs/potentialrisks_specs_refillableecigarettes.pdf

¹⁰⁸ <https://www.gov.uk/government/publications/e-cigarettes-and-heated-tobacco-products-evidence-review>

Where data is available, Figure 129 shows the trend of increased current retail selling price (RSP) of e-cigarettes. The crude EU average of price increases was of four-fold between 2012 and 2017. The most significant increases occurred in Portugal, Sweden and Latvia.

Figure 129 Price dynamics of electronic cigarettes (in retail selling prices, RSP): 2012-2017



Note: The EU average (in red) was calculated for 24 EU MS (excluding Cyprus, Luxembourg, Malta & Slovenia – as there was no available data for the countries). It is the sum of all current Retail Sale Price (RSP) (in euro) in all countries over the number of countries for which the data was available for 2012 and for 2017 separately. These values were then used in the current price growth formula to establish the growth between these two years.

Source: OE elaboration of Passport, 2017

Several determinants of e-cigarette use, including age, gender and place of residence have been explored. Tarrazo (2017, Spain, N=16,000) found that age was a significant determinant of e-cigarette use, with young people (aged 15-24) more likely to use e-cigarettes than young adults (aged 25-39) or adults (aged 40-54). Kristjansson (2017, EU) found that 15 to 17-year-olds reported having experimented with e-cigarettes at least once in their lifetime and 20% had used them three times or more. On gender and marital status, Vardavas (2015, EU, N=26,566) found no significant differences in 'ever users' of e-cigarettes. Several studies found that underage males were more likely to use or to try them compared to females of similar age (Lindström, 2018, Sweden, N=17,092; Kotz et al., 2018, Germany, N=12,273; Kinnunen, 2015, Finland, N=3,535; Coleman, 2015, USA, N=4,310). E-cigarette adult users were also more likely to live in urban rather than rural areas (Vardavas, 2015; Tarrazo, 2016). Furthermore, e-cigarette users were more likely to be living in Northern or Eastern Europe as opposed to Western Europe (Vardavas, 2015). In terms of the type of e-cigarette, Kotz et al. (2018, Germany) found that 66% of German e-cigarette users had a tank refilled by the user, i.e. an open system (third generation of e-cigarettes).

Vardavas (2015) reported a positive correlation between e-cigarette use and previous conventional tobacco smoking among European youth and adults. Furthermore on dual use has been reported in other studies (ASH, 2018; Strong, 2019, USA, N=14,287). However, Action on Smoking and Health (ASH)¹⁰⁹ reported that current e-cigarette users were more likely to be former consumers of tobacco products (1.7 million) rather than dual users with other tobacco products (1.4 million) in the UK. The results suggested that the main reason for e-cigarette use in the UK was to quit other tobacco

¹⁰⁹ A public health charity in the UK. <https://ash.org.uk/wp-content/uploads/2019/09/Use-of-e-cigarettes-among-adults-2019.pdf>

products. Pokhrel (2015, USA) provided a framework comprising three broad contexts of e-cigarette use among dual users.

- **Activities.** This refers to the types of activity associated with e-cigarette use instead of conventional cigarettes. Two activities fall under this category (i) before or during work-out or physical activity and (ii) when working - users tend to prefer using e-cigarettes when doing physical activity as they perceive that it does not affect their performance.
- **Places/situations.** Certain places or situations were perceived to be more conducive to e-cigarette use. It was viewed as more suitable for indoor use (e.g. no smell, reduced risk of passive smoking). Participants perceived e-cigarette use to be a more personal (they would not share their e-cigarette with other people) and a solitary activity – as compared to waterpipe tobacco use.
- **Tobacco substitution.** Three situations were reported to be drivers of e-cigarette use as a tobacco substitute, namely; (i) when cigarettes were not available; (ii) when a hookah was not available; and (iii) when smoking was not allowed. E-cigarette use as a substitute for conventional cigarette smoking was associated with the temporary reduction of cravings.

Schneider & Diehl (2016, USA) proposed a model to explain the dynamics of e-cigarette use patterns, and notably the association between e-cigarette use and the likelihood of subsequent conventional tobacco consumption. The model identified causalities involved in the transition from no consumption to tobacco consumption in two trajectories: from abstinence to e-cigarette use, and from there to the consumption of tobacco cigarettes. To do so, drawing on the literature they extracted the following hypotheses (see Table 51).

Table 51. Hypotheses proposed in the Schneider & Diehl (2016) model

Category	Name	Description
Explanations for the initiation of e-cigarette use	The flavour hypothesis	The multitude of e-cigarette flavours available, such as candy, chocolate or strawberry, make the initial use of e-cigarettes more likely than starting to smoke tobacco, especially among adolescents
	The health hypothesis	The assumption that e-cigarettes pose fewer health risks than tobacco cigarettes makes the initial use of e-cigarettes more likely than starting to smoke tobacco, especially among adolescents.
	The role model hypothesis	The assumption that e-cigarette users are more likely to be seen as role models by their peers makes the initial use of e-cigarettes more likely than starting to smoke tobacco, especially among adolescents.
	The concealment hypothesis	The fact that e-cigarettes are easier to conceal makes the initial use of e-cigarettes more likely than starting to smoke tobacco, especially among adolescents.
	The acceptance hypothesis	The perception that e-cigarettes are more socially accepted makes the initial use of e-cigarettes more likely than starting to smoke tobacco, especially among adolescents.
	The addiction hypothesis	The fact that nicotine is addictive makes e-cigarette users more likely than non-users to start smoking tobacco, especially among adolescents.
	The accessibility hypothesis	The fact that e-cigarettes and tobacco cigarettes are often available through the same commercial and social sources makes users more likely than nonusers to start smoking tobacco, especially among adolescents.

Category	Name	Description
Explanations for any transition to tobacco cigarette consumption	The experience hypothesis	Experience with the use of e-cigarettes makes users more likely than non-users to start smoking tobacco, especially among adolescents.
Explanations outside the model	The liability hypothesis	Individual predisposition and specific socialization make the initiation of e-cigarette and tobacco use more likely, especially among adolescents.
	The renormalization hypothesis	The increasing rate of e-cigarette use leads to the renormalization of smoking and to more people initiating tobacco smoking, especially among adolescents.

Source: Own elaboration from Schneider & Diehl (2016)

The model proposed by Schneider & Diehl has been supported by several studies (Leventhal et al., 2015, USA, N=2,530; East et al., 2018, UK, N=932; Kinnunen et al., 2015). Findings have established a correlation between e-cigarette ever use and the likelihood of initiating conventional tobacco use during early adolescence – especially for nicotine e-cigarettes – as under aged users may develop an addiction which they eventually satisfy through conventional cigarettes.

Several studies have explored the outreach of the industry to attract consumers. The multi-billion Euro industry has used a variety of channels to target current tobacco users, former tobacco users, and young non-tobacco users. Notably, the availability and online marketing of e-cigarettes has been considered a driver for the market development. In a study on Finnish teenagers, Kinnunen (2015, Finland, N=3,535) reported that the internet was the second most common source for e-cigarette acquisition. A study by Martinez (2015, Spain, N=736) found that most of participants discovered e-cigarettes through media marketing strategies promoting them as a safer option than conventional cigarettes. The TPD has regulated marketing of e-cigarettes since its transposition in 2016 across Member States.¹¹⁰ However, the globalised internet is a challenge national legislation. While different advertising regulations apply, studies in the USA identified the use of online tools and mobile applications to target youth (O'Brien, 2018; Krugman, 2016). Krugman (2016) argued that marketing communications on e-cigarettes were 'normalising' the product category and cultivating favourable perceptions among young people. Through the internet, European youth also has access to these advertisements, despite efforts to limit exposure to them. A 2016 EC-commissioned study¹¹¹ found examples of content originating from producers or retailers which appeared to promote tobacco, such as prize draws, news items about tobacco retailing, and 'corporate' content. The study also found examples of user-generated content appearing to promote specific tobacco brands and products on Facebook which, in effect, could be considered as tobacco advertising.

El-Toukhy et al. (2018, USA, N=5,766) analysed the impact of marketing practices that claimed lower risks to health harms for alternative tobacco products (i.e. e-cigarettes and heated tobacco products) on the perceived risks. The study concluded that:

- The public inferred that both risk and exposure to harmful chemicals was lower when they 'learned' that either one was lower;
- Participants perceived lower risk from alternative tobacco products due to lower quantities of harmful chemicals and less harm to health;
- Reduced quantities of chemicals were associated with lower perceived harm;
- Perceived risk and perceived chemical quantity were highly correlated;

¹¹⁰ https://ec.europa.eu/health/sites/health/files/tobacco/docs/citizensexposure_tobaccomarketing_en.pdf

¹¹¹ https://ec.europa.eu/health/sites/health/files/tobacco/docs/citizensexposure_tobaccomarketing_en.pdf

- Perceived risk was insensitive to pictorial warnings and other persuasive approaches;
- Lower exposure claims influenced the assessment of alternative tobacco products harm;

The marketing activities of e-cigarette companies have increased dramatically over recent years, notably in social media.¹¹² While most of the marketing or pseudo-marketing content originates from the USA, globalisation threatens efforts to minimise the exposure of younger generations to advertisements and marketing. In the 2014 Eurobarometer,¹¹³ young respondents were more exposed than any other age category to advertisements or promotions for e-cigarettes: 12% of them reported having seen them often vs. 5% of those aged 55+.

Consumer perception. Laverty et al. (2016, EU, N=2,430) found that 'ever users' of tobacco and e-cigarettes were likely to be influenced by taste, price and nicotine content of e-cigarettes. In comparison to older people, those aged 15-24 years were more likely to cite packaging and design features as important, as well as being influenced by the price of e-cigarettes. Kotz, Böckmann and Kastaun (2018, Germany) suggested that the major reasons stated for the use of e-cigarettes among young users were the array of flavours, followed by the potential to help in reducing their tobacco consumption. In a recent systematic review Meernik et al. (2019, global) found consistent evidence that flavours (some 7000) were attractive to both youth and adults.

Table 52 presents findings from the Eurobarometer 458 survey (2017) on the factors that influence consumers' choice of e-cigarettes. The survey found that consumers' decision to start using e-cigarettes was mainly driven by health rather than social or aesthetics reasons. A large majority of respondents (61%) cited "'to stop or reduce their tobacco consumption'. as a reason for initiating e-cigarette use. The second most stated reason (31%) was the belief that the product was less harmful than other tobacco products. A lower price, relative to other tobacco products was cited by 25% of respondents. Other studies have found a significant positive correlation between attempts to quit traditional tobacco consumption and e-cigarette use (Chapman, 2014, USA; Giovenco, 2018, USA, N=15,532).

¹¹² https://ec.europa.eu/health/sites/health/files/tobacco/docs/potentialrisks_specs_refillableecigarettes.pdf

¹¹³ Special Eurobarometer 429. Attitudes of Europeans towards Tobacco and Electronic Cigarettes.
http://ec.europa.eu/public_opinion/archives/ebs/ebs_429_en.pdf

Table 52. Reasons for initiating use e-cigarettes

(% - of current and past users)

	To stop or reduce your tobacco consumption, %	They were cool or attractive, %	So you could vape in places where tobacco smoking is not allowed, %	They were cheaper than tobacco, %	Your friends used e-cigarettes, %	You liked the flavours of e-cigarettes, %	You believed that vaping was less harmful than using tobacco, %	Other N/A, %
EU28	61	6	15	25	11	12	31	5
AT	50	13	31	26	23	25	38	3
BE	55	12	14	25	20	21	22	7
BG	39	7	37	7	8	11	23	4
CY	63	8	24	18	14	11	40	0
CZ	48	7	27	19	23	14	28	3
DE	50	8	25	32	5	28	36	8
DK	74	-	19	25	12	13	34	6
EE	51	1	8	11	22	20	32	13
EL	69	5	19	11	17	-	39	0
ES	88	8	10	19	7	6	20	3
FI	59	4	8	18	14	9	31	3
FR	79	-	17	31	5	4	34	2
HR	61	10	27	32	9	5	42	5
HU	34	7	18	20	13	15	38	3
IE	65	2	5	33	11	-	33	0
IT	33	25	10	26	24	6	25	2
LT	37	5	7	32	32	21	12	6
LU	61	-	6	12	17	2	34	6
LV	47	7	22	18	26	24	16	12
MT	60	12	14	12	15	3	33	8
NL	72	-	26	21	17	9	30	8
PL	29	11	7	18	28	22	18	9
PT	65	11	26	15	3	12	37	8
RO	24	11	22	11	13	7	10	28
SE	80	2	26	16	6	32	33	6
SI	19	13	28	17	26	8	39	4
SK	18	12	14	38	25	-	44	0
UK	64	2	5	18	6	15	32	6

Base: respondents who currently use or used e-cigarettes, N=1,565

Question: QB14 of Eurobarometer 458 – "Which of the following factors, if any, were important in your decision to start using e-cigarettes?"

Note: Respondents were able to select up to 3 reasons. The share of respondents that selected each reason is provided in the table, per country.

Source: Eurobarometer 458, 2017

While e-cigarettes have been promoted as an aid to cessation, a majority (55%) of EU respondents perceived them as harmful to health (see Table 53). The table below shows striking variation in the perception of harmfulness, from 85% in the Netherlands to 34% in Italy

Table 53. Perception of harmfulness e-cigarettes

	Yes %	No %	I don't know %
EU28	55	28	17
AT	47	40	13
BE	62	34	4
BG	45	16	39
CY	76	11	13
CZ	56	30	14
DE	54	32	14
DK	79	13	8
EE	75	11	14
EL	65	19	16
ES	60	19	21
FI	81	11	8
FR	62	21	17
HR	51	32	17
HU	49	39	12
IE	59	27	14
IT	34	46	20
LT	80	11	9
LU	76	12	12
LV	80	11	9
MT	67	20	13
NL	85	7	8
PL	56	28	16
PT	58	26	16
RO	48	37	15
SE	65	14	21
SI	65	22	13
SK	48	36	16
UK	50	29	21

Base: all respondents, N=27,901

Question: question QB9 of the Eurobarometer – "In recent years electronic cigarettes or e-cigarettes have been increasingly marketed in Europe. Do you think that they are harmful or not to the health of those who use them?"

Source: Eurobarometer 458, 2017

Meernik et al. (2019) found that the perception of harm of e-cigarettes was dependent on the product flavouring, notably amongst the youth.¹¹⁴ Specifically, fruit and candy-flavoured e-cigarettes were perceived as less harmful than tobacco-flavoured e-cigarettes. In similar vein Czoli (2016) (Canada, N=915) found that e-cigarette flavour significantly predicted lower perceptions of product harm. Several studies have found that both young (aged 12-17) and adult (aged <65) populations perceived e-cigarettes¹¹⁵ as less harmful than conventional cigarettes in Europe and the Western region (Martinez, 2015, Spain, N=736; Strong, 2019, USA, N=13,620). Other studies¹¹⁶ conducted outside of the EU have found that the awareness of e-cigarette has increased over the years (McNeill et al., 2018, UK; Huerta, 2017, USA, N=10,492¹¹⁷; Huang et al., 2019, USA, N=24,442¹¹⁸), and that while it may be relatively less harmful than conventional tobacco consumption, it still carried potential risks for users.

Impacts. Results from the literature review suggested that health risks associated with e-cigarettes were still uncertain and inconclusive (PHE, 2018; RCP, 2016; Hua, 2016; Meo, 2014; Callahan-Lyon, 2014; Kim, 2016) as previously suggested in section 6.1. However, the long-term harms of these products remained unknown (Czoli, 2017; Andler, 2015, France; NASEM, 2018). The evidence showed that the level of exposure to nicotine from e-cigarettes is variable, depending on the characteristics of the device and the e-liquid, as well as on how the device is operated. Callahan-Lyon (2014) reported that differences in product engineering, components and potential toxicities hinder the study of e-cigarettes as a single device.

In the USA, a lung ailment reportedly killed and afflicted several patients in the last few years.¹¹⁹ While this disease highlighted the need for further investigations on the product and its composition, differences in the regulatory arrangement of the US and the EU, protected EU citizens from the risk as further explained in section 6.2.3.¹²⁰

Pisinger (2014) outlined some methodological challenges that must be acknowledged in the research on e-cigarette health impacts. First, with research covering some hundreds of brands that had been subject to frequent modifications, it is a challenge to arrive at valid generalisations. Furthermore, many studies have compared the health impacts of e-cigarettes with conventional cigarettes and also with non-users, whereas health hazards for each product may be different. For example, e-cigarette use topography is quite different to conventional smoking as use of e-cigarettes involves sucking harder with longer puffs. Finally, some experimental studies with human participants may not reflect real-life use of e-cigarettes.

Many recent studies have argued for the need to better codify and understand the health impacts of e-cigarettes in order to have evidence-based policies for tobacco control (PHE, 2018; RCP, 2016; Hua, 2016; Kim, 2016). Levy (2017) proposed a framework describing and incorporating patterns of vaporised nicotine products (VNPs), such as e-cigarettes and HTPs, to determine the total amount of toxic exposure effects on a population health. While the framework identified some critical information, it is expected that there will be further product innovations and changes in patterns of use such that

¹¹⁴ The finding from Meernik et al. (2019) stems from studies conducted in 2016.

¹¹⁵ As well as waterpipe tobacco, which was also under the scope of some of the cited studies.

¹¹⁶ External referring to outside of the EU.

¹¹⁷ N=3,630 in 2012; N=3,185 in 2013; N=3,677 in 2014

¹¹⁸ N=2,800 in 2012; N=5,668 in 2014; N=5,372 in 2015; N=5,245 in 2016; N=5,357 in 2017

¹¹⁹ "Outbreak of Lung Injury Associated with E-Cigarette Use, or Vaping". Centers for Disease Control and Prevention. 28 January 2020. This article incorporates text from this source, which is in the public domain. Available at: https://www.cdc.gov/tobacco/basic_information/e-cigarettes/severe-lung-disease.html

¹²⁰ There have been suspected cases (e.g. Belgium) but the underlying cause has not yet been clarified. https://www.europarl.europa.eu/doceo/document/P-9-2019-003949-ASW_EN.html

health risks may change as well. Kalkhoran (2015, UK & USA) developed a model to evaluate potential health effects of varying scenarios of increasing promotion and use of e-cigarettes. The outcome, supported by results from Polosa (2013)¹²¹ was that greater e-cigarette use could have different health effects for different population segments.

Controversy surrounds the public health response to the health impacts of e-cigarettes as exemplified by the contrasting conclusions and policy recommendations in reports from Public Health England (PHE) and the Food and Drugs Administration (FDA) who had commissioned a report on e-cigarettes from the National Academies of Science, Engineering and Medicine in the United States (NASEM). The issue was, and still is, whether e-cigarettes were an aid to quitting conventional cigarettes or at minimum a means to reduce health risks (the cessation scenario) or whether young people would take up the use of e-cigarettes and progress to conventional cigarettes (the gateway scenario).

Fairchild et al, (2019) point to the crux of the controversy “whose risk should be given priority”. For the US Authorities the protection of children is the paramount goal, while for PHE the goal is to protect current smokers of traditional tobacco products.

PHE reported that “the current best estimate is that e-cigarettes are around 95% less harmful” than traditional cigarettes. PHE reported concern that “Nearly half the population don’t realise e-cigarettes are much less harmful than smoking”. Their conclusion was that if current smokers of traditional cigarettes switched to e-cigarettes the number of smoking related illnesses and deaths would be significantly reduced.

On the evidence regarding the gateway scenario, e-cigarettes acting as a route for young people to use conventional cigarettes, PHE states that “despite some experimentation with these devices among never smokers, e-cigarettes are attracting very few young people who have never smoked into regular use.

The NASEM reported that e-cigarettes are “not safe but safer” and acknowledged that e-cigarettes “might be useful as a cessation aid to smokers who use e-cigarettes exclusively”. The report highlighted the risks to young people and to bystanders. On the gateway scenario “youth who begin with e-cigarettes are more likely to transition to combustible tobacco cigarette use and become smokers who may be at risk to suffer the known health burdens of combustible tobacco cigarettes”.

Fairchild et al, (2019) noted the fear of “addicting a generation of youth” in the USA and cited statistics from the National Youth Risk Behavior Survey that showed “that vaping in the past 30 days increased from 0.6% to 4.9% among middle-school students and 1.5% to 20.8% among high-school students between 2011 and 2018. During that period, regulators were most alarmed by a spike between 2017 and 2018: past-30-day e-cigarette use increased 48% among middle schoolers and 78% among high schoolers”.

PHE cited Action on Smoking and Health data that show similar trends: between 2014 and 2018, the percentage of youths reporting that they had ever tried an e-cigarette increased from 6.5% to 11.7%; during that same period, the percentage of youths who were current vapers increased from 1.6% to 3.4%.

On the use of e-cigarettes as an aid to quitting conventional cigarettes Hajek et al, 2019, reported a randomised control trial that supports PHE’s recommendation of a combined strategy of access to e-cigarettes supported by behavioural measures. The randomised control trial showed that compared to nicotine replacement therapy, e-cigarettes were more effective for smoking cessation for those who attended a stop smoking programme. After 12 months, 18.0% of the e-cigarette group and 9.9% of

¹²¹ Disclaimer: the publication was supported by the electronic cigarettes industry.

the nicotine replacement group had quit all tobacco products. But, for those who were unsuccessful, dual use was evident.

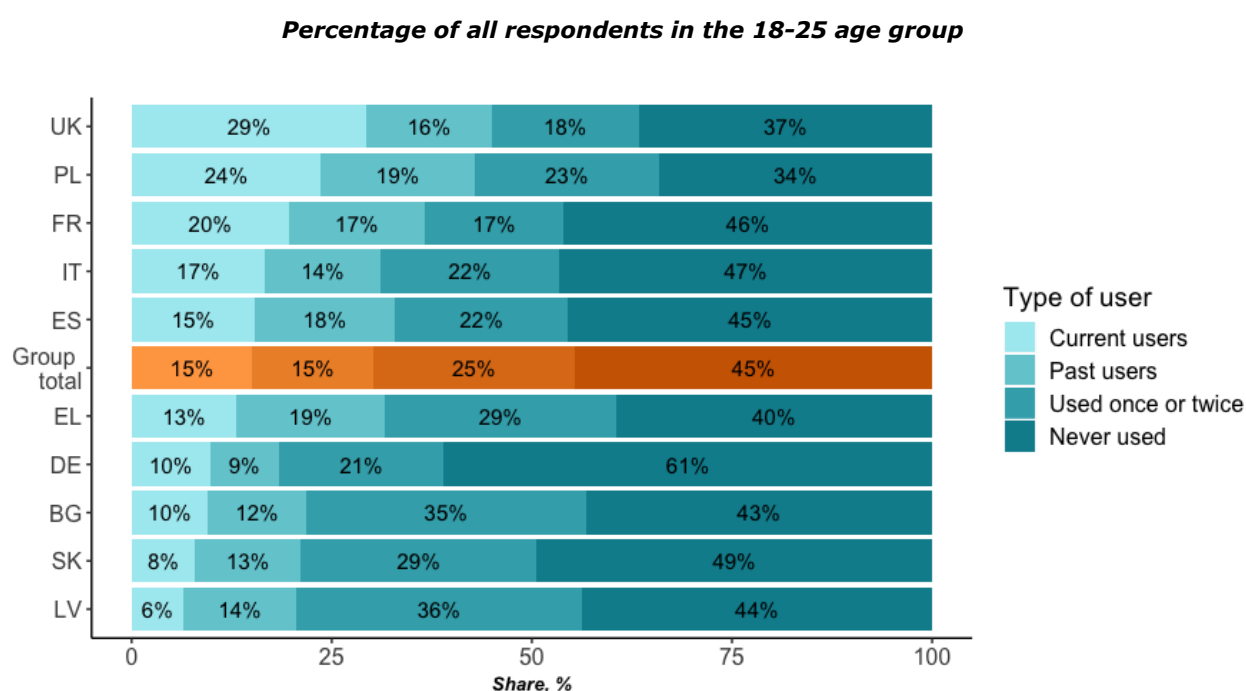
The NASEM report identifies unique risks associated with e-cigarettes including emissions of aerosol metals, other potentially toxic substances and device explosions and draws the conclusion that “more and better research on short- and long-term health effects of e-cigarettes, as well as their effects on initiation and cessation of combustible tobacco product use, will bring clarity to the question of whether e-cigarettes will prove to reduce harm or induce harm at the individual and the population levels”.

6.2 Market developments

6.2.1 Socio-demographic characteristics

The online survey found that the majority (55%) of the **18-25 age group** had tried e-cigarettes and that 15% were current users (see Figure 130). The highest proportion of current e-cigarette users was found in the United Kingdom (29%), followed by Poland (24%), France (20%) and Italy (17%). The largest share of never users was found in Germany, at 61%.

Figure 130. Use of e-cigarettes: country breakdown (18-25 age group)



Base: all respondents in the 18-25 age group (N=6,090)

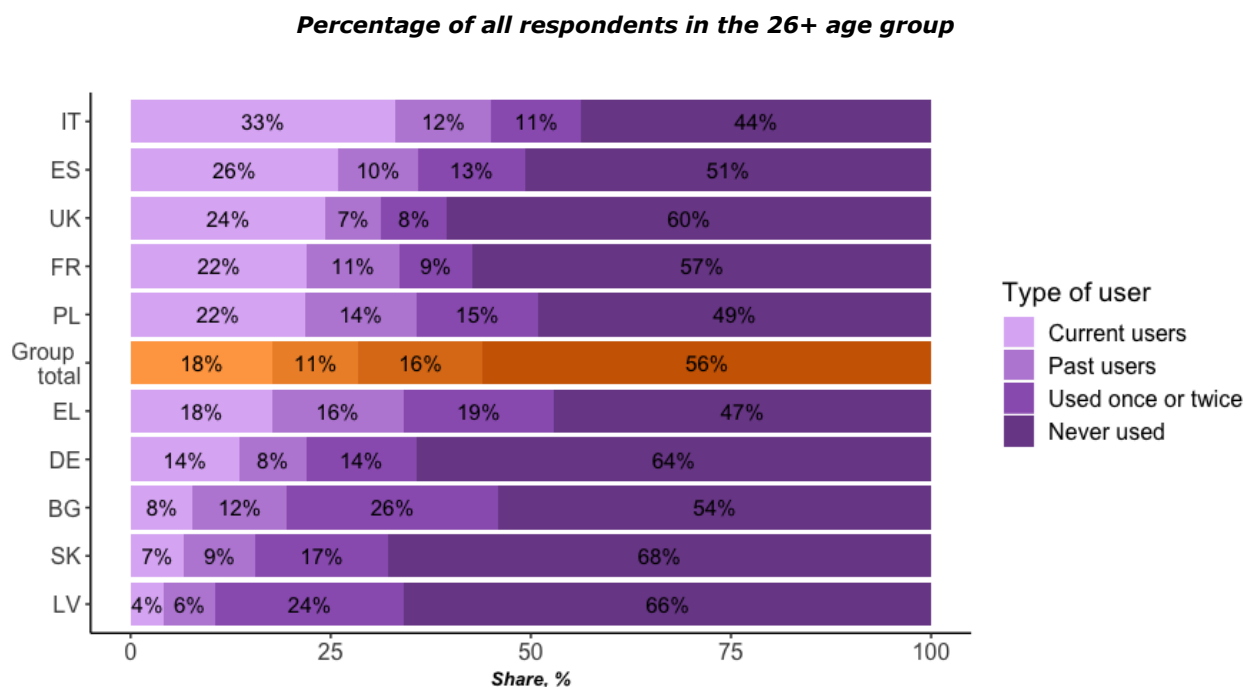
Question: Q.1.1. Have you ever used e-cigarettes?

Note: The group total category refers to all respondents within the age group.

Source: OE elaboration from online survey data

Figure 131 reports the pattern of use for the **26+ age group**. 45% had tried e-cigarettes and the majority (56%) had never used them. 18% were current and 11% past users. Italy was identified as the largest consumer of electronic cigarettes with 33% of current users, followed by Spain (26%), the United Kingdom (24%) and France (22%). In contrast, Bulgaria, Slovakia and Latvia had relatively small shares of current users, but higher percentages of respondents having used the product once or twice.

Figure 131. Use of e-cigarettes: country breakdown (26+ age group)



Base: all respondents in the 26+ age group (N=5,910)

Question: Q.1.1. Have you ever used e-cigarettes?

Note: The group total category refers to all respondents within the age group.

Source: OE elaboration from online survey data

The socio-demographic breakdown for the **18-25 age group** is shown in Table 54, and those of the **26+ age group** in Table 55. Both age groups evidenced similar profiles. The age groups were relatively gender balanced; users of e-cigarettes were predominately male in both age groups. Females in the 18-25 age group were more likely to have experimented with the product or to never have used it, whereas no differences in gender were observed for the 26+ age group.

In the 18-25 age group, the majority of the respondents had a medium level of education – at least high school-level education – on account of the fact that 42% of respondents reported that they were students (see section 1.3.3). The education levels were relatively consistent across different types of users. In terms of place of residency, a large majority of respondents resided in urban areas (e.g. 66% in the 26+ age group). Never users had a higher share of rural residents than current users or the overall group breakdown. In terms of economic status, most respondents in the 18-25 age group and the 26+ age group self-reported a medium status, but the current users had a greater share of self-reported high economic status.

Table 54. E-cigarette use socio-demographic breakdown (18-25 age group)

Percentage and number of respondents in the 18-25 age group

	Group total % (n)	Current users %	Past users %	Used once or twice %	Never users %
Use of product	100 (6,090)	15	15	25	45
Gender					
Male	47 (2,869)	19	19	23	38
Female	53 (3,221)	11	13	26	49
Education level *					
1-2 (low)	5 (335)	20	20	20	40
3-4 (medium)	63 (3,829)	14	16	27	44
5-6 (high)	32 (1,926)	16	13	22	47
Residency					
Urban	65 (3,949)	15	15	26	43
Suburban	19 (1,173)	16	16	21	42
Rural	16 (968)	13	13	25	56
Self-reported socio-economic status					
1-4 (low)	18 (1,126)	11	17	28	50
5-7 (medium)	65 (3,954)	14	14	28	45
8-10 (high)	17 (1,010)	24	18	18	41

Base: all respondents in the 18-25 age group (N=6,090)

Notes: The group total column refers to all respondents within the age group, i.e. it is a within column percentage. It provides the breakdown of each socio-demographic variable, including absolute numbers. The white cells in the four right columns represent the breakdown by user type, i.e. they are within row percentages (white cells in each row sum to 100).

* The sub-categories of the education level were the following: (1) No official education; (2) Primary school; (3) High school; (4) Some years of university (not completed); (5) University degree completed; (6) Post-graduate (master, PhD, other)

Source: OE elaboration from online survey data

Table 55. E-cigarette use socio-demographic breakdown (26+ age group)

Percentage and number of respondents in the 26+ age group)

	Group total % (n)	Current users %	Past users %	Used once or twice %	Never used %
Use of product	100 (5910)	18	11	16	56
Gender					
Male	52 (3074)	21	12	15	54
Female	48 (2836)	15	10	17	58
Education level *					
1-2 (low)	5 (308)	20	0	20	60
3-4 (medium)	46 (2704)	15	11	15	37
5-6 (high)	49 (2898)	20	10	16	53
Residency					
Urban	66 (3879)	20	12	17	52
Suburban	17 (1010)	18	12	12	59
Rural	17 (1021)	12	6	12	71
Self-reported socio-economic status					
1-4 (low)	23 (1374)	13	9	17	61
5-7 (medium)	64 (3769)	17	11	16	56
8-10 (high)	13 (767)	31	15	15	46

Base: all respondents in the 26+ age group (N=5,910)

Notes: The group total column refers to all respondents within the age group, i.e. it is a within column percentage. It provides the breakdown of each socio-demographic variable, including absolute numbers. The white cells in the four right columns represent the breakdown by user type, i.e. they are within row percentages (white cells in each row sum to 100).

* The sub-categories of the education level were the following: (1) No official education; (2) Primary school; (3) High school; (4) Some years of university (not completed); (5) University degree completed; (6) Post-graduate (master, PhD, other)

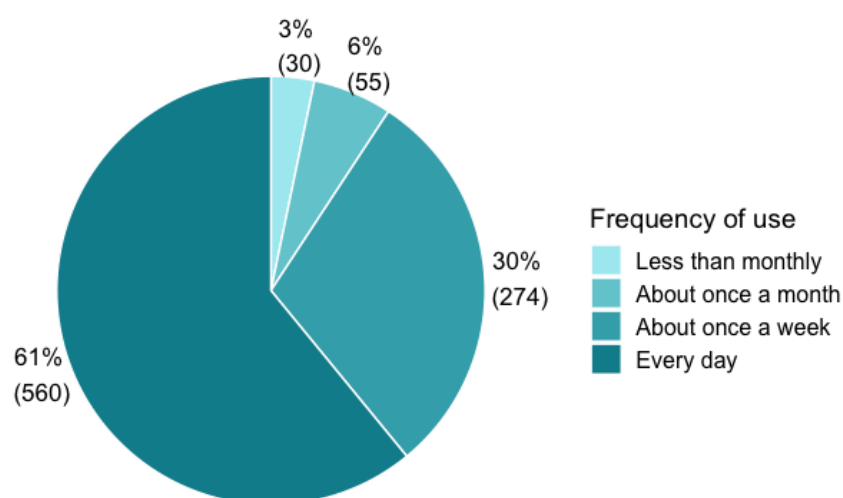
Source: OE elaboration from online survey data

6.2.2 Pattern of use

In the online survey, a majority of current e-cigarette users in the **18-25 age group** (see Figure 132.) and the **26+ age group** (see Figure 133) used the product daily, with 61% and 68% providing this response respectively. When considered relative to the total sample, these shares become much smaller but nevertheless considerable: 9% of respondents in the 18-25 age group were daily users of the e-cigarettes, and 12% in the 26+ age group. Over a quarter reported doing so about once a week in both group samples. In the 18-25 age group, less than one in ten used them once a month or less (9%). This share was lower in the 26+ age group at 6% using it once a month or less.

Figure 132. Frequency of use of current e-cigarette users (18-25 age group)

Percentage and number of current users in the 18-25 age group



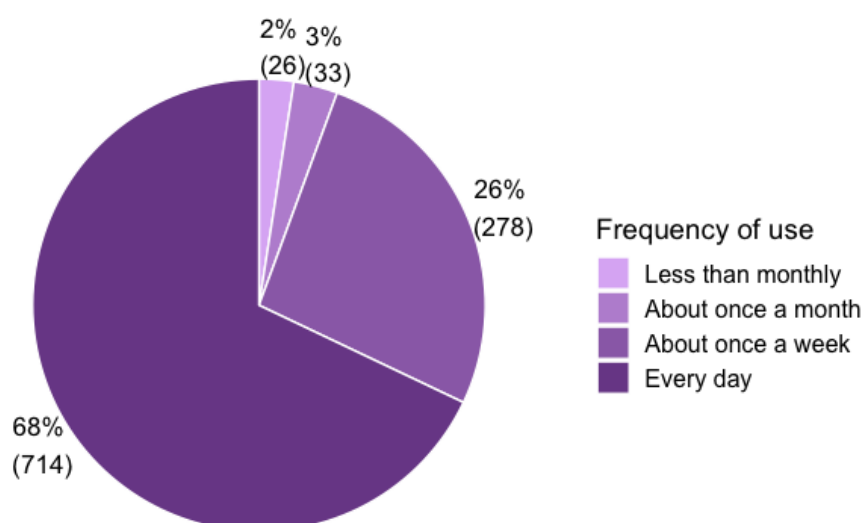
Base: current users of electronic cigarettes in the 18-25 age group (N=919)

Question: Q.1.3. How often do you use e-cigarettes?

Source: OE elaboration from online survey data

Figure 133. Frequency of use of current e-cigarette users (26+ age group)

Percentage and number of current users in the 26+ age group



Base: current users of electronic cigarettes in the 26+ age group (N=1,051)

Question: Q.1.3. How often do you use e-cigarettes?

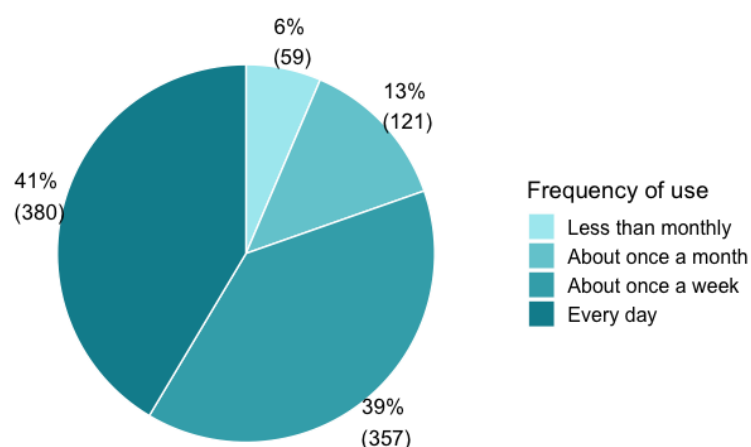
Source: OE elaboration from online survey data

The consumption patterns of past e-cigarette users in the **18-25 age group** were different to those of current users (see Figure 134). The percentage of past daily users was 41% compared to 61% for current users. In the **26+ age group**, 55% were past daily users compared to 68% for current users. This trend is consistent with the commentary from the focus groups. Participants identified e-cigarettes as an aid to reduce or quit another tobacco product. The level of nicotine intake can be progressively decreased, until users were able to get rid of the addiction or the craving for tobacco. It suggests that participants mainly use 'open-system' e-cigarettes as this feature is not available in 'closed systems'. However, it may not be an effective tool for all users:

"And the search for a sensation that is stronger and stronger! It's a crescendo! Because we can start with the e-cigarette at zero nicotine and then we will put some nicotine in it as we don't find what we had was enough and then we keep going higher." (F, 49, FR)

Figure 134. Frequency of use of past e-cigarette users (18-25 age group)

Percentage and number of past users in the 18-25 age group



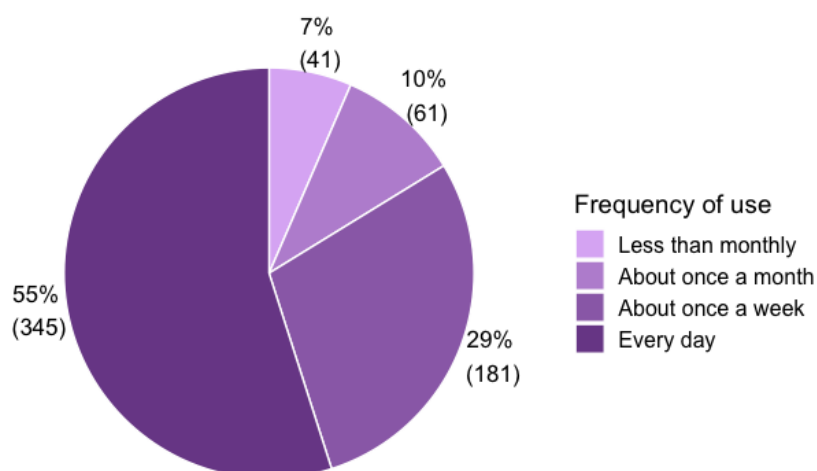
Base: past users of electronic cigarettes in the 18-25 age group (N=917)

Question: Q.1.6. How often did you use e-cigarettes?

Source: OE elaboration from online survey data

Figure 135. Frequency of use of past e-cigarette users (26+ age group)

Percentage and number of past users in the 26+ age group



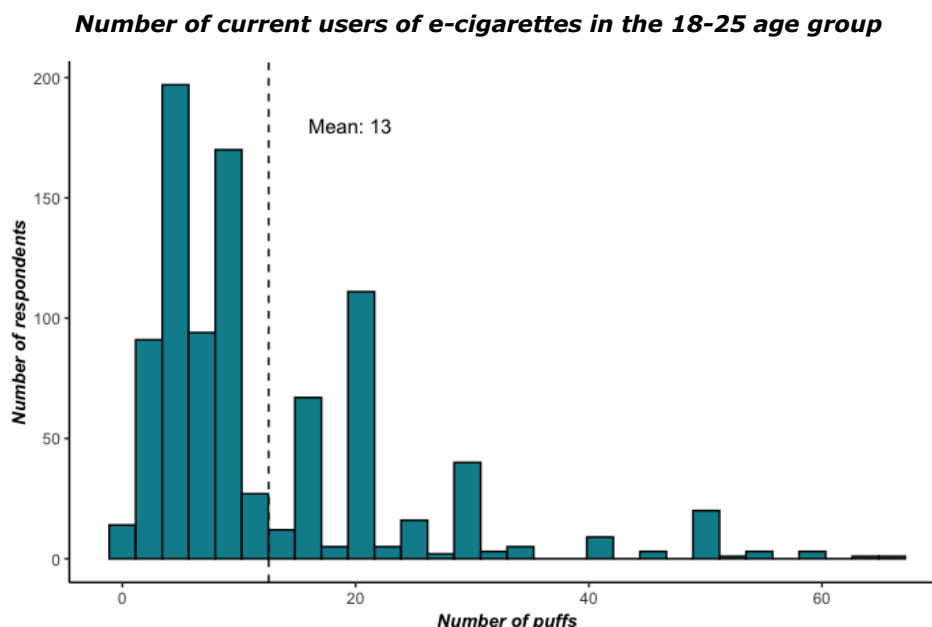
Base: past users of electronic cigarettes in the 26+ age group (N=628)

Question: Q.1.6. How often did you use e-cigarettes?

Source: OE elaboration from online survey data

Current users were also asked about the number of puffs per session. The average number of puffs taken per session in the **18-25 age group** was 13 puffs (see Figure 136). The histogram is skewed to the right, indicating that consumers usually inhale less than 10 puffs per session. Current users in **the 26+ age group** followed the same pattern (see Figure 137), with 12 puffs per session on average.

Figure 136. Average number of puffs per session (18-25 age group)

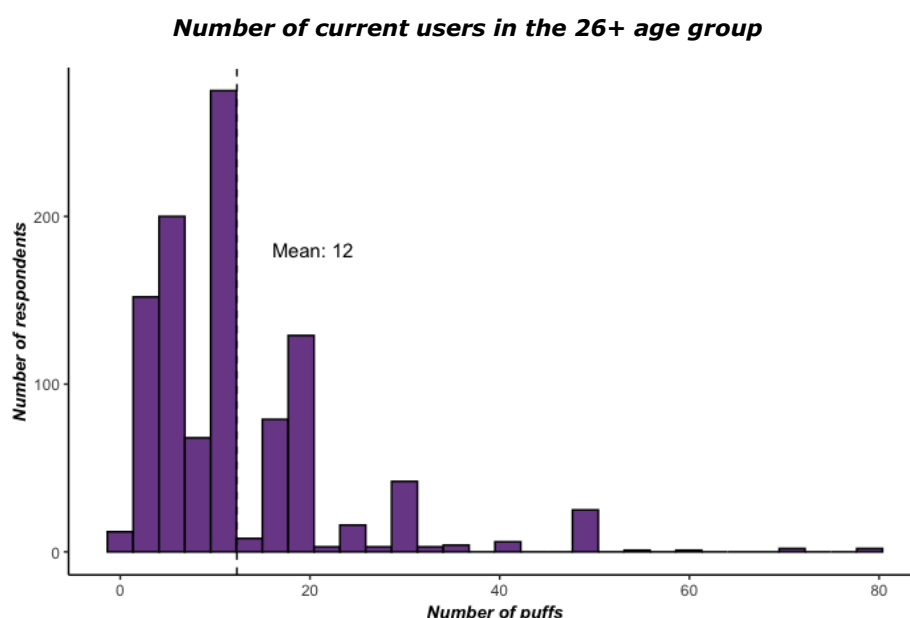


Base: current users of electronic cigarettes in the 18-25 age group (N=919)

Question: Q.1.4.b. On average, how many puffs do you take during a vaping session?

Source: OE elaboration from online survey data

Figure 137. Average number of puffs per session (26+ age group)



Base: current users of electronic cigarettes in the 26+ age group (N=1,051)

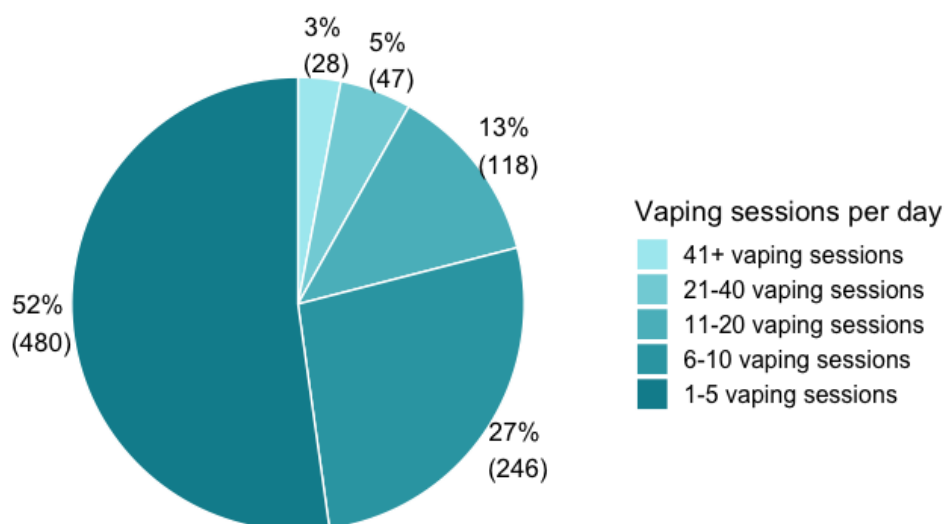
Question: Q.1.4.b. On average, how many puffs do you take during a vaping session?

Source: OE elaboration from online survey data

Figure 138 and Figure 139 show the number of sessions per day for current users. 52% of current users in the 18-25 age group reported 1 to 5 sessions per day; 45% of the 26+ age group reported similarly.¹²²

Figure 138. Average number of sessions per day of use (18-25 age group)

Percentage and number of current users in the 18-25 age group



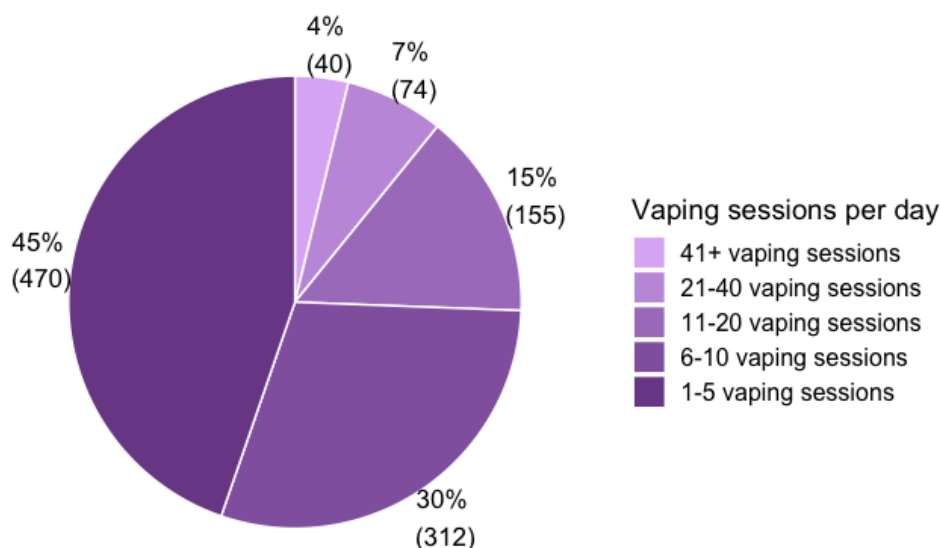
Base: current users of electronic cigarettes in the 18-25 age group (N=919)

Question: Q.1.4.c. On the days that you vape, about how many vaping sessions do you have?

Source: OE elaboration from online survey data

Figure 139. Average number of sessions per day (26+ age group)

Percentage and number of current users in the 26+ age group



Base: current users of electronic cigarettes in the 26+ age group (N=1051)

Question: Q.1.4.c. On the days that you vape, about how many vaping sessions do you have?

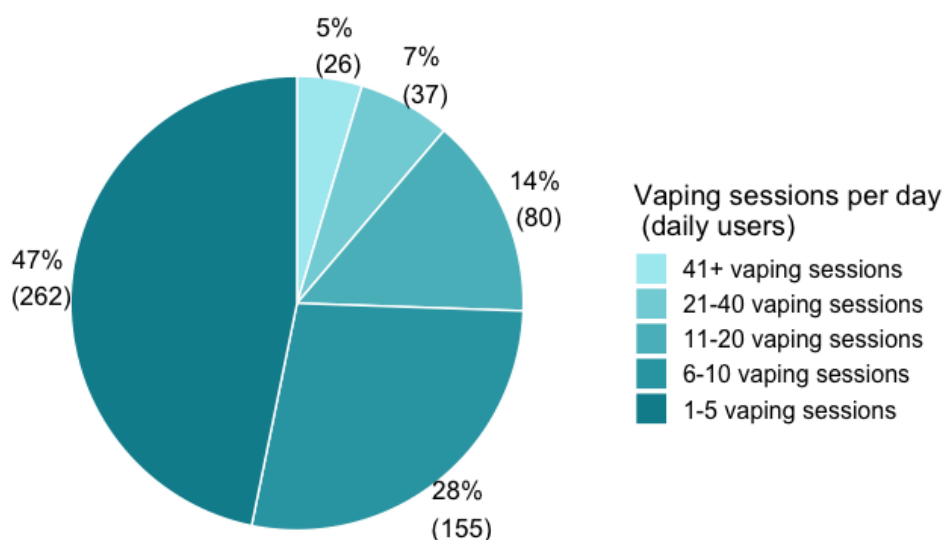
Source: OE elaboration from online survey data

¹²² The correlation between the number of puffs and the number of sessions for the 18-25 age group at 0.21 (p-value <0.01) was statistically significant. In comparison, the correlation for the respondents in the 26+ age group was smaller and insignificant (corr= 0.05, p-value=0.07). The findings suggested that the intensity of use of vaping sessions (i.e. the number of puffs) increased with the number of sessions of vaping per day of use, particularly for the 18-25 age group.

Figure 140 and Figure 141 show the reported number of sessions for daily e-cigarette users. While in both age groups 1-5 sessions daily is the highest percentage, 42% of the **18-25 age group**, and 47% of the **26+ age group** reported between 6 and 20 sessions daily – an indication of addiction?

Figure 140. Number of sessions per day, for daily users (18-25 age group)

Percentage and number of current daily users in the 18-25 age group)



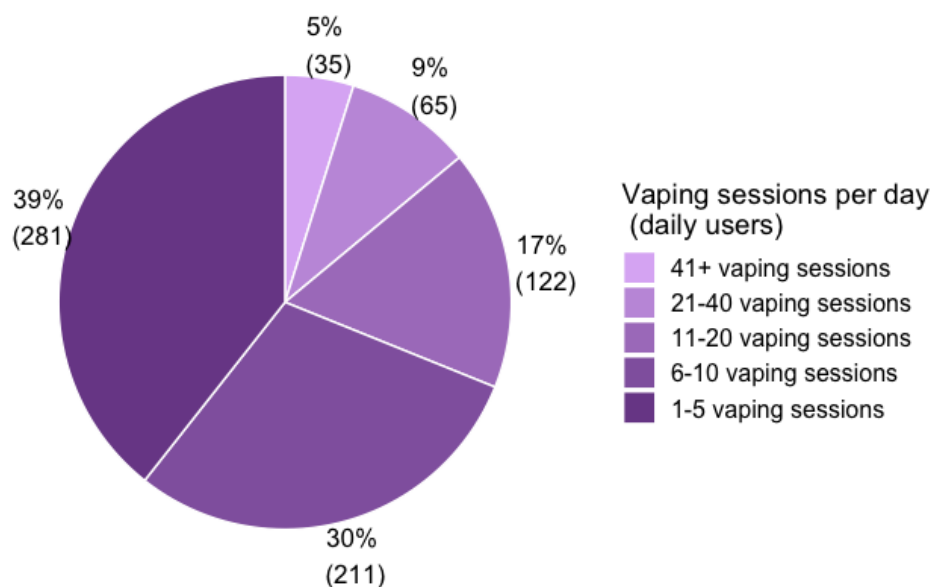
Base: daily current users of electronic cigarettes in the 18-25 age group (N=560)

Question: Q.1.4.c. On the days that you vape, about how many vaping sessions do you have?

Source: OE elaboration from online survey data

Figure 141. Number of sessions per day, for daily users (26+ age group)

Percentage and number of current daily users in the 26+ age group)



Base: daily current users of electronic cigarettes in the 26+ age group (N=714)

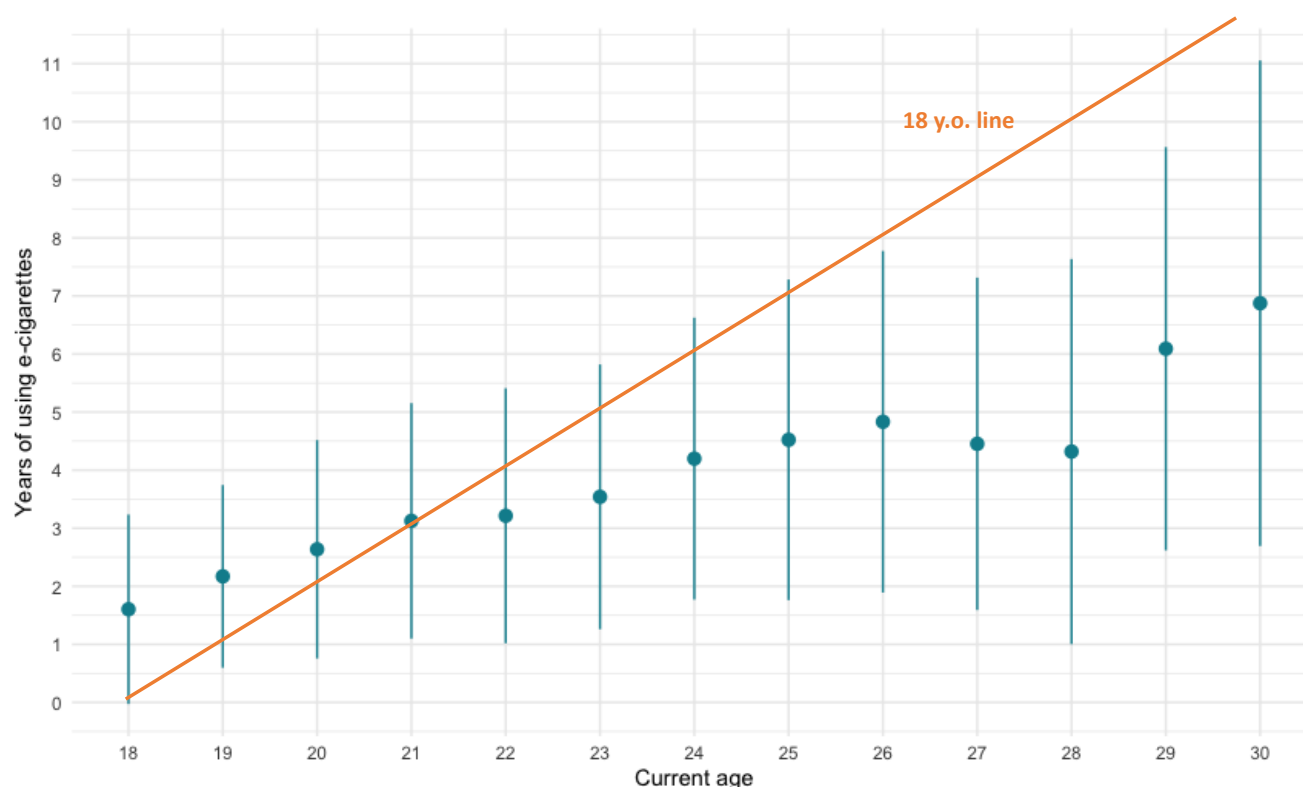
Question: Q.1.4.c. On the days that you vape, about how many vaping sessions do you have?

Source: OE elaboration from online survey data

On average, respondents that were 18 years old had used e-cigarettes for approximately 1.5 years – since they were 16 years old. Indeed, for respondents between the ages of 18 and 21, the average starting age was below 18 years. E-cigarettes were introduced to Europe in 2006 and the survey data showed that most respondents over the age of 30 had begun using e-cigarettes around 5 or 6 years ago, suggesting that the market reached maturity around 2014.

Figure 142. Initiation of use for current and past e-cigarette users (total sample)

(number of current and past users in the overall sample)



Base: current and past users of electronic cigarettes in the total sample (N=3,221¹²³)

Question: Q.1.2.a. At about what age did you start using the products below? & Q.5.2. Age

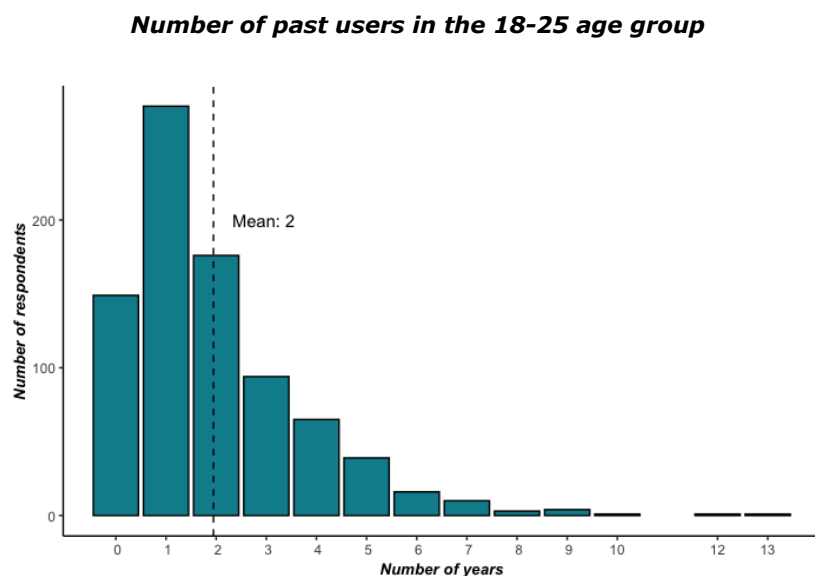
Note: The figure illustrates the approximate starting age of e-cigarette use among all respondents of the online survey, where the x-axis represents the current age of the respondent (Q.5.2) and the y-axis is the average numbers of years of product consumption (calculated by subtracting the current age from the starting age). The orange line represents the 45° or 18 y.o. line. Each blue point represents the average starting age for respondents of that age; points above the 18 y.o. line reveal those initiating consumption of e-cigarettes before the age of 18, while those below the line reveal age brackets starting consumption after the age of 18 on average. The blue lines represent the mean standard deviation, i.e. it does not show the full range of values for each age.

Source: OE elaboration from online survey data

Figure 143 and Figure 144 show the average duration of use of e-cigarettes for past users. Two years for the **18-25 age group** and three years for the **26+ age group**. For both groups, the histograms are skewed towards the lower number of years, indicating a short period of use of the product. Whether this evidenced successful cessation of e-cigarettes and other tobacco products or a switch back to conventional cigarettes is discussed in Table 58 and Table 59.

¹²³ NA and misreported values are excluded.

Figure 143. Duration of use of past e-cigarette users (18-25 age group)



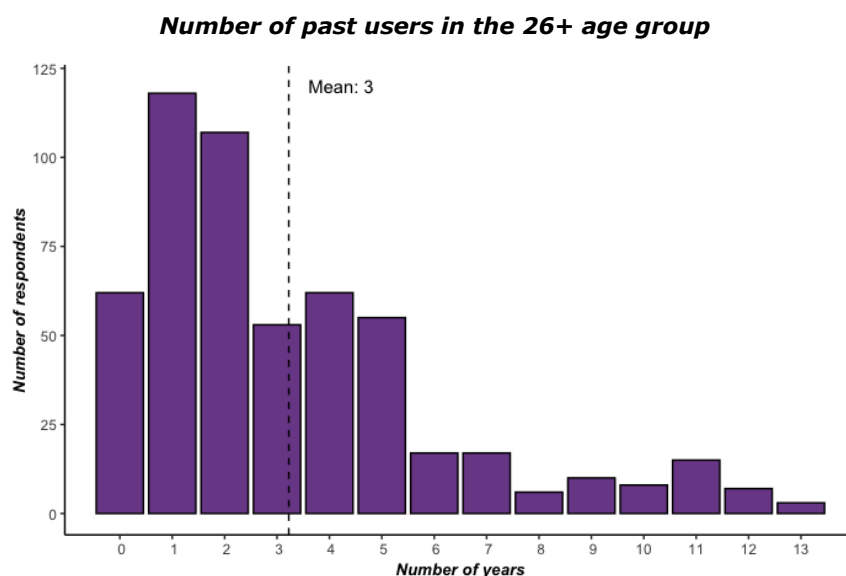
Base: past users of electronic cigarettes in the 18-25 age group (N=837¹²⁴)

Question: Q.1.2.a. At about what age did you start using e-cigarettes? & Q.1.2.b. At about what age did you stop using e-cigarettes?

Note: The duration was calculated by subtracting Q.1.2.a. from Q.1.2.b for each individual that was a past user of the product. The mean shows the average duration of use. We cut off the data at 13 years, since the product was introduced in 2006.

Source: OE elaboration from online survey data

Figure 144. Duration of use of past e-cigarette users (26+ age group)



Base: past users of electronic cigarettes in the 26+ age group (N=564¹²⁵)

Question: Q.1.2.a. At about what age did you start using e-cigarettes? & Q.1.2.b. At about what age did you stop using e-cigarettes?

Note: The duration was calculated by subtracting Q.1.2.a. from Q.1.2.b for each individual that was a past user of the product. The mean shows the average duration of use. We cut off the data at 13 years, since the product was introduced in 2006.

Source: OE elaboration from online survey data

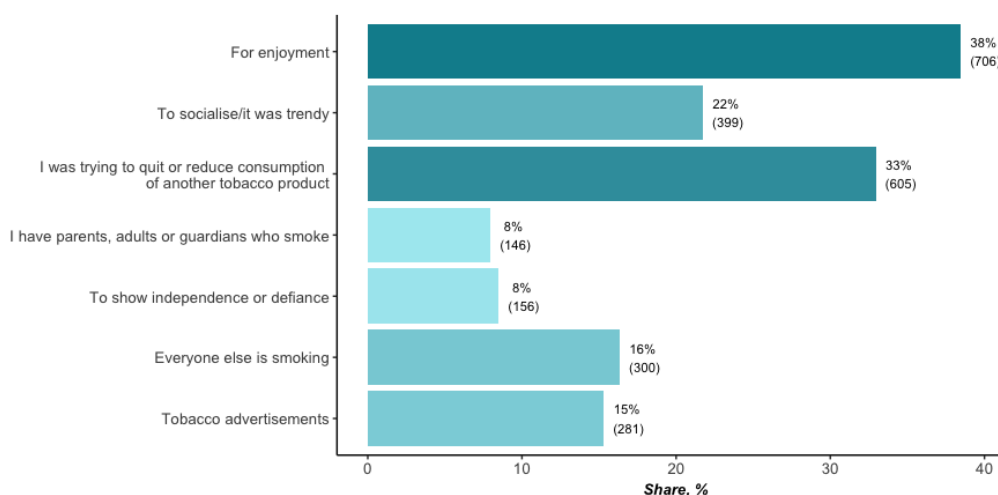
¹²⁴ NA and misreported values are excluded.

¹²⁵ NA and misreported values are excluded.

Current and former e-cigarette users were asked about the reasons for starting to use the product. In the **18-25 age group** (see Figure 145) 38% cited "for enjoyment," 33% because they were trying to quit or reduce the consumption of another tobacco product and 22% 'to socialise and its trendy". Furthermore, 15% reported starting due to advertisements. In the **26+ age group** (see Figure 146) 54% reported that they has begun to use e-cigarettes to stop or reduce the consumption of another tobacco product, 28% cited for 'enjoyment' and 19% attributed it to advertisements. The key driver for young people was enjoyment while for older respondents it was health.

Figure 145. Initiation of e-cigarette use: reasons (18-25 age group)

Percentage and number of current and past users in the 18-25 age group)



Base: current and past users of electronic cigarettes in the 18-25 age group (N=1,836)

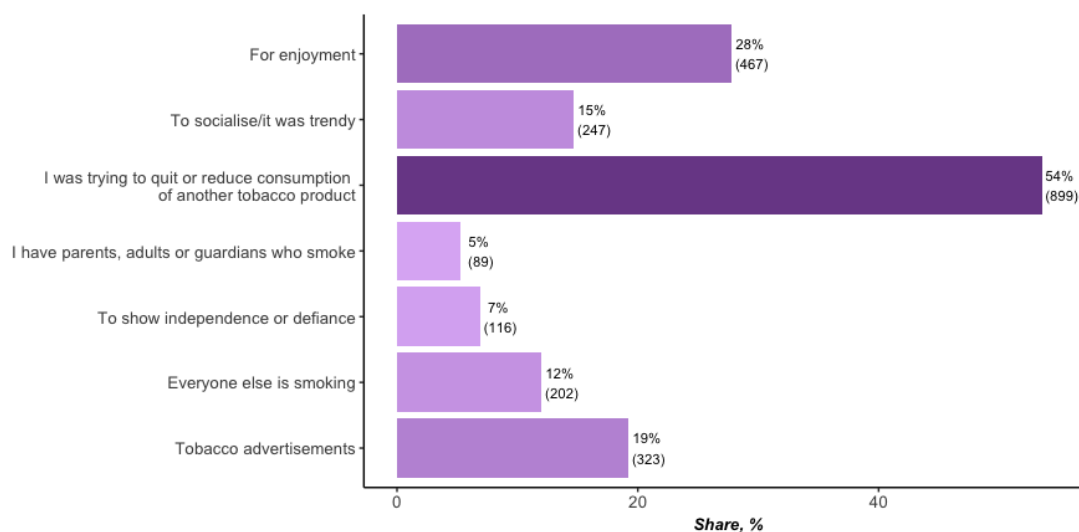
Question: Q.2.1.a. Why did you start using e-cigarettes? Please select up to 2 options.

Note: Each bar represents the share of respondents that selected the option within the age group. The percentages do not add up to 100% because respondents could select more than one term.

Source: OE elaboration from online survey data

Figure 146. Initiation of e-cigarette use: reasons (26+ age group)

Percentage and number of current & past users in 26+ age group



Base: current and past users of electronic cigarettes in the 26+ age group (N=1,679)

Question: Q.2.1.a. Why did you start using e-cigarettes? Please select up to 2 options.

Note: Each bar represents the share of respondents that selected the option within the age group. The percentages do not add up to 100% because respondents could select more than one term.

Source: OE elaboration from online survey data

The same question about reasons for trying e-cigarettes was asked of those who had only tried them once or twice. For **the 18-25 age group** (see Figure 147) 41% cited enjoyment, 30% to socialise or that it was trendy and 17% because everyone else was using e-cigarettes – essentially social reasons. **The 26+ age group** (see Figure 148) health was the dominant motivation with 47% saying were trying to quit or reduce the consumption of another tobacco product. Around one in five cited enjoyment, socialising and advertisements.

These findings resonate with comments in focus groups. Several features of e-cigarettes allowed the product to be viewed as an aid to quitting or cutting down. The open-tank system e-cigarettes allow users to customise the amount of nicotine intake and the number of puffs taken for each session. This to progressively decrease the nicotine content and reduce the craving for tobacco consumption. The sense of control over the product may translate into a feeling of reduced harmfulness. Yet, it may not always appear to be an effective tool.

"I myself decide on the taste and the level of nicotine and prepare the liquid. A bit like a chemist." (M, 24, BG)

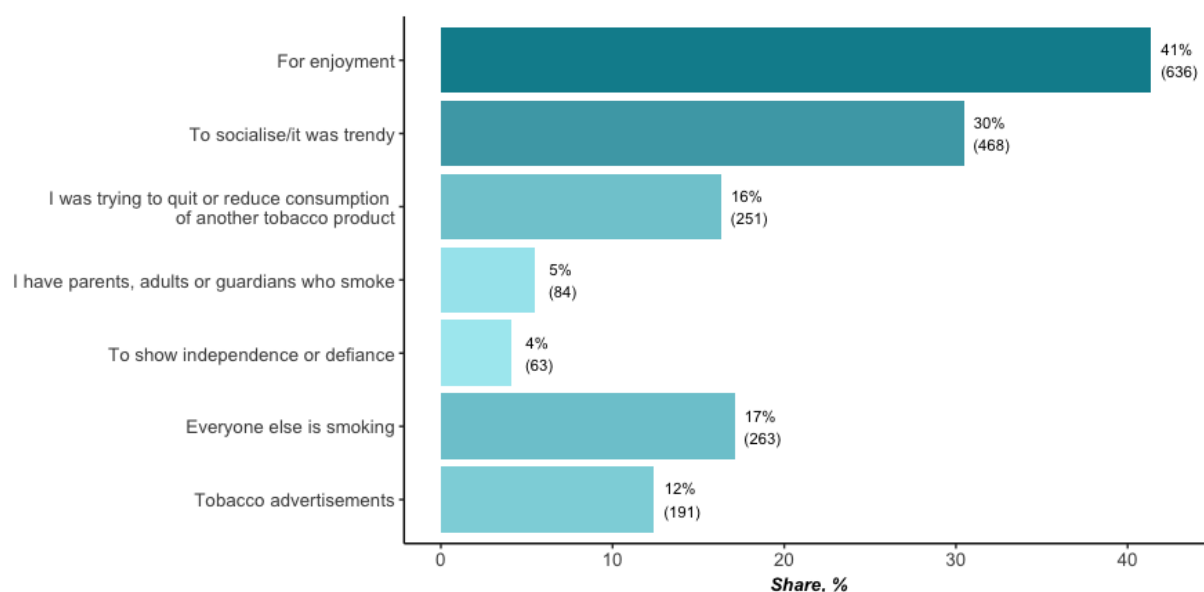
"You can control the dose more easily. [...] you can give three puffs and stop. For example, in my case... I go to the terrace and give this a few puffs... You can control the nicotine intake much more." (M, 34, ES)

"This is to get rid of the craving... That's if you want to stop smoking, instead of a patch, you give this a few puffs... because it also helps you with the habit of inhaling and exhaling smoke." (M, 28, ES)

"Mainly because it was somewhat trendy, I was infected by young people and also wanted electronic, I wanted to change and see if I could stop. On the contrary, I returned to cigarettes." (M, 20, EL)

Figure 147. Initiation of e-cigarette use: reasons (18-25 age group)

Percentage and number of those who tried once or twice in the 18-25 age group



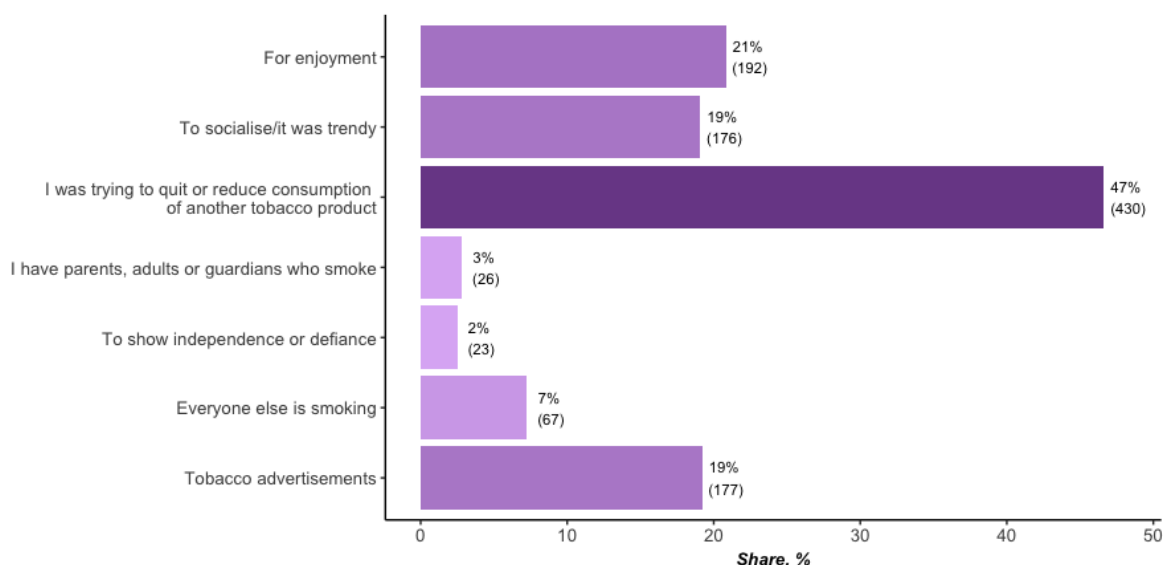
Base: those who tried to use electronic cigarettes in the 18-25 age group (N=1,537)

Question: Q.2.1.b. Why did you try e-cigarettes? Please select up to 2 options.

Note: Each bar represents the share of respondents that selected the option within the age group. The percentages do not add up to 100% because of the fact that respondents could select more than one term – therefore, the bars must be interpreted separately.

Source: OE elaboration from study survey data

Figure 148. Initiation of e-cigarette use: reasons (26+ age group)

Percentage and number of those who tried once or twice in the 26+ age group

Base: those who tried to use electronic cigarettes once or twice in the 26+ age group (N=922)

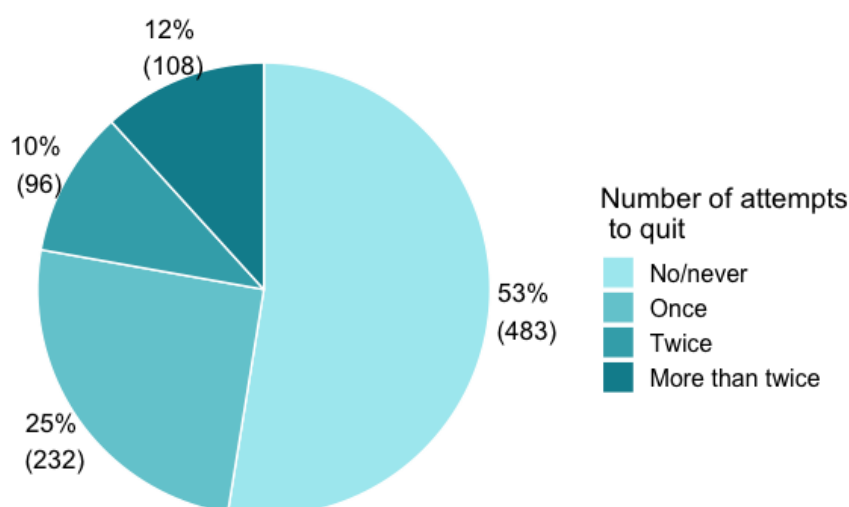
Question: Q.2.1.b. Why did you try e-cigarettes? Please select up to 2 options.

Note: Each bar represents the share of respondents that selected the option within the age group. The percentages do not add up to 100% because of the fact that respondents could select more than one term – therefore, the bars must be interpreted separately.

Source: OE elaboration from online survey data

E-cigarette users were asked whether they had tried to quit. In the **18-25 age group**, 53% of current users reported having never attempted to quit e-cigarette use. 25% had tried to quit once and the remaining 22% had tried twice or more times. (see Figure 149). 64% of current users in the **26+ age group** had never attempted to quit, 17% reported that they had tried to quit once and 20% had attempted to quit at least twice in the past (see Figure 150).

Figure 149. Cessation: attempts to quit e-cigarettes (18-25 age group)

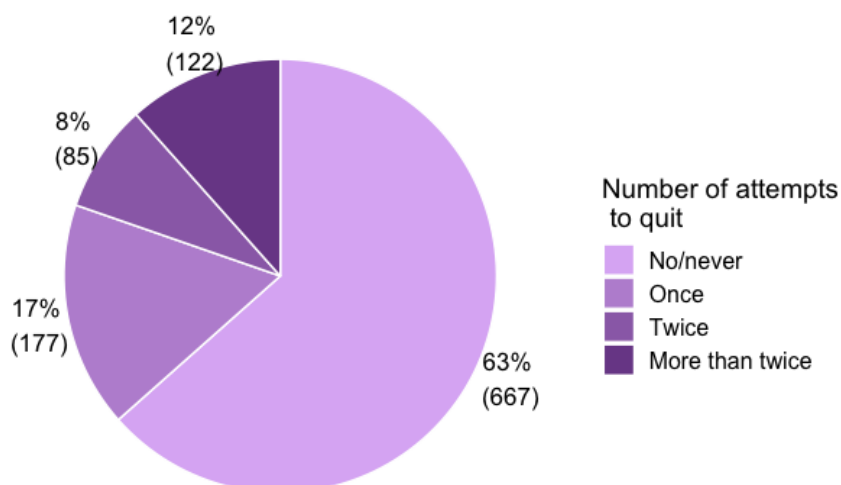
Percentage and number of current users in the 18-25 age group

Base: current users of electronic cigarettes in the 18-25 age group (N=919)

Question: Q.1.5. Have you ever tried to quit e-cigarettes? If yes, how many times?

Source: OE elaboration from online survey data

Figure 150. Cessation: attempts to quit e-cigarettes (26+ age group)

Percentage and number of current users in the 26+ age group

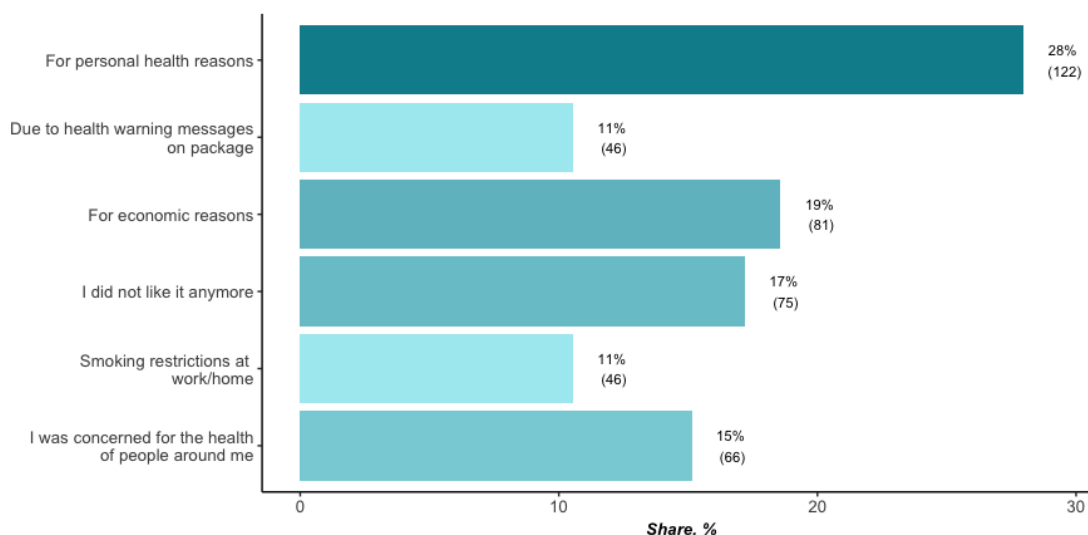
Base: current users of electronic cigarettes in the 26+ age group (N=1,051)

Question: Q.1.5. Have you ever tried to quit e-cigarettes? If yes, how many times?

Source: OE elaboration from online survey data

For the **18-25 age group** (see Figure 151) health was the dominant reason for trying to quit. 28% cited health, 11% health warning messages and 15% were concerned about 'the health of people around me'. About one in five cited economic reasons and 'not liking it anymore'. A similar pattern emerged in the **26+ age group** (see Figure 152): 41% cited health reasons and health warning messages, and 10% concerns about passive smoking.

Figure 151. Cessation: reasons for trying to quit e-cigarette use (18-25 age group)

Percentage and number of current users in the 18-25 age group

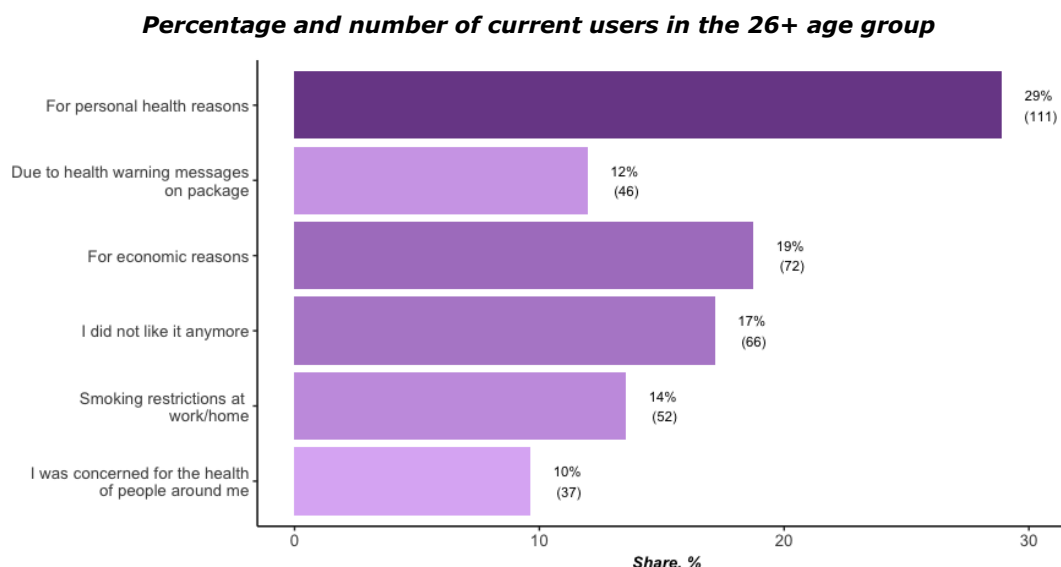
Base: current users of electronic cigarettes that tried to quit in the 18-25 age group (N=436)

Question: What made you try to quit e-cigarettes? Please select the most fitting answer.

Note: Respondents were asked to provide only one option. Therefore, each bar represents the share of respondents that selected the option within the age group.

Source: OE elaboration from online survey data

Figure 152. Cessation: reasons for trying to quit e-cigarette use (26+ age group)



Base: current users of electronic cigarettes that tried to quit in the 26+ age group (N=384)

Question: What made you try to quit e-cigarettes? Please select the most fitting answer.

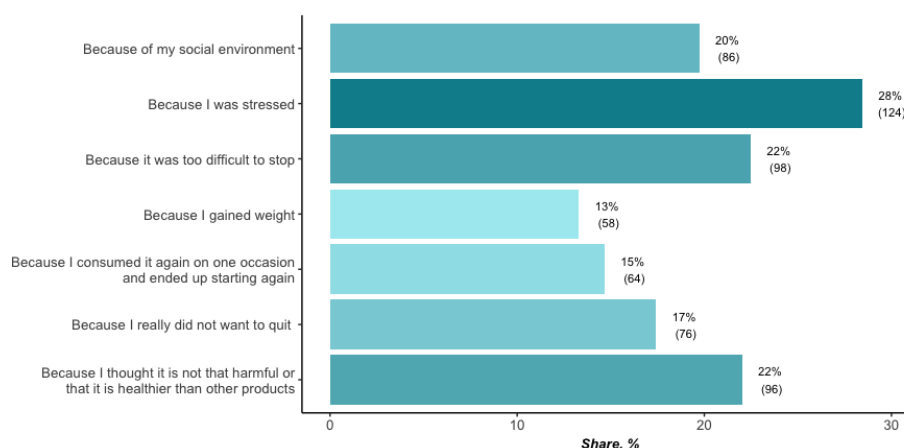
Note: Respondents were asked to provide only one option. Therefore, each bar represents the share of respondents that selected the option within the age group.

Source: OE elaboration from online survey data

A number of reasons for re-starting e-cigarette use after an attempt to quit were reported. In the **18-25 age group** (see Figure 153) 28% cited stress, 22% that it was difficult to stop and 22% that it was healthier than other tobacco products. Almost one in five said they really did not want to stop or that it was due to their social environment. The **26+ age group** (see Figure 154) cited the same issues. 'Stress 25%, difficulty 24% and healthier than other tobacco products 22%. As with the younger group about one in five said that they really did not want to stop or that the social environment was the reason.

Figure 153. Reasons for starting to use e-cigarettes again (18-25 age group)

Percentage and number of current users that had attempted to quit in the 18-25 age group



Base: current users of electronic cigarettes that tried to quit in the 18-25 age group (N=436)

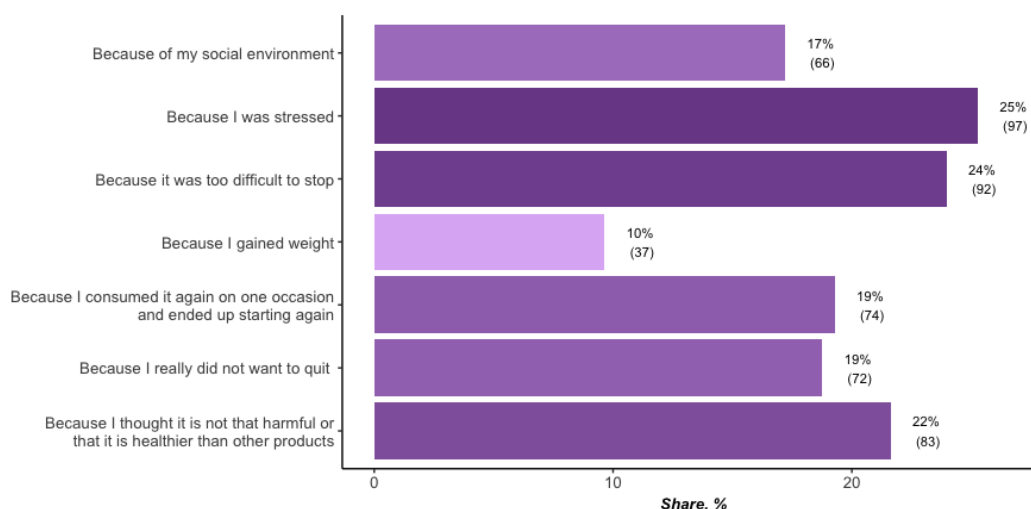
Question: Q.2.3. Why did you start to use e-cigarettes again? Please select up to 2 options.

Note: Each bar represents the share of respondents that selected the option within the age group. The percentages do not add up to 100% because of the fact that respondents could select more than one term – therefore, the bars must be interpreted separately.

Source: OE elaboration from online survey data

Figure 154. Reasons for starting to use e-cigarettes again (26+ age group)

Percentage and number of current users that had attempted to quit in the 26+ age group



Base: current users of electronic cigarettes that tried to quit in the 26+ age group (N=384)

Question: Q.2.3. Why did you start to use e-cigarettes again? Please select up to 2 options.

Note: Each bar represents the share of respondents that selected the option within the age group. The percentages do not add up to 100% because of the fact that respondents could select more than one term – therefore, the bars must be interpreted separately.

Source: OE elaboration from online survey data

6.2.3 Parallel use with other tobacco and related products

In the **18-25 age group**, 74% of current e-cigarette users were dual users, i.e. they also used other tobacco and related products in parallel (see Table 56). The greater majority were also smoking conventional cigarettes. Close to one fifth of current users (18%) indicated that they had previously used other products but were now only using e-cigarettes. Less than one in ten (8%) had only ever used e-cigarettes. The same pattern was found for the **26+ age group** (see Table 57). Here 78% were dual users and 90% reported smoking traditional cigarettes.

These findings on dual use resonate with the focus group discussions. People said that the main purpose for using e-cigarettes was to quit conventional cigarettes; to alternate e-cigarettes with conventional tobacco products in order to reduce consumption of the latter; to use the e-cigarette in places where smoking was not allowed but e-cigarettes were tolerated (at some work places, bars, malls, etc. , also depending on the country); or, to use in places where smoking is not forbidden, but it was preferable not to smoke in order to avoid harming others (e.g. at home) – the latter was particularly relevant for those with children.

The online survey findings suggested that the e-cigarettes as a tool for cessation is not a magic bullet. Only 18% of respondents managing to consume only e-cigarettes after having used other tobacco products in the past.

Table 56. Career flows of use for current e-cigarette users (18-25 age group)

Number and percentage of current users in the 18-25 age group

			Number	%
Total			919	100
Never users of other products			77	8
Exclusive current user of e-cigarettes & past user of other product(s)			164	18
Out of them, switching from (one or more products) *				
Cigarettes	(n=143)	87% *		
Slim cigarettes	(n=51)	31% *		
Waterpipe tobacco	(n=50)	30% *		
Small cigarillos	(n=32)	19% *		
HTPs	(n=29)	18% *		
Current user of e-cigarettes & both current and past user of other product(s)			678	74
Other products consumed in parallel now (one or more products) *				
Cigarettes	(n=555)	82% *		
Slim cigarettes	(n=274)	40% *		
Waterpipe tobacco	(n=281)	41% *		
Small cigarillos	(n=218)	32% *		
HTPs	(n=277)	41% *		

Base: current users of e-cigarettes in the 18-25 age group (N=919)

Question: Q.1.1. Have you ever used e-cigarettes?

Note: Patterns of use and consumption careers are explored for different consumer profiles of current users. A breakdown is provided for the two variables of interest, i.e. current parallel use patterns. The last row only provides the breakdown of current dual use as this is the variable of interest; it does not include a breakdown that takes account past consumption of products to avoid confusion (from too many permutations).

(*) Percentages add up to more than 100% on account of the prevalence of dual use.

Source: OE elaboration from online survey data

Table 57. Career flows of current e-cigarette users (26+ age group)

Number and percentage of current users in the 26+ age group

	Number	%
Total	1051	100
Never users of other products	41	4
Exclusive current user of e-cigarettes & past user of other product(s)	189	18
Out of them, switching from (one or more products)		
Cigarettes (n=177)	94% *	
Slim cigarettes (n=43)	25% *	
Waterpipe tobacco (n=47)	23% *	
Small cigarillos (n=26)	16% *	
HTPs (n=31)	14% *	
Current user of e-cigarette & both current and past user of other product(s)	821	78
Other products consumed in parallel now (one or more products) *		
Cigarettes (n=736)	90% *	
HTPs (n=362)	44% *	
Slim cigarettes (n=284)	35% *	
Small cigarillos (n=231)	28% *	
Waterpipe tobacco (n=202)	25% *	

Base: current users of electronic cigarettes in the 26+ age group (N=1,051)

Question: Q.1.1. Have you ever used e-cigarettes?

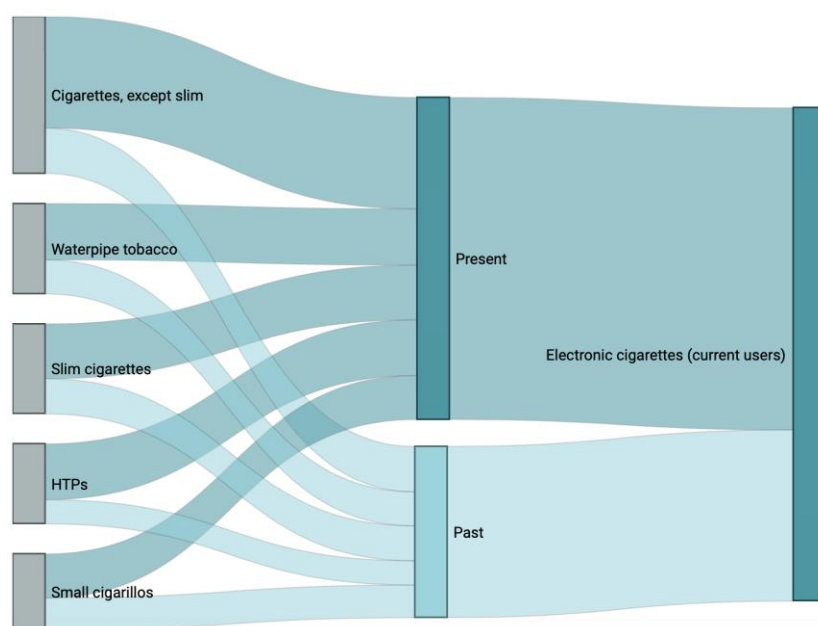
Note: Patterns of use and consumption careers are explored for different consumer profiles of current users. A breakdown is provided for the two variables of interest, i.e. current parallel use patterns. The last row only provides the breakdown of current dual use as this is the variable of interest; it does not include a breakdown that takes account past consumption of products to avoid confusion (from too many permutations).

(*) Percentages add up to more than 100% on account of the prevalence of dual use.

Source: OE elaboration from online survey data

Figure 155 and Figure 156 provide a visual representation of the career flows for current users of e-cigarettes. The objective of the figures is to highlight dual use patterns, and as such exclusive e-cigarette users and those not using other products were excluded. In the visualisation the thickness of the tails is only indicative. As previously noted, a majority of current e-cigarette users in the **18-25 age group** were dual users of tobacco products; in the main conventional cigarettes (see Figure 155). The visualisation for the **26+ age group** (see Figure 156) showed that the majority of current e-cigarette users had also used other tobacco products in the past, but there is a lower level of current dual use. The finding supports the hypothesis that e-cigarettes were perceived more strongly as a cessation tool among the 26+ age group, while the younger generations may be influenced by trends, potentially supporting the gateway theory (see Figure 145 and Figure 146 on reasons for initiation of use).

Figure 155. Representation of career flows for current e-cigarette users (18-25 age group)



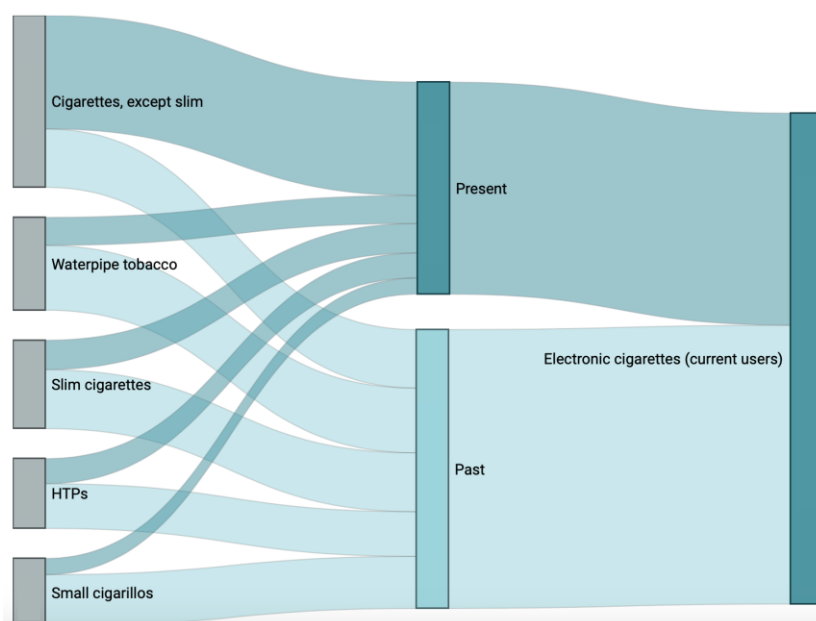
Base: current users of e-cigarettes (excluding never users of other products) in the 18-25 age group (N=842)

Question: Q.1.1. Have you ever used e-cigarettes?

Note: The diagram is a simplified version of the career flows of dual e-cigarette users. The thickness/width of the tails, as well as the percentages included, are only indicative. Some individuals may be represented twice (or more) if they use/have used more than one product at a time. The diagram highlights dual use patterns as well as transitions to/from other products; therefore, never users of other products were excluded.

Source: OE elaboration from online survey data

Figure 156. Representation of career flows for current e-cigarette users (26+ age group)



Base: current users of e-cigarettes (excluding never users of other products) in the 26+ age group (N=1,010)

Question: Q.1.1. Have you ever used e-cigarettes?

Note: The diagram is a simplified version of the career flows of dual e-cigarette users. The thickness/width of the tails, as well as the percentages included, are only indicative. Some individuals may be represented twice (or more) if they use/have used more than one product at a time. The diagram highlights dual use patterns as well as transitions to/from other products; therefore, never users of other products were excluded. Source:

OE elaboration from online survey data

Table 58 and Table 59 show the career flows of past users of e-cigarettes. In the **18-25 age group**, 24% of past e-cigarette users appeared to have used the product as a cessation aid tool successfully, having stopped consuming other tobacco products at the time of the survey. There were 23% exclusive past e-cigarette users that had transitioned to other tobacco products, with most switching to conventional cigarettes. In the **26+ age group**, of former users of e-cigarettes 23% had stopped consuming tobacco products while 43% of had switched to other tobacco products – mainly conventional cigarettes. It is also of particular note that in the **18-25 age group**, 23% exclusive past users of e-cigarettes (i.e. never users of other tobacco products) were now consuming other products. Of these 87% reported smoking traditional cigarettes. In the **26+ age group**, 31% exclusive past e-cigarette users were now dual users of whom 96% reported smoking traditional cigarettes. These findings point to the conclusion that for some people at any age e-cigarettes may serve as a gateway product. The focus groups hint at a possible explanation. It was said that e-cigarettes are not very satisfying and did not give the pleasures associated with smoking a conventional cigarette. They are not a substitute for a traditional cigarette and as such have limited appeal as an aid to quitting (see section 6.3). The findings of the survey appear to confirm the view that e-cigarettes may not always be an effective aid to quitting. 23% of past users managed to quit all tobacco and related products in the 26+ age group.

Table 58. Career flow of past e-cigarette users (18-25 age group)

Number and percentage of past users in the 18-25 age group

	Number	%
Total	917	100
Never user of other products	63	7
Dual past user & not using any products now	223	24
Exclusive past user of e-cigarettes & current user of other products (but not e-cigarettes)	215	23
<i>Out of them, switching to (one or more products) *</i>		
Cigarettes (n=188)	87% *	
Slim cigarettes (n=43)	20% *	
Waterpipe tobacco (n=36)	17% *	
Small cigarillos (n=24)	11% *	
HTPs (n=17)	8% *	
Dual past user & current user of other products (but not e-cigarettes)	416	45
<i>Other products currently consumed in parallel (one or more products) *</i>		
Cigarettes (n=314)	75% *	
Slim cigarettes (n=95)	23% *	
HTPs (n=87)	21% *	
Waterpipe tobacco (n=82)	20% *	
Small cigarillos (n=55)	13% *	

Base: past users of e-cigarettes (excluding never users of other products) in the 18-25 age group (N=917)

Question: Q.1.1. Have you ever used e-cigarettes?

Note: Patterns of use and consumption careers are explored for different consumer profiles of past users. A breakdown is provided for the variables of interest, i.e. those who continued using other tobacco products after using e-cigarettes. The last row only provides the breakdown of current consumption because this is the variable of interest; it does not include a breakdown that takes account past consumption of products to avoid confusion (from too many permutations).

(*) Percentages add up to more than 100% on account of the prevalence of dual use.

Source: OE elaboration from online survey data

Table 59. Career flows of past e-cigarette users (26+ age group)
Number and percentage of past users in the 26+ age group

	Number	%
Total	628	100
Never users of other products	16	2
Dual past user & not using any products now	144	23
Exclusive past user of e-cigarettes & current user of other products (but not e-cigarettes)	198	31
<i>Out of them, switching to (one or more products) *</i>		
Cigarettes (n=191)	96% *	
Slim cigarettes (n=40)	20% *	
Waterpipe tobacco (n=38)	19% *	
Small cigarillos (n=19)	10% *	
HTPs (n=11)	6% *	
Dual past user & current user of other products (but not e-cigarettes)	270	43
<i>Other products currently consumed (one or more products) *</i>		
Cigarettes (n=202)	75% *	
Slim cigarettes (n=70)	26% *	
HTPs (n=73)	27% *	
Waterpipe tobacco (n=45)	17% *	
Small cigarillos (n=37)	14% *	

Base: past users of e-cigarettes in the 26+ age group (N=628)

Question: Q.1.1. Have you ever used e-cigarettes?

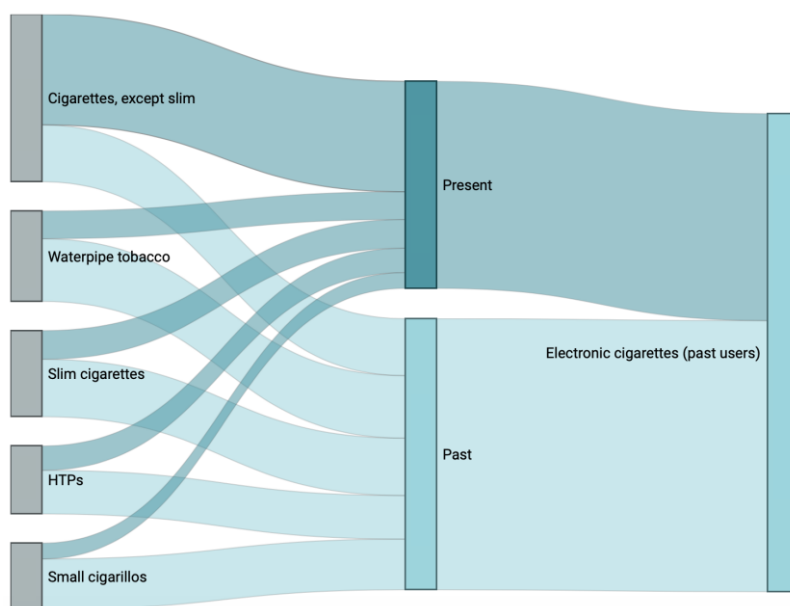
Note: Patterns of use and consumption careers are explored for different consumer profiles of past users. A breakdown is provided for the variables of interest, i.e. those who continued using other tobacco products after using e-cigarettes. The last row only provides the breakdown of current consumption because this is the variable of interest; it does not include a breakdown that takes account past consumption of products to avoid confusion (from too many permutations).

(*) Percentages add up to more than 100% on account of the prevalence of dual use.

Source: OE elaboration from online survey data

Figure 157 and Figure 158 provide a visual representation of dual product use behaviours. Most past users of e-cigarettes in both the **18-25 age group** and the **26+ age group** were dual users. It is important to note that the 'past' tail in the figures includes both dual users and those that switched between products. A majority of the past users still using other tobacco and related products had transitioned to cigarettes, as displayed by the thicker "present" tails in both age groups. Again, this resonates with comments from the focus group participants who noted that e-cigarettes were not always very satisfying, that the consumption of tobacco by methods other than burning were not expected to calm one down, and that the smoking ritual was experienced as "colder" emotionally-speaking (see section 6.3).

Figure 157. Representation of career flows for past e-cigarette users (18-25 age group)



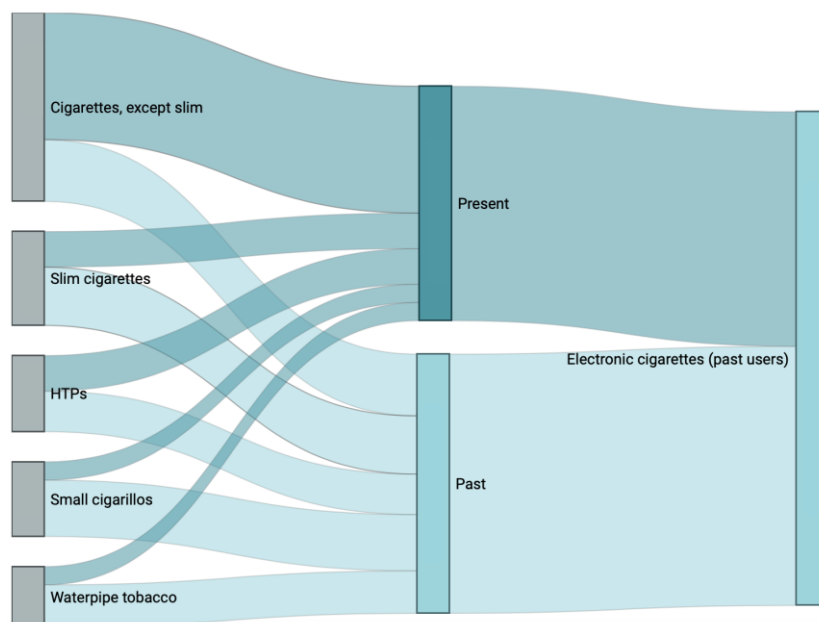
Base: past users of e-cigarettes (excluding never users of other products) in the 18-25 age group (N=854)

Question: Q.1.1. Have you ever used e-cigarettes?

Note: The diagram is a simplified version of the career flows of dual e-cigarette users. The thickness/width of the tails, as well as the percentages included, are only indicative. Some individuals may be represented twice (or more) if they use/have used more than one product at a time. The diagram highlights dual use patterns as well as transitions to/from other products; therefore, never users of other products were excluded.

Source: OE elaboration from online survey data

Figure 158. Representation of career flows for past e-cigarette users (26+ age group)



Base: past users of e-cigarettes (excluding never users of other products) in the 26+ age group (N=612)

Question: Q.1.1. Have you ever used e-cigarettes?

Note: The diagram is a simplified version of the career flows of dual e-cigarette users. The thickness/width of the tails, as well as the percentages included, are only indicative. Some individuals may be represented twice (or more) if they use/have used more than one product at a time. The diagram highlights dual use patterns as well as transitions to/from other products; therefore, never users of other products were excluded.

Source: OE elaboration from online survey data

6.3 Consumer perception

From the focus group discussions eight words or word combinations that people used to describe e-cigarettes were selected; four were positive associations and four negative. The associations were complicated, unhealthy, costly, addictive, practical, modern/technological, elegant/trendy/fashionable and enjoyable.

In the online survey respondents were asked which of the words best captured what they thought about e-cigarettes. The results are shown in Table 60 and Table 61. In both age groups the most frequent associations were 'unhealthy', 'addictive' and 'modern/technological'. A striking finding is the difference in perceptions of the current users and never users of e-cigarettes. For the **18-25 age group** a comparison between current and never users showed that 52% of the never users selected 'unhealthy' and 43% more chose 'addictive'. Similar differences were found in the **26+ age group** in which 64% more never users selected unhealthy and 54% more selected addictive. In terms of positive associations, there was a curious finding for 'elegant/trendy/fashionable': 10% more **18-25 age group** never users chose this association as did 23% more in the **26+ age group**.

Table 60. Perception of e-cigarettes (18-25 age group)
Percentage and number of respondents in the 18-25 age group

	Group total %(n)	Current users %	Past users %	Used once or twice %	Never used %
Use of product	100 (6090)	15	15	25	45
Complicated	11 (640)	16	16	25	42
Unhealthy	41 (2515)	6	11	26	58
Costly	26 (1608)	10	53	14	23
Addictive	37 (2281)	10	12	25	53
Practical	24 (1435)	22	17	26	35
Modern/technological	41 (2507)	15	15	29	42
Elegant/trendy/fashionable	31 (1907)	16	14	29	42
Enjoyable	25 (1518)	25	20	30	25

Base: all respondents in the 18-25 age group (N=6,090)

Question: Q.4.10. Which of the following words best captures what you think about vaping electronic cigarettes? Please tick as many boxes as you would like.

Note: Respondents could provide multiple answers. The group total category refers to the total respondents in that age group. The orange column represents the proportion of respondents in the age group that selected the response option (i.e. 11% of respondents in the 18-25 age group responded "complicated"). The white cells (in the four right columns) represent the breakdown by user type, i.e. they are within row percentages (white cells in each row sum to 100).

Source: OE elaboration from online survey data

Table 61 Perception of e-cigarettes (26+ age group)
Percentage and number of respondents in the 26+ age group

	Group total %(n)	Current users %	Past users %	Used once or twice %	Never used %
Use of product	100 (5910)	18	11	16	56
Complicated	11 (643)	12	14	17	56
Unhealthy	46 (2744)	7	7	15	71
Costly	26 (1526)	9	9	18	64
Addictive	36 (2136)	12	8	14	66
Practical	20 (1205)	35	13	18	33
Modern/technological	36 (2134)	22	11	17	49
Elegant/trendy/fashionable	23 (1353)	23	11	20	46
Enjoyable	16 (937)	46	15	15	24

Base: 26+ age group (N=5,910)

Question: Q.4.10. Which of the following words best captures what you think about vaping electronic cigarettes? Please tick as many boxes as you would like.

Note: Respondents could provide multiple answers. The group total category refers to the total respondents in that age group. The orange column represents the proportion of respondents in the age group that selected the response option (i.e. 11% of respondents in the 18-25 age group responded "complicated"). The white cells (in the four right columns) represent the breakdown by user type, i.e. they are within row percentages (white cells in each row sum to 100).

Source: OE elaboration from online survey data

The focus group participants, users of various tobacco products, expected e-cigarettes to be better tolerated in society because they "do not produce smoke, nor does it smell bad". Not causing nuisance to others was stated as a key factor for social acceptance. It also allowed users not to identify themselves as "real" smokers.

"I see loads of young people that want to differentiate themselves, that want to be a little Avant Garde (...)." (M, 33, FR)

"You were no longer a smoker but a vaper. You wanted to get away from the group of smokers and no longer belong to them. You wanted to be a vaper." (M, 22, DE)

"Lately I use vape and I think this group has more positives than the other. I do not smell that much, my teeth, fingers, nails do not become yellow, I do not cough and do not produce coughing secretions." (M, 24, BG)

"Too much technology (referring to using electronic devices to smoke) ... I prefer to smoke and that's it, (but) you don't bother the person next to you (e-cigarettes)." (M, 52, ES)

"I think that the electronic cigarette is accepted, it's not the same. I find that the vapour can smell pleasant compared to that of the cigarette." (M, 28, FR)

"I've tried the e-cigarette because of the price, but I was too lazy to charge the battery all the time and go out and buy the liquids...Now I'm switching between roll-your-own tobacco and cigarettes" (M, 22, DE)

"My need of cigarettes is not satisfied by the liquid as I have to wait for some time for it to recharge in order to be able to smoke. This is a disadvantage." (F, 50, BG)

Participants in the focus group argued that marketing techniques had exploited the flavours and design features of the product to target younger generations (see also section 6.2). Participants also suggested that using the product might encourage young people to start using other tobacco and related products; in other words, they believed there may be a gateway effect associated with the product (see section 6.1).

"The flavours were made to attract kids, so they start smoking a lot sooner, so they like the flavour compared to having their first cigarette." (M, 33, DE)

"It has penetrated young people so much, I see it with my son, they put up things in Instagram and such, [...] it is very trendy." (F, 43, EL)

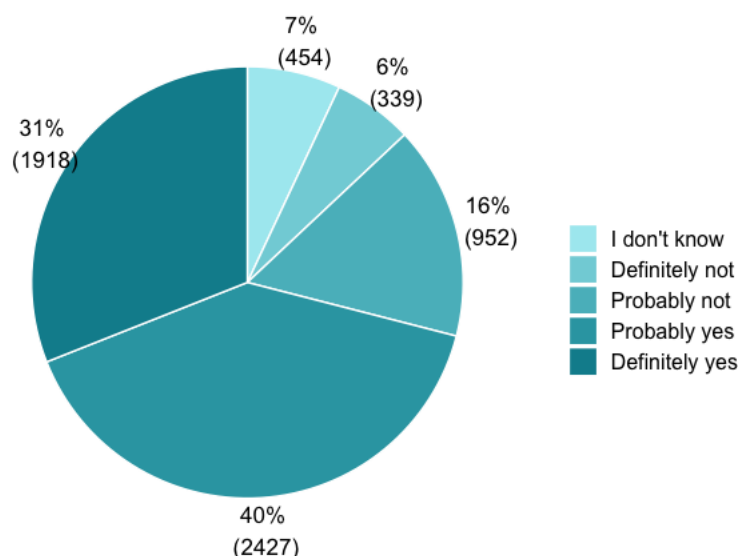
"Maybe very young people, who are 16 or 17. Some start with this, they don't all start with conventional tobacco." (M, 31, ES)

"It's innovative so I wanted to try it out and see what it's like." (M, 22, DE)

In the survey 71% of the 18-25 age group agreed that using e-cigarettes might encourage people to start smoking/using other tobacco and related products (see Figure 159). On this issue there is a striking difference between the current and never users: 28% more of the never users said 'definitely yes' to the gateway effect. In the 26+ age group (see Figure 160) 67% agreed about the possibility of the gateway effect and 38% more of the never users than current users said 'definitely yes' to the gateway effect.

Figure 159. Perceived gateway effect of e-cigarettes (18-25 age group)

Percentage and number of respondents in the 18-25 age group

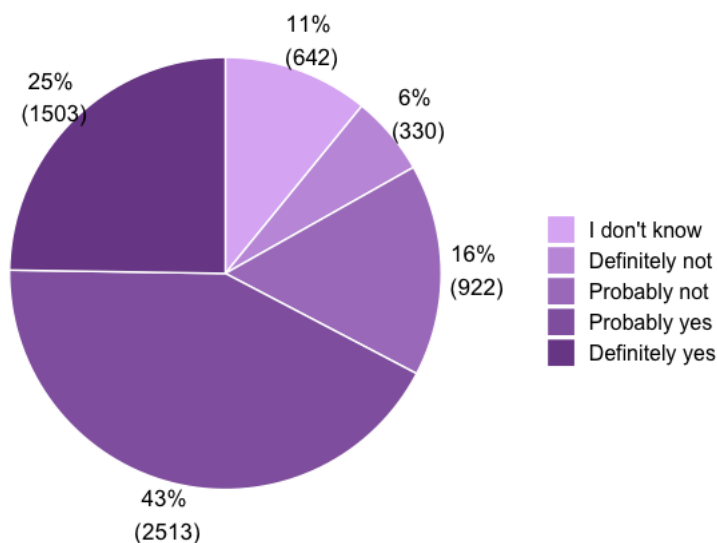


Base: respondents in the 18-25 age group (N=6,090)

Question: Q.4.9. Do you think that vaping electronic cigarettes might encourage people to start smoking/using other tobacco products, or not?

Source: OE elaboration from online survey data

Figure 160. Perceived gateway effect of e-cigarettes (26+ age group)

Percentage and number of 26+ age group

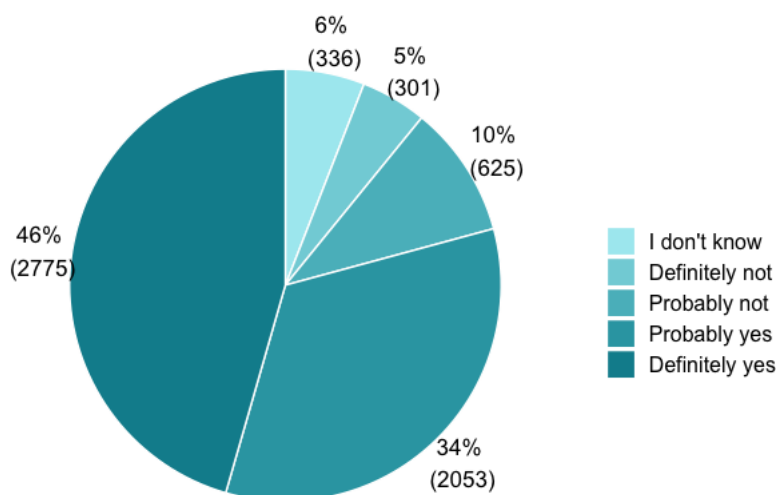
Base: respondents in the 26+ age group (N=5,910)

Question: Q.4.9. Do you think that vaping electronic cigarettes might encourage people to start smoking/using other tobacco products, or not?

Source: OE elaboration from online survey data

In the online survey 80% of the **18-25 age group** believed that young people would be particularly attracted to e-cigarettes (i.e. both "definitely yes" and "probably yes"), matched by 76% in the **26+ age group** (see Figure 161 and Figure 162). As with other aspects of e-cigarettes the current users and never used held very different opinions. Furthermore, in the focus groups, the range of designs and flavours was mentioned as a factors in attracting users of all age groups. Participants reported being particularly worried about the growing popularity among the younger generations.

Figure 161. Perceived appeal of e-cigarettes to youth (18-25 age group)

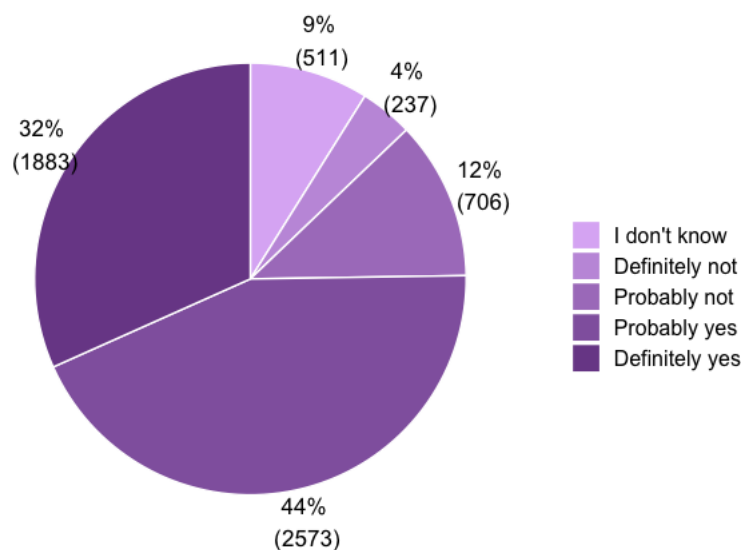
Percentage and number of 18-25 age group

Base: respondents in the 18-25 age group (N=6,090)

Question: Question: Q.4.11. Do you think that young people might be particularly attracted to electronic cigarettes (vaping), or not?

Source: OE elaboration from online survey data.

Figure 162. Perceived appeal of e-cigarettes to youth (26+ age group)

Percentage and number of 26+ age group

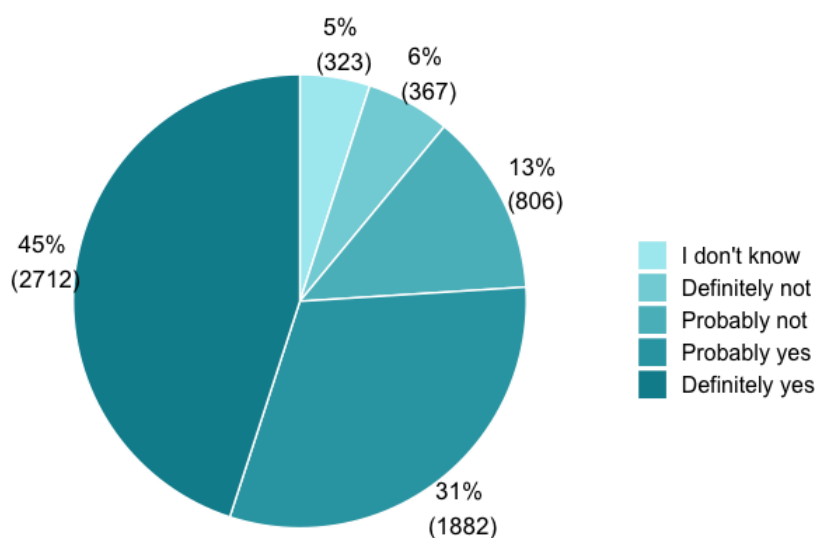
Base: respondents in the 26+ age group (N=5,910)

Question: Q.4.11. Do you think that young people might be particularly attracted to electronic cigarettes (vaping), or not?

Source: OE elaboration from online survey data

The concerns about e-cigarette use among young people led on to the view that they underestimate the risks to health. In the survey 76% of the **18-25 age group** and 77% of the **26+ age group** agreed that young people might underestimate the risks of e-cigarette use (see Figure 163 and Figure 164).

Figure 163 Extent to which respondents believed that youth underestimate risks of using e-cigarettes

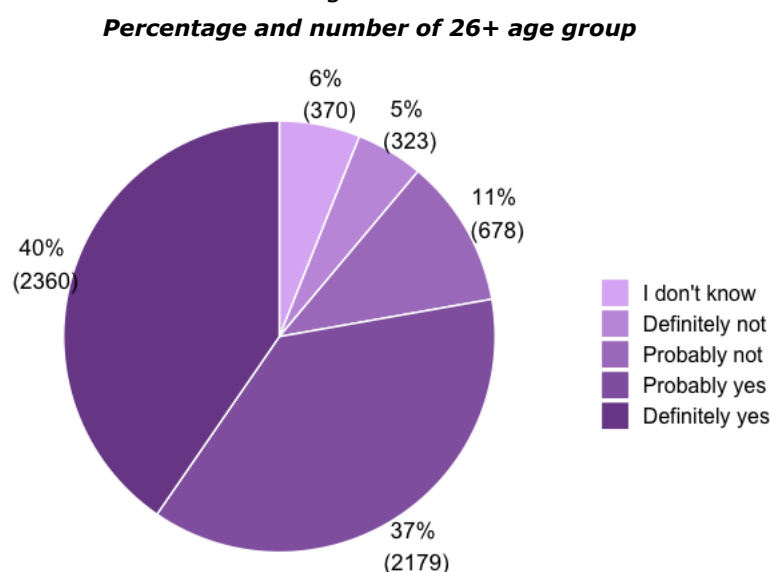
Percentage and number of respondents in the 18-25 age group

Base: respondents in the 18-25 age group (N=6,090)

Question: Q.4.12. Do you think that young people might underestimate the risk of using electronic cigarettes (vaping), or not?

Source: OE elaboration from online survey data

Figure 164 Extent to which respondents believed that youth underestimated risks of using e-cigarettes



Base: respondents in the 26+ age group (N=5,910)

Question: Q.4.12. Do you think that young people might underestimate the risk of using electronic cigarettes (vaping), or not?

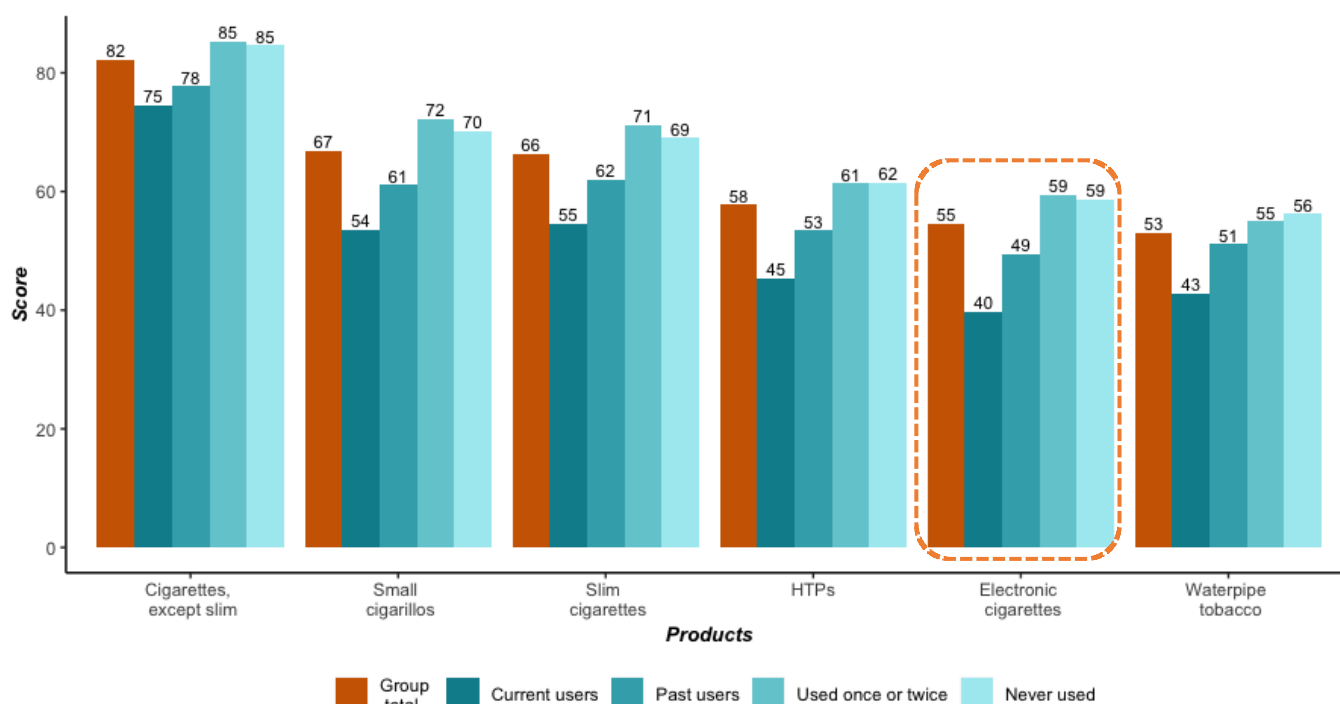
Source: OE elaboration from online survey data

Relative health risks of tobacco products. Figures 151 and 152 show the results of the 'swing method' to assess risk perceptions of the five tobacco and related products included in this study plus conventional cigarettes. The swing method is explained in 1.3.3. In brief, the respondent is asked which of the six products they think is most damaging to a user's health. This product is given 100 points on a scale from 100 (most harmful) to 0 (not harmful). Then using this scale, the respondent is asked to rate each of the other five products on its harmfulness to health of users.

In the **18-25 age group**, conventional cigarettes at a score of 82 are seen as the most harmful to health of the six products. E-cigarettes ranked second to lowest in terms of harmfulness to the users' health marginally more harmful than waterpipe tobacco (see Figure 165). Current users perceived e-cigarette as less harmful than 'past' 'once or twice' and 'never' users. Current users gave a score of 40 out of 100 (where 100 represents the most damage to the users' health) to e-cigarettes, in contrast to 60 for 'never' users.

The findings on perceived harmfulness in the **26+ age group** are broadly similar to those of the 18-25 age group (see Figure 166). Conventional cigarettes at 82 were seen as most harmful. E-cigarettes at 52 are seen as equally harmful to HTPs and marginally more than waterpipe tobacco. Current e-cigarette users of gave a score of 36 compared to 57 for never users.

Figure 165. Perceived harmfulness of e-cigarettes to users (18-25 age group)
(average score)



Base: respondents in the 18-25 age group (N=6,090)

Question: Q.3.1.a.¹²⁶ & Q.3.1.b. You have chosen (e.g.; cigarettes, except slims¹²⁷) as likely to be the most damaging to the health of users. We will give (e.g.; cigarettes, except slims) 100 points on a scale of 0-100 for damaging users' health. Please assess the other products' damage on a scale of 0 to 100. If you think a product is about half as damaging to the health of users, you would give that product 50 points. If you think another product is just as damaging as (e.g.; cigarettes, except slims), you would give it 100 points.

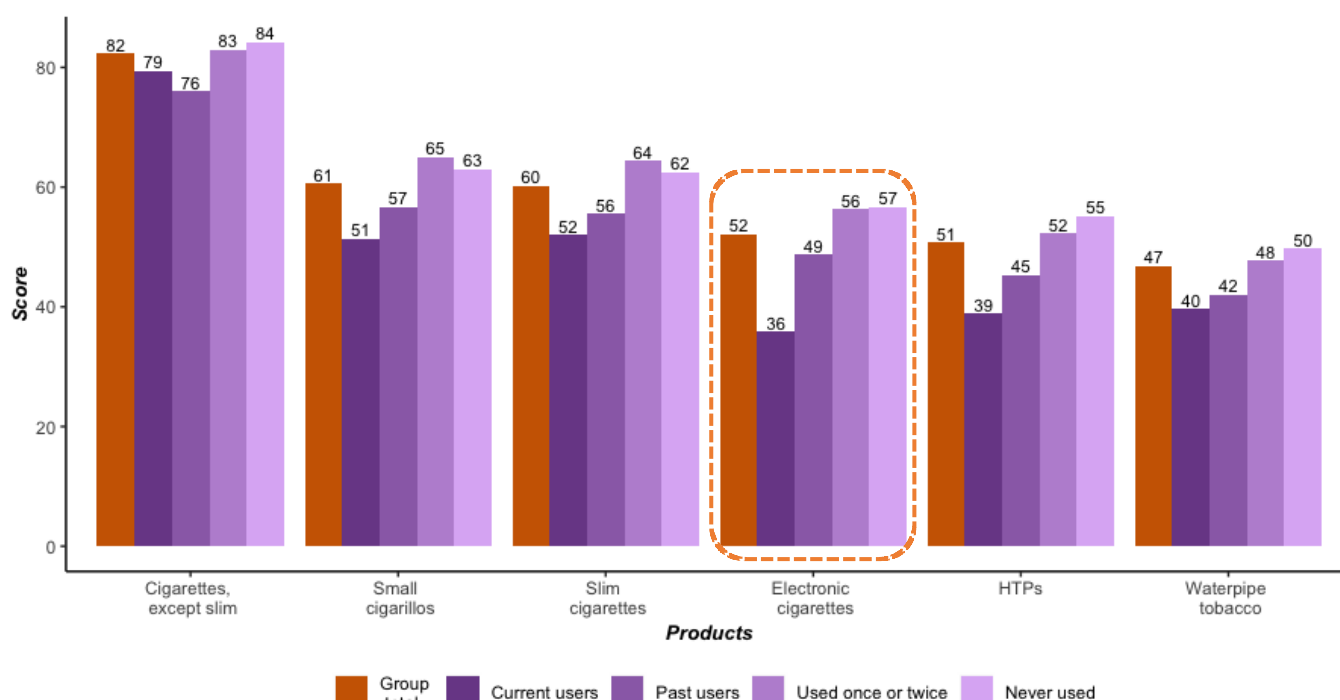
Note: The graph is based on the swing method. The scores were constructed using the responses to the two abovementioned questions. The bars in the graph are the average score of harm to the user provided by respondents for products under the scope of this study. The products are sorted in descending order in terms of the score in the "group total," i.e. amongst all respondents of the age group.

Source: OE elaboration from online survey data

¹²⁶ Q.3.1.a. Which of the six tobacco products listed below do you think is likely to be most damaging to the health of users?

¹²⁷ The product in parentheses would change in the online survey based on individual responses to Q.3.1.a.

Figure 166. Perceived harmfulness of e-cigarettes to users (26+ age group)
(average score)



Base: respondents in the 26+ age group (N=5,910)

Question: Q.3.1.a.¹²⁸ & Q.3.1.b. You have chosen (e.g.; cigarettes, except slims¹²⁹) as likely to be the most damaging to the health of users. We will give (e.g.; cigarettes, except slims) 100 points on a scale of 0-100 for damaging users' health. Please assess the other products' damage on a scale of 0 to 100. If you think a product is about half as damaging to the health of users, you would give that product 50 points. If you think another product is just as damaging as (e.g.; cigarettes, except slims), you would give it 100 points.

Note: The graph is based on the swing method. The scores were constructed using the responses to the two abovementioned questions. The bars in the graph are the average score of harm to the user provided by respondents for products under the scope of this study. The products are sorted in descending order in terms of the score in the "group total," i.e. amongst all respondents of the age group.

Source: OE elaboration from online survey data

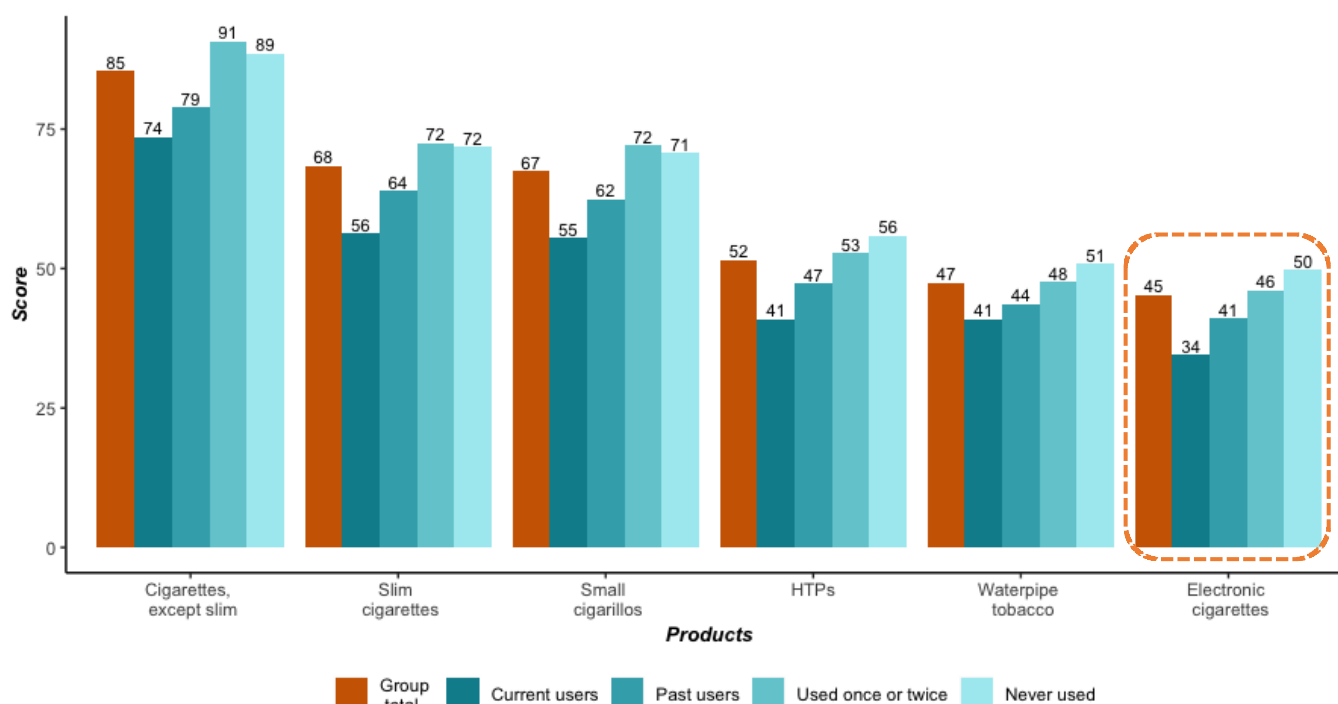
Passive use of e-cigarettes – harmfulness to others. In terms of harmfulness to passive users, electronic cigarettes scored as the least harmful product of the six products for both age groups. E-cigarette users in the **18-25 age group** (Figure 167) gave the lowest rating of 34 compared to 50 for never users. The results are similar in the **26+ age group** (see Figure 168); respondents rated e-cigarettes as the least harmful product for others. Current users gave a score of 30, never users 45. Furthermore, 13% of current e-cigarette users in 18-25 age group gave the maximum score of 100 to e-cigarettes in relation to passive use harms. In the 26+ age group, 9% of current users gave a maximum score of 100 to e-cigarettes, suggesting that the older cohort of e-cigarette users may underestimate the risks associated to e-cigarettes more so than the younger cohort. The focus group participants were unclear on whether the emissions of e-cigarettes affected bystanders, but the impact on their health was expected to be much smaller than with other products.

¹²⁸ Q.3.1.a. Which of the six tobacco products listed below do you think is likely to be most damaging to the health of users?

¹²⁹ The product in parentheses would change in the online survey based on individual responses to Q.3.1.a.

"I've noticed that there were people with e-cigarettes, and nobody said anything, but then five meters next to it there were smokers and people were complaining about them" (M, 30, DE)

Figure 167. Perceived harmfulness of e-cigarettes to bystanders(18-25 age group)
(average score)



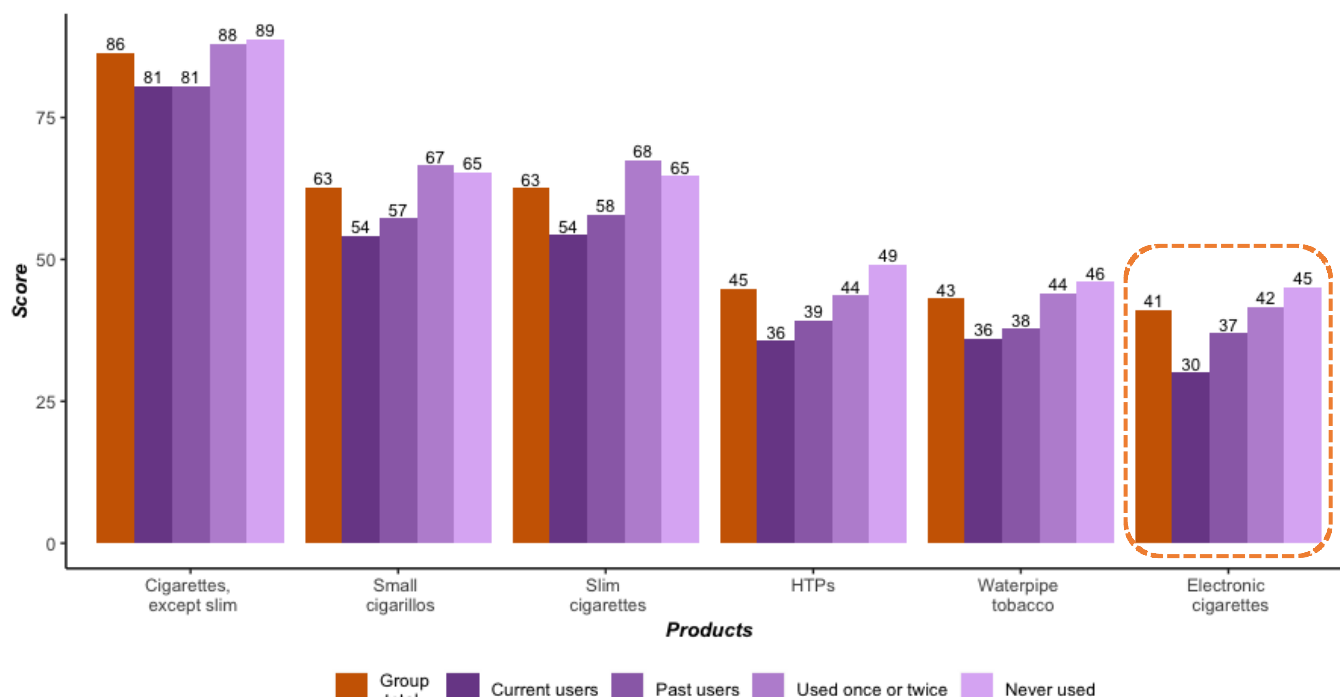
Base: respondents in the 18-25 age group (N=6,090)

Question: Q.3.2.a. & Q.3.2.b. Q.3.2.b. You have chosen (e.g.; cigarettes, except slims) as likely to be the most damaging to the health of passive users. We will give (e.g.; cigarettes, except slims) 100 points on a scale of 0-100 for damaging passive users' health. Please assess the other products' damage on the health of passive users on a scale of 0 to 100. If you think a product is about half as damaging to the health of passive users, you would give it 50 points. But if you think another product is just as damaging as (e.g.; cigarettes, except slims), you would give it 100 points

Note: The graph is based on the swing method. The scores were constructed using the responses to the two abovementioned questions. The bars in the graph are the average score of harm to passive users provided by respondents for products under the scope of this study. The products are sorted in descending order of the score in the "group total," i.e. all respondents of the age group.

Source: OE elaboration from online survey data

Figure 168. Perceived harmfulness of e-cigarettes to bystanders(26+group)
(average score)



Base: respondents in the 26+ age group (N=5,910)

Question: Q.3.2.a.¹³⁰ & Q.3.2.b. Q.3.2.b. You have chosen (e.g.; cigarettes, except slims¹³¹) as likely to be the most damaging to the health of passive users. We will give (e.g.; cigarettes, except slims) 100 points on a scale of 0-100 for damaging passive users' health. Please assess the other products' damage on the health of passive users on a scale of 0 to 100. If you think a product is about half as damaging to the health of passive users, you would give it 50 points. But if you think another product is just as damaging as (e.g.; cigarettes, except slims), you would give it 100 points

Note: The graph is based on the swing method. The scores were constructed using the responses to the two abovementioned questions. The bars in the graph are the average score of harm to passive users provided by respondents for products under the scope of this study. The products are sorted in descending order of the score in the "group total," i.e. all respondents of the age group.

Source: OE elaboration from online survey data

¹³⁰ Q.3.2.a. Which of the six tobacco products listed below is likely to be the most damaging to the health of other people from passive use?

¹³¹ The product in parentheses would change in the online survey based on individual responses to Q.3.1.a.

Table 62 and Table 63 shows those settings in which respondents thought that e-cigarette use should be restricted or prohibited. In both age groups prohibition at work, in public transports, and in restaurants and bars (indoors) is supported by one in two respondents. Furthermore, 15% of the **18-25 age group** and 22% of the **26+ age group** wanted prohibition in all of the settings included in the questionnaire.

Table 62 Views on settings where e-cigarettes should be prohibited (18-25 age group)

Percentage and number of respondents in the 18-25 age group

	Group total %(n)	Current users %	Past users %	Used once or twice %	Never used %
Use of product	100 (6090)	15	15	25	45
Open outdoor spaces (e.g. bus stops, playgrounds)	17 (1032)	17	13	24	46
Work	46 (2789)	12	13	27	49
Public transport	55 (3342)	13	14	28	46
Restaurants/ bars indoors	49 (2966)	12	14	27	47
Restaurants/ bars outdoor terraces	14 (831)	15	17	23	45
Private transportation	26 (1572)	11	13	27	48
All of the above	15 (943)	7	10	21	62
None	8 (513)	21	16	27	36
Other	3 (162)	7	12	15	65

Base: respondents in the 18-25 age group (N=6,090)

Question: Q.4.14. In which settings, if any, do you think that the consumption of electronic cigarettes (vaping) should be prohibited?

Note: Respondents could provide multiple answers, except if they selected "All the above" or "None". Hence, a special consideration must be made when analysing the responses. Although 17% reported "open outdoor spaces", 32% of respondents in the 18-25 age group were in favour of prohibiting the product use in open outdoor spaces (i.e. 17% + 15% that answered "all of the above"). The group total category refers to the total respondents in that age group. The orange column represents the proportion of respondents in the age group that selected the response option (i.e. 3% of respondents in the 18-25 age group responded "other"). The white cells (in the four right columns) represent the breakdown by user type, i.e. they are within row percentages (white cells in each row sum to 100).

Source: OE elaboration from online survey data

Table 63 Views on settings where e-cigarettes should be prohibited (26+ age group)

Percentage and number of respondents in the 26+ age group

	Group total %(n)	Current users %	Past users %	Used once or twice %	Never used %
Use of product	100 (5910)	18	11	16	56
Open outdoor spaces (e.g. bus stops, playgrounds)	17 (989)	14	10	15	61
Work	46 (2723)	16	10	16	59
Public transport	56 (3307)	17	10	17	56
Restaurants/ bars indoors	47 (2766)	16	10	16	58
Restaurants/ bars outdoor terraces	14 (826)	17	10	13	60
Private transportation	23 (1377)	16	10	15	59
All of the above	22 (1291)	8	5	12	75
None	10 (592)	28	14	15	43
Other	1 (84)	6	11	23	61

Base: respondents in the 26+ age group (N=5,910)

Question: Q.4.14. In which settings, if any, do you think that the consumption of electronic cigarettes (vaping) should be prohibited?

Note: Respondents could provide multiple answers, except if they selected "All the above" or "None". Hence, a special consideration must be made when analysing the responses. Although 17% reported "open outdoor spaces", 39% of respondents in the age group were in favour of prohibiting product use in open outdoor spaces (i.e. 17% + 22% that answered "all of the above"). The group total category refers to the total respondents in that age group. The orange column represents the proportion of respondents in the age group that selected the response option (i.e. 1% of respondents in the 26+ age group responded "other"). The white cells (in the four right columns) represent the breakdown by user type, i.e. they are within row percentages (white cells in each row sum to 100).

Source: OE elaboration from online survey data

In the focus group, it was mentioned that while banning the product could be useful in order to change social norms and perception, campaigns and price changes of the product were viewed as more effective tools. Participants believed that when products had no restrictive-use regulations or were tolerated in places where consuming conventional tobacco was forbidden or frowned upon (e.g. waterpipe tobacco is allowed in bars) would indicate to users that these are less harmful than conventional tobacco.

"You can smoke that at home (electronic cigarettes) if you don't want to go out to smoke."
(M, 28, ES)

"I think you were no longer allowed to smoke in restaurants, and then the e-cigarettes came out as a solution." (M, 22, DE)

"In my workplace we are allowed to smoke electronic cigarettes, while regular cigarettes are not allowed." (F, 43, EL)

6.4 Health, social and economic impacts

It has been well-documented that heavy marketing practices were adopted by manufacturers of e-cigarettes to appeal to a range of potential users, including young people, even under-age, and to promote the product as a cessation aid (e.g. Kinnunen, 2015; Martinez, 2015). The systematic literature review concluded that specific health risks associated with e-cigarettes were still uncertain and the long-term impact remained unknown. (PHE, 2018; RCP, 2016; Hua, 2016; Meo, 2014; Callahan-Lyon, 2014; Kim, 2016). The evidence showed that the level of exposure to nicotine from e-cigarettes is variable, depending on the characteristics of the device and the e-liquid, as well as on how the device is operated. Callahan-Lyon (2014) reported that differences in product engineering, components and potential toxicities hinder the study of e-cigarettes as a single device. Pisinger (2014) outlined some methodological challenges that must be acknowledged in the research on e-cigarette health impacts. First, with research covering some 460 brands that had been subject to frequent modifications, it is a challenge to arrive at valid generalisations. Furthermore, many studies have compared the health impacts of e-cigarettes with conventional cigarettes, whereas health hazards for each product may be different, and also include non-smokers. For example, e-cigarette use topography is quite different to conventional smoking as the use of e-cigarettes involves sucking harder with longer puffs. In addition, experimental studies with human participants may not reflect real-life use of e-cigarettes.

Gotts, Jordt, McConnell, and Tarran (2019) provided an extensive assessment of the evidence on the pulmonary effects of e-cigarettes, covering the population, clinical, animal, and cellular evidence. The research found that studies funded by the tobacco industry, unlike non-tobacco industry funded literature, did not find lung problems with e-cigarettes. More recently, in the USA, a lung ailment reportedly killed at least 68 people and afflicted nearly 2,807 patients, as of February 2020 (Center for Disease Control and Prevention).¹³² While this disease highlighted the need for further investigations on the product and its composition, differences in the regulatory arrangement of the US and the EU, protected EU citizens from the risk.¹³³ Specifically, the outbreak of illness was linked to the refill capsules, and laboratory studies identified vitamin E acetate, an additive in some THC-containing e-cigarette or related products, as strongly linked to what has been coined the EVALI outbreak.¹³⁴ The TPD, which has been applicable in the EU Member States since May 2016, regulates the sale and merchandising of e-cigarettes in the EU. The TPD (2014/40/EU) does not permit the addition of types of vitamins. The Directive also requires e-cigarette manufacturers to notify Member States of all products placed on the market and to report annually on sales volumes.

Many recent studies have argued for the need to better codify and understand the health impacts of e-cigarettes in order to have evidence-based policies for tobacco control (PHE, 2018; RCP, 2016; Hua, 2016; Kim, 2016). Levy (2017) proposed a framework describing and incorporating patterns of vaporised nicotine products (VNP, such as e-cigarettes and HTPs) to determine the total amount of toxic exposure effects on a population health. While the framework identified some critical information, it is expected that there will be further product innovations and changes in patterns of use such that health risks may change as well. Kalkhoran (2015, UK & USA) developed a model to evaluate potential health effects of increasing promotion and use of e-cigarettes. The outcome, supported by results

¹³² "Outbreak of Lung Injury Associated with E-Cigarette Use, or Vaping". Centers for Disease Control and Prevention. 28 January 2020. This article incorporates text from this source, which is in the public domain. Available at: https://www.cdc.gov/tobacco/basic_information/e-cigarettes/severe-lung-disease.html

¹³³ There have been suspected cases (e.g. Belgium) but the underlying cause has not yet been clarified.

¹³⁴ https://www.europarl.europa.eu/doceo/document/P-9-2019-003949-ASW_EN.html

¹³⁴ <https://www.bloomberg.com/news/articles/2019-09-13/europe-says-thank-you-for-vaping-as-u-s-tells-consumers-to-stop>

from Polosa (2013)¹³⁵ was that greater e-cigarette use could have different health effects for different population segments.

What have the survey findings contributed to the debate on whether e-cigarettes were an aid to quitting conventional cigarettes (the cessation scenario) or whether young people would take up the use of e-cigarettes and progress to conventional cigarettes (the gateway scenario). Before looking at the survey evidence a note of caution is needed. The online survey was a cross-sectional study and, as such, cannot lead to causal claims.

Just below 1 in 5 in both age groups said they had previously used other products but were now only using e-cigarettes. 1 in 12 of the younger age group and 1 in 25 of the older age group were exclusive e-cigarette users and had not used other products. Given that 'cutting down or quitting other tobacco products' was cited as a reason for taking up e-cigarettes, these findings were suggestive of the cessation scenario. However, over 7 in 10 of the 18-25 age group and 8 in 10 of 26+ age group were dual users of e-cigarettes and other products, mainly conventional cigarettes.

Of past e-cigarette users, 1 in 4 in both age groups had been dual users and now were not using any tobacco and related products. This is suggestive of the cessation scenario. Suggestive of the gateway scenario, 1 in 4 of the 18-25 age group and 3 in 10 of the 26+ age group had been exclusive e-cigarette users and now reported using other tobacco and related products, with some 90% reporting the use of conventional cigarettes. Marginally more than 4 in 10 of both age groups were past e-cigarette users and reported current dual use of other tobacco and related products. The use of e-cigarettes had not made an impact on their use of tobacco.

Overall, the survey findings are suggestive of both cessation and gateway scenarios. This in the context of dual use reported by a substantial majority of the current e-cigarette users and by significant numbers of past e-cigarette users.

Finally, in terms of policy, some Member States have taken additional measures to regulate the consumption of e-cigarettes. In Germany, the DKFZ (2018b) have in the past advised schools to extend smoking bans to all e-products via the house right, as there were no legal regulations regarding e-cigarettes and e-hookahs in place at that time. This recommendation was justified on the basis that schools are a place of living and learning and that they should be a protected space in which legal and illegal drugs have no place. An ongoing study by Kimber (2018-19, EU) compares the effects of the EU TPD, e-cigarette health warnings and a comparative harm message (developed specifically for that study) on smokers' and non-smokers' perceptions of harm, addictiveness and social acceptability of e-cigarettes. Additionally, the potential effects of the TPD warnings and the impact on tobacco consumers' purchase and use intentions regarding e-cigarettes are being explored (Kimber, 2018-19, EU). Health warnings, such as those implemented by the TPD, may help to prevent non-smokers from e-cigarette use, but it is also possible that they may inadvertently deter tobacco consumers from switching from conventional tobacco products to e-cigarettes.

¹³⁵ Disclaimer: the paper was supported by the electronic cigarettes industry.

6.5 Concluding remarks on electronic cigarettes

The main findings on e-cigarettes emerging from the literature review and empirical work conducted under the present study are summarised below. The systematic review of secondary sources revealed:

- E-cigarettes are a heterogeneous class of products that use an electrically powered coil to heat and turn a liquid into an aerosol, which is inhaled by the user.
- E-cigarettes are not harmless. Dual use of electronic cigarettes with cigarettes and other tobacco products does not provide any significant reduction in exposure to tobacco-related toxicants compared to those smoking conventional cigarettes.
- While there is limited evidence that e-cigarettes may be effective aids to promote smoking cessation, there is moderate evidence from observational studies that more frequent use of e-cigarettes is associated with an increased likelihood of cessation among older smokers. However, e-cigarettes may act as a gateway product encouraging youth who use them to transition to use of conventional cigarettes or develop lasting habits through addiction to nicotine.
- The prevalence of current e-cigarette use in the EU is 2% among adults (15+). According to Passport data, electronic cigarette sales were 4.2 billion euros. The leading markets were UK and Italy.

The empirical work (focus groups and online survey) revealed the following user profile:

- A typical current e-cigarette user has between 1 and 10 sessions a day, started as a minor or young adult, and started either for enjoyment (in the 18-25 group) or to try to quit or reduce smoking of conventional tobacco products (in the 26+ group). In the 18-25 age group 38% started using e-cigarettes for 'enjoyment' and 33% were 'trying to quit or reduce consumption of another tobacco product'. The 26+ age group cited the same reasons. But 54% cited 'trying to quit or reduce consumption of another tobacco product' and 28% said for 'enjoyment'.
- About three in four current e-cigarette users were dual users of other tobacco products. 53% of the 18-25 age groups had never attempted to quit. Of those who had tried 54% did so on account of 'personal health', 'health warning labels' or 'concern for bystanders' health'. 63% of the 26+ age group had never attempted to quit and of those who had tried 51% cited the same reasons as the younger group.
- The UK and Italy had the largest current consumption of e-cigarettes. Respondents in the 18-25 age group reported experimenting with the product (i.e. 'used once or twice') more often than respondents in the 26+ age group. The 18-25 age group reported using the product mainly for 'enjoyment'.
- Current users were evenly split between those that had never attempted quitting the use of e-cigarettes and those who tried at least once and failed in the 18-25 group. In the 26+ group, almost two in three had never tried to quit. Among failed quitters, the main reason to try was for personal health. Other reasons frequently mentioned were economic reasons and not liking it anymore. The main reason cited for failing to quit was being stressed. Other important reasons mentioned were that they found quitting electronic cigarettes difficult and that electronic cigarettes were not that harmful compared to other products anyway.
- In the 18-25 age group, 74% of current e-cigarette users were dual user of tobacco and/or other tobacco related products: 82% reported using conventional cigarettes. 18% indicated that they had previously used other products but were now only using e-cigarettes. 8% had only ever used e-cigarettes. The same pattern was found for the 26+ age group. 78% were

dual users and 90% of these reported using conventional cigarettes. 18% had used other products and were now only using e-cigarettes and 4% were exclusive e-cigarette users.

- Evidence for a strong positive relationship between use of e-cigarettes and later cigarette consumption amongst youth has continued to accumulate in observational studies. The present research has findings that support both the cessation and gateway scenarios.
- Suggestive of the cessation scenario in both age groups 18% reported using other tobacco and related products in the past and now only using e-cigarettes. Furthermore, of current e-cigarette users 8% of the 18-25 age group and 4% of the 26+ age group reported never using other products. However, that 74% of the 18-25 age group were dual users and 82% reported using conventional cigarettes chimes with the gateway scenario.
- Also suggestive of the gateway scenario, 23% of the 18-25 age group and 31% of the 26+ age group were exclusive e-cigarette users and now reported using other tobacco and related products, with some 90% reporting the use of conventional cigarettes.
- Concerning the past users of e-cigarettes, 7% of the 18-25 age group and 2% of the 26+ age group were never users of other tobacco and related products. 24% of the 18-25 age group and 23% of the 26+ age group were dual users in the past and now were not using any tobacco and related products – suggestive of the cessation scenario.
- These quantitative findings on dual use were illustrated in the focus group discussions. People said that the main purpose for using e-cigarettes was to quit conventional cigarettes; to alternate e-cigarettes with conventional tobacco products in order to reduce consumption of the latter; to use the e-cigarette in places where smoking was not allowed but e-cigarettes were tolerated (at some work places, bars, malls, etc. , also depending on the country); or, to use in places where smoking is not forbidden, but it was preferable not to smoke in order to avoid harming others (e.g. at home) – the latter was particularly relevant for those with children.
- The perception of e-cigarettes was varied in terms of general awareness, but especially in terms of the awareness of some danger to the health of the user. The literature revealed similar misperceptions and confusion, and further explained that evidence on the health impacts was still inconclusive and at times even controversial.

Lastly, the following paragraphs highlight the *Product and Regulatory perceptions remarks*

- While the overall sample viewed e-cigarettes in a more negative than a positive light, this view varied according to the level of use of the product. Current users of the product were much more likely to view the product in a favourable light than never users. E-cigarettes are seen as unhealthy, addictive and modern/technological. Users of e-cigarettes considered them more socially acceptable than smoking because they consider that it does not produce “smoke”, smell bad, or a nuisance to others.
- A majority of e-cigarette users and ‘never users’ raised concern about the possibility that use of e-cigarettes might appeal to youth and encourage them to start smoking/using other tobacco and related products. The survey data revealed that the younger generations were of particular target group for the product and a large majority of respondents believed that youth may be particularly attracted to using the product.
- The range of designs and flavours were mentioned as key factors in attracting users of all age groups and particularly the younger generations. Flavour and trends could be important influences as well as the marketing practices, notably on social media (e.g. Instagram). The findings were consistent with the literature.

- Perception of marketing practices. Participants in the focus groups argued that marketing techniques had exploited the flavours and design features of the product to target young people.
- There was support for regulations to prohibit the use of e-cigarettes in at work, in public transport and in restaurants and bars. Passive exposure of e-cigarettes was viewed as the least harmful of the tobacco products in this study.
- The marketing practices of the new tobacco products should be closely monitored as the overall perception of e-cigarettes appeared to be marked by social factors; tobacco advertising claims were also cited quite frequently in the online survey. The literature indicated that youth was more likely to be influenced by the marketing practices and design, which have established the device as both a trend and as a smoking cessation aid.

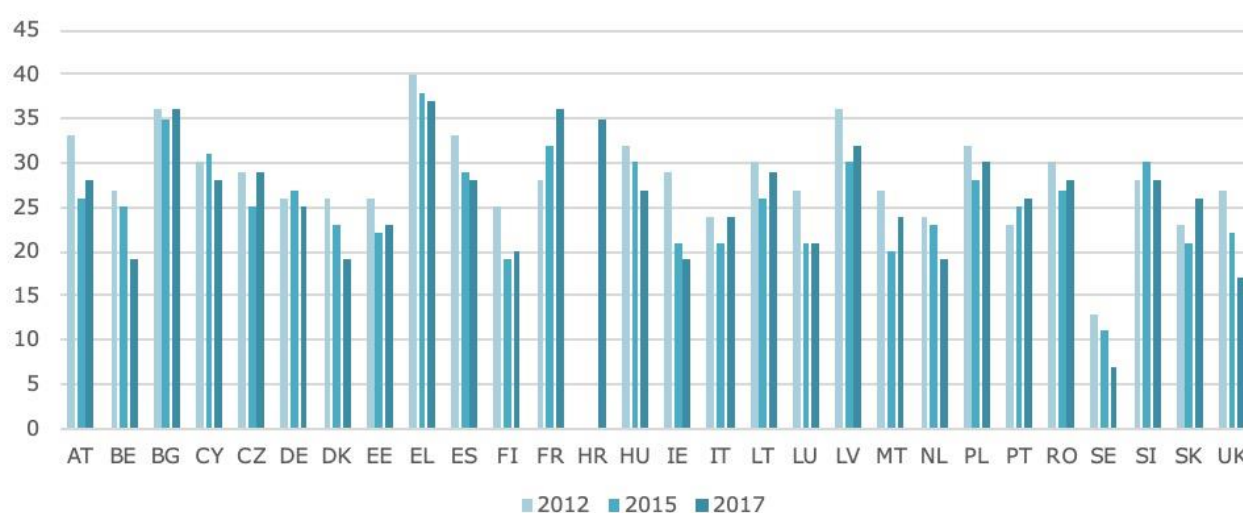
7 Comparative analysis of tobacco and related products

In this chapter, we describe the average profiles of users of the five tobacco and related products within the scope of this study with occasional reference to users of conventional cigarettes, highlighting potential similarities and differences in terms of age, gender, education level, socioeconomic status and regional distribution. We also explore and compare the patterns of use, the perception and the impacts of the five tobacco and related products.

7.1 Market developments

The last decade has seen an overall decrease in smoking prevalence. Tobacco products have been the focus of tightening regulations for decades across the EU region to ensure consumer protection as well as smooth functioning of the internal market. Measures include higher taxation and increasing health awareness across society, through for example the establishment of mandatory health warnings on packaging. However, concurrently, the tobacco industry has developed new alternative products in order to maintain its customer base as well as attract new consumers. There has been a general decrease in the use of tobacco products in the EU between 2012 and 2017. As displayed in Figure 169, the share of smokers decreased in 20 of the 28 EU countries between 2012 and 2017, while it either increased or remained the same in 7 countries.¹³⁶

Figure 169 Share of smokers by countries



Source: Eurobarometer (2012, 2015, 2017)

The figure below summarises the market share of specific tobacco and related products across the different national markets according to Passport data in 2017.¹³⁷ Western EU countries reported larger markets for e-cigarettes and waterpipe tobacco products. The UK (77%), Poland (4.2%) and Denmark (4.2%) had the highest share of electronic cigarettes in the tobacco and related products' market. Waterpipe tobacco had high shares of the market in Denmark (2%) and Germany (1.4%). In comparison, small cigarillos and slim cigarettes tend to have larger markets in eastern Member States. Small cigarillos were more prevalent in Finland (5%), Lithuania (4%), and Latvia (2.4%). Slim cigarettes¹³⁸ were most widely used in Bulgaria (37.4%), Poland (32%), Lithuania (22.3%), Latvia

¹³⁶ Note that Croatia does not have available data prior to 2017 and therefore a trend cannot be implied.

¹³⁷ It is important to note that the data for the waterpipe market are aggregated at a higher level including both waterpipe and pipe tobacco. Acknowledging this, for simplicity, we will refer to waterpipe market data

¹³⁸ Slim and superslim cigarettes are combined in this section.

(21.3%) and Greece (14%). Among the countries for which data was available, the highest share of HTP was found in Romania (2.2%) and Slovakia (1.4%).

Table 64. Market size share of selected tobacco and related products by countries
Percentage of total tobacco market in the country in 2017¹³⁹

	Small cigarillos	Waterpipe	Electronic cigarettes	HTP	Slim cigarettes
AT	1.1	0.1	0.7	-	2.6
BE	1.4	0.1	1.7	-	0.1
BG	0.2	0.1	1.2	-	37.4
CY	-	-	-	-	-
CZ	0.2	0.3	1.5	0.1	8.1
DE	2.5	1.4	2.0	0.0	1.3
DK	0.9	1.9	4.2	0.3	3.0
EE	1.4	0.4	2.2	-	9.3
EL	0.6	0.0	2.5	0.8	14.0
ES	2.1	0.5	0.7	0.2	0.1
FI	5.1	0.2	0.3	-	0.1
FR	2.6	0.3	2.2	0.0	2.9
HR	0.5	0.1	1.8	-	7.0
HU	2.3	0.2	0.5	-	12.6
IE	0.2	0.2	3.9	-	0.7
IT	0.6	0.2	3.0	0.6	6.9
LT	3.0	0.4	1.4	0.4	22.3
LU	-	-	-	-	-
LV	2.5	0.5	1.5	0.2	21.3
MT	-	-	-	-	-
NL	1.3	0.3	1.8	0.1	1.4
PL	0.1	0.3	4.2	0.2	32.0
PT	2.1	0.2	0.9	0.8	3.6
RO	0.1	0.1	2.7	2.2	13.1
SE	0.8	0.2	2.1	-	4.8
SK	0.3	0.0	2.3	1.4	6.0
SL	1.9	0.3	2.3	0.2	5.9
UK	0.7	0.4	7.0	0.0	0.8
EU	1.1	0.1	0.7	-	2.6

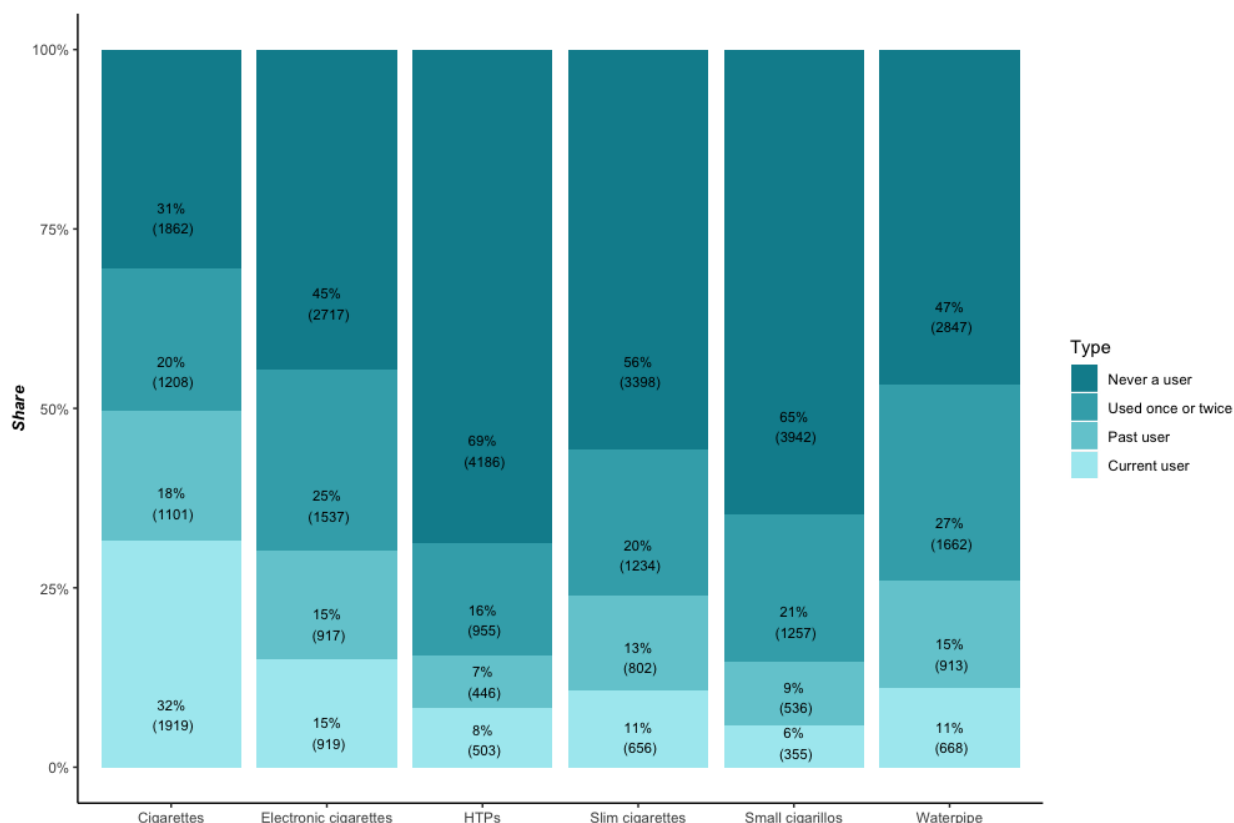
Note: EU average is calculated for 25 EU members (excluding Cyprus, Luxembourg and Malta)

Source: Passport, 2017

¹³⁹ Note: Data on heated tobacco products is not available for Austria, Croatia, Czech Republic, Hungary, Ireland, Latvia, Poland, Slovakia, Sweden. Share of slim cigarettes is available only for retail volume in million sticks, in order to make a comparable graph we used it as a share of the retail volume RSP for cigarettes in general. Slim cigarettes include slim and super slim cigarettes classification.

The comparative findings from the online survey are presented in the figures below. In the **18-25 age group**, 32% were current users of cigarettes, 15% of e-cigarettes, 11% of waterpipe, 11% of slim cigarettes, 8% of HTPs and 6% of small cigarillos. Although current users of HTPs had a marginally higher share than small cigarillo users, 69% of respondents had never used HTPs compared to 65% of respondents for small cigarillos. While HTPs were a relatively new product, there were 8% of current users in the 18-25 age group. This suggests that the market is growing and can be expected to continue growing.

Figure 170. Overview of the consumption of the five products (18-25 age group)



Base: all respondents in the 18-25 age group (N=6,090)

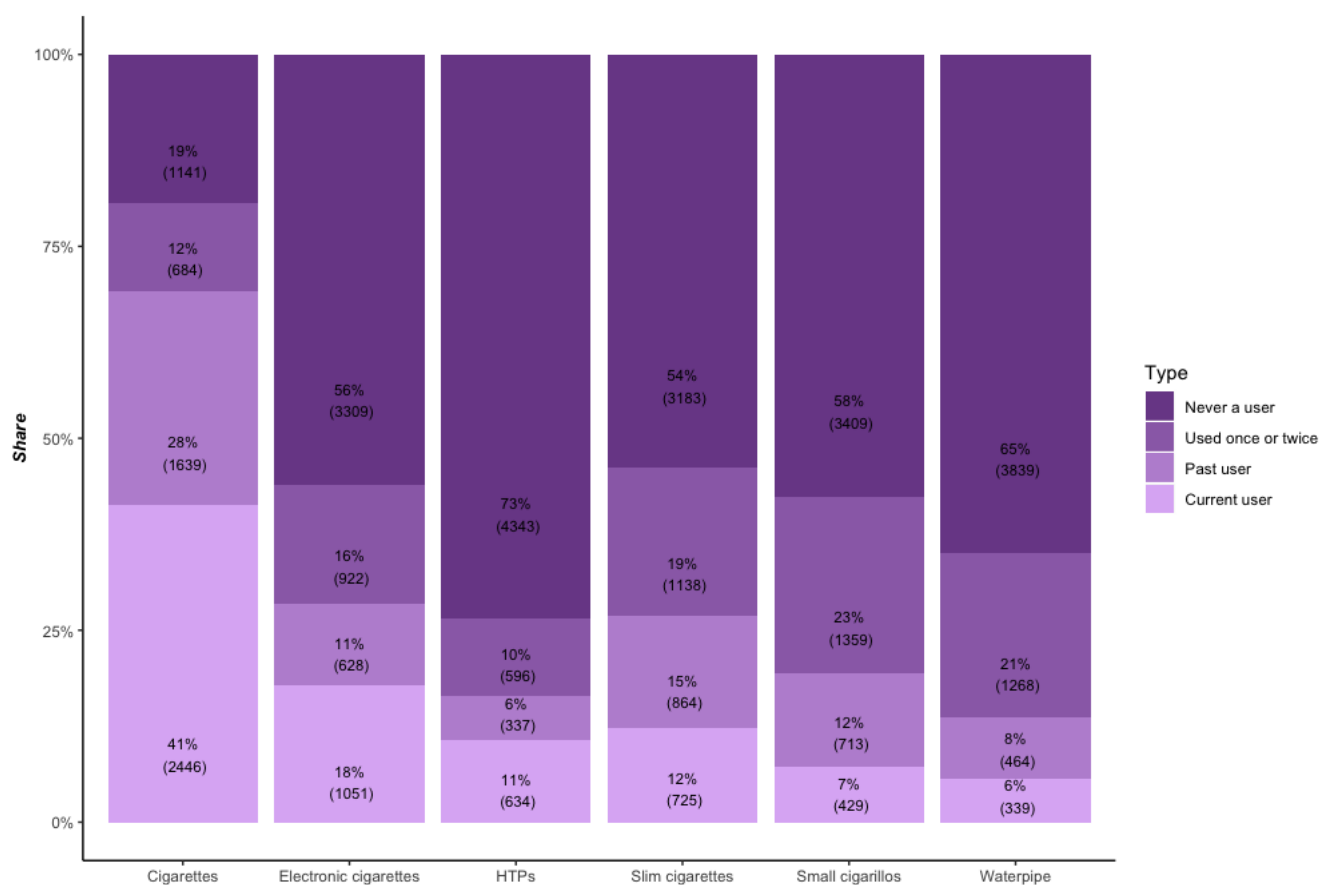
Question: Q.1.1. Have you ever used the following products?

Note: The group total category refers to all respondents within the age group. Note that there is a positive sampling bias so these numbers may appear higher than they would in a representative sample.

Source: OE elaboration from online survey data

For the **26+ age group** (see Figure 171), there were higher levels of current consumption for electronic cigarettes (18%), followed by slim cigarettes (12%) and HTPs (11%) across the EU markets under study. The share of HTP current users appeared particularly high given the novelty of the product.

Figure 171. Overview of the consumption of the five products (26+ age group)



Base: all respondents in the 26+ age group (N=5,910)

Question: Q.1.1. Have you ever used the following products?

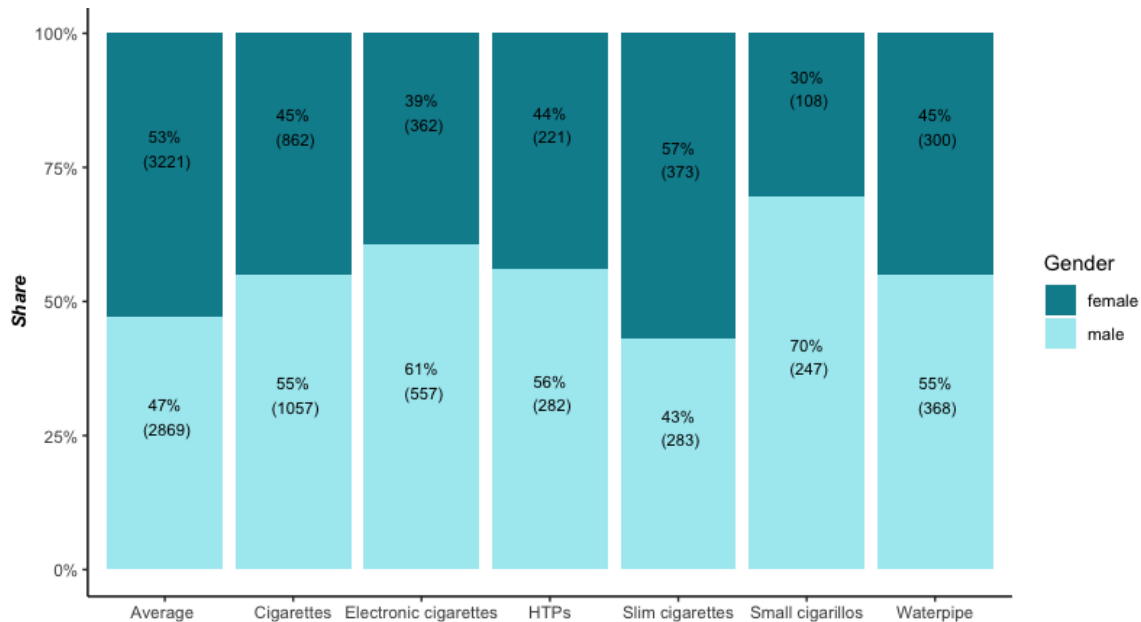
Note: The group total category refers to all respondents within the age group. Note that there is a positive sampling bias so these numbers may appear higher than they would in a representative sample.

Source: OE elaboration from online survey data

The sample of the online survey did not reveal significant differences in education levels across the products users for the 18-25 age group, except for small cigarillos (see Figure 200. in Annex). One in ten small cigarillo users (10%) had a low education level, compared to 6% in the total age group. For respondents in the 26+ age group (see Figure 201. in Annex), users of HTPs and waterpipe tobacco appeared to have a relatively higher level of education - with 64% and 62% respectively - when compared to the overall sample (49%).

Figure 172 and Figure 173 report gender differences in the use of the tobacco and related products for the two age groups. The large majority of small cigarillo users were male (70%) in the 18-25 age group. The same trend was observed in the 26+ age group with 68% of small cigarillo users being males. Furthermore, 58% of slim cigarette users were females and 62% of waterpipe tobacco users were male. The gender division were much more balanced for these products and others amongst the 18-25 age group.

Figure 172. Gender of consumers by product: 18+25 age group

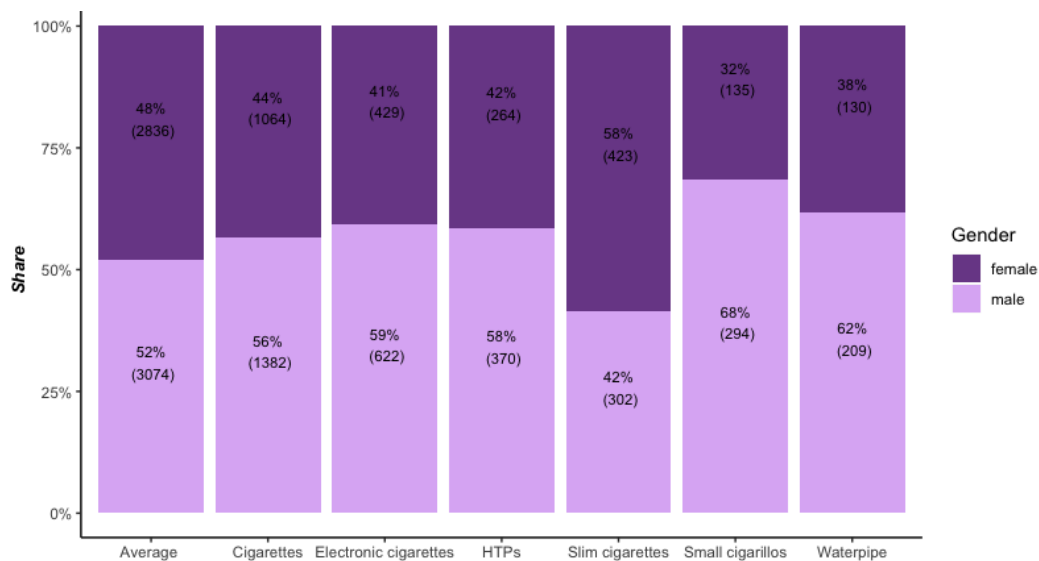


Base: all respondents in the 18-25 age group (N=6,090) & respondents that were current users for each product (see numbers in parentheses)

Note: The group total category refers to all respondents within the age group. Note that there is a positive sampling bias so these numbers may appear higher than they would in a representative sample.

Source: OE elaboration from online survey data

Figure 173. Gender of consumers by product: 26+ age group



Base: all respondents in the 26+ age group (N=5,910) & respondents that were current users for each product (see numbers in parentheses)

Note: The group total category refers to all respondents within the age group. Note that there is a positive sampling bias so these numbers may appear higher than they would in a representative sample.

Source: OE elaboration from online survey data

On self-reported economic status, users of small cigarillos in the 18-25 age group reported belonging to a high socio-economic class more often than no users of this product. This trend was also observed to a lesser degree for HTPs, electronic cigarettes and waterpipe tobacco. In the 26+ age group, waterpipe users were more likely to self-report a higher socio-economic status as did users of small cigarillos and HTPs. In both age groups users of HTPs, slim cigarettes and waterpipe tobacco were more likely to come from urban areas. (see section 10.6).

Over three quarters of current users of HTPs, slim cigarettes and small cigarillos were dual users of other tobacco and related products. The great majority reported smoking conventional cigarettes. While HTPs and e-cigarettes have been marketed as an aid to quitting conventional tobacco products the percentages of the current users of HTPs and e-cigarettes who are (i) using no other tobacco products and (ii) are dual smokers are shown in the following table:

Table 65. Career flows of use for current users of HTPs and e-cigarettes (18-25 & 26+ age groups)

Percentage of current users in the 18-25 & 26+ age groups

Research questions	HTPs (%)	e-cigarettes (%)
Exclusive users of either product		
18-25 age group	16%	26%
26+ age group	14%	22%
Users of either e-cigarettes or HTPs and parallel users (dual users)		
18-25 age group	84%	74%
26+ age group	86%	78%

Base: current users of HTPs in the 18-25 age group (N=503); current users of HTPs in the 26+ age group (N=634); current users of e-cigarettes in the 18-25 age group (N=919); current users of electronic cigarettes in the 26+ age group (N=1,051). This includes dual use with any tobacco and related product

Question: Q.1.1. Have you ever used HTPs?; Q.1.1. Have you ever used e-cigarettes?

Source: OE elaboration from online survey data

Among the users of either HTPs or e-cigarettes who are using no other tobacco products, more than three quarters were past users of other products. The percentage is particularly high in the case of HTPs, almost 90%. In the case of e-cigarettes 32% of people in the 18-25 age group never used other products. The percentage declines to 18% for the 26+ age group as summarised in Table 66

Table 66. History of current exclusive users of HTPs and e-cigarettes (18-25 & 26+ age groups)

Percentage of current users in the 18-25 & 26+ age groups

Research questions	HTPs (%)	e-cigarettes (%)
Never users of other products		
18-25 age group	6%	32%
26+ age group	8%	18%
Exclusive current user & past user of other product(s)		
18-25 age group	94%	68%
26+ age group	92%	82%

Base: current users of HTPs in the 18-25 age group (N=503); current users of HTPs in the 26+ age group (N=634); current users of e-cigarettes in the 18-25 age group (N=919); current users of electronic cigarettes in the 26+ age group (N=1,051)

Question: Q.1.1. Have you ever used HTPs?; Q.1.1. Have you ever used e-cigarettes?

Source: OE elaboration from online survey data

7.2 Consumer perception

The survey elicited a wide range of perceptions on the five tobacco and related products, with some emerging trends. Respondents who had tried or used the products had typically done so for enjoyment. Specifically, the two main reasons cited for using waterpipe tobacco, slim cigarettes and small cigarillos were '*for enjoyment*' followed by '*to socialise / it was trendy*' consistently across both age groups and user types (current users, past users and tried the product once or twice). Furthermore, the range of flavours was part of the appeal of waterpipe tobacco, as it was for e-cigarettes and small cigarillos. Flavours make the smoke more palatable and lead some to think that use is less harmful.

Small cigarillos were used by marginally over 5% in both age groups and mainly by males. The survey identified higher than average consumption in Poland which had relatively lower prices for the product compared to the EU average. The distribution of prices for small cigarillos in 2017 showed some significant variation across the Member States, an issue that is worthy of monitoring.

HTPs and e-cigarette users had somewhat different profiles. In the 18-25 age group, for 33% the dominant reason for using either e-cigarettes or HTPs was 'enjoyment'. Other reasons included trying to quit and '*to socialise / it was trendy*'. While HTPs and e-cigarettes were promoted and are viewed as cessation tools, the main driver for the younger respondents were trends and the social aspect.

By contrast, for the 26+ age group the main reason for using these products was 'to try to quit or to reduce consumption of another tobacco product'. This pattern was particularly pronounced among e-cigarette users with 47% reporting this as their main reason for using the product. 41% of HTP users also stated 'concern for their health' as their main reason for using the product closely followed by '*for enjoyment*' (34%).

Respondents in the online survey were asked where they had heard about HTPs first, if they knew the product prior to the survey. A logit regression was performed to determine if respondents were more likely to become users after hearing about HTPs through a particular information channel. There was a positive correlation between being a user of HTPs and having heard about HTPs for the first time through Facebook, Instagram or other social networks.

Table 67. Influence of channels of information on use

User (1) ^ψ		
Predictors	Estimates	CI
Facebook, Instagram or other social media networks	0.67 *	1.11 – 3.67
Other online advertisement	0.48	0.92 – 3.09
Traditional media (print, TV, radio, billboards)	0.39	0.83 – 2.81
Marketing in shops or tobacco stores	0.44	0.89 – 2.89
Observed in the street	0.22	0.71 – 2.35
Friends and family	0.25	0.75 – 2.38
Product promotions in stores, stations or airports etc.	0.49	0.93– 3.07
Control variables	Yes	-
(Intercept)	-0.99 *	0.17 – 0.78
Observations	7945	
R ² Tjur	0.060	

Base: respondents of the online survey that had heard of HTPs prior to the survey (N=7,945)

Note: *** refers to the 0.01 level, ** refers to the 0.05 level & * refers to the 0.1 level, the latter not being considered as significant for the purpose of this study.

^ψ User is a dummy variable where users have a value of 1 and non-users a value of 0.

Source: OE elaboration from online survey data

A study commissioned by the EC (2015) found that there was no evidence of paid advertising for tobacco on the internet prior to the TPD. However, the focus group participants mentioned, and the survey charted, some evidence of commercial practices that could be interpreted as promoting tobacco products. These included prize draws, news items about tobacco retailing, and product promotions in stores and airports. User-generated content promoting specific tobacco brands and products was reported to have been seen on Facebook. Given that younger people are frequent users of the social media they may be exposed to tobacco-related content whether genuine advertising or not.

Despite the fact that most respondents claimed their decision to smoke e-cigarettes and HTPs was based on health reasons, the majority of respondents across EU countries were aware that there were potentially harmful health effects linked to product use – as consistent with findings from the Eurobarometer 458. However, users of products systematically ranked their preferred product as less harmful than did past users, those who had tried the product once or twice and never users. Use creates a tendency to deny the risks. Users of waterpipe tobacco perceived it as less harmful than all the other products, while the evidence has shown that waterpipe tobacco use can be more toxic than conventional cigarettes.

Although nicotine was widely identified as the key chemical compound that causes and sustains addiction to tobacco and related products, other indicators have been found to also affect dependence. A study analysing the indicators of dependence on e-cigarettes, waterpipe and small cigarillos reported that appeal, ease of use, use frequency and the manner in which nicotine is delivered were factors that influenced their use (Strong, 2017, USA, N=14.287). Furthermore, dual use of tobacco and related products was found to be a driver of higher tobacco dependence and of higher use of tobacco and related products. For waterpipe tobacco in particular, the present study found that consumers largely associated it with a socialising ritual and that it was not necessarily considered “smoking” in

the traditional sense and dissociated from conventional tobacco products. Small cigarillos were associated with special days and family events and slim cigarettes carried connotations of femininity, elegance and sophistication.

As illustrated in the tobacco and related product chapters, both age groups systematically perceived the five products in this study as less harmful than conventional cigarettes. The 18-25 age group gave higher scores of harmfulness for all the products than did the 26+ age group. The same held for the perception of the harmfulness to bystanders. Respondents in the online survey were asked to select which of the six products that they considered most harmful to the health of users. They were asked to give that product a score of 100 on a harmfulness scale from 100 (most harmful) to 0 (not harmful at all). They then scored each of the other tobacco products on the scale. In the table below two average scores for the two age groups are presented, one for all respondents in each age group and the other for current users of the product. The scores show that (i) the products divide into three categories, (ii) while users of a particular product assess the health risk to be lower than the age groups as a whole, they are still rating the products as harmful to health and (iii) the older age group report lower risk perception than the younger age group. Conventional cigarettes are the most harmful, followed by small cigarillos and slim cigarettes rated about 20% less harmful, and in the third category e-cigarettes, HTPs and waterpipe tobacco approximately another 15% less harmful. While there have been claims that e-cigarettes are 95% less harmful than conventional cigarettes, those surveyed in this study report much high health risk perception. A related and significant finding is the perception of risk of passive smoking. For e-cigarettes, for example, the harm to others scored only about 10 points lower than the perceived risk to users.

Table 68. Relative risk perception of the products

Age	Cigarettes		Small cigarillos		Slim cigarettes		E-cigarettes		HTPs		Waterpipe	
	All	Users	All	Users	All	Users	All	Users	All	Users	All	Users
18-25	82	75	67	62	66	61	55	48	58	52	53	45
26+	82	73	61	52	60	52	52	45	51	44	47	41

Source: Own elaboration

A logistic regression was performed to see how the perception of harm to passive users was related to the level of agreement with the proposal that e-cigarettes and HTPs should fall under the same regulations as conventional cigarettes (see Table 91 in the annex). Perception of harmfulness to others is not correlated with the support for regulation, as confirmed by the findings presented in the previous chapters. Only being a user of the product impacted the desire to regulate for e-cigarettes. In other words, a user of e-cigarettes was less likely to support the idea that the product should be regulated like conventional cigarettes

7.3 Health, social and economic impacts

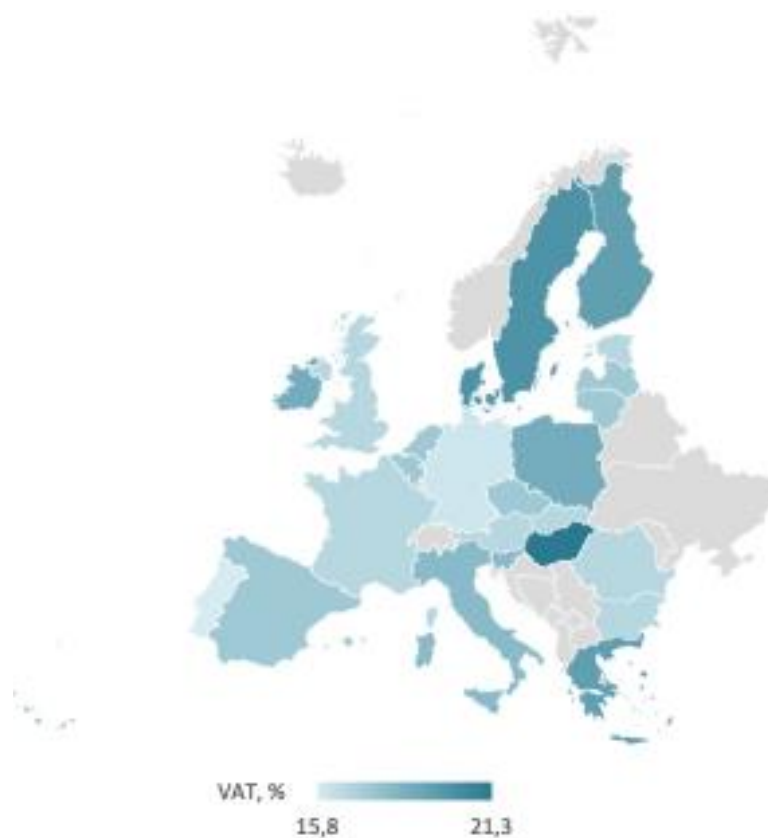
Tobacco consumption has negative health, social and economic impact not only at the individual level but also on society as a whole. Smoking and other forms of tobacco use remain one of the largest avoidable causes of morbidity and premature death in the EU. The negative health impacts of tobacco use are well established, and smoking has been linked to several forms of cancer, respiratory diseases, vascular diseases, negative reproductive effects and a wide range of other negative health impacts such as increased risks of cataracts and adverse surgical outcomes related to poor wound healing. Tobacco-related diseases incur considerable direct and indirect costs for society, including direct healthcare costs and indirect costs such as productivity losses (absenteeism, lost skills, unemployment), welfare provision costs (sickness and unemployment benefits) and fire and other

accidents (property losses, wild fires), as well as intangible costs such as pain and suffering that result from loss of life or illnesses brought on by tobacco use.

Devices for novel tobacco products (and so HTPs) are not explicitly stated in the TPD, except for e-cigarettes (liquids and devices). Emissions of e-cigarettes and HTPs are however an interplay of the device, the sticks, and/or the liquids.

The economic costs of tobacco use are substantial and include significant healthcare costs for treating tobacco-related diseases as well as the loss of human capital due to morbidity and mortality attributable to tobacco use. It has been suggested that about 15% of the aggregate health care expenditure in high-income countries can be attributed to smoking (Epku & Brown, 2015). Smoking also poses considerable indirect costs to society through passive smoking, it can cost employers in the form of loss of productivity and absenteeism of smokers due to illnesses, it pollutes the environment, and more (Epku & Brown, 2015). In terms of economic impact, the main source of data was Passport data on taxes. Significant increases in the taxes and prices of tobacco and related products is a strong measure to reduce their use. There are certain disparities in taxation levels of tobacco and related products across countries and within different product types. The first figure represents the geographical scope of VAT in the year 2017 applied to all tobacco products. The highest VAT was applied in Hungary, Denmark and Sweden.

Figure 174. VAT values on tobacco products by countries in 2017



Note: Percentage share of slim cigarettes were estimated with respect to the whole tobacco market in the country in million euros.

Source: Passport

In terms of excise duty applied to manufactured tobacco, tax rates differed across Member States and across specific product categories. In 2011, the Council adopted the Directive 2011/64 on the structure and rates of excise duty applied to manufactured tobacco (also known as 'Tobacco Excise Directive' – TED), which updated the harmonised rules for the structure and rates of excise duty applied to manufactured tobacco products on tobacco products in the EU. Following the directive, the tax provisions applicable to manufactured tobacco should follow the structure for each main category. The products within scope of this study fall under the 4 main category defined by the Directive 2011/64 . The rates or amount defined in this Directive shall be effective for all products belonging to the group without distinction within each group.

Table 69. Tax provisions applicable to manufactured tobacco

Product	Tax structure
Cigarettes (including slim cigarettes)	• Mandatory mixed structure including both an ad valorem excise duty and a specific excise duty, which must be b/w 7.5% and 76.5% of total tax burden (since Jan. 2014). • At least 60% of WAP and no less than EUR 90 per 1,000 cigarettes; or EUR 115 per 1,000 cigarettes. • A minimum excise duty (MED) may apply (i.e. a fixed monetary amount per quantity applicable if the amount of the excise duty falls below a minimum floor).
Cigars and cigarillos (including small cigarillos)	• 5% or more of the retail selling price or EUR 12 per 1,000 items or Kg. • A minimum amount of excise duty can be established
Other Smoking Tobacco category (including waterpipe)	• 20% or more of the retail selling price or EUR 22 per Kg. • A minimum amount of excise duty can be established.

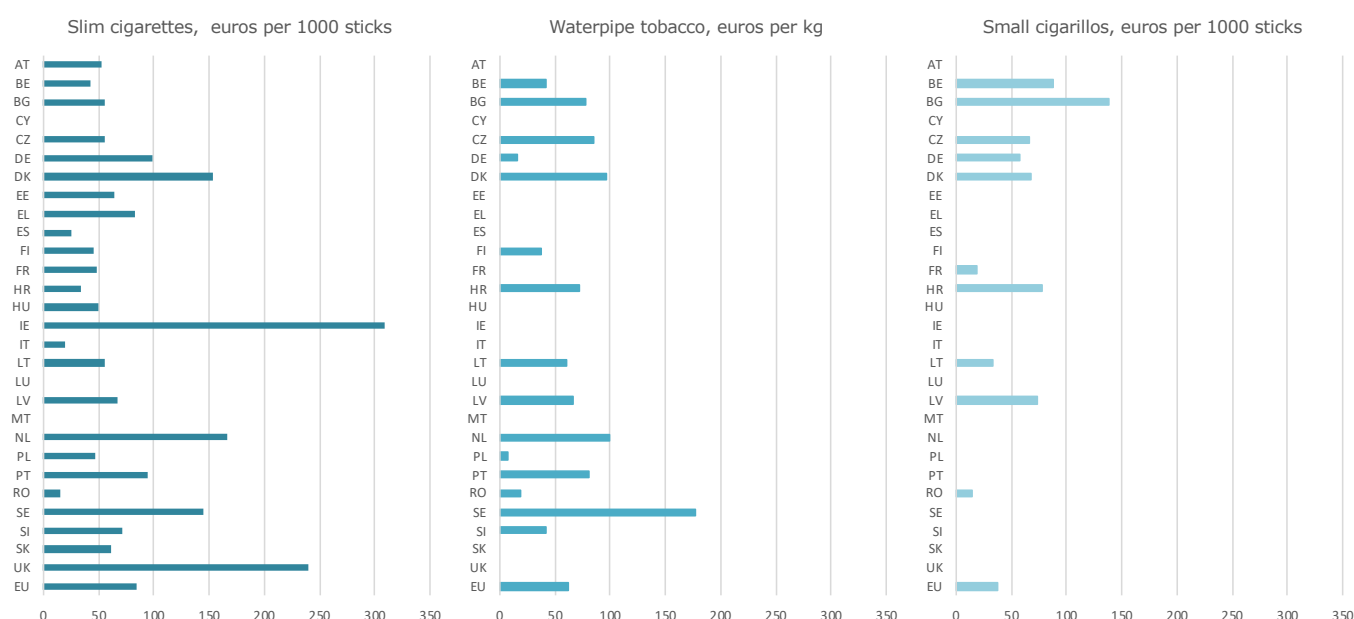
Source: Own elaboration

Due to their novelty, the categorisation of HTPs under the Directive was not clear and various Member States adopted *ad-hoc* and special tax regimes. Similarly, e-cigarettes were not included in the Directive and some Member States independently adopted specific tax regimes. The heterogenous tax regime in the EU has created uncertainties, burden and potential distortions in the commercialisation and movement of tobacco and related products.

Using Passport data, information on excise duties on waterpipe tobacco, small cigarillos and slim cigarettes were retrieved, as presented in the separate chapters for products individually.¹⁴⁰ Time series data was also analysed but no significant changes in tax policies were detected and no relevant findings emerged. Hence, the following analysis focusses on the most recent available data, i.e. 2017. The figure below represents the differences across the EU countries in terms of taxation of the specific categories of the tobacco products.

¹⁴⁰ Slim cigarettes have the same rate as cigarettes.

Figure 175 Taxes on tobacco products by countries in 2017



Note: EU average is calculated for the countries where data was available. Where there is no bar, data was not available.

Source: Passport, 2017

The data shows that the highest excise tax on slim cigarettes is applied in Ireland, the UK and the Netherlands¹⁴⁰. It is worth mentioning that slim cigarettes do not have their own excise tax but that they are the same as that for cigarettes. The data on taxation of small cigarillos and waterpipe tobacco was not available for all the EU countries, but among those with available information, the highest taxes on waterpipe tobacco were found to be in Sweden and Denmark. The EU average shows that small cigarillos were the least taxed product, with the notable exceptions of in Bulgaria, Belgium and Latvia. However, there were some differences across the countries. In Bulgaria and Belgium, the tax on small cigarillos was more than twice that of other tobacco products, and in Germany tax on waterpipe tobacco was less than a third of that applied to small cigarillos and slim cigarettes. This may favour the formation of a black market (see section 2.2). Indeed, tax evasion has been found to be particularly high in the case of waterpipe tobacco, resulting in revenue loss and in the creation of a black market for the product, as reported by the European Commission.¹⁴¹

According to this same study, even if e-cigarettes do not qualify for being taxed under Directive 2011/64, some Member States started to adopt ad-hoc consumption taxes on e-cigarettes since 2014. The precursors were Italy (2013) and Portugal (2014), later joined by Romania, Slovenia and Latvia. As of today, some nine Member States have applied ad-hoc taxes as summarised in the table below. The same study also noted that some countries have adopted various definitions of HTPs in regard to the tax scheme, resulting in many interpretations of how to implement the directive. The table below summarises these results.

¹⁴¹ European Commission (2017): Study on Council Directive 2011/64/EU on the structure and rates of excise duty applied to manufactured tobacco.

Table 70. Overview of non-harmonised taxes for e-cigarettes and heated tobacco products

Country	Electronic Cigarettes	Heated Tobacco Products
PT	€0.60/ml nicotine liquid, reduced to €0.30 since 01.01.2017	Ad valorem: 20% of RSP (reduced to 16% in 2017) Specific: € 78 /kg (€80/kg in 2017) Minimum Excise: €169/kg (as smoking tobacco)
IT	€0.385/ml on all liquids, but taxation of non-nicotine liquids has been suspended after Constitutional Court's ruling 83/2015	Depending on SKU: From € 63.25 to €63.36 per 1,000 sticks (as 50% of cigarettes excise)
RO	RON 0.5/ml nicotine liquid (ca. €0.11/ml)	RON 384/kg (ca. €85) (as smoking tobacco)
SI	€0.18/ml nicotine liquid	€ 88/kg (as FCT – minimum duty)
LV	€0.01/ml of e-liquid + €0.005/mg of nicotine	€ 62/kg (as smoking tobacco)
HU	HUF 55/ml (ca. €0.18/ml) since 01.01.2017 (a different legislation may enter into force since April 2017, bringing the tax rate to HUF 65/ml)	HUF 10,000 per 1,000 sticks (€ 32.46)
FI	€0.30/ml	NA
EL	€0.10/ml	€ 156.7/kg (as smoking tobacco)
HR	HRK 0.00/ml	HRK 600 / kg (ca. € 80.7) (as fine-cut tobacco)
SK	NA	€ 73.9 / kg (as smoking tobacco)

Source: European Commission

The focus group results highlighted that users of different products were often mentioning price. They believed that the prices of tobacco and related products were high, but this was not considered a major drawback. Yet price-based policy measures such as increase in tobacco and related products taxes are unarguably the most effective means of reducing the consumption of tobacco. A 10% tax-induced cigarette price increase anywhere in the world was expected to reduce smoking prevalence by between 4% and 8% (Epku & Brown, 2015). A publication by WHO (2018) highlighted that the large price increases in France were followed by declines in smoking prevalence and lung cancer deaths. In line with this, the United Kingdom successfully raised taxes above the inflation level, increased the revenues from tobacco taxes despite consistent drops in smoking prevalence, and reduced illicit trade at the same time. A meta-analysis review by Jawad et al. (2018) found that a 10% price increase would reduce demand, e.g. by 8.3% for cigars. It was suggested that this impact would mostly affect youth (Levy et al., 2018) and seem to be much more responsive to price changes of the product. A recent study found that increases in tobacco tax, warning signs on packaging, restriction of tobacco product advertisements, national law to discourage young people, and peer-based approaches to quit tobacco are popular policies (Ranabhat et al., 2019). Indeed, an often-cited reason for attempts to quit was “*for economic reasons*” in our survey, although rarely as the main reason.

In terms of health impacts, the WHO was an important source of information for many indicators, especially mortality. As a main indicator, compared to the rest of the world, the WHO European

Region¹⁴² had one of the highest percentages of deaths attributable to tobacco use. The WHO estimated that tobacco use was responsible for 16% of all deaths in adults over 30 in the region, many of these deaths being premature. This contrasts with the global average of 12% as well as with surrounding regions.¹⁴³ Many studies have suggested that use of e-cigarettes is less harmful than conventional smoking, but no clinical trials have proven that this product can help to quit tobacco consumption as opposed to other pharmaceutical products available on the market. In regard to HTPs, there was a great deal of uncertainty around the market size at the time of the study due to insufficient data, but the market appeared to be growing and the effects to health remained unknown. The European Commission has closely followed the market developments of the newer products as they gained grip on the market. For example, the transposition of the new TPD in 2016 regulated e-cigarette marketing across the Union.

The finding that younger generations are particularly susceptible to influences of design and prices of tobacco and related products (Laverty et al., 2016; Meernik et al., 2019) has raised concerns regarding youth pick-up rates of e-cigarettes and waterpipe tobacco consumption. These findings are consistent with our results, in which the youth consistently stated trends and social aspects as the appealing factor that motivated their initiation to the use all products under the scope of this study. The design appeal is also observed for adults, but at a lesser degree. For example, our study found that the slim cigarette packaging was viewed as more convenient (e.g. *"easier to transport"*) than conventional cigarettes, while the designs of the non-conventional tobacco products was perceived as appealing (e.g. *"very cute and nice"*, *"fun packaging"*). Indeed, Laverty et al. (2016) found that younger people (15-24 years old) were more likely to cite external packaging and design as important features. HTPs have also been marketed and designed around eight themes specific to the device, namely, clinical purity, a closed system, premium design, sensory invitations, nostalgia for conventional tobacco smoking, stability, familiar technology and normalisation (Hair, 2018).

Plain packaging policies have been suggested as a measure to reduce the attractiveness of smoking, particularly among youths. France introduced this policy in 2017 – which has provided a natural setting to test the impact on perceptions of branding and packaging. El-Khoury Lesueur et al. (2018) examined changes in smoking-related perceptions and behaviours among French adolescents prior to (2016) and one-year post implementation (2017). He found that youth was more likely to report negative views of smoking (e.g. fear of the consequences). They also seemed to perceive it as less socially acceptable. Furthermore, smoking initiation had significantly decreased, with a less significant drop in current tobacco use. This finding is consistent with our study, during which participants in France reported that the neutral packaging was less *"attractive"* or *"fun"* and more *"austere"*, whereas other products that had not been regulated in the same manner due to their novelty were subsequently and implicitly positively highlighted. These findings suggest that plain packaging may also be an interesting measure to reduce the marketing advantage of certain tobacco and related products. Indeed, consumption of slim cigarettes was found to be particularly low in France in our study – ranking second to last in terms of consumption of the product, after Germany. Several reviews have concluded that plain packaging may reduce smoking prevalence, with the most consistent evidence for its role in reducing the appeal of the products. This finding is consistent across many studies on the impact of plain packaging (Greenhalgh, 2018). Furthermore, our study found that images on the

¹⁴² Albania, Andorra, Armenia, Austria, Azerbaijan, Belarus, Belgium, Bosnia and Herzegovina, Bulgaria, Croatia, Cyprus, Czechia, Denmark, Estonia, Finland, France, Georgia, Germany, Greece, Hungary, Iceland, Ireland, Israel, Italy, Kazakhstan, Kyrgyzstan, Latvia, Lithuania, Luxembourg, Malta, Monaco, Montenegro, Netherlands, North Macedonia, Norway, Poland, Portugal, Republic of Moldova, Romania, Russian Federation, San Marino, Serbia, Slovakia, Slovenia, Spain, Sweden, Switzerland, Tajikistan, Turkey, Turkmenistan, Ukraine, United Kingdom of Great Britain and Northern Ireland and Uzbekistan

¹⁴³ https://apps.who.int/iris/bitstream/handle/10665/44815/9789241564434_eng.pdf;jsessionid=888154B027B96ED6455C90F3BDE539FB?sequence=1

packaging were not so influential among smokers – potentially because they had reportedly become accustomed to visualising these pictures. Rather, it was believed to possibly prevent new consumption of products.

8 Conclusion

The available evidence gathered during the systematic review and empirical research suggested the following points in regard to the main drivers of market developments and consumer perceptions for the five products under the scope of this study:

- Slim cigarettes were marginally more used by females (58% of slim cigarette users in 18-25 age group), while small cigarillos were mainly used by males (e.g. 70% of current consumers in 18-25 age group). Waterpipe tobacco had 62% of male users. The consumer base for other products were relatively balanced in terms of gender. The focus group discussions revealed that slim cigarettes were perceived as feminine as well as business-like, especially in eastern Member States, and small cigarillos were associated with masculinity.
- It is not easy to establish the reasons behind tobacco and related products patterns of use. Enjoyment, stress reduction, sociability, being trendy all feature in users accounts of their probably addiction to nicotine. Reduced health risk is one of the multiple drivers in the choice of certain products, even when there is no evidence of health risk reduction and the use of the product may lead to behaviours that are more damaging to health. As an example, some slim cigarette users said that they sometimes needed to consume more sticks to feel as satiated as they would with conventional cigarettes.
- Small cigarillos were reported as being associated with social gatherings and special occasions. However, they were popular in some markets in part due to the flavouring of the product which has been banned for cigarettes and RYO in the TPD. This warrants the monitoring of market developments in the product.
- Products were experienced differently by users. HTPs and e-cigarettes were viewed as a colder experience, more modern and technological, but without the sense of satisfaction provided by conventional cigarettes. Slim cigarettes and small cigarillos were generally perceived as old-fashioned (e.g. authentic, gender-oriented, etc.). Waterpipe tobacco was typically dissociated from other tobacco and related products as it is used in special premises and within a group. This is of concern as it may limit exposure of users to package health warnings.
- Respondents in the 18-25 age group were flagged as an important consumer group for waterpipe tobacco, with the majority (53%) having at least tried the product across the surveyed countries. Waterpipe was also typically perceived to be mainly consumed by youth. The data indicated that respondents in the 18-25 age group began consumption of waterpipe tobacco products before the age of 18 years.
- The online survey identified prevalent dual use among users with conventional cigarettes the second product of choice for the greater majority. Only a minority reported quitting all tobacco and related products and the greater majority of those who were using e-cigarettes or HTPs were using conventional cigarettes in parallel.
- What have the survey findings contributed to the debate on whether e-cigarettes were an aid to quitting conventional cigarettes (the cessation scenario) or whether young people would take up the use of e-cigarettes and progress to conventional cigarettes (the gateway scenario). Before looking at the survey evidence a note of caution is needed. The online survey was a cross-sectional study and, as such, cannot lead to causal claims.

- Just below 1 in 5 in both age groups said they had previously used other products but were now only using e-cigarettes. 1 in 12 of the younger age group and 1 in 25 of the older age group were exclusive e-cigarette users and had not used other products. Given that 'cutting down or quitting other tobacco products' was cited as a reason for taking up e-cigarettes, these findings were suggestive of the cessation scenario. However, over 7 in 10 of the 18-25 age group and 8 in 10 of 26+ age group were dual users of e-cigarettes and other products, mainly conventional cigarettes.
- Of past e-cigarette users, 1 in 4 in both age groups had been dual users and now were not using any tobacco and related products. This is suggestive of the cessation scenario. Suggestive of the gateway scenario, 1 in 4 of the 18-25 age group and 3 in 10 of the 26+ age group had been exclusive e-cigarette users and now reported using other tobacco and related products, with some 90% reporting the use of conventional cigarettes. Marginally more than 4 in 10 of both age groups were past e-cigarette users and reported current dual use of other tobacco and related products. The use of e-cigarettes had not made an impact on their use of conventional tobacco.
- Overall, the survey findings are suggestive of both cessation and gateway scenarios. This in the context of dual use reported by a substantial majority of the current e-cigarette users and by significant numbers of past e-cigarette users.
- Since younger generations are increasingly less likely to smoke conventional tobacco products, tobacco companies have developed new products which are marketed as "*safer*" alternatives. There has been a proliferation of novel tobacco products entering the EU market. Of particular significance to the TPD are heated tobacco products (HTPs). They have been marketed in over 40 countries worldwide as of July 2019, with at least 22 countries from the WHO European region. According to the few existing population surveys, the prevalence of HTP current use among adults is less than 0.5% in the general population of European Union MS.
- Heated tobacco products (HTPs) are tobacco products that produce an emission containing nicotine and other chemicals, which is then inhaled by users. They have been marketed as so-called potentially reduced risk exposure products, or even as modified-risk tobacco products. Currently, there is insufficient evidence to conclude that HTPs are less harmful than conventional cigarettes. There are concerns that while they may expose users to lower levels of some toxicants than conventional cigarettes, they also expose users to higher levels of other toxicants. It is still unclear how this toxicological profile translates into short- and long-term health effects.
- According to Passport data (2017), HTP sales were 769 million euros. The leading markets were Italy and Romania. Although there are no time series trend data on HTP sales or HTP use, the market size for HTP is expected to grow soon but there is still a big gap before it reaches the size of the market for conventional tobacco cigarettes (EC, 2017).
- Marketing and promotion of HTPs has included alleviating health concerns through claims that HTPs are reduced-risk products; designing HTP systems to boost the user's social image; using "bait and hook" pricing strategy; employing social marketing techniques and brand ambassadors; selling and marketing via multiple channels, including dedicated stores, retail establishments and e-commerce websites.
- Member States have adopted different definitions of HTPs with respect to taxes, resulting in substantial fragmentation of the EU internal market and resulting in a large variation in prices across European countries

- The empirical research (focus groups and online survey) revealed that HTPs were the least known product among the products under study. HTPs and electronic cigarettes were sometimes confused, considered to be similar products, and believed to not be as satisfying as conventional tobacco products.
- Everyday use of HTPs with 1 to 10 sticks was typical for current users. Initiation of use was motivated by either enjoyment or trying to quit or reduce smoking of conventional .About 85% were dual users.
- Current HTP users were evenly split between those than had never attempted quitting and those who tried at least once and failed in the 18-25 group. In the 26+ group, almost two out of three had never tried to quit. Among failed quitters, the main reasons to try quitting were for 'personal health' and 'economic reasons'. In the 26+ group, another reason frequently mentioned was the 'concern for the health of the people around them'. The main reason cited for failing to quit was being 'stressed'. In the 26+ group, another important reason was that they found quitting HTPs 'too difficult'.
- The majority of past users of HTPs were still using other tobacco products at the time of the survey.
- Health risk perception of the five tobacco and related products analysed in this study plus conventional cigarettes found the following. Conventional cigarettes were perceived to be the most harmful to the health of users and bystanders. Comparing the risk perception of users of conventional cigarettes with the users of HTPs, the 18-25 age group rated HTPs as 35% less harmful than conventional cigarettes to users and 46% less harmful to bystanders. In the 26+ age group HTPs were perceived to be 52% less harmful to users and 58% less harmful to bystanders.
- About 75% of respondents agree that HTPs should be regulated as conventional cigarettes and a many support restrictions in use in workplaces, public transport and the hospitality sector.
- Social media sites have enabled indirect product promotion (e.g. through user-generated content) which influences public opinions and perception and runs the risk of re-normalising smoking in the public. As a result, around one quarter of respondents for each age group [23-25%] reported advertisements as a factor in their decision to start using HTPs. The overall perception of e-cigarettes and HTPs appeared to be marked by social factors, including marketing practices.
- Focus groups participants raised concern over the influence of marketing practices and product designs on attracting the youth. In the online survey, a majority of respondents of both age groups strongly believed that the youth was a vulnerable target group for HTPs (70% 18-25 age group; 66% 26+ age group) and electronic cigarettes (80% 18-25 age group; 76% 26+ age group). Packaging and design were found to influence consumption prevalence, especially in younger generations (Lavery et al., 2016; Meernik et al., 2019). El-Khoury Lesueur et al. (2018) demonstrated that the implementation of standardised cigarette appearance reduced the appeal of cigarettes in adolescents and reduced the opportunity for stick design to mislead young smokers in terms of harm. In addition, flavours have been marketed in products for which the TPD has not banned flavours. Accordingly, the focus group discussions revealed that flavours were an important feature to consumers of e-cigarettes, waterpipe tobacco and small cigarillos. Flavour and trends were stated as important influences, also on social media (e.g. Instagram). The focus group findings warrant for close monitoring of the market developments, especially of these features.

- The 18-25 age group had a tendency to perceive higher health risks from tobacco and related products than the 26+ age group. Current research has concluded that none of the products under study were risk-free and while HTPs and e-cigarettes are considered relatively less harmful than conventional cigarettes, the longer-term health impacts of these new products have yet to be ascertained. Research provided no indication that slim cigarettes, waterpipe tobacco and small cigarillos should be considered less harmful than conventional cigarettes. It is of concern that users of these products consider them to 'safer' than conventional cigarettes.

In addition to the main drivers of market developments and consumer perceptions, this study also considered the health, social and economic impacts¹⁴⁴. Emerging new products are distinguishable by the nature of their emissions and increasingly being used by those who share the perception that they are "healthier options" compared to traditional tobacco products such as cigarettes. This despite the absence of evidence that these products are 'healthier' and that most regulators in the EU and beyond have taken a precautionary approach to these products. E-cigarette companies have enhanced the appeal of their offerings by advancing the efficiency of nicotine delivery, but also by adding appealing flavourings.¹⁴⁵ Further studies are needed to clarify health, social and economic impacts.

Novel and emerging tobacco products, above all, HTPs, pose particular challenges to the application of TAPS bans and restrictions. The devices are aggressively marketed in many Member States' providing the appearance of a healthier product and normalising the promotion and use of tobacco products. Challenges arise in particular from the increasing use of the internet and social media platforms for covert and indirect advertising, which, it has been suggested, the industry increasingly uses to exploit legal loopholes to circumvent prohibitions.

Despite the identified gaps in evidence and the dynamic of evolving patterns of the use of e-cigarettes and novel tobacco products, policy options are needed that ensure protection of the public at large, both users and potential users. A major challenge from a public health perspective appears to be balancing potential "harm reduction" claims of certain products, where the evidence is still mixed, and the ethical issue of substituting an addictive and health harmful product with a new addictive and putatively less health harmful product. For example, the potential of e-cigarettes to act as a gateway to smoking or as an attractive alternative to conventional tobacco products in particular for young people has been evidenced in the literature and in the current research. E-cigarettes were found to simulate smoking behaviour and can lead to further experimentation with other nicotine-containing products. Recent studies suggest that e-cigarettes are increasingly used by non-smokers and young people. For example, a French study of 2013 revealed that the number of Parisian students experimenting with e-cigarettes has doubled in one year reaching 18%.¹⁴⁶ Governments across Member States have become increasingly concerned over the appeal of e-cigarette use among young people. The balance between risks and potential harm reduction benefits of e-cigarette use depends on its gateway impacts among youth and users at large as well as the actual harm of those products. It is clear that the assessment of 'actual harm' and longer-term impacts of the use has yet to be established.

Overall, policy efforts related to tobacco control have been advancing worldwide, and the EU and Member States have implemented a number of legal and financial measures to reduce the consumption of tobacco and related products. The extensive list of adopted tobacco control laws and policies across countries (e.g. Andersen, Becher & Winkler, 2016) supports data related to recent

¹⁴⁴ See sections 2.4, 3.4, 4.4, 5.4 and 6.4.

¹⁴⁵ National Center for Chronic Disease Prevention and Health Promotion (US) Office on Smoking and Health. E-Cigarette Use Among Youth and Young Adults: A Report of the Surgeon General [Internet]. Atlanta (GA): Centers for Disease Control and Prevention (US); 2016. Chapter 5, E-Cigarette Policy and Practice Implications.

¹⁴⁶ https://ec.europa.eu/health/sites/health/files/tobacco/docs/tobacco_mythbuster_en.pdf

trends that prove increased efforts to ratify and implement the Framework Convention on Tobacco Control (FCTC), which legally binds signatory states to implement its articles. However, in order to determinate the critical balance for public health between the harms and potential benefits of e-cigarettes and other novel tobacco related products, more evidence is needed.

Article 28 of the Directive specifies that the Commission should pay special attention to novel tobacco products and electronic cigarettes and refill containers, cigarettes with a diameter of less than 7,5 mm and waterpipe tobacco. This study has investigated the market developments, initiation of use of products by young and older people, cessation efforts, consumer perceptions, as well as considerations around use of flavours in some of these products. The results reported are intended to support the Commission by assessing how specific product categories are perceived by the public, what impacts they may have, and mapping consumer preferences and use patterns of these products. The evidence gathered here will help inform policymakers for potential next steps in tobacco control policies with the ultimate aim of protecting public health.

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10 Annexes

10.1 Background

Table 71. online survey sample description (18-25 age group)
(number & (percentage) – of all respondents in the 18-25 age group)

18-25 age group	Group total	BG	DE	EL	ES	FR	IT	LV	PL	SK	UK
Total (n)	6,090	610	603	610	605	606	601	603	609	643	600
Gender											
Male	2,869	290	262	298	298	305	300	234	301	280	301
%	47.1	47.5	43.4	48.9	49.3	50.3	49.9	38.8	49.4	43.5	50.2
Female	3,221	320	341	312	307	301	301	369	308	363	299
%	52.9	52.5	56.6	51.1	50.7	49.7	50.1	61.2	50.6	56.5	49.8
Education											
No official education	84	9	8	8	6	20	5	11	6	2	9
%	1.4	1.5	1.3	1.3	1.0	3.3	0.8	1.8	1.0	0.3	1.5
Primary school	251	19	13	6	14	34	21	48	21	54	21
%	4.1	3.1	2.2	1.0	2.3	5.6	3.5	8.0	3.4	8.4	3.5
High school	2,049	190	209	35	145	169	272	200	315	302	212
%	33.6	31.1	34.7	5.7	24.0	27.9	45.3	33.2	51.7	47.0	35.3
Some years of university	1,780	195	228	305	211	107	150	159	132	125	168
%	29.2	32.0	37.8	50.0	34.9	17.7	25.0	26.4	21.7	19.4	28.0
University degree	1,509	136	119	213	160	184	119	142	122	156	158
%	24.8	22.3	19.7	34.9	26.4	30.4	19.8	23.5	20.0	24.3	26.3
Post-graduate	417	61	26	43	69	92	34	43	13	4	32
%	6.8	10.0	4.3	7.0	11.4	15.2	5.7	7.1	2.1	0.6	5.3
Employment Status											
Self-employed	543	83	24	79	65	56	44	68	37	30	57
%	8.9	13.6	4.0	13.0	10.7	9.2	7.3	11.3	6.1	4.7	9.5

18-25 group	age	Group total	BG	DE	EL	ES	FR	IT	LV	PL	SK	UK
Managerial position		358	49	23	13	20	31	12	48	40	37	85
%		5.9	8.0	3.8	2.1	3.3	5.1	2.0	8.0	6.6	5.8	14.2
Other collar	white	1,084	123	105	94	123	93	90	153	119	79	105
%		17.8	20.2	17.4	15.4	20.3	15.3	15.0	25.4	19.5	12.3	17.5
Manual worker		779	96	134	23	63	80	33	52	104	87	107
%		12.8	15.7	22.2	3.8	10.4	13.2	5.5	8.6	17.1	13.5	17.8
Unemployed		528	51	18	82	57	28	72	61	43	51	65
%		8.7	8.4	3.0	13.4	9.4	4.6	12.0	10.1	7.1	7.9	10.8
Retired		36	4	8	3	2	3	1	5	2	6	2
%		0.6	0.7	1.3	0.5	0.3	0.5	0.2	0.8	0.3	0.9	0.3
Student		2,545	191	276	304	269	257	328	187	254	329	150
%		41.8	31.3	45.8	49.8	44.5	42.4	54.6	31.0	41.7	51.2	25.0
Non-employed		217	13	15	12	6	58	21	29	10	24	29
%		3.6	2.1	2.5	2.0	1.0	9.6	3.5	4.8	1.6	3.7	4.8
Residency												
Urban		3,949	532	327	462	444	354	330	440	408	378	274
%		64.8	87.2	54.2	75.7	73.4	58.4	54.9	73.0	67.0	58.8	45.7
Suburban		1,173	49	113	96	104	125	177	97	101	52	259
%		19.3	8.0	18.7	15.7	17.2	20.6	29.5	16.1	16.6	8.1	43.2
Rural		968	29	163	52	57	127	94	66	100	213	67
%		15.9	4.8	27.0	8.5	9.4	21.0	15.6	10.9	16.4	33.1	11.2
Self-reported socio-economic status												
1 lowest		53	8	4	7	5	6	1	6	2	5	9
%		0.9	1.3	0.7	1.1	0.8	1.0	0.2	1.0	0.3	0.8	1.5
2		122	11	5	8	2	21	13	18	16	5	23
%		2.0	1.8	0.8	1.3	0.3	3.5	2.2	3.0	2.6	0.8	3.8
3		351	36	14	39	17	46	26	32	42	39	60

18-25 group	age	Group total	BG	DE	EL	ES	FR	IT	LV	PL	SK	UK
%		5.8	5.9	2.3	6.4	2.8	7.6	4.3	5.3	6.9	6.1	10.0
4		600	79	43	69	46	53	43	66	62	78	61
%		9.9	13.0	7.1	11.3	7.6	8.7	7.2	10.9	10.2	12.1	10.2
5		1,145	139	80	118	100	96	105	85	155	146	121
%		18.8	22.8	13.3	19.3	16.5	15.8	17.5	14.1	25.5	22.7	20.2
6		1,439	140	124	165	164	127	142	160	146	154	117
%		23.6	23.0	20.6	27.0	27.1	21.0	23.6	26.5	24.0	24.0	19.5
7		1,370	120	185	123	150	122	157	142	123	141	107
%		22.5	19.7	30.7	20.2	24.8	20.1	26.1	23.5	20.2	21.9	17.8
8		618	43	98	48	74	73	68	58	38	54	64
%		10.1	7.0	16.3	7.9	12.2	12.0	11.3	9.6	6.2	8.4	10.7
9		209	13	31	19	25	32	27	14	17	11	20
%		3.4	2.1	5.1	3.1	4.1	5.3	4.5	2.3	2.8	1.7	3.3
10 highest		183	21	19	14	22	30	19	22	8	10	18
%		3.0	3.4	3.2	2.3	3.6	5.0	3.2	3.6	1.3	1.6	3.0

Source: OE elaboration from online survey data

Table 72. online survey sample description (26+ age group)
(number & (percentage) – of all respondents in the 26+ age group)

26+ age group	Group total	BG	DE	EL	ES	FR	IT	LV	PL	SK	UK
Total	5,910	590	597	590	595	594	599	597	591	557	600
Gender											
Male	3,074	301	338	298	299	279	294	371	287	315	292
%	52.0	51.0	56.6	50.5	50.3	47.0	49.1	62.1	48.6	56.6	48.7
Female	2,836	289	259	292	296	315	305	226	304	242	308
%	48.0	49.0	43.4	49.5	49.7	53.0	50.9	37.9	51.4	43.4	51.3
Education											
No official education	37	1	2	5	0	17	3	3	2	1	3
%	0.6	0.2	0.3	0.8	0.0	2.9	0.5	0.5	0.3	0.2	0.5
Primary school	271	7	82	3	12	66	49	24	11	10	7
%	4.6	1.2	13.7	0.5	2.0	11.1	8.2	4.0	1.9	1.8	1.2
High school	2,041	123	265	87	159	179	248	193	221	303	263
%	34.5	20.8	44.4	14.7	26.7	30.1	41.4	32.3	37.4	54.4	43.8
Some years of university	663	67	99	100	77	41	53	84	31	49	62
%	11.2	11.4	16.6	16.9	12.9	6.9	8.8	14.1	5.2	8.8	10.3
University degree	2,005	199	111	288	251	203	139	210	246	179	179
%	33.9	33.7	18.6	48.8	42.2	34.2	23.2	35.2	41.6	32.1	29.8
Post-graduate	893	193	38	107	96	88	107	83	80	15	86
%	15.1	32.7	6.4	18.1	16.1	14.8	17.9	13.9	13.5	2.7	14.3
Employment status											
Self-employed	819	105	57	111	96	71	110	90	63	60	56
%	13.9	17.8	9.5	18.8	16.1	12.0	18.4	15.1	10.7	10.8	9.3
Managerial position	749	87	75	43	80	25	43	93	97	62	144
%	12.7	14.7	12.6	7.3	13.4	4.2	7.2	15.6	16.4	11.1	24.0

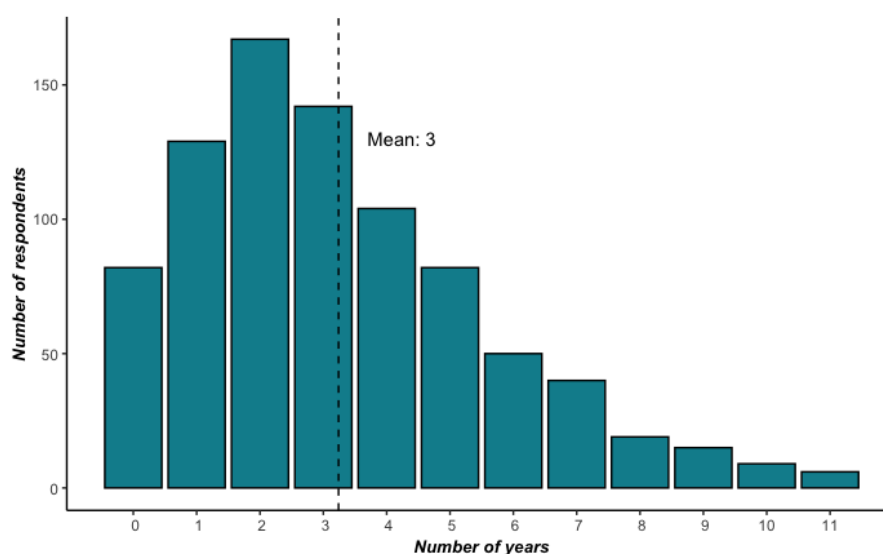
26+ group	age	Group total	BG	DE	EL	ES	FR	IT	LV	PL	SK	UK
Other white collar		1,790	219	132	240	214	160	209	167	205	134	110
%		30.3	37.1	22.1	40.7	36.0	26.9	34.9	28.0	34.7	24.1	18.3
Manual worker		941	98	105	40	104	122	44	101	115	140	72
%		15.9	16.6	17.6	6.8	17.5	20.5	7.3	16.9	19.5	25.1	12.0
Unemployed		400	23	40	96	37	33	58	39	16	21	37
%		6.8	3.9	6.7	16.3	6.2	5.6	9.7	6.5	2.7	3.8	6.2
Retired		718	36	136	36	31	102	38	55	65	106	113
%		12.1	6.1	22.8	6.1	5.2	17.2	6.3	9.2	11.0	19.0	18.8
Student		106	8	15	12	10	4	25	9	10	5	8
%		1.8	1.4	2.5	2.0	1.7	0.7	4.2	1.5	1.7	0.9	1.3
Non-employed		387	14	37	12	23	77	72	43	20	29	60
%		6.5	2.4	6.2	2.0	3.9	13.0	12.0	7.2	3.4	5.2	10.0
Residency												
Urban		3,879	553	313	474	476	272	385	429	459	354	164
%		65.6	93.7	52.4	80.3	80.0	45.8	64.3	71.9	77.7	63.6	27.3
Suburban		1,010	21	116	79	69	101	131	79	65	47	302
%		17.1	3.6	19.4	13.4	11.6	17.0	21.9	13.2	11.0	8.4	50.3
Rural		1,021	16	168	37	50	221	83	89	67	156	134
%		17.3	2.7	28.1	6.3	8.4	37.2	13.9	14.9	11.3	28.0	22.3
Self-reported socio-economic status												
1 lowest		71	8	5	5	6	10	11	15	0	4	7
%		1.2	1.4	0.8	0.8	1.0	1.7	1.8	2.5	0.0	0.7	1.2
2		205	29	15	8	11	28	17	32	12	21	32
%		3.5	4.9	2.5	1.4	1.8	4.7	2.8	5.4	2.0	3.8	5.3
3		433	49	32	28	24	61	30	49	56	45	59
%		7.3	8.3	5.4	4.7	4.0	10.3	5.0	8.2	9.5	8.1	9.8
4		665	74	59	77	42	57	55	76	65	81	79
%		11.3	12.5	9.9	13.1	7.1	9.6	9.2	12.7	11.0	14.5	13.2

26+ group	age	Group total	BG	DE	EL	ES	FR	IT	LV	PL	SK	UK
5		1,254	159	102	128	114	134	108	120	139	127	123
%		21.2	26.9	17.1	21.7	19.2	22.6	18.0	20.1	23.5	22.8	20.5
6		1,365	136	131	138	161	130	162	134	132	117	124
%		23.1	23.1	21.9	23.4	27.1	21.9	27.0	22.4	22.3	21.0	20.7
7		1,150	86	122	135	146	93	128	115	117	101	107
%		19.5	14.6	20.4	22.9	24.5	15.7	21.4	19.3	19.8	18.1	17.8
8		528	30	93	48	59	49	60	43	46	46	54
%		8.9	5.1	15.6	8.1	9.9	8.2	10.0	7.2	7.8	8.3	9.0
9		131	8	21	11	22	16	11	5	16	10	11
%		2.2	1.4	3.5	1.9	3.7	2.7	1.8	0.8	2.7	1.8	1.8
10 highest		108	11	17	12	10	16	17	8	8	5	4
%		1.8	1.9	2.8	2.0	1.7	2.7	2.8	1.3	1.4	0.9	0.7

Source: OE elaboration from online survey data

10.2 Waterpipe tobacco products

Figure 176. Duration of use of past users (waterpipe; 18-25 age group)
(number of past users in the 18-25 age group)



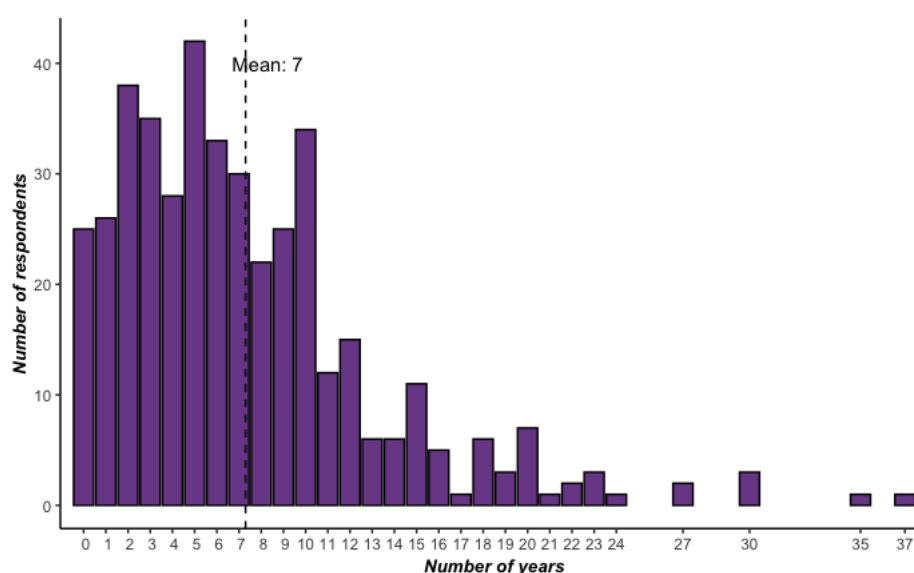
Base: past users of waterpipe tobacco in the 18-25 age group (N=845¹⁴⁷)

Question: Q.1.2.a. At about what age did you start using waterpipe tobacco? & Q.1.2.b. At about what age did you stop using waterpipe tobacco?

Note: The duration was calculated by subtracting Q.1.2.a. from Q.1.2.b for each individual that was a past user of the product. The mean shows the average duration of use.

Source: OE elaboration from online survey data

Figure 177. Duration of use of past users (waterpipe; 26+ age group)
(number of past users in the 26+ age group)



Base: past users of waterpipe tobacco in the 26+ age group (N=424¹⁴⁸)

¹⁴⁷ NA and misreported values are excluded.

¹⁴⁸ NA and misreported values are excluded.

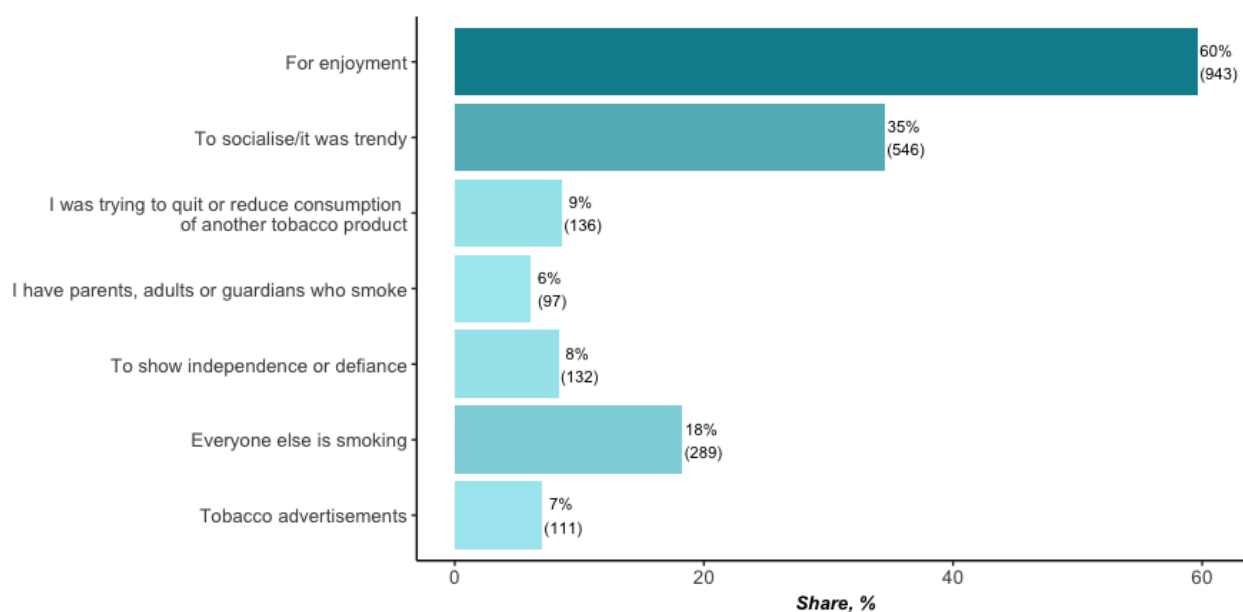
Question: Q.1.2.a. At about what age did you start using waterpipe tobacco? & Q.1.2.b. At about what age did you stop using waterpipe tobacco?

Note: The duration was calculated by subtracting Q.1.2.a. from Q.1.2.b for each individual that was a past user of the product. The mean shows the average duration of use.

Source: OE elaboration from online survey data

Figure 178. Initiation of waterpipe use: reasons (18-25 age group)

(% (number)- of users in the 18-25 age group)



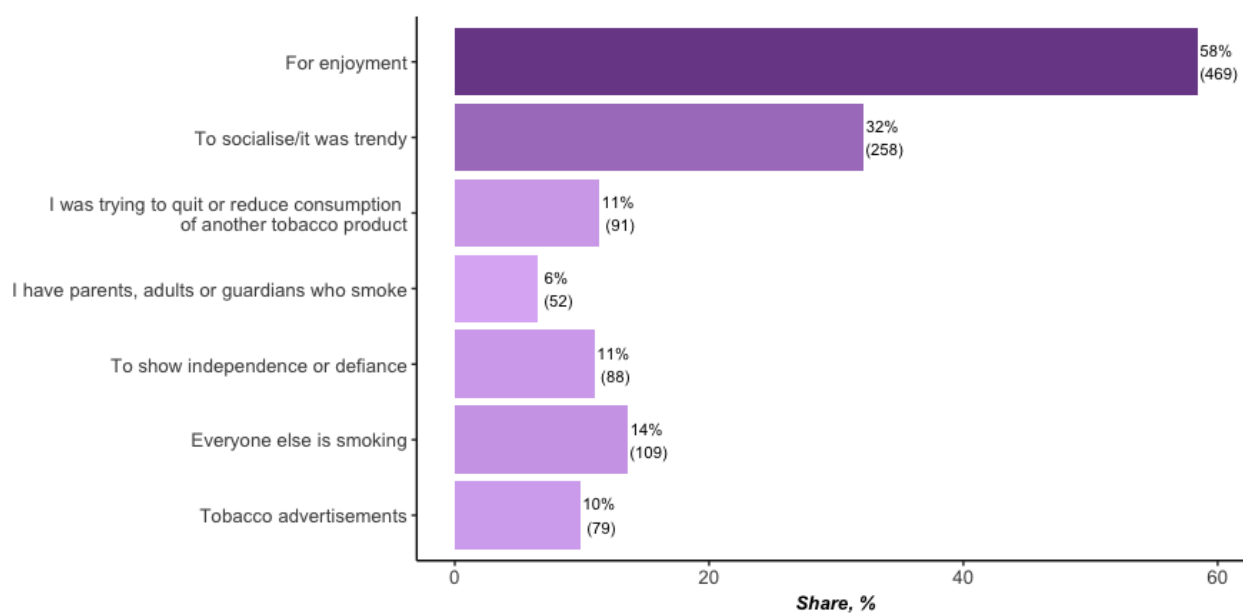
Base: current and past users of waterpipe tobacco in the 18-25 age group (N=1,581)

Question: Q.2.1.a. Why did you start using waterpipe tobacco? Please select up to 2 options.

Note: Each bar represents the share of respondents that selected the option within the age group. The percentages do not add up to 100% because respondents could select more than one term.

Source: OE elaboration from online survey data

Figure 179. Initiation of waterpipe use: reasons (26+ age group)
(% (number)- of current & past users in 26+ age group)



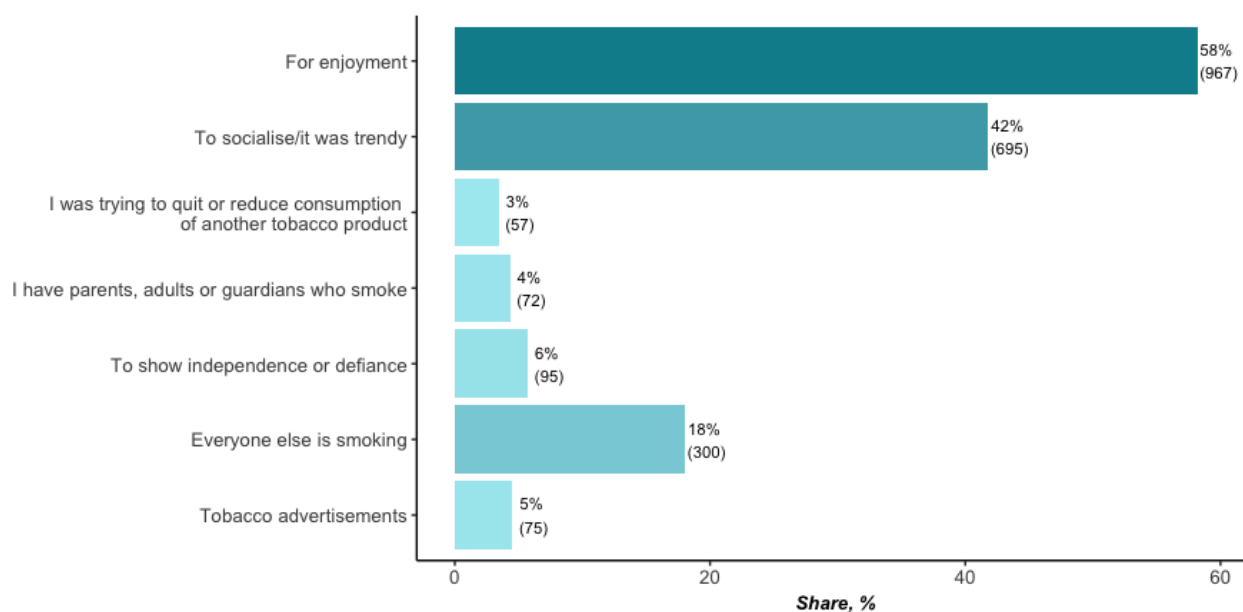
Base: current and past users of waterpipe tobacco in the 26+ age group (N=803)

Question: Q.2.1.a. Why did you start using waterpipe tobacco? Please select up to 2 options.

Note: Each bar represents the share of respondents that selected the option within the age group. The percentages do not add up to 100% because respondents could select more than one term.

Source: OE elaboration from online survey data

Figure 180. Initiation of waterpipe use: reasons for trying (waterpipe; 18-25 age group)
 (% (number)- of those who tried once or twice in the 18-25 age group)



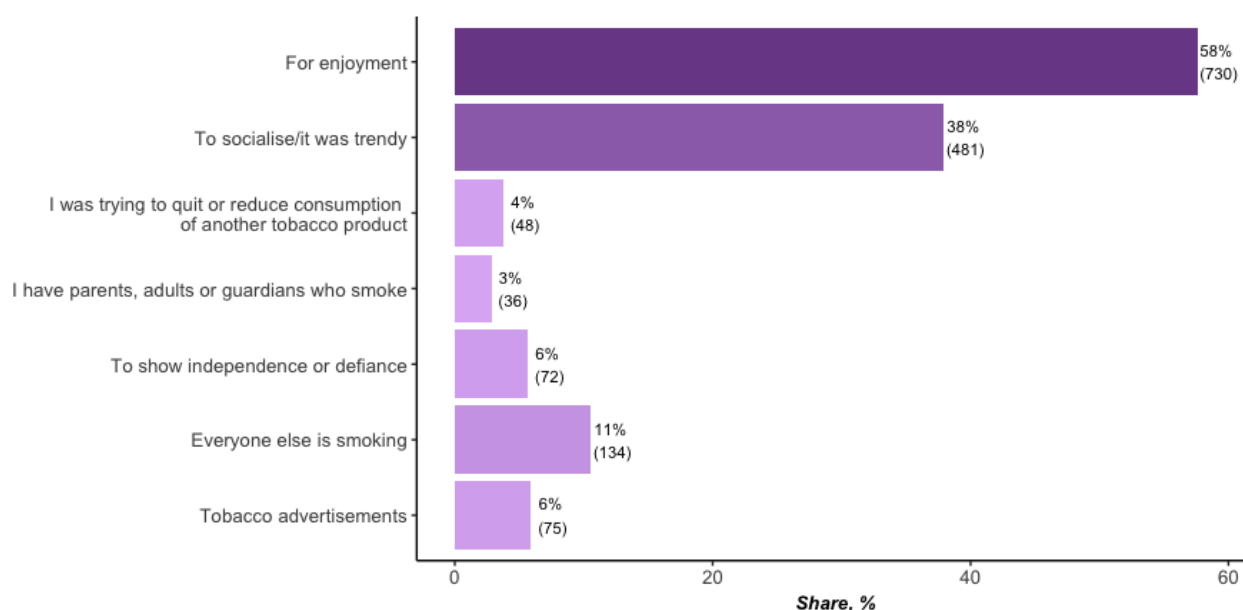
Base: those who tried to use waterpipe tobacco in the 18-25 age group (N=1,662)

Question: Q.2.1.b. Why did you try waterpipe tobacco? Please select up to 2 options.

Note: Each bar represents the share of respondents that selected the option within the age group. The percentages do not add up to 100% because of the fact that respondents could select more than one term – therefore, the bars must be interpreted separately.

Source: OE elaboration from study survey data

Figure 181. Initiation of waterpipe use: reasons for trying (waterpipe; 26+ age group)
 (% (number)- of those who tried once or twice in the 26+ age group)



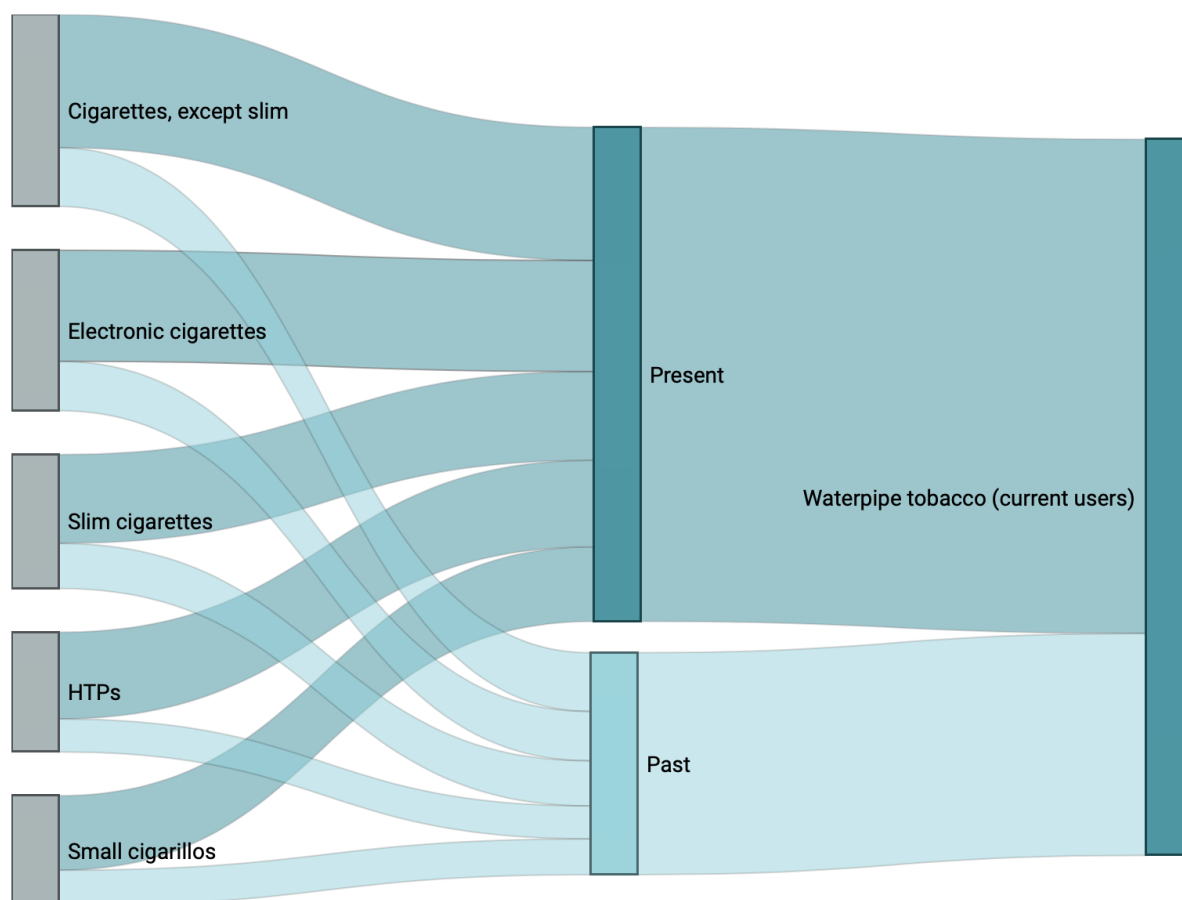
Base: those who tried to use waterpipe tobacco once or twice in the 26+ age group (N=1,268)

Question: Q.2.1.b. Why did you try waterpipe tobacco? Please select up to 2 options.

Note: Each bar represents the share of respondents that selected the option within the age group. The percentages do not add up to 100% because of the fact that respondents could select more than one term – therefore, the bars must be interpreted separately.

Source: OE elaboration from online survey data

Figure 182. Representation of career flows for current user (waterpipe tobacco; 18-25 age group)



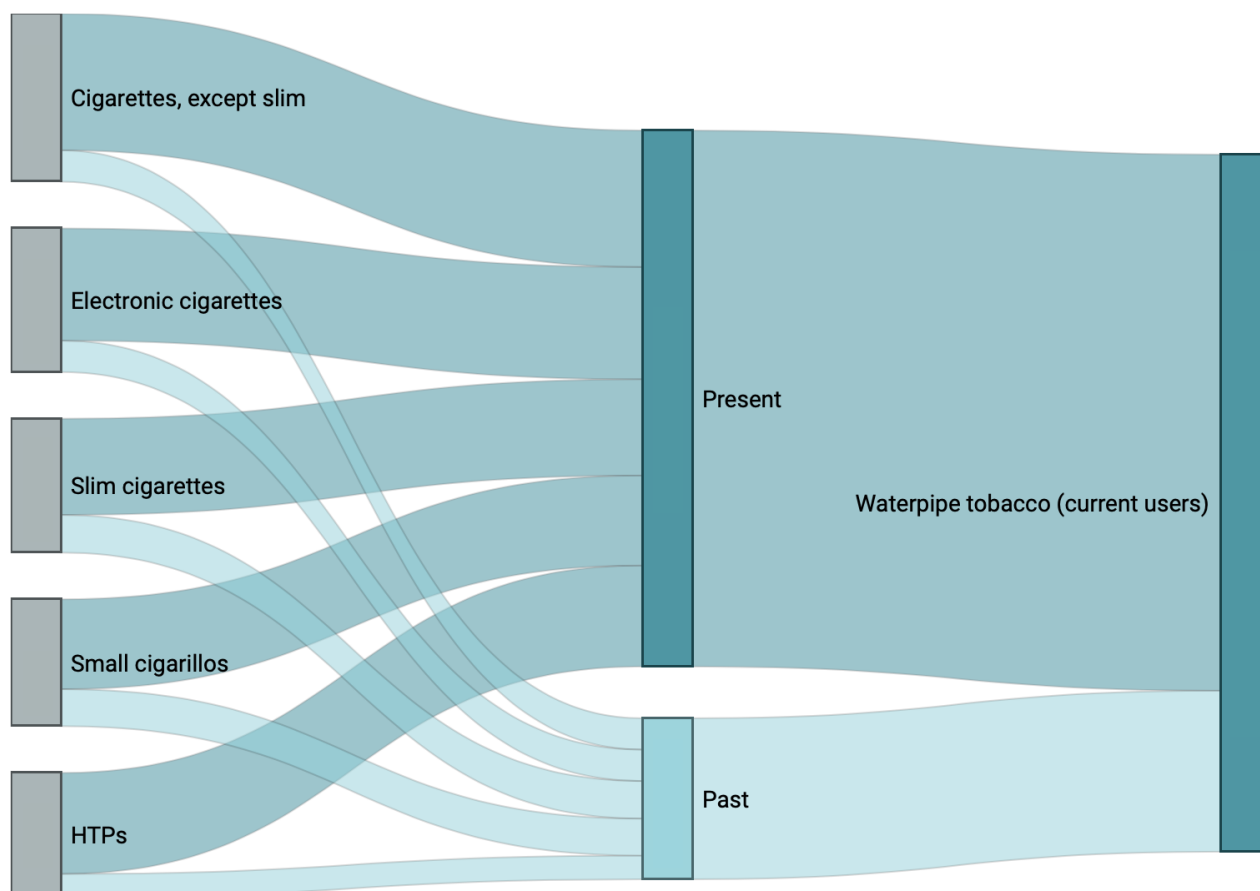
Base: current users of waterpipe tobacco (excluding never users of other products) in the 18-25 age group (N=668)

Question: Q.1.1. Have you ever used waterpipe tobacco?

Note: The diagram is a simplified version of the career flows of dual waterpipe tobacco users, i.e. it highlights dual use patterns as well as transitions to/from other products; therefore, never users of other products were excluded. The thickness/width of the tails, as well as the percentages included, are only indicative. Some individuals may be represented twice (or more) if they use/have used more than one product at a time.

Source: OE elaboration from online survey data

Figure 183. Representation of career flows for current user (waterpipe tobacco; 26+ age group)



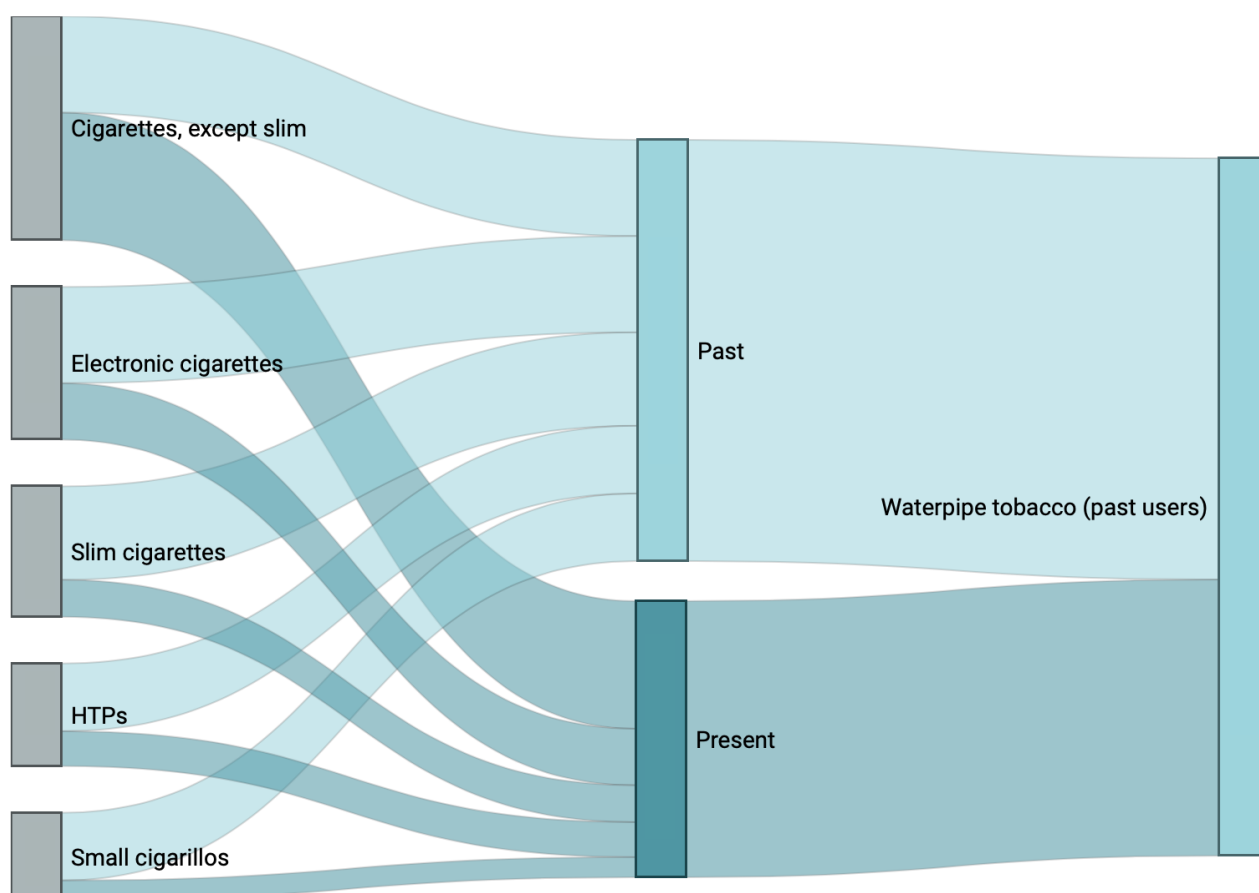
Base: current users of waterpie tobacco (excluding never users of other products) in the 26+ age group (N=339)

Question: Q.1.1. Have you ever used waterpipe tobacco?

Note: The diagram is a simplified version of the career flows of dual waterpipe tobacco users, i.e. it highlights dual use patterns as well as transitions to/from other products; therefore, never users of other products were excluded. The thickness/width of the tails, as well as the percentages included, are only indicative. Some individuals may be represented twice (or more) if they use/have used more than one product at a time.

Source: OE elaboration from online survey data

Figure 184 Representation of career flows for past user (waterpipe tobacco; 18-25 age group)



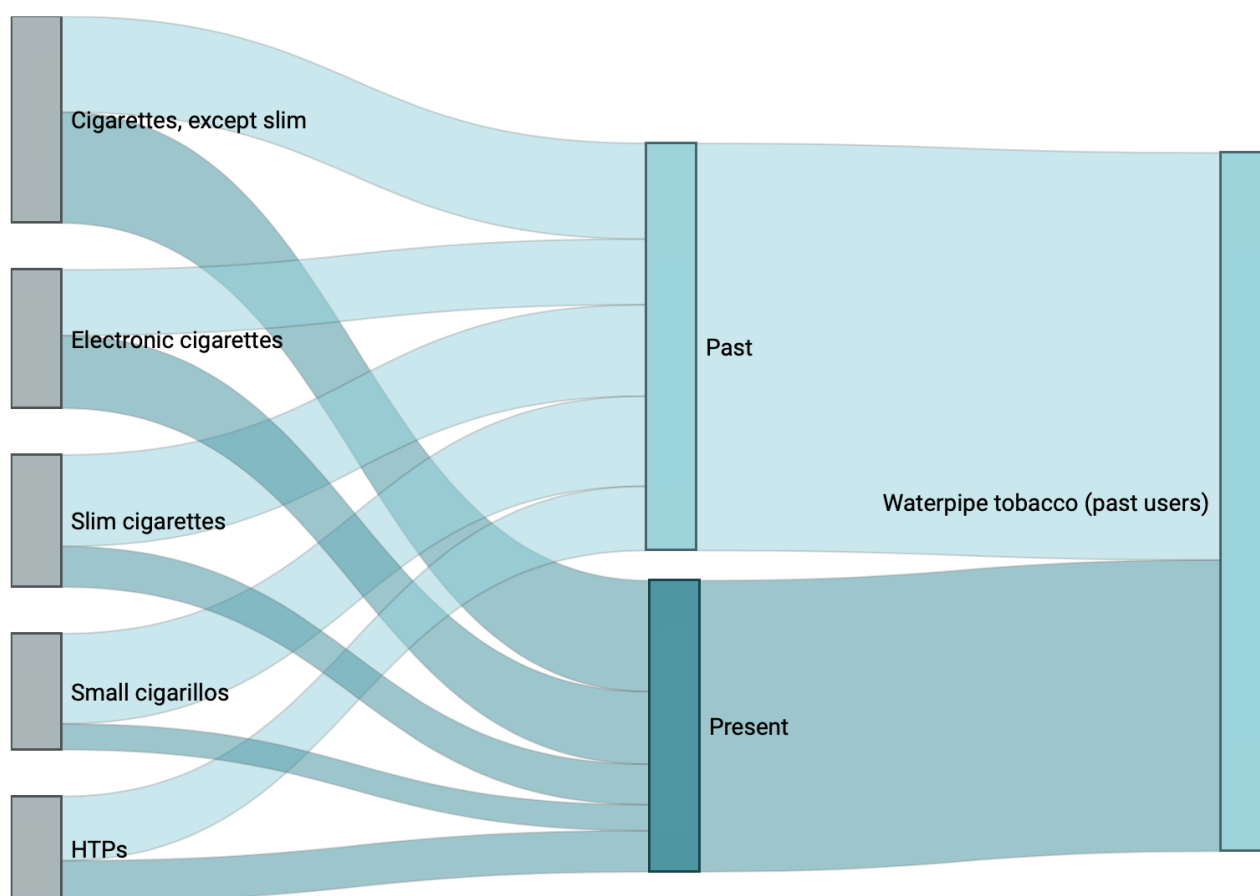
Base: past users of waterpipe tobacco (excluding never users of other products) in the 18-25 age group (N=913)

Question: Q.1.1. Have you ever used waterpipe tobacco?

Note: The diagram is a simplified version of the career flows of dual waterpipe tobacco users, i.e. it highlights dual use patterns as well as transitions to/from other products; therefore, never users of other products were excluded. The thickness/width of the tails, as well as the percentages included, are only indicative. Some individuals may be represented twice (or more) if they use/have used more than one product at a time.

Source: OE elaboration from online survey data

Figure 185 Representation of career flows for past user (waterpipe tobacco; 26+ age group)



Base: past users of waterpipe tobacco (excluding never users of other products) in the 26+ age group (N=464)

Question: Q.1.1. Have you ever used waterpipe tobacco?

Note: The diagram is a simplified version of the career flows of dual waterpipe tobacco users, i.e. it highlights dual use patterns as well as transitions to/from other products; therefore, never users of other products were excluded. The thickness/width of the tails, as well as the percentages included, are only indicative. Some individuals may be represented twice (or more) if they use/have used more than one product at a time.

Source: OE elaboration from online survey data

Table 73. Perceived appeal of product to youth (waterpipe; 18-25 age group)
(% - of respondents in the 18-25 age group)

	Group total	Current users	Past users	Used once or twice	Never used
Use of product	100	11	15	27	47
Definitely yes	42	14	16	31	39
Probably yes	30	9	15	29	48
Probably not	14	9	14	24	53
Definitely not	6	11	19	18	52
I don't know	7	6	7	16	72

Base: respondents in the 18-25 age group (N=6,090)

Question: Q.4.16. Do you think that young people might be particularly attracted to waterpipe (e.g. shisha & narghile) tobacco, or not?

Note: The orange column represents the share of respondents in the age group that selected the response. The white cells in the four columns on the right represent the breakdown by type of users for that response in particular (i.e. the sum of the white cells in each row should add up to 100). The group total category refers to the total respondents in the 18-25 age group.

Source: OE elaboration from online survey data

Table 74. Perceived appeal of product to youth (waterpipe; 26+ age group)
(% - of respondents in the 26+ age group)

	Group total	Current users	Past users	Used once or twice	Never used
Use of product	100	6	8	21	65
Definitely yes	29	8	9	24	59
Probably yes	35	6	9	25	60
Probably not	18	4	6	20	69
Definitely not	7	5	6	12	77
I don't know	11	2	4	12	83

Base: respondents in the 26+ age group (N=5,910)

Question: Q.4.16. Do you think that young people might be particularly attracted to waterpipe (e.g. shisha & narghile) tobacco, or not?

Note: The orange column represents the share of respondents in the age group that selected the response. The white cells in the four columns on the right represent the breakdown by type of users for that response in particular (i.e. the sum of the white cells in each row should add up to 100). The group total column refers to the total respondents in the 26+ age group.

Source: OE elaboration from online survey data

Table 75. Extent to which respondents believed that youth underestimate risks of using waterpipe tobacco (18-25 age group)

(%- of respondents in the 18-25 age group)

	Group total	Current users	Past users	Used once or twice	Never used
Use of product	100	11	15	27	47
Definitely yes	37	12	14	27	47
Probably yes	33	10	16	28	46
Probably not	15	11	19	30	40
Definitely not	7	12	17	30	41
I don't know	7	6	8	18	68

*Base: respondents in the 18-25 age group (N=6,090)**Question: Q.4.17. Do you think that young people might underestimate the risk of using to waterpipe (e.g. shisha & narghile) tobacco, or not?**Note: The orange column represents the share of respondents in the age group that selected the response. The white cells in the four columns on the right represent the breakdown by type of users for that response in particular (i.e. the sum of the white cells in each row should add up to 100). The group total category refers to the total respondents in the 18-25 age group.**Source: OE elaboration from online survey data*

Table 76. Extent to which respondents believed that youth underestimate risks of using waterpipe tobacco (26+ age group)

(%- of respondents in the 26+ age group)

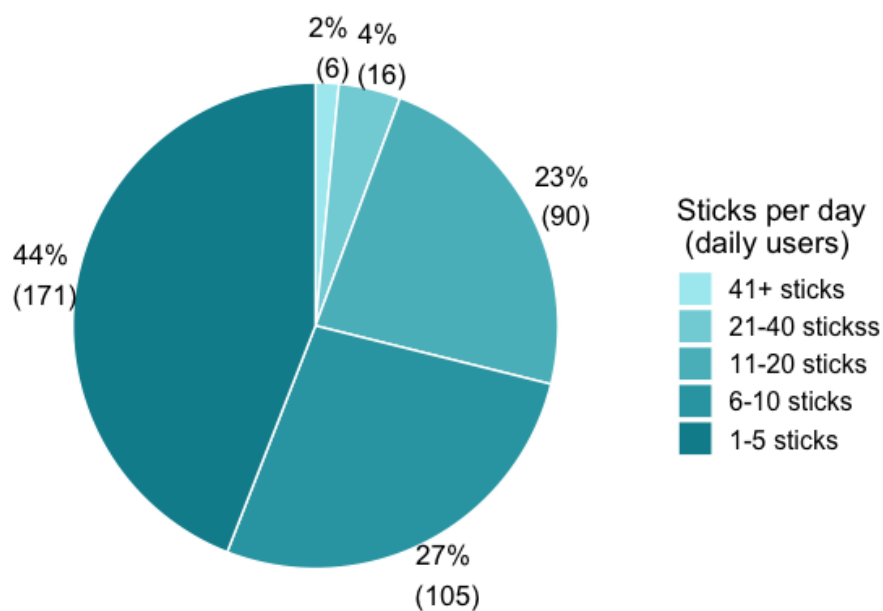
	Group total	Current users	Past users	Used once or twice	Never used
Use of product	100	6	8	21	65
Definitely yes	33	6	7	18	68
Probably yes	38	6	8	22	64
Probably not	14	6	11	29	54
Definitely not	6	5	10	25	59
I don't know	9	3	4	14	79

*Base: respondents in the 26+ age group (N=5,910)**Question: Q.4.17. Do you think that young people might underestimate the risk of using to waterpipe (e.g. shisha & narghile) tobacco, or not?**Note: The orange column represents the share of respondents in the age group that selected the response. The white cells in the four columns on the right represent the breakdown by type of users for that response in particular (i.e. the sum of the white cells in each row should add up to 100). The group total category refers to the total respondents in the 26+ age group.**Source: OE elaboration from online survey data*

10.3 Slim cigarettes

Figure 186. Average number of slim cigarette sticks per day of use, daily users (18-25 age group)

(% (number)-of current daily users in the 18-25 age group)

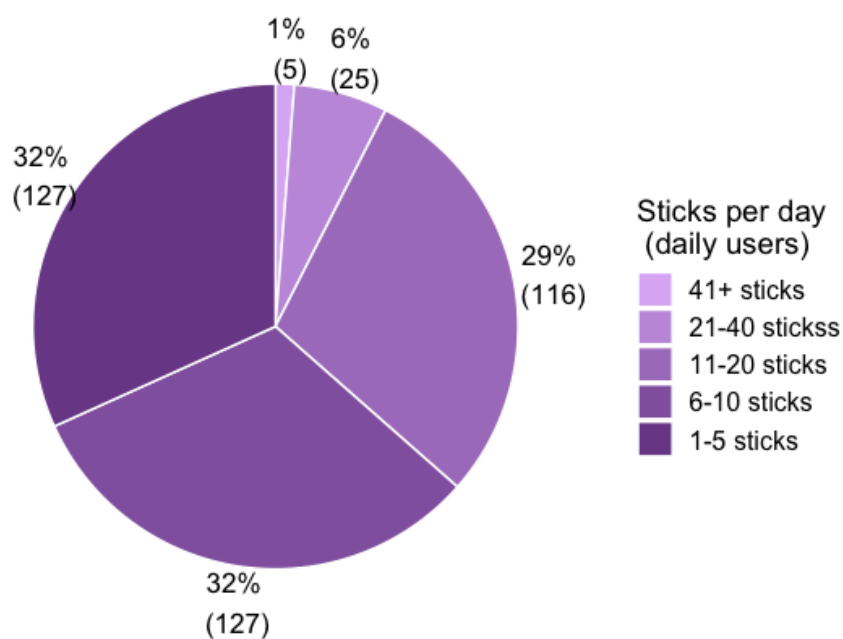


Base: daily current users of slim cigarettes in the 18-25 age group (N=388)

Question: Q.1.4.a. On the days that you use slim cigarettes, about how much do you use?

Source: OE elaboration from online survey data

Figure 187. Average number of slim cigarettes sticks per day, for daily users (26+ age group)
 (% (number)-of current daily users in the 26+ age group)

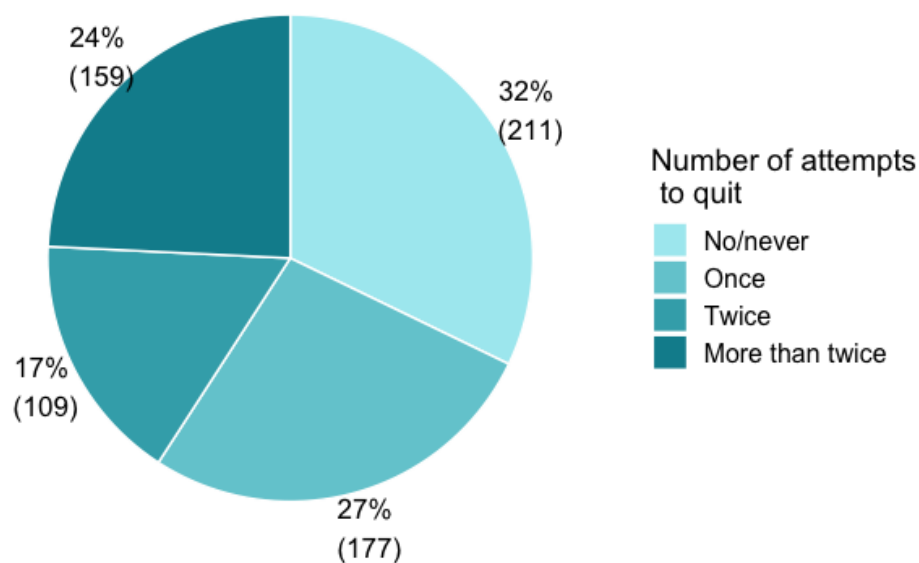


Base: daily current users of slim cigarettes in the 26+ age group (N=400)

Question: Q.1.4.a. On the days that you use slim cigarettes, about how much do you use?

Source: OE elaboration from online survey data

Figure 188. Cessation: slim cigarettes quit attempts (slim cigarettes; 18-25 age group)
 (% (number)- of current users in the 18-25 age group)

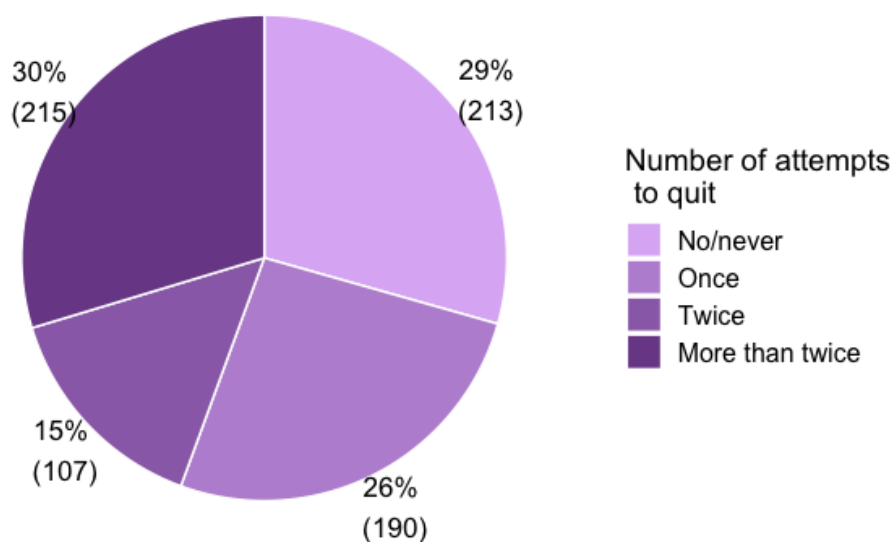


Base: current users of slim cigarettes in the 18-25 age group (N=656)

Question: Q.1.5. Have you ever tried to quit slim cigarettes? If yes, how many times?

Source: OE elaboration from online survey data

Figure 189. Cessation: slim cigarettes quit attempts (slim cigarettes; 26+ age group)
 (% (number)- of current users in the 26+ age group)

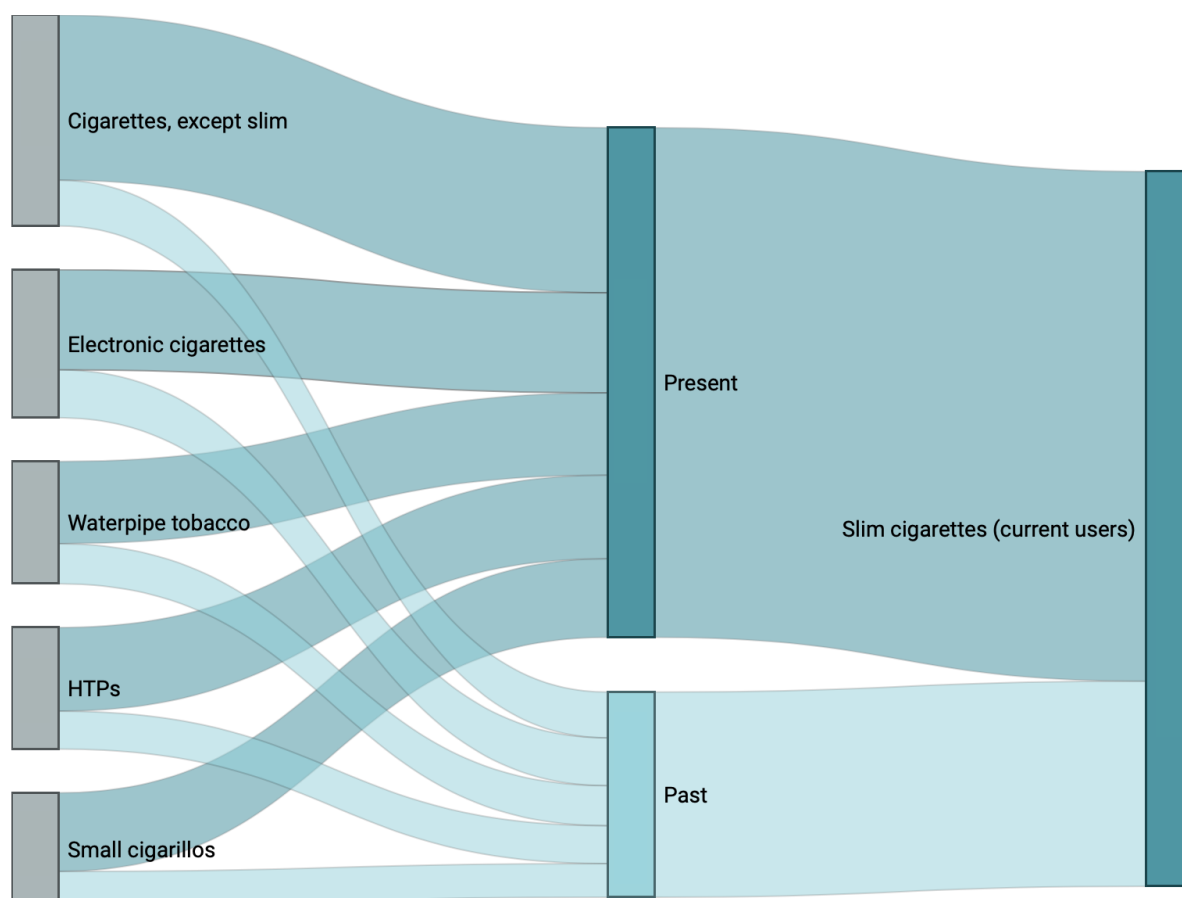


Base: current users of slim cigarettes in the 26+ age group (N=725)

Question: Q.1.5. Have you ever tried to quit slim cigarettes? If yes, how many times?

Source: OE elaboration from online survey data

Figure 190. Representation of career flows for current users (slim cigarettes; 18-25 age group)



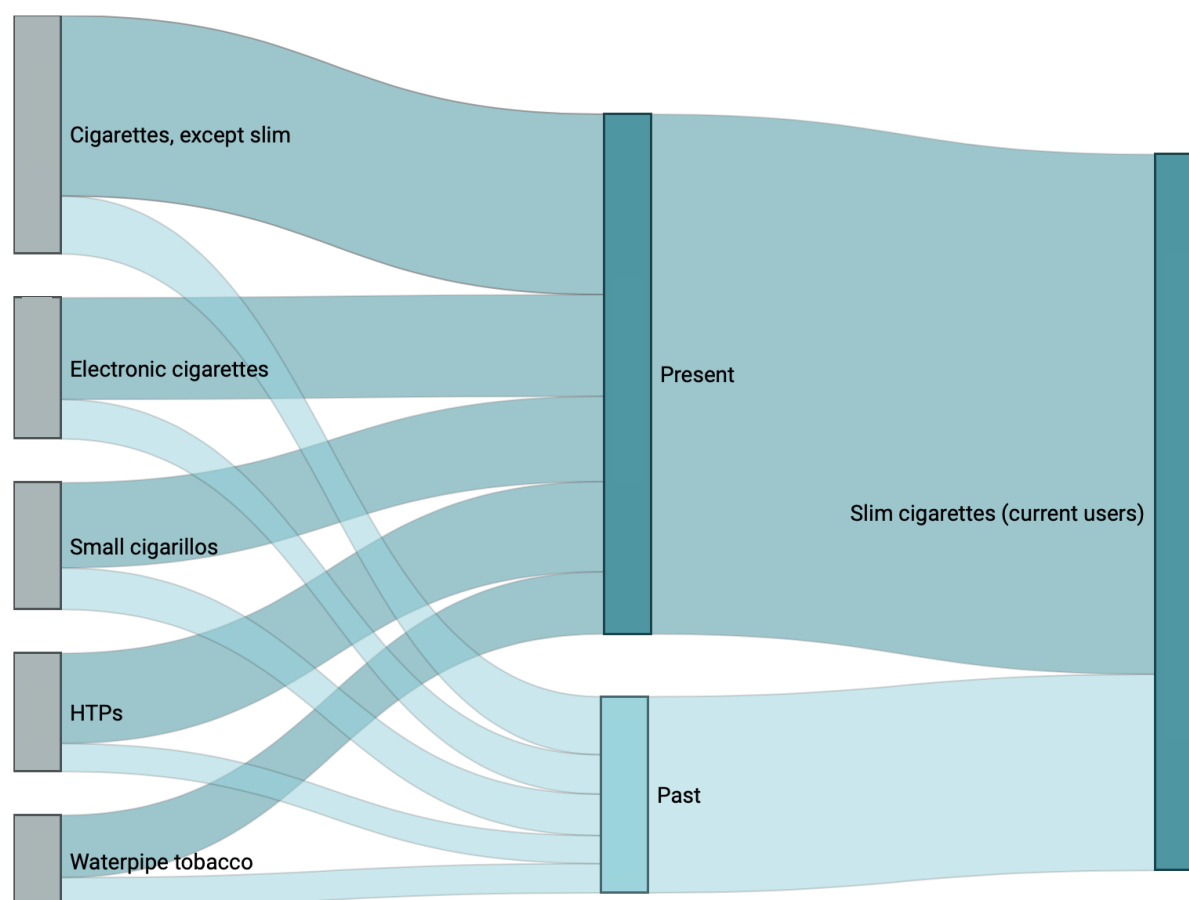
Base: current users of slim cigarettes (excluding never users of other products) in the 18-25 age group (N=622)

Question: Q.1.1. Have you ever used slim cigarettes?

Note: The diagram is a simplified version of the career flows of dual slim cigarette users, i.e. it highlights dual use patterns as well as transitions to/from other products; therefore, never users of other products were excluded. The thickness/width of the tails, as well as the percentages included, are only indicative. Some individuals may be represented twice (or more) if they use/have used more than one product at a time.

Source: OE elaboration from online survey data

Figure 191. Representation of career flows for current users (slim cigarettes; 26+ age group)



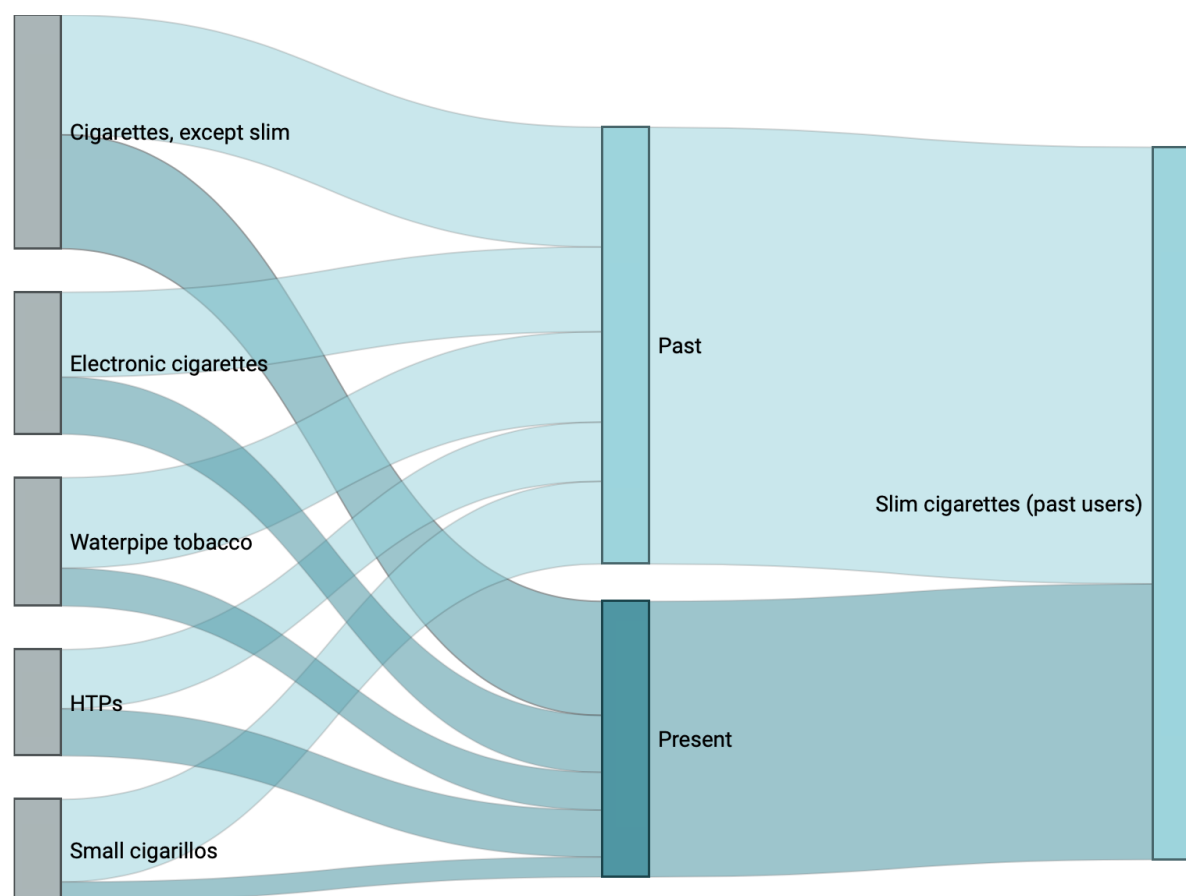
Base: current users of slim cigarettes (excluding never users of other products) in the 26+ age group (N=691)

Question: Q.1.1. Have you ever used slim cigarettes?

Note: The diagram is a simplified version of the career flows of dual slim cigarette users, i.e. it highlights dual use patterns as well as transitions to/from other products; therefore, never users of other products were excluded. The thickness/width of the tails, as well as the percentages included, are only indicative. Some individuals may be represented twice (or more) if they use/have used more than one product at a time.

Source: OE elaboration from online survey data

Figure 192. Representation of career flows for past users (slim cigarettes; 18-25 age group)



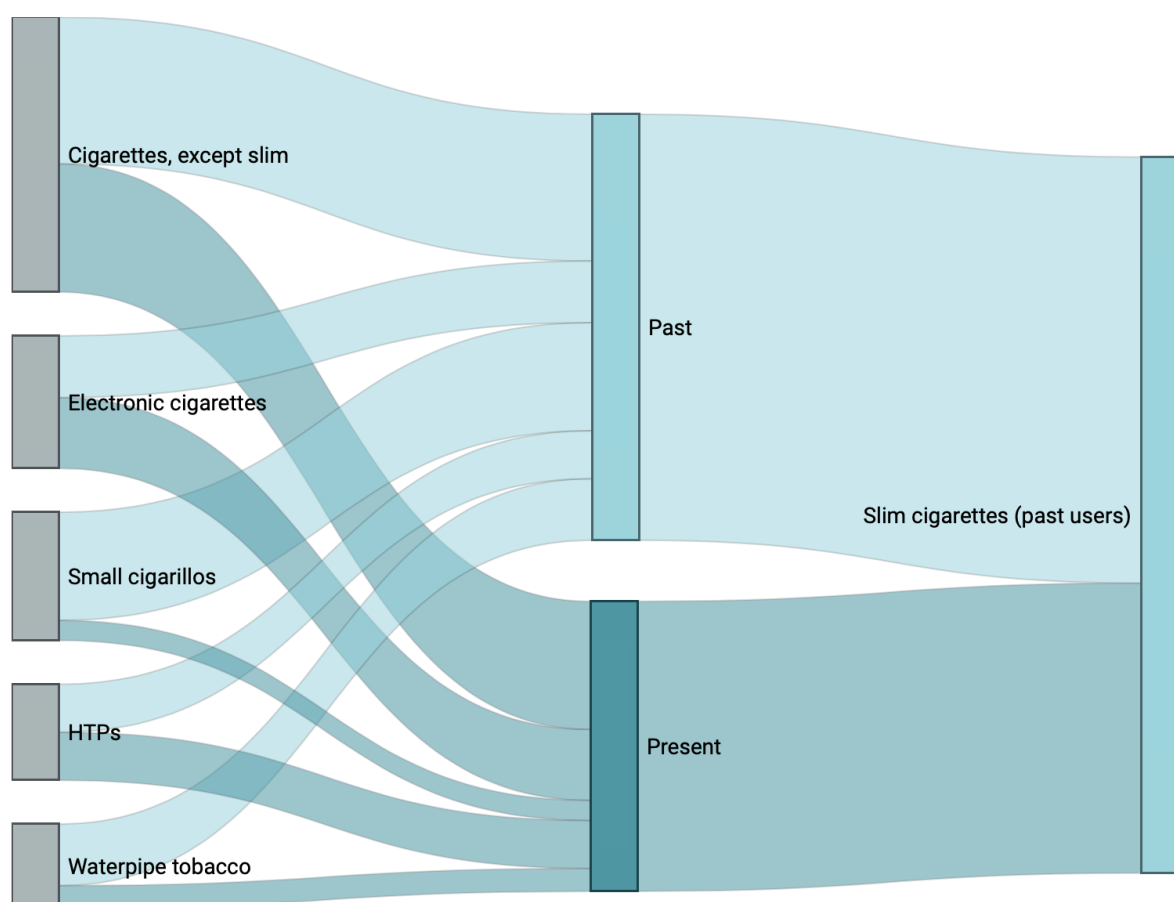
Base: past users of slim cigarettes (excluding never users of other products) in the 18-25 age group (N=771)

Question: Q.1.1. Have you ever used slim cigarettes?

Note: The diagram is a simplified version of the career flows of dual slim cigarette users, i.e. it highlights dual use patterns as well as transitions to/from other products; therefore, never users of other products were excluded. The thickness/width of the tails, as well as the percentages included, are only indicative. Some individuals may be represented twice (or more) if they use/have used more than one product at a time.

Source: OE elaboration from online survey data

Figure 193. Representation of career flows for past users (slim cigarettes; 26+ age group)



Base: past users of slim cigarettes (excluding never users of other products) in the 26+ age group (N=840)

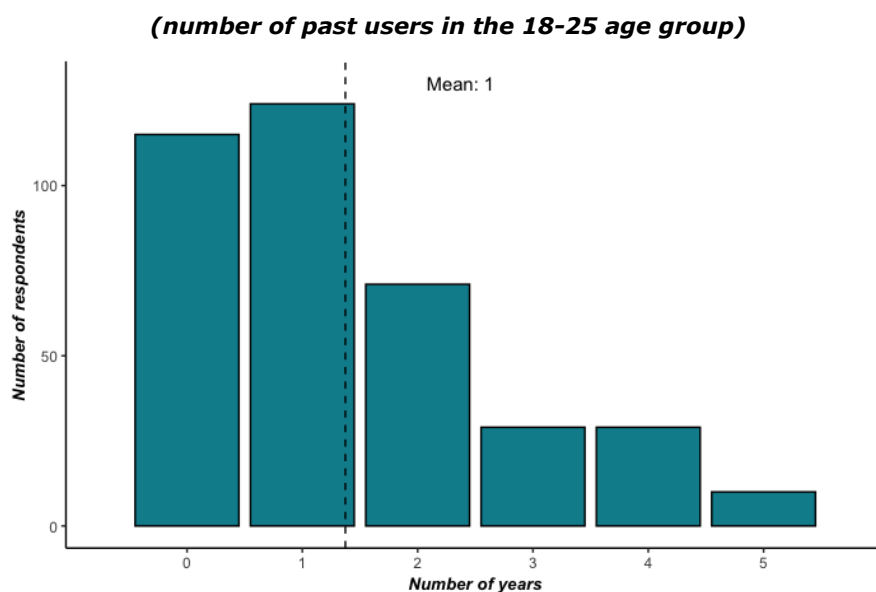
Question: Q.1.1. Have you ever used slim cigarettes?

Note: The diagram is a simplified version of the career flows of dual slim cigarette users, i.e. it highlights dual use patterns as well as transitions to/from other products; therefore, never users of other products were excluded. The thickness/width of the tails, as well as the percentages included, are only indicative. Some individuals may be represented twice (or more) if they use/have used more than one product at a time.

Source: OE elaboration from online survey data

10.4 Novel tobacco products

Figure 194. Duration of use of past users (HTPs; 18-25 age group)



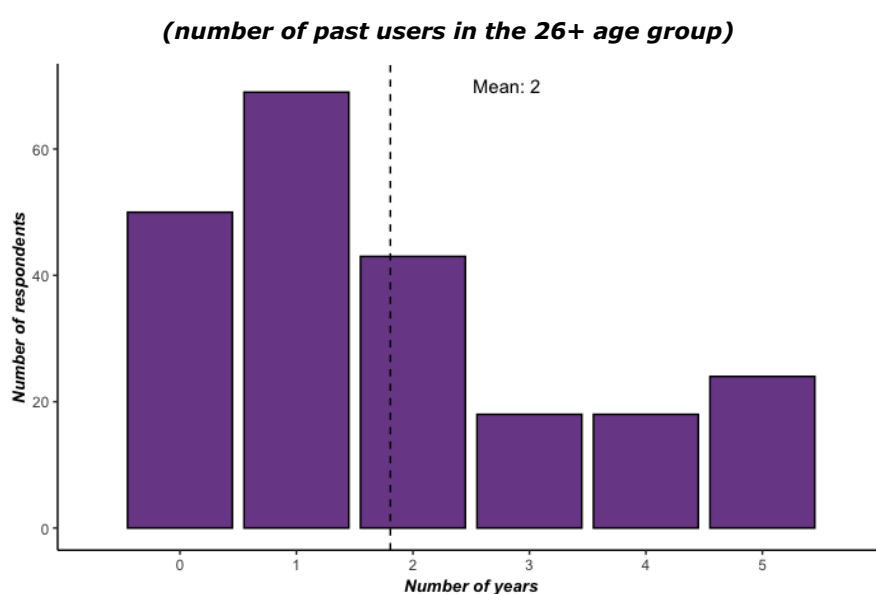
Base: past HTP users in the 18-25 age group (N=378¹⁴⁹)

Question: Q.1.2.a. At about what age did you start using HTPs? & Q.1.2.b. At about what age did you stop using HTPs?

Note: The duration was calculated by subtracting Q.1.2.a. from Q.1.2.b for each individual that was a past user of the product. The mean shows the average duration of use. We cut off the data at 5 years, since the product was introduced in 2014.

Source: OE elaboration from online survey data

Figure 195. Duration of use of past users (HTPs; 26+ age group)



Base: past HTP users in the 26+ age group (N=222¹⁵⁰)

¹⁴⁹ NA and misreported values are excluded.

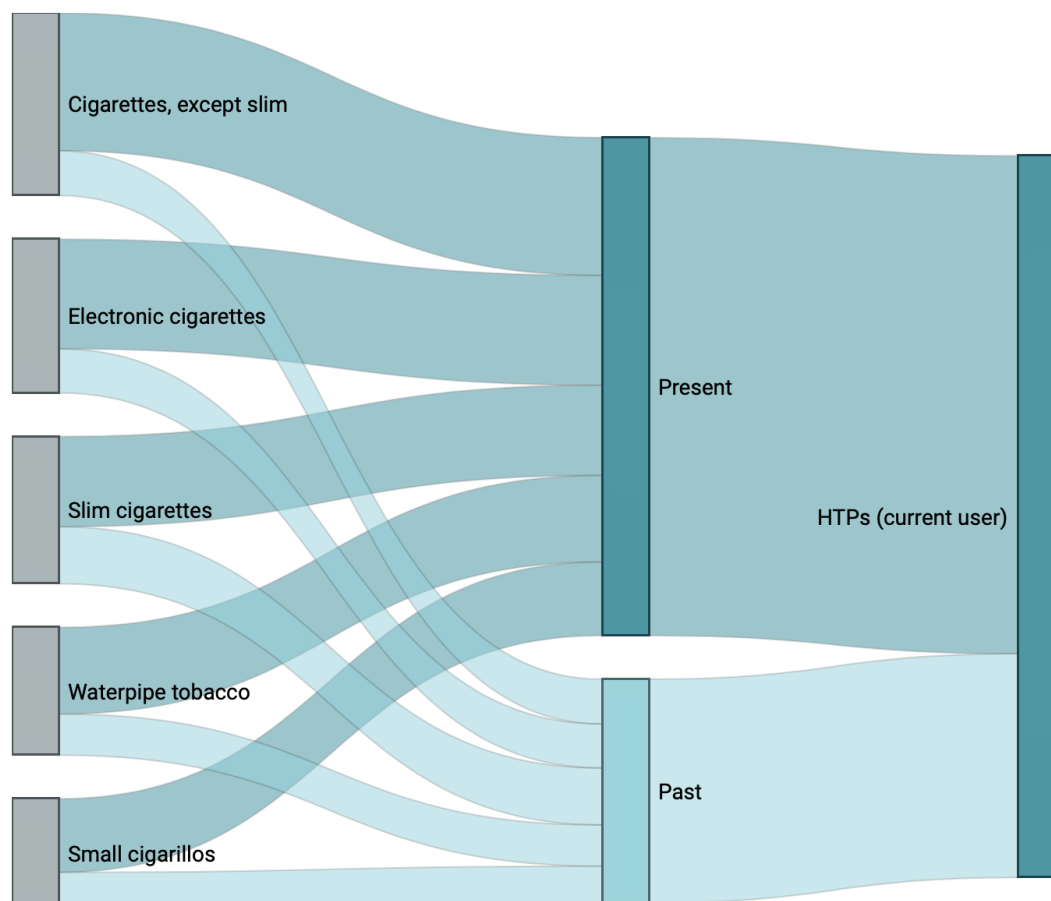
¹⁵⁰ NA and misreported values are excluded.

Question: Q.1.2.a. At about what age did you start using HTPs? & Q.1.2.b. At about what age did you stop using HTPs?

Note: The duration was calculated by subtracting Q.1.2.a. from Q.1.2.b for each individual that was a past user of the product. The mean shows the average duration of use. We cut off the data at 5 years, since the product was introduced in 2014 in Italy.

Source: OE elaboration from online survey data

Figure 196. Representation of career flows for current users (HTPs; 18-25 age group)



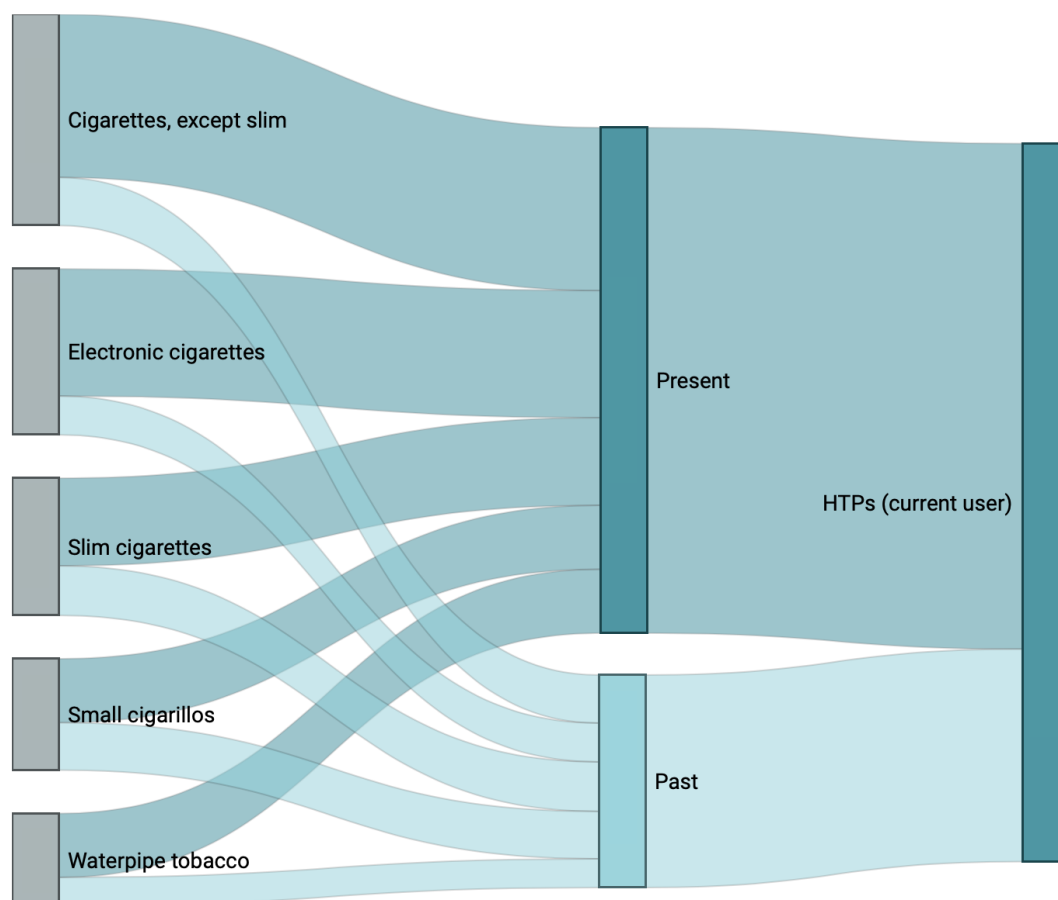
Base: current users of HTPs (excluding never users of other products) in the 18-25 age group (N=498)

Question: Q.1.1. Have you ever used HTPs?

Note: The diagram is a simplified version of the career flows of dual HTP users, i.e. it highlights dual use patterns as well as transitions to/from other products; therefore, never users of other products were excluded. The thickness/width of the tails, as well as the percentages included, are only indicative. Some individuals may be represented twice (or more) if they use/have used more than one product at a time.

Source: OE elaboration from online survey data

Figure 197. Representation of career flows for current users (HTPs; 26+ age group)



Base: current users of HTPs (excluding never users of other products) in the 26+ age group (N=627)

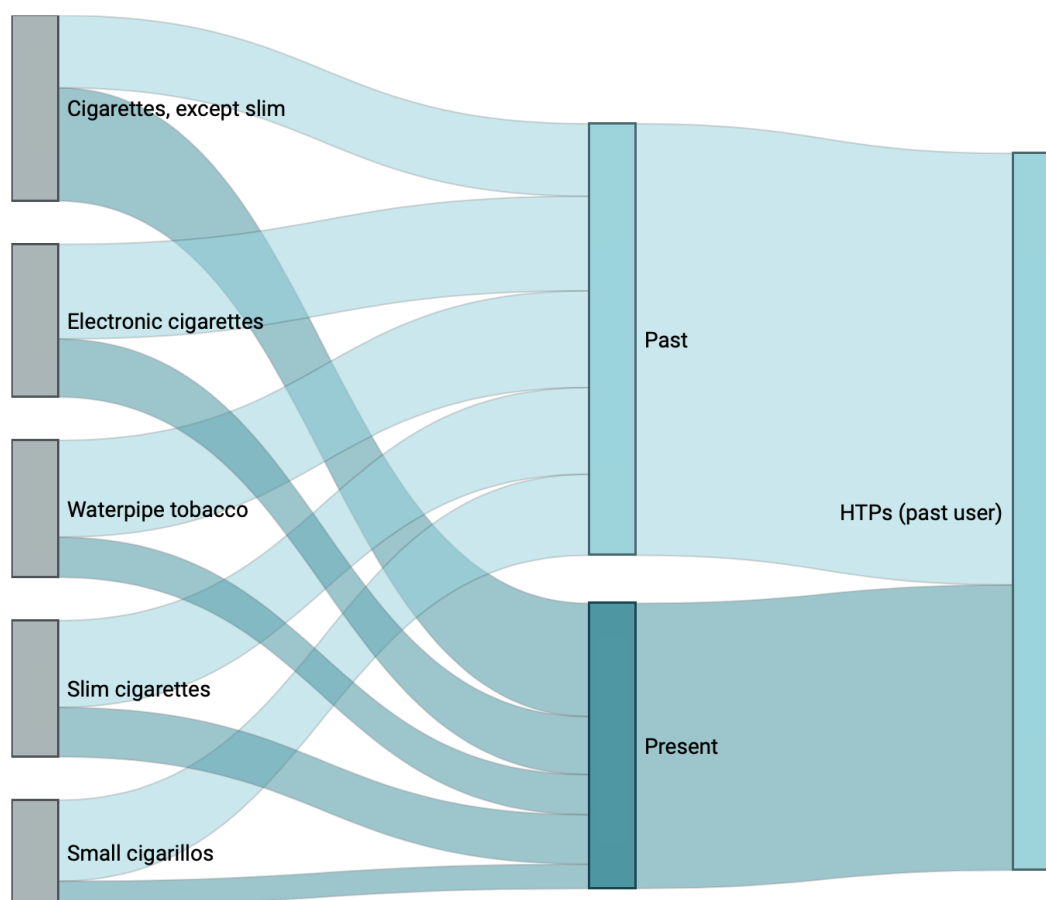
Question: Q.1.1. Have you ever used HTPs?

Note: The diagram is a simplified version of the career flows of dual HTP users, i.e. it highlights dual use patterns as well as transitions to/from other products; therefore, never users of other products were excluded. The thickness/width of the tails, as well as the percentages included, are only indicative. Some individuals may be represented twice (or more) if they use/have used more than one product at a time.

Source: OE elaboration from online survey data¹⁵¹

¹⁵¹ <http://sankey-diagram-generator.acquireprocure.com>

Figure 198. Representation of career flows for past users (HTPs; 18-25 age group)



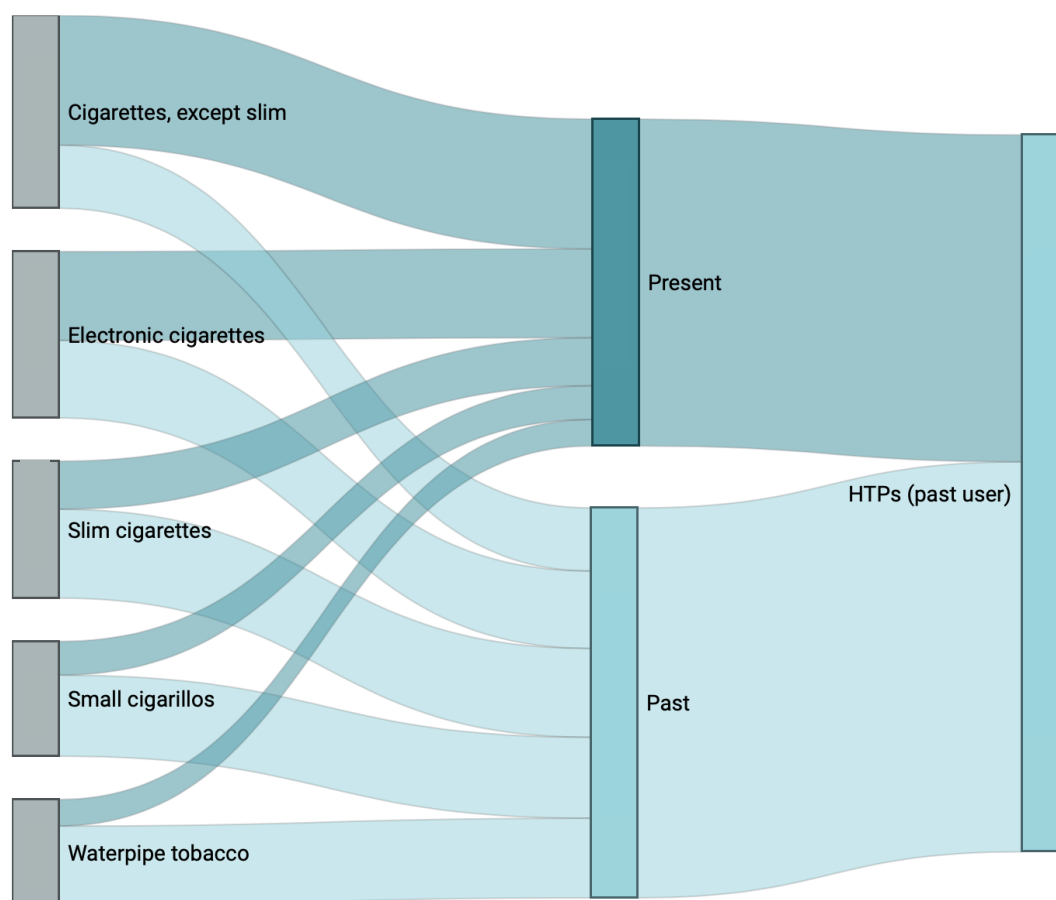
Base: past users of HTPs (excluding never users of other products) in the 18-25 age group (N=443)

Question: Q.1.1. Have you ever used HTPs?

Note: The diagram is a simplified version of the career flows of dual HTP users, i.e. it highlights dual use patterns as well as transitions to/from other products; therefore, never users of other products were excluded. The thickness/width of the tails, as well as the percentages included, are only indicative. Some individuals may be represented twice (or more) if they use/have used more than one product at a time.

Source: OE elaboration from online survey data

Figure 199. Representation of career flows for past users (HTPs; 26+ age group)



Base: current users of HTPs (excluding never users of other products) in the 26+ age group (N=334)

Question: Q.1.1. Have you ever used HTPs?

Note: The diagram is a simplified version of the career flows of dual HTP users, i.e. it highlights dual use patterns as well as transitions to/from other products; therefore, never users of other products were excluded. The thickness/width of the tails, as well as the percentages included, are only indicative. Some individuals may be represented twice (or more) if they use/have used more than one product at a time.

Source: OE elaboration from online survey data¹⁵²

¹⁵² The diagram was developed using the Sankey Diagram Generator online tool.
<http://sankey-diagram-generator.acquireprocure.com>

Table 77. Perceived gateway effect of product (HTPs; 18-25 age group)
 (% (number)- of 18-25 age group)

Answer	Group total %(n)	Current users %	Past users %	Used once or twice %	Never used %
Use of product	100 (6090)	8	7	16	69
Definitely yes	22 (1313)	15	9	16	60
Probably yes	44 (2703)	7	8	17	68
Probably not	16 (966)	8	7	18	67
Definitely not	7 (425)	6	5	12	77
I don't know	11 (683)	3	2	9	86

Base: respondents in the 18-25 age group (N=6,090)

Question: Q4.3. Do you think using heated tobacco products (e.g. iQOS, glo) might encourage people to start smoking/using other tobacco products, or not? [SINGLE ANSWER]

Note: The orange column represents the share of respondents in the age group that selected the response. The white cells in the four columns on the right represent the breakdown by type of users for that response in particular (i.e. the sum of the white cells in each row should add up to 100). The group total category refers to the total respondents in the 18-25 age group.

Source: OE elaboration from online survey data

Table 78. Perceived gateway effect of product (HTPs; 26+ age group)
 (% (number) - of respondents in the 26+ age group)

Answer	Group total %(n)	Current users %	Past users %	Used once or twice %	Never used %
Use of product	100 (5,910)	11	6	10	73
Definitely yes	19 (1114)	19	6	11	64
Probably yes	41 (2429)	12	8	12	69
Probably not	15 (910)	10	4	11	74
Definitely not	8 (466)	4	4	10	82
I don't know	17 (991)	3	2	4	90

Base: respondents in the 26+ age group (N=5,910)

Question: Question: Q4.3. Do you think using heated tobacco products (e.g. iQOS, glo) might encourage people to start smoking/using other tobacco products, or not? [SINGLE ANSWER]

Note: The orange column represents the share of respondents in the age group that selected the response. The white cells in the four columns on the right represent the breakdown by type of users for that response in particular (i.e. the sum of the white cells in each row should add up to 100). The group total category refers to the total respondents in the 26+ age group.

Source: OE elaboration from online survey data

Table 79. Perceived appeal of product to youth (HTPs; 18-25 age group)

(% (number) - of respondents in the 18-25 age group)

Answer	Group total %(n)	Current users %	Past users %	Used once or twice %	Never used %
Use of product	100 (6090)	8	7	16	69
Definitely yes	28 (1712)	11	9	19	61
Probably yes	42 (2535)	8	7	16	70
Probably not	16 (969)	6	8	14	72
Definitely not	6 (356)	12	10	13	65
I don't know	9 (518)	3	3	9	84

Base: respondents in the 18-25 age group (N=6,090)

Question: Q.4.5. Do you think that young people might be particularly attracted to heated tobacco products (e.g. iQOS, glo), or not? [SINGLE ANSWER]

Note: The orange column represents the share of respondents in the age group that selected the response. The white cells in the four columns on the right represent the breakdown by type of users

for that response in particular (i.e. the sum of the white cells in each row should add up to 100). The group total category refers to the total respondents in the 18-25 age group.
Source: OE elaboration from online survey data

Table 80. Perceived appeal of product to youth (HTPs; 26+ age group)
(% (number) - of respondents in the 26+ age group)

Answer	Group total %(n)	Current users %	Past users %	Used once or twice %	Never used %
Use of product	100 (5910)	11	6	10	73
Definitely yes	21 (1252)	18	7	13	62
Probably yes	45 (2679)	12	6	11	71
Probably not	15 (859)	6	6	10	78
Definitely not	5 (288)	7	5	10	79
I don't know	14 (832)	3	2	4	92

Base: respondents in the 26+ age group (N=5,910)

Question: Q.4.5. Do you think that young people might be particularly attracted to heated tobacco products (e.g. iQOS, glo), or not? [SINGLE ANSWER]

Note: The orange column represents the share of respondents in the age group that selected the response. The white cells in the four columns on the right represent the breakdown by type of users for that response in particular (i.e. the sum of the white cells in each row should add up to 100). The group total column refers to the total respondents in the 26+ age group.

Source: OE elaboration from online survey data

Table 81. Extent to which respondents believed that youth underestimated risks of using HTPs
(% (number)- of 18-25 age group)

Answer	Group total %(n)	Current users %	Past users %	Used once or twice %	Never used %
Use of product	100 (6090)	8	7	16	69
Definitely yes	42 (2556)	7	6	15	72
Probably yes	35 (2134)	9	8	16	67
Probably not	13 (765)	10	11	18	62
Definitely not	5 (309)	12	7	19	61
I don't know	5 (326)	8	4	10	78

Base: respondents in the 18-25 age group (N=6,090)

Question: Q.4.6. Do you think that young people might underestimate the risk of consuming heated tobacco products (e.g. iQOS, glo), or not? [SINGLE ANSWER]

Note: The orange column represents the share of respondents in the age group that selected the response. The white cells in the four columns on the right represent the breakdown by type of users for that response in particular (i.e. the sum of the white cells in each row should add up to 100). The group total category refers to the total respondents in the 18-25 age group.

Source: OE elaboration from online survey data

Table 82. Extent to which respondents believed that youth underestimate risks of HTPs
(% (number) - of respondents in the 26+ age group)

Answer	Group total %(n)	Current users %	Past users %	Used once or twice %	Never used %
Use of product	100 (5910)	11	6	10	73
Definitely yes	36 (2149)	9	5	9	76
Probably yes	40 (2378)	13	7	11	69
Probably not	11 (663)	13	8	11	69
Definitely not	5 (271)	6	4	10	81
I don't know	8 (449)	5	3	7	85

Base: respondents in the 26+ age group (N=5,910)

Question: Q.4.6. Do you think that young people might underestimate the risk of consuming heated tobacco products (e.g. iQOS, glo), or not? [SINGLE ANSWER]

Note: The orange column represents the share of respondents in the age group that selected the response. The white cells in the four columns on the right represent the breakdown by type of users for that response in particular (i.e. the sum of the white cells in each row should add up to 100). The group total category refers to the total respondents in the 26+ age group.

Source: OE elaboration from online survey data

Table 83. Extent of agreement with regulating the product like cigarettes (HTPs; 18-25 age group)

(% - of respondents in the 18-25 age group)

Answer	Group total %	Current users	Past users	Used once or twice	Never used
Use of product	100 (6090)	8	7	16	69
Definitely yes	44 (2662)	6	5	15	74
Probably yes	30 (1825)	10	8	16	65
Probably not	12 (716)	12	11	21	56
Definitely not	6 (351)	15	13	16	57

I don't know	9 (536)	5	4	27	79
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Base: respondents in the 18-25 age group (N=6,090)

Question: Q.4.7. Do you think that heated tobacco products should be regulated like cigarettes, or not? [SINGLE ANSWER]

Note: The orange column represents the share of respondents in the age group that selected the response. The white cells in the four columns on the right represent the breakdown by type of users for that response (i.e. the sum of the white cells in each row should add up to 100). The group total category refers to the total respondents in the 18-25 age group.

Source: OE elaboration from online survey data

Table 84. Extent of agreement with regulating the product like cigarettes (HTPs; 26+ age group)

(% - of respondents in the 26+ age group)

Answer	Group total	Current users	Past users	Used once or twice	Never used
Use of product	100 (5910)	11	6	10	73
Definitely yes	48 (2824)	8	4	9	79
Probably yes	31 (1836)	15	8	11	67
Probably not	9 (506)	14	10	15	60
Definitely not	5 (288)	15	7	12	65
I don't know	8 (456)	6	4	5	85

Base: respondents in the 26+ age group (N=5,910)

Question: Q.4.7. Do you think that heated tobacco products should be regulated like cigarettes, or not? [SINGLE ANSWER]

Note: The orange column represents the share of respondents in the age group that selected the response. The white cells in the four columns on the right represent the breakdown by type of users for that response (i.e. the sum of the white cells in each row should add up to 100). The group total category refers to the total respondents in the 26+ age group.

Source: OE elaboration from online survey data

10.5 Electronic cigarettes

Table 85. Perceived gateway effect of product (e-cigarettes; 18-25 age group)
(% (number) - of respondents in the 18-25 age group)

Answer	Group total %(n)	Current users %	Past users %	Used once or twice %	Never used %
Use of product	100 (6090)	15	15	25	45
Definitely yes	31 (1918)	16	16	24	44
Probably yes	40 (2427)	15	15	27	43
Probably not	16 (952)	14	17	27	42
Definitely not	6 (339)	18	16	22	44
I don't know	7 (454)	11	9	19	61

Base: respondents in the 18-25 age group (N=6,090)

Question: Q.4.9. Do you think that vaping electronic cigarettes might encourage people to start smoking/using other tobacco products, or not?

Note: The orange column represents the share of respondents in the age group that selected the response. The white cells in the four columns on the right represent the breakdown by type of users for that response in particular (i.e. the sum of the white cells in each row should add up to 100). The group total category refers to the total respondents in the 18-25 age group.

Source: OE elaboration from online survey data

Table 86. Perceived gateway effect of product (e-cigarettes; 26+ age group)
(% (number) - of respondents in the 26+ age group)

Answer	Group total % %(n)	Current users %	Past users %	Used once or twice %	Never used %
Use of product	100 (5910)	18	11	16	56
Definitely yes	25 (1503)	18	11	15	56
Probably yes	42 (2513)	18	11	17	54
Probably not	16 (922)	20	12	19	49
Definitely not	6 (330)	22	10	10	58

I don't know	11 (642)	9	8	10	73
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Base: respondents in the 26+ age group (N=5,910)

Question: Q.4.9. Do you think that vaping electronic cigarettes might encourage people to start smoking/using other tobacco products, or not?

Note: The orange column represents the share of respondents in the age group that selected the response. The white cells in the four columns on the right represent the breakdown by type of users for that response in particular (i.e. the sum of the white cells in each row should add up to 100). The group total category refers to the total respondents in the 26+ age group.

Source: OE elaboration from online survey data

Table 87. Perceived appeal of product to youth (e-cigarettes; 18-25 age group)
(% (number) - of respondents in the 18-25 age group)

Answer	Group total %(n)	Current users %	Past users %	Used once or twice %	Never used %
Use of product	100 (6090)	15	15	25	45
Definitely yes	46 (2775)	16	14	29	42
Probably yes	34 (2053)	15	15	24	46
Probably not	10 (625)	18	18	23	41
Definitely not	5 (301)	18	22	18	42
I don't know	6 (336)	14	10	15	61

Base: respondents in the 18-25 age group (N=6,090)

Question: Question: Q.4.11. Do you think that young people might be particularly attracted to electronic cigarettes (vaping), or not?

Note: The orange column represents the share of respondents in the age group that selected the response. The white cells in the four columns on the right represent the breakdown by type of users for that response in particular (i.e. the sum of the white cells in each row should add up to 100). The group total category refers to the total respondents in the 18-25 age group.

Source: OE elaboration from online survey data

Table 88. Perceived appeal of product to youth (e-cigarettes; 26+ age group)
 (% (number) - of respondents in the 26+ age group)

Answer	Group total %(n)	Current users %	Past users %	Used once or twice %	Never used %
Use of product	100 (5910)	18	11	16	56
Definitely yes	32 (1883)	20	11	16	54
Probably yes	44 (2573)	19	10	16	55
Probably not	12 (706)	16	16	17	51
Definitely not	4 (237)	14	13	11	62
I don't know	9 (511)	9	6	12	73

Base: respondents in the 26+ age group (N=5,910)

Question: Q.4.11. Do you think that young people might be particularly attracted to electronic cigarettes (vaping), or not?

Note: The orange column represents the share of respondents in the age group that selected the response. The white cells in the four columns on the right represent the breakdown by type of users for that response in particular (i.e. the sum of the white cells in each row should add up to 100). The group total column refers to the total respondents in the 26+ age group.

Source: OE elaboration from online survey data

Table 89. Extent to which respondents believed that youth underestimate risks of using e-cigarettes

(% (number) - of respondents in the 18-25 age group)

Answer	Group total %(n)	Current users %	Past users %	Used once or twice %	Never used %
Use of product	100 (6090)	15	15	25	45
Definitely yes	45 (2712)	12	13	27	48
Probably yes	31 (1882)	18	16	24	42
Probably not	13 (806)	18	21	25	36
Definitely not	6 (367)	17	14	30	38
I don't know	5 (323)	14	11	17	59

Base: respondents in the 18-25 age group (N=6,090)

Question: Q.4.12. Do you think that young people might underestimate the risk of using electronic cigarettes (vaping), or not?

Note: The orange column represents the share of respondents in the age group that selected the response. The white cells in the four columns on the right represent the breakdown by type of users for that response in particular (i.e. the sum of the white cells in each row should add up to 100). The group total category refers to the total respondents in the 18-25 age group.

Source: OE elaboration from online survey data

Table 90. Extent to which respondents believed that youth underestimated risks of using e-cigarettes

(% (number) - of respondents in the 26+ age group)

Answer	Group total %(n)	Current users %	Past users %	Used once or twice %	Never used %
Use of product	100 (5910)	18	11	16	56
Definitely yes	40 (2360)	14	9	15	62
Probably yes	37 (2179)	22	12	17	50
Probably not	11 (678)	20	14	16	49
Definitely not	5 (323)	15	10	18	56
I don't know	6	15	7	11	68

	(370)				
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Base: respondents in the 26+ age group (N=5,910)

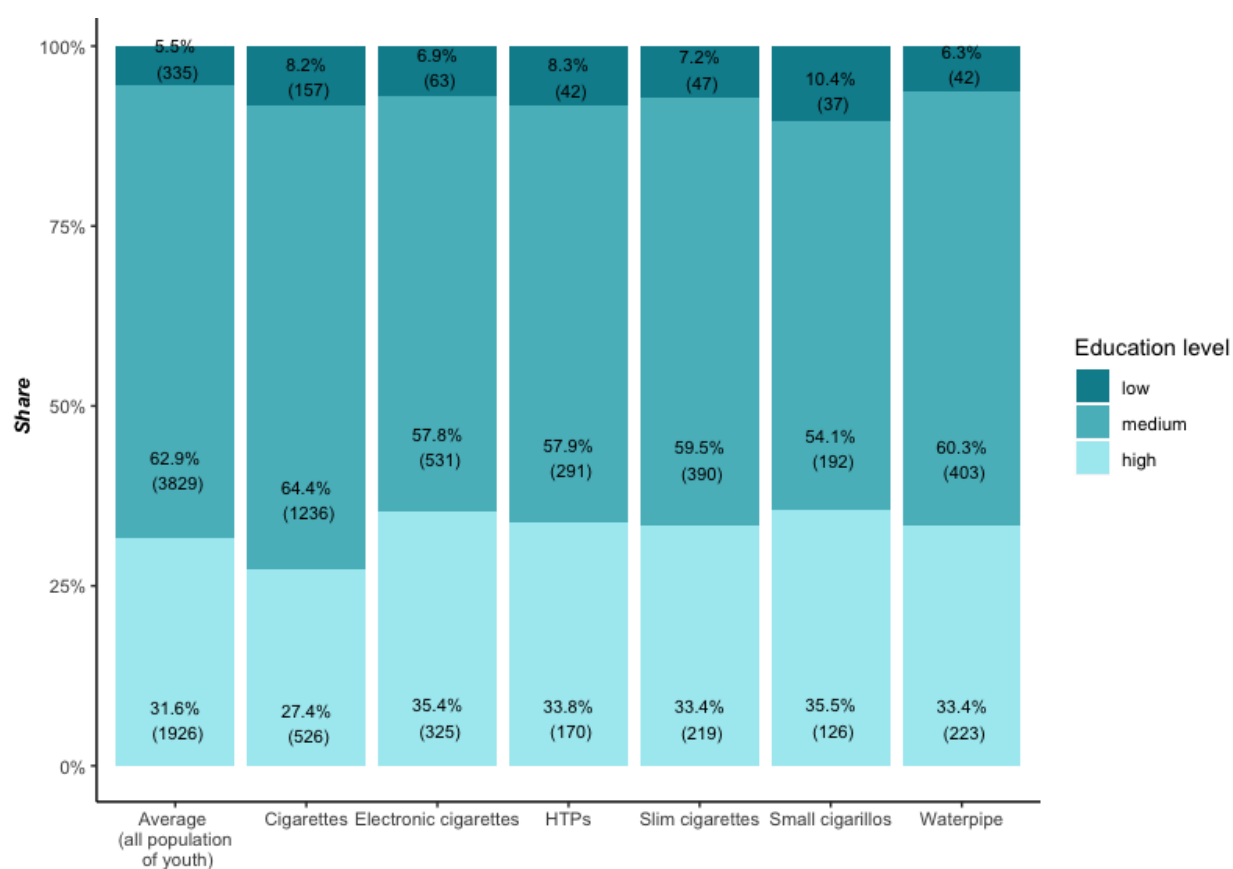
Question: Q.4.12. Do you think that young people might underestimate the risk of using electronic cigarettes (vaping), or not?

Note: The orange column represents the share of respondents in the age group that selected the response. The white cells in the four columns on the right represent the breakdown by type of users for that response in particular (i.e. the sum of the white cells in each row should add up to 100). The group total category refers to the total respondents in the 26+ age group.

Source: OE elaboration from online survey data

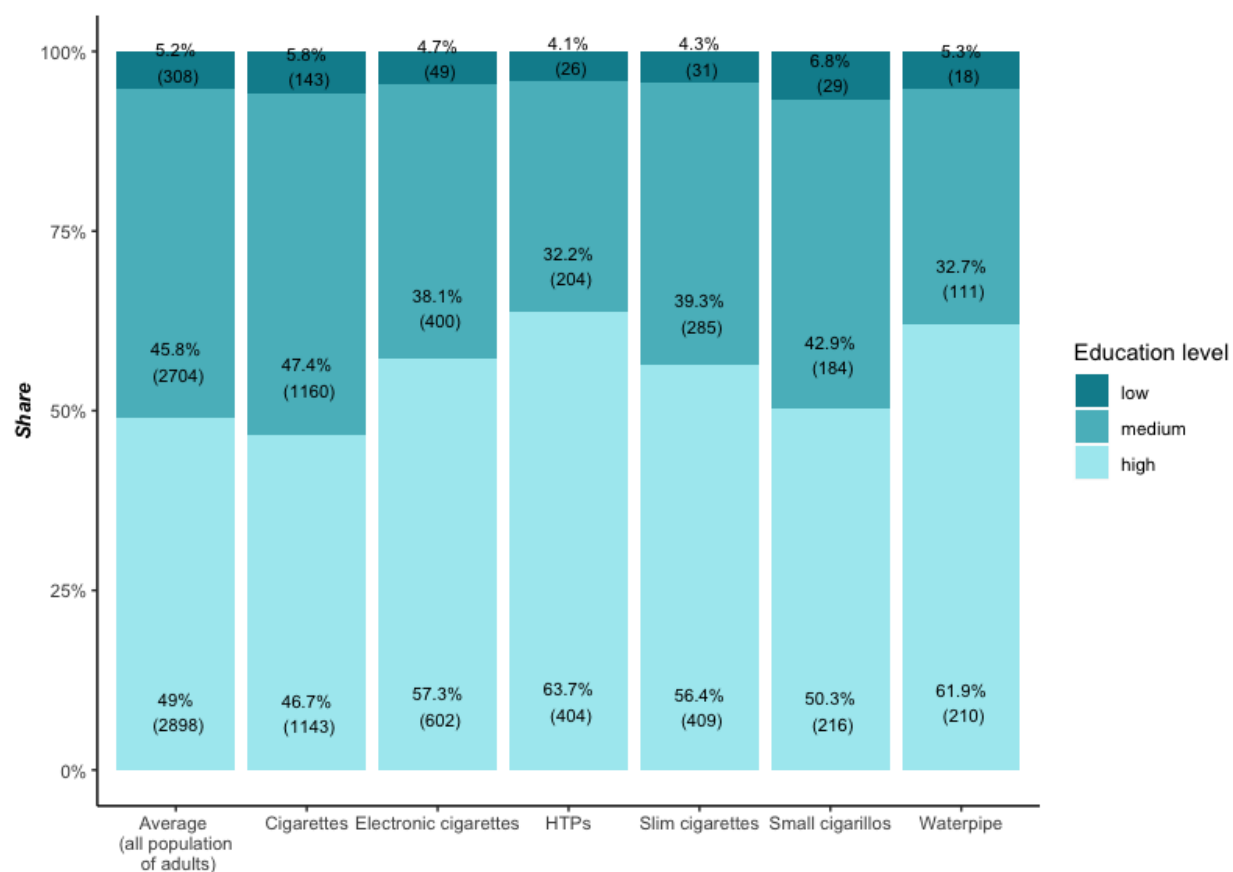
10.6 Comparative Analysis

Figure 200. Education level of product consumers by product: 18-25 age group



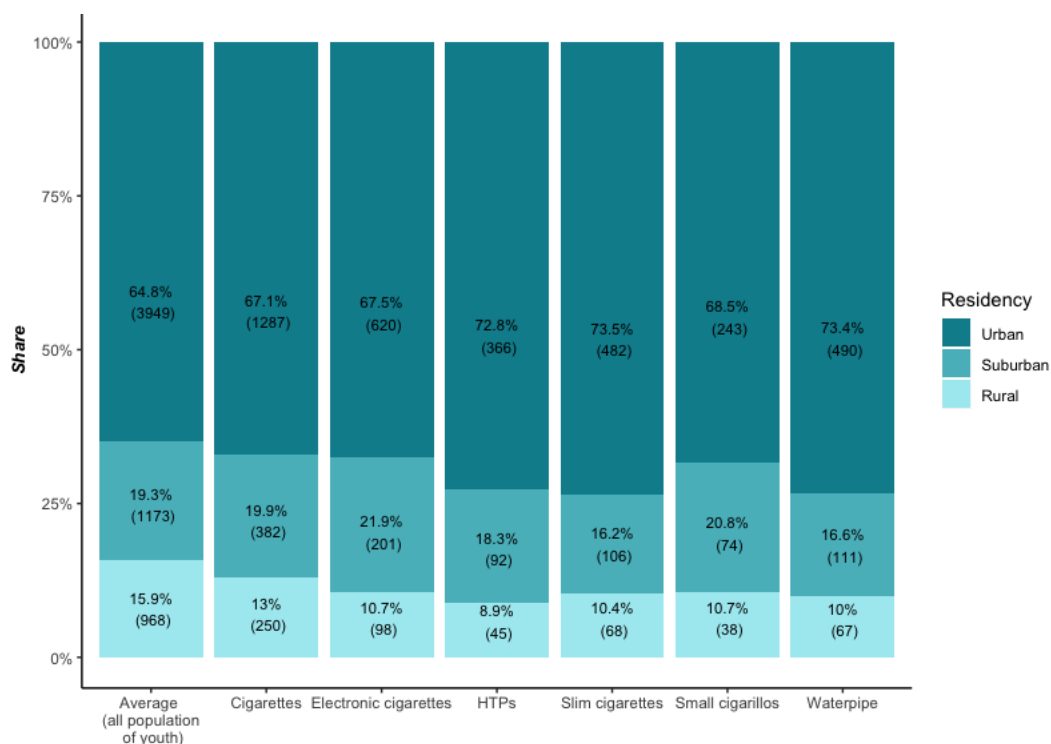
Source: OE elaboration from study data

Figure 201. Education level of product consumers by product: 26+ age group



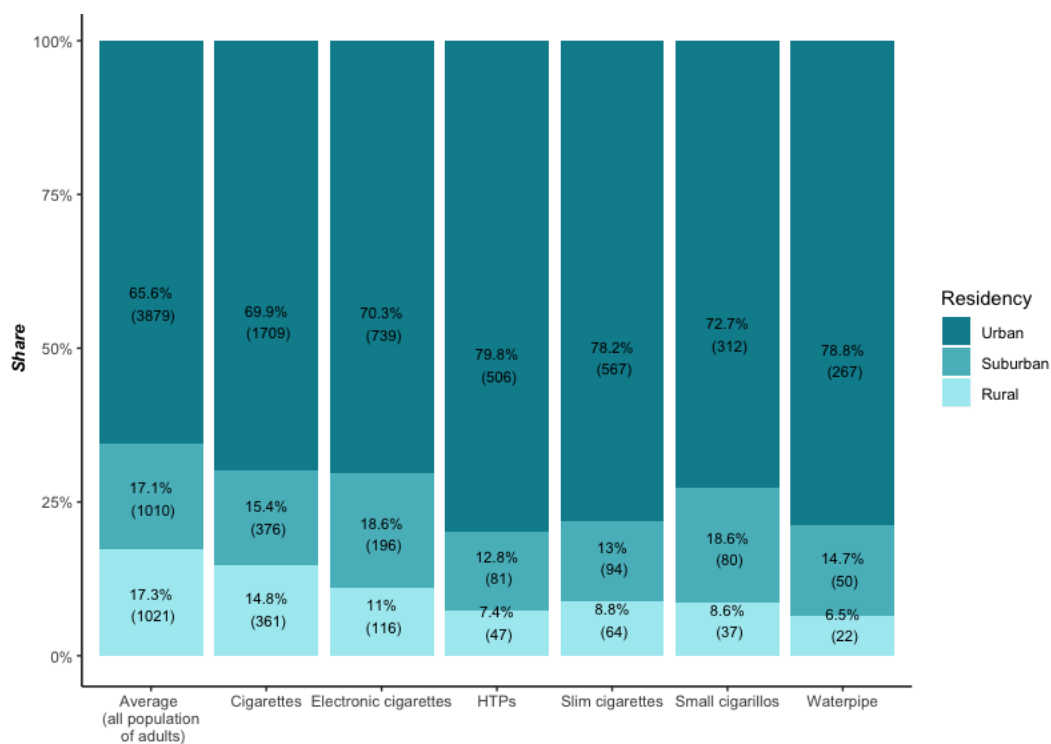
Source: OE elaboration from study data

Figure 202. Place of residency of consumers by product: 18-25 age group



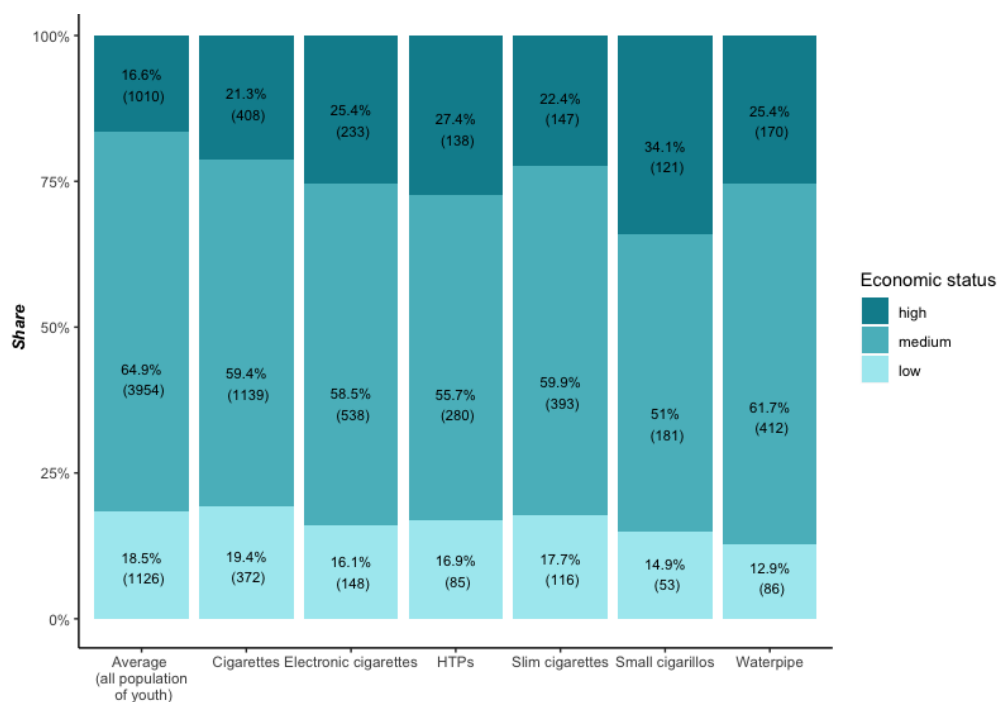
Source: OE elaboration from study data

Figure 203. Place of residency of consumers by product: 26+ age group



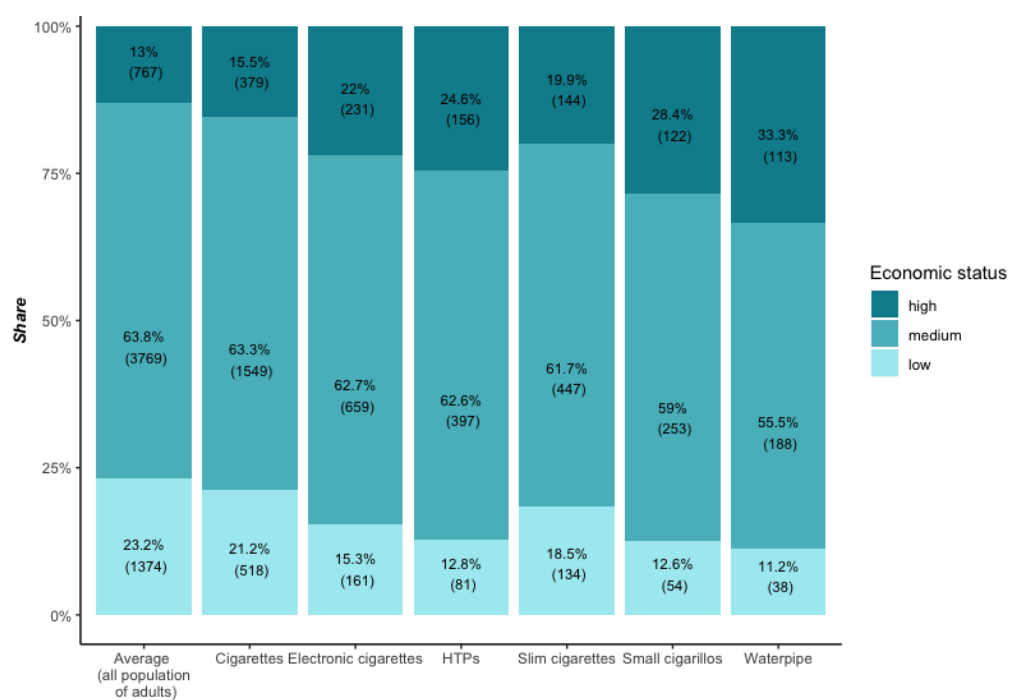
Source: OE elaboration from study data

Figure 204. Self-reported economic status by product: 18-25 age group



Source: OE elaboration from study data

Figure 205. Self-reported economic status by product: 26+ age group



Source: OE elaboration from study data

Table 91. Regulating like cigarettes v. harm to passive users

<i>Predictors</i>	Regulation for electronic cigarettes (1- Definitely not, 2- Probably not, 3- Probably yes, 4- Definitely yes)			Regulation for HTPs (1- Definitely not, 2- Probably not, 3- Probably yes, 4- Definitely yes)		
	<i>Estimates</i>	<i>CI</i>	<i>p</i>	<i>Estimates</i>	<i>CI</i>	<i>p</i>
Harmfulness for others (scores 0-100)	0.00	0.00 – 0.01	<0.001	0.00	0.00 – 0.00	<0.001
Dummy user (1-user)	-0.35	-0.40 – -0.31	<0.001	-0.17	-0.23 – 0.12	<0.001
Gender	0.12	0.09 – 0.16	<0.001	0.13	0.10 – 0.16	<0.001
Age	0.01	0.01 – 0.01	<0.001	0.01	0.00 – 0.01	<0.001
Education level	0.04	0.02 – 0.05	<0.001	0.02	0.01 – 0.03	<0.001
Residency	-0.00	-0.02 – 0.02	0.975	-0.01	-0.03 – 0.01	0.282
Employment status	0.00	-0.01 – 0.01	0.741	0.01	0.01 – 0.02	0.001
Socio-economic status	0.00	-0.01 – 0.01	0.700	-0.00	-0.01 – 0.01	0.518
(Intercept)	2.40	2.27 – 2.53	<0.001	2.18	2.01 – 2.34	<0.001
Observations	11176			11007		
R ² / R ² adjusted	0.076 / 0.075			0.046 / 0.045		

<i>Predictors</i>	Regulation for electronic cigarettes (1- Definitely not, 2- Probably not, 3- Probably yes, 4- Definitely yes)			Regulation for HTPs (1- Definitely not, 2- Probably not, 3- Probably yes, 4- Definitely yes)		
	<i>Estimates</i>	<i>CI</i>	<i>p</i>	<i>Estimates</i>	<i>CI</i>	<i>p</i>
Harmfulness for others (scores 0-100)	0.00	0.00 – 0.01	<0.001	0.00	0.00 – 0.00	<0.001
Dummy user (1-user)	-0.35	-0.40 – -0.31	<0.001	-0.17	-0.23 – 0.12	<0.001
Gender	0.12	0.09 – 0.16	<0.001	0.13	0.10 – 0.16	<0.001
Age	0.01	0.01 – 0.01	<0.001	0.01	0.00 – 0.01	<0.001
Education level	0.04	0.02 – 0.05	<0.001	0.02	0.01 – 0.03	<0.001
Residency	-0.00	-0.02 – 0.02	0.975	-0.01	-0.03 – 0.01	0.282
Employment status	0.00	-0.01 – 0.01	0.741	0.01	0.01 – 0.02	0.001
Socio-economic status	0.00	-0.01 – 0.01	0.700	-0.00	-0.01 – 0.01	0.518
(Intercept)	2.40	2.27 – 2.53	<0.001	2.18	2.01 – 2.34	<0.001
Observations	11176			11007		
R ² / R ² adjusted	0.076 / 0.075			0.046 / 0.045		

Base: all respondents in the online survey (N=12,000)

Note: *** refers to the 0.01 level, ** refers to the 0.05 level & * refers to the 0.1 level, the latter not being considered as significant for the purpose of this study.

Source: OE elaboration from online survey data

11 List of figures

Figure 1. Scope of the study	18
Figure 2. Overall approach.....	19
Figure 3. Market size of waterpipe & pipe tobacco products in million EUR in 2017.....	31
Figure 4. Price of waterpipe and pipe tobacco products across EU in 2017, euros per kg	32
Figure 5. Market size trends of waterpipe & pipe tobacco products: 2012-2017	33
Figure 6. Use of waterpipe: country breakdown (18-25 age group).....	35
Figure 7. Use of waterpipe: country breakdown (26+ age group)	36
Figure 8. Frequency of use of current waterpipe users (18-25 age group).....	39
Figure 9. Frequency of use of current waterpipe users (26+ age group)	39
Figure 10. Frequency of use of past waterpipe users (18-25 age group)	40
Figure 11. Frequency of use of past waterpipe users (26+ age group).....	40
Figure 12. Average number of minutes per day of waterpipe use (18-25 age group)	41
Figure 13. Average number of minutes per day of waterpipe use (26+ age group)	41
Figure 14. Average number of minutes per day for daily waterpipe users (18-25 age group). ..	42
Figure 15. Average number of minutes per day, for daily waterpipe users (26+ age group) ..	42
Figure 16. Initiation of use for current and past users (waterpipe; total sample)	43
Figure 17. Cessation: attempts to quit waterpipe (18-25 age group).....	45
Figure 18. Cessation: attempts to quit waterpipe (26+ age group)	45
Figure 19. Cessation: reasons for trying to quit waterpipe use (18-25 age group).....	46
Figure 20. Cessation: reasons for trying to quit waterpipe use (26+ age group)	46
Figure 21. Reasons for starting to use waterpipe again (18-25 age group).....	47
Figure 22. Reasons for starting to use waterpipe again (26+ age group)	47
Figure 23. Perceived appeal of waterpipe to youth (waterpipe; 18-25 age group)	54
Figure 24. Perceived appeal of waterpipe to youth (waterpipe; 26+ age group).....	54
Figure 25. Extent to which respondents believed that youth underestimate risks of using waterpipe tobacco (18-25 age group)	55
Figure 26. Extent to which respondents believed that youth underestimate risks of using waterpipe tobacco (26+ age group).....	56
Figure 27. Perceived harmfulness of waterpipe tobacco to users (18-25 age group).....	57
Figure 28. Perceived harmfulness of waterpipe to users (26+ age group)	58
Figure 29. Perceived harmfulness of waterpipe to bystanders (18-25 age group).....	59
Figure 30. Perceived harmfulness of waterpipe to bystanders (26+group)	60
Figure 31. Market size of slim cigarettes in million units in 2017	65
Figure 32. Share of slim cigarettes within the national total tobacco market, per country (2017)	66

Figure 33. Market size growth of slim cigarettes, in retail volume in 2017 as a baseline of 2012	66
Figure 34. Share of slim cigarettes in total market of cigarettes, in retail volume: 2012-2017	67
Figure 35. Use of slim cigarettes: country breakdown (18-25 age group)	70
Figure 36. Use of slim cigarettes: country breakdown (26+ age group)	71
Figure 37. Frequency of use of current slim cigarette users (18-25 age group).....	74
Figure 38. Frequency of use of current slim cigarette users (26+ age group)	74
Figure 39. Frequency of use of past slim cigarette users (18-25 age group).....	75
Figure 40. Frequency of use of past slim cigarette users (26+ age group)	75
Figure 41. Average number of slim cigarettes per day of use (18-25 age group)	76
Figure 42. Average number of slim cigarettes per day of use (26+ age group)	76
Figure 43. Initiation of use for current and past users (slim cigarettes; total sample)	77
Figure 44. Duration of use of past slim cigarette users (18-25 age group)	78
Figure 45. Duration of use of past slim cigarette users (26+ age group)	78
Figure 46. Reasons for initiation of slim cigarette use (18-25 age group)	79
Figure 47. Reasons for initiation of slim cigarette use (26+ age group)	79
Figure 48. Initiation of slim cigarette use: reasons for trying (18-25 age group)	80
Figure 49. Initiation of slim cigarette use: reasons for trying (26+ age group)	80
Figure 50. Reasons for trying to quit slim cigarette use (18-25 age group)	81
Figure 51. Reasons for trying to quit slim cigarette use (26+ age group)	81
Figure 52. Reasons for starting to use slim cigarettes again (18-25 age group)	82
Figure 53. Reasons for starting to use slim cigarettes again (26+ age group)	82
Figure 54. Perceived harmfulness of slim cigarettes to users (18-25 age group)	89
Figure 55. Perceived harmfulness of slim cigarettes to users (26+ age group)	90
Figure 56. Perceived harmfulness of slim cigarettes to bystanders (18-25 age group)	91
Figure 57. Perceived harmfulness of slim cigarettes to bystanders (26+ group)	92
Figure 58. Market size of small cigarillos in million euro in 2017	97
Figure 59. Share of small cigarillos within the national total tobacco market, per country (2017)	98
Figure 60. Price of small cigarillos in 2017, euros per unit (stick)	98
Figure 61. Market size trends of small cigarillos over 2012-2017	99
Figure 62. Use of small cigarillos: country breakdown (18-25 age group)	100
Figure 63. Use of small cigarillos: country breakdown (26+ age group)	101
Figure 64. Frequency of use of current small cigarillo users (18-25 age group)	104
Figure 65. Frequency of use of current small cigarillo users (26+ age group)	104
Figure 66. Frequency of use of past small cigarillo users (18-25 age group)	105

Figure 67 Frequency of use of past small cigarillo users (26+ age group)	105
Figure 68. Average number of small cigarillo sticks per day of use (18-25 age group)	106
Figure 69. Average number of small cigarillo sticks per day of use (26+ age group)	106
Figure 70. Average number of small cigarillo sticks per day, for daily users (18-25 age group)	107
Figure 71. Average number of small cigarillo sticks per day, for daily users (26+ age group)	107
Figure 72. Initiation of use for current and past small cigarillo users (total sample)	108
Figure 73. Duration of use of past small cigarillo users (18-25 age group)	109
Figure 74. Duration of use of past small cigarillo users (26+ age group).....	109
Figure 75. Initiation of small cigarillo use: reasons (18-25 age group).....	110
Figure 76. Initiation of small cigarillo use: reasons (26+ age group)	110
Figure 77. Initiation of small cigarillo use: reasons for trying small cigarillos (18-25 age group)	111
Figure 78. Initiation of small cigarillo use: reasons for trying small cigarillos (26+ age group)	111
Figure 79. Cessation: attempts to quit small cigarillos (18-25 age group)	112
Figure 80. Cessation: attempts to quit small cigarillos (26+ age group)	112
Figure 81. Cessation: reasons for trying to quit small cigarillos (18-25 age group).....	113
Figure 82. Cessation: reasons for trying to quit small cigarillos (26+ age group)	113
Figure 83. Reasons for starting to use small cigarillos again (18-25 age group).....	114
Figure 84. Reasons for starting to use small cigarillos again (26+ age group)	114
Figure 85. Representation of career flows for current small cigarillo users (18-25 age group)	117
Figure 86. Representation of career flows for current small cigarillo users (26+ age group) .	117
Figure 87. Representation of career flows for past small cigarillo users (18-25 age group)...	120
Figure 88. Representation of career flows for past small cigarillo users (26+ age group)	120
Figure 89. Perceived harmfulness of small cigarillos to users (18-25 age group).....	124
Figure 90. Perceived harmfulness of small cigarillos to users (26+ age group)	125
Figure 91. Perceived harmfulness of small cigarillos to bystanders (18-25 age group).....	126
Figure 92. Perceived harmfulness of small cigarillos to bystanders (26+ age group)	127
Figure 93. Market size of heated tobacco products (HTPs) in million euros in 2017	131
Figure 94. Share of HTPs within the national total tobacco market, per country (2017)	132
Figure 95. Use of HTPs: country breakdown (18-25 age group).....	135
Figure 96. Use of HTPs: country breakdown (26+ age group)	136
Figure 97. Frequency of use of current HTP users (18-25 age group)	139

Figure 98. Frequency of use of current HTP users (26+ age group).....	139
Figure 99. Frequency of use of past HTP users (18-25 age group)	140
Figure 100. Frequency of use of past HTP users (26+ age group)	140
Figure 101. Average number of HTP sticks a day for current users (18-25 age group).....	141
Figure 102. Average number of HTP sticks a day for current users (26+ age group)	141
Figure 103. Average number of HTP sticks a day, for daily users (18-25 age group)	142
Figure 104. Average number of HTP sticks a day, for daily users (26+ age group).....	142
Figure 105. Initiation of HTP use: reasons (18-25 age group)	143
Figure 106. Initiation of HTP use: reasons (26+ age group)	143
Figure 107. Initiation of HTP use: reasons for trying (18-25 age group)	144
Figure 108. Initiation of HTP use: reasons for trying (26+ age group)	145
Figure 109. Cessation: attempts to quit HTPs (18-25 age group).....	146
Figure 110. Cessation: attempts to quit HTPs (26+ age group)	146
Figure 111. Cessation: reasons for trying to quit HTP use (18-25 age group)	147
Figure 112. Cessation: reasons for trying to quit HTP use (26+ age group).....	147
Figure 113. Reasons for starting to use HTPs again (18-25 age group).....	148
Figure 114. Reasons for starting to use HTPs again (26+ age group)	148
Figure 115. Perceived gateway effect of HTPs (18-25 age group)	158
Figure 116 Perceived gateway effect of HTPs (26+ age group)	158
Figure 117 Perceived appeal of HTPs to youth (18-25 age group)	159
Figure 118 Perceived appeal of HTPs to youth (26+ age group).....	159
Figure 119 Extent to which respondents believed that youth underestimated risks of using HTPs	160
Figure 120 Extent to which respondents believed that youth underestimate risks of HTPs....	161
Figure 121. Perceived harmfulness of HTPs to users (18-25 age group).....	163
Figure 122. Perceived harmfulness of HTPs to users (26+ age group)	164
Figure 123. Perceived harmfulness of HTPs to bystanders (18-25 age group).....	165
Figure 124. Perceived harmfulness of HTPs to bystanders (26+group)	166
Figure 125. Extent of agreement with regulating HTPs like cigarettes (18-25 age group).....	167
Figure 126. Extent of agreement with regulating HTPs like cigarettes (26+ age group)	167
Figure 127. Market size of electronic cigarettes in million euros in 2017	174
Figure 128. Share of electronic cigarettes within the national total tobacco market, per country (2017)	175
Figure 129 Price dynamics of electronic cigarettes (in retail selling prices, RSP): 2012-2017	176
Figure 130. Use of e-cigarettes: country breakdown (18-25 age group)	184
Figure 131. Use of e-cigarettes: country breakdown (26+ age group).....	185

Figure 132. Frequency of use of current e-cigarette users (18-25 age group).....	188
Figure 133. Frequency of use of current e-cigarette users (26+ age group)	188
Figure 134. Frequency of use of past e-cigarette users (18-25 age group).....	189
Figure 135. Frequency of use of past e-cigarette users (26+ age group)	189
Figure 136. Average number of puffs per session (18-25 age group).....	190
Figure 137. Average number of puffs per session (26+ age group)	190
Figure 138. Average number of sessions per day of use (18-25 age group)	191
Figure 139. Average number of sessions per day (26+ age group)	191
Figure 140. Number of sessions per day, for daily users (18-25 age group)	192
Figure 141. Number of sessions per day, for daily users (26+ age group).....	192
Figure 142. Initiation of use for current and past e-cigarette users (total sample).....	193
Figure 143. Duration of use of past e-cigarette users (18-25 age group)	194
Figure 144. Duration of use of past e-cigarette users (26+ age group).....	194
Figure 145. Initiation of e-cigarette use: reasons (18-25 age group).....	195
Figure 146. Initiation of e-cigarette use: reasons (26+ age group)	195
Figure 147. Initiation of e-cigarette use: reasons (18-25 age group).....	196
Figure 148. Initiation of e-cigarette use: reasons (26+ age group)	197
Figure 149. Cessation: attempts to quit e-cigarettes (18-25 age group)	197
Figure 150. Cessation: attempts to quit e-cigarettes (26+ age group).....	198
Figure 151. Cessation: reasons for trying to quit e-cigarette use (18-25 age group)	198
Figure 152. Cessation: reasons for trying to quit e-cigarette use (26+ age group).....	199
Figure 153. Reasons for starting to use e-cigarettes again (18-25 age group)	199
Figure 154. Reasons for starting to use e-cigarettes again (26+ age group)	200
Figure 155. Representation of career flows for current e-cigarette users (18-25 age group).....	203
Figure 156. Representation of career flows for current e-cigarette users (26+ age group) ...	203
Figure 157. Representation of career flows for past e-cigarette users (18-25 age group)	206
Figure 158. Representation of career flows for past e-cigarette users (26+ age group)	206
Figure 159. Perceived gateway effect of e-cigarettes (18-25 age group).....	209
Figure 160. Perceived gateway effect of e-cigarettes (26+ age group)	210
Figure 161. Perceived appeal of e-cigarettes to youth (18-25 age group)	210
Figure 162. Perceived appeal of e-cigarettes to youth (26+ age group).....	211
Figure 163 Extent to which respondents believed that youth underestimate risks of using e-cigarettes	211
Figure 164 Extent to which respondents believed that youth underestimated risks of using e-cigarettes	212
Figure 165. Perceived harmfulness of e-cigarettes to users (18-25 age group).....	213

Figure 166. Perceived harmfulness of e-cigarettes to users (26+ age group)	214
Figure 167. Perceived harmfulness of e-cigarettes to bystanders(18-25 age group)	215
Figure 168. Perceived harmfulness of e-cigarettes to bystanders(26+group).....	216
Figure 169 Share of smokers by countries	224
Figure 170. Overview of the consumption of the five products (18-25 age group).....	226
Figure 171. Overview of the consumption of the five products (26+ age group)	227
Figure 172. Gender of consumers by product: 18+25 age group	228
Figure 173. Gender of consumers by product: 26+ age group	228
Figure 174. VAT values on tobacco products by countries in 2017.....	233
Figure 175 Taxes on tobacco products by countries in 2017	235
Figure 176. Duration of use of past users (waterpipe; 18-25 age group)	257
Figure 177. Duration of use of past users (waterpipe; 26+ age group)	257
Figure 178. Initiation of waterpipe use: reasons (18-25 age group)	258
Figure 179. Initiation of waterpipe use: reasons (26+ age group).....	259
Figure 180. Initiation of waterpipe use: reasons for trying (waterpipe; 18-25 age group)	260
Figure 181. Initiation of waterpipe use: reasons for trying (waterpipe; 26+ age group).....	261
Figure 182. Representation of career flows for current user (waterpipe tobacco; 18-25 age group)	262
Figure 183. Representation of career flows for current user (waterpipe tobacco; 26+ age group)	263
Figure 184 Representation of career flows for past user (waterpipe tobacco; 18-25 age group)	264
Figure 185 Representation of career flows for past user (waterpipe tobacco; 26+ age group)	265
Figure 186. Average number of slim cigarette sticks per day of use, daily users (18-25 age group)	268
Figure 187. Average number of slim cigarettes sticks per day, for daily users (26+ age group)	269
Figure 188. Cessation: slim cigarettes quit attempts (slim cigarettes; 18-25 age group)	270
Figure 189. Cessation: slim cigarettes quit attempts (slim cigarettes; 26+ age group)	270
Figure 190. Representation of career flows for current users (slim cigarettes; 18-25 age group)	271
Figure 191. Representation of career flows for current users (slim cigarettes; 26+ age group)	272
Figure 192. Representation of career flows for past users (slim cigarettes; 18-25 age group)	273
Figure 193. Representation of career flows for past users (slim cigarettes; 26+ age group)	274
Figure 194. Duration of use of past users (HTPs; 18-25 age group)	275

Figure 195. Duration of use of past users (HTPs; 26+ age group)	275
Figure 196. Representation of career flows for current users (HTPs; 18-25 age group)	276
Figure 197. Representation of career flows for current users (HTPs; 26+ age group)	277
Figure 198. Representation of career flows for past users (HTPs; 18-25 age group)	278
Figure 199. Representation of career flows for past users (HTPs; 26+ age group)	279
Figure 200. Education level of product consumers by product: 18-25 age group	289
Figure 201. Education level of product consumers by product: 26+ age group	290
Figure 202. Place of residency of consumers by product: 18-25 age group	291
Figure 203. Place of residency of consumers by product: 26+ age group	291
Figure 204. Self-reported economic status by product: 18-25 age group	292
Figure 205. Self-reported economic status by product: 26+ age group	292

12 List of tables

Table 1. Current Consumption by product and by Member State.....	16
Table 2. Mixed methods approach	20
Table 3. Literature review information found by product	21
Table 4. Results by tobacco product category	22
Table 5. Quantitative information found by product.....	22
Table 6. Summary of the main information gaps	23
Table 7. Focus groups geographical scope	24
Table 8. Data gaps addressed by the focus group	24
Table 9. Online survey: Geographic scope and quota breakdown by country (N=12,000)	25
Table 10. Data gaps addressed by the Online survey	26
Table 11. Mapping of survey questionnaire by research question	27
Table 12. Socio-demographic breakdown of online survey (18-25 age group)	28
Table 13. Socio-demographic breakdown of online survey (26+ age group)	29
Table 14. Waterpipe use socio-demographic breakdown (18-25 age group)	37
Table 15. Waterpipe use socio-demographic breakdown (26+ age group).....	38
Table 16. Reasons for starting to use or trying waterpipe (18-25 age group)	44
Table 17. Reasons for starting to use or trying waterpipe (26+ age group)	44
Table 18. Career flows of use for current waterpipe users (18-25 age group)	48
Table 19. Career flows of current waterpipe users (26+ age group)	49
Table 20. Career flow of past waterpipe users (18-25 age group)	50
Table 21. Career flows of past waterpipe users (26+ age group)	51
Table 22. Perception of waterpipe tobacco (18-25 age group)	52
Table 23. Perception of waterpipe tobacco; 26+ age group).....	53
Table 24. "Which of the following statements, if any, do you think apply to slim cigarettes?" (2017)	69
Table 25. Slim cigarette use socio-demographic breakdown (18-25 age group)	72
Table 26. Slim cigarette use socio-demographic breakdown (26+ age group)	73
Table 27. Career flows of slim cigarette use for current users (18-25 age group).....	83
Table 28. Career flows of current slim cigarette users (26+ age group)	84
Table 29 Career flow of past slim cigarette users (18-25 age group)	85
Table 30 Career flows of past slim cigarette users (26+ age group)	86
Table 31. Tax provisions applicable to manufactured tobacco	94
Table 32. Small cigarillo use socio-demographic breakdown (18-25 age group)	102
Table 33. Small cigarillo use socio-demographic breakdown (26+ age group)	103

Table 34. Career flows of small cigarillo use for current users (18-25 age group).....	115
Table 35. Career flows of current small cigarillo users for current users (26+ age group)	116
Table 36. Career flow of past small cigarillo users (18-25 age group).....	118
Table 37. Career flows of past small cigarillo users (26+ age group).....	119
Table 38. HTP use socio-demographic breakdown (18-25 age group)	137
Table 39. HTP use socio-demographic breakdown (26+ age group)	138
Table 40. Career flows of use for current HTP users (18-25 age group)	149
Table 41. Career flows of current HTP users (26+ age group)	150
Table 42. Career flow of past HTP users (18-25 age group)	151
Table 43 Career flows of past HTP users (26+ age group)	152
Table 44. Perception of HTPs (18-25 age group).....	153
Table 45 Perception of HTPs (26+ age group)	154
Table 46. Perception of the HTPs (18-25 age group)	155
Table 47 Perception of HTPs (26+ age group)	156
Table 48. Views on settings where HTPs should be prohibited (18-25 age group)	168
Table 49. Views on settings where HTPs should be prohibited (26+ age group)	169
Table 50. Overview of non-harmonised taxes for heated tobacco products.....	172
Table 51. Hypotheses proposed in the Schneider & Diehl (2016) model.....	177
Table 52. Reasons for initiating use e-cigarettes.....	180
Table 53. Perception of harmfulness e-cigarettes.....	181
Table 54. E-cigarette use socio-demographic breakdown (18-25 age group)	186
Table 55. E-cigarette use socio-demographic breakdown (26+ age group).....	187
Table 56. Career flows of use for current e-cigarette users (18-25 age group)	201
Table 57. Career flows of current e-cigarette users (26+ age group).....	202
Table 58. Career flow of past e-cigarette users (18-25 age group).....	204
Table 59. Career flows of past e-cigarette users (26+ age group).....	205
Table 60. Perception of e-cigarettes (18-25 age group).....	207
Table 61 Perception of e-cigarettes (26+ age group)	208
Table 62 Views on settings where e-cigarettes should be prohibited (18-25 age group).....	217
Table 63 Views on settings where e-cigarettes should be prohibited (26+ age group)	218
Table 64. Market size share of selected tobacco and related products by countries	225
Table 65. Career flows of use for current users of HTPs and e-cigarettes (18-25 & 26+ age groups)	229
Table 66. History of current exclusive users of HTPs and e-cigarettes (18-25 & 26+ age groups)	229
Table 67. Influence of channels of information on use	231

Table 68. Relative risk perception of the products.....	232
Table 69. Tax provisions applicable to manufactured tobacco	234
Table 70. Overview of non-harmonised taxes for e-cigarettes and heated tobacco products	236
Table 71. online survey sample description (18-25 age group)	251
Table 72. online survey sample description (26+ age group)	254
Table 73. Perceived appeal of product to youth (waterpipe; 18-25 age group)	266
Table 74. Perceived appeal of product to youth (waterpipe; 26+ age group)	266
Table 75. Extent to which respondents believed that youth underestimate risks of using waterpipe tobacco (18-25 age group)	267
Table 76. Extent to which respondents believed that youth underestimate risks of using waterpipe tobacco (26+ age group)	267
Table 77. Perceived gateway effect of product (HTPs; 18-25 age group).....	280
Table 78. Perceived gateway effect of product (HTPs; 26+ age group)	281
Table 79. Perceived appeal of product to youth (HTPs; 18-25 age group)	281
Table 80. Perceived appeal of product to youth (HTPs; 26+ age group).....	282
Table 81. Extent to which respondents believed that youth underestimated risks of using HTPs	282
Table 82. Extent to which respondents believed that youth underestimate risks of HTPs.....	283
Table 83. Extent of agreement with regulating the product like cigarettes (HTPs; 18-25 age group)	283
Table 84. Extent of agreement with regulating the product like cigarettes (HTPs; 26+ age group)	284
Table 85. Perceived gateway effect of product (e-cigarettes; 18-25 age group).....	285
Table 86. Perceived gateway effect of product (e-cigarettes; 26+ age group)	285
Table 87. Perceived appeal of product to youth (e-cigarettes; 18-25 age group)	286
Table 88. Perceived appeal of product to youth (e-cigarettes; 26+ age group)	287
Table 89. Extent to which respondents believed that youth underestimate risks of using e-cigarettes	288
Table 90. Extent to which respondents believed that youth underestimated risks of using e-cigarettes	288
Table 91. Regulating like cigarettes v. harm to passive users	293

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